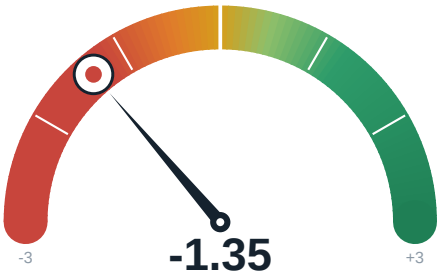
SHOTGUN METAGENOMICS · STOOL DNA · 8.0 MILLION READ PAIRS, EVERY ONE ANALYSED

Gut Health Test Metagenomic Report

What your gut bacteria can produce, who is living there, how healthy the community looks, and which published disease patterns it resembles — read straight from the DNA in your sample.



BROADLY DISTURBED

16 healthy · 15 disease markers

GMWI2 gut-health index

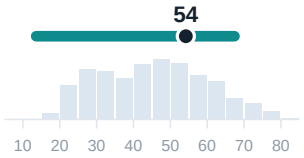
WHAT STOOD OUT

- **General gut health** — broadly disturbed (index -1.35).
- **Overgrown** — *Mediterraneibacter gnavus* (6.9%, ×110 vs typical carrier), *Anaerotignum lactatifermentans* (0.78%, ×17 vs typical carrier) and 3 more at a level of concern. *Mediterraneibacter gnavus* is an opportunist: normal at low levels, associated with inflammation when it expands.
- **Missing** — 6 species most reference adults carry were not detected: *Anaerostipes hadrus*, *Dorea longicatena*, *Eubacterium eligens* and 3 more.
- **Depleted** — *Faecalibacterium prausnitzii*, *Agathobaculum butyriciproducens* and 2 more well below the reference carriers.
- **Secondary bile acids** — notably high, 95th percentile.
- **Butyrate** — notably low, 5th percentile.
- **Folate** — notably high in the favourable direction, 95th percentile.
- **Parkinson's disease** — notably high resemblance, >99th percentile.

Biological Age Estimate

section 8

54
ESTIMATED AGE (YRS)
90% interval 14–67



Biofilm Score

section 11

Biofilm-forming (bad)		60/100
Protective gut-lining (good)		55/100

Mycobiome

section 12

Myco-Score (experimental)		75/100

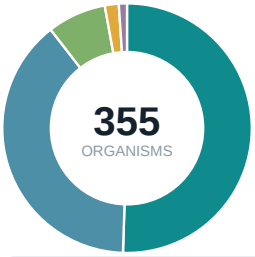
Skin

section 13

Gut-skin axis (beta)		50/100

Microbiome Diversity

sections 3 to 5, 15, 16



- Firmicutes 51%
- Bacteroidota 39%
- Actinobacteria 8%
- Proteobacteria 2%
- Verrucomicrobia 1%

3.17
DIVERSITY
Shannon index · 73rd pct

0.70
EVENNESS
0 to 1 · 72nd pct

6
EXPECTED SPECIES MISSING
in most reference adults, absent here

2 / 6
DEPLETED / EXPANDED
2 need qualification

355
ALL ORGANISMS DETECTED
198 genera, every catalogue pooled

54
RESOLVED TO STRAIN
read at single-base resolution

Disease Signatures

sections 10, 20

Parkinson's disease	>99th
Type 2 diabetes	>99th
Ulcerative colitis	>99th
IBD (composite)	>99th
Long COVID	>99th
Ankylosing spondylitis	>99th
Preclinical amyloid	>99th

Top 7 of 33 scored (33 in the library). Resemblance is not disease — section 10

Microbial Functions

sections 9, 19

Secondary bile acids ▲	95th
Butyrate ▼	5th
Propionate ▲	92nd
BCAAs ▲	91st
Glucuronidase activity ▲	85th
TMAO ▲	76th
Fermentation routes	95th

▲ worse ▼ better · **38 more** (27 notable) in section 9

1 disease-causing organism found

Clostridioides difficile detected

3 traces, unconfirmed | 8 opportunists

Section 7

Possible solutions

211 studied options for the **54** findings in this report

Section 14 →

Table of contents

This report is interactive. Click a section for results at a glance, then a reading for its detailed explanation. Use ‘Back to overview’ to return.

1 Your results at a glance
every score on one page

2 What stood out
the findings worth your attention first

3 Your gut community
diversity, composition and community type

4 Your microbial groups
curated sets of species, each summed to one reading

5 Organisms that need attention
the ones flagged, one by one

6 Every organism found
the full inventory, searchable

7 Pathogen screening
what was searched for, and what was found

8 Estimated biological age of biota
how your community compares by age

9 Your Microbial Functions
what your microbes can make, break down and transform

10 Resemblance to published disease patterns
how your community compares with published patterns

11 Biofilm-related potential
organisms that build and degrade the mucus layer

12 Your gut mycobiome
the fungal side of the community

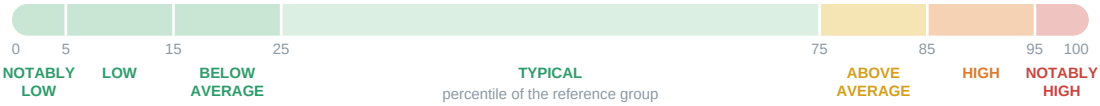
13 Gut-skin axis
one score for how your gut leans across published skin panels

14 What you can do about it
options with the evidence behind each one

24 Your information & report context
what was supplied, and what was assumed

How to read this report

One scale, everywhere. Every reading — organism, group, metabolic function, disease pattern — is a **percentile against a reference group**, translated into the same seven words. The 60th percentile means 60% of the reference had a lower value than you; the reference’s middle half (25th to 75th) is **typical**.

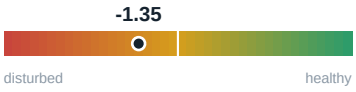


Colour is the judgement; the word is the position. Where research links higher with worse outcomes (▲) the high end is red; where higher is favourable (▼) the bar is mirrored and high is green; where the evidence is mixed the bar is grey and no verdict is given. Disease-pattern resemblance always reads higher as less favourable. Everything here is *capacity* or *composition* read from DNA — never a measured amount of a compound in your body, and never a diagnosis.

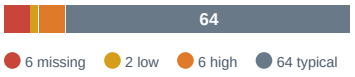
PART A — AT A GLANCE

2 What stood out

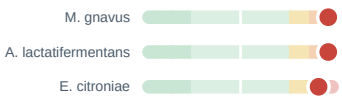
The findings that matter most, each with the graphic it comes from. Everything here is expanded in the sections named at the end of each line.



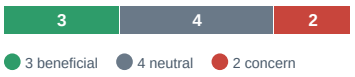
General gut health: -1.35, broadly disturbed. The most important number here. A community in this state resembles several disease patterns at once, so read those as reflecting the broad disturbance rather than any one condition. Section 3.



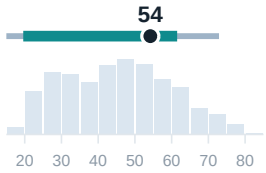
6 expected species missing (*Anaerostipes hadrus*, *Dorea longicatena*, *Eubacterium eligens*, +3); **2 depleted** (*Agathobaculum butyriciproducens*, *Faecalibacterium prausnitzii*); **6 expanded** (*Anaerotignum lactatifermentans*, *Barnesiella intestinihominis*); 64 typical. A further **24** expected species were not placed against the reference at all (*Bacillus coagulans*, *Bacillus subtilis group*, *Blautia wexlerae*, +21), so nothing here says whether they are present or absent. Sections 5 and 16.



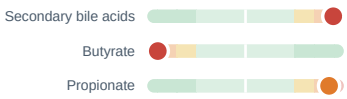
Overgrown: *Mediterraneibacter gnavus* at 6.9% of your community, ×110 against the typical carrier, 99th percentile among reference adults who carry it — opportunist; mucin-degrading Lachnospiraceae member present in most healthy adults at low level. Also above range: *Anaerotignum lactatifermentans*, *Enterocloster citroniae* and 5 more. Section 5 lists every overgrown organism; section 16 has a page on each.



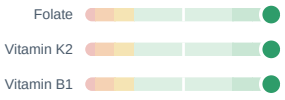
7 uncommon in the reference, 3 of them health-associated; **2 need qualification** (*Clostridium innocuum*, *Enterococcus faecium*) — species DNA is not a strain, toxin or infection. Sections 5 and 16.



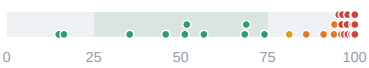
Your gut community reads like an adult in their fifties — 54 years. Built by an ensemble of a transformer neural network and a linear carriage model reading species composition, then moved by 21 disease patterns scoring high, 10 patterns in the typical range or better, a disturbed general gut-health index (-1.35). Communities carrying fewer disease-associated patterns and a healthier index read younger; this is an estimate of the community, not of you. Section 8.



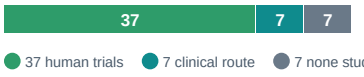
6 of 45 metabolic functions sit outside the typical band in the less favourable direction: **Secondary bile acids** (95th), **Butyrate (SCFA)** (5th), **Propionate (SCFA)** (92nd), +3 more. Sections 9 and 19.



Favourably placed: Folate (vitamin B9) (95th), **Vitamin K2 (menaquinone, MK-n)** (95th), **Vitamin B1 (thiamine)** (95th). Section 9.



23 of 33 disease patterns above typical, 18 notably high. That many rising together reads as **general disturbance**, not a specific condition: the patterns share features and one disturbed community lights them all. Sections 10 and 20.



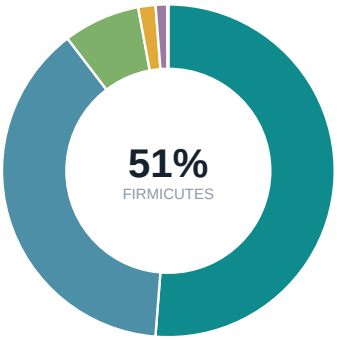
47 evidence cards for the readings that fired: 37 with human trial or guideline evidence for an exact product, diet or drug; 7 pointing to a conventional clinical route; 7 stating that nothing studied exists for that reading. Section 14, and under every reading in Part B.

PART A — MICROBIOME DIVERSITY: WHAT'S THERE AND WHAT'S MISSING

3 Your gut community

Every reference catalogue available was searched, so an organism one catalogue cannot name is still found by another. These are **measured identifications**, not closest matches. Human DNA had already been removed by the sequencing provider.

355	3.17	0.70	1.33
ORGANISMS DETECTED	DIVERSITY	EVENNESS	FIRMICUTES : BACTEROIDETES
92 on the reference catalogue	Shannon index, reference-comparable	0 to 1, reference-comparable	two dominant groups



COMPOSITION BY PHYLUM

Firmicutes	51.2%
Bacteroidota	38.4%
Actinomycetota	7.5%
Pseudomonadota	1.7%
Verrucomicrobiota	1.1%
Thermodesulfobacteriota	0.1%

Most abundant organisms

Phocaeicola vulgatus	11.7%	*Bifidobacterium longum*	1.8%
Gemmiger sp963557315	8.4%	*Collinsella aerofaciens*	1.5%
Mediterraneibacter gnavus	6.9%	*Bacteroides stercoris*	1.2%
Phocaeicola plebeius	3.7%	*Parabacteroides distasonis*	1.4%
Bacteroides uniformis	3.1%	*Bifidobacterium breve*	1.4%
Anaerostipes sp900756035	2.4%	*Alistipes onderdonkii*	1.2%
Phocaeicola massiliensis	2.0%	*Barnesiella intestinihominis*	1.2%
Alistipes putredinis	1.9%	*Faecimonas sp900548695*	1.2%

General gut-health index



BROADLY DISTURBED
16 healthy · 15 disease markers

GMWI2 is an independent, published index of general gut health trained on 8,069 stool metagenomes across 54 studies. Positive values lean healthy, negative lean disturbed. It is repeated beside every pattern score in the next section because **a community that is generally disturbed will resemble many disease patterns at once**; the index tells you whether that is what is happening here.

The general dysbiosis index is strongly negative (-1.35). This community is broadly disturbed, so a high score on any profile most likely reflects that general disturbance rather than anything condition-specific.

What is driving this index

the organisms present, weighted as the published model weights them

Pulling the index down		Holding it up	
Lachnoclostridium (genus)	-0.68	*Bifidobacterium adolescentis*	+0.18
Lactobacillus salivarius	-0.48	*Dorea formicigenerans*	+0.17
Clostridium sp CAG 242	-0.37	*Ruminococcus torques*	+0.15
Bacteroides stercoris	-0.24	*Butyricimonas synergistica*	+0.13
Eisenbergiella (genus)	-0.23	*Alistipes shahii*	+0.11
Erysipelatoclostridium ramosum	-0.16	*Bifidobacterium animalis*	+0.11

The single largest downward pull is *Lachnoclostridium* (genus) at -0.68. Each weight is the model's coefficient for that organism when present; the index is their sum. These are the organisms whose change would move the number.

What your community is made of

genus shares of 220 organisms, as non-overlapping partitions

Phocaeicola	**19.9%**
GGB45596 an unnamed genome bin	**9.7%**
Mediterraneibacter	**9.0%**
Bacteroides	**7.8%**
Blautia	**7.5%**
Bifidobacterium	**5.1%**
Alistipes	**4.3%**
GGB81342 an unnamed genome bin	**2.8%**
Parabacteroides	**2.3%**
Collinsella	**1.8%**
155 other genera	29.9%

Shares of unified inventory (marker + whole-genome lanes) (organism_inventory.primary). These are partitions of one total, so they add to 100%: a genus is not counted twice and nothing is dropped. 12% of your community sits in genome bins that have been assembled and counted but not yet given a name — a real part of your community, not a gap in the measurement.

Which research community type yours resembles

a resemblance, not a diagnosis

Your community is closest to the **Bacteroides-dominant** reference group, the group distinguished in that cohort by *Bacteroides*, *Alistipes*, *Ruminococcus*. It is 82% closer to that group than to the next, which is a clear rather than a borderline placement.

REFERENCE GROUP	DISTANCE	RESEMBLANCE
Bacteroides-dominant	0.350	68%
Prevotella-dominant	0.635	32%

Groups were fixed once on 3,027 reference samples from 22 studies, 2 groups chosen by held-out silhouette among those that survived study-by-study resampling. Your sample is compared with that frozen reference and never with the other samples analysed alongside it. The reference cohort predates several genus reclassifications, so your organisms were restated under its names to be comparable: *Phocaeicola* as *Bacteroides* (20%); *Mediterraneibacter* as *Ruminococcus* (9%); *Lactiplantibacillus* as *Lactobacillus* (1%). Same organisms, older names. The comparison uses the 71% of your community that the reference's genus list names, against 99% for a typical reference sample. The gap is newer unnamed genome bins that the older reference catalogue has no entry for; they are excluded from the comparison rather than forced into a neighbouring genus.

A community type is a resemblance to a published composition, not a diagnosis and not a probability of any disease. A *Bacteroides*-dominant result does not establish a high-protein or high-fat diet. Relative composition cannot establish how much microbial mass you carry; that needs an absolute-load assay.

Diversity, in species you can picture

companion views; the index above is unchanged

45.2
EFFECTIVE SPECIES
how many equally-common species…

21.0
INVERSE SIMPSON
weighted toward the common organis…

13.4%
LARGEST SINGLE ORGANISM
share of the named community

42.5%
TOP FIVE TOGETHER
of 220 organisms counted

Counted over 220 named organisms from the organism_inventory.primary lane. Natural logarithms. These describe the same community as the index above and do not replace it; comparing them between samples needs the same catalogue and the same floor.

Composition ratios

descriptive; no healthy cut-point is defined for any of these

RATIO	VALUE	WHAT WENT INTO IT	READING
Proteobacteria (Pseudomonadota) : Actinobacteriota	**0.571**	Proteobacteria against Actinobacteriota (phylum level)	Descriptive. No established healthy cut-point in stool.
Prevotella : Bacteroides	**0.006**	Prevotella against Bacteroides (genus level)	Diet-response context only (E17). Genus membership follows the declared taxonomy: Segatella copri was Prevotella copri.
Fusobacterium : Faecalibacterium	below the floor	none found against Faecalibacterium (genus level)	The numerator was searched for and not detected, so the ratio is below what this depth can resolve rather than measured at zero.

Aerotolerance balance

an ecological proxy, not a measurement of oxygen in your gut

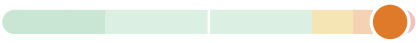
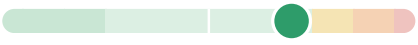
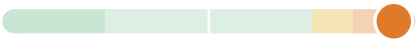
Of the organisms whose oxygen requirement is established, **95.6%** of the abundance is strictly anaerobic and **4.4%** tolerates oxygen. A gut community is normally dominated by strict anaerobes. Oxygen requirements are established for **92%** of the abundance here; the rest is counted as unknown rather than assigned to either

side.

Oxygen-tolerant organisms here: *Lactiplantibacillus plantarum* (1.31%), *Sutterella SGB9281* (0.91%), *Sutterella wadsworthensis* (0.73%), *Bacillus velezensis* (0.30%), *Lacticaseibacillus rhamnosus* (0.21%).

4 Your microbial groups at a glance

Curated sets of species that share a job or an origin, each summed to one reading and placed against 3,027 matched reference adults. ▼ higher is favourable, ▲ higher is adverse, no arrow: it depends on context. Under each reading are the organisms carrying it and the common members that were not found.

GROUP	WHERE YOU SIT	READING
FUNCTIONAL GROUPS		
Butyrate producers ▼ 0.82% of reads · 7 of 19 species		4th NOTABLY LOW
● *Odoribacter splanchnicus* **44%** · ● *Allocopropcoccus comes* **40%** · ● *Agathobacter rectalis* **12%** · ● *Agathobaculum butyriciproducens* **4%** +1 more not detected: *Faecalibacterium prausnitzii*, *Anaerostipes hadrus*, *Eubacterium hallii* +7		
Mucin degraders ▲ 10.29% of reads · 7 of 8 species		94th HIGH
● *Mediterraneibacter gnavus* **67%** · ● *Barnesiella intestinihominis* **11%** · ● *Akkermansia muciniphila* **8%** · ● *Bacteroides caccae* **7%** +2 more not detected: *Mediterraneibacter torques*, *Bacteroides fragilis*		
Polyphenol metabolisers (urolithin and equol) ▼ 0.15% of reads · 2 of 5 species		80th ABOVE AVERAGE
● *Eggerthella lenta* **67%** · ● *Gordonibacter pamelaeae* **33%** not detected: *Adlercreutzia equolifaciens*		
Sulfate-reducing bacteria ▲ 0.09% of reads · 1 of 4 species		70th TYPICAL
● *Bilophila wadsworthia* **100%**		
Methanogens (archaea) ▲ 0.00% of reads · 0 of 3 species		35th NOT DETECTED · TYPICAL
none of this group's species were detected		
ORGANISMS FROM ELSEWHERE IN THE BODY		
Oral-origin organisms found in stool ▲ 0.00% of reads · 0 of 33 species		8th NOT DETECTED · LOW
none detected — usually present: *Streptococcus salivarius*, *Streptococcus parasanguinis*		
COMMONLY BENEFICIAL ORGANISMS		
Supplement and fermented-dairy species 3.88% of reads · 8 of 14 species		99th NOTABLY HIGH
● *Bifidobacterium breve* **36%** · ● *Lactiplantibacillus plantarum* **29%** · ● *Bifidobacterium animalis* **20%** · ● *Lacticaseibacillus rhamnosus* **5%** +4 more		
Resident bifidobacteria ▼ 2.29% of reads · 3 of 5 species		89th HIGH
● *Bifidobacterium longum* **80%** · ● *Bifidobacterium adolescentis* **19%** · ● *Bifidobacterium bifidum* **1%**		
Health-associated core species ▼ 6.43% of reads · 10 of 25 species		51st TYPICAL
● *Alistipes putredinis* **30%** · ● *Bifidobacterium longum* **29%** · ● *Akkermansia muciniphila* **13%** · ● *Alistipes shahii* **9%** +4 more not detected: *Faecalibacterium prausnitzii*, *Anaerostipes hadrus*, *Eubacterium hallii* +9		
OPPORTUNISTIC ORGANISMS		
Opportunistic pathogens and pathobionts ▲ 7.08% of reads · 4 of 20 species		95th HIGH
● *Mediterraneibacter gnavus* **98%** · ● *Bilophila wadsworthia* **1%** · ● *Escherichia coli* **0%** · ● *Enterococcus faecium* **0%**		
CELL-SURFACE CHARACTERISTICS		
Hexa-acylated LPS carriers ▲ 0.12% of reads · 2 of 17 species		40th TYPICAL
● *Bilophila wadsworthia* **73%** · ● *Escherichia coli* **27%**		

5 Organisms that need attention: overgrown, missing, low

Your community, read in both directions. The full inventory of everything detected is the section before this; this page is the part that inventory cannot show — **what is expanded past its range**, **what is not there that usually is**, and what is present but low. Every organism here has its own page in section 16: what it does, why it matters, and what has been shown to move it.

5	6	4	4
OVERGROWN	MISSING	DEPLETED	WORTH WATCHING
opportunists or very high	usually carried, not found	beneficial, well below range	expanded, not yet a concern

Reference: 3,027 adult stool metagenomes from 22 studies. 3,027 matched reference adults 355 organisms detected in all; percentiles are against the adults who carry each organism. An organism with no percentile is one the reference set has no measurement for, and is flagged on presence and class alone.

OVERGROWN — at a level that is a concern 5

Opportunists above the 90th percentile among reference adults who carry them, any organism above the 97th, and opportunists that few adults carry but that are several times above the level of the people who do. These are the readings to act on first.

ORGANISM	CLASS	SHARE	YOUR LEVEL	READING
Mediterraneibacter gnavus A mucus-degrading resident of most healthy adults at low levels, typically well under 1%	Opportunist	6.93% carried by 47% · typical 0.121%	×110 99th pct	OVERGROWN
Anaerotignum lactatifermentans Ferments lactate to propionate and acetate	Conditional	0.776% carried by 13% · typical 0.096%	×17 99th pct	OVERGROWN
Enterocloster citroniae A group of bile-tolerant Lachnospiraceae, formerly filed under Clostridium	Opportunist	0.023% carried by 36% · typical 0.015%	+381% 90th pct	OVERGROWN
Clostridium innocuum A vancomycin-resistant Clostridium found at low levels in many adults	Opportunist	0.041% carried by 29% · typical 0.018%	+372% 81st pct	OVERGROWN
Flavonifractor plautii A resident that breaks down dietary flavonoids such as quercetin and catechins, which sounds useful but means it degra…	Opportunist	0.442% carried by 76% · typical 0.106%	+811% 91st pct	OVERGROWN

MISSING — commonly carried, not detected here 6

Species most reference adults carry that were not found at a depth where they should have been. The organisms that should be there.

ORGANISM	CLASS	SHARE	YOUR LEVEL	READING
Dorea longicatena Gas producers (hydrogen, carbon dioxide) that break down mucus-associated sugars	Conditional	<0.0001% not detected · carried by 88%	~0	MISSING
Roseburia inulinivorans Flagellated butyrate producers that thrive on dietary fibre, especially beta-glucans and inulin an unconfirmed single-method hit at 0.062% is not counted as a find	Beneficial	<0.0001% not detected · carried by 86%	~0	MISSING
Anaerostipes hadrus Converts lactate and acetate into butyrate, which keeps lactate from accumulating (lactate build-up is itself pro-infl… a relative is present: *Anaerostipes sp900756035* at 2.44%	Beneficial	<0.0001% not detected · carried by 85%	~0	MISSING
Roseburia faecis Flagellated butyrate producers that thrive on dietary fibre, especially beta-glucans and inulin a relative is present: *Agathobacter rectalis* at 0.097%	Beneficial	<0.0001% not detected · carried by 83%	~0	MISSING

ORGANISM	CLASS	SHARE	YOUR LEVEL	READING
Eubacterium eligens A pectin specialist that produces acetate and the precursors of butyrate; one of the organisms most associated with pl… a relative is present: *Lachnospira pectinoschiza* at 0.164%	Beneficial	<0.0001% not detected · carried by 79%		~0 MISSING
Oscillibacter sp 57 20 Valerate-producing Oscillospiraceae associated with leanness and healthy ageing in several cohorts; many members are k… a relative is present: *Dysosmobacter welbionis* at 0.113%	Beneficial	<0.0001% not detected · carried by 74%		~0 MISSING

DEPLETED — well below the reference 4

Beneficial organisms present but below the 10th percentile of people who carry them.

ORGANISM	CLASS	SHARE	YOUR LEVEL	READING
Faecalibacterium prausnitzii The single most abundant butyrate producer in a healthy adult gut, often 5-15% of the community counted within *Faecalibacterium sp900540455* · whole-genome reading 0.010%	Beneficial	no share of its own carried by 98% · typical 5.40%		−100% 0th pct LOW
Agathobaculum butyriciproducens A butyrate producer studied as a next-generation probiotic: in mouse models it reduced colitis and improved cognition …	Beneficial	0.029% carried by 89% · typical 0.214%		−98% 2nd pct LOW
Intestinimonas massiliensis Produces butyrate from both sugars and amino acids; lower in some metabolic cohorts counted within *Intestinimonas butyriciproducens* · whole-genome reading 0.045%	Beneficial	no share of its own carried by 18% · typical 0.082%		−46% 6th pct LOW
Butyricicoccus pullicaecorum A member of the genus *Butyricicoccus* for which no human evidence has been curated	Beneficial	0.001% carried by 3% · typical 0.049%		−95% 5th pct LOW

WORTH WATCHING — above the reference range, or uncommon to carry 4

Conditional organisms between the 90th and 97th percentile among carriers - normal residents that have expanded, a signal about diet and balance more than a problem in themselves - and opportunists that fewer than one in ten reference adults carry at all, whatever their level. A beneficial organism that is high is not listed: a supplement being taken is not an overgrowth.

ORGANISM	CLASS	SHARE	YOUR LEVEL	READING
Mediterraneibacter glycyrrhizinilyticus Formerly Ruminococcus gnavus and R. torques: mucus degraders that are normal residents at low levels but among the mos…	Conditional	0.348% carried by 18% · typical 0.084%		+327% 94th pct HIGH
Clostridium sp AM29 11AC A large, much-divided genus	Conditional	0.226% carried by 13% · typical 0.026%		×16 95th pct HIGH
Phocaeicola vulgatus Until 2019 these were Bacteroides, and P. vulgatus and P. dorei are often the two most abundant organisms in adult stool	Conditional	11.75% carried by 89% · typical 4.29%		+420% 93rd pct HIGH
Enterococcus faecium A hardy low-level resident that survives antibiotics, heat and bile	Opportunist	0.021% carried by 5% · typical 0.040%		+78% 66th pct UNCOMMON

PART A — EVERY ORGANISM DETECTED

6.1 Your organisms, classified

355 organisms were found in this sample, pooling every reference catalogue available. Each is placed in one of four classes from the human literature, and — where a reference population exists for it — its level is compared with 3,027 matched reference adults. The bar is the community by share of reads: a dominant organism is a wide band, a trace one is invisible here and listed below. The grey tail is the 12% of sequence that matched nothing on record — shown so the parts add to the whole.

Detection pools **9 methods**, each with its own reference catalogue, so a species one catalogue has no genome for is still found when another can name it. Counted separately: **188** named species and **160** unnamed species-level clusters — organisms known only from assembled genomes, real and counted, whose names will change when they are formally described; **5** populations that the catalogues place in sibling species and that are listed once; **2** placed above species. Each organism has one share of the whole, on the scale of the marker catalogue that measures the composition; for **48** organisms that catalogue has no entry, so their shares (2.0% together, marked °) are estimated from the whole-genome search and drawn from the unclassified band; **135** populations the whole-genome search tells apart within a relative's share are listed with that relative and carry no share of their own; for **2** calls the whole-genome competition rejected, the marker share went to the relative the reads belong to (marked §). **136** organisms rest on a single detection method, said so beside their identifiers.



Beneficial 22%

58 organisms · 4 flagged

Higher levels are associated with better outcomes in human studies. Butyrate producers, mucus-layer keepers and the classic probiotic genera live here.

Conditional 47%

145 organisms · 4 flagged

Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed. Most of the gut is this.

Opportunist 9%

24 organisms · 5 flagged

Normal at low levels. Expansion is what the research associates with inflammation, infection or a disturbed community. Not a pathogen label — that is the pathogen screen's decision.

Unknown 9%

128 organisms

Not enough human evidence to say. Many are known only from genomes and have never been grown in a laboratory.

Unclassified 12%

sequence, not organisms

DNA that matched nothing in any catalogue. Not counted in the four classes; shown so the bar adds to 100%.

7 organisms at a level that is a concern and **6 worth watching**. The most consequential is ***Mediterraneibacter gnavus*** — opportunist, 6.93% of the community, notably high — 99th percentile among reference adults who carry it (×110 against the typical carrier). Flagged rows are tinted throughout this section.

Class is a property of the organism, from the literature; the flag is a property of this sample, from its percentile. **vs typical carrier** is how far the level sits from the median of reference adults who carry the organism at all (that median is the figure in brackets): +400% is five times their level, –90% a tenth of it, ×110 a hundred and ten times. Where fewer than ten reference adults carry an organism, no rank is stated. An organism with no reference was measured on a catalogue the reference population was not, so there is nothing to compare it against; the detection stands on its own.

Furthest from the reference

Against the typical carrier in 3,027 matched reference adults. Left: the largest excesses among organisms at 0.1% or more. Right: the deepest shortfalls among organisms carried by at least half of the reference adults.

MOST ABOVE THE REFERENCE	SHARE	VS TYPICAL
Lactiplantibacillus plantarum	1.14%	**×131** >99th pct
Mediterraneibacter gnavus	6.93%	**×110** 99th pct
Bifidobacterium breve	1.40%	**×63** 91st pct
Lacticaseibacillus rhamnosus	0.185%	**×32** >99th pct
Bifidobacterium animalis	0.767%	**×24** 93rd pct
Anaerotignum lactatifermentans	0.776%	**×17** 99th pct
Clostridium sp AM29 11AC	0.226%	**×16** 95th pct
Thomasclavelia spiroformis	0.188%	**×14** 98th pct

MOST BELOW THE REFERENCE	SHARE	VS TYPICAL
Agathobaculum butyriciproducens	0.029%	**–98%** 2nd pct
Agathobacter rectalis	0.097%	**–92%** 16th pct
Blautia obeum	0.022%	**–76%** 21st pct

6.2 The 30 most abundant, one by one

Together these make up **78%** of the composition. Each has what it is, what the research says about it, its class, and whether the amount found is a concern.

#	ORGANISM	WHAT IT IS	CLASS · SHARE	LEVEL
1	***Phocaeicola vulgatus*** formerly Bacteroides vulgatus SGB 1814 · GTDB GCF_964248265.1 · mOTU 000655 · 6 methods	Very common Bacteroides species (now Phocaeicola vulgatus); implicated in some IBD and metabolic cohorts and protective in others. No fixed reading.	Conditional **11.75%**	▲ high **+420%** vs typical carrier (4.29%) 93rd percentile
2	**Gemmiger sp963557315** not yet formally named · known from genomes SGB 63306 · GlobDB GCA_963557315 · mOTU 016981 · 6 methods	Butyrate producer related to Faecalibacterium, reported lower in IBD and obesity cohorts. Several members are known only from genomes and carry placeholder names. (genus-level)	Beneficial **8.45%**	only 2 reference adults carry it: too few to rank against
3	***Mediterraneibacter gnavus*** formerly Ruminococcus gnavus SGB 4584 · GlobDB GCF_008121495 · mOTU 000538 · 8 methods	Mucin-degrading Lachnospiraceae member present in most healthy adults at low level. Blooms are one of the most consistent features of active Crohn's disease and it produces an inflammatory glucorhamnan polysaccharide. It is an ecological pathobiont, not an infectious agent, and must not share an infection rule with Enterobacterales.	Opportunist **6.93%**	▲ notably high **×110** vs typical carrier (0.121%) 99th percentile
4	***Blautia caecimuris*** also Blautia_A caecimuris includes reads the marker catalogue read as Blautia massiliensis ex Durand et al 2017 SGB 4862 · GlobDB GCF_024622975 · mOTU 001287 · 9 methods	One of the most abundant gut genera, producing acetate and some members butyrate. Reported protective against visceral fat and some infections, but also expanded in obesity, type 2 diabetes and IBS cohorts. A common sign of a Western diet rather than a problem in itself. (genus-level)	Conditional **6.36%**§	reference catalogue reads this species differently
5	***Phocaeicola plebeius*** formerly Bacteroides plebeius SGB 1903 · GlobDB GCF_003437535 · mOTU 000043 · 8 methods	Until 2019 these were classed as Bacteroides, and P. vulgatus and P. dorei are often the single most abundant organisms in adult stool. Fibre degraders and propionate producers; strain-dependent associations with both metabolic health and inflammation. (genus-level)	Conditional **3.75%**	**+229%** vs typical carrier (2.04%) 70th percentile
6	***Bacteroides uniformis*** SGB 1836 · GlobDB GCF_025147485 · mOTU 000312 · 9 methods	Among the most prevalent Bacteroides in Western adults; level varies with diet and has no established directional meaning on its own.	Conditional **3.11%**	**+51%** vs typical carrier (4.38%) 62nd percentile
7	**Anaerostipes sp900756035** not yet formally named · known from genomes SGB 47944 · GlobDB GCA_900756035 · mOTU 048563 · 7 methods	Convert lactate and acetate into butyrate, which keeps lactate from accumulating and feeds the colon lining. Lower in some inflammatory cohorts. (genus-level)	Beneficial **2.44%**	only 3 reference adults carry it: too few to rank against
8	***Phocaeicola massiliensis*** formerly Bacteroides massiliensis SGB 1812 · GTDB GCF_000382445.1 · mOTU 002531 · 5 methods	Until 2019 these were classed as Bacteroides, and P. vulgatus and P. dorei are often the single most abundant organisms in adult stool. Fibre degraders and propionate producers; strain-dependent associations with both metabolic health and inflammation. (genus-level)	Conditional **1.96%**	reference catalogue reads this species differently
9	***Alistipes putredinis*** SGB 2318 · GlobDB GCF_000154465 · mOTU 000454 · 9 methods	Bile-tolerant Rikenellaceae member; higher in some depression and CRC cohorts, protective in others (liver fibrosis, colitis models). Reviewed as a genus with both faces.	Conditional **1.95%**	**+20%** vs typical carrier (3.30%) 56th percentile
10	***Bifidobacterium longum*** also Bifidobacterium infantis SGB 17248 · GlobDB GCF_000196555 · mOTU 000164 · 9 methods	Widespread across life stages (subspecies infantis in infants, longum in adults). Endogenous level does not indicate whether any B. longum-containing product would colonise or help.	Beneficial **1.84%**	**×10** vs typical carrier (0.412%) 92nd percentile
11	***Collinsella aerofaciens*** also Collinsella sp023656745 SGB 14535_group · SGB 14535 · 3 methods	Common Actinobacterium enriched in rheumatoid arthritis and symptomatic atherosclerosis cohorts and responsive to dietary fibre reduction; common in healthy people too.	Opportunist **1.53%**	**+131%** vs typical carrier (0.841%) 72nd percentile
12	***Parabacteroides distasonis*** SGB 1934 · GlobDB GCF_000012845 · mOTU 000211 · 9 methods	Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans.	Conditional **1.42%**	**+217%** vs typical carrier (0.950%) 85th percentile
13	***Bifidobacterium breve*** SGB 17247 · GlobDB GCF_001025175 · mOTU 000269 · 8 methods	Infant-associated Bifidobacterium; in adults usually supplement-derived.	Beneficial **1.40%**	**×63** vs typical carrier (0.028%) 91st percentile
14	***Alistipes onderdonkii*** SGB 2303 · GTDB GCF_025145285.1 · mOTU 000215 · 6 methods	Common Alistipes species; lower in a small two-year Long-COVID cohort (unreplicated) and variable elsewhere.	Conditional **1.22%**	no reference

#	ORGANISM	WHAT IT IS	CLASS · SHARE	LEVEL
15	***Bacteroides stercoris*** SGB 1830 · GlobDB GCF_025147325 · mOTU 000479 · 9 methods	Common Bacteroides species with no fixed directional interpretation.	Conditional **1.18%**†	**+115%** vs typical carrier (1.31%) 63rd percentile
16	***Barnesiella intestinihominis*** SGB 1965 · GlobDB GCF_000296465 · mOTU 000336 · 9 methods	Common Bacteroidales member; listed among mucin utilisers. Context-dependent, with no fixed directional reading.	Beneficial **1.16%**	**+181%** vs typical carrier (0.844%) 79th percentile
17	**Faecimonas sp900548695** not yet formally named · known from genomes SGB 47368 · GlobDB GCA_900548695 · mOTU 059012 · 7 methods	An organism of the genus *Faecimonas* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described.	Unknown **1.16%**	no reference
18	***Lactiplantibacillus plantarum*** formerly Lactobacillus plantarum SGB 7202 · GlobDB GCF_014131735 · mOTU 004675 · 9 methods	Usually food-derived in stool (fermented vegetables, supplements); transient. Trial evidence exists only for specific strains and products.	Beneficial **1.14%**	**×131** vs typical carrier (0.017%) >99th percentile
19	***Fusicatenibacter saccharivorans*** also Fusicatenibacter sp. (SGB4874) SGB 4874 · GTDB GCF_027667425.1 · mOTU 000133 · 8 methods	Butyrate producer reported lower in ulcerative colitis and depression cohorts. (genus-level)	Beneficial **1.02%**	**−11%** vs typical carrier (0.820%) 47th percentile
20	**Brotaphodocola sp900538475** not yet formally named · known from genomes SGB 4746 · GlobDB GCF_964238185 · mOTU 002751 · 7 methods	An organism of the genus *Brotaphodocola* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described.	Unknown **0.987%**	**+392%** vs typical carrier (0.210%) >99th percentile‡
21	***Akkermansia muciniphila*** SGB 9226 · GTDB GCF_000020225.1 · mOTU 000414 · 7 methods	Mucin-degrading Verrucomicrobium. Higher levels are associated with metabolic health in several cohorts and a pasteurised strain (MucT) has human trial evidence, yet the same species is enriched in multiple sclerosis and Parkinson's disease cohorts. Endogenous abundance does not predict response to any Akkermansia product.	Beneficial **0.846%**†	**+166%** vs typical carrier (0.535%) 66th percentile
22	***Sutterella SGB9281*** also Duodenibacillus intestinavium SGB 9281 · GlobDB GCA_019117135 · mOTU 000431 · 8 methods	Bile-tolerant Proteobacteria common in healthy adults; expanded in some autism and IBD cohorts, with mixed evidence. (genus-level)	Conditional **0.796%**	only 5 reference adults carry it: too few to rank against
23	***Paraprevotella clara*** SGB 1798 · GlobDB GCF_000233955 · mOTU 000277 · 9 methods	Fibre degrader related to Prevotella; little condition-specific evidence. (genus-level)	Conditional **0.791%**	**+185%** vs typical carrier (0.295%) 90th percentile‡
24	***Anaerotignum lactatifermentans*** SGB 5193 · GlobDB GCF_900142265 · mOTU 000201 · 8 methods	Lactate fermenters producing propionate; A. lactatifermentans is reported expanded in some IBD cohorts. (genus-level)	Conditional **0.776%**	**▲ notably high** **×17** vs typical carrier (0.096%) 99th percentile
25	***Bifidobacterium animalis*** also Bifidobacterium animalis_B SGB 17278 · GlobDB MOTU40_005954 · GTDB GCF_000260715.1 · 9 methods	Almost always supplement- or yogurt-derived (subsp. lactis strains); transient. Strain evidence does not transfer to the species.	Beneficial **0.767%**	**×24** vs typical carrier (0.072%) 93rd percentile
26	***Pseudoflavonifractor capillosus*** also Enterenecus sp902461475 SGB 15161 · GlobDB GCA_902461475 · UHGG 662 · 5 methods	A member of the genus *Enterenecus* for which no human evidence has been curated. Its presence is recorded; nothing here says it is helpful or harmful.	Unknown **0.766%**	only 1 reference adult carry it: too few to rank against
27	***Bacteroides caccae*** SGB 1877 · GlobDB GCF_002222615 · mOTU 001775 · 8 methods	Common mucin-utilising Bacteroides; context-dependent.	Conditional **0.759%**	**+83%** vs typical carrier (0.858%) 69th percentile
28	**Dialister complex (sp021633965, sp000434475)** not yet formally named · known from genomes sibling species, one population SGB 5809 · 2 methods: Jun23, Jan26	Propionate producers; reported lower in ankylosing spondylitis and depression, expanded in some periodontal and IBS contexts. (genus-level)	Conditional **0.722%**	no reference
29	***Clostridium sp AT4*** also Enterocloster sp001517625 SGB 4753 · GlobDB GCF_001517625 · mOTU 003249 · 7 methods	A large and much-split genus. Most gut members are harmless fermenters; a few named species (C. perfringens, C. difficile) are pathogens and are handled by the pathogen screen. Unnamed members are read as neutral residents. (genus-level)	Conditional **0.716%**	**+392%** vs typical carrier (0.155%) 86th percentile‡

#	ORGANISM	WHAT IT IS	CLASS · SHARE	LEVEL
30	***Sutterella wadsworthensis*** SGB 9283 · GlobDB GCF_003315195 · mOTU 000497 · 9 methods	Bile-tolerant Proteobacteria common in healthy adults; expanded in some autism and IBD cohorts, with mixed evidence. (genus-level)	Conditional **0.643%**	only 9 reference adults carry it: too few to rank against

6.3 Every organism detected

All 355 organisms, largest share first, colour-coded by class. **Level** is how far the organism sits from the typical carrier in 3,027 matched reference adults, with its percentile; **carried by** is the share of that group in which the organism is found at all. Rows tinted red are at a level that is a concern; amber, worth watching. **54** were also read to strain level, marked in the last column.

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Phocaeicola vulgatus formerly Bacteroides vulgatus SGB 1814 · GTDB GCF_964248265.1 · mOTU 000655 · 6 methods	Conditional	**11.75%**	▲ high **+420%** vs typical carrier (4.29%)	89%	93rd	30 markers; nearest GCA_001412315.1 (0.306)
Phocaeicola SPECIV4_34405 not yet formally named · known from genomes GlobDB SPECIV4_34405 · 1 method: GlobDB sketch	Conditional	within *Phocaeicola* above	**+229%** vs typical carrier (3.42%)	37%	92nd‡	—
Phocaeicola MGYG000000243 not yet formally named · known from genomes GlobDB MGYG000000243 · UHGG 517 · 3 methods	Conditional	within *Phocaeicola* above	**+93%** vs typical carrier (1.13%)	27%	78th‡	—
Phocaeicola sp900760795 not yet formally named · known from genomes GlobDB GCA_900760795 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	within *Phocaeicola* above	**–9%** vs typical carrier (1.92%)	15%	47th‡	—
Phocaeicola coprocola formerly Bacteroides coprocola UHGG 2005 · 1 method: Kraken	Conditional	within *Phocaeicola* above	no reference	—	not in reference set	—
Phocaeicola sp900546645 not yet formally named · known from genomes UHGG 4213 · 2 methods: Kraken, Kraken panel	Conditional	within *Phocaeicola* above	no reference	—	not in reference set	—
Phocaeicola sp000432735 not yet formally named · known from genomes 1 method: Kraken panel	Conditional	within *Phocaeicola* above	no reference	—	not in reference set	—
Phocaeicola mediterraneensis_A UHGG 5682 · 1 method: Kraken	Conditional	within *Phocaeicola* above	no reference	—	not in reference set	—
Phocaeicola SPECIV4_00061 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Phocaeicola* above	no reference	—	not in reference set	—
Gemmiger sp963557315 not yet formally named · known from genomes SGB 63306 · GlobDB GCA_963557315 · mOTU 016981 · 6 methods	Beneficial	**8.45%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	196 markers
Gemmiger sp900545545 not yet formally named · known from genomes UHGG 4094 · 1 method: Kraken	Beneficial	within *Gemmiger* above	no reference	—	not in reference set	—
Gemmiger variabilis UHGG 2026 · 1 method: Kraken	Beneficial	within *Gemmiger* above	no reference	—	not in reference set	—
Gemmiger sp900539695 not yet formally named · known from genomes 1 method: Kraken panel	Beneficial	within *Gemmiger* above	no reference	—	not in reference set	—
Gemmiger formicilis UHGG 265 · 1 method: Kraken	Beneficial	within *Gemmiger* above	no reference	—	not in reference set	—
Mediterraneibacter gnavus formerly Ruminococcus gnavus SGB 4584 · GlobDB GCF_008121495 · mOTU 000538 · 8 methods	Opportunist	**6.93%**	▲ notably high **×110** vs typical carrier (0.121%)	47%	99th	200 markers; nearest GCF_008121495.1 (0.420)
Mediterraneibacter HRGMV2_2200 not yet formally named · known from genomes GlobDB HRGMV2_2200 · 3 methods	Conditional	within *Mediterraneibacter* above	**×24** vs typical carrier (0.200%)	16%	>99th‡	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Mediterraneibacter torques formerly Ruminococcus torques GTDB GCF_000153925.1 · mOTU 002160 · UHGG 2093 · 5 methods	Opportunist	within *Mediterraneibacter* above	**+191%** vs typical carrier (0.441%)	78%	80th	—
Mediterraneibacter sp900544395 not yet formally named · known from genomes UHGG 4113 · 3 methods	Conditional	within *Mediterraneibacter* above	no reference	—	not in reference set	—
Blautia caecimuris also Blautia_A caecimuris includes reads the marker catalogue read as Blautia massiliensis ex Durand et al 2017 SGB 4862 · GlobDB GCF_024622975 · mOTU 001287 · 9 methods	Conditional	**6.36%**§	reference catalogue reads this species differently	21%	not comparable	163 markers; nearest GCA_013 302405.1 (0.283)
Blautia_A fusiformis GlobDB GCF_020687435 · mOTU 000153 · UHGG 371 · 6 methods	Conditional	within *Blautia* above	**+764%** vs typical carrier (0.666%)	82%	98th‡	—
Blautia massiliensis also Blautia_A complex (fusiformis, luti_A, sp005848555) SGB 4826 · 1 method: Jun23	Conditional	within *Blautia* above	no reference	—	not in reference set	193 markers
Blautia sp958352855 not yet formally named · known from genomes GlobDB GCA_958352855 · mOTU 008599 · UHGG 3776 · 5 methods	Conditional	within *Blautia* above	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Blautia_A wexlerae mOTU 000158 · 2 methods: mOTUs, Kraken panel	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Ruminococcus sp AF13 28 also Blautia_A sp003471165 SGB 4834 · GTDB GCA_003471165.1 · mOTU 001615 · 4 methods	Conditional	within *Blautia* above	no reference	—	not in reference set	96 markers
Blautia_A GWHBMLB00000000 not yet formally named · known from genomes GlobDB GWHBMLB00000000 · 1 method: GlobDB sketch	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia sp N6H1 15 not yet formally named · known from genomes 1 method: MP3	Conditional	within *Blautia* above	**+354%** vs typical carrier (0.059%)	2%	76th	—
Blautia sp900555025 not yet formally named · known from genomes GlobDB GCA_905201915 · mOTU 014988 · UHGG 6124 · 4 methods	Conditional	within *Blautia* above	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Blautia_A intestinihominis UHGG 3260 · 2 methods: Kraken, Kraken panel	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia sp900556555 not yet formally named · known from genomes UHGG 1528 · 2 methods: Kraken, Kraken panel	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia_A sp003474435 not yet formally named · known from genomes UHGG 474 · 1 method: Kraken	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia_A sp900066335 not yet formally named · known from genomes UHGG 544 · 1 method: Kraken	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia_A intestinalis UHGG 529 · 1 method: Kraken	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia_A schinkii_B UHGG 5466 · 1 method: Kraken	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia sp900753905 not yet formally named · known from genomes UHGG 5963 · 1 method: Kraken	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia_A acetigignens UHGG 455 · 1 method: Kraken	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Blautia_A hydrogenotrophica 1 method: Kraken panel	Conditional	within *Blautia* above	no reference	—	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Blautia_A luti_A 1 method: SingleM	Conditional	within *Blautia* above	no reference	—	not in reference set	—
Phocaeicola plebeius formerly Bacteroides plebeius SGB 1903 · GlobDB GCF_003437535 · mOTU 000043 · 8 methods	Conditional	**3.75%**	**+229%** vs typical carrier (2.04%)	24%	70th	127 markers; nearest GCA_003512905.1 (0.360)
Bacteroides uniformis SGB 1836 · GlobDB GCF_025147485 · mOTU 000312 · 9 methods	Conditional	**3.11%**	**+51%** vs typical carrier (4.38%)	92%	62nd	56 markers; nearest GCA_002320565.1 (0.475)
Bacteroides sp900755095 not yet formally named · known from genomes GTDB GCA_900755095.1 · 1 method: GTDB sketch	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Bacteroides GWHFDIJ00000000 not yet formally named · known from genomes GlobDB GWHFDIJ00000000 · 1 method: GlobDB sketch	Conditional	within *Bacteroides* above	**+415%** vs typical carrier (0.140%)	12%	83rd‡	—
Bacteroides eggerthii UHGG 2015 · 1 method: Kraken	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Bacteroides sp900556625 not yet formally named · known from genomes GlobDB GCA_900556625 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	within *Bacteroides* above	≈ **typical** vs typical carrier (0.353%)	10%	50th‡	—
Bacteroides nordii UHGG 546 · 1 method: Kraken	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Bacteroides sp900761785 not yet formally named · known from genomes UHGG 4320 · 1 method: Kraken	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Bacteroides faecis UHGG 3243 · 1 method: Kraken	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Bacteroides sp947071285 not yet formally named · known from genomes UHGG 3753 · 1 method: Kraken	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Bacteroides MOTU40_003245 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Bacteroides* above	no reference	—	not in reference set	—
Anaerostipes sp900756035 not yet formally named · known from genomes SGB 47944 · GlobDB GCA_900756035 · mOTU 048563 · 7 methods	Beneficial	**2.44%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	183 markers; nearest GCA_900756035.1 (2.937)
Anaerostipes sp900066705 not yet formally named · known from genomes UHGG 219 · 1 method: Kraken	Beneficial	within *Anaerostipes* above	no reference	—	not in reference set	—
Anaerostipes sp905213305 not yet formally named · known from genomes 1 method: SingleM	Beneficial	within *Anaerostipes* above	no reference	—	not in reference set	—
Phocaeicola massiliensis formerly Bacteroides massiliensis SGB 1812 · GTDB GCF_000382445.1 · mOTU 002531 · 5 methods	Conditional	**1.96%**	reference catalogue reads this species differently	33%	not comparable	193 markers; nearest GCA_000438395.1 (0.513)
Alistipes putredinis SGB 2318 · GlobDB GCF_000154465 · mOTU 000454 · 9 methods	Conditional	**1.95%**	**+20%** vs typical carrier (3.30%)	75%	56th	191 markers; nearest GCA_003541835.1 (0.272)
Alistipes sp047233565 not yet formally named · known from genomes GTDB GCA_047233565.1 · 2 methods: GTDB sketch, SingleM	Conditional	within *Alistipes* above	no reference	—	not in reference set	—
Alistipes GCMETA_00080716 not yet formally named · known from genomes GlobDB GCMETA_00080716 · 2 methods: GlobDB sketch, SingleM	Conditional	within *Alistipes* above	**+136%** vs typical carrier (0.517%)	50%	78th‡	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Alistipes finegoldii UHGG 5868 · 3 methods	Conditional	within *Alistipes* above	**+647%** vs typical carrier (0.229%)	71%	85th	—
Bifidobacterium longum also Bifidobacterium infantis SGB 17248 · GlobDB GCF_000196555 · mOTU 000164 · 9 methods	Beneficial	**1.84%**	**×10** vs typical carrier (0.412%)	67%	92nd	185 markers; nearest GCA_018369245.1 (0.951)
Bifidobacterium MGYG000002469 not yet formally named · known from genomes GlobDB MGYG000002469 · UHGG 3473 · 4 methods	Beneficial	within *Bifidobacterium* above	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Collinsella aerofaciens also Collinsella sp023656745 SGB 14535_group · SGB 14535 · 3 methods	Opportunist	**1.53%**	**+131%** vs typical carrier (0.841%)	86%	72nd	—
Collinsella aerofaciens_G GlobDB GCF_005844325 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	within *Collinsella* above	no reference	—	not in reference set	—
Collinsella sp900541235 not yet formally named · known from genomes 1 method: Kraken panel	Conditional	within *Collinsella* above	no reference	—	not in reference set	—
Collinsella sp900545605 not yet formally named · known from genomes UHGG 4066 · 1 method: Kraken	Conditional	within *Collinsella* above	no reference	—	not in reference set	—
Collinsella MOTU40_014808 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Collinsella* above	no reference	—	not in reference set	—
Parabacteroides distasonis SGB 1934 · GlobDB GCF_000012845 · mOTU 000211 · 9 methods	Conditional	**1.42%**	**+217%** vs typical carrier (0.950%)	85%	85th	179 markers; nearest GCF_000012845.1 (0.340)
Parabacteroides sp003473295 not yet formally named · known from genomes UHGG 484 · 1 method: Kraken	Conditional	within *Parabacteroides* above	no reference	—	not in reference set	—
Parabacteroides intestinavium UHGG 695 · 1 method: Kraken	Conditional	within *Parabacteroides* above	no reference	—	not in reference set	—
Bifidobacterium breve SGB 17247 · GlobDB GCF_001025175 · mOTU 000269 · 8 methods	Beneficial	**1.40%**	**×63** vs typical carrier (0.028%)	5%	91st	197 markers; nearest GCF_001025175.1 (0.214)
Alistipes onderdonkii SGB 2303 · GTDB GCF_025145285.1 · mOTU 000215 · 6 methods	Conditional	**1.22%**	no reference	—	not in reference set	183 markers; nearest GCA_001405655.1 (0.280)
Bacteroides stercoris SGB 1830 · GlobDB GCF_025147325 · mOTU 000479 · 9 methods	Conditional	**1.18%**†	**+115%** vs typical carrier (1.31%)	53%	63rd	191 markers; nearest GCA_003463895.1 (0.762)
Barnesiella intestinihominis SGB 1965 · GlobDB GCF_000296465 · mOTU 000336 · 9 methods	Beneficial	**1.16%**	**+181%** vs typical carrier (0.844%)	51%	79th	190 markers; nearest GCA_003489685.1 (1.059)
Faecimonas sp900548695 not yet formally named · known from genomes SGB 47368 · GlobDB GCA_900548695 · mOTU 059012 · 7 methods	Unknown	**1.16%**	no reference	—	not in reference set	187 markers; nearest GCA_900548695.1 (3.981)
Faecimonas MGYG000002024 not yet formally named · known from genomes 1 method: Kraken panel	Unknown	within *Faecimonas* above	no reference	—	not in reference set	—
Lactiplantibacillus plantarum formerly Lactobacillus plantarum SGB 7202 · GlobDB GCF_014131735 · mOTU 004675 · 9 methods	Beneficial	**1.14%**	**×131** vs typical carrier (0.017%)	2%	>99th	178 markers; nearest GCA_000604145.1 (0.003)
Fusicatenibacter saccharivorans also Fusicatenibacter sp. (SGB4874) SGB 4874 · GTDB GCF_027667425.1 · mOTU 000133 · 8 methods	Beneficial	**1.02%**	**–11%** vs typical carrier (0.820%)	92%	47th	179 markers

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Fusicatenibacter GCMETA_00500418 not yet formally named · known from genomes GlobDB GCMETA_00500418 · 2 methods: GlobDB sketch, SingleM	Beneficial	within *Fusicatenibacter* above	**+12%** vs typical carrier (1.02%)	77%	56th‡	—
Brotaphodocola sp900538475 not yet formally named · known from genomes SGB 4746 · GlobDB GCF_964238185 · mOTU 002751 · 7 methods	Unknown	**0.987%**	**+392%** vs typical carrier (0.210%)	16%	>99th‡	180 markers; nearest GCF_964238185.1 (0.080)
Akkermansia muciniphila SGB 9226 · GTDB GCF_000020225.1 · mOTU 000414 · 7 methods	Beneficial	**0.846%**†	**+166%** vs typical carrier (0.535%)	51%	66th	168 markers; nearest GCA_000436395.1 (0.768)
Akkermansia MGYG000002454 not yet formally named · known from genomes GlobDB MGYG000002454 · UHGG 3456 · 3 methods	Beneficial	within *Akkermansia* above	≈ **typical** vs typical carrier (0.956%)	18%	50th‡	—
Sutterella SGB9281 also Duodenibacillus intestinavium SGB 9281 · GlobDB GCA_019117135 · mOTU 000431 · 8 methods	Conditional	**0.796%**	only 5 reference adults carry it: too few to rank against	5%	not in reference set	159 markers; nearest GCA_041333695.1 (0.382)
Duodenibacillus intestinigallinarum mOTU 001088 · UHGG 489 · 2 methods: Kraken, mOTUs	Conditional	within *Duodenibacillus* above	no reference	—	not in reference set	—
Paraprevotella clara SGB 1798 · GlobDB GCF_000233955 · mOTU 000277 · 9 methods	Conditional	**0.791%**	**+185%** vs typical carrier (0.295%)	20%	90th‡	153 markers; nearest GCA_003462815.1 (0.190)
Anaerotignum lactatifermentans SGB 5193 · GlobDB GCF_900142265 · mOTU 000201 · 8 methods	Conditional	**0.776%**	▲ **notably high** **×17** vs typical carrier (0.096%)	13%	99th	84 markers; nearest GCA_047014905.1 (0.897)
Anaerotignum sp001304995 not yet formally named · known from genomes mOTU 001883 · 2 methods: mOTUs, Kraken panel	Conditional	within *Anaerotignum* above	no reference	—	not in reference set	—
Bifidobacterium animalis also Bifidobacterium animalis_B SGB 17278 · GlobDB MOTU40_005954 · GTDB GCF_000260715.1 · 9 methods	Beneficial	**0.767%**	**×24** vs typical carrier (0.072%)	6%	93rd	40 markers; nearest GCA_018363555.1 (0.001)
Pseudoflavonifractor capillosus also Enterenecus sp902461475 SGB 15161 · GlobDB GCA_902461475 · UHGG 662 · 5 methods	Unknown	**0.766%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	82 markers
Enterenecus faecium mOTU 000486 · 2 methods: mOTUs, SingleM	Unknown	within *Enterenecus* above	no reference	—	not in reference set	—
Enterenecus GCMETA_00079489 not yet formally named · known from genomes 1 method: SingleM	Unknown	within *Enterenecus* above	no reference	—	not in reference set	—
Enterenecus GCMETA_00141408 not yet formally named · known from genomes 1 method: SingleM	Unknown	within *Enterenecus* above	no reference	—	not in reference set	—
Bacteroides caccae SGB 1877 · GlobDB GCF_002222615 · mOTU 001775 · 8 methods	Conditional	**0.759%**	**+83%** vs typical carrier (0.858%)	61%	69th	180 markers; nearest GCA_000433575.1 (0.802)
Dialister complex (sp021633965, sp000434475) not yet formally named · known from genomes sibling species, one population SGB 5809 · 2 methods: Jun23, Jan26	Conditional	**0.722%**	no reference	—	not in reference set	33 markers
Dialister sp000434475 not yet formally named · known from genomes GlobDB GCF_042846925 · mOTU 000013 · UHGG 1311 · 6 methods	Conditional	within *Dialister* above	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Dialister MOTU40_020935 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Dialister* above	no reference	—	not in reference set	—

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Dialister sp003447275 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Dialister* above	no reference	—	not in reference set	—
Dialister sp022722495 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Dialister* above	no reference	—	not in reference set	—
Clostridium sp AT4 also Enterocloster sp001517625 SGB 4753 · GlobDB GCF_001517625 · mOTU 003249 · 7 methods	Conditional	**0.716%**	**+392%** vs typical carrier (0.155%)	29%	86th‡	180 markers; nearest GCA_018055585.1 (0.242)
Enterocloster MGYG000329419 not yet formally named · known from genomes GlobDB MGYG000329419 · 1 method: GlobDB sketch	Opportunist	within *Enterocloster* above	**+126%** vs typical carrier (0.054%)	42%	79th‡	—
Enterocloster GCMETA_00004575 not yet formally named · known from genomes GlobDB GCMETA_00004575 · 1 method: GlobDB sketch	Opportunist	within *Enterocloster* above	**+23%** vs typical carrier (0.034%)	17%	59th‡	—
Enterocloster MGYG000001606 not yet formally named · known from genomes UHGG 2449 · 2 methods: Kraken, Kraken panel	Opportunist	within *Enterocloster* above	no reference	—	not in reference set	—
Enterocloster MRGM_0886 not yet formally named · known from genomes GlobDB MRGM_0886 · 1 method: GlobDB sketch	Opportunist	within *Enterocloster* above	**–13%** vs typical carrier (0.034%)	24%	42nd‡	—
Enterocloster GWHGMID00000000 not yet formally named · known from genomes GlobDB GWHGMID00000000 · 1 method: GlobDB sketch	Opportunist	within *Enterocloster* above	**–72%** vs typical carrier (0.093%)	11%	18th‡	—
Enterocloster excrementipullorum GlobDB GCA_019119575 · mOTU 003749 · 3 methods	Opportunist	within *Enterocloster* above	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Sutterella wadsworthensis SGB 9283 · GlobDB GCF_003315195 · mOTU 000497 · 9 methods	Conditional	**0.643%**	only 9 reference adults carry it: too few to rank against	9%	not in reference set	137 markers; nearest GCA_003522065.1 (0.007)
Dorea_D sp900543415 not yet formally named · known from genomes SGB 4574 · GTDB GCA_900543415.1 · mOTU 048565 · 7 methods	Conditional	**0.630%**	no reference	—	not in reference set	132 markers; nearest GCA_026169395.1 (0.240)
Dorea_D GCMETA_00065684 not yet formally named · known from genomes GlobDB GCMETA_00065684 · 2 methods: GlobDB sketch, SingleM	Conditional	within *Dorea* above	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Dorea GWHBMXR00000000 not yet formally named · known from genomes GlobDB GWHBMXR00000000 · 1 method: GlobDB sketch	Conditional	within *Dorea* above	**–9%** vs typical carrier (0.244%)	28%	46th‡	—
Dorea_D sp000433535 not yet formally named · known from genomes mOTU 001257 · UHGG 2548 · 3 methods	Conditional	within *Dorea* above	**+152%** vs typical carrier (0.084%)	17%	68th	—
Dorea_D GWHBNUO00000000 not yet formally named · known from genomes 1 method: SingleM	Conditional	within *Dorea* above	no reference	—	not in reference set	—
Parabacteroides merdae SGB 1949 · GlobDB GCF_025151215 · mOTU 000708 · 9 methods	Conditional	**0.621%**	≈ **typical** vs typical carrier (0.868%)	65%	49th	53 markers; nearest GCA_002319995.1 (0.263)
Frisingicoccus sp900757195 not yet formally named · known from genomes SGB 4671 · GlobDB GCA_900757195 · mOTU 010966 · 7 methods	Unknown	**0.587%**	**+269%** vs typical carrier (0.148%)	12%	>99th‡	125 markers; nearest GCA_902496395.1 (0.103)
Bacteroides thetaiotaomicron SGB 1861 · GlobDB GCF_000011065 · mOTU 000293 · 8 methods	Conditional	**0.574%**	**+136%** vs typical carrier (0.544%)	79%	71st	140 markers; nearest GCA_003439845.1 (0.338)
Alistipes shahii SGB 2295 · GlobDB GCF_025145845 · mOTU 000521 · 8 methods	Conditional	**0.571%**	**+88%** vs typical carrier (0.407%)	49%	86th‡	136 markers; nearest GCF_025145845.1 (0.591)

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Mediterraneibacter faecis SGB 4563 · GlobDB GCA_001312505 · mOTU 000056 · 7 methods	Conditional	**0.566%**	≈ **typical** vs typical carrier (0.687%)	77%	51st‡	125 markers; nearest GCA_000 437035.1 (0.284)
GGB52088 SGB72791 also Anaerofilum excrementigallinarum SGB 72791 · GlobDB GCA_019115305 · mOTU 011804 · 5 methods	Unknown	**0.562%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	156 markers; nearest GCA_017 889025.1 (0.475)
Anaerofilum GCMETA_00141151 not yet formally named · known from genomes 1 method: SingleM	Unknown	within *Anaerofilum* above	no reference	—	not in reference set	—
Fimivicinus sp900544375 not yet formally named · known from genomes SGB 14874 · GlobDB GCA_900544375 · mOTU 003598 · 8 methods	Unknown	**0.546%**	**–58%** vs typical carrier (1.26%)	29%	17th‡	123 markers; nearest GCA_030 844365.1 (0.070)
Eubacterium ramulus also Eubacterium_I ramulus SGB 4959 · GlobDB GCF_000469345 · mOTU 000772 · 9 methods	Conditional	**0.477%**	**×11** vs typical carrier (0.080%)	68%	98th	115 markers; nearest GCA_031 293675.1 (0.581)
Eubacterium sp CAG 38 not yet formally named · known from genomes 1 method: MP3	Conditional	within *Eubacterium* above	**+96%** vs typical carrier (0.198%)	61%	64th	—
Bacteroides ovatus also Bacteroides sp019412865 SGB 1871 · GlobDB GCF_001314995 · mOTU 000239 · 8 methods	Conditional	**0.451%**	**+160%** vs typical carrier (0.339%)	79%	71st	89 markers; nearest GCA_048 511295.1 (1.515)
Blautia_A sp900541345 not yet formally named · known from genomes SGB 4859 · GlobDB GCA_900541345 · mOTU 002374 · 6 methods	Conditional	**0.448%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	63 markers; nearest GCA_048 362375.1 (0.256)
Flavonifractor plautii SGB 15132 · GlobDB GCF_000239295 · mOTU 000255 · 9 methods	Opportunist	**0.442%**	▲ **high** **+811%** vs typical carrier (0.106%)	76%	91st	88 markers; nearest GCA_000 508885.1 (0.767)
Bifidobacterium adolescentis SGB 17244 · GTDB GCF_000010425.1 · mOTU 000016 · 7 methods	Beneficial	**0.435%**	≈ **typical** vs typical carrier (0.957%)	51%	50th	111 markers; nearest GCA_000 829865.1 (0.005)
Bacillota bacterium also Blautia sp000432195 SGB 4791 · mOTU 002129 · UHGG 330 · 7 methods	Conditional	**0.418%**	**+269%** vs typical carrier (0.083%)	5%	77th	25 markers; nearest GCA_018 380135.1 (2.014)
Coprococcus_A intestinihominis 1 method: Kraken panel	Beneficial	**0.408%**◦	no reference	—	not in reference set	—
Coprococcus phoceensis also Faecalimonas sp029983955 SGB 4588 · 2 methods: Jan26, SingleM	Beneficial	**0.390%**	no reference	—	not in reference set	59 markers; nearest GCA_023 454305.1 (1.045)
Faecalimonas phoceensis GlobDB GCF_900104635 · mOTU 001061 · UHGG 380 · 5 methods	Unknown	within *Faecalimonas* above	only 6 reference adults carry it: too few to rank against	6%	not in reference set	—
Tyzzerella nexilis also Faecalimonas complex (phoceensis, sp029983955) SGB 4588 · 2 methods: Jun23, MP3	Unknown	within *Faecalimonas* above	**–20%** vs typical carrier (0.016%)	11%	46th	59 markers; nearest GCA_023 454305.1 (1.045)
Odoribacter splanchnicus SGB 1790 · GlobDB GCF_000190535 · mOTU 002115 · 8 methods	Beneficial	**0.364%**	**+62%** vs typical carrier (0.442%)	74%	70th	69 markers; nearest GCA_003 448135.1 (0.471)
Mediterraneibacter glycyrrhizinilyticus also Schaedlerella glycyrrhizinilytica_A SGB 4606 · GlobDB GCF_042852765 · mOTU 002019 · 7 methods	Conditional	**0.348%**	▲ **high** **+327%** vs typical carrier (0.084%)	18%	94th‡	74 markers; nearest GCA_000 218385.1 (0.374)
Schaedlerella hominis UHGG 551 · 1 method: Kraken	Unknown	within *Schaedlerella* above	no reference	—	not in reference set	—
Schaedlerella sp900765975 not yet formally named · known from genomes UHGG 6017 · 1 method: Kraken	Unknown	within *Schaedlerella* above	no reference	—	not in reference set	—

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Schaedlerella sp905203555 not yet formally named · known from genomes UHGG 2912 · 1 method: Kraken	Unknown	within *Schaedlerella* above	no reference	—	not in reference set	—
Bacteroides stercoris_A GlobDB GCA_958432865 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	**0.341%**†	**–11%** vs typical carrier (1.73%)	31%	48th‡	—
Allocoprococcus comes also Allocoprococcus comes / comes_A SGB 4577 · GlobDB GCF_025149785 · mOTU 000371 · 9 methods	Unknown	**0.330%**	**+72%** vs typical carrier (0.415%)	79%	70th	● 44 markers; nearest GCA_934 358615.1 (1.912)
Bacteroides finegoldii also Bacteroides zhangwenhongii SGB 1862 · GlobDB GCF_009193325 · mOTU 002746 · 7 methods	Conditional	**0.317%**	**–62%** vs typical carrier (0.256%)	27%	33rd	● 74 markers; nearest GCA_937 983295.1 (0.588)
Thomasclavelia ramosa SGB 6744 · GlobDB GCF_014131695 · mOTU 003629 · 9 methods	Unknown	**0.316%**	**×13** vs typical carrier (0.056%)	23%	92nd	● 59 markers; nearest GCF_014 131695.1 (0.455)
Faecalibacterium sp900540455 not yet formally named · known from genomes SGB 15313 · GlobDB GCA_900540455 · mOTU 003270 · 7 methods	Beneficial	**0.314%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	● 51 markers; nearest GCA_047 234385.1 (0.561)
Faecalibacterium longum mOTU 000007 · 2 methods: mOTUs, Kraken panel	Beneficial	within *Faecalibacterium* above	no reference	—	not in reference set	—
Faecalibacterium prausnitzii 1 method: MP3	Beneficial	within *Faecalibacterium* above	▼ notably low **–100%** vs typical carrier (5.40%)	98%	0th	—
Faecalibacterium GCMETA_00500036 not yet formally named · known from genomes 1 method: SingleM	Beneficial	within *Faecalibacterium* above	no reference	—	not in reference set	—
Bacillus velezensis SGB 7785 · GlobDB GCF_001461825 · mOTU 006346 · 6 methods	Unknown	**0.264%**	no reference	—	not in reference set	—
Bacillus subtilis group 1 method: MP3	Unknown	within *Bacillus* above	no reference	—	not in reference set	—
Neobittarella massiliensis UHGG 3567 · 1 method: Kraken	Unknown	**0.249%**◦	no reference	—	not in reference set	—
Clostridium sp AM29 11AC also Otoolea saccharolyticum_A SGB 4701 · GTDB GCF_020736365.1 · mOTU 009024 · 7 methods	Conditional	**0.226%**	▲ high **×16** vs typical carrier (0.026%)	13%	95th	● 34 markers
Otoolea MGYG000313663 not yet formally named · known from genomes GlobDB MGYG000313663 · 1 method: GlobDB sketch	Unknown	within *Otoolea* above	only 6 reference adults carry it: too few to rank against	6%	not in reference set	—
Dorea formicigenerans SGB 4575 · GTDB GCF_025150245.1 · mOTU 000271 · 8 methods	Conditional	**0.206%**	**+69%** vs typical carrier (0.212%)	85%	70th	—
Thomasclavelia spiroformis SGB 6747 · GlobDB GCF_025149465 · mOTU 005156 · 8 methods	Unknown	**0.188%**	**×14** vs typical carrier (0.025%)	20%	98th	—
Lacticaseibacillus rhamnosus formerly Lactobacillus rhamnosus SGB 7144 · GlobDB GCF_900636965 · mOTU 000909 · 9 methods	Beneficial	**0.185%**	**×32** vs typical carrier (0.012%)	2%	>99th	—
Lacticaseibacillus MGYG000002388 not yet formally named · known from genomes GlobDB MGYG000002388 · UHGG 3375 · 2 methods: GlobDB sketch, Kraken	Beneficial	within *Lacticaseibacillus* above	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Clostridium leptum also Clostridium_A leptum SGB 14853 · GlobDB GCA_000154345 · mOTU 001285 · 8 methods	Conditional	**0.184%**	**+700%** vs typical carrier (0.028%)	55%	86th	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Clostridium_AQ sp003481775 not yet formally named · known from genomes GlobDB GCA_003481775 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	within *Clostridium* above	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Clostridium sp900759995 not yet formally named · known from genomes UHGG 5422 · 1 method: Kraken	Conditional	within *Clostridium* above	no reference	—	not in reference set	—
Evtepia gabavorous SGB 15120 · GlobDB GCF_003425665 · mOTU 002738 · 9 methods	Unknown	**0.183%**	**–11%** vs typical carrier (0.208%)	68%	35th‡	—
Streptococcus thermophilus includes reads the marker catalogue read as Streptococcus anginosus SGB 8002 · GlobDB GCF_010120595 · mOTU 000815 · 9 methods	Conditional	**0.175%**§	**+290%** vs typical carrier (0.076%)	36%	76th	—
Streptococcus SPIREOTU_03006347 not yet formally named · known from genomes GlobDB SPIREOTU_03006347 · 1 method: GlobDB sketch	Conditional	within *Streptococcus* above	**–38%** vs typical carrier (0.275%)	45%	31st‡	—
Streptococcus vaginalis GlobDB GCF_017315345 · mOTU 011008 · 3 methods	Conditional	within *Streptococcus* above	no reference	—	not in reference set	—
Senegalimassilia anaerobia SGB 14824 · SGB 14824_group · GlobDB GCF_000236865 · 7 methods	Conditional	**0.174%**	**–8%** vs typical carrier (0.229%)	11%	46th‡	—
CAG-273 sp000438355 not yet formally named · known from genomes SGB 6305 · GlobDB GCA_000438355 · mOTU 003257 · 7 methods	Unknown	**0.173%**	**–17%** vs typical carrier (0.220%)	29%	41st‡	—
Butyricimonas paravirosa SGB 1786 · mOTU 000736 · UHGG 430 · 3 methods	Beneficial	**0.168%**	no reference	—	not in reference set	—
Butyricimonas faecihominis SGB 1786 · GlobDB GCF_033096445 · 4 methods	Beneficial	within *Butyricimonas* above	**+56%** vs typical carrier (0.141%)	17%	94th‡	—
Butyricimonas synergistica 1 method: MP3	Beneficial	within *Butyricimonas* above	**+298%** vs typical carrier (0.031%)	32%	79th	—
Lachnospira pectinoschiza also Lachnospira intestinalis SGB 5089 · GlobDB GCF_040095895 · mOTU 000439 · 8 methods	Beneficial	**0.164%**	**+41%** vs typical carrier (0.133%)	52%	62nd‡	—
Blautia argi SGB 4797 · GlobDB GCF_003287895 · mOTU 026327 · 8 methods	Conditional	**0.155%**	only 7 reference adults carry it: too few to rank against	7%	not in reference set	—
GGB3678 SGB4991 also Eisenbergiella intestinigallinarum SGB 4991 · GlobDB GCA_019120115 · mOTU 001130 · 7 methods	Conditional	**0.153%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Choladousia sp900757145 not yet formally named · known from genomes SGB 4784 · GlobDB GCA_900757145 · mOTU 036957 · 6 methods	Unknown	**0.148%**	only 9 reference adults carry it: too few to rank against	9%	not in reference set	—
Heritagella_A sp905215105 not yet formally named · known from genomes SGB 105428 · GlobDB GCA_905215105 · mOTU 001042 · 5 methods	Unknown	**0.145%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Jilunia laotingensis also Bacteroides laotingensis SGB 1802 · GlobDB GCF_014385165 · mOTU 023556 · 7 methods	Conditional	**0.136%**	no reference	—	not in reference set	—
Ruthenibacterium lactatiformans SGB 15271 · GTDB GCF_000949455.1 · mOTU 002162 · 7 methods	Unknown	**0.131%**	**+33%** vs typical carrier (0.074%)	82%	57th	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Ruthenibacterium GCMETA_00001082 not yet formally named · known from genomes GlobDB GCMETA_00001082 · 1 method: GlobDB sketch	Unknown	within *Ruthenibacterium* above	**+64%** vs typical carrier (0.073%)	89%	72nd‡	—
Ruthenibacterium GWHGLNR00000000 not yet formally named · known from genomes GlobDB GWHGLNR00000000 · 1 method: GlobDB sketch	Unknown	within *Ruthenibacterium* above	only 7 reference adults carry it: too few to rank against	7%	not in reference set	—
Coprobacter fastidiosus SGB 1963 · GlobDB GCF_030296935 · mOTU 001330 · 7 methods	Conditional	**0.131%**	**+114%** vs typical carrier (0.101%)	26%	68th	—
Coprobacter SPIREOTU_02584761 not yet formally named · known from genomes GlobDB SPIREOTU_02584761 · 1 method: GlobDB sketch	Conditional	within *Coprobacter* above	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Sellimonas intestinalis SGB 4617 · GTDB GCF_001280875.1 · mOTU 004016 · 6 methods	Conditional	**0.126%**	**+128%** vs typical carrier (0.038%)	15%	69th	—
Sellimonas GCMETA_00003065 not yet formally named · known from genomes GlobDB GCMETA_00003065 · 2 methods: GlobDB sketch, SingleM	Conditional	within *Sellimonas* above	**+200%** vs typical carrier (0.041%)	18%	>99th‡	—
Sellimonas HRGMV2_4332 not yet formally named · known from genomes 1 method: Kraken panel	Conditional	within *Sellimonas* above	no reference	—	not in reference set	—
CAG-273 sp900752095 not yet formally named · known from genomes SGB 6298 · GlobDB GCA_900752095 · mOTU 062279 · 7 methods	Unknown	**0.123%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
CAG-217 MGYG000001029 not yet formally named · known from genomes UHGG 1666 · 2 methods: Kraken, Kraken panel	Unknown	**0.117%**◦	no reference	—	not in reference set	—
Oliverpabstia sp000432335 not yet formally named · known from genomes SGB 4870 · GlobDB GCA_000432335 · mOTU 015007 · 9 methods	Unknown	**0.117%**	**+512%** vs typical carrier (0.028%)	1%	72nd	—
CAG-269 MOTU40_008527 not yet formally named · known from genomes GlobDB MOTU40_008527 · mOTU 008527 · 2 methods: GlobDB sketch, mOTUs	Unknown	**0.116%**◦	no reference	—	not in reference set	—
Akkermansia muciniphila_A UHGG 2775 · 2 methods: Kraken, SingleM	Beneficial	**0.115%**†	no reference	—	not in reference set	—
Ligilactobacillus salivarius formerly Lactobacillus salivarius SGB 7082 · GlobDB GCF_001435955 · mOTU 001881 · 7 methods	Beneficial	**0.114%**	**×10** vs typical carrier (0.022%)	5%	78th	—
Amedibacillus dolichus also Merdibacter merdigallinarum SGB 4030 · GlobDB GCA_019118705 · mOTU 005952 · 6 methods	Unknown	**0.114%**	**+142%** vs typical carrier (0.052%)	11%	73rd‡	—
Dysosmobacter welbionis also Oscillibacter welbionis SGB 15078 · GlobDB GCF_005121165 · mOTU 000902 · 8 methods	Conditional	**0.113%**	**–34%** vs typical carrier (0.190%)	83%	19th‡	—
Oscillibacter sp CAG 241 not yet formally named · known from genomes 1 method: MP3	Conditional	within *Oscillibacter* above	**–99%** vs typical carrier (0.170%)	61%	7th	—
Dorea sp AF36 15AT also Claveliimonas ammoniilytica SGB 4552 · GlobDB GCF_025567415 · mOTU 000335 · 6 methods	Conditional	**0.113%**	**+27%** vs typical carrier (0.118%)	37%	70th‡	—
Claveliimonas acetigenes UHGG 223 · 1 method: Kraken	Unknown	within *Claveliimonas* above	no reference	—	not in reference set	—

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Eisenbergiella massiliensis also Eisenbergiella porci SGB 4987 · GlobDB GCF_009696275 · mOTU 002293 · 8 methods	Conditional	**0.113%**	**×11** vs typical carrier (0.022%)	23%	88th	—
Faecalicatena orotica also Faecalicatena_A orotica SGB 4616 · UHGG 422 · 2 methods: Jun23, Kraken	Conditional	**0.110%**°	no reference	—	not in reference set	—
Eggerthella lenta SGB 14809 · GlobDB GCF_000024265 · mOTU 000459 · 9 methods	Opportunist	**0.104%**	**+189%** vs typical carrier (0.047%)	47%	75th	—
Eggerthella MOTU40_078089 not yet formally named · known from genomes GlobDB MOTU40_078089 · mOTU 078089 · 2 methods: GlobDB sketch, mOTUs	Opportunist	within *Eggerthella* above	no reference	—	not in reference set	—
Eggerthella sp CAG 209 not yet formally named · known from genomes 1 method: MP3	Opportunist	within *Eggerthella* above	no reference	—	not in reference set	—
Anaerobutyricum sp900754855 not yet formally named · known from genomes SGB 4680 · GlobDB GCA_900754855 · mOTU 062590 · 6 methods	Beneficial	**0.103%**	no reference	—	not in reference set	—
Anaerobutyricum GCMETA_00500231 not yet formally named · known from genomes GlobDB GCMETA_00500231 · 1 method: GlobDB sketch	Beneficial	within *Anaerobutyricum* above	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Enterocloster bolteae formerly Clostridium bolteae SGB 4758 · GTDB GCF_002234575.2 · mOTU 000552 · 7 methods	Opportunist	**0.102%**	**+126%** vs typical carrier (0.025%)	41%	65th	—
Lawsonibacter sp902363045 not yet formally named · known from genomes SGB 15158 · GlobDB GCA_902363045 · mOTU 002554 · 5 methods	Conditional	**0.101%**	**–37%** vs typical carrier (0.210%)	27%	33rd‡	—
Lawsonibacter MOTU40_007028 not yet formally named · known from genomes GlobDB MOTU40_007028 · mOTU 007028 · 2 methods: GlobDB sketch, mOTUs	Conditional	within *Lawsonibacter* above	no reference	—	not in reference set	—
Lawsonibacter sp019424865 not yet formally named · known from genomes GlobDB GCA_019424865 · mOTU 007829 · 3 methods	Conditional	within *Lawsonibacter* above	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Lawsonibacter sp949288405 not yet formally named · known from genomes GlobDB GCA_949288405 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	within *Lawsonibacter* above	no reference	—	not in reference set	—
Agathobacter rectalis formerly Eubacterium rectale SGB 4933 · GlobDB GCF_000020605 · mOTU 000006 · 8 methods	Beneficial	**0.097%**	**–92%** vs typical carrier (2.15%)	88%	16th	—
Roseburia inulinivorans 1 method: Kraken panel	Beneficial	within *Roseburia* above	no reference	—	not in reference set	—
Roseburia sp905209075 not yet formally named · known from genomes UHGG 1842 · 1 method: Kraken	Beneficial	within *Roseburia* above	no reference	—	not in reference set	—
Metalachnospira sp900546215 not yet formally named · known from genomes SGB 5183 · GTDB GCA_900546215.1 · mOTU 011788 · 5 methods	Unknown	**0.091%**	no reference	—	not in reference set	—
Metalachnospira GCMETA_00083624 not yet formally named · known from genomes GlobDB GCMETA_00083624 · 1 method: GlobDB sketch	Unknown	within *Metalachnospira* above	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Metalachnospira sp959604475 not yet formally named · known from genomes mOTU 026255 · 1 method: mOTUs	Unknown	within *Metalachnospira* above	no reference	—	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Bilophila wadsworthia SGB 15452 · GlobDB GCF_000701705 · mOTU 000928 · 8 methods	Opportunist	**0.089%**	**–13%** vs typical carrier (0.045%)	61%	46th	—
Ruminococcaceae bacterium KLE1738 not yet formally named · known from genomes 1 method: MP3	Conditional	**0.088%**◦	no reference	—	not in reference set	—
Anaerofustis stercorihominis SGB 4116 · GlobDB GCF_000154825 · mOTU 009021 · 7 methods	Unknown	**0.078%**	**×25** vs typical carrier (0.008%)	2%	>99th	—
Clostridiaceae bacterium also Alitiscatomonas aceti an earlier catalogue listed this population as Enterocloster hominis SGB 4721 · SGB 4770 · GlobDB GCF_025567195 · 8 methods	Conditional	**0.078%**	**–24%** vs typical carrier (0.121%)	48%	43rd	—
Alitiscatomonas SPIREOTU_02333767 not yet formally named · known from genomes GlobDB SPIREOTU_02333767 · 1 method: GlobDB sketch	Unknown	within *Alitiscatomonas* above	**–39%** vs typical carrier (0.114%)	58%	28th‡	—
Allofournierella massiliensis SGB 15280 · GTDB GCF_004345265.1 · mOTU 001819 · 5 methods	Unknown	**0.075%**	no reference	—	not in reference set	—
Allofournierella GCMETA_00520113 not yet formally named · known from genomes GlobDB GCMETA_00520113 · 2 methods: GlobDB sketch, SingleM	Unknown	within *Allofournierella* above	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Eubacterium maltosivorans SGB 4078 · GlobDB GCF_002441855 · mOTU 005163 · 6 methods	Conditional	**0.069%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Agathobaculum faecis also Agathobaculum ammoniilyticum SGB 14995 · GlobDB GCF_025567315 · mOTU 008954 · 6 methods	Beneficial	**0.069%**	**+37%** vs typical carrier (0.064%)	11%	82nd‡	—
Lactobacillus paragasseri an earlier catalogue listed this population as Lactobacillus gasseri SGB 7038 · GlobDB GCF_003584685 · mOTU 002235 · 8 methods	Beneficial	**0.069%**	**+302%** vs typical carrier (0.012%)	2%	71st	—
Lactobacillus casei group 1 method: MP3	Beneficial	within *Lactobacillus* above	no reference	—	not in reference set	—
Bacteroides xylanisolvens SGB 1867 · GlobDB GCF_000210075 · mOTU 000341 · 8 methods	Conditional	**0.065%**	**–41%** vs typical carrier (0.193%)	65%	40th	—
Lawsonibacter asaccharolyticus SGB 15154 · GlobDB MOTU40_001350 · GTDB GCA_003112755.1 · 8 methods	Conditional	**0.064%**	**+87%** vs typical carrier (0.050%)	66%	66th	—
Acutalibacter sp900548545 not yet formally named · known from genomes SGB 14842 · GlobDB GCA_900548545 · mOTU 002814 · 7 methods	Unknown	**0.061%**	**+80%** vs typical carrier (0.045%)	23%	74th‡	—
Acutalibacter sp900755895 not yet formally named · known from genomes UHGG 1729 · 1 method: Kraken	Unknown	within *Acutalibacter* above	no reference	—	not in reference set	—
Finegoldia magna mOTU 000404 · 2 methods: mOTUs, MP3	Unknown	**0.058%**◦	**+294%** vs typical carrier (0.003%)	1%	80th	—
Hominilimicola fabiformis GTDB GCF_020687385.1 · mOTU 000168 · 3 methods	Unknown	**0.058%**◦	no reference	—	not in reference set	—
Limosilactobacillus reuteri formerly Lactobacillus reuteri SGB 7095 · UHGG 2043 · 5 methods	Beneficial	**0.057%**	**+76%** vs typical carrier (0.138%)	1%	52nd	—
Limosilactobacillus reuteri L also Limosilactobacillus reuteri_E / reuteri_L GlobDB GCA_000236455 · mOTU 000611 · 4 methods	Beneficial	within *Limosilactobacillus* above	no reference	—	not in reference set	—

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Candidatus Borkfalkia ceftriaxoniphila also Borkfalkia ceftriaxoniphila SGB 14027 · GlobDB GCF_004134775 · mOTU 003253 · 6 methods	Unknown	**0.057%**	**+68%** vs typical carrier (0.041%)	22%	77th‡	—
Anaerotardibacter sp000436075 not yet formally named · known from genomes SGB 14322 · GlobDB GCA_000436075 · mOTU 000070 · 7 methods	Unknown	**0.052%**	**–64%** vs typical carrier (0.163%)	20%	10th‡	—
Gordonibacter pamelaeae SGB 14807 · GlobDB GCF_042665095 · mOTU 001297 · 8 methods	Beneficial	**0.051%**	**+232%** vs typical carrier (0.029%)	62%	77th	—
Heyndrickxia coagulans also Weizmannia coagulans SGB 7729 · UHGG 442 · 4 methods	Unknown	**0.051%**	no reference	—	not in reference set	—
Weizmannia faecalis GlobDB GCF_018101145 · mOTU 003472 · 3 methods	Unknown	within *Weizmannia* above	no reference	—	not in reference set	—
Gordonibacter urolithinfaciens mOTU 001343 · 1 method: mOTUs	Beneficial	within *Gordonibacter* above	no reference	—	not in reference set	—
Gordonibacter sp003339845 not yet formally named · known from genomes UHGG 3445 · 1 method: Kraken	Beneficial	within *Gordonibacter* above	no reference	—	not in reference set	—
UBA3402 sp003478355 not yet formally named · known from genomes UHGG 561 · 1 method: Kraken	Unknown	**0.049%**◦	no reference	—	not in reference set	—
Flintibacter hominis also Lawsonibacter aceti SGB 15143 · GlobDB GCF_025567565 · mOTU 003708 · 5 methods	Conditional	**0.045%**	**–34%** vs typical carrier (0.114%)	57%	28th‡	—
UBA1417 MGYG000004815 not yet formally named · known from genomes UHGG 6033 · 1 method: Kraken	Unknown	**0.042%**◦	no reference	—	not in reference set	—
Clostridium innocuum also Clostridium_AQ innocuum SGB 4037 · GlobDB GCF_012317185 · mOTU 001874 · 8 methods	Opportunist	**0.041%**	**▲ expanded and uncommon** **+372%** vs typical carrier (0.018%)	29%	81st	—
Lactobacillus acidophilus SGB 7044 · GlobDB GCF_034298135 · mOTU 008678 · 7 methods	Beneficial	**0.040%**	**+110%** vs typical carrier (0.039%)	3%	69th	—
Lachnospira sp900552795 not yet formally named · known from genomes SGB 5086 · GlobDB GCA_900552795 · mOTU 059060 · 7 methods	Beneficial	**0.040%**	no reference	—	not in reference set	—
Anaerotruncus colihominis GlobDB GCF_025146135 · mOTU 002391 · UHGG 79 · 4 methods	Conditional	**0.039%**◦	**–81%** vs typical carrier (0.091%)	11%	<1st‡	—
CAG-508 mOTUv4.0_058998 not yet formally named · known from genomes mOTU 058998 · 1 method: mOTUs	Unknown	**0.039%**◦	no reference	—	not in reference set	—
Massiliimalia massiliensis mOTU 027815 · 1 method: mOTUs	Unknown	**0.039%**◦	no reference	—	not in reference set	—
UMGS1975 sp959600905 not yet formally named · known from genomes GlobDB GCA_959600905 · mOTU 007360 · 3 methods	Unknown	**0.039%**◦	**–69%** vs typical carrier (0.034%)	17%	<1st‡	—
UMGS2037 sp900557385 not yet formally named · known from genomes UHGG 5353 · 1 method: Kraken	Unknown	**0.039%**◦	no reference	—	not in reference set	—
Holdemania massiliensis also Holdemania massiliensis_A SGB 4044 · GTDB GCF_000327285.1 · mOTU 003283 · 5 methods	Conditional	**0.039%**	no reference	—	not in reference set	—

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Holdemania GWHFFRE00000000 not yet formally named · known from genomes GlobDB GWHFFRE00000000 · 1 method: GlobDB sketch	Conditional	within *Holdemania* above	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Holdemania MGYG000000192 not yet formally named · known from genomes GlobDB MGYG000000192 · 1 method: GlobDB sketch	Conditional	within *Holdemania* above	**–63%** vs typical carrier (0.036%)	13%	<1st‡	—
UMGS1071 sp900542375 not yet formally named · known from genomes SGB 4121 · GlobDB GCA_900542375 · mOTU 003371 · 6 methods	Unknown	**0.037%**	**–18%** vs typical carrier (0.082%)	57%	39th‡	—
Mediterraneibacter SGB4598 also Mediterraneibacter intestinigallinarum SGB 4598 · GlobDB GCA_019119265 · mOTU 009175 · 6 methods	Conditional	**0.037%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Oscillibacter sp900548505 not yet formally named · known from genomes SGB 15070 · GlobDB GCA_900548505 · mOTU 003300 · 6 methods	Conditional	**0.034%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Escherichia coli 1 method: Kraken panel	Opportunist	**0.033%**◦	no reference	—	not in reference set	—
Prevotella sp900770395 not yet formally named · known from genomes 1 method: Kraken panel	Conditional	**0.032%**◦	no reference	—	not in reference set	—
Blautia sp An46 also Blautia ornithocaccae SGB 4800 · GlobDB GCA_019120495 · mOTU 009025 · 6 methods	Conditional	**0.030%**	no reference	—	not in reference set	—
Hominisplanchenecus faecis 1 method: Kraken panel	Unknown	**0.030%**◦	no reference	—	not in reference set	—
Eubacterium_G sp904420085 not yet formally named · known from genomes SGB 47610 · GlobDB GCA_904420085 · mOTU 006186 · 5 methods	Conditional	**0.030%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Wansuia hejianensis also Scatomonas hejianensis SGB 25431 · GlobDB GCF_014337215 · mOTU 026276 · 7 methods	Unknown	**0.030%**	≈ **typical** vs typical carrier (0.043%)	28%	57th‡	—
Agathobaculum butyriciproducens SGB 14993 · GlobDB GCF_003096535 · mOTU 000033 · 8 methods	Beneficial	**0.029%**	▼ **notably low** **–98%** vs typical carrier (0.214%)	89%	2nd	—
Alistipes indistinctus also Alistipes_A indistinctus SGB 2325 · GlobDB GCF_025144995 · mOTU 002384 · 7 methods	Conditional	**0.029%**	**–34%** vs typical carrier (0.058%)	41%	41st	—
Negativibacillus massiliensis SGB 14912 · GlobDB GCF_900148495 · mOTU 000248 · 7 methods	Unknown	**0.029%**	**–40%** vs typical carrier (0.088%)	17%	37th	—
Novisyntrophococcus sp900539175 not yet formally named · known from genomes UHGG 4246 · 1 method: Kraken	Unknown	**0.028%**◦	no reference	—	not in reference set	—
Eubacteriaceae bacterium Marseille Q4139 also CABSIM01 sp900541255 SGB 4718 · GlobDB GCA_018223415 · mOTU 005135 · 6 methods	Conditional	**0.028%**	no reference	—	not in reference set	—
Enterenecus sp900549885 not yet formally named · known from genomes SGB 15164 · GlobDB GCA_900549885 · mOTU 005092 · 6 methods	Unknown	**0.028%**	**–36%** vs typical carrier (0.084%)	33%	15th‡	—
Clostridiales bacterium also Lawsonibacter sp900066825 SGB 15159 · SGB 15158 · GlobDB GCA_003573975 · 6 methods	Conditional	**0.027%**	**–75%** vs typical carrier (0.349%)	12%	<1st‡	—

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bacterium 210917 DFI 7 65 also Pseudoscillospira sp018369915 SGB 14999 · GlobDB GCA_020555865 · mOTU 009017 · 6 methods	Unknown	**0.026%**	**–31%** vs typical carrier (0.055%)	34%	26th‡	—
Veillonella SPIREOTU_01978711 not yet formally named · known from genomes UHGG 6115 · 1 method: Kraken	Conditional	**0.026%**◦	no reference	—	not in reference set	—
UMGS775 sp900545325 not yet formally named · known from genomes SGB 14043 · GlobDB GCA_900545325 · mOTU 023385 · 6 methods	Unknown	**0.026%**	only 4 reference adults carry it: too few to rank against	4%	not in reference set	—
Oscillibacter valericigenes also Oscillibacter avistercoris SGB 15076 · GlobDB GCA_019119105 · mOTU 000776 · 6 methods	Conditional	**0.025%**	only 4 reference adults carry it: too few to rank against	4%	not in reference set	—
Enterocloster citroniae formerly Clostridium citroniae SGB 4761 · GTDB GCF_040545455.1 · mOTU 015990 · 6 methods	Opportunist	**0.023%**	**▲ expanded and uncommon** **+381%** vs typical carrier (0.015%)	36%	90th	—
GGB9469 SGB14863 also Hominenteromicrobium gallistercoris SGB 14863 · GlobDB GCA_019119145 · mOTU 001150 · 6 methods	Unknown	**0.023%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
MOTU40_062277 not yet formally named · known from genomes GlobDB MOTU40_062277 · mOTU 062277 · 2 methods: GlobDB sketch, mOTUs	Conditional	**0.022%**◦	no reference	—	not in reference set	—
Limiplasma sp023404345 not yet formally named · known from genomes SGB 73066 · GlobDB GCA_023404345 · mOTU 011840 · 4 methods	Unknown	**0.022%**	no reference	—	not in reference set	—
Blautia obeum formerly Ruminococcus obeum SGB 4811 · GlobDB GCF_025147765 · mOTU 000471 · 7 methods	Conditional	**0.022%**	**–76%** vs typical carrier (0.173%)	83%	21st	—
Hominisplanchenecus_A sp937965795 not yet formally named · known from genomes UHGG 1883 · 1 method: Kraken	Unknown	**0.022%**◦	no reference	—	not in reference set	—
Lysinibacillus fusiformis_D UHGG 5944 · 1 method: Kraken	Unknown	**0.021%**◦	no reference	—	not in reference set	—
Enterococcus faecium also Enterococcus_B faecium SGB 7967 · GlobDB GCF_029023785 · mOTU 001919 · 8 methods	Opportunist	**0.021%**	**■ uncommon** **+78%** vs typical carrier (0.040%)	5%	66th	—
Enterococcus_A devriesei UHGG 4845 · 1 method: Kraken	Opportunist	within *Enterococcus* above	no reference	—	not in reference set	—
Anaerococcus tetradius SGB 6674 · mOTU 006044 · 2 methods: Jun23, mOTUs	Unknown	**0.019%**◦	no reference	—	not in reference set	—
CAKTFW01 MOTU40_018619 not yet formally named · known from genomes GlobDB MOTU40_018619 · mOTU 018619 · 2 methods: GlobDB sketch, mOTUs	Unknown	**0.019%**◦	**–75%** vs typical carrier (0.033%)	13%	<1st‡	—
Finegoldia magna_H GlobDB GCF_000010185 · mOTU 000702 · 3 methods	Unknown	**0.019%**◦	no reference	—	not in reference set	—
JAAWCD01 MOTU40_058831 not yet formally named · known from genomes GlobDB MOTU40_058831 · mOTU 058831 · 2 methods: GlobDB sketch, mOTUs	Unknown	**0.019%**◦	no reference	—	not in reference set	—
Lachnoclostridium_A stercoravium mOTU 027561 · 1 method: mOTUs	Unknown	**0.019%**◦	no reference	—	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Neoanaerotignum_A sp944377425 not yet formally named · known from genomes mOTU 006348 · 1 method: mOTUs	Unknown	**0.019%**°	no reference	—	not in reference set	—
Ruminiclostridium_E siraeum mOTU 000039 · 1 method: mOTUs	Unknown	**0.019%**°	no reference	—	not in reference set	—
UBA3818 sp904397845 not yet formally named · known from genomes GlobDB GCA_904397845 · mOTU 011969 · 3 methods	Unknown	**0.019%**°	**–74%** vs typical carrier (0.051%)	20%	<1st‡	—
Limisoma sp000437795 not yet formally named · known from genomes 1 method: Kraken panel	Unknown	**0.019%**°	no reference	—	not in reference set	—
Pseudoflavonifractor phocaeensis also Lawsonibacter porci SGB 15149 · GlobDB GCF_041925195 · mOTU 000403 · 6 methods	Conditional	**0.019%**	**–75%** vs typical carrier (0.173%)	14%	<1st‡	—
Lentihominibacter faecis SGB 3957 · GlobDB GCF_014306095 · mOTU 000249 · 5 methods	Unknown	**0.018%**	**–69%** vs typical carrier (0.084%)	45%	9th‡	—
Lentihominibacter hominis GlobDB GCF_014385065 · 2 methods: GTDB sketch, GlobDB sketch	Unknown	within *Lentihominibacter* above	only 7 reference adults carry it: too few to rank against	7%	not in reference set	—
GGB42688 SGB59891 also Scatomonas merdavium SGB 59891 · GlobDB GCA_018716575 · mOTU 006182 · 4 methods	Unknown	**0.018%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Agathobaculum sp900557315 not yet formally named · known from genomes SGB 53508 · GlobDB GCA_900557315 · mOTU 009022 · 5 methods	Beneficial	**0.018%**	no reference	—	not in reference set	—
Pseudoflavonifractor gallinarum also Pseudoflavonifractor_A gallinarum SGB 29328 · GlobDB GCF_014982855 · mOTU 004160 · 5 methods	Unknown	**0.018%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Anaeromassilibacillus senegalensis also Anaeromassilibacillus stercoravium SGB 14894 · GTDB GCA_019118845.1 · mOTU 002815 · 7 methods	Unknown	**0.018%**	**–23%** vs typical carrier (0.019%)	31%	43rd	—
Anaeromassilibacillus MGYG000309509 not yet formally named · known from genomes GlobDB MGYG000309509 · 1 method: GlobDB sketch	Unknown	within *Anaeromassilibacillus* above	**–27%** vs typical carrier (0.042%)	23%	39th‡	—
Butyribacter sp000436755 not yet formally named · known from genomes UHGG 2716 · 1 method: Kraken	Unknown	**0.017%**°	no reference	—	not in reference set	—
Sellimonas catena SGB 4619 · GlobDB GCF_045063725 · mOTU 001706 · 6 methods	Conditional	**0.017%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Zhenpiania hominis also Hominibacterium hominis SGB 14854 · GlobDB GCF_014333425 · mOTU 008955 · 5 methods	Unknown	**0.017%**	**–43%** vs typical carrier (0.073%)	12%	8th‡	—
Hominibacterium sp019420305 not yet formally named · known from genomes GlobDB GCA_019420305 · 2 methods: GTDB sketch, GlobDB sketch	Unknown	within *Hominibacterium* above	only 6 reference adults carry it: too few to rank against	6%	not in reference set	—
Caccosoma sp900762575 not yet formally named · known from genomes UHGG 740 · 1 method: Kraken	Unknown	**0.017%**°	no reference	—	not in reference set	—
JBKXSZ01 sp047017005 not yet formally named · known from genomes GlobDB GCA_047017005 · 2 methods: GTDB sketch, GlobDB sketch	Unknown	**0.015%**°	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Bifidobacterium bifidum SGB 17256 · GTDB GCF_001025135.1 · mOTU 000343 · 6 methods	Beneficial	**0.015%**	**–90%** vs typical carrier (0.335%)	22%	19th	—
CAKTXU01 sp934615635 not yet formally named · known from genomes SGB 53517 · GlobDB GCA_934615635 · mOTU 026268 · 6 methods	Unknown	**0.015%**	**–13%** vs typical carrier (0.035%)	35%	26th‡	—
Pusillibacter faecalis also Oscillibacter faecalis SGB 15124 · GlobDB GCF_018408705 · mOTU 010912 · 5 methods	Conditional	**0.015%**	**–44%** vs typical carrier (0.068%)	39%	10th‡	—
UMGS1251 sp900549995 not yet formally named · known from genomes UHGG 4003 · 1 method: Kraken	Unknown	**0.014%**◦	no reference	—	not in reference set	—
GGB3534 SGB4722 also Enterocloster excrementigallinarum SGB 4722 · GlobDB GCA_019119955 · mOTU 005311 · 5 methods	Opportunist	**0.014%**	only 5 reference adults carry it: too few to rank against	5%	not in reference set	—
Erysipelotrichaceae bacterium LH1062 also Massiliomicrobiota sp. (SGB109603) SGB 109603 · 1 method: Jun23	Conditional	**0.014%**◦	no reference	—	not in reference set	—
Allocoprobacillus halotolerans SGB 109603 · GlobDB GCF_024399475 · mOTU 106389 · 4 methods	Unknown	**0.013%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Scybalocola_A sp902489925 not yet formally named · known from genomes SGB 4667 · GlobDB GCA_902489925 · mOTU 026357 · 5 methods	Unknown	**0.013%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Prevotella melaninogenica UHGG 4361 · 1 method: Kraken	Conditional	**0.013%**◦	no reference	—	not in reference set	—
Vescimonas sp000435975 not yet formally named · known from genomes SGB 15090 · GlobDB GCA_000435975 · mOTU 000210 · 5 methods	Unknown	**0.013%**	**–96%** vs typical carrier (0.929%)	17%	<1st‡	—
HGM13634 sp905198755 not yet formally named · known from genomes UHGG 2749 · 1 method: Kraken	Unknown	**0.012%**◦	no reference	—	not in reference set	—
Lacticaseibacillus paracasei formerly Lactobacillus paracasei SGB 7142 · GTDB GCF_000829035.1 · mOTU 004541 · 4 methods	Beneficial	**0.012%**	no reference	—	not in reference set	—
Egerieimonas GCMETA_00081355 not yet formally named · known from genomes GlobDB GCMETA_00081355 · 1 method: GlobDB sketch	Unknown	**0.012%**◦	no reference	—	not in reference set	—
HGM12998 sp900756495 not yet formally named · known from genomes SGB 15123 · GlobDB GCA_958477305 · mOTU 031558 · 6 methods	Unknown	**0.011%**	only 7 reference adults carry it: too few to rank against	7%	not in reference set	—
Hydrogeniiclostridium mannosilyticum SGB 14890 · GlobDB GCF_003268275 · mOTU 011796 · 6 methods	Unknown	**0.011%**	**–53%** vs typical carrier (0.056%)	12%	<1st‡	—
Alangreenwoodia sp905206455 not yet formally named · known from genomes SGB 3964 · GlobDB GCA_905206455 · mOTU 011126 · 5 methods	Unknown	**0.010%**	**–59%** vs typical carrier (0.038%)	34%	<1st‡	—
Parachristensenella GWHENXG00000000 not yet formally named · known from genomes GlobDB GWHENXG00000000 · 1 method: GlobDB sketch	Unknown	**0.010%**◦	only 5 reference adults carry it: too few to rank against	5%	not in reference set	—
Lachnospiraceae bacterium AF02 47 also Pullilachnospira_A gallistercoris SGB 4687 · GlobDB GCA_018712125 · mOTU 003373 · 6 methods	Conditional	**0.010%**	only 7 reference adults carry it: too few to rank against	7%	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Hydrogenoanaerobacterium saccharovorans also Avimicrobium caecorum SGB 15350 · GlobDB GCA_904394705 · mOTU 000421 · 6 methods	Unknown	**0.010%**	**–65%** vs typical carrier (0.063%)	49%	6th‡	—
Ruthenibacterium sp018380115 not yet formally named · known from genomes SGB 15273 · SGB 53514 · GTDB GCA_018380115.1 · 4 methods	Unknown	**0.010%**	no reference	—	not in reference set	—
Scatosoma sp900552625 not yet formally named · known from genomes SGB 13981 · GlobDB GCA_905209085 · mOTU 005923 · 6 methods	Unknown	**0.009%**	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Finegoldia magna_K GlobDB GCA_032548025 · 2 methods: GTDB sketch, GlobDB sketch	Unknown	**0.009%**◦	no reference	—	not in reference set	—
Clostridiaceae bacterium AF18 31LB also Hominiventricola aquisgranensis SGB 4767 · GTDB GCF_040096225.1 · mOTU 000512 · 5 methods	Conditional	**0.009%**	no reference	—	not in reference set	—
Hominiventricola GEMOTU_42051 not yet formally named · known from genomes GlobDB GEMOTU_42051 · 1 method: GlobDB sketch	Unknown	within *Hominiventricola* above	**–71%** vs typical carrier (0.134%)	15%	<1st‡	—
GGB140948 SGB47555 also Anaerobutyricum stercoripullorum SGB 47555 · GlobDB GCA_019114885 · mOTU 003732 · 4 methods	Beneficial	**0.009%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Lachnoclostridium sp An131 also Laedolimicola sp002160825 SGB 4775 · GlobDB GCF_002160825 · 4 methods	Unknown	**0.008%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Laedolimicola ammoniilytica GlobDB GCF_025567165 · mOTU 001713 · UHGG 197 · 4 methods	Unknown	within *Laedolimicola* above	**–66%** vs typical carrier (0.114%)	24%	<1st‡	—
Laedolimicola MOTU40_078016 not yet formally named · known from genomes GlobDB MOTU40_078016 · mOTU 078016 · 2 methods: GlobDB sketch, mOTUs	Unknown	within *Laedolimicola* above	no reference	—	not in reference set	—
Laedolimicola sp963559965 not yet formally named · known from genomes GlobDB GCA_963559965 · mOTU 026288 · 3 methods	Unknown	within *Laedolimicola* above	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Neoanaerotignum sp944378105 not yet formally named · known from genomes GlobDB GCA_944378105 · 2 methods: GTDB sketch, GlobDB sketch	Unknown	**0.008%**◦	no reference	—	not in reference set	—
Parasutterella excrementihominis GlobDB GCF_000205025 · 2 methods: GTDB sketch, GlobDB sketch	Conditional	**0.007%**◦	**–91%** vs typical carrier (0.106%)	17%	<1st‡	—
GGB3000 SGB3991 also Fimisoma avicola SGB 3991 · GlobDB GCA_018714945 · 4 methods	Unknown	**0.007%**	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Fimisoma sp900545135 not yet formally named · known from genomes UHGG 2598 · 1 method: Kraken	Unknown	within *Fimisoma* above	no reference	—	not in reference set	—
Hominilimicola GCMETA_00500345 not yet formally named · known from genomes GlobDB GCMETA_00500345 · 1 method: GlobDB sketch	Unknown	**0.006%**◦	**–90%** vs typical carrier (0.081%)	23%	<1st‡	—
Latilactobacillus sakei formerly Lactobacillus sakei GlobDB GCF_002370355 · 2 methods: GTDB sketch, GlobDB sketch	Unknown	**0.006%**◦	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—
Claveliimonas bilis SGB 4621 · GlobDB GCF_030296775 · mOTU 011576 · 5 methods	Unknown	**0.006%**	only 2 reference adults carry it: too few to rank against	2%	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Intestinimonas butyriciproducens SGB 15126 · GlobDB GCF_004154955 · mOTU 004129 · 7 methods	Beneficial	**0.005%**	**+14%** vs typical carrier (0.021%)	44%	53rd	—
Faecimorpha sp937892735 not yet formally named · known from genomes SGB 62935 · GlobDB GCA_937892735 · mOTU 078054 · 5 methods	Unknown	**0.005%**	**–41%** vs typical carrier (0.037%)	13%	8th‡	—
Intestinimonas massiliensis GlobDB GCF_001244995 · 2 methods: GTDB sketch, GlobDB sketch	Beneficial	within *Intestinimonas* above	**▼ low** **–46%** vs typical carrier (0.082%)	18%	6th‡	—
Intestinimonas MOTU40_028427 not yet formally named · known from genomes GlobDB MOTU40_028427 · mOTU 028427 · 2 methods: GlobDB sketch, mOTUs	Beneficial	within *Intestinimonas* above	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Enterocloster aldenensis SGB 4762 · GTDB GCF_042849935.1 · mOTU 002994 · 6 methods	Opportunist	**0.005%**	no reference	—	not in reference set	—
CAJFTX01 sp904420075 not yet formally named · known from genomes SGB 63362 · GlobDB GCA_904420075 · mOTU 027749 · 4 methods	Unknown	**0.005%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Fimisoma sp000435715 not yet formally named · known from genomes SGB 3996 · GlobDB GCA_000435715 · mOTU 000907 · 5 methods	Unknown	**0.004%**	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Enterocloster clostridioformis formerly Clostridium clostridioforme SGB 4760 · GTDB GCF_020297485.1 · mOTU 001557 · 7 methods	Opportunist	**0.004%**	**–60%** vs typical carrier (0.038%)	15%	31st	—
Blautia hansenii SGB 4794 · 3 methods	Conditional	**0.003%**	no reference	—	not in reference set	—
Woodwardiibium sp900760275 not yet formally named · known from genomes SGB 14954 · GlobDB GCA_900760275 · mOTU 001872 · 5 methods	Unknown	**0.003%**	**–70%** vs typical carrier (0.085%)	10%	<1st‡	—
Blautia marasmi also Blautia celeris an earlier catalogue listed this population as Blautia producta SGB 4786 · GlobDB GCF_014287615 · mOTU 002740 · 7 methods	Conditional	**0.003%**	**–71%** vs typical carrier (0.063%)	14%	7th‡	—
Gallintestinimicrobium sp900539715 not yet formally named · known from genomes SGB 4994 · GlobDB GCA_900539715 · mOTU 005100 · 6 methods	Unknown	**0.003%**	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Egerieisoma sp900552695 not yet formally named · known from genomes SGB 14151 · GlobDB GCA_900761125 · mOTU 014962 · 5 methods	Unknown	**0.003%**	only 8 reference adults carry it: too few to rank against	8%	not in reference set	—
Eisenbergiella sp900555195 not yet formally named · known from genomes SGB 4990 · GlobDB GCA_902492555 · 3 methods	Conditional	**0.003%**	only 3 reference adults carry it: too few to rank against	3%	not in reference set	—
Longicatena caecimuris SGB 4041 · GlobDB GCF_004341945 · mOTU 005141 · 4 methods	Unknown	**0.002%**	**–85%** vs typical carrier (0.111%)	28%	<1st‡	—
Clostridium methylpentosum also Massilioclostridium methylpentosum SGB 14962 · GlobDB GCA_000158655 · mOTU 011560 · 6 methods	Conditional	**0.002%**	**–46%** vs typical carrier (0.003%)	2%	33rd	—
GGB85112 SGB118983 also Ruthenibacterium komagatae SGB 118983 · mOTU 027394 · 2 methods: Jan26, mOTUs	Unknown	**0.002%**	no reference	—	not in reference set	—

ORGANISM	CLASS	SHARE	LEVEL	CARRIED BY	PERCENTILE	STRAIN
Butyricicoccus pullicaecorum also Butyricicoccus sp959018045 SGB 14985 · GTDB GCA_959018045.1 · mOTU 000600 · 5 methods	Beneficial	**0.001%**	▼ low **–95%** vs typical carrier (0.049%)	3%	5th	—
Butyricicoccus GCMETA_00500439 not yet formally named · known from genomes GlobDB GCMETA_00500439 · 1 method: GlobDB sketch	Unknown	within *Butyricicoccus* above	only 1 reference adult carry it: too few to rank against	1%	not in reference set	—
Waltera acetigignens also Acetatifactor acetigignens SGB 4914 · GTDB GCF_025567015.1 · 2 methods: GTDB sketch, Jan26	Unknown	**0.001%**	no reference	—	not in reference set	—
Acetatifactor SPIREOTU_02871567 not yet formally named · known from genomes GlobDB SPIREOTU_02871567 · 1 method: GlobDB sketch	Unknown	within *Acetatifactor* above	**–98%** vs typical carrier (0.506%)	71%	<1st‡	—
Enterocloster lavalensis SGB 4725 · GlobDB GCF_003024655 · mOTU 003061 · 5 methods	Opportunist	**0.001%**	**–81%** vs typical carrier (0.076%)	21%	5th‡	—
Merdisoma sp. (SGB105947) not yet formally named · known from genomes SGB 105947 · 1 method: Jun23	Unknown	**0.000%**°	no reference	—	not in reference set	—
Dysosmobacter sp. (SGB15073) not yet formally named · known from genomes SGB 15073 · 1 method: Jun23	Unknown	**0.000%**°	no reference	—	not in reference set	—

Share is the organism's percentage of the whole community, on one scale. † the marker catalogue measured these populations as one species; its total is divided among them in proportion to the reads that mapped uniquely to each genome. ° the marker catalogue has no entry for this organism; the share is its whole-genome reading brought onto the same scale, and comes out of the unclassified band. § includes a share the marker catalogue had read under a relative; the whole-genome competition showed those reads belong to this organism, so the share came with them (the relative's name is given beneath). Where that makes the share differ fivefold from what the reference catalogue read under this name, the percentile is left out. *within*: a population the whole-genome search tells apart inside its genus, whose reads the marker catalogue counted under the relatives listed above it; it carries no share of its own. *not comparable*: the reference catalogue's reading of this species differs more than fivefold from the share shown - it draws the species' boundary or its markers differently - so its percentile is not a statement about the share and is left out. Where an organism has been reclassified, the current name is shown with the previous one beneath it; *also* gives its name in the Genome Taxonomy Database. A name made of a genus and a genome identifier marks an organism known only from assembled genomes.

Organisms people ask about

The organisms most often named in gut-health writing, each looked up in your results. **23 of 34** were found in your sample. A blank here means this test searched and did not find it, which is not the same as the organism being absent from you — a low-abundance organism can sit below what one stool sample resolves.

ORGANISM	IN YOUR SAMPLE	LEVEL	NOTE
Akkermansia muciniphila	found	**0.846%** 92th pct	
Bifidobacterium	found	**4.453%**	genus total over 6 species
Bifidobacterium adolescentis	found	**0.435%** 85th pct	
Bifidobacterium longum	found	**1.838%** 95th pct	
Blautia	found	**0.611%**	genus total over 10 species
Blautia obeum	found	**0.022%** 50th pct	
Blautia wexlerae	found	**0.634%**	listed here as *Blautia A wexlerae*
Faecalibacterium	found	**0.314%**	genus total over 4 species
Faecalibacterium prausnitzii	found	**0.005%** 2th pct	
Lactobacillus	found	**0.109%**	genus total over 3 species
Roseburia	found	**0.097%**	genus total over 3 species
Ruminococcus	not found	—	searched, not detected
Ruminococcus bromii	not found	—	searched, not detected
Agathobacter rectalis/Eubacterium rectale	found	**0.097%** 35th pct	
Alistipes finegoldii	found	**0.859%** 95th pct	
Alistipes putredinis	found	**1.945%** 90th pct	
Anaerobutyricum hallii/Eubacterium hallii	not found	—	searched, not detected
Bacteroides	found	**6.929%**	genus total over 18 species

ORGANISM	IN YOUR SAMPLE	LEVEL	NOTE
Bacteroides fragilis	not found	—	searched, not detected
Bacteroides thetaiotaomicron	**found**	**0.574%** 86th pct	
Oxalobacter formigenes	not found	—	searched, not detected
Phocaeicola vulgatus	**found**	**11.748%** 93th pct	
Prevotella	**found**	**0.045%**	genus total over 2 species
Prevotella/Segatella copri	**found**	**0.045%**	genus total over 2 species
Streptococcus thermophilus	**found**	**0.175%** 96th pct	
Enterococcus faecalis	not found	—	searched, not detected
Escherichia coli	**found**	**0.033%**	
Klebsiella pneumoniae	not found	—	searched, not detected
Ruminococcus/Mediterraneibacter gnavus	**found**	**6.934%** 98th pct	
Bilophila wadsworthia	**found**	**0.089%** 83th pct	
Desulfovibrio	not found	—	searched, not detected
Fusobacterium nucleatum	not found	—	searched, not detected
Proteobacteria/Pseudomonadota	not found	—	
Methanobrevibacter smithii	not found	—	searched, not detected

What kind of organisms make up your community

46.8% conditional · **22.3%** beneficial · **9.3%** opportunist · **9.0%** unknown. Counted over 355 organisms in 162 genera. Organisms with no assigned class are shown as unclassified rather than dropped, and “unknown” describes the evidence, not the organism.

PART A — PATHOGENS: DISEASE-CAUSING ORGANISMS

7 Pathogens

A direct search of your sequencing reads for known disease-causing organisms — bacteria, protozoa, parasitic worms, microsporidia, fungi and viruses — against curated reference genomes. This is a **sequence** question, not a diagnosis: it reports what DNA is present, at what resolution, and how strong the evidence is. Section 18 takes every finding one at a time, and lists what could *not* be assessed in this sample.

BACTERIAL PATHOGENS	PROTOZOA PATHOGENS	PARASITIC WORMS	MICROSPORIDIA PATHOGENS	FUNGAL PATHOGENS	VIRAL PATHOGENS
1	2	1	none	none	none
found	traces only	trace only	nameable	nameable	found
7 normal or minor	2 normal or minor	118 of 125 searched	all 15 searched	5 normal or minor	30 of 32 searched

Disease-causing organisms — 1 named, 3 trace

Recognised causes of gut illness. A finding here is worth acting on, and worth showing to a doctor.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Clostridioides difficile	Bacterial pathogens	enough sequence to name it	**faint signal** 1.2 per million · 0.0001% of analysed DNA · 10 fragments in 9 regions	0.03% of its genome covered · 96.5% identity disease genes not found
Cryptosporidium canis	Protozoa pathogens	trace, not enough to confirm	**clear signal** 169 per million · 0.02% of analysed DNA · 1,357 fragments in 1 regions	0.04% of its genome covered · 99.0% identity
Entamoeba nuttalli	Protozoa pathogens	trace, not enough to confirm	**faint signal** 3.1 per million · 0.0003% of analysed DNA · 25 fragments in 1 regions	0.00% of its genome covered · 97.2% identity
Enterobius vermicularis	Parasitic worms	trace, not enough to confirm	**at the detection floor** 0.62 per million · well under a thousandth of a percent · 5 fragments in 1 regions	0.00% of its genome covered · 97.9% identity

Sequence that belongs to relatives, not to these organisms —

The screen aligns reads to its own 485 reference genomes. Where a relative that is present has no reference of its own here, the sequence the two organisms share lands on the target and reads as a signal. These rows failed that check: the organism inventory, which maps the same reads competitively against every relative it found, did not find them, and the coverage is piled into the regions relatives share rather than spread across the genome. The relative that carries the reads is named on each row.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Enterotoxigenic Bacteroides fragilis (ETBF)	Bacterial pathogens	shared sequence only — cannot tell apart shared sequence from relatives found in this sample; species not found by the organism inventory	**sequence from relatives** 19,839 fragments, shared with relatives · from Bacteroides uniformis (3.11%), Bacteroides stercoris (1.18%) · 19,839 fragments in 123 regions	2.82% of its genome covered · 98.0% identity disease genes not assessed

Opportunists — harmless in most people, a problem in some — 5 named, 3 trace

These live in many healthy guts and cause disease mainly when something else has changed: recent antibiotics, a weakened immune system, surgery, or a large overgrowth. Presence alone is not illness.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Burkholderia cepacia complex	Bacterial pathogens	enough sequence to name it	**faint signal** 1.2 per million · 0.0001% of analysed DNA · 10 fragments in 5 regions	0.01% of its genome covered · 95.4% identity
Candida krusei (Pichia kudriavzevii)	Fungal pathogens	enough sequence to name it	**at the detection floor** 0.62 per million · well under a thousandth of a percent · 5 fragments in 5 regions	0.01% of its genome covered · 100.0% identity
Clostridium innocuum	Bacterial pathogens	enough sequence to name it	**large signal** 1,170 per million · 0.12% of analysed DNA · 9,399 fragments in 469 regions	15.75% of its genome covered · 100.0% identity
Enterococcus faecium	Bacterial pathogens	enough sequence to name it	**clear signal** 319 per million · 0.03% of analysed DNA · 2,560 fragments in 256 regions	11.75% of its genome covered · 100.0% identity

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Streptococcus agalactiae	Bacterial pathogens	enough sequence to name it	**faint signal** 2.2 per million · 0.0002% of analysed DNA · 18 fragments in 11 regions	0.12% of its genome covered · 98.7% identity
Apophysomyces, human-infecting members	Fungal pathogens	trace, not enough to confirm	**at the detection floor** 0.25 per million · well under a thousandth of a percent · 2 fragments in 1 regions	0.00% of its genome covered · 97.1% identity
Brachyspira pilosicoli	Bacterial pathogens	trace, not enough to confirm	**small signal** 36 per million · 0.0036% of analysed DNA · 291 fragments in 2 regions	0.07% of its genome covered · 100.0% identity
Saksenaea, human-infecting members	Fungal pathogens	trace, not enough to confirm	**small signal** 28 per million · 0.0028% of analysed DNA · 227 fragments in 1 regions	0.00% of its genome covered · 98.7% identity

Common residents whose role is still unsettled — 1 named

Frequently found in healthy people. Research links some of them to disease, but whether they cause it is not established. Reported for completeness, not as a problem to fix.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Parvimonas micra	Bacterial pathogens	enough sequence to name it	**faint signal** 1.9 per million · 0.0002% of analysed DNA · 15 fragments in 8 regions	0.12% of its genome covered · 98.0% identity

Organisms that cause disease outside the gut — 1 trace

Stool is the wrong specimen for these. Sequence here does not indicate infection of another organ, and its absence does not rule one out.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Toxoplasma gondii	Protozoa pathogens	trace, not enough to confirm	**faint signal** 2.5 per million · 0.0002% of analysed DNA · 20 fragments in 1 regions	0.00% of its genome covered · 99.3% identity

Normal residents, food and environmental organisms — 2 named, 2 trace

Not pathogens. These are ordinary members of a healthy gut, or they arrive with food. They are listed because the screen searches for them by name, and finding them is expected rather than concerning.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Mediterraneibacter gnavus	Bacterial pathogens	enough sequence to name it	**large signal** 40,682 per million · 4.07% of analysed DNA · 326,669 fragments in 339 regions	85.13% of its genome covered · 100.0% identity
Saccharomyces cerevisiae	Fungal pathogens	enough sequence to name it	**small signal** 31 per million · 0.0031% of analysed DNA · 245 fragments in 100 regions	0.29% of its genome covered · 100.0% identity
Entamoeba dispar	Protozoa pathogens	trace, not enough to confirm	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 100.0% identity
Geotrichum candidum	Fungal pathogens	trace, not enough to confirm	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 100.0% identity

How to read the two strengths. Enough sequence to name it means distinct, organism-specific DNA passed every gate: the organism really is in the sample. **Trace** means matching DNA was found but too little of it to be sure of the species — it is neither a finding nor a clean negative, and at these amounts a close relative or a small amount of carry-over would look the same. **How much** is that organism's share of all the DNA analysed here; the bands describe the size of the signal in this sample and are not clinical thresholds, because no validated cut-off exists for stool sequencing.

154 further targets matched only sequence shared with relatives, so no organism can be named from it. Listed in section 18.

PART A — ESTIMATED BIOLOGICAL AGE OF BIOTA

8 Estimated biological age of biota

This page runs a **state-of-the-art machine learning ensemble** for microbiome age estimation: a linear carriage model over species presence and a **transformer neural network** applying multi-head attention across robust principal components, combined at fixed weights and trained on 4,070 healthy adults reconstructed from curatedMetagenomicData across 33 studies. **The interval, not the point, is the result.** It is the age this community reads as from its species patterns — a descriptive measure of the biota, not a health grade.

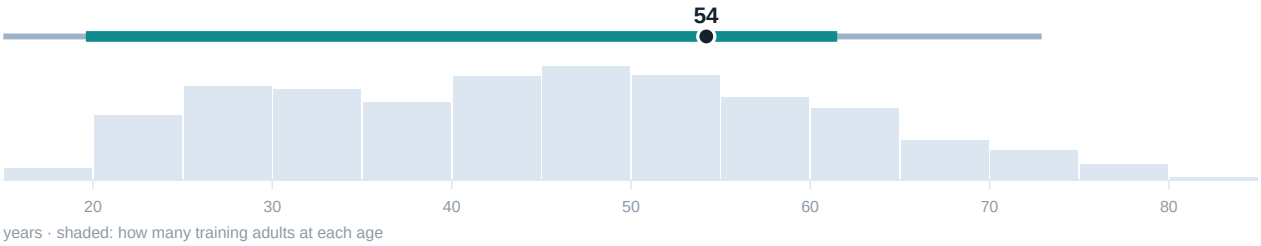
54
ESTIMATED AGE OF BIOTA (YEARS)

14–67
90% INTERVAL
the range the estimate sits in

—
DIFFERENCE FROM ACTUAL AGE
shown when actual age is on file

SUPPORTED

all gates passed · out-of-distribution check: in distribution · 83 of 92 detected species are in the model's namespace · mode: research



Dark bar: 80% interval. Light bar: 95% interval. Dot: point estimate. Red line: your actual age, when supplied. The shaded histogram is where the training adults sit; an estimate outside the well-populated range is less supported.

What moved this estimate

Local contributions of species to *this* prediction, restricted to features that were stable across validation folds. They describe the arithmetic of the estimate, not age biology, causality or anything to act on.

PUSHED THE ESTIMATE OLDER		PUSHED THE ESTIMATE YOUNGER	
Dorea sp CAG 317	+2.2 y	*Enterococcus faecium*	-1.7 y
Lactobacillus salivarius	+1.1 y	*Clostridium spiroforme*	-1.3 y
Firmicutes bacterium CAG 145	+1.1 y	*Finegoldia magna*	-1.0 y
Lawsonibacter asaccharolyticus	+1.1 y	*Roseburia intestinalis*	-0.8 y
Clostridium bolteae	+1.1 y	*Eubacterium eligens*	-0.8 y
Blautia wexlerae	+1.0 y	*Clostridium disporicum*	-0.7 y

How the estimate was built

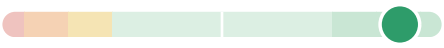
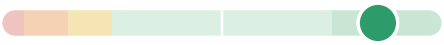
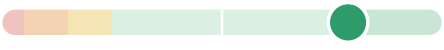
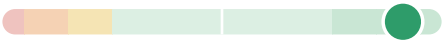
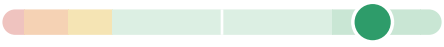
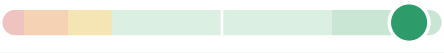
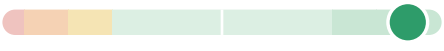
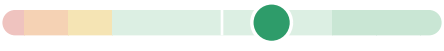
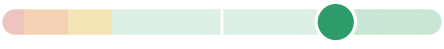
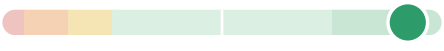
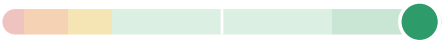
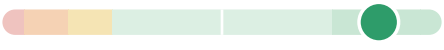
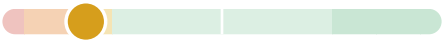
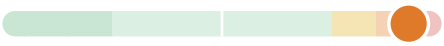
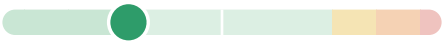
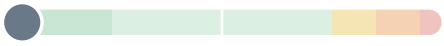
ALGORITHM	ESTIMATE	WEIGHT
Linear carriage model ridge regression over standardised species presence/absence	38.7 y	80%
Transformer transformer-based robust PCA: multi-head attention over robust principal components, with a linear regression head	48.0 y	20%
Model ensemble	**40.5 y**	
Microbiome health adjustment	**+13.7 y**	
Estimated age of biota	**54 y**	

PART A — WHAT YOUR BACTERIA CAN MAKE AND BREAK DOWN

9 Your Microbial Functions at a Glance

What your microbes can make, break down and transform.

Where your community's gene capacity for each function falls among 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort), grouped by what the function does and most notable first within each group. The bar is the same seven-band scale used throughout this report; the centre band is the middle half of the reference group. Every reading is *capacity* — the genes are present — not a measured amount of the compound.

FUNCTION	WHERE YOU SIT	READING	
FIBRE & DIETARY SUBSTRATES			
Dietary fibre and carbohydrate breakdown ▼		90th	HIGH
Your community carries enzymes for a wider range of dietary carbohydrates than most samples.			
Cellulose ▼		5th	NOTABLY LOW
General starch ▼		86th	HIGH
Resistant starch ▼		79th	ABOVE AVERAGE
Chitin ▼		91st	HIGH
Pectin ▼		84th	ABOVE AVERAGE
Inulin ▼		93rd	HIGH
Fructooligosaccharides ▼		93rd	HIGH
Galactooligosaccharides ▼		92nd	HIGH
Xylooligosaccharides ▼		61st	TYPICAL
Isomaltooligosaccharides ▼		76th	ABOVE AVERAGE
Lactose ▼		92nd	HIGH
Beta-glucans ▼		95th	NOTABLY HIGH
Arabinoxylan ▼		86th	HIGH
Galactomannan and guar ▼		19th	BELOW AVERAGE
FERMENTATION & CROSS-FEEDING			
Butyrate (SCFA) ▼		5th	NOTABLY LOW
Fewer of your gut bacteria carry the butyrate-producing genes than in most people, which is the less favourable direction here.			
Propionate (SCFA) ▲		92nd	HIGH
One genetic study reported that higher gut propionate production associates with increased Alzheimer's likelihood — the opposite direction to butyrate — and your capacity sits above most people's.			
Fermentation routes and cross-feeding		95th	NOTABLY HIGH
More fermentation-route capacity than most samples carry, across the acetate and lactate chemistry measured here.			
Hydrogen sulfide (H2S) ▲		29th	TYPICAL
Your sulfide-producing capacity is unremarkable. Some is normal and useful — sulfide is a signalling molecule in the gut wall.			
Methane (CH4) ▲		0	NOT DETECTED
No reads matched this pathway in your sample.			

Acetate production	95th	NOTABLY HIGH
Acetate made from hydrogen and CO2	62nd	TYPICAL
D-lactate production	95th	NOTABLY HIGH
L-lactate production	27th	TYPICAL
Turning lactate into butyrate ▼	20th	BELOW AVERAGE
Turning lactate into propionate ▲	70th	TYPICAL
Succinate production ▲	95th	NOTABLY HIGH
Turning succinate into propionate ▲	95th	NOTABLY HIGH
Turning propanediol into propionate ▲	85th	HIGH
Hydrogen production	48th	TYPICAL
Hydrogen used to make sulfide ▲	42nd	TYPICAL
Turning taurine into sulfide ▲	94th	HIGH

VITAMINS & NUTRIENTS

Folate (vitamin B9) ▼	95th	NOTABLY HIGH

Your gut community has more folate synthesis capacity than most people's, and unlike B12, some colonic folate is absorbed.

Vitamin K2 (menaquinone, MK-n) ▼	95th	NOTABLY HIGH

Your gut community has more menaquinone synthesis capacity than most people's.

Vitamin B1 (thiamine) ▼	95th	NOTABLY HIGH

More of your bacteria carry a complete route to thiamine than in most samples.

Vitamin B12 (cobalamin) ▼	93rd	HIGH

Your gut community has more B12 synthesis capacity than most people's.

Vitamin B3 (niacin and NAD precursors) ▼	93rd	HIGH

More of your bacteria carry the aspartate route to the niacin ring than in most samples.

Biotin (vitamin B7) ▼	89th	HIGH

Your gut community has more biotin synthesis capacity than most people's.

Riboflavin (vitamin B2) ▼	76th	ABOVE AVERAGE

Your gut community has more riboflavin synthesis capacity than most people's.

Vitamin B6 (pyridoxal phosphate) ▼	75th	ABOVE AVERAGE

More of your bacteria carry a complete B6 route than in most samples, by one or other of the two chemistries.

Vitamin B5 (pantothenate) ▼	74th	TYPICAL

Mid-range pantothenate synthesis capacity, which is the usual finding.

PROTEIN & NITROGEN METABOLISM

Branched-chain amino acids (valine, leucine, isoleucine) ▲ — 91st — HIGH

Your gut community has more branched-chain amino acid synthesis capacity than most people's.

Trimethylamine / TMAO ▲ — 76th — ABOVE AVERAGE

More of your gut bacteria can convert choline and carnitine into TMA, which your liver turns into TMAO.

Protein and nitrogen metabolism — 95th — NOTABLY HIGH

More protein-fermentation capacity than most samples. This chemistry yields branched-chain fatty acids, ammonia and phenols rather than butyrate, and it rises when fermentable fibre runs short.

Ammonia (from urea) ▲ — 8th — LOW

Less urease capacity than most people, meaning less bacterial breakdown of urea to ammonia in the colon.

Indole-3-propionate (IPA) ▼ — 86th — HIGH

Your gut has more indole-3-propionate capacity than most people, which is the favourable direction.

Peptide uptake — 77th — ABOVE AVERAGE

More peptide-uptake capacity than most samples carry. That can mean peptides are being used for growth rather than fermented, so it does not point in a single direction on its own.

p-Cresol / p-cresyl sulfate ▲ — 67th — TYPICAL

Your capacity to make p-cresol is unremarkable. This compound is cleared by the kidneys, so it matters most when kidney function is reduced; at typical capacity and normal kidney function there is little here to note.

Imidazole propionate (ImP) ▲ — 40th — TYPICAL

Nothing here points toward the insulin-signalling effects associated with high imidazole propionate.

Indole / indoxyl sulfate ▼ — 41st — TYPICAL

Tryptophan can be routed down several different paths and this test cannot tell which one dominates, so read this number together with indole-3-propionate below.

Protein and peptide breakdown — 84th — ABOVE AVERAGE

Amino-acid fermentation, including Stickland pairs — 42nd — TYPICAL

Aromatic precursor routes — 88th — HIGH

GUT-BRAIN & METABOLIC SIGNALLING

Tryptamine — 95th — NOTABLY HIGH

You carry more tryptamine-producing capacity than most people, which is unusual — the enzyme is rare.

GABA breakdown — 92nd — HIGH

More GABA-degrading capacity than most samples. Shown beside your GABA production reading, which is a separate measurement.

gamma-aminobutyric acid (GABA) ▼ — 85th — ABOVE AVERAGE

Your gut community has more GABA-producing capacity than most people's.

P9 protein (experimental) — 77th — ABOVE AVERAGE

Reads map to the experimentally tested sequence. That is a gene being present, not a protein being made or secreted, and it says nothing about your own GLP-1 or about GLP-1 medication.

Histamine ▲ — 77th — ABOVE AVERAGE

Your gut carries more histamine-producing capacity than most people's.

Dopamine (from L-DOPA) and tyramine (from tyrosine) — 62nd — TYPICAL

Your tyrosine decarboxylase capacity sits in the middle of the reference range.

GABA degradation	91st	HIGH
4-aminobutyrate to butyrate	66th	TYPICAL
Short-chain fatty acid signalling ▼	95th	NOTABLY HIGH

PLANT-COMPOUND CONVERSION

Urolithin A formation	83rd	PRESENT

Reads map to the published dehydroxylase. The gene is present, which is not the same as a measured urolithin A response to pomegranate or walnuts.

Glucosinolate to isothiocyanate conversion	73rd	TYPICAL

Mid-range presence of these enzymes against this reference. The route is carried by particular strains rather than being widespread, so a middling value is the ordinary finding.

Equol formation	0	NOT DETECTED

No reads matched this pathway in your sample.

MUCUS, BILE & OTHER TRANSFORMATIONS

Secondary bile acids ▲	95th	NOTABLY HIGH

More of your gut bacteria can convert your liver's bile acids into secondary bile acids like DCA than in most people.

Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin) ▲	85th	ABOVE AVERAGE

Your gut carries more beta-glucuronidase capacity than most people's, meaning more capacity to reverse the liver's glucuronide tagging and send compounds back into circulation.

Mucin glycans and lipid A	95th	NOTABLY HIGH

More mucin-foraging and lipid A capacity than most samples. Foraging capacity is not barrier damage, and lipid A genes are not an endotoxin level.

Oxalate (degraded, not produced) ▼	84th	ABOVE AVERAGE

Your gut carries more oxalate-degrading capacity than most people's, which is the favourable direction: more dietary oxalate broken down in the gut means less reaching the kidney.

Unconjugated bile acids	17th	BELOW AVERAGE

Your bile-salt-deconjugating capacity is on the low side. Since this step sits immediately upstream of secondary bile acid production, low capacity here combined with high capacity there would be biologically odd and would point to a problem with the run rather than a real finding.

Mucin glycan foraging	95th	NOTABLY HIGH
Lipid A acylation and modification	94th	HIGH
Hydroxysteroid dehydrogenase transformations	67th	TYPICAL

URATE & PURINES

Urate degradation	80th	ABOVE AVERAGE

More of the anaerobic urate-degrading route than most samples carry.

POLYAMINES

Putrescine formation	92nd	HIGH

More putrescine-forming capacity than most samples. Polyamines support gut lining renewal and are also elevated in some disease states, so this reading does not point in a single direction on its own.

Agmatine formation	87th	HIGH

More arginine decarboxylase capacity than most samples carry.

Spermidine formation 85th HIGH

More spermidine-forming capacity than most samples carry, counted across both routes.

GLUTATHIONE & ETHANOL

Glutathione synthesis 81st ABOVE AVERAGE

More bacterial glutathione synthesis capacity than most samples carry.

Microbial ethanol formation 77th ABOVE AVERAGE

More ethanol-forming gene capacity than most samples carry. Carrying adhE is common and is not the same as being a high-producing strain, which is what the fatty liver disease studies identified.

TOXINS & VIRULENCE

Colibactin and bacterial toxins (virulence factors, not metabolites) 76th PRESENT

▲

One or more of these virulence genes was detected in your sample.

▲ research associates higher levels with worse outcomes · ▼ research associates higher levels with better outcomes · no arrow: the evidence is mixed, so the reading is shown in neutral grey. Section 19 has the findings, the references and the research on changing each pathway; every row above is a link to its card.

PART A — RESEMBLANCE TO PUBLISHED DISEASE PATTERNS

10 Resemblance to published disease patterns at a glance

For every pattern in the library: how closely this community resembles the group-level pattern published for people with that condition, as a percentile of the reference adults — the same scale as everything else. Resemblance is not a diagnosis and not a probability of disease. Every scored pattern has a full page, and the research on what has been tried for that condition, in Part B.

The shape of the result

BROADLY RAISED

A general-disturbance picture, not a specific pattern.

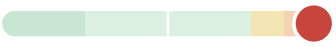
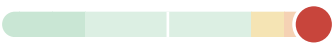
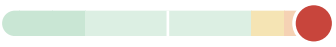
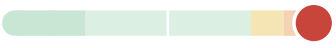
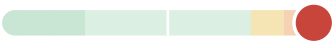
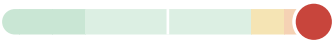
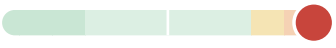
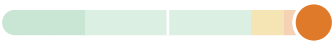
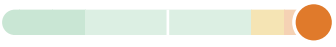
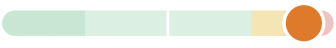
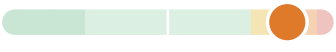
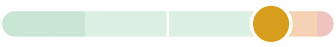
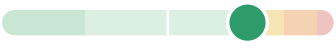
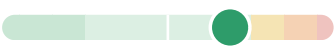
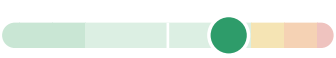
23 of 33 scored profiles sit above the typical band, and the features they rest on are shared across many conditions (mean specificity 0.34). When most profiles score high at once, the honest reading is that the community is disturbed in the way many illnesses disturb it — the general dysbiosis index is the headline, not any one disease. GMWI2 general dysbiosis index: -1.35 (strongly dysbiotic).

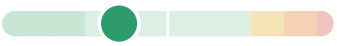
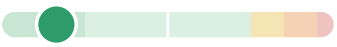
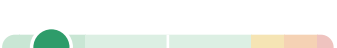
33	23	0.34	0	-1.35
PATTERNS SCORED	ABOVE TYPICAL	SPECIFICITY OF THOSE	NOT SCORED	GENERAL GUT-HEALTH INDEX
of 33 in the library	above 75th pct · 18 notably…	part from challenge-cohort…	0 declined · 0 unmeasurable	broadly disturbed

Every pattern in the library, most resemblant first

Maturity is how far the underlying research has come (P0 preclinical → P5 clinical-grade); **evidence** lists which kinds of measurement the pattern uses. A pattern marked *declined* did not meet its own conditions for a meaningful score; *not measurable* means it needs an engine this pipeline does not yet run. Every scored pattern has a full page in Part B, and its row here is a link to it.

PATTERN	MATURITY · EVIDENCE	RESEMBLANCE	READING
Parkinson's disease — gut pattern parkinsons_disease competes with t2d, uc	P3 replicated species · genes · carriers		>99th NOTABLY HIGH
Type 2 diabetes — gut pattern (metformin-gated) type_2_diabetes competes with parkinsons, uc	P3 replicated species · genes · carriers		>99th NOTABLY HIGH
Ulcerative colitis — gut pattern ulcerative_colitis competes with parkinsons, t2d	P3 replicated species · diversity · genes · carriers		>99th NOTABLY HIGH
Inflammatory bowel disease (composite) — gut pattern inflammatory_bowel_disease competes with parkinsons, t2d	P3 replicated species · diversity · genes · carriers		>99th NOTABLY HIGH
Long COVID chronic-state community similarity post_acute_sequelae_of_covid19 competes with parkinsons, t2d	P1 one small study species · genes · diversity		>99th NOTABLY HIGH
Ankylosing spondylitis — gut pattern ankylosing_spondylitis competes with parkinsons, t2d	P2 human cohort species · genes · carriers		>99th NOTABLY HIGH
Preclinical amyloid positivity — gut pattern alzheimers_disease competes with parkinsons, t2d	P2 human cohort species · genes		>99th NOTABLY HIGH
Colorectal adenoma — gut pattern colorectal_adenoma competes with parkinsons, t2d	P3 replicated species · genes		>99th NOTABLY HIGH
Chronic kidney disease — gut pattern chronic_kidney_disease competes with parkinsons, uc	P2 human cohort species · genes · carriers		99th NOTABLY HIGH
Crohn's disease — gut pattern crohns_disease competes with parkinsons, t2d	P3 replicated species · diversity · genes · carriers		99th NOTABLY HIGH

PATTERN	MATURITY · EVIDENCE	RESEMBLANCE	READING
Irritable bowel syndrome — gut pattern irritable_bowel_syndrome competes with parkinsons, t2d	P2 human cohort species · diversity · genes		98th NOTABLY HIGH
Androgenetic alopecia — gut pattern (sex-stratified) androgenetic_alopecia competes with parkinsons, t2d	P1 one small study species · genes		98th NOTABLY HIGH
Rheumatoid arthritis — gut pattern rheumatoid_arthritis competes with parkinsons, t2d	P3 replicated species · strain · genes		98th NOTABLY HIGH
Mild cognitive impairment — gut pattern alzheimers_disease competes with parkinsons, t2d	P2 human cohort species · pathways		97th NOTABLY HIGH
Systemic lupus erythematosus — shared inflammatory gut pattern (not lupus-specific) systemic_lupus_erythematosus competes with parkinsons, t2d	P1 one small study species · strain		96th NOTABLY HIGH
Clinical Alzheimer's disease — stage-specific gut pattern alzheimers_disease competes with parkinsons, t2d	P2 human cohort species · pathways · genes · carriers		96th NOTABLY HIGH
IBS with diarrhoea — gut pattern irritable_bowel_syndrome competes with parkinsons, t2d	P1 one small study species · genes		96th NOTABLY HIGH
Major depressive disorder — gut pattern major_depressive_disorder competes with parkinsons, t2d	P2 human cohort species · genes		95th NOTABLY HIGH
ME/CFS-like gut microbiome similarity me_cfs competes with parkinsons, t2d	P2 human cohort species · genes · diversity		94th HIGH
Atherosclerotic cardiovascular disease — gut pattern atherosclerotic_cardiovascular_disease competes with parkinsons, uc	P3 replicated species · genes · carriers		94th HIGH
Multiple sclerosis — gut pattern multiple_sclerosis competes with parkinsons, t2d	P3 replicated species · genes · carriers		91st HIGH
IBS mixed type — gut pattern irritable_bowel_syndrome competes with parkinsons, t2d	P1 one small study species · genes		86th HIGH
Symptomatic dermographism (chronic inducible hives from skin stroking) — translated 16S pattern symptomatic_dermographism competes with parkinsons, t2d	P1 one small study species		81st ABOVE AVERAGE
IBS with constipation — gut pattern irritable_bowel_syndrome competes with parkinsons, t2d	P1 one small study species · genes		74th TYPICAL
Chronic spontaneous urticaria (chronic hives) gut-microbiome similarity chronic_spontaneous_urticaria competes with parkinsons, t2d	P2 human cohort species · carriers		69th TYPICAL
Metabolic dysfunction-associated steatotic liver disease — gut pattern masld competes with parkinsons, t2d	P2 human cohort species · diversity · genes · carriers		68th TYPICAL

PATTERN	MATURITY · EVIDENCE	RESEMBLANCE	READING
Hypertension — gut pattern hypertension competes with parkinsons, uc	P2 human cohort species · diversity · genes		57th TYPICAL
Liver cirrhosis — gut pattern liver_cirrhosis competes with parkinsons, t2d	P3 replicated species · diversity · genes · carriers		52nd TYPICAL
Coeliac disease — gut pattern celiac_disease competes with parkinsons, t2d	P2 human cohort species · CAZymes · genes		51st TYPICAL
Type 1 diabetes — gut pattern type_1_diabetes competes with parkinsons, t2d	P2 human cohort species · diversity · genes · carriers		46th TYPICAL
Alopecia areata — gut pattern alopecia_areata competes with parkinsons, t2d	P1 one small study species · pathways · genes		35th TYPICAL
Obesity — ecological pattern obesity competes with parkinsons, uc	P3 replicated diversity · species · genes · carriers		16th BELOW AVERAGE
Colorectal cancer-associated community pattern colorectal_cancer · validation control competes with parkinsons, t2d	P3 replicated species · genes · carriers		15th LOW

PART A — BIOFILM-RELATED POTENTIAL

11 Biofilm-related potential

Biofilms are organised communities of microbes held together by a shared matrix. They are a normal way for microbes to live, not a disease in themselves. Some support stable colonisation and a protective relationship with the intestinal lining; others help organisms persist, grow too close to tissue, or take part in inflammation. **This page reports concerning and protective readings separately, so that one never cancels the other.**

Concerning biofilm-forming (bad)

H-C · BIOFILM-ASSOCIATED COMMUNITY PATTERN

60 / 100 percentile

◆ middle reference range

Scope: limited to Biofilm-associated community pattern. Reference: 1,127 adults. Panel coverage: 1 of 2 features detected.
Reference interval 58–63 (200 resamples of the reference cohort).

What drove this reading

ORGANISM	LEVEL	RANK
Mediterraneibacter gnavus	13.29%	99th
Escherichia coli and its close relatives pooling every method: Escherichia coli 0.033% (section 6)	0% not detected by the reference catalogue	10th

Protective gut lining (good)

P-E · PROTECTIVE ECOLOGICAL SUPPORT (PARTIAL PANEL)

55 / 100 percentile

◆ middle reference range

Scope: limited to Protective ecological support (partial panel). Reference: 1,127 adults. Panel coverage: 3 of 3 features detected.
Reference interval 52–58 (200 resamples of the reference cohort).

What contributed to this score

ORGANISM	LEVEL	RANK
Blautia species B. sp CAG 257 1.6%, B. sp N6H1 15 0.27%, B. obeum 0.042%, B. producta 0.0054%	1.95%	91st
Coprococcus species	0.71%	64th
Faecalibacterium prausnitzii	0.01%	4th

What these numbers mean

60 / 100 — **Biofilm-associated community pattern**
You are around the middle of the reference range: 60 means roughly 60 in every 100 reference adults measure lower than you, and the rest measure higher. Higher means more of your community is made up of organisms that take part in biofilm growth.
Made up of ***Mediterraneibacter gnavus*** 13.3% (99th); ***Escherichia coli and its close relatives*** 0%, not detected by the reference catalogue (10th) — the organisms found are what lift the reading. pooling every method: Escherichia coli 0.033%

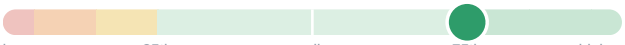
55 / 100 — **Protective ecological support (partial panel)**
You are around the middle of the reference range: 55 means roughly 55 in every 100 reference adults measure lower than you, and the rest measure higher. Higher means more of the community that keeps the gut lining intact and well fed.
Made up of ***Blautia species*** 1.9% (91st); ***Coprococcus species*** 0.7% (64th); ***Faecalibacterium prausnitzii*** 0.0% (4th).

A middle reading is the common one and is not a finding. These two numbers are never added, subtracted or averaged into a single biofilm score: a high concerning reading and a high protective reading can both be true at once, and collapsing them would hide exactly the case worth seeing. Both are read from which organisms are present and in what proportion, against 1,127 reference adults.

12 Your gut mycobiome

The fungi detected in your stool — and what the evidence says about them.

Myco-Score (experimental)



75/100

Higher is healthier. From a neutral 50: +20 opportunistic fungi absent or very little; +5 Saccharomyces present. Policy weights, versioned (myco-score-v2); the evidence rules and every contribution are traced in the detail section.

0.0096%	17	0.0001%	961
FUNGAL DNA	FUNGI FOUND	OPPORTUNISTIC FUNGI	SPECIES SEARCHED
773 of 8.0M read pairs	2 confirmed, 6 likely, 9 trace	of all read pairs, 1 found	1,696 genomes, 3 methods

Every one of the 8,029,909 quality-passed read pairs was searched against 961 fungal species (1,696 genomes, 321 genera) plus a single-copy-marker search and the targeted pathogen screen. Fungal signal is the share of all pairs confidently assigned to fungi after competition against near neighbours and bacterial decoys.

What was searched for, and what was found

GROUP	SPECIES SEARCHED	FOUND IN YOUR SAMPLE
Candida and relatives	45 (408 genomes)	2 · *Pichia kudriavzevii* (provisional) 5, *Debaryomyces* (trace) 2
Saccharomyces and brewing/baking yeasts	23 (86 genomes)	3 · *Saccharomyces cerevisiae* 239, *Brettanomyces bruxellensis* 41, *Lachancea* (trace) 4
Malassezia and other skin yeasts	41 (68 genomes)	none
Aspergillus	137 (254 genomes)	1 · *Aspergillus* (trace) 238
Penicillium	113 (223 genomes)	3 · *Penicillium* (provisional) 5, *Penicillium* (trace) 15, *Talaromyces* (trace) 2
Mucor, Rhizopus and relatives	16 (31 genomes)	3 · *Mucor* (genus) 76, *Rhizopus* (provisional) 100, *Rhizomucor* (provisional) 29
Edible mushrooms	3 (3 genomes)	none
Other moulds	73 (79 genomes)	none
All other fungi	510 (544 genomes)	5 · *Schizothecium* (provisional) 11, *[Candida] boidinii* (trace) 3, *Saksenaea, human-infecting members* (trace) 227, *Apophysomyces, human-infecting members* (trace) 2, *Geotrichum candidum* (trace) 1

Is this a normal amount of fungal DNA? context, not a target

In the largest survey of healthy adults (HMP, 147 volunteers), about **0.01%** of shotgun reads mapped to fungal genomes. Yours: **0.0096%** — within tenfold of the HMP figure. Fungi are a small fraction of stool DNA in everyone; there is no validated healthy range, and less is not better or worse. Genome size, cell walls and yesterday's meal all move this number.

GENUS	HEALTHY ADULTS	USUALLY MEANS	YOU
Saccharomyces	in 97% of samples	bread, beer, wine, probiotics	**found**
Malassezia	in 88% of samples	skin yeast; stool carriage usually from skin	not detected
Candida	in 81% of samples	common resident; C. albicans most often	not detected
Penicillium	commonly reported	mould on cheese and cured food; also inhaled from air	trace
Debaryomyces	commonly reported	salt-tolerant yeast of cheese rind and cured meat	trace
Cladosporium	commonly reported	one of the commonest airborne moulds; swallowed daily	not detected
Aspergillus	commonly reported	airborne mould; stool reads usually mean exposure, not colonisation	trace
Pichia	commonly reported	fermentation yeast; competes with Candida in the gut	provisional
Kluyveromyces	commonly reported	dairy yeast of kefir and some cheeses	not detected
Cyberlindnera	commonly reported	fermentation yeast, often from plant food	not detected
Geotrichum	commonly reported	rind yeast of soft cheese; a normal transient	trace
Rhodotorula	commonly reported	pink environmental yeast of damp surfaces and produce	not detected

HMP prevalence is ITS2 amplicon on 317 stool samples from 147 healthy US adults; the shotgun read fraction is method-specific. Neither is a healthy range or a target: less fungal DNA is not better, and more is not worse. Source: Nash et al. 2017, Microbiome.

Fungi detected

8 findings; each has a page in section 22

FUNGUS	WHAT IT IS	DETECTION	FRAGMENTS	SHARE	STRAIN	EVIDENCE
Saccharomyces cerevisiae	species with studied probiotic strains	SUPPORTED	239 both lanes	86%	unresolved insufficient discriminatory coverage	●● ●●
Brettanomyces bruxellensis	food-associated	SUPPORTED	41 genome	14%	species only reference panel missing	—
Mucor	environmental or food mould	GENUS ONLY	76 genome	—	not assessed	●
Rhizopus	environmental or food mould	PROVISIONAL	100 genome	—	not assessed	●
Rhizomucor	environmental or food mould	PROVISIONAL	29 genome	—	not assessed	—
Schizothecium reported as Schizothecium sp. R5324	environmental or food mould	PROVISIONAL	11 genome	—	not assessed	—
Pichia kudriavzevii	common resident, opportunistic potential	PROVISIONAL	5 genome	—	not assessed	—
Penicillium reported as Penicillium sp. B98-03	environmental or food mould	PROVISIONAL	5 genome	—	not assessed	—

Traces, below the rule and not interpreted: *Aspergillus* 238 fragments (whole genome); *Penicillium* 15 fragments (whole genome); *Lachancea* 4 fragments (whole genome); *[Candida] boidinii* 3 fragments (whole genome); *Debaryomyces* 2 fragments (whole genome); *Talaromyces* 2 fragments (whole genome); *Saksenaea, human-infecting members* 227 fragments (pathogen screen); *Apophysomyces, human-infecting members* 2 fragments (pathogen screen); *Geotrichum candidum* 1 fragment (pathogen screen). One or two reads consistent with a species is a trace, not a detection.

What kind of fungi

share of fungal DNA by organism class



● food & drink yeasts 55% ● opportunistic yeasts 1% ● environmental & food moulds 44%

Composition of the quantified fungal community

genome-coverage shares among supported species; not cell counts



Plus 493 fungal fragments placed only at genus level or below the support rule — counted in the signal, not inserted into the shares.

Potentially beneficial evidence

None activated under the current policy. Unscored means unscored, not benign.

Potential concerns

None activated under the current policy. Unscored means unscored, not benign.

Bacteria that hold Candida in check

from your bacterial results; mouse and laboratory evidence

● *Bacteroides thetaiotaomicron* (0.57%, 71th pct) · ● *Blautia producta* (<0.01%, 7th pct) · ● *Lacticaseibacillus rhamnosus* (0.18%, 100th pct). Relevant here because an opportunistic yeast was found. Depleting these bacteria (antibiotics especially) is the best-described route to Candida expansion; their presence is context, not protection proven in you.

What to do next

• Food- or probiotic-associated yeasts were found. Recording yeast probiotic and fermented-food intake and the time since the last dose separates a transient passenger from a persistent resident; a repeat sample after a break is the standard way to tell. Do not stop a beneficial therapy on a sequence result alone.

• This module reads stool DNA. It does not establish small-intestinal fungal overgrowth, mucosal invasion, fungemia, viability, growth phase or toxin production. No eradication target for all fungi follows from any result here, and no antifungal is selected by relative abundance.

Section 22 has each fungus in full: lanes, strain loci searched, evidence with its label, and the complete score trace.

PART A — SKIN-CONDITION RESEARCH PATTERNS

13 Gut-skin axis

Gut-skin axis (beta)



50/100

Higher is the better direction. The weighted middle of 8 published panels, each weighted by how much of its own marker list this sample could measure. Carried mostly by Psoriatic disease, skin and joint pooled (52), Hidradenitis suppurativa (52), Severe plaque psoriasis (50).

Researchers have compared the stool communities of people with several inflammatory skin diseases against healthy controls. This section asks, for each published list of organisms: **does this sample move in the direction that study reported?** There is no established, uniquely diagnostic stool signature for any of these conditions. A high reading is not a diagnosis and a low one does not rule anything out.

The scale is **pattern concordance** from low to high, drawn in neutral greys because it is agreement with a study's reported directions and not a health grade. It carries no per cent sign, because it is not a percentage or a probability. Where some organisms could not be measured, two separate figures appear: the index over the features that *were* measured, and the range the whole declared panel could occupy. The second is a coverage range, not a margin of error on the first.

CONDITION AND COMPARISON	PANEL	INDEX	FEATURES	PANEL RANGE	EVIDENCE
Plaque psoriasis psoriasis vs controls	Plaque psoriasis — Deng 2026 single-marker SGB4348	— (no usable features)	0/1 single-marker result	0–100	tested in the same centre, not externally validated
Plaque psoriasis psoriasis vs healthy	Plaque psoriasis — Chang 2022 seven-taxon panel	**73** of 100	1/7 one feature only	10–96	one cohort, selected subset of features
Plaque psoriasis severe psoriasis vs healthy	Severe plaque psoriasis — Sá 2026 three-taxon panel	**49** of 100	2/3	33–66	small cohort with heavy treatment confounding
Psoriatic disease (skin and joint forms pooled) pooled plaque psoriasis and psoriatic arthritis vs healthy	Psoriatic disease, skin and joint pooled — Xiao 2024 three species	**52** of 100	3/3	52–52	pooled psoriasis and psoriatic arthritis contrast
Psoriatic arthritis treatment naive psoriatic arthritis vs healthy	Psoriatic arthritis — Liu 2024 nine gene families	— (none of the organisms could be measured)	0/9	0–100	one cohort, nominal significance only
Psoriatic arthritis psoriatic arthritis vs skin only psoriasis	Psoriatic arthritis versus skin-only psoriasis — KEGG 00072 capacity	— (none of the organisms could be measured)	0/1 single-marker result	0–100	comparison between two disease subtypes
Atopic dermatitis, adult adult atopic dermatitis vs healthy	Adult atopic dermatitis (eczema) — Wang 2023 thirteen slots, translated from 16S	**48** of 100	5/13	18–80	16S-derived pattern applied to shotgun measurements
Non-segmental vitiligo advanced nonsegmental vitiligo vs healthy	Advanced non-segmental vitiligo — Luan 2023 eleven species	**38** of 100	2/11	7–89	one cohort, selected subset of features
Non-segmental vitiligo active spreading vitiligo vs external controls	Actively spreading vitiligo — Ju 2025 four species, exploratory	**24** of 100	1/4 one feature only	6–81	cases and controls from different studies
Hidradenitis suppurativa hidradenitis suppurativa vs healthy	Hidradenitis suppurativa — Ogut 2022 single genus, translated from 16S	**52** of 100	1/1 single-marker result	52–52	one feature at nominal significance

Panels for the same condition are shown side by side and never averaged into one condition score: they come from different cohorts, they disagree in places, and averaging would hide that. One organism appearing in several panels is one measurement, not several agreeing results.

14 What you can do about it

This report found **54** things worth doing something about. For **48** of them there is published research on changing them — **211** findings in all, each one named below with the exact food, compound, supplement, probiotic, herb, medication or procedure that was studied, where the finding comes from, and what was actually seen. Each problem is set out the same way: what was found, what might move it the right way, and what has been shown to move it the wrong way.

WHAT WAS FOUND	WHERE	OPTIONS ON RECORD
1. Clostridioides difficile was detected by the pathogen screen	PATHOGEN SCREEN · section 18	**52** · 2 to avoid
2. Mediterraneibacter gnavus is overgrown	ORGANISM · section 16	**13** · 12 to avoid
3. Dorea longicatena was not detected	ORGANISM · section 16	**3** · 2 to avoid
4. Roseburia inulinivorans was not detected	ORGANISM · section 16	**9** · 3 to avoid
5. Anaerostipes hadrus was not detected	ORGANISM · section 16	**5**
6. Roseburia faecis was not detected	ORGANISM · section 16	**9** · 3 to avoid
7. Eubacterium eligens was not detected	ORGANISM · section 16	**4** · 2 to avoid
8. Oscillibacter sp 57 20 was not detected	ORGANISM · section 16	**2** · 1 to avoid
9. Anaerotignum lactatifermentans is overgrown	ORGANISM · section 16	**2**
10. Enterocloster citroniae is overgrown	ORGANISM · section 16	**5** · 1 to avoid
11. Clostridium innocuum is overgrown	ORGANISM · section 16	**5** · 1 to avoid
12. Faecalibacterium prausnitzii is well below the usual level	ORGANISM · section 16	**12** · 6 to avoid
13. Flavonifractor plautii is overgrown	ORGANISM · section 16	**2** · 1 to avoid
14. Clostridium sp AM29 11AC needs qualification	ORGANISM · section 16	**3**
15. Phocaeicola vulgatus needs qualification	ORGANISM · section 16	**7** · 9 to avoid
16. Agathobaculum butyriciproducens is well below the usual level	ORGANISM · section 16	**2**
17. Enterococcus faecium needs qualification	ORGANISM · section 16	**8** · 4 to avoid
18. Mediterraneibacter glycyrrhizinilyticus needs qualification	ORGANISM · section 16	nothing on record
19. Intestinimonas massiliensis is well below the usual level	ORGANISM · section 16	nothing on record
20. Butyricicoccus pullicaecorum is well below the usual level	ORGANISM · section 16	nothing on record
21. Butyrate (SCFA) capacity is notably low	METABOLITE PATHWAY · section 19	**17** · 12 to avoid
22. Branched-chain amino acids (valine, leucine, isoleucine) capacity is high	METABOLITE PATHWAY · section 19	**1**
23. Colibactin and bacterial toxins (virulence factors, not metabolites) capacity is above average	METABOLITE PATHWAY · section 19	**1** · 1 to avoid
24. Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin) capacity is above average	METABOLITE PATHWAY · section 19	**2**
25. Histamine capacity is above average	METABOLITE PATHWAY · section 19	**2** · 2 to avoid
26. Mucin glycans and lipid A capacity is high	METABOLITE PATHWAY · section 19	**2** · 5 to avoid
27. Oxalate (degraded, not produced) capacity is above average	METABOLITE PATHWAY · section 19	**2** · 2 to avoid
28. Secondary bile acids capacity is high	METABOLITE PATHWAY · section 19	**4** · 1 to avoid
29. Trimethylamine / TMAO capacity is above average	METABOLITE PATHWAY · section 19	**1**
30. Unconjugated bile acids capacity is below average	METABOLITE PATHWAY · section 19	**1** · 2 to avoid
31. Microbial ethanol formation capacity is above average	METABOLITE PATHWAY · section 19	nothing on record
32. Putrescine formation capacity is high	METABOLITE PATHWAY · section 19	nothing on record
33. Androgenetic alopecia pattern: notably high	DISEASE PATTERN · section 20	**1**
34. Ankylosing spondylitis pattern: notably high	DISEASE PATTERN · section 20	**1**
35. Chronic kidney disease pattern: notably high	DISEASE PATTERN · section 20	**1**
36. Clinical Alzheimer's disease pattern: notably high	DISEASE PATTERN · section 20	**2**
37. Colorectal adenoma pattern: notably high	DISEASE PATTERN · section 20	**2**

WHAT WAS FOUND	WHERE	OPTIONS ON RECORD
38. Crohn's disease pattern: notably high	DISEASE PATTERN · section 20	**2**
39. IBS with diarrhoea pattern: notably high	DISEASE PATTERN · section 20	**3**
40. Inflammatory bowel disease (composite) pattern: notably high	DISEASE PATTERN · section 20	**3**
41. Irritable bowel syndrome pattern: notably high	DISEASE PATTERN · section 20	**2**
42. Long COVID chronic-state community similarity pattern: notably high	DISEASE PATTERN · section 20	**1**
43. Major depressive disorder pattern: notably high	DISEASE PATTERN · section 20	**1**
44. Mild cognitive impairment pattern: notably high	DISEASE PATTERN · section 20	**2**
45. Parkinson's disease pattern: notably high	DISEASE PATTERN · section 20	**1**
46. Preclinical amyloid positivity pattern: notably high	DISEASE PATTERN · section 20	**2**
47. Rheumatoid arthritis pattern: notably high	DISEASE PATTERN · section 20	**1**
48. Type 2 diabetes pattern: notably high	DISEASE PATTERN · section 20	**3**
49. Ulcerative colitis pattern: notably high	DISEASE PATTERN · section 20	**2**
50. Atherosclerotic cardiovascular disease pattern: high	DISEASE PATTERN · section 20	**1**
51. IBS mixed type pattern: high	DISEASE PATTERN · section 20	**2**
52. ME/CFS-like gut microbiome similarity pattern: high	DISEASE PATTERN · section 20	**1**
53. Multiple sclerosis pattern: high	DISEASE PATTERN · section 20	**1**
54. Systemic lupus erythematosus pattern: notably high	DISEASE PATTERN · section 20	nothing on record

PATHOGEN SCREEN

1. Clostridioides difficile was detected by the pathogen screen

carriage_common · see section 18

WHAT MIGHT BRING IT DOWN 10 of 52 on record

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Limosilactobacillus reuteri DSM 17938 + glycerol; paper: Lactobacillus reuteri 17938; precursor…	probiotic	human cohorts	Clostridioides difficile CD2015 ribotype 027 growth/invasion. in vitro and antibiotic-treated human fecal minibioreactor community. positive conditional on glycerol substrate and reuterin pathway.
Limosilactobacillus reuteri DSM17938 + glycerol	probiotic	human cohorts	Reuterin pathway inhibited C. difficile CD2015 ribotype027 in culture and antibiotic-disrupted human fecal minibioreactors DSM17938 commercially available; glycerol and pathway dependence mandatory; not evidence of stand-alone human CDI cure
Clostridium butyricum MIYAIRI588 in clindamycin/CDI mouse context†	spore probiotic	animal study taken by mouth in the study	Community modulation → less gut succinate → reduced C. difficile proliferation; parallel IL-17A / plasma-B-cell / IgA responses protect tissue. Do not label all IL-17 increases harmful: direction was protective in this infection model. Not every C. butyricum is MIYAIRI588. Succinate change is ecological, not dire
Carbohydrate fermentation plus antibiotic or motility disturbance†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	B. thetaiotaomicron succinate → C. difficile reduction to butyrate → pathogen growth advantage
Host mucin glycans after antibiotics; not a supplement†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	B. thetaiotaomicron sialidase liberates sialic acid → C. difficile uptake → expansion
HST012 — Alanyl-glutamine dipeptide with vancomycin; AR009†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	Epithelial-cell work and mouse infection. Host-injury benefit demonstrated; not direct decolonization. Alanyl-glutamine → recovery of proliferation/migration and less apoptosis after toxin-B injury `M`; antibiotic plus dipeptide → improved histology/survival compared with
HST017 — Excess dietary zinc in an antibiotic/CDI context†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	Mouse nutritional/infection study. Adverse ecological/disease cascade experimentally supported. Excess dietary zinc → altered gut community `M` → greater susceptibility to antibiotic-associated C. difficile infection `M`; zinc exposure → increased toxin activity and
Inulin as the sole microbiota-accessible carbohydrate, or complex MAC diet†	prebiotic fibre	animal study reaches the gut unaided	Community/metabolite shift → lower C. difficile burden
Saccharomyces boulardii culture-derived 54-kDa protease, Biocodex-supplied study preparation†	spore probiotic	animal study reaches the gut unaided	Secreted protease → C. difficile toxins/receptor interactions reduced → less mucosal damage and secretion. Toxin neutralization is not pathogen eradication. Exact culture-deposit strain was not established in the retrieved paper; do not silently label it CNCMI745 or extrapolat
B. clausii UBBC07 / MTCC5472, and separately purified clausin‡	spore probiotic	human tissue or stool, outside the body reaches the gut unaided	Spores survived simulated GI passage; clausin inhibited tested Gram-positive bacteria including C. difficile, with protease/pH stability tested Commercial UBBC07 spore ingredient; extracted clausin effect not guaranteed at human luminal concentration

SHOWN TO PUSH IT THE OTHER WAY 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Excess dietary zinc carbonate in defined mouse chow; Mouse experiments with calprotectin mechan…†	mineral or protein supplement	animal study taken by mouth in the study	Luminal zinc exposure increased → Community changes lowered colonization resistance in an antibiotic context → C. difficile toxin activity and disease severity increased Dietary zinc load, baseline community, antibiotic perturbation and host metal sequestration. Animal high-exposure signal; not a reason to deny correction of genuine zinc
Clostridium scindens ATCC35704 alone or defined four-isolate consortium†	probiotic	animal study reaches the gut unaided	7α-dehydroxylation-capable organism → secondary bile acids → increased resistance to C. difficile after antibiotics. Depends on upstream bile substrates and ecosystem. Deoxycholate is not uniformly beneficial and an abundance increase is not a universal goal; isolate remains a research

ORGANISM

2. Mediterraneibacter gnavus is overgrown

6.9% of your community; 110× the typical carrier; 99th percentile · see section 16

WHAT MIGHT BRING IT DOWN 10 of 13 on record

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Exclusive enteral nutrition (in Crohn's disease)	way of eating	human trial	Clinically supervised; reduces R. gnavus alongside inflammation in paediatric Crohn's.
Dietary fibre, especially inulin and resistant starch	prebiotic fibre	human trial	Fibre feeds competitors that outgrow it and gives it an alternative to host mucus. Inulin supplementation reduced R. gnavus relative abundance in adults in controlled feeding studies.
Bifidobacterium longum subsp. infantis	probiotic	human trial	In infants, B. infantis colonisation reduced R. gnavus and other mucin-degrading Lachnospiraceae; the mechanism is competition for milk oligosaccharides and mucin sugars and applies less directly in adults.
A studied high-fiber dietary intervention; whole intervention, not a claim for every isolated f…	fibre	human trial	In a T2D dietary-intervention dataset, fecal tryptamine and phenethylamine decreased alongside suppression of an R. gnavus genome carrying the relevant decarboxylase. The high-fiber group had 27 participants and the comp Evidence-supported food-pattern candidate for reducing these amines, with human measured-fecal-endpoint support. Preserve the acarbose co-treatment and original WTP-diet
Mediterranean-pattern diet (high fibre, low processed food)	way of eating	human cohorts	Higher fibre intake is consistently associated with lower R. gnavus abundance across cohorts.
Oral berberine†	plant compound	animal study taken by mouth in the study	Reduced R. gnavus representation in one mouse study Indirect ecosystem shift, not selective MIC or universal response; alias TM30
Acid-hydrolyzed agaro-oligosaccharides; R. gnavus / F. nucleatum suppression†	prebiotic fibre	animal study reaches the gut unaided	Culture inhibition of R. gnavus and F. nucleatum; tested bifidobacteria/Lactobacillales were spared. B. longum dominated an R. gnavus co-culture under AOS exposure.
Lactobacillus rhamnosus GG‡	probiotic	laboratory culture taken by mouth in the study	Co-culture and biofilm assays show LGG suppresses R. gnavus growth on mucin.
Acid-hydrolyzed agaro-oligosaccharides‡	prebiotic fibre	laboratory culture reaches the gut unaided	R. gnavus inhibition plus spared B. longum → competitive B. longum advantage
Rosemary aqueous culinary extract‡	herb or botanical extract	laboratory culture reaches the gut unaided	Inhibited an R. gnavus isolate Mandatory option for matched finding; not equivalent to oil or pure carnosic acid; alias FIX-RGN-01

SHOWN TO PUSH IT THE OTHER WAY 12

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Broad-spectrum antibiotics	medication - a prescriber's decision prescriber decides	human cohorts	Blooms after antibiotic courses as competitors are cleared.
Dietary L-serine → competitive advantage for AIEC LF82 in the inflamed gut†	way of eating	animal study taken by mouth in the study	Mechanistic bacterial and mouse work: inflammation changed metabolic programs; removing serine blunted blooms in the tested setting. Keep strain, inflammatory context, and nutrient separate. This does not establish that glutamine feeds R. gnavus or justify general amino-acid/protein restriction.
Low-fibre, high-fat Western diet†	way of eating	animal study taken by mouth in the study	Fibre deprivation drives mucus-degrader expansion in gnotobiotic mice.
Berberine in renal-injury model†	plant compound	animal study reaches the gut unaided	Increased R. gnavus in a different context Contrary direction must remain visible; no averaging across incomparable host models

ORGANISM

3. Dorea longicatena was not detected

WHAT MIGHT BRING IT UP 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Fermentable fibre	prebiotic fibre	human trial	—
Oligonol, litchi-derived low-molecular-weight polyphenol preparation	plant compound	human trial	Dorea genus Genus-level exploratory community endpoint; not exact D. formicigenerans, D. longicatena or CAG 317. No basis to label all Dorea pathogenic. Liver changes within one arm

SHOWN TO PUSH IT THE OTHER WAY 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Low-FODMAP diet (for gas symptoms)	way of eating	human trial	Reduces fermentable substrate for gas producers; clinically supervised.
TOTUM-070 proprietary plant-extract blend: olive leaf, artichoke leaf, chrysanthellum aerial pa…	herb or botanical extract	human trial	Dorea genus Microbiome was a secondary endpoint, species not resolved; cannot assign blend effects to any individual ingredient. Multiple authors employed by or financially linked to

ORGANISM

4. Roseburia inulinivorans was not detected

WHAT MIGHT BRING IT UP 9

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fibre versus high-fermented-food diet (Wastyk 2021)	way of eating	human trial	10-week high-fibre or high-fermented-food diet Studied in people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36. Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with d
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Whole grains	food	human trial	—
Inulin and fructo-oligosaccharides	prebiotic fibre	human trial	—
Oat and barley beta-glucan	prebiotic fibre	human trial	—
Resistant starch	prebiotic fibre	human trial	—
Psyllium husk; stool water / ecological support	prebiotic fibre	human trial	In constipated subjects, increased stool water accompanied changes in Lachnospira, Roseburia and Faecalibacterium.
Therapeutic ferrous sulfate with or without vitamin E; Small double-blind randomized US infant/…	supplement or compound	human trial	Iron deficiency was corrected in both groups → Roseburia relative abundance increased with vitamin E; Escherichia fell over time overall → No persistent inflammatory-marker increase; clinical benefit from Roseburia chang Documented iron deficiency; baseline diet measured; different setting from African fortification trials. No untreated-iron comparison establishing all effects of iron. An
Arabinoxylan + galactomannan, 1:1‡	prebiotic fibre	laboratory culture reaches the gut unaided	Dominates fiber-enriched consortium

SHOWN TO PUSH IT THE OTHER WAY 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Low-fibre diet, antibiotics	way of eating	human cohorts	—
Curcumin in a 481-compound/47-gut-strain screen, followed by obese-mouse and donor-community-tr…†	plant compound	animal study taken by mouth in the study	M: isolate responses differed: E. lenta and B. thetaiotaomicron were inhibited, whereas Roseburia intestinalis and Eubacterium rectale could be stimulated under studied conditions. [S18] M: mouse curcumin exposure shifted the community, including decreased Lactobacillus; some insulin-resistance effects transferred with donor communities. [S18]
Chitosan / low-MW chitosan / chitooligosaccharides‡¶	supplement or compound	laboratory culture route not stated	Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes.

ORGANISM

5. Anaerostipes hadrus was not detected

WHAT MIGHT BRING IT UP 5

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fibre versus high-fermented-food diet (Wastyk 2021)	way of eating	human trial	10-week high-fibre or high-fermented-food diet Studied in people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36. Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with d
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Inulin	prebiotic fibre	human trial	—
Resistant starch	prebiotic fibre	human trial	—
Lactate-producing probiotics (Lactobacillus, Bifidobacterium)‡	probiotic	laboratory culture taken by mouth in the study	Supply its substrate; cross-feeding shown in culture.

ORGANISM

6. Roseburia faecis was not detected

WHAT MIGHT BRING IT UP 9

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fibre versus high-fermented-food diet (Wastyk 2021)	way of eating	human trial	10-week high-fibre or high-fermented-food diet Studied in people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36. Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with d
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Whole grains	food	human trial	—
Inulin and fructo-oligosaccharides	prebiotic fibre	human trial	—
Oat and barley beta-glucan	prebiotic fibre	human trial	—
Resistant starch	prebiotic fibre	human trial	—
Psyllium husk; stool water / ecological support	prebiotic fibre	human trial	In constipated subjects, increased stool water accompanied changes in Lachnospira, Roseburia and Faecalibacterium.
Therapeutic ferrous sulfate with or without vitamin E; Small double-blind randomized US infant/…	supplement or compound	human trial	Iron deficiency was corrected in both groups → Roseburia relative abundance increased with vitamin E; Escherichia fell over time overall → No persistent inflammatory-marker increase; clinical benefit from Roseburia chang Documented iron deficiency; baseline diet measured; different setting from African fortification trials. No untreated-iron comparison establishing all effects of iron. An
Arabinoxylan + galactomannan, 1:1‡	prebiotic fibre	laboratory culture reaches the gut unaided	Dominates fiber-enriched consortium

SHOWN TO PUSH IT THE OTHER WAY 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Low-fibre diet, antibiotics	way of eating	human cohorts	—
Curcumin in a 481-compound/47-gut-strain screen, followed by obese-mouse and donor-community-tr…[†]	plant compound	animal study taken by mouth in the study	M: isolate responses differed: E. lenta and B. thetaiotaomicron were inhibited, whereas Roseburia intestinalis and Eubacterium rectale could be stimulated under studied conditions. [S18] M: mouse curcumin exposure shifted the community, including decreased Lactobacillus; some insulin-resistance effects transferred with donor communities. [S18]
Chitosan / low-MW chitosan / chitooligosaccharides[‡¶]	supplement or compound	laboratory culture route not stated	Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes.

ORGANISM

7. Eubacterium eligens was not detected

WHAT MIGHT BRING IT UP 4

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Pectin (apples, citrus, carrots)	prebiotic fibre	human trial	The organism's defining substrate.
Fruit and vegetable intake	food	human cohorts	—
Crystalline maize RS4[‡¶]	supplement or compound	laboratory culture route not stated	Enrichment linked to butyrate shift

SHOWN TO PUSH IT THE OTHER WAY 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Curcumin in a 481-compound/47-gut-strain screen, followed by obese-mouse and donor-community-tr…[†]	plant compound	animal study taken by mouth in the study	M: isolate responses differed: E. lenta and B. thetaiotaomicron were inhibited, whereas Roseburia intestinalis and Eubacterium rectale could be stimulated under studied conditions. [S18] M: mouse curcumin exposure shifted the community, including decreased Lactobacillus; some insulin-resistance effects transferred with donor communities. [S18]
Quercetin[†]	plant compound	animal study taken by mouth in the study	Reduced representation of these taxa and altered the community Rat diet context; sequencing shift is not direct selective killing; phylum ratio is not a clinical treatment target

ORGANISM

8. Oscillibacter sp 57 20 was not detected

WHAT MIGHT BRING IT UP 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Dietary fibre, Mediterranean diet	way of eating	human trial	—

SHOWN TO PUSH IT THE OTHER WAY 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Same cayenne-pepper food intervention	probiotic	human trial	Bifidobacterium increased; Oscillibacter and Phascolarctobacterium decreased Most taxon findings did not persist across BMI strata; a decrease is not inherently favorable

ORGANISM

9. Anaerotignum lactatifermentans is overgrown

0.78% of your community; 17× the typical carrier; 99th percentile · see section 16

WHAT MIGHT BRING IT DOWN 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Uncommon or relatively high organism — no supported targeted intervention	supplement or compound	human cohorts	This organism is uncommon in the reference or higher than the reference range here. Unless it is named in one of the specific cards above, no human study shows that reduc
Dietary fibre	prebiotic fibre	human cohorts	—

ORGANISM

10. Enterocloster citroniae is overgrown

0.02% of your community; 5× the typical carrier; 90th percentile · see section 16

WHAT MIGHT BRING IT DOWN 5

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Dietary fibre (inulin, pectin)	prebiotic fibre	human trial	Fibre shifts the Lachnospiraceae balance away from bile-tolerant members toward butyrate producers.
Lower saturated-fat intake	way of eating	human cohorts	Bile-acid load falls with dietary fat, and with it the bile-tolerant genera.
Bifidobacterium longum strains†	probiotic	animal study taken by mouth in the study	—
DSLNT / other HMOs; strain-specific clostridial metabolism‡	supplement or compound	human tissue or stool, outside the body taken by mouth in the study	pfoA-negative infant C. perfringens metabolized HMOs and produced supernatants that suppressed pathobionts/inflammation and supported commensals in experimental systems.
HMO and GOS structures; direct inhibition and collateral growth‡§	prebiotic fibre	laboratory culture needs an enteric-protected form	2026 culture screen found strong inhibitory effects of 3′-galactosyllactose/GOS against tested E. coli, S. enterica and Y. enterocolitica, but growth stimulation of C. perfringens under some oligosaccharide conditions.

SHOWN TO PUSH IT THE OTHER WAY 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fat, animal-based diet	way of eating	human trial	—

ORGANISM

11. Clostridium innocuum is overgrown

0.04% of your community; 5× the typical carrier; 81th percentile · see section 16

WHAT MIGHT BRING IT DOWN 5

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Uncommon or relatively high organism — no supported targeted intervention	supplement or compound	human cohorts	This organism is uncommon in the reference or higher than the reference range here. Unless it is named in one of the specific cards above, no human study shows that reduc
Dietary fibre	prebiotic fibre	human cohorts	—
Saccharomyces boulardii†	probiotic	animal study taken by mouth in the study	Yeast probiotic; competes with Clostridia during and after antibiotics and is not itself affected by them.
DSLNT / other HMOs; strain-specific clostridial metabolism‡	supplement or compound	human tissue or stool, outside the body taken by mouth in the study	pfoA-negative infant C. perfringens metabolized HMOs and produced supernatants that suppressed pathobionts/inflammation and supported commensals in experimental systems.
HMO and GOS structures; direct inhibition and collateral growth‡§	prebiotic fibre	laboratory culture needs an enteric-protected form	2026 culture screen found strong inhibitory effects of 3′-galactosyllactose/GOS against tested E. coli, S. enterica and Y. enterocolitica, but growth stimulation of C. perfringens under some oligosaccharide conditions.

SHOWN TO PUSH IT THE OTHER WAY 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Vancomycin and other antibiotics	medication - a prescriber's decision prescriber decides	human cohorts	—

ORGANISM

12. Faecalibacterium prausnitzii is well below the usual level

0.00% of your community; 100% below the typical carrier; 0th percentile · see section 16

WHAT MIGHT BRING IT UP 10 of 12 on record

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fibre versus high-fermented-food diet (Wastyk 2021)	way of eating	human trial	10-week high-fibre or high-fermented-food diet Studied in people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36. Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with d
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Mediterranean diet	way of eating	human trial	—
Apple pectin and other pectins	prebiotic fibre	human trial	—
Inulin and fructo-oligosaccharides (chicory, Jerusalem artichoke, onion)	prebiotic fibre	human trial	The best-replicated prebiotic effect on F. prausnitzii; 10-15 g/day over 2-4 weeks in several RCTs.
Resistant starch (cooled potatoes/rice, green banana, high-amylose maize)	prebiotic fibre	human trial	—
Standardized oral pomegranate extract	herb or botanical extract	human trial	Increased relative abundance of selected species; increases also included M. smithii and C. aerofaciens Mixed ecological response; do not label every increased organism beneficial or infer lower methane
Polyphenols: pomegranate, cranberry, cocoa flavanols	supplement or compound	human trial	Small trials; pomegranate extract and cocoa flavanols each raised Faecalibacterium.
Berberine†	herb or botanical extract	animal study taken by mouth in the study	Poorly absorbed, so it reaches the colon; raised Faecalibacterium in rodent models. Human data are for metabolic outcomes, not this organism.
Bifidobacterium longum and Lactobacillus strains (cross-feeding)‡	probiotic	laboratory culture taken by mouth in the study	Bifidobacteria release acetate that F. prausnitzii uses; cross-feeding shown in co-culture.

SHOWN TO PUSH IT THE OTHER WAY 6

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Oral ferrous sulfate versus intravenous iron sucrose; Open-label randomized IBD/non-inflamed ir…	mineral or protein supplement	human trial	Both routes improved iron deficiency → Oral route selectively reduced some F. prausnitzii/R. bromii-associated OTUs and altered metabolites → Clinical disease activity did not differ by route IBD subtype, pre-existing inflammation and person-specific baseline communities. IV also changed some communities; route is not an absolute microbiome-safety category.
Low-fibre Western diet	way of eating	human cohorts	—
Antibiotics	medication - a prescriber's decision prescriber decides	human cohorts	—
Berberine-containing arms of PREMOTE; adults with newly diagnosed type 2 diabetes, after antibi…‡	medication - a prescriber's decision prescriber decides	laboratory culture taken by mouth in the study	M: treatment depleted several carbohydrate-fermenting taxa, including R. bromii, Roseburia, F. prausnitzii and some Bifidobacterium. [S07] M/A: circulating deoxycholic-acid species and FGF19 decreased; BA changes correlated with glycaemic improvement. Laboratory AF25-7 showed BA conversion and berberine grow

ORGANISM

13. Flavonifractor plautii is overgrown

0.44% of your community; 9× the typical carrier; 91th percentile · see section 16

WHAT MIGHT BRING IT DOWN 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Bifidobacterium multi-strain probiotics	probiotic	human trial	Small trial; Flavonifractor fell alongside a rise in bifidobacteria.
Dietary fibre	prebiotic fibre	human cohorts	—

SHOWN TO PUSH IT THE OTHER WAY 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High polyphenol intake without fibre‡	way of eating	laboratory culture taken by mouth in the study	Its substrate; more relevant is that it is fed by what it degrades.

ORGANISM

14. Clostridium sp AM29 11AC needs qualification

0.23% of your community; 16× the typical carrier; 95th percentile · see section 16

WHAT MIGHT BRING IT DOWN 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Dietary fibre diversity	prebiotic fibre	human cohorts	General competition; no organism-specific evidence for unnamed bins.
DSLNT / other HMOs; strain-specific clostridial metabolism‡	supplement or compound	human tissue or stool, outside the body taken by mouth in the study	pfoA-negative infant C. perfringens metabolized HMOs and produced supernatants that suppressed pathobionts/inflammation and supported commensals in experimental systems.
HMO and GOS structures; direct inhibition and collateral growth‡§	prebiotic fibre	laboratory culture needs an enteric-protected form	2026 culture screen found strong inhibitory effects of 3′-galactosyllactose/GOS against tested E. coli, S. enterica and Y. enterocolitica, but growth stimulation of C. perfringens under some oligosaccharide conditions.

ORGANISM

15. Phocaeicola vulgatus needs qualification

11.7% of your community; 5× the typical carrier; 93th percentile · see section 16

WHAT MIGHT BRING IT DOWN 7

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Increased fibre diversity, especially inulin and resistant starch	prebiotic fibre	human trial	—
Chitosan / low-MW chitosan / chitooligosaccharides‡¶	supplement or compound	laboratory culture route not stated	Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes.
Curcumin in a 481-compound/47-gut-strain screen, followed by obese-mouse and donor-community-tr…†	plant compound	animal study taken by mouth in the study	M: isolate responses differed: E. lenta and B. thetaiotaomicron were inhibited, whereas Roseburia intestinalis and Eubacterium rectale could be stimulated under studied conditions. [S18] M: mouse curcumin exposure shifted the community, including decreased Lactobacillus; some insulin-resistance effects transferred with donor communities. [S18]
Carbohydrate fermentation plus antibiotic or motility disturbance†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	B. thetaiotaomicron succinate → C. difficile reduction to butyrate → pathogen growth advantage
Host mucin glycans after antibiotics; not a supplement†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	B. thetaiotaomicron sialidase liberates sialic acid → C. difficile uptake → expansion
Curcumin, isolated compound†	plant compound	animal study reaches the gut unaided	E. lenta; collateral Bacteroides thetaiotaomicron Direct exact-species laboratory observation; no human E. lenta depletion trial. The same study found context-dependent inhibition of other commensals and growth promotion
0.5% inulin in nutrient-rich BHIS; six-species community‡	prebiotic fibre	laboratory culture reaches the gut unaided	Community competition shifts → R. gnavus relative enrichment; B. uniformis, B. thetaiotaomicron and A. muciniphila decline

SHOWN TO PUSH IT THE OTHER WAY 9

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fat, high-protein diet	way of eating	human trial	—
Sugar-beet arabinan‡	supplement or compound	laboratory culture taken by mouth in the study	Increased abundance
HST028 — NAC; microbiota-dependent absorption, AB024	supplement or compound	human cohorts	Rat perfusion and mechanistic microbial work plus a human predictive cohort. Drug-disposition pathway supported; antibiofilm efficacy not measured. NAC exposure / derived cysteine → Bacteroides sphingolipid-related metabolism `M`; B. thetaiotaomicron/B. fragilis cysteine palmitoylation → altered NAC intestinal permea
Pea fiber†	fibre	animal study reaches the gut unaided	Increased abundance

ORGANISM

16. Agathobaculum butyriciproducens is well below the usual level

0.03% of your community; 98% below the typical carrier; 2th percentile · see section 16

WHAT MIGHT BRING IT UP 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Expected organism not detected or low — no supported way to 'add it back'	supplement or compound	human cohorts	A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, n
Dietary fibre (inulin, resistant starch)†	prebiotic fibre	animal study taken by mouth in the study	—

ORGANISM

17. Enterococcus faecium needs qualification

0.02% of your community; 78% above the typical carrier; 66th percentile · see section 16

WHAT MIGHT BRING IT DOWN 8

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Uncommon or relatively high organism — no supported targeted intervention	supplement or compound	human cohorts	This organism is uncommon in the reference or higher than the reference range here. Unless it is named in one of the specific cards above, no human study shows that reduc
Medication-context card for the prescriber (levodopa / MAO inhibitor and bacterial tyrosine dec…	clinical route prescriber decides	human cohorts	information card to the prescriber; no dose change from this report Studied in ages 18–99. Enterococci expand after antibiotics and with lower resident biomass. The only clinically relevant fact is that E. faecalis tyrosine decarboxylase can metabolise levodopa
Dietary fibre†	prebiotic fibre	animal study taken by mouth in the study	Fibre-fed mice clear VRE faster; SCFA and competition.
Bifidobacterium longum†	probiotic	animal study taken by mouth in the study	Reduced VRE colonisation density in mice.
Lactobacillus rhamnosus and L. plantarum‡	probiotic	laboratory culture taken by mouth in the study	Bacteriocin and lactic-acid inhibition of Enterococcus in co-culture; consistent across many strains.
Ascorbic acid, conventional oral supplement; Uncontrolled healthy-adult before/after pilot, n=1…	supplement or compound	human trial	Ascorbate exposure increased → Lachnospiraceae increased; Enterococcus and some other taxa decreased relatively → Functional or clinical benefit was not established Baseline supplement exclusion, short follow-up. Relative abundance and no placebo; loss of a taxon such as Gemmiger is not automatically favorable.
(S)-alpha-fluoromethyltyrosine, AFMT; research inhibitor, not a commercial supplement recommend…‡	supplement or compound	laboratory culture taken by mouth in the study	Biochemical and microbiota experiments identified Enterococcus faecalis TyrDC conversion of levodopa to dopamine and Eggerthella lenta conversion to m-tyramine. AFMT inhibited bacterial levodopa decarboxylation. The host Show as proof that the bacterial pathway is potentially modifiable and as an investigational candidate. Do not offer AFMT as a consumer supplement, substitute carbidopa,
Berberine hydrochloride‡	plant compound	laboratory culture reaches the gut unaided	Reduced E. faecalis biofilm development; dispersion endpoint also reported Separate prevention and dispersion; no unqualified killing claim; aliases TM08–TM09

SHOWN TO PUSH IT THE OTHER WAY 4

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Antibiotics, especially cephalosporins and vancomycin	medication - a prescriber's decision prescriber decides	human cohorts	—
Copper sulfate in pig feed; Controlled pig feeding experiment; oral feed‡	way of eating	laboratory culture taken by mouth in the study	Copper selected copper-resistant enterococci → tcr-associated E. faecium increased → Macrolide/glycopeptide-resistant organisms were co-selected in this setting Baseline resistant population and linked resistance determinants. Animal-production exposure, not a quantified human supplement risk; metal resistance and antibiotic resi
Red-wine polyphenol beverage arms, including dealcoholized wine	plant compound	human trial	Polyphenol exposure increased several measured groups, including Bifidobacterium, Enterococcus and E. lenta Retain beverage arm in any quantitative extraction; this observation is not a recommendation to drink alcohol
Mixed human-milk HMOs in infant fecal consortia‡	food	laboratory culture taken by mouth in the study	Community context without B. infantis → R. gnavus / Enterococcus enrichment in selected samples

ORGANISM

18. Mediterraneibacter glycyrrhizinilyticus needs qualification

0.35% of your community; 4× the typical carrier; 94th percentile · see section 16

Nothing on record. No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it down. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

ORGANISM

19. Intestinimonas massiliensis is well below the usual level

0.05% of your community; 46% below the typical carrier; 6th percentile · see section 16

Nothing on record. No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it up. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

ORGANISM

20. Butyricicoccus pullicaecorum is well below the usual level

0.00% of your community; 95% below the typical carrier; 5th percentile · see section 16

Nothing on record. No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it up. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

METABOLITE PATHWAY

21. Butyrate (SCFA) capacity is notably low

5th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT UP 10 of 17 on record

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
CoreBiome tributyrin simulated delivery and butyrate, 2025	supplement or compound	human trial	Reactor butyrate increased. Barrier and immune readouts changed in cell assays exposed to reactor supernatants. In vitro delivery estimates, not human colonic bioavailability. Three donors, no placebo-controlled human efficacy from this paper, manufacturer funding. Useful direct pr
Riboflavin/B2 in RIBOGUT; Randomized placebo-controlled healthy-adult trial; oral	supplement or compound	human trial	Vitamin exposure changed microbial activity → Pooled active groups showed a butyrate increase without Faecalibacterium abundance gain → Clinical benefit and a change in stool butyrate genes were not demonstrated Existing fermenters, two studied exposure groups and short follow-up. Individual active arms had limited effects; pooled finding is not a guaranteed producer-expansion ef
Arabinoxylan oligosaccharides (AXOS); bifidobacteria / butyrate producers	probiotic	human trial	Increased bifidobacteria and butyrate-producing species in adults with metabolic-syndrome features; lipid/glucose markers were not improved.
A. soehngenii CH106 next-generation strain, 2025	supplement or compound	human trial	Glycemic variability and diastolic pressure improved; HbA1c difference emerged after 16 weeks including follow-up. Exact strain produces butyrate in culture from sugars or lactate plus acetate; no direct demonstrated hum Geographic response heterogeneity, modest clinical effect sizes, multiple mechanisms plausible. Manufacturer/developer involvement. CH106 has 37 genome mutations relative
A. soehngenii CH106	supplement or compound	human trial	Glycemic-variability and some cardiometabolic signals in 98-person trial; no established direct human butyrate increase Freeze-dried trial product; strain derivative of L2-7, not same organism ID
HST032 — Slippery-elm bark powder, study-specific preparation	herb or botanical extract	human cohorts	Anaerobic community-culture study. Taxa observed; metabolite/host cascade not demonstrated. Herbal substrate → changed pooled human fecal-community composition `M`, including expansion of reported butyrate-associated taxa; taxa → greater SCFA output and epitheli
Latilactobacillus sakei LZ217	probiotic	human cohorts	Identified folate product as 5-methyltetrahydrofolate; simulated gastrointestinal-juice resistance and increased butyrate in fecal-slurry fermentation were reported. The human material was cultured fecal slurry, not an intervention in people. Gastric-juice tolerance does not establish durable engraftment. No retail strain substitution
Same standardized citrus extract‡	herb or botanical extract	human tissue or stool, outside the body reaches the gut unaided	Increased cumulative SCFA production Different response from healthy-donor AX017; host phenotype and community matter
B. pullicaecorum25-3T‡¶	supplement or compound	human tissue or stool, outside the body route not stated	Single dose transiently increased abundance; only two of eight communities increased butyrate. Butyrate response depended on receiving community; not universal functional restoration.
Apiin (apigenin-7-O-apiosylglucoside)‡	way of eating	laboratory culture taken by mouth in the study	Positive: host-digestion-resistant glycoside underwent distal-intestinal conversion, with barrier/inflammation and butyrate-associated improvements. Strong delivery/mechanism candidate; exact human conversion remains individual and unvalidated. Distinct from apigenin alone.

SHOWN TO PUSH IT THE OTHER WAY 12

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
MicrobiomeX standardized citrus extract	herb or botanical extract	human trial	Butyrate fraction increased; calprotectin decrease was only a trend Relative SCFA profile, not established absolute butyrate production or clinical remission
PHGG preparation, human fecal fermentation	prebiotic fibre	human cohorts	Mixed-community response → acetate/butyrate increase and lower Klebsiella / Escherichia-Shigella signals
PHGG; lower Escherichia-Shigella / Klebsiella alongside support	prebiotic fibre	human cohorts	In 41 human fecal inocula, PHGG fermentation enriched Bifidobacterium/Faecalibacterium and reduced Escherichia-Shigella, Klebsiella and Dorea; acetate/butyrate increased.
CB-a barrier / SCFA study, 2026†	supplement or compound	animal study taken by mouth in the study	Measured fecal SCFA restoration, particularly butyrate ($p<0.05$), alongside reduced inflammatory markers and restoration of ZO-1/occludin, altered GPR41/43/109A and HDAC1/2 measures. This supports a barrier-and-metabolite Mouse study, strain isolated from chicken feces, no verified retail formulation. Correlated pathway measurements do not by themselves prove butyrate is the sole necessary

METABOLITE PATHWAY

22. Branched-chain amino acids (valine, leucine, isoleucine) capacity is high

91th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Higher microbial BCAA-synthesis capacity — association only‡	supplement or compound	laboratory culture taken by mouth in the study	Microbial BCAA biosynthesis potential correlated with serum BCAA and insulin resistance in one cohort, with causal work in mice. No human trial targets it and protein int

METABOLITE PATHWAY

23. Colibactin and bacterial toxins (virulence factors, not metabolites) capacity is above average

76th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Age-appropriate colorectal cancer screening (USPSTF)	clinical route prescriber decides	human trial	screening per USPSTF (stool-based test or colonoscopy at guideline intervals) Studied in ages 45–75. These genes mark the potential to make colibactin, fragilysin or FadA; carriage is common and is not toxin expression or cancer. No trial shows that removing these organi

SHOWN TO PUSH IT THE OTHER WAY 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
E. coli Nissle1917 / NC101 pks-positive with tested oligosaccharides‡	prebiotic fibre	laboratory culture taken by mouth in the study	Substrate context → increased colibactin-pathway expression → greater epithelial DNA damage in culture. Specific oligosaccharides and concentrations matter; this does not prove every fiber feeds every pks-positive strain or causes cancer. This is a collateral interaction, n

METABOLITE PATHWAY

24. Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin) capacity is above average

85th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Higher β-glucuronidase capacity — preclinical only‡	supplement or compound	laboratory culture taken by mouth in the study	Bacterial β-glucuronidase inhibitors protected mice from irinotecan toxicity; there is no human trial, and no 'estrogen detox' or calcium-D-glucarate claim is supported.
Standard/functional kimchi; additional arm with heat-killed L. plantarum nF1	food	human trial	Within-group enzyme activity and symptoms decreased; functional kimchi increased B. adolescentis signal. All groups received kimchi despite title calling study placebo-controlled; fecal enzymes were measured, not inferred from genes.

METABOLITE PATHWAY

25. Histamine capacity is above average

77th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation)	way of eating	human trial	three-week low-FODMAP versus high-FODMAP diet Studied in people with clinically confirmed IBS, ages 18–70, in Canada, n = 37. Stool hdc genes are not a histamine measurement. In diagnosed IBS, a three-week randomized low-FODMAP diet lowered urinary histamine and improved symptoms; the causal Kle
L. reuteri ATCC PTA6475, hdc-positive; hdc mutants and H2-receptor blockade†	supplement or compound	animal study route not stated	Histidine → histamine → H2 receptor / cAMP-PKA → lower TNF signaling and less experimental colitis. Histamine can have different effects through other receptors/compartments; this is not evidence that histamine intolerance improves. H2-blocking medication is a mechanist

SHOWN TO PUSH IT THE OTHER WAY 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Oral diamine oxidase(DAO) → dietary histamine handling‡	way of eating	laboratory culture taken by mouth in the study	Open-label human histamine-intolerance study reported symptom improvement during supplement intake.
Lactiplantibacillus plantarum LP115 → host epithelialDAO response‡	probiotic	laboratory culture reaches the gut unaided	2025 cell-model study foundLP115 stimulated epithelialDAO secretion in the presence of exogenous histamine.

METABOLITE PATHWAY

26. Mucin glycans and lipid A capacity is high

95th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
HST018 — Zinc gluconate in antibiotic/peptidoglycan injury†	medication - a prescriber's decision prescriber decides	animal study taken by mouth in the study	August 2026 mouse intervention after antibiotic disruption and peptidoglycan challenge; study-specific administration. Parallel host/community outcomes; not infection clearance. Zinc gluconate → restoration of junction/mucin proteins, altered macrophage markers and less inflammatory signaling `M`; zinc gluconate → altered community structure/dive
0.5% inulin in nutrient-rich BHIS; six-species community‡	prebiotic fibre	laboratory culture reaches the gut unaided	Community competition shifts → R. gnavus relative enrichment; B. uniformis, B. thetaiotaomicron and A. muciniphila decline

SHOWN TO PUSH IT THE OTHER WAY 5

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
HST032 — Slippery-elm bark powder, study-specific preparation	herb or botanical extract	human cohorts	Anaerobic community-culture study. Taxa observed; metabolite/host cascade not demonstrated. Herbal substrate → changed pooled human fecal-community composition `M`, including expansion of reported butyrate-associated taxa; taxa → greater SCFA output and epitheli
Apigenin†	prebiotic fibre	animal study taken by mouth in the study	Positive: improved mucus/junction and inflammatory endpoints with microbiota/SCFA changes. Do not label each changed taxon as a direct antimicrobial target without separate susceptibility evidence.
Dietary fiber deprivation†	fibre	animal study reaches the gut unaided	Increased microbial mucin foraging → thinner mucus barrier → greater Citrobacter rodentium susceptibility
L-serine restriction + mucolytic community → AIEC access to host nutrients/epithelial niche†	probiotic	animal study reaches the gut unaided	Follow-up mouse work found A. muciniphila-mediated mucus degradation permitted AIEC to obtain host-derived nutrients during serine restriction; restriction could increase E. coli in that setting. A documented direction reversal from community context. The right model is conditional rules, not permanent “serine bad” or “Akkermansia good” scores.

METABOLITE PATHWAY

27. Oxalate (degraded, not produced) capacity is above average

84th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Tested probiotic/synbiotic products → urinary oxalate; negative finding‡	probiotic	laboratory culture reaches the gut unaided	Controlled diet lowered urinary oxalate, whereas the probiotic/synbiotic preparations tested did not add the expected benefit.
Calcium carbonate taken with food → intestinal oxalate binding‡	mineral or protein supplement	laboratory culture reaches the gut unaided	Human comparative schedule study examined meal versus bedtime calcium; meal timing reduced urinary oxalate through limiting absorption.

SHOWN TO PUSH IT THE OTHER WAY 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Ascorbic-acid supplement with controlled dietary oxalate; Randomized controlled-diet crossover;…	way of eating	human trial	Ascorbate contributed to endogenous oxalate and altered oxalate absorption → Urinary oxalate increased in a responder subset → Calcium-oxalate crystallization risk index increased, not incident stone events Background oxalate, renal/stone history and response heterogeneity. A beneficial vitamin-C microbiome edge can coexist with this non-microbial collateral pathway. Not eve
Oxalobacter formigenes HC-1(OxabactOC5) → oxalate disposal‡§	probiotic	laboratory culture needs an enteric-protected form	ePHex trial: oral enteric-coated preparation established gut presence; primary plasma-oxalate difference at52weeks did not reach statistical significance(p0.064).

METABOLITE PATHWAY

28. Secondary bile acids capacity is high

95th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 4

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Five-day animal-based versus plant-based diet (David 2014)	way of eating	human cohorts	five days animal-based then plant-based diet (or reverse) Studied in people with clinically confirmed healthy adults, ages 21–33, in USA, n = 10. Bile-acid 7α-dehydroxylation capacity rose within days on an animal-based diet and fell on a plant-based diet in a small feeding study. Actual bile-acid levels require ta
Walnuts‡	food	laboratory culture reaches the gut unaided	Reduced fecal secondary bile acids
Konjac glucomannan; β-glucuronidase / secondary bile acids‡	prebiotic fibre	laboratory culture reaches the gut unaided	In a 30-adult controlled trial, fecal β-glucuronidase activity and secondary bile acids decreased; bifidobacteria/lactobacilli increased.
Clostridium scindens ATCC35704, secreted1-acetyl-beta-carboline‡	probiotic	laboratory culture reaches the gut unaided	Secreted compound inhibited C. difficile; secondary bile acids enhanced activity. Also affected other gut bacteria; C. sordellii comparator is not a suggested probiotic.

SHOWN TO PUSH IT THE OTHER WAY 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Clostridium scindens ATCC35704 alone or defined four-isolate consortium†	probiotic	animal study reaches the gut unaided	7α-dehydroxylation-capable organism → secondary bile acids → increased resistance to C. difficile after antibiotics. Depends on upstream bile substrates and ecosystem. Deoxycholate is not uniformly beneficial and an abundance increase is not a universal goal; isolate remains a research

METABOLITE PATHWAY

29. Trimethylamine / TMAO capacity is above average

76th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT DOWN 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Controlled red-meat versus white-meat versus non-meat protein feeding (Wang 2019)	way of eating	human trial	three isocaloric four-week diets (red meat, white meat, non-meat protein) in random order Studied in people with clinically confirmed healthy adults, ages 21–65, in USA, n = 113. cutC gene capacity is not a plasma TMAO measurement. In a controlled feeding trial, red meat raised plasma TMAO about three-fold and switching protein source reversed it

METABOLITE PATHWAY

30. Unconjugated bile acids capacity is below average

17th percentile against the reference cohort · see section 19

WHAT MIGHT BRING IT UP 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Limosilactobacillus reuteri NCIMB 30242 (bile salt hydrolase-active)	probiotic	human trial	2.9 × 10^9 CFU capsule twice daily for nine weeks Studied in people with clinically confirmed hypercholesterolaemia, ages 20–75, in Canada, n = 127. Total stool BSH gene capacity is not the same as one BSH-active strain. L. reuteri NCIMB 30242 lowered LDL by 9–12% in two trials in people with measured high cholesterol

SHOWN TO PUSH IT THE OTHER WAY 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Berberine; high-fat-fed mice, intestinal-FXR-deficient comparison, and complementary microbial …†	plant compound	animal study taken by mouth in the study	M: berberine reduced BSH activity and increased tauro-conjugated BAs, especially taurocholic acid. [S06a], [S06b] M: intestinal FXR signalling increased, with downstream hepatic lipid-handling changes. [S06a]
Resveratrol; atherosclerosis-prone mice with microbiota-depletion and BA-receptor perturbation …†	plant compound	animal study reaches the gut unaided	M: resveratrol changed the microbiota, reduced microbial trimethylamine/TMAO formation, and increased BSH-related BA deconjugation/excretion. [S25] M: ileal BA/FXR–FGF15 signalling decreased and hepatic BA synthesis increased. [S25]

METABOLITE PATHWAY

31. Microbial ethanol formation capacity is above average

77th percentile against the reference cohort · see section 19

Nothing on record. No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it down. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

METABOLITE PATHWAY

32. Putrescine formation capacity is high

92th percentile against the reference cohort · see section 19

Nothing on record. No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it down. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

DISEASE PATTERN

33. Androgenetic alopecia pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Lactiplantibacillus plantarum DCn_07 (CECT 30102) + DCn_06 (CECT 30103) + Lactiplantibacillus p…	probiotic	human trial	three-strain capsule daily for 16 weeks (source-reported dose/formulation) Studied in people with clinically confirmed androgenetic alopecia, ages 18–65, in Spain, n = 136. A single industry trial of three exact Lactiplantibacillus strains reported mixed hair-cycle endpoints in diagnosed androgenetic alopecia. Other probiotics are not covere

DISEASE PATTERN

34. Ankylosing spondylitis pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Ankylosing-spondylitis pattern — no supported targeted intervention	supplement or compound	human trial	The one randomized probiotic trial in ankylosing spondylitis was null and FMT is case-level. The condition is diagnosed and treated by rheumatology.

DISEASE PATTERN

35. Chronic kidney disease pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
CKD evaluation and management per KDIGO 2024	clinical route prescriber decides	human trial	eGFR, albuminuria, electrolytes and management per KDIGO 2024 Studied in ages 18–99. CKD is staged by eGFR and albuminuria. In established CKD, synbiotic and resistant-starch trials changed some uraemic-toxin surrogates without hard renal outcomes; any di

DISEASE PATTERN

36. Clinical Alzheimer's disease pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. ferment…	probiotic	human trial	200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks Studied in people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60. One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level on
FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk mo…	lifestyle	human trial	two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring) Studied in people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognit Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition

DISEASE PATTERN

37. Colorectal adenoma pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Clostridium butyricum MIYAIRI 588 (CBM588)	probiotic	human trial	CBM588 tablets daily as studied (source-reported dose) Studied in people with clinically confirmed history of colorectal adenoma after polypectomy, ages 40–80, in Japan. Early, conflicting evidence for the exact Clostridium butyricum MIYAIRI 588 product after polypectomy; pooled adenoma probiotic trials are inconclusive and surveillance i
Age-appropriate colorectal cancer screening (USPSTF)	clinical route prescriber decides	human trial	screening per USPSTF (stool-based test or colonoscopy at guideline intervals) Studied in ages 45–75. A stool pattern resembling colorectal-cancer or adenoma cohorts does not detect cancer. Guideline screening (FIT or colonoscopy, ages 45–75) is the established route, and

DISEASE PATTERN

38. Crohn's disease pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Crohn's disease exclusion diet with partial enteral nutrition (CDED + PEN)	way of eating	human trial	CDED + 50% PEN for 6 weeks, then CDED + 25% PEN to week 12, versus exclusive enteral nutrition then free diet + 25% PEN Studied in people with clinically confirmed mild-to-moderate paediatric Crohn's disease, ages 4–18, The Crohn's Disease Exclusion Diet with partial enteral nutrition induced sustained remission in children with confirmed mild-to-moderate Crohn's disease. It is delivered
Gastroenterology evaluation with fecal calprotectin when symptoms warrant	clinical route prescriber decides	human trial	symptom-driven gastroenterology evaluation with fecal calprotectin Studied in ages 0–99. Resemblance to an IBD-cohort pattern is not a diagnosis. Persistent diarrhoea, blood in stool, weight loss or pain warrant gastroenterology evaluation with fecal calprote

DISEASE PATTERN

39. IBS with diarrhoea pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation)	way of eating	human trial	restriction ordinarily 4–6 weeks, then structured reintroduction and personalisation with dietitian support Studied in people with clinically confirmed IBS (clinically diagnosed), ages 18–99. IBS is a symptom-based clinical diagnosis. For people with that diagnosis, ACG 2021 and the AGA diet update support soluble fibre, peppermint oil and a limited dietitian-
Heat-inactivated Bifidobacterium bifidum MIMBb75 (non-viable)	probiotic	human trial	10^9 non-viable cells once daily for eight weeks Studied in people with clinically confirmed IBS (Rome III), ages 18–70, in Germany, n = 443. Heat-inactivated B. bifidum MIMBb75 (443 people) and B. infantis 35624 at one specific dose are the exact products with positive randomized IBS trials; guidelines still p
Rifaximin	medication - a prescriber's decision prescriber decides	human trial	studied protocol on file (not shown in participant mode) Studied in people with clinically confirmed IBS-D (clinically diagnosed), ages 18–99. Rifaximin is guideline-listed for global IBS-D symptoms after clinical diagnosis. It is a prescription decision made by a clinician, never a response to organisms in stoo

DISEASE PATTERN

40. Inflammatory bowel disease (composite) pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Crohn's disease exclusion diet with partial enteral nutrition (CDED + PEN)	way of eating	human trial	CDED + 50% PEN for 6 weeks, then CDED + 25% PEN to week 12, versus exclusive enteral nutrition then free diet + 25% PEN Studied in people with clinically confirmed mild-to-moderate paediatric Crohn's disease, ages 4–18, The Crohn's Disease Exclusion Diet with partial enteral nutrition induced sustained remission in children with confirmed mild-to-moderate Crohn's disease. It is delivered
Fecal microbiota transplantation for a non-CDI indication (investigational)	procedure prescriber decides	human trial	studied protocol on file (not shown in participant mode) Studied in people with clinically confirmed active ulcerative colitis, ages 18–75, in Australia, n = 81. Intensive multidonor FMT produced remission in 27% versus 8% in one trial of active ulcerative colitis; guidelines still limit FMT for UC to clinical trials.
Gastroenterology evaluation with fecal calprotectin when symptoms warrant	clinical route prescriber decides	human trial	symptom-driven gastroenterology evaluation with fecal calprotectin Studied in ages 0–99. Resemblance to an IBD-cohort pattern is not a diagnosis. Persistent diarrhoea, blood in stool, weight loss or pain warrant gastroenterology evaluation with fecal calprote

DISEASE PATTERN

41. Irritable bowel syndrome pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation)	way of eating	human trial	restriction ordinarily 4–6 weeks, then structured reintroduction and personalisation with dietitian support Studied in people with clinically confirmed IBS (clinically diagnosed), ages 18–99. IBS is a symptom-based clinical diagnosis. For people with that diagnosis, ACG 2021 and the AGA diet update support soluble fibre, peppermint oil and a limited dietitian-
Heat-inactivated Bifidobacterium bifidum MIMBb75 (non-viable)	probiotic	human trial	10^9 non-viable cells once daily for eight weeks Studied in people with clinically confirmed IBS (Rome III), ages 18–70, in Germany, n = 443. Heat-inactivated B. bifidum MIMBb75 (443 people) and B. infantis 35624 at one specific dose are the exact products with positive randomized IBS trials; guidelines still p

DISEASE PATTERN

42. Long COVID chronic-state community similarity pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
SIM01 synbiotic (Bifidobacterium adolescentis, B. bifidum, B. longum with galacto-, xylo- oligo…	probiotic	human trial	SIM01 10 billion CFU in sachets twice daily for six months Studied in people with clinically confirmed post-acute COVID-19 syndrome (PACS), ages 18–99, in Hong Kong, n = 463. Resemblance to a published Long-COVID stool signature is a research observation. The only randomized evidence with a clinical endpoint is for one exact synbiotic (SIM01)

DISEASE PATTERN

43. Major depressive disorder pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
ModiMedDiet dietary support (SMILES trial) as adjunct in major depression	way of eating	human trial	ModiMedDiet with seven dietitian sessions over 12 weeks versus social support Studied in people with clinically confirmed major depressive episode (moderate–severe), ages 18–99, in Australia, n = 67. Depression is a clinical diagnosis and this pattern is not a mood measurement. As adjuncts to treatment, dietitian-supported Mediterranean-style dietary change (SMILES) a

DISEASE PATTERN

44. Mild cognitive impairment pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. ferment…	probiotic	human trial	200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks Studied in people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60. One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level on
FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk mo…	lifestyle	human trial	two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring) Studied in people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognit Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition

DISEASE PATTERN

45. Parkinson's disease pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Multistrain probiotic capsule used in Tan et al. 2021 (Parkinson disease constipation)	probiotic	human trial	multistrain capsule for four weeks (full formula/dose from source) Studied in people with clinically confirmed Parkinson disease with constipation, ages 18–99, in Malaysia, n = 72. Parkinson disease is a neurological diagnosis. In people who have it, one exact multistrain probiotic improved constipation (not motor symptoms), and FMT trials are split

DISEASE PATTERN

46. Preclinical amyloid positivity pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. ferment…	probiotic	human trial	200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks Studied in people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60. One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level on
FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk mo…	lifestyle	human trial	two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring) Studied in people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognit Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition

DISEASE PATTERN

47. Rheumatoid arthritis pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Lacticaseibacillus casei 01 (Alipour 2014, rheumatoid arthritis)	probiotic	human trial	10^8 CFU/day L. casei 01 for eight weeks Studied in people with clinically confirmed rheumatoid arthritis, ages 20–80, female, in Iran, n = 46. Rheumatoid arthritis is diagnosed by rheumatology with serology and examination. Two small probiotic trials were positive on disease activity, one was null; many RA patie

DISEASE PATTERN

48. Type 2 diabetes pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 3

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
High-fibre diet (whole grains, traditional Chinese medicinal foods, prebiotics) in type 2 diabe…	way of eating	human trial	high-fibre WTP diet (whole grains, TCM foods, prebiotics) for 12 weeks; acarbose in both arms Studied in people with clinically confirmed type 2 diabetes, ages 30–70, in China, n = 43. Resemblance to a type-2-diabetes cohort is not a glucose measurement. HbA1c and fasting glucose confirm or exclude diabetes. In diagnosed diabetes, a 12-week high-fibre d
Pasteurised Akkermansia muciniphila MucT (ATCC BAA-835)	probiotic	human trial	10^10 pasteurised cells/day for three months Studied in people with clinically confirmed overweight/obesity with insulin resistance, ages 18–70, in Belgium, n = 32. The Akkermansia evidence is for one pasteurised strain at one dose; positive pilot results and null primary endpoints exist side by side, and stool Akkermansia abundance
Fecal microbiota transplantation for a non-CDI indication (investigational)	procedure prescriber decides	human trial	studied protocol on file (not shown in participant mode) Studied in people with clinically confirmed metabolic syndrome (male), ages 21–65, male, in Netherlands, n = 18. Lean-donor FMT transiently improved insulin sensitivity in one small trial; the follow-up trial found the effect only in a subset and lost by 18 weeks, and encapsulated F

DISEASE PATTERN

49. Ulcerative colitis pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Fecal microbiota transplantation for a non-CDI indication (investigational)	procedure prescriber decides	human trial	studied protocol on file (not shown in participant mode) Studied in people with clinically confirmed active ulcerative colitis, ages 18–75, in Australia, n = 81. Intensive multidonor FMT produced remission in 27% versus 8% in one trial of active ulcerative colitis; guidelines still limit FMT for UC to clinical trials.
Gastroenterology evaluation with fecal calprotectin when symptoms warrant	clinical route prescriber decides	human trial	symptom-driven gastroenterology evaluation with fecal calprotectin Studied in ages 0–99. Resemblance to an IBD-cohort pattern is not a diagnosis. Persistent diarrhoea, blood in stool, weight loss or pain warrant gastroenterology evaluation with fecal calprote

DISEASE PATTERN

50. Atherosclerotic cardiovascular disease pattern: high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Mediterranean diet supplemented with extra-virgin olive oil or nuts (PREDIMED)	way of eating	human trial	Mediterranean diet supplemented with extra-virgin olive oil or mixed nuts; median 4.8 years Studied in people with clinically confirmed high cardiovascular risk without CVD, ages 55–80, in Spain, n = 7447. Cardiovascular risk is assessed with blood pressure, lipids, glucose, smoking and family history. A Mediterranean dietary pattern has randomized event-reduction evidence;

DISEASE PATTERN

51. IBS mixed type pattern: high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 2

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation)	way of eating	human trial	restriction ordinarily 4–6 weeks, then structured reintroduction and personalisation with dietitian support Studied in people with clinically confirmed IBS (clinically diagnosed), ages 18–99. IBS is a symptom-based clinical diagnosis. For people with that diagnosis, ACG 2021 and the AGA diet update support soluble fibre, peppermint oil and a limited dietitian-
Heat-inactivated Bifidobacterium bifidum MIMBb75 (non-viable)	probiotic	human trial	10^9 non-viable cells once daily for eight weeks Studied in people with clinically confirmed IBS (Rome III), ages 18–70, in Germany, n = 443. Heat-inactivated B. bifidum MIMBb75 (443 people) and B. infantis 35624 at one specific dose are the exact products with positive randomized IBS trials; guidelines still p

DISEASE PATTERN

52. ME/CFS-like gut microbiome similarity pattern: high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
Lacticaseibacillus casei Shirota (Rao 2009, chronic fatigue syndrome)	probiotic	human trial	24 billion CFU L. casei Shirota daily for eight weeks Studied in people with clinically confirmed chronic fatigue syndrome, ages 18–65, in Canada, n = 39. ME/CFS is a clinical diagnosis of exclusion. The microbiome classifier evidence is strong on association; intervention evidence is one small probiotic pilot and a null FM

DISEASE PATTERN

53. Multiple sclerosis pattern: high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

WHAT HAS BEEN TRIED IN THIS CONDITION 1

WHAT	KIND	WHERE THIS COMES FROM	WHAT WAS SEEN
VSL#3 eight-strain probiotic (Tankou 2018, multiple sclerosis)	probiotic	human trial	VSL#3 3600 billion CFU/day for two months Studied in people with clinically confirmed relapsing-remitting multiple sclerosis, ages 18–65, in USA, n = 9. MS is diagnosed by neurology with MRI and CSF. Probiotic and FMT studies in MS are small and report immune or microbiome surrogates, not disease activity.

DISEASE PATTERN

54. Systemic lupus erythematosus pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis · see section 20

Nothing on record. No study in any setting — trial, animal or laboratory — has been recorded here for how to change it. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

† Shown in animals, not yet in people. ‡ Shown in laboratory culture rather than in a living body: it tells you the organism is susceptible to the substance, not that eating it will do this in you. § Would need an enteric-coated or delayed-release preparation to reach the colon intact; the plain form is absorbed or destroyed before it arrives. ¶ The source does not say how it was delivered.

Every row is a description of published research, not a recommendation, and not a claim that it is safe or right for you. Nothing here is a prescription; anything marked as a prescriber's decision is one. Talk to a qualified clinician before acting on any of it.

Personalised synbiotic options: what the model predicts

Your own community is rebuilt as a metabolic model from the organisms found in your sample and run four ways for each fibre — as it is, with a five-species probiotic consortium added, with the fibre added, and with both — so the question "is the combination worth more than either part alone?" can be asked directly, and the question "who would this fibre actually feed in *my* gut?" can be answered organism by organism. These are **experimental predictions from a model**, not measurements, and not a guarantee of how you would respond.

COMPONENT	STATE	DETAIL
Simulation engine	ready	MICOM 0.37.0; community growth held at 0.99 of its maximum while each output is bounded
Metabolic model library	ready	AGORA2 species models, checksum verified
Numerical solver	exact	every value is the optimum of a linear programme; each arm's growth maximum was recomputed by independent worker processes and agreed to every digit (0 programmes solved, 790 served from cache)

The consortium added in the model is *Akkermansia muciniphila*, *Clostridium beijerinckii*, *Clostridium butyricum*, *Bifidobacterium longum* and *Anaerobutyricum hallii*. These are species added to a metabolic model following the published protocol — not a named product, not a dose, and not a recommendation to take any of them.

50 of the 213 organisms found in your sample have a metabolic model in the library, covering 63% of the bacterial abundance they represent. The model library predates some renames, so two of your organisms were matched *M. gnavus* under its earlier name *R. gnavus*; *A. rectalis* under its earlier name *E. rectale*. They are the same organism, not an extra one. In this diet definition the model's total growth is capped by **cobalt** supply, so growth is identical in every arm and only what the community *makes* is compared.

Butyrate potential under each option mmol/gDW/h in the model; ranked by the combined option

FIBRE	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH	INTERACTION	VERDICT
Arabinoxylan* Wheat and rye bran, psyllium	57.4	171.6	57.4	**220.3**	+48.7	**pair gains**
Inulin Chicory root, alliums, artichoke	57.4	171.6	57.4	**220.3**	+48.7	**pair gains**
Maltodextrin Defined maltodextrin preparations	57.4	171.6	57.4	**216.8**	+45.3	**pair gains**
Starch Cooked and cooled starchy foods	57.4	171.6	57.4	**216.8**	+45.3	**pair gains**
Pectin Apple, citrus peel, vegetables	57.4	171.6	57.4	**216.8**	+45.3	**pair gains**
Cellulose* Hemp seed, intact plant material	57.4	171.6	57.4	**171.6**	0.0	**consortium alone suffices**

Interaction is the combined option minus both single options plus the baseline (F11 – F10 – F01 + F00): above zero, the pair does more than the sum of its parts in the model; below zero, less. A fibre that adds nothing on its own is still shown, and a verdict of "fibre alone suffices" means the consortium adds little on top of it. **Every fibre alone leaves butyrate unchanged in your community**, while the same fibre with the consortium raises it: your present butyrate producers are already at capacity in the model, so extra fibre has nowhere to go until producers are added. That is the interaction these rows are measuring, and it is the case where a combination is worth more than either part. * Arabinoxylan: An arabinoxylan exchange standing in for psyllium husk. Psyllium is a gel-forming mucilage whose physical behaviour the model does not represent. Cellulose: A cellulose exchange standing in for hemp seed. The model sees purified cellulose, not the whole seed's protein, fat and other fibre.

Who each fibre would feed in your community share of the fibre each organism can take up when the community makes the most butyrate it can

FIBRE	BENEFICIAL OR ADDED ORGANISMS	ORGANISMS THIS REPORT FLAGGED	OTHER ORGANISMS
Arabinoxylan 5 of your organisms can use it	*B. longum* ≤30%; *B. adolescentis* ≤3%	*P. vulgatus* ≤81% (p93)	*B. thetaiotaomicron* ≤4%; *B. ovatus* ≤3%
Inulin 13 of your organisms can use it	*B. longum* ≤97%; *A. hallii* ≤33%; *B. breve* ≤31%; *B. adolescentis* ≤10%; *F. saccharivorans* ≤8%; +1 more	*P. vulgatus* ≤100% (p93)	*B. stercoris A* ≤34%; *B. uniformis* ≤30%; *B. caccae* ≤17%; *B. thetaiotaomicron* ≤13%; *C. aerofaciens G* ≤12%; +1 more
Maltodextrin 22 of your organisms can use it	*B. longum* ≤20%; *C. butyricum* ≤20%; *B. breve* ≤6%; *L. plantarum* ≤5%; *F. saccharivorans* ≤5%; +4 more	*P. vulgatus* ≤53% (p93)	*A. putredinis* ≤9%; *B. stercoris A* ≤7%; *P. distasonis* ≤6%; *P. capillosus* ≤3%; *B. caccae* ≤3%; +7 more
Starch 11 of your organisms can use it	*C. beijerinckii* ≤36%; *C. butyricum* ≤36%; *B. breve* ≤11%; *B. adolescentis* ≤4%	none in the model	*B. stercoris A* ≤12%; *P. distasonis* ≤12%; *B. caccae* ≤6%; *P. merdae* ≤5%; *B. thetaiotaomicron* ≤5%; +2 more

FIBRE	BENEFICIAL OR ADDED ORGANISMS	ORGANISMS THIS REPORT FLAGGED	OTHER ORGANISMS
Pectin 7 of your organisms can use it	*C. beijerinckii* ≤42%; *C. butyricum* ≤42%; *L. pectinoschiza* ≤1%	none in the model	*B. stercoris A* ≤15%; *B. caccae* ≤7%; *B. thetaiotaomicron* ≤6%; *B. ovatus* ≤4%
Cellulose	no organism in your modelled community carries a transporter for it	—	—

"≤ 40%" means that in some butyrate-optimal state of your community this organism takes up to 40% of the fibre supplied; **required** means no such state exists without it. A flagged organism listed here has the transporter and the community state permits the uptake — a specific, testable hypothesis about your sample, which is why it is shown rather than assumed either way.

None of these fibres can feed *A. lactatifermentans*, *C. sp. AM29_11AC*, *F. plautii*, *M. gnavus* — they carry no transporter for any of them in the model, so no amount of any of these fibres reaches them directly. *P. vulgatus* can use only Arabinoxylan, Inulin, Maltodextrin of the fibres tested. This concerns direct uptake only: an organism can still gain from products another organism releases, which the model does allow and which the detail section shows.

Who would make the butyrate each organism's share of the community's potential; a range is what it must and what it can contribute

SCENARIO	REQUIRED PRODUCERS
As you are 57.4 mmol/gDW/h	*C. sp. AM29_11AC* 29%–30%; *A. lactatifermentans* 24%–25%; *F. plautii* 12%–14%; *E. ramulus* 11%–14%; *O. splanchnicus* 11%–12%; *A. dolichus* 3%–4%; *A. rectalis* 2%–3%; +1 more
Arabinoxylan + consortium 220.3 mmol/gDW/h	*A. hallii* 26%–27% (added); *C. beijerinckii* 26%–27% (added); *C. butyricum* 26%–27% (added); *C. sp. AM29_11AC* 5%–6%; *A. lactatifermentans* 4%–5%; *F. plautii* 2%–3%; *E. ramulus* 2%–3%; +1 more (optional: *A. dolichus*)

Your present producers' ranges are narrow — each must contribute almost exactly what it can — which means they are already at capacity in the model. That is why a fibre on its own may raise acetate and propionate without raising butyrate: the extra carbon has no spare butyrate-making capacity to flow into, until producers are added.

How much of the goal set these options cover

Goal coverage (C)	100%	share of modelled goals with a favourable edge
Pair-supported coverage (P)	83%	share where the pair, not either part alone, carries it

Over 6 modelled goals, each weighted equally. A candidate whose model did not solve is left out of the set rather than counted against it. These are percentages of a modelled goal set — not a response rate, and not a share of people helped.

On high-fibre diet, no fibre supplement changes the prediction. Every candidate reaches the same output as the consortium alone, because that diet already supplies what the supplement would have added. If your own diet is already high in fibre, the model puts the benefit in the organisms rather than in the fibre. This is a range across predeclared dietary assumptions, not a confidence interval and not a probability of response.

The method is validated against published predictions. Run against the released predictions of the study it follows, this pipeline reproduces every value to 1e-08, including the result that a fibre alone can outrank a fibre-plus-probiotic combination, and that ranking by total output and by output per unit of growth picks different winners. That replay and your solved model are different things, and the data file labels them differently.

What the published synbiotic trials actually show

4 of 17	**4** of 17	**9** of 17	**8** of 17
demonstrated synergy	no synergy demonstrated	undetermined	missing a component-only arm

Four of these are negative results: a strain, a substrate and the two together, with the combination no better than the strain alone. Eight lack a component-only arm and cannot show synergy in either direction. Neither fact is an absence of data.

Units are mmol per gram of community dry weight per hour in the model: a production potential, not a stool concentration, an absorbed amount or a dose. Fibre caps in the model are exchange limits, not grams to eat. The consortium is a species-level perturbation following the published protocol, not a named product. Organisms the model library lacks are counted as unmapped, never substituted. Feeding of a flagged organism is a specific model edge — it carries the transporter and the community state permits the uptake — not a rule that fibre feeds pathogens. In ISAPP terms every row is a *candidate combination*; a positive contrast in the model is evidence about the model, not a demonstrated host benefit.

Measurements not in this edition

Your sample could not answer these. Plant-compound conversion (a declared negative control matched for Urolithin C to urolithin A (9-dehydroxylation); the step is withheld because the panel cannot currently tell the two enzymes apart in this sample); Beta-glucuronidase substrate classes (The structural class of a beta-glucuronidase is defined by the length of a loop at one of two active-site positions, measured in a curated alignment guided by solved structures. Reproducing that from a pairwise alignment to the E. coli enzyme works for enzymes close to it and fails for the divergent gut ones: in testing, the Bacteroides fragilis enzyme came out as Loop 1 where the published structure is mini-Loop 1, and the Bacteroides uniformis enzyme came out as No Loop where it is the published example of Loop 2. Those are the gut organisms the breakdown would exist to describe, so no break); Change over time (only one sample is on file for this participant, so nothing can be compared with it yet); Return toward a pre-event community (a recovery estimate needs a recorded perturbation and at least 3 eligible samples before it); External laboratory results (None are needed: every reading in this report is produced from your sequencing data. A supplied result would be shown beside the finding it relates to, not used to replace it.).

PART B

In detail

Every reading from Part A, one at a time and in the same order: what it is, exactly where you sit, what that means, the research behind the reading — and, under each one, **what the human research says about changing it**: the exact products, diets, drugs or clinical routes that have been studied, in whom, with the evidence for and against side by side and the safety rules that apply. Where nothing has been shown to work, the card says so. Where fecal microbiota transplantation is the only route with evidence, the card says that and gives its regulatory status.

IN THIS PART

- 15 · Your microbial groups in detail, and every species detected
- 16 · Organisms that need attention, one by one
- 18 · Pathogens, one at a time, and what was not assessed
- 19 · Your metabolic functions in detail (45 pathways)
- 20 · Disease-pattern resemblance, every scored pattern
- 23 · Gut-skin axis, panel by panel
- 24 · Your information, and what this report assumed
- 25 · What none of this can tell you · 26 · Sequencing quality and reference coverage
- 27 · Measured accuracy · 28 · Technical record

PART B — WHO IS THERE

15 Your microbial groups in detail

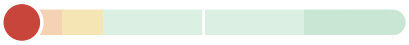
Curated sets of species that share a job or an origin — the fibre fermenters, the mouth bacteria that should not be in stool in quantity, the organisms that make methane or sulfide — each summed to one reading and placed against 3,027 matched reference adults. The arrow says which direction the research reads as favourable; no arrow means it depends on context. Under each group are the species that made up your reading.

Functional groups

‹ at a glance

Butyrate producers ▼

0.82% of classified reads · **–92%** vs typical · 7 of 19 species detected · 2 in the literature not nameable here



2nd NOTABLY LOW

The organisms that make butyrate, the short-chain fatty acid your colon cells burn for fuel. Depletion of this group is the most common finding in gut disease research and the least specific: it accompanies inflammatory bowel disease, recent antibiotics, low-fibre eating and general ill health alike.

Carriage of the pathway is not production. Butyrate output depends on fermentable fibre reaching the colon, and a person with a rich butyrate- producer community on a low-fibre diet produces little of it.

What is driving this reading

5 of 19 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Odoribacter splanchnicus	Beneficial	44%	0.364%	**+62%** vs typical carrier **70th** percentile carried by 74% of adults
Allocoprococcus comes	Unknown	40%	0.330%	**+72%** vs typical carrier **70th** percentile carried by 79% of adults
Agathobacter rectalis	Beneficial	12%	0.097%	**–92%** vs typical carrier **16th** percentile carried by 88% of adults
Agathobaculum butyriciproducens full page →	Beneficial	4%	0.029%	**–98%** vs typical carrier **2nd** percentile carried by 89% of adults
Butyricicoccus pullicaecorum full page →	Beneficial	0%	0.001%	**–95%** vs typical carrier **5th** percentile carried by 3% of adults

Why. This reading is low because **10** species most reference adults carry were not detected: *Faecalibacterium prausnitzii* (98%), *Anaerostipes hadrus* (87%), *Eubacterium hallii* (86%), *Roseburia inulinivorans* (85%), and 6 more. What is present, led by *Odoribacter splanchnicus*, is not enough to make up the sum.

Not detected (14 of 19): *Faecalibacterium prausnitzii* (98%), *Anaerostipes hadrus* (87%), *Eubacterium hallii* (86%), *Roseburia inulinivorans* (85%), *Roseburia hominis* (83%), *Roseburia faecis* (81%), *Eubacterium eligens* (81%), *Roseburia intestinalis* (73%), *Coprococcus catus* (69%), *Eubacterium ventriosum* (52%), *Coprococcus eutactus* (36%), *Butyrivibrio crossotus* (18%), *Anaerostipes caccae* (2%), *Subdoligranulum variabile* (0%). Percentages are how many reference adults carry each.

Two measurements agree

This agrees with the butyrate production capacity reading (5th percentile), which measures the same function by counting the pathway's genes across the whole community instead of these species.

What your reading means

notably low

This group is 0.82% of your classified reads, counted over 7 of the 19 species this group is defined by. That places you at the 2nd percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Odoribacter splanchnicus* (0.36% of your community); *Allocoprococcus comes* (0.33% of your community); *Agathobacter rectalis* (0.10% of your community); *Agathobaculum butyriciproducens* (0.03% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

The organisms that make butyrate, the short-chain fatty acid your colon cells burn for fuel. Depletion of this group is the most common finding in gut disease research and the least specific: it accompanies inflammatory bowel disease, recent antibiotics, low-fibre eating and general ill health alike.

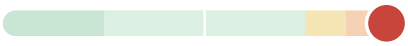
Carriage of the pathway is not production. Butyrate output depends on fermentable fibre reaching the colon, and a person with a rich butyrate- producer community on a low-fibre diet produces little of it.

Odoribacter splanchnicus is included as a characterised butyrate producer but is also elevated in the Alzheimer's continuum panel. Group membership is about metabolic capacity, not about whether an organism is desirable.

‹ at a glance

Mucin degraders ▲

10.29% of classified reads · **+397%** vs typical · 7 of 8 species detected



96th NOTABLY HIGH

Bacteria that feed on the mucus layer lining your gut. This is normal and part of how the mucus layer renews itself. When dietary fibre is scarce, these organisms turn to mucin as their main carbon source and the barrier thins — an effect demonstrated cleanly in gnotobiotic mice.

Akkermansia muciniphila carries the strongest and most contradictory literature of any species here. It is depleted in obesity and metabolic syndrome and improved insulin sensitivity in a human trial when given as a pasteurised supplement; it is elevated in Parkinson's disease and multiple sclerosis. Whether it is desirable depends on the question being asked, which is why this group is not scored as favourable.

What is driving this reading

6 of 8 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Mediterraneibacter gnavus full page →	Opportunist	67%	6.93%	**×110** vs typical carrier **99th** percentile carried by 47% of adults
Barnesiella intestinihominis	Beneficial	11%	1.16%	**+181%** vs typical carrier **79th** percentile carried by 51% of adults
Akkermansia muciniphila	Beneficial	8%	0.846%	**+166%** vs typical carrier **66th** percentile carried by 51% of adults
Bacteroides caccae	Conditional	7%	0.759%	**+83%** vs typical carrier **69th** percentile carried by 61% of adults
Bacteroides thetaiotaomicron	Conditional	6%	0.574%	**+136%** vs typical carrier **71st** percentile carried by 79% of adults
Bifidobacterium bifidum	Beneficial	0%	0.015%	**–90%** vs typical carrier **19th** percentile carried by 22% of adults

Why. This reading is carried by *Mediterraneibacter gnavus*, which alone is **67%** of the group's reads and sits at the **99th** percentile among adults who carry it. Changing this reading means changing them.

Not detected (2 of 8): *Mediterraneibacter torques* (78%), *Bacteroides fragilis* (52%). Percentages are how many reference adults carry each.

What your reading means

notably high

This group is 10.29% of your classified reads, counted over 7 of the 8 species this group is defined by. That places you at the 96th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Mediterraneibacter gnavus* (6.93% of your community); *Barnesiella intestinihominis* (1.16% of your community); *Akkermansia muciniphila* (0.85% of your community); *Bacteroides caccae* (0.76% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

Bacteria that feed on the mucus layer lining your gut. This is normal and part of how the mucus layer renews itself. When dietary fibre is scarce, these organisms turn to mucin as their main carbon source and the barrier thins — an effect demonstrated cleanly in gnotobiotic mice.

Akkermansia muciniphila carries the strongest and most contradictory literature of any species here. It is depleted in obesity and metabolic syndrome and improved insulin sensitivity in a human trial when given as a pasteurised supplement; it is elevated in Parkinson's disease and multiple sclerosis. Whether it is desirable depends on the question being asked, which is why this group is not scored as favourable.

Ruminococcus gnavus is a mucin degrader and is enriched in inflammatory bowel disease, where its glucorhamnan polysaccharide induces TNF-alpha. Same functional group, very different clinical company.

What the research says about changing this

Mucin-degrading species (Akkermansia, R. gnavus) — a context-dependent group LANE C HUMAN TRIAL · LIMITED

Fired by: mucin degraders relative abundance: high

Akkermansia is linked to metabolic health in some cohorts and enriched in MS and Parkinson cohorts in others; R. gnavus blooms in Crohn's flares. The only trials are of one exact pasteurised Akkermansia product with mixed primary endpoints. Group abundance is not a target.

Studied: Pasteurised Akkermansia muciniphila MucT (ATCC BAA-835) (live biotherapeutic) · **in:** people with clinically confirmed overweight/obesity with insulin resistance, ages 18–70, in Belgium, n = 32 · **studied protocol (a source fact, not an instruction):** 10^10 pasteurised cells/day for three months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In a 32-completer exploratory pilot, pasteurised A. muciniphila MucT (10^10 cells/day for three months) improved insulin sensitivity and lowered insulinaemia and cholesterol in overweight/obese insulin-resistant adults. [Depommier C et al. Nat Med 2019;25:1096] · lane C · randomized double-blind placebo-controlled exploratory pilot · n = 40 · benefit
- **Against or limiting —** A 2025 randomized trial of live A. muciniphila AKK-WST01 in obese type 2 diabetes found no overall weight or HbA1c benefit, only exploratory subgroup signals. [Zhang et al. 2025 — AKK-WST01 in obese type 2 diabetes] · lane X · randomized placebo-controlled · null
- **Against or limiting —** A 2026 pasteurised-Akkermansia trial missed its primary insulin-sensitivity endpoint. [Suenaert et al. 2026 — pasteurised A. muciniphila, primary insulin-sensitivity endpoint missed] · lane X · randomized placebo-controlled · null

Applicability: The study required a clinical diagnosis of insulin resistance. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “mucus-eating bacteria damaging your gut” · “boost Akkermansia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Sulfate-reducing bacteria ▲

0.09% of classified reads · **–50%** vs typical · 1 of 4 species detected · 2 in the literature not nameable here



33rd TYPICAL

Bacteria that breathe sulfate instead of oxygen, releasing hydrogen sulfide as they do. They are normal members of most healthy communities. Sustained high sulfide interferes with how colon cells use butyrate for fuel, and sulfate-reducer expansion is reported in ulcerative colitis.

A small group in a low-abundance niche, so this reading is noisier than the larger groups and more sensitive to sequencing depth.

What is driving this reading

1 of 4 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Bilophila wadsworthia	● Opportunist	**100%**	0.089%	**–13%** vs typical carrier **46th** percentile carried by 61% of adults

Why. Led by *Bilophila wadsworthia* at **100%** of the group, the 46th percentile among carriers. The group as a whole sits within the reference range.

Not detected (3 of 4): *Desulfovibrio piger* (20%), *Desulfovibrio fairfieldensis* (2%), *Desulfovibrio legallii* (0%). Percentages are how many reference adults carry each.

Two measurements of the same function disagree

Read this beside the hydrogen-sulfide capacity reading. The sulfate-reducing species reading sits at the 70th percentile; the hydrogen-sulfide capacity reading is at the 29th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not. Of the two, the gene reading is the better guide here, because this capacity does not depend on any one set of species. Hydrogen sulfide has two independent sources: dissimilatory sulfate reduction by Desulfovibrio and relatives, and cysteine degradation, which is widespread in ordinary gut bacteria. The gene panel sees both; the group sees only the sulfate reducers, which are a minority of stool even when sulfide capacity is high.

What your reading means

typical

This group is 0.09% of your classified reads, counted over 1 of the 4 species this group is defined by. That places you at the 33rd percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bilophila wadsworthia* (0.09% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

Bacteria that breathe sulfate instead of oxygen, releasing hydrogen sulfide as they do. They are normal members of most healthy communities. Sustained high sulfide interferes with how colon cells use butyrate for fuel, and sulfate-reducer expansion is reported in ulcerative colitis.

A small group in a low-abundance niche, so this reading is noisier than the larger groups and more sensitive to sequencing depth.

Bilophila wadsworthia rose sharply on an animal-based diet in the David et al. dietary intervention, making this group unusually diet-responsive over days rather than months.

‹ at a glance

Methanogens (archaea) ▲

0.00% of classified reads · 0 of 3 species detected · 1 in the literature not nameable here



35th NOT DETECTED · TYPICAL

The methane-producing archaea in your gut. Roughly a third to a half of adults carry them in quantity and the rest carry almost none — this is a normal split, not a spectrum of health. Methane slows intestinal transit, and high carriage is consistently associated with constipation.

Bimodal by nature. A near-zero reading is the normal state for a large fraction of healthy adults and means the hydrogen goes elsewhere — typically to acetogens or sulfate reducers instead.

What is driving this reading 0 of 3 species detected; the reading is their sum

Why. None of this group's species were detected in your sample.

Not detected (3 of 3): *Methanobrevibacter smithii* (30%), *Methanosphaera stadtmanae* (4%), *Methanobrevibacter woesei* (0%). Percentages are how many reference adults carry each.

Two measurements of the same function disagree

Read this beside the methane capacity (mcrA) reading. The methanogen abundance reading sits at the 35th percentile; the methane capacity (mcrA) reading is at the <1st. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not; the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Neither reading is preferred: a split this size means one of the two engines is at its limit for this function. mcrA is exclusive to methanogenic archaea, so these two should track each other tightly. A split means one of the two engines is at its limit: methanogens are archaea, they are a small fraction of stool DNA, and the bacterial profiler under-reports them.

What your reading means not detected · typical

This group is 0.00% of your classified reads, counted over 0 of the 3 species this group is defined by. That places you at the 35th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

The methane-producing archaea in your gut. Roughly a third to a half of adults carry them in quantity and the rest carry almost none — this is a normal split, not a spectrum of health. Methane slows intestinal transit, and high carriage is consistently associated with constipation.

Bimodal by nature. A near-zero reading is the normal state for a large fraction of healthy adults and means the hydrogen goes elsewhere — typically to acetogens or sulfate reducers instead.

Carriage is partly heritable and increases with age and with slow transit, so cause and effect run both ways with constipation.

‹ at a glance

Polyphenol metabolisers (urolithin and equol) ▼

0.15% of classified reads · **+35%** vs typical · 2 of 5 species detected · 4 in the literature not nameable here



59th TYPICAL

The bacteria that turn compounds in pomegranate, berries, nuts and soy into the forms your body can actually use. Whether you benefit from these foods depends partly on whether you carry the converters, and a substantial share of people do not.

Eggerthella lenta is a versatile metaboliser: it also inactivates the heart drug digoxin via its cgr operon, and it is enriched in several disease cohorts (rheumatoid arthritis, Parkinson's). It is counted here for its urolithin activity, not as a health marker.

What is driving this reading

2 of 5 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Eggerthella lenta	● Opportunist	67%	0.104%	**+189%** vs typical carrier **75th** percentile carried by 47% of adults
Gordonibacter pamelaeae	● Beneficial	33%	0.051%	**+232%** vs typical carrier **77th** percentile carried by 62% of adults

Why. Led by *Eggerthella lenta* at **67%** of the group, the 75th percentile among carriers. The group as a whole sits within the reference range.

Not detected (3 of 5): *Adlercreutzia equolifaciens* (64%), *Slackia isoflavoniconvertens* (37%), *Slackia exigua* (0%). Percentages are how many reference adults carry each.

Two measurements of the same function disagree

Read this beside the urolithin and equol capacity reading. The polyphenol-metabolising species reading sits at the 80th percentile; the urolithin and equol capacity reading is at the 40th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Of the two, the species reading is the better guide here. Urolithin and equol production is strain-level: two isolates of the same species differ, and only a handful of characterised strains do it. The curated species are the better guide; a high gene reading without them is most likely a related enzyme.

What your reading means

typical

This group is 0.15% of your classified reads, counted over 2 of the 5 species this group is defined by. That places you at the 59th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Eggerthella lenta* (0.10% of your community); *Gordonibacter pamelaeae* (0.05% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

The bacteria that turn compounds in pomegranate, berries, nuts and soy into the forms your body can actually use. Whether you benefit from these foods depends partly on whether you carry the converters, and a substantial share of people do not.

Eggerthella lenta is a versatile metaboliser: it also inactivates the heart drug digoxin via its cgr operon, and it is enriched in several disease cohorts (rheumatoid arthritis, Parkinson's). It is counted here for its urolithin activity, not as a health marker.

Presence of the converters does not guarantee the conversion happens at a useful rate — the metabotype is determined by the whole community and by diet, and the gene-level detail (the specific dehydroxylases) is below the resolution of species-level profiling.

Organisms from elsewhere in the body

‹ at a glance

Oral-origin organisms found in stool ▲

0.00% of classified reads · 0 of 33 species detected



8th NOT DETECTED · LOW

Bacteria that normally live in your mouth, measured in stool. Ordinarily stomach acid and bile stop them; when they arrive in quantity it usually means one of those barriers is reduced. Acid-suppressing medication is the most common everyday cause by far.

Proton pump inhibitors raise this group substantially and are extremely common. Before reading anything else into a high value, check whether the person takes one.

What is driving this reading

0 of 33 species detected; the reading is their sum

Why. This reading is low because **2** species most reference adults carry were not detected: *Streptococcus salivarius* (70%), *Streptococcus parasanguinis* (57%).

Not detected (33 of 33): *Streptococcus salivarius* (70%), *Streptococcus parasanguinis* (57%), *Haemophilus parainfluenzae* (41%), *Veillonella parvula* (39%), *Veillonella atypica* (30%), *Veillonella dispar* (29%), *Rothia mucilaginosa* (23%), *Streptococcus mitis* (22%), *Actinomyces odontolyticus* (21%), *Gemella sanguinis* (19%), *Streptococcus oralis* (16%), *Streptococcus australis* (15%), *Streptococcus gordonii* (12%), *Streptococcus vestibularis* (12%), *Streptococcus sanguinis* (11%), *Streptococcus anginosus group* (9%), *Gemella haemolysans* (6%), *Veillonella rogosae* (6%), *Actinomyces graevenitzii* (5%), *Parvimonas micra* (4%), *Solobacterium moorei* (3%), *Atopobium parvulum* (3%), *Gemella morbillorum* (3%), *Atopobium rimae* (2%), *Streptococcus cristatus* (2%), *Rothia dentocariosa* (2%), *Peptostreptococcus stomatis* (2%), *Fusobacterium nucleatum* (1%), *Actinomyces naeslundii* (1%), *Fusobacterium periodonticum* (1%), *Porphyromonas gingivalis* (0%), *Rothia aeria* (0%), *Porphyromonas endodontalis* (0%). Percentages are how many reference adults carry each.

What your reading means

not detected · low

This group is 0.00% of your classified reads, counted over 0 of the 33 species this group is defined by. That places you at the 8th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

Bacteria that normally live in your mouth, measured in stool. Ordinarily stomach acid and bile stop them; when they arrive in quantity it usually means one of those barriers is reduced. Acid-suppressing medication is the most common everyday cause by far.

Proton pump inhibitors raise this group substantially and are extremely common. Before reading anything else into a high value, check whether the person takes one.

Periodontal disease increases the oral load being swallowed, which raises this group without any change in gut barrier function.

Commonly beneficial organisms

‹ at a glance

Supplement and fermented-dairy species

3.88% of classified reads · **×56** vs typical · 8 of 14 species detected · 2 in the literature not nameable here



98th NOTABLY HIGH

The bacteria found in yoghurt, kefir and probiotic capsules. These are mostly transient in the adult gut, so a high reading usually means you are currently eating or taking them rather than that they have settled in — and it can also mean the opposite of good news, because the same organisms rise when stomach acid is low or when bacteria overgrow the small intestine. A high or low reading is not a health verdict on its own, which is why this scale is drawn in neutral grey.

A high Lactobacillus reading in an adult is most often diet or a supplement passing through; colonisation is uncommon. It is also raised in small-intestinal bacterial overgrowth and in people with low stomach acid (proton pump inhibitor use), where oral and upper-gut organisms reach the stool in larger numbers.

What is driving this reading

8 of 14 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Bifidobacterium breve	● Beneficial	**36%**	1.40%	**×63** vs typical carrier **91st** percentile carried by 5% of adults
Lactiplantibacillus plantarum	● Beneficial	**29%**	1.14%	**×131** vs typical carrier **>99th** percentile carried by 2% of adults
Bifidobacterium animalis	● Beneficial	**20%**	0.767%	**×24** vs typical carrier **93rd** percentile carried by 6% of adults
Lacticaseibacillus rhamnosus	● Beneficial	**5%**	0.185%	**×32** vs typical carrier **>99th** percentile carried by 2% of adults
Streptococcus thermophilus	● Conditional	**5%**	0.175%	**+290%** vs typical carrier **76th** percentile carried by 36% of adults
Ligilactobacillus salivarius	● Beneficial	**3%**	0.114%	**×10** vs typical carrier **78th** percentile carried by 5% of adults
Limosilactobacillus reuteri	● Beneficial	**1%**	0.057%	**+76%** vs typical carrier **52nd** percentile carried by 1% of adults
Lactobacillus acidophilus	● Beneficial	**1%**	0.040%	**+110%** vs typical carrier **69th** percentile carried by 3% of adults

Why. This reading is carried by *Bifidobacterium breve*, which alone is **36%** of the group's reads and sits at the **91st** percentile among adults who carry it. *Lactiplantibacillus plantarum* contributes another **29%**. Changing this reading means changing them.

Not detected (6 of 14): *Lactobacillus ruminis* (13%), *Bifidobacterium dentium* (12%), *Lactobacillus delbrueckii* (5%), *Lactobacillus fermentum* (3%), *Lactobacillus gasseri* (1%), *Lactobacillus johnsonii* (0%). Percentages are how many reference adults carry each.

Two measurements of the same function disagree

Read this beside the bile-salt deconjugation capacity reading. The probiotic-genus species reading sits at the 99th percentile; the bile-salt deconjugation capacity reading is at the 17th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Of the two, the gene reading is the better guide here, because this capacity does not depend on any one set of species. Bile salt hydrolase is not a probiotic trait: it is carried across Lactobacillaceae, Bifidobacterium, Clostridium, Bacteroides and Enterococcus alike. A high capacity with few probiotic-genus species simply means the ordinary residents are doing it.

What your reading means

notably high

This group is 3.88% of your classified reads, counted over 8 of the 14 species this group is defined by. That places you at the 98th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Bifidobacterium breve* (1.40% of your community); *Lactiplantibacillus plantarum* (1.14% of your community); *Bifidobacterium animalis* (0.77% of your community); *Lacticaseibacillus rhamnosus* (0.18% of your community). Anything that changes this reading

has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

The bacteria found in yoghurt, kefir and probiotic capsules. These are mostly transient in the adult gut, so a high reading usually means you are currently eating or taking them rather than that they have settled in — and it can also mean the opposite of good news, because the same organisms rise when stomach acid is low or when bacteria overgrow the small intestine. A high or low reading is not a health verdict on its own, which is why this scale is drawn in neutral grey.

A high Lactobacillus reading in an adult is most often diet or a supplement passing through; colonisation is uncommon. It is also raised in small-intestinal bacterial overgrowth and in people with low stomach acid (proton pump inhibitor use), where oral and upper-gut organisms reach the stool in larger numbers.

Bifidobacterium dentium is an oral species and is counted here for completeness; it is not a probiotic and is also a member of the oral-origin group.

What the research says about changing this

Probiotic-type species — stool abundance reflects transit as much as colonisation LANE D HUMAN · NON-RANDOMISED

Fired by: probiotics relative abundance: notably high

Lactobacilli and bifidobacteria in stool largely reflect what passed through (fermented foods, supplements) rather than residents. Colonisation is person- and strain-specific, probiotic effects are strain- and indication-specific, and after antibiotics a probiotic slowed native recovery.

- An 11-strain probiotic colonised the gut mucosa in a person-, strain- and region-specific way; stool abundance of probiotic species reflects transit as much as colonisation. [Zmora N et al. Cell 2018;174:1388] · lane D · human intervention with endoscopic sampling · mixed
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

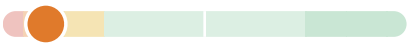
This report will not say: “you need a probiotic” · “probiotic deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Health-associated core species ▼

6.43% of classified reads · **–60%** vs typical · 10 of 25 species detected



11th LOW

The organisms that healthy people consistently have more of. Most ferment fibre into short-chain fatty acids, most are oxygen-intolerant, and most are lost first when the gut is disturbed. This reading tracks the "core" of a well-functioning adult community.

"Health-associated" is a statistical description of who carries more, not a mechanism, and the set was derived from mostly Western industrialised cohorts. Some members (Akkermansia in particular) reverse direction in specific diseases, which is why disease profiles score species individually while this group provides the overview.

What is driving this reading 8 of 25 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Alistipes putredinis	Conditional	30%	1.95%	**+20%** vs typical carrier **56th** percentile carried by 75% of adults
Bifidobacterium longum	Beneficial	29%	1.84%	**×10** vs typical carrier **92nd** percentile carried by 67% of adults
Akkermansia muciniphila	Beneficial	13%	0.846%	**+166%** vs typical carrier **66th** percentile carried by 51% of adults
Alistipes shahii	Conditional	9%	0.571%	**+88%** vs typical carrier **86th** percentile carried by 49% of adults
Bifidobacterium adolescentis	Beneficial	7%	0.435%	**≈ typical** vs typical carrier **50th** percentile carried by 51% of adults
Odoribacter splanchnicus	Beneficial	6%	0.364%	**+62%** vs typical carrier **70th** percentile carried by 74% of adults
Allocoprococcus comes	Unknown	5%	0.330%	**+72%** vs typical carrier **70th** percentile carried by 79% of adults
Agathobacter rectalis	Beneficial	2%	0.097%	**–92%** vs typical carrier **16th** percentile carried by 88% of adults

Why. This reading is low because **12** species most reference adults carry were not detected: *Faecalibacterium prausnitzii* (98%), *Anaerostipes hadrus* (87%), *Eubacterium hallii* (86%), *Roseburia inulinivorans* (85%), and 8 more. What is present, led by *Alistipes putredinis*, is not enough to make up the sum.

Not detected (17 of 25): *Faecalibacterium prausnitzii* (98%), *Anaerostipes hadrus* (87%), *Eubacterium hallii* (86%), *Roseburia inulinivorans* (85%), *Roseburia hominis* (83%), *Eubacterium eligens* (81%), *Roseburia intestinalis* (73%), *Oscillibacter sp 57 20* (73%), *Coprococcus catus* (69%), *Ruminococcus bromii* (62%), *Eubacterium siraeum* (56%), *Eubacterium ventriosum* (52%), *Bacteroides cellulosilyticus* (49%), *Ruminococcus lactaris* (47%), *Coprococcus eutactus* (36%), *Butyrivibrio crossotus* (18%), *Lachnospiraceae bacterium 2 1 46FAA* (1%). Percentages are how many reference adults carry each.

What your reading means low

This group is 6.43% of your classified reads, counted over 10 of the 25 species this group is defined by. That places you at the 11th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Alistipes putredinis* (1.95% of your community); *Bifidobacterium longum* (1.84% of your community); *Akkermansia muciniphila* (0.85% of your community); *Alistipes shahii* (0.57% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

The organisms that healthy people consistently have more of. Most ferment fibre into short-chain fatty acids, most are oxygen-intolerant, and most are lost first when the gut is disturbed. This reading tracks the "core" of a well-functioning adult community.

"Health-associated" is a statistical description of who carries more, not a mechanism, and the set was derived from mostly Western industrialised cohorts. Some members (Akkermansia in particular) reverse direction in specific diseases, which is why disease profiles score species individually while this group provides the overview.

This group overlaps heavily with the butyrate-producer group by design: fibre fermenters are health-associated because fibre fermentation is what a healthy colon does.

‹ at a glance

Resident bifidobacteria ▼

2.29% of classified reads · **+291%** vs typical · 3 of 5 species detected



79th ABOVE AVERAGE

The bifidobacteria that actually live in an adult colon, rather than the ones passing through from food or a supplement. They are consistently more abundant in healthy adults, and they feed the butyrate producers that line your gut wall. More of these is generally the direction you want.

A low reading is common and is not a diagnosis. Bifidobacterial abundance falls with age, with low-fibre eating and after antibiotics, and a single stool sample cannot separate those causes.

What is driving this reading

3 of 5 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Bifidobacterium longum	● Beneficial	80%	1.84%	**×10** vs typical carrier **92nd** percentile carried by 67% of adults
Bifidobacterium adolescentis	● Beneficial	19%	0.435%	≈ **typical** vs typical carrier **50th** percentile carried by 51% of adults
Bifidobacterium bifidum	● Beneficial	1%	0.015%	**–90%** vs typical carrier **19th** percentile carried by 22% of adults

Why. This reading is carried by *Bifidobacterium longum*, which alone is **80%** of the group's reads and sits at the **92nd** percentile among adults who carry it. Changing this reading means changing them.

Not detected (2 of 5): *Bifidobacterium pseudocatenulatum* (36%), *Bifidobacterium catenulatum* (10%). Percentages are how many reference adults carry each.

What your reading means

above average

This group is 2.29% of your classified reads, counted over 3 of the 5 species this group is defined by. That places you at the 79th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bifidobacterium longum* (1.84% of your community); *Bifidobacterium adolescentis* (0.44% of your community); *Bifidobacterium bifidum* (0.01% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

The bifidobacteria that actually live in an adult colon, rather than the ones passing through from food or a supplement. They are consistently more abundant in healthy adults, and they feed the butyrate producers that line your gut wall. More of these is generally the direction you want.

A low reading is common and is not a diagnosis. Bifidobacterial abundance falls with age, with low-fibre eating and after antibiotics, and a single stool sample cannot separate those causes.

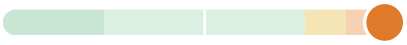
Bifidobacterium animalis and Bifidobacterium breve are deliberately not counted here. Both are predominantly supplement and fermented-dairy organisms in adults, so they are grouped with the other transient probiotic species where a high reading is read as intake rather than as colonisation.

Opportunistic organisms

‹ at a glance

Opportunistic pathogens and pathobionts ▲

7.08% of classified reads · **×37** vs typical · 4 of 20 species detected · 2 in the literature not nameable here



95th HIGH

Bacteria that most healthy adults carry at trace levels or not at all, and which take over when the normal community is disrupted — after antibiotics, during illness, or in a low-diversity gut. A high reading is a general sign of disturbance, not a diagnosis of any specific infection.

Escherichia coli is the largest term in this sum and is itself a mix of harmless commensal strains and pathogenic ones; species-level profiling does not separate them. Read a high E. coli-driven total as "Enterobacteriaceae expansion", which is meaningful on its own.

What is driving this reading

4 of 20 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Mediterraneibacter gnavus full page →	● Opportunist	98%	6.93%	**×110** vs typical carrier **99th** percentile carried by 47% of adults
Bilophila wadsworthia	● Opportunist	1%	0.089%	**–13%** vs typical carrier **46th** percentile carried by 61% of adults
Escherichia coli	● Opportunist	0%	0.033%	**19th** percentile carried by 62% of adults
Enterococcus faecium full page →	● Opportunist	0%	0.021%	**+78%** vs typical carrier **66th** percentile carried by 5% of adults

Why. This reading is carried by *Mediterraneibacter gnavus*, which alone is **98%** of the group's reads and sits at the **99th** percentile among adults who carry it. Changing this reading means changing them.

Not detected (16 of 20): *Haemophilus parainfluenzae* (41%), *Klebsiella pneumoniae* (21%), *Klebsiella variicola* (18%), *Streptococcus anginosus group* (9%), *Enterobacter cloacae complex* (7%), *Fusobacterium mortiferum* (5%), *Enterococcus faecalis* (5%), *Clostridium perfringens* (4%), *Citrobacter freundii* (4%), *Fusobacterium varium* (3%), *Klebsiella oxytoca* (3%), *Fusobacterium nucleatum* (1%), *Pseudomonas aeruginosa group* (1%), *Proteus mirabilis* (1%), *Clostridioides difficile* (0%), *Staphylococcus aureus* (0%). Percentages are how many reference adults carry each.

What your reading means

high

This group is 7.08% of your classified reads, counted over 4 of the 20 species this group is defined by. That places you at the 95th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Mediterraneibacter gnavus* (6.93% of your community); *Bilophila wadsworthia* (0.09% of your community); *Escherichia coli* (0.03% of your community); *Enterococcus faecium* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

Bacteria that most healthy adults carry at trace levels or not at all, and which take over when the normal community is disrupted — after antibiotics, during illness, or in a low-diversity gut. A high reading is a general sign of disturbance, not a diagnosis of any specific infection.

Escherichia coli is the largest term in this sum and is itself a mix of harmless commensal strains and pathogenic ones; species-level profiling does not separate them. Read a high E. coli-driven total as "Enterobacteriaceae expansion", which is meaningful on its own.

Detection of Clostridioides difficile in a stool sample is not the same as C. difficile infection. Asymptomatic carriage runs around 5-15% of healthy adults and is higher in infants.

What the research says about changing this

More opportunistic organisms than the reference — what the human evidence does and does not support LANE A CLINICAL ROUTE

Fired by: opportunistic pathogens relative abundance: high

Enterobacterales and other opportunists live in most healthy guts; a higher relative share is common after antibiotics, with proton-pump inhibitors, or when resident biomass falls. If there are symptoms, clinical stool testing (IDSA) is the route. No human trial shows that berberine, oregano oil, allicin, black-seed oil or multi-ingredient botanical products reduce stool E. coli or Klebsiella; the frequently cited herbal study was a retrospective breath-test chart review.

Studied: Clinical stool pathogen testing and assessment per IDSA infectious-diarrhoea guideline when symptomatic (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** stool culture/PCR and clinical assessment when symptomatic (IDSA 2017)

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: strong for.

- Enteric pathogen findings are confirmed with clinical culture/PCR and managed per IDSA guidance when symptoms are present; antibiotic choice is a clinical decision, never a report output. [Shane AL et al. IDSA Clinical Practice Guidelines for the Diagnosis and Management of Infectious Diarrhea. Clin Infect Dis 2017:65:e45] · lane A · guideline
- Relative enrichment of oral bacteria in stool can arise because resident gut biomass falls, not because oral organisms invaded; absolute load and oral-health context are needed. [Liao C et al. Nat Microbiol 2024 — oral bacteria in stool track loss of gut biomass] · lane D · longitudinal cohort with absolute quantification
- **Against or limiting —** The often-cited 2014 herbal SIBO report was a retrospective, non-randomized, patient-choice chart review with a breath-test endpoint (46% vs 34% normalisation, not significant). It does not establish that any botanical eradicates stool E. coli or Klebsiella. [Chedid V et al. Glob Adv Health Med 2014;3:16] · lane X · retrospective, non-randomized, patient-choice chart review · n = 104 · null
- **Against or limiting —** Multi-ingredient botanical antimicrobial products (e.g. GI Microb-X) have no human trial for stool-pathobiont reduction; evidence cannot be split across ingredients and composition is not verified by any regulator before sale. [FDA — dietary supplements are not reviewed for efficacy before sale] · lane X · regulatory fact
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Conventional route: Clinical stool culture/PCR only when symptomatic.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “kill the bad bacteria” · “natural antibiotics eradicate” · “dysbiosis cured”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

Cell-surface characteristics

‹ at a glance

Hexa-acylated LPS carriers ▲

0.12% of classified reads · **–87%** vs typical · 2 of 17 species detected · 1 in the literature not nameable here



16th BELOW AVERAGE

Bacteria whose cell-surface lipopolysaccharide strongly activates the innate immune receptor TLR4. Not all Gram-negative bacteria do this — the Bacteroides that dominate most healthy guts carry a weakly-activating variant, which is why counting them all together would be misleading.

Carriage of hexa-acylated LPS is normal. Enterobacteriaceae are ordinary members of a healthy gut community at low abundance; the concern in the literature is expansion, not presence.

What is driving this reading 2 of 17 species detected; the reading is their sum

ORGANISM	CLASS	SHARE OF GROUP	IN YOUR SAMPLE	YOUR LEVEL
Bilophila wadsworthia	● Opportunist	73%	0.089%	–13% vs typical carrier **46th** percentile carried by 61% of adults
Escherichia coli	● Opportunist	27%	0.033%	**19th** percentile carried by 62% of adults

Why. Every common member is present; the reading is low because they are scarce. *Bilophila wadsworthia* is the largest at 0.089% of your community, the **46th** percentile among carriers.

Not detected (15 of 17): *Klebsiella pneumoniae* (21%), *Desulfovibrio piger* (20%), *Klebsiella variicola* (18%), *Enterobacter cloacae complex* (7%), *Citrobacter freundii* (4%), *Fusobacterium varium* (3%), *Klebsiella aerogenes* (3%), *Klebsiella oxytoca* (3%), *Fusobacterium nucleatum* (1%), *Citrobacter koseri* (1%), *Escherichia fergusonii* (1%), *Morganella morganii* (1%), *Proteus mirabilis* (1%), *Escherichia albertii* (0%), *Salmonella enterica* (0%). Percentages are how many reference adults carry each.

What your reading means below average

This group is 0.12% of your classified reads, counted over 2 of the 17 species this group is defined by. That places you at the 16th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bilophila wadsworthia* (0.09% of your community); *Escherichia coli* (0.03% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

The research behind it

Bacteria whose cell-surface lipopolysaccharide strongly activates the innate immune receptor TLR4. Not all Gram-negative bacteria do this — the Bacteroides that dominate most healthy guts carry a weakly-activating variant, which is why counting them all together would be misleading.

Carriage of hexa-acylated LPS is normal. Enterobacteriaceae are ordinary members of a healthy gut community at low abundance; the concern in the literature is expansion, not presence.

This is not a blood endotoxin measurement. Whether LPS reaches the circulation depends on gut barrier integrity, which is not measured here and for which no validated stool test exists.

16 Organisms that need attention, one by one

One page per organism flagged in Part A. Each says what the organism is and does in a gut, why it is flagged in your sample, what has been shown to move it in the direction you would want — with the strength of that evidence stated on every line — and what competes with it. Laboratory and animal findings are listed and labelled as such; they are leads, not recommendations. Nothing here is a prescription.

Human trial	Human cohorts	Animal model	Laboratory culture
given to people; the organism measured before and after	associations in people; nothing was given	a mouse or rat gut, not a human one	a culture dish; no gut at all

OVERGROWN — at a level that is a concern

‹ at a glance

Mediterraneibacter gnavus ● Opportunist

OVERGROWN

What it is and what it does

A mucus-degrading resident of most healthy adults at low levels, typically well under 1%. It lives in the mucus layer coating the gut wall and eats the sugars in it, which at low abundance is normal turnover. When it expands it does two things the literature keeps returning to: it thins the protective mucus faster than the lining rebuilds it, and it produces a glucorhamnan polysaccharide that triggers TNF-alpha release from immune cells. It is among the most consistently expanded organisms in Crohn's disease and in spondyloarthritis flares, and it blooms after antibiotics and on a low-fibre diet because it can live on host mucus when dietary fibre is scarce.

Why it is flagged in your sample

YOUR LEVEL	**6.93%** of the community
POSITION	**99th** percentile among reference adults who carry it — **×110** against their typical level of 0.121%
HOW COMMON	carried by **47%** of reference adults
CLASS	Opportunist — Normal at low levels.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, MetaPhlAn 3, SingleM
IDENTIFIERS	scoring Ruminococcus_gnavus, SGB4584, GCF_008121495, mOTU 000538, UHGG species 2101; GTDB R232 Mediterraneibacter gnavus
DETECTION	**supported** — seen by 5 independent methods: genome_sketch, marker, marker_window, read_classification, universal_marker
REFERENCE STRAINS	12 in BacDive: culture collection DSM 108212; anaerobe; grows at 37 °C; isolated from patient, blood. Traits of reference strains, not measured in your sample.
STRAIN PLACEMENT	nearest reference genome GCF_008121495.1.fna, distance 0.420148; polymorphic sites at 1.59%; a consensus can hide co-resident strains. A placement, not proof of an identical strain.

What the level means. Presence is normal. Abundance is the finding: above a few percent it is one of the strongest single-organism markers of an inflamed, mucus-depleted gut.

What has been shown to lower it

5 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre, especially inulin and resistant starch	prebiotic fibre	**human trial**	Fibre feeds competitors that outgrow it and gives it an alternative to host mucus. Inulin supplementation reduced R. gnavus relative abundance in adults in controlled feeding studies. Vandeputte D et al. Gut 2017;66:1968; Holscher HD, Gut Microbes 2017
Bifidobacterium longum subsp. infantis	probiotic	**human trial**	In infants, B. infantis colonisation reduced R. gnavus and other mucin-degrading Lachnospiraceae; the mechanism is competition for milk oligosaccharides and mucin sugars and applies less directly in adults. Henrick BM et al. Cell 2021;184:3884
Exclusive enteral nutrition (in Crohn's disease)	diet pattern	**human trial**	Clinically supervised; reduces R. gnavus alongside inflammation in paediatric Crohn's. Quince C et al. Am J Gastroenterol 2015;110:1718
Mediterranean-pattern diet (high fibre, low processed food)	diet pattern	**human cohorts**	Higher fibre intake is consistently associated with lower R. gnavus abundance across cohorts. Bolte LA et al. Gut 2021;70:1287
Lactobacillus rhamnosus GG	probiotic	**laboratory culture**	Co-culture and biofilm assays show LGG suppresses R. gnavus growth on mucin. Crost EH et al. FEMS Microbiol Rev 2023 (review of mucin degraders)

Competes with: *Faecalibacterium prausnitzii*, *Bifidobacterium longum*, *Akkermansia muciniphila*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: Broad-spectrum antibiotics (human cohorts); Low-fibre, high-fat Western diet (animal model).

‹ at a glance

Anaerotignum lactatifermentans ● Conditional

OVERGROWN

What it is and what it does

Ferments lactate to propionate and acetate. Reported expanded in some IBD cohorts; otherwise an ordinary low-level resident.

Why it is flagged in your sample

YOUR LEVEL	**0.776%** of the community
POSITION	**99th** percentile among reference adults who carry it — **×17** against their typical level of 0.096%
HOW COMMON	carried by **13%** of reference adults
CLASS	Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, MetaPhlAn 3, SingleM
IDENTIFIERS	scoring Anaerotignum_lactatifermentans, SGB5193, GCF_900142265, mOTU 000201, UHGG species 2705; GTDB R232 Anaerotignum lactatifermentans
DETECTION	**supported** — seen by 5 independent methods: genome_sketch, marker, marker_window, read_classification, universal_marker
REFERENCE STRAINS	2 in BacDive: culture collection DSM 109233, DSM 14214; anaerobe, obligate anaerobe; grows at 37 °C; isolated from large intestine, chicken; produces acetate, butyrate, isobutyrate, isovalerate. Traits of reference strains, not measured in your sample.
STRAIN PLACEMENT	nearest reference genome GCA_047014905.1.fna, distance 0.897207; polymorphic sites at 0.06%; a consensus can hide co-resident strains. A placement, not proof of an identical strain.

What the level means. High levels suggest lactate is accumulating in the gut, which usually means lactate producers (Lactobacillus, Bifidobacterium, Streptococcus) are abundant — often a probiotic or fermented-food signal.

What has been shown to lower it 1 item, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre	prebiotic fibre	**human cohorts**	Bolte LA et al. Gut 2021

What the intervention registry holds for this organism

Uncommon or relatively high organism — no supported targeted intervention LANE D NO SUPPORTED INTERVENTION

Fired by: Anaerofustis stercorihominis: unexpected present; Anaerotignum lactatifermentans: high; Barnesiella intestinihominis: high; +12 more

This organism is uncommon in the reference or higher than the reference range here. Unless it is named in one of the specific cards above, no human study shows that reducing it changes a health outcome.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “dangerous bacteria” · “must be eliminated”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Enterocloster citroniae ● Opportunist · genus-level knowledge

OVERGROWN

What it is and what it does

A group of bile-tolerant Lachnospiraceae, formerly filed under Clostridium. They are ordinary low-level residents that thrive on bile acids and are expanded in autism-spectrum, IBD and post-antibiotic cohorts. E. bolteae in particular has been isolated from bloodstream and abdominal infections.

Why it is flagged in your sample

YOUR LEVEL	**0.023%** of the community
POSITION	**90th** percentile among reference adults who carry it — **+381%** against their typical level of 0.015%
HOW COMMON	carried by **36%** of reference adults
CLASS	Opportunist — Normal at low levels.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, MetaPhlAn 3
IDENTIFIERS	scoring Clostridium_citroniae, SGB4761, mOTU 015990, UHGG species 452, GCF_040545455.1; GTDB R232 Enterocloster citroniae
DETECTION	**supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker
REFERENCE STRAINS	3 in BacDive: culture collection DSM 108162, DSM 19261, DSM 6877; anaerobe; grows at 28–37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample.

What the level means. Bile tolerance means a high-fat diet feeds them; expansion is a common signal of a fat-rich, fibre-poor pattern and of a recently disturbed community.

What has been shown to lower it 3 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre (inulin, pectin)	prebiotic fibre	**human trial**	Fibre shifts the Lachnospiraceae balance away from bile-tolerant members toward butyrate producers. So D et al. Am J Clin Nutr 2018;107:965 (meta-analysis)
Lower saturated-fat intake	diet pattern	**human cohorts**	Bile-acid load falls with dietary fat, and with it the bile-tolerant genera. David LA et al. Nature 2014;505:559
Bifidobacterium longum strains	probiotic	**animal model**	Wang Y et al. Front Microbiol 2020 (Clostridium cluster XIVa modulation)

Competes with: *Faecalibacterium prausnitzii*, *Roseburia intestinalis*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: High-fat, animal-based diet (human trial).

‹ at a glance

Clostridium innocuum ● Opportunist

OVERGROWN

What it is and what it does

A vancomycin-resistant Clostridium found at low levels in many adults. It is an emerging cause of antibiotic-associated diarrhoea that mimics C. difficile infection, and it is enriched in Crohn's disease, where it drives fat accumulation in the gut wall ("creeping fat"). A trace is ordinary; expansion after antibiotics is the concern.

Why it is flagged in your sample

YOUR LEVEL	**0.041%** of the community
POSITION	**81st** percentile among reference adults who carry it — **+372%** against their typical level of 0.018%
HOW COMMON	carried by **29%** of reference adults
CLASS	Opportunist — Normal at low levels.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue, MetaPhlAn 3
IDENTIFIERS	scoring Clostridium_innocuum, SGB4037, GCF_012317185, mOTU 001874, UHGG species 2133, panel BackhedF_2015_ERR525836_bin.6; GTDB R232 Clostridium_AQ innocuum
DETECTION	**supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker
REFERENCE STRAINS	11 in BacDive: culture collection DSM 100998, DSM 107457, DSM 108163; anaerobe; grows at 37 °C; isolated from human, blood. Traits of reference strains, not measured in your sample.

What the level means. Its intrinsic vancomycin resistance means it blooms precisely when C. difficile is being treated.

What has been shown to lower it 2 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre	prebiotic fibre	**human cohorts**	Ha CWY et al. Cell 2020;183:666 (C. innocuum in Crohn's)
Saccharomyces boulardii	probiotic	**animal model**	Yeast probiotic; competes with Clostridia during and after antibiotics and is not itself affected by them. McFarland LV, World J Gastroenterol 2010;16:2202 (review)

Competes with: *Lactobacillus rhamnosus*, *Bifidobacterium longum*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: Vancomycin and other antibiotics (human cohorts).

What the intervention registry holds for this organism

Uncommon or relatively high organism — no supported targeted intervention LANE D NO SUPPORTED INTERVENTION

Fired by: Anaerofustis stercorihominis: unexpected present; Anaerotignum lactatifermentans: high; Barnesiella intestinihominis: high; +12 more

This organism is uncommon in the reference or higher than the reference range here. Unless it is named in one of the specific cards above, no human study shows that reducing it changes a health outcome.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “dangerous bacteria” · “must be eliminated”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Flavonifractor plautii ● Opportunist

OVERGROWN

What it is and what it does

A resident that breaks down dietary flavonoids such as quercetin and catechins, which sounds useful but means it degrades the very polyphenols that feed other beneficial organisms. It is expanded in depression, IBD and colorectal cancer cohorts and lower in some others; the evidence is mixed but leans adverse at high abundance.

Why it is flagged in your sample

YOUR LEVEL	**0.442%** of the community
POSITION	**91st** percentile among reference adults who carry it — **+811%** against their typical level of 0.106%
HOW COMMON	carried by **76%** of reference adults
CLASS	Opportunist — Normal at low levels.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue, MetaPhlAn 3, SingleM
IDENTIFIERS	scoring Flavonifractor_plautii, SGB15132, GCF_000239295, mOTU 000255, UHGG species 289, panel CokerMO_2019_SRR8692221_bin.16; GTDB R232 Flavonifractor plautii
DETECTION	**supported** — seen by 5 independent methods: genome_sketch, marker, marker_window, read_classification, universal_marker
REFERENCE STRAINS	1 in BacDive: culture collection DSM 6740; anaerobe; grows at 37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample.
STRAIN PLACEMENT	nearest reference genome GCA_000508885.1.fna, distance 0.76672; polymorphic sites at 0.26%; a consensus can hide co-resident strains. A placement, not proof of an identical strain.

What the level means. High levels may blunt the benefit of a polyphenol-rich diet by consuming the polyphenols first.

What has been shown to lower it 2 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Bifidobacterium multi-strain probiotics	probiotic	**human trial**	Small trial; Flavonifractor fell alongside a rise in bifidobacteria. Ohsawa K et al. Int J Food Sci Nutr 2015 (B. breve and gut flora)
Dietary fibre	prebiotic fibre	**human cohorts**	Bolte LA et al. Gut 2021

Competes with: *Bifidobacterium adolescentis*, *Akkermansia muciniphila*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: High polyphenol intake without fibre (laboratory culture).

MISSING — commonly carried, not detected here

‹ at a glance

Dorea longicatena ● Conditional · genus-level knowledge

MISSING

What it is and what it does

Gas producers (hydrogen, carbon dioxide) that break down mucus-associated sugars. Expanded in IBS and some multiple sclerosis cohorts, lower in others; D. longicatena is common in healthy adults. Contributes to bloating when abundant.

Why it is flagged in your sample

YOUR LEVEL	**<0.0001%** — not detected; this sample resolves organisms below that level
HOW COMMON	carried by **88%** of reference adults
CLASS	Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed.

What the level means. High Dorea with bloating is a plausible link; missing Dorea is usually unremarkable.

What has been shown to raise it 1 item, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Fermentable fibre	prebiotic fibre	**human trial**	Vandeputte D et al. Gut 2017

Known to push it the other way: Low-FODMAP diet (for gas symptoms) (human trial).

What the intervention registry holds for this organism

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Roseburia inulinivorans ● Beneficial · genus-level knowledge MISSING

What it is and what it does

Flagellated butyrate producers that thrive on dietary fibre, especially beta-glucans and inulin. Lower levels are reported in ulcerative colitis, type 2 diabetes and atherosclerosis cohorts. R. intestinalis has direct anti-inflammatory effects on the gut lining in cell and mouse work.

Why it is flagged in your sample

YOUR LEVEL	**<0.0001%** — not detected; this sample resolves organisms below that level. In the wider inventory, an unconfirmed single-method hit at 0.062% is not counted as a find
HOW COMMON	carried by **86%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.

What the level means. Missing or low Roseburia is a fibre-intake signal as much as a health one: these are among the first organisms to fall on a low-fibre diet and the first to recover on a high-fibre one.

What has been shown to raise it 4 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Oat and barley beta-glucan	prebiotic fibre	human trial	Martinez I et al. ISME J 2013;7:269
Inulin and fructo-oligosaccharides	prebiotic fibre	human trial	Vandeputte D et al. Gut 2017
Whole grains	food	human trial	Vanegas SM et al. Am J Clin Nutr 2017;105:635
Resistant starch	prebiotic fibre	human trial	Walker AW et al. ISME J 2011

Known to push it the other way: Low-fibre diet, antibiotics (human cohorts).

What the intervention registry holds for this organism

Lower or absent butyrate producers — what has been tried to raise them LANE C HUMAN TRIAL · LIMITED

Fired by: Anaerostipes hadrus: absent expected; Faecalibacterium prausnitzii: low; Roseburia faecis: absent expected; +1 more

Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with diet, antibiotics and transit in healthy people. Fermented-food and diverse-fibre diets changed communities in some people; riboflavin supplementation did not reliably raise F. prausnitzii. No live F. prausnitzii product is approved.

Studied: High-fibre versus high-fermented-food diet (Wastyk 2021) (diet) · **in:** people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36 · **studied protocol (a source fact, not an instruction):** 10-week high-fibre or high-fermented-food diet

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 36 healthy adults, ten weeks of high fermented-food intake increased diversity and lowered several inflammatory markers, while high-fibre responses depended on the baseline community — responses are heterogeneous. [Wastyk HC et al. Cell 2021;184:4137] · lane C · randomized, two-arm dietary · n = 36 · mixed
- Titrated resistant potato starch (up to 48 g/day) raised fecal butyrate only in people who already carried the starch-degrader Ruminococcus bromii — a heterogeneous surrogate response, not a guaranteed correction. [Venkataraman A et al. Microbiome 2016;4:33] · lane D · single-arm dietary intervention · n = 20 · mixed
- Two weeks of riboflavin (50 or 100 mg/day) in 105 healthy adults did not significantly increase F. prausnitzii at either dose; pooled groups showed higher butyrate — a surrogate result only. [RIBOGUT trial (2022) — riboflavin 50 or 100 mg/day in healthy adults] · lane C · randomized double-blind placebo-controlled · n = 105 · mixed
- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** Three weeks of riboflavin 100 mg/day in Crohn's disease did not materially alter metagenomic diversity, taxonomy or pathways. [RISE-UP (2020) — riboflavin 100 mg/day in Crohn's disease] · lane X · single-arm · n = 70 · null

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “butyrate producer deficiency” · “take Faecalibacterium”

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Anaerostipes hadrus ● Beneficial

MISSING

What it is and what it does

Converts lactate and acetate into butyrate, which keeps lactate from accumulating (lactate build-up is itself pro-inflammatory) and feeds the lining. One of the most common organisms in healthy adults and reduced in some inflammatory and metabolic cohorts.

Why it is flagged in your sample

YOUR LEVEL	**<0.0001%** — not detected; this sample resolves organisms below that level. In the wider inventory, a relative is present: *Anaerostipes sp900756035* at 2.44%
HOW COMMON	carried by **85%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.

What the level means. Missing Anaerostipes removes a key link in the chain that turns fibre fermentation into butyrate.

What has been shown to raise it 3 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Inulin	prebiotic fibre	human trial	Vandeputte D et al. Gut 2017
Resistant starch	prebiotic fibre	human trial	Baxter NT et al. mBio 2019;10:e02566
Lactate-producing probiotics (Lactobacillus, Bifidobacterium)	probiotic	laboratory culture	Supply its substrate; cross-feeding shown in culture. Duncan SH et al. Appl Environ Microbiol 2004;70:5810

What the intervention registry holds for this organism

Lower or absent butyrate producers — what has been tried to raise them LANE C HUMAN TRIAL · LIMITED

Fired by: Anaerostipes hadrus: absent expected; Faecalibacterium prausnitzii: low; Roseburia faecis: absent expected; +1 more

Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with diet, antibiotics and transit in healthy people. Fermented-food and diverse-fibre diets changed communities in some people; riboflavin supplementation did not reliably raise F. prausnitzii. No live F. prausnitzii product is approved.

Studied: High-fibre versus high-fermented-food diet (Wastyk 2021) (diet) · **in:** people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36 · **studied protocol (a source fact, not an instruction):** 10-week high-fibre or high-fermented-food diet

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 36 healthy adults, ten weeks of high fermented-food intake increased diversity and lowered several inflammatory markers, while high-fibre responses depended on the baseline community — responses are heterogeneous. [Wastyk HC et al. Cell 2021;184:4137] · lane C · randomized, two-arm dietary · n = 36 · mixed
- Titrated resistant potato starch (up to 48 g/day) raised fecal butyrate only in people who already carried the starch-degrader Ruminococcus bromii — a heterogeneous surrogate response, not a guaranteed correction. [Venkataraman A et al. Microbiome 2016;4:33] · lane D · single-arm dietary intervention · n = 20 · mixed
- Two weeks of riboflavin (50 or 100 mg/day) in 105 healthy adults did not significantly increase F. prausnitzii at either dose; pooled groups showed higher butyrate — a surrogate result only. [RIBOGUT trial (2022) — riboflavin 50 or 100 mg/day in healthy adults] · lane C · randomized double-blind placebo-controlled · n = 105 · mixed
- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022:604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** Three weeks of riboflavin 100 mg/day in Crohn's disease did not materially alter metagenomic diversity, taxonomy or pathways. [RISE-UP (2020) — riboflavin 100 mg/day in Crohn's disease] · lane X · single-arm · n = 70 · null

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “butyrate producer deficiency” · “take Faecalibacterium”

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Roseburia faecis ● Beneficial · genus-level knowledge MISSING

What it is and what it does

Flagellated butyrate producers that thrive on dietary fibre, especially beta-glucans and inulin. Lower levels are reported in ulcerative colitis, type 2 diabetes and atherosclerosis cohorts. R. intestinalis has direct anti-inflammatory effects on the gut lining in cell and mouse work.

Why it is flagged in your sample

YOUR LEVEL	**<0.0001%** — not detected; this sample resolves organisms below that level. In the wider inventory, a relative is present: *Agathobacter rectalis* at 0.097%
HOW COMMON	carried by **83%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.

What the level means. Missing or low Roseburia is a fibre-intake signal as much as a health one: these are among the first organisms to fall on a low-fibre diet and the first to recover on a high-fibre one.

What has been shown to raise it 4 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Oat and barley beta-glucan	prebiotic fibre	**human trial**	Martinez I et al. ISME J 2013;7:269
Inulin and fructo-oligosaccharides	prebiotic fibre	**human trial**	Vandeputte D et al. Gut 2017
Whole grains	food	**human trial**	Vanegas SM et al. Am J Clin Nutr 2017;105:635
Resistant starch	prebiotic fibre	**human trial**	Walker AW et al. ISME J 2011

Known to push it the other way: Low-fibre diet, antibiotics (human cohorts).

What the intervention registry holds for this organism

Lower or absent butyrate producers — what has been tried to raise them LANE C HUMAN TRIAL · LIMITED

Fired by: Anaerostipes hadrus: absent expected; Faecalibacterium prausnitzii: low; Roseburia faecis: absent expected; +1 more

Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with diet, antibiotics and transit in healthy people. Fermented-food and diverse-fibre diets changed communities in some people; riboflavin supplementation did not reliably raise F. prausnitzii. No live F. prausnitzii product is approved.

Studied: High-fibre versus high-fermented-food diet (Wastyk 2021) (diet) · **in:** people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36 · **studied protocol (a source fact, not an instruction):** 10-week high-fibre or high-fermented-food diet

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 36 healthy adults, ten weeks of high fermented-food intake increased diversity and lowered several inflammatory markers, while high-fibre responses depended on the baseline community — responses are heterogeneous. [Wastyk HC et al. Cell 2021;184:4137] · lane C · randomized, two-arm dietary · n = 36 · mixed
- Titrated resistant potato starch (up to 48 g/day) raised fecal butyrate only in people who already carried the starch-degrader Ruminococcus bromii — a heterogeneous surrogate response, not a guaranteed correction. [Venkataraman A et al. Microbiome 2016;4:33] · lane D · single-arm dietary intervention · n = 20 · mixed
- Two weeks of riboflavin (50 or 100 mg/day) in 105 healthy adults did not significantly increase F. prausnitzii at either dose; pooled groups showed higher butyrate — a surrogate result only. [RIBOGUT trial (2022) — riboflavin 50 or 100 mg/day in healthy adults] · lane C · randomized double-blind placebo-controlled · n = 105 · mixed
- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** Three weeks of riboflavin 100 mg/day in Crohn's disease did not materially alter metagenomic diversity, taxonomy or pathways. [RISE-UP (2020) — riboflavin 100 mg/day in Crohn's disease] · lane X · single-arm · n = 70 · null

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “butyrate producer deficiency” · “take Faecalibacterium”

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Eubacterium eligens ● Beneficial MISSING

What it is and what it does

A pectin specialist that produces acetate and the precursors of butyrate; one of the organisms most associated with plant-rich diets and lower in asthma and IBD cohorts. In cell studies it strongly induces IL-10.

Why it is flagged in your sample

YOUR LEVEL	**<0.0001%** — not detected; this sample resolves organisms below that level. In the wider inventory, a relative is present: *Lachnospira pectinoschiza* at 0.164%
HOW COMMON	carried by **79%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.

What the level means. Its absence is a fruit-and-vegetable intake signal; its presence at high levels is good news.

What has been shown to raise it 2 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Pectin (apples, citrus, carrots)	prebiotic fibre	**human trial**	The organism's defining substrate. Chung WSF et al. BMC Biol 2016;14:3
Fruit and vegetable intake	food	**human cohorts**	Bolte LA et al. Gut 2021

What the intervention registry holds for this organism

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Oscillibacter sp 57 20 ● Beneficial · genus-level knowledge MISSING

What it is and what it does

Valerate-producing Oscillospiraceae associated with leanness and healthy ageing in several cohorts; many members are known only from genomes.

Why it is flagged in your sample

YOUR LEVEL	**<0.0001%** — not detected; this sample resolves organisms below that level. In the wider inventory, a relative is present: *Dysosmobacter welbionis* at 0.113%
HOW COMMON	carried by **74%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.

What the level means. Missing Oscillibacter is a mild depletion signal shared with other fibre-associated organisms.

What has been shown to raise it 1 item, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre, Mediterranean diet	diet pattern	**human trial**	Ghosh TS et al. Gut 2020

What the intervention registry holds for this organism

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

DEPLETED — well below the reference

‹ at a glance

Faecalibacterium prausnitzii ● Beneficial

LOW

What it is and what it does

The single most abundant butyrate producer in a healthy adult gut, often 5-15% of the community. Butyrate is the main fuel of the colon lining, tightens the barrier, and promotes regulatory T cells; F. prausnitzii also secretes an anti-inflammatory protein (MAM) of its own. Reduced F. prausnitzii is one of the most replicated findings in Crohn's disease and appears across ulcerative colitis, IBS, obesity and coeliac disease.

Why it is flagged in your sample

YOUR LEVEL	**no share of its own** — its reads are counted within *Faecalibacterium sp900540455* in the composition; the whole-genome lane reads it at **0.010%** of its own scale
POSITION	**0th** percentile among reference adults who carry it, on that lane — –**100%** against their typical level of 5.40%
HOW COMMON	carried by **98%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.
DETECTED BY	MetaPhlAn 3
IDENTIFIERS	scoring Faecalibacterium_prausnitzii
DETECTION	**provisional** — one method (scoring) at 0.00; presence rests on competitive confirmation; confirmation pending: no reference genome identifier for this organism
REFERENCE STRAINS	4 in BacDive: culture collection DSM 107838, DSM 107839, DSM 107840; grows at 37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample.

What the level means. Low means the gut lining is short of its main fuel and its main anti-inflammatory resident. It is strictly anaerobic and hard to supplement directly, so restoring it means feeding it.

What has been shown to raise it 7 items, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Inulin and fructo-oligosaccharides (chicory, Jerusalem artichoke, onion)	prebiotic fibre	human trial	The best-replicated prebiotic effect on F. prausnitzii; 10-15 g/day over 2-4 weeks in several RCTs. Ramirez-Farias C et al. Br J Nutr 2009;101:541; Vandeputte D et al. Gut 2017
Resistant starch (cooled potatoes/rice, green banana, high-amylose maize)	prebiotic fibre	human trial	Venkataraman A et al. Microbiome 2016;4:33
Apple pectin and other pectins	prebiotic fibre	human trial	Chung WSF et al. BMC Biol 2016;14:3
Polyphenols: pomegranate, cranberry, cocoa flavanols	compound	human trial	Small trials; pomegranate extract and cocoa flavanols each raised Faecalibacterium. Li Z et al. Food Funct 2015;6:2487; Tzounis X et al. Am J Clin Nutr 2011;93:62
Mediterranean diet	diet pattern	human trial	Ghosh TS et al. Gut 2020;69:1218 (NU-AGE trial)
Berberine	herb	animal model	Poorly absorbed, so it reaches the colon; raised Faecalibacterium in rodent models. Human data are for metabolic outcomes, not this organism. Zhang X et al. Sci Rep 2015;5:14405
Bifidobacterium longum and Lactobacillus strains (cross-feeding)	probiotic	laboratory culture	Bifidobacteria release acetate that F. prausnitzii uses; cross-feeding shown in co-culture. Rios-Covian D et al. FEMS Microbiol Lett 2015

Known to push it the other way: Antibiotics (human cohorts); Low-fibre Western diet (human cohorts).

What the intervention registry holds for this organism

Lower or absent butyrate producers — what has been tried to raise them LANE C HUMAN TRIAL · LIMITED

Fired by: Anaerostipes hadrus: absent expected; Faecalibacterium prausnitzii: low; Roseburia faecis: absent expected; +1 more

Reduced Faecalibacterium, Roseburia and Eubacterium rectale are among the most replicated findings in inflammatory bowel disease cohorts, and they also vary widely with diet, antibiotics and transit in healthy people. Fermented-food and diverse-fibre diets changed communities in some people; riboflavin supplementation did not reliably raise F. prausnitzii. No live F. prausnitzii product is approved.

Studied: High-fibre versus high-fermented-food diet (Wastyk 2021) (diet) · **in:** people with clinically confirmed healthy adults, ages 18–70, in USA, n = 36 · **studied protocol (a source fact, not an instruction):** 10-week high-fibre or high-fermented-food diet

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 36 healthy adults, ten weeks of high fermented-food intake increased diversity and lowered several inflammatory markers, while high-fibre responses depended on the baseline community — responses are heterogeneous. [Wastyk HC et al. Cell 2021;184:4137] · lane C · randomized, two-arm dietary · n = 36 · mixed
- Titrated resistant potato starch (up to 48 g/day) raised fecal butyrate only in people who already carried the starch-degrader Ruminococcus bromii — a heterogeneous surrogate response, not a guaranteed correction. [Venkataraman A et al. Microbiome 2016;4:33] · lane D · single-arm dietary intervention · n = 20 · mixed
- Two weeks of riboflavin (50 or 100 mg/day) in 105 healthy adults did not significantly increase F. prausnitzii at either dose; pooled groups showed higher butyrate — a surrogate result only. [RIBOGUT trial (2022) — riboflavin 50 or 100 mg/day in healthy adults] · lane C · randomized double-blind placebo-controlled · n = 105 · mixed
- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** Three weeks of riboflavin 100 mg/day in Crohn's disease did not materially alter metagenomic diversity, taxonomy or pathways. [RISE-UP (2020) — riboflavin 100 mg/day in Crohn's disease] · lane X · single-arm · n = 70 · null

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “butyrate producer deficiency” · “take Faecalibacterium”

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Agathobaculum butyriciproducens ● Beneficial LOW

What it is and what it does

A butyrate producer studied as a next-generation probiotic: in mouse models it reduced colitis and improved cognition and Parkinson's-like features. Common in healthy adults, lower in some disease cohorts.

Why it is flagged in your sample

YOUR LEVEL	**0.029%** of the community
POSITION	**2nd** percentile among reference adults who carry it — –**98%** against their typical level of 0.214%
HOW COMMON	carried by **89%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue, MetaPhlAn 3
IDENTIFIERS	scoring Agathobaculum_butyriciproducens, SGB14993, GCF_003096535, mOTU 000033, UHGG species 432, panel CokerMO_2019_SRR8692218_bin.2; GTDB R232 Agathobaculum butyriciproducens
DETECTION	**supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker
REFERENCE STRAINS	1 in BacDive: culture collection DSM 100391; obligate anaerobe, anaerobe; grows at 30–37 °C; isolated from female, feces (stool); produces butyrate, ethanol, butanol, acetate. Traits of reference strains, not measured in your sample.

What the level means. One more butyrate producer in the depleted set; not yet available as a supplement.

What has been shown to raise it 1 item, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre (inulin, resistant starch)	prebiotic fibre	animal model	Go J et al. Int J Mol Sci 2021 (A. butyriciproducens in mice)

What the intervention registry holds for this organism

Expected organism not detected or low — no supported way to 'add it back' LANE D NO SUPPORTED INTERVENTION

Fired by: Agathobaculum butyriciproducens: low; Anaerostipes hadrus: absent expected; Dorea longicatena: absent expected; +5 more

A species that most reference adults carry is missing or low here. Large cohorts show there is no universal healthy core, so this is a position relative to a reference, not a deficiency. No approved product supplies most gut commensals, and after antibiotics a probiotic delayed native recovery.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “deficiency” · “replace with a probiotic”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Intestinimonas massiliensis ● Beneficial · genus-level knowledge

LOW

What it is and what it does

Produces butyrate from both sugars and amino acids; lower in some metabolic cohorts.

Why it is flagged in your sample

YOUR LEVEL	**no share of its own** — its reads are counted within *Intestinimonas butyriciproducens* in the composition; the whole-genome lane reads it at **0.045%** of its own scale
POSITION	**6th** percentile among reference adults who carry it, on that lane — –**46%** against their typical level of 0.082%
HOW COMMON	carried by **18%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.
DETECTED BY	GTDB whole-genome sketch, GlobDB whole-genome sketch
IDENTIFIERS	GCF_001244995; GTDB R232 Intestinimonas massiliensis
DETECTION	**supported** — competitive confirmation: 2,296 uniquely mapping fragments over 10.8% of the genome at 98.9% identity, spread evenly (evenness 0.34), with 2 relatives competing
REFERENCE STRAINS	1 in BacDive: culture collection DSM 100417; grows at 37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample.

What the level means. No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

What has been shown to raise it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to raise this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

‹ at a glance

Butyricicoccus pullicaecorum ● Beneficial

LOW

What it is and what it does

A member of the genus *Butyricicoccus* for which no human evidence has been curated. Its presence is recorded; nothing here says it is helpful or harmful.

Why it is flagged in your sample

YOUR LEVEL	**0.001%** of the community
POSITION	**5th** percentile among reference adults who carry it — –**95%** against their typical level of 0.049%
HOW COMMON	carried by **3%** of reference adults
CLASS	Beneficial — Higher levels are associated with better outcomes in human studies.
DETECTED BY	GTDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, MetaPhlAn 3
IDENTIFIERS	scoring Butyricicoccus_pullicaecorum, SGB14985, mOTU 000600, UHGG species 2199, GCA_959018045.1; GTDB R232 Butyricicoccus sp959018045
DETECTION	**supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker
REFERENCE STRAINS	1 in BacDive: culture collection DSM 23266; anaerobe; grows at 37–42 °C; isolated from chicken, caecal content. Traits of reference strains, not measured in your sample.

What the level means. No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

What has been shown to raise it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to raise this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

WORTH WATCHING — above the reference range, or uncommon to carry

‹ at a glance

Mediterraneibacter glycyrrhizinilyticus ● Conditional · genus-level knowledge HIGH

What it is and what it does

Formerly Ruminococcus gnavus and R. torques: mucus degraders that are normal residents at low levels but among the most consistently expanded organisms in Crohn's disease, and producers of an inflammatory polysaccharide. Level matters more than presence.

Why it is flagged in your sample

YOUR LEVEL	**0.348%** of the community
POSITION	**94th** percentile among reference adults who carry it — **+327%** against their typical level of 0.084%
HOW COMMON	carried by **18%** of reference adults
CLASS	Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, SingleM
IDENTIFIERS	SGB4606, GCF_042852765, mOTU 002019, UHGG species 2099; GTDB R232 Schaedlerella glycyrrhizinilytica_A
DETECTION	**supported** — seen by 5 independent methods: genome_sketch, marker, marker_window, read_classification, universal_marker
REFERENCE STRAINS	2 in BacDive: culture collection DSM 17593; anaerobe; grows at 37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample.
STRAIN PLACEMENT	nearest reference genome GCA_000218385.1.fna, distance 0.373674; polymorphic sites at 0.03%; a consensus can hide co-resident strains. A placement, not proof of an identical strain.

What the level means. No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

What has been shown to lower it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to lower this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

‹ at a glance

Clostridium sp AM29 11AC ● Conditional · genus-level knowledge

HIGH

What it is and what it does

A large, much-divided genus. The unnamed members reported here are known only from genomes assembled out of stool and are mostly fibre fermenters of the Lachnospiraceae and Clostridiaceae with no specific human evidence. The dangerous Clostridia (C. difficile, C. perfringens) are named species handled by the pathogen screen and are not what an unnamed bin means.

Why it is flagged in your sample

YOUR LEVEL	**0.226%** of the community
POSITION	**95th** percentile among reference adults who carry it — **×16** against their typical level of 0.026%
HOW COMMON	carried by **13%** of reference adults
CLASS	Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, MetaPhlAn 3, SingleM
IDENTIFIERS	SGB4701, mOTU 009024, UHGG species 3490, GCF_020736365.1, scoring Clostridium_saccharolyticum; GTDB R232 Otoolea saccharolyticum_A
DETECTION	**supported** — seen by 5 independent methods: genome_sketch, marker, marker_window, read_classification, universal_marker

What the level means. Expansion of unnamed Clostridia is a marker of disturbance rather than a specific risk; there is too little evidence to say more.

What has been shown to lower it 1 item, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Dietary fibre diversity	prebiotic fibre	**human cohorts**	General competition; no organism-specific evidence for unnamed bins. Bolte LA et al. Gut 2021

Competes with: *Faecalibacterium prausnitzii*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

‹ at a glance

Phocaeicola vulgatus ● Conditional · genus-level knowledge

HIGH

What it is and what it does

Until 2019 these were Bacteroides, and P. vulgatus and P. dorei are often the two most abundant organisms in adult stool. They are fibre degraders and propionate producers with strain-dependent associations: some strains are linked to lower cardiovascular risk and immune regulation, others to inflammation in IBD models. P. plebeius carries the enzymes to digest seaweed polysaccharides and is common in Japan.

Why it is flagged in your sample

YOUR LEVEL	**11.75%** of the community
POSITION	**93rd** percentile among reference adults who carry it — **+420%** against their typical level of 4.29%
HOW COMMON	carried by **89%** of reference adults
CLASS	Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), mOTUs 4, MetaPhlAn 3, SingleM
IDENTIFIERS	scoring Bacteroides_vulgatus, SGB1814, mOTU 000655, GCF_964248265.1; GTDB R232 Phocaeicola vulgatus
DETECTION	**supported** — seen by 4 independent methods: genome_sketch, marker, marker_window, universal_marker
REFERENCE STRAINS	12 in BacDive: culture collection DSM 107446, DSM 108222, DSM 108228; anaerobe; grows at 37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample.
STRAIN PLACEMENT	nearest reference genome GCA_001412315.1.fna, distance 0.306338; polymorphic sites at 0.51%; a consensus can hide co-resident strains. A placement, not proof of an identical strain.

What the level means. Very high Phocaeicola is common in Western guts. It is a concern mainly when it dominates so completely that diversity falls and butyrate producers are squeezed out.

What has been shown to lower it 1 item, strongest evidence first

WHAT	KIND	EVIDENCE	NOTE
Increased fibre diversity, especially inulin and resistant starch	prebiotic fibre	human trial	Vandeputte D et al. Gut 2017

Competes with: *Faecalibacterium prausnitzii*, *Prevotella copri*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: High-fat, high-protein diet (human trial).

‹ at a glance

Enterococcus faecium ● Opportunist

UNCOMMON

What it is and what it does

A hardy low-level resident that survives antibiotics, heat and bile. It is the source of vancomycin-resistant Enterococcus (VRE) in hospitals, and its expansion in stool predicts bloodstream infection in immunocompromised patients. Some strains are sold as probiotics in animal feed; in humans the species is watched rather than welcomed.

Why it is flagged in your sample

YOUR LEVEL	**0.021%** of the community
POSITION	**66th** percentile among reference adults who carry it — **+78%** against their typical level of 0.040%
HOW COMMON	carried by **5%** of reference adults
CLASS	Opportunist — Normal at low levels.
DETECTED BY	MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue, MetaPhlAn 3
IDENTIFIERS	scoring Enterococcus_faecium, SGB7967, GCF_029023785, mOTU 001919, UHGG species 3327, panel BrooksB_2017_SRR5405979_bin.5; GTDB R232 Enterococcus_B faecium
DETECTION	**supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker
REFERENCE STRAINS	12 in BacDive: culture collection DSM 13589, DSM 13590, DSM 17050; microaerophile, facultative anaerobe; grows at 10–37 °C; isolated from feces (stool), human; produces acetoin. Traits of reference strains, not measured in your sample.

What the level means. Blooms after antibiotics mark a cleared-out community; at trace levels it is unremarkable.

Nothing to lower

This is a beneficial organism at a high level. That is listed so you know, not so you can change it; nothing that raises it is a thing to avoid. If it is very high alongside a low-fibre diet the organism may be living on your gut's mucus for want of anything else, and the answer to that is fibre, not less of this organism.

What sustains it: Antibiotics, especially cephalosporins and vancomycin (human cohorts).

What the intervention registry holds for this organism

Higher Enterococcus share — common after antibiotics; one medication fact LANE D HUMAN · NON-RANDOMISED

Fired by: Enterococcus faecium: concern

Enterococci expand after antibiotics and with lower resident biomass. The only clinically relevant fact is that E. faecalis tyrosine decarboxylase can metabolise levodopa; that is information for a prescriber of levodopa, not an action.

Studied: Medication-context card for the prescriber (levodopa / MAO inhibitor and bacterial tyrosine decarboxylase) (clinical route) · **in:** ages 18–99 · **studied protocol (a source fact, not an instruction):** information card to the prescriber; no dose change from this report

Evidence lane D: non-randomized human, pilot, case-series or observational evidence. Replication: single study. Strongest design: observational study.

- Bacterial tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — a medication-context fact for the prescriber; there is no approved microbiome inhibitor or eradication protocol. [van Kessel SP et al. Nat Commun 2019;10:310] · lane D · human association + in vitro/rat
- Relative enrichment of oral bacteria in stool can arise because resident gut biomass falls, not because oral organisms invaded; absolute load and oral-health context are needed. [Liao C et al. Nat Microbiol 2024 — oral bacteria in stool track loss of gut biomass] · lane D · longitudinal cohort with absolute quantification

Conventional route: If on levodopa, information card to the prescriber.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “VRE infection” · “eradicate Enterococcus”

Uncommon or relatively high organism — no supported targeted intervention LANE D NO SUPPORTED INTERVENTION

Fired by: Anaerofustis stercorihominis: unexpected present; Anaerotignum lactatifermentans: high; Barnesiella intestinihominis: high; +12 more

This organism is uncommon in the reference or higher than the reference range here. Unless it is named in one of the specific cards above, no human study shows that reducing it changes a health outcome.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- In 8,208 Dutch participants, geography, age, stool form, medications and protocol shaped the microbiome and there was no universal healthy core. A low reading of 'reference-associated' taxa is a position, not a deficiency. [Gacesa R et al. Nature 2022;604:732 (Dutch Microbiome Project)] · lane D · population cohort, n = 8208

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “dangerous bacteria” · “must be eliminated”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

PART B — PATHOGENS, ONE AT A TIME

17 Which strain, not just which species

Two people can carry the same species and a different **strain** of it, and the difference often matters more than the species does: whether an *E. coli* is a harmless resident or carries a toxin operon is a strain-level question. This section reports, for each organism with enough coverage, the fingerprint of the population actually living in this sample — read from the marker genes, at single-base resolution.

ORGANISMS RESOLVED	CALLABLE BASES	DETECTED, UNRESOLVED
52	**5.7M**	**45**
population fingerprinted	compared at base level	too little coverage

What a fingerprint can and cannot tell you

It is this sample's own population. Not a match to a named laboratory strain. Naming one would require comparing the whole genome against that isolate, which marker genes cannot do.

It is the dominant population. Where two populations of one species are present, this reports the majority one. A less abundant second population can be hidden by it entirely.

It enables comparison. Because the fingerprint is read at single-base resolution, two samples can be compared organism by organism — which is how donor and recipient populations are told apart in the matching report.

It does not attribute genes. Resolving an organism's population does not show which organism carries a separately detected toxin or resistance gene. That needs physical linkage.

Every organism resolved to population level

Ordered by how much sequence was callable, which is how much of the population could actually be characterised. **Markers** is how many marker genes resolved; **breadth** is how much of each was covered; **depth** is the average read depth over them.

ORGANISM	SGB	MARKERS	CALLABLE BASES	BREADTH	DEPTH
GGB81342 SGB47944	SGB47944	183	295,457	98.5%	12.83×
GGB47148 SGB47368	SGB47368	187	281,326	96.6%	5.84×
Parabacteroides distasonis	SGB1934	179	206,456	97.1%	7.04×
GGB45596 SGB63306	SGB63306	196	182,877	99.5%	43.39×
Bifidobacterium breve	SGB17247	197	180,878	97.0%	7.73×
Bifidobacterium longum	SGB17248	185	180,637	97.2%	11.08×
Blautia massiliensis	SGB4826	193	177,602	99.3%	27.79×
Paraprevotella clara	SGB1798	153	174,038	94.5%	4.16×
Ruminococcus gnavus	SGB4584	200	170,314	97.8%	32.46×
Bacteroides caccae	SGB1877	180	167,636	93.8%	3.72×
Alistipes putredinis	SGB2318	191	160,363	98.0%	9.72×
Bacteroides stercoris	SGB1830	191	152,995	96.3%	7.04×
Alistipes onderdonkii	SGB2303	183	151,916	95.2%	5.74×
Fusicatenibacter saccharivorans	SGB4874	179	149,681	95.6%	5.11×
Phocaeicola massiliensis	SGB1812	193	149,469	97.7%	9.35×
Barnesiella intestinihominis	SGB1965	190	149,284	95.9%	5.67×
Clostridium sp AT4	SGB4753	180	135,110	93.7%	3.73×
Blautia caecimuris	SGB4862	163	127,789	93.6%	4.17×

ORGANISM	SGB	MARKERS	CALLABLE BASES	BREADTH	DEPTH
Phocaeicola plebeius	SGB1903	127	126,932	98.8%	19.15×
Lactiplantibacillus plantarum	SGB7202	178	124,501	95.5%	4.92×
GGB3550 SGB4746	SGB4746	180	123,286	94.2%	4.77×
Bacteroides thetaiotaomicron	SGB1861	140	122,171	91.3%	3.32×
Akkermansia muciniphila	SGB9226	168	115,275	93.5%	4.22×
Alistipes shahii	SGB2295	136	114,859	91.1%	3.0×
Mediterraneibacter faecis	SGB4563	125	112,025	89.3%	2.82×
Sutterella SGB9281	SGB9281	159	109,040	92.7%	3.72×
Sutterella wadsworthensis	SGB9283	137	101,799	91.8%	3.34×
GGB3433 SGB4574	SGB4574	132	100,679	91.6%	3.33×
Bifidobacterium adolescentis	SGB17244	111	99,372	89.4%	2.4×
Eubacterium ramulus	SGB4959	115	98,186	90.2%	2.6×
GGB3496 SGB4671	SGB4671	125	95,595	91.8%	3.14×
Ruminococcus sp AF13 28	SGB4834	96	85,837	90.0%	2.4×
GGB9480 SGB14874	SGB14874	123	85,216	90.9%	2.83×
Bacteroides ovatus	SGB1871	89	83,976	90.3%	2.44×
Bacteroides finegoldii	SGB1862	74	75,944	86.5%	1.99×
Bacteroides uniformis	SGB1836	56	72,046	99.2%	17.18×
Flavonifractor plautii	SGB15132	88	64,275	88.9%	2.56×
Anaerotignum lactatifermentans	SGB5193	84	61,025	92.7%	3.83×
Mediterraneibacter glycyrrhizinilyticus	SGB4606	74	56,975	87.3%	2.26×
Pseudoflavonifractor capillosus	SGB15161	82	55,682	91.4%	4.04×
Tyzzerella nexilis	SGB4588	59	53,260	88.7%	2.48×
Odoribacter splanchnicus	SGB1790	69	51,802	87.5%	2.1×
Parabacteroides merdae	SGB1949	53	45,607	91.8%	3.64×
GGB3600 SGB4859	SGB4859	63	44,625	89.0%	2.74×
Erysipelatoclostridium ramosum	SGB6744	59	43,386	89.5%	2.24×
Bifidobacterium animalis	SGB17278	40	41,045	95.8%	4.54×
Coprococcus comes	SGB4577	44	38,060	88.3%	2.08×
GGB9739 SGB15313	SGB15313	51	36,662	86.8%	2.02×
Phocaeicola vulgatus	SGB1814	30	28,011	99.3%	65.79×
Clostridium sp AM29 11AC	SGB4701	34	26,849	86.1%	1.81×
GGB4266 SGB5809	SGB5809	33	26,444	91.6%	4.11×
Firmicutes bacterium	SGB4791	25	19,021	89.1%	3.66×

45 further organism(s) were detected but carried too few resolvable marker genes to characterise a population. They are present; their strain is simply unresolved at this sequencing depth, which is a limit of the data rather than a finding about the organism.

This is the dominant population's consensus across marker genes. A second, less abundant population of the same species can be concealed by it, so 'one population' here means 'one detectable dominant population', not proof of a single strain.

Marker genes are a conserved subset of the genome. Agreement across them supports a related population; it does not establish whole-genome identity to a named isolate, and it cannot attribute an unlinked toxin or resistance gene to this organism.

Method: metaphlan 4.1.1 marker consensus (sample2markers) · database mpa_vJun23_CHOCOPhlAnSGB_202403.

Placed among the reference genomes of its species

For every organism with enough reconstructed marker sequence, its consensus markers were aligned with the marker sequences of member genomes of the same species (GTDB R232) and a tree was inferred. The **nearest reference** is the genome the sample's population sits closest to in that tree and the **distance** is the tree's branch length between them — comparable within one tree, not a genome-wide identity. A placement is not proof of an identical resident strain: it says which known genome is closest, not that they are the same. **Polymorphic sites** in the consensus are the mixture evidence — a high rate suggests more than one population of the species is present, and the consensus then describes neither exactly. No pathogenicity or resistance is inferred from a neighbour.

PLACED	UNRESOLVED	DEFERRED
46	51	0
of 51 eligible organisms	too few markers or references	left for the next run

ORGANISM	NEAREST REFERENCE	DISTANCE	MARKERS	POLYMORPHIC	GENOMES
Phocaeicola vulgatus SGB1814	GCA_001412315.1	0.306	30	0.51%	6
Mediterraneibacter gnavus SGB4584	GCF_008121495.1	0.420	191	1.59%	6
Blautia caecimuris SGB4862	GCA_013302405.1	0.283	155	0.62%	6
Phocaeicola plebeius SGB1903	GCA_003512905.1	0.360	143	0.11%	6
Bacteroides uniformis SGB1836	GCA_002320565.1	0.475	125	0.15%	6
GGB81342 SGB47944 SGB47944	GCA_900756035.1	2.937	197	0.03%	3
Phocaeicola massiliensis SGB1812	GCA_000438395.1	0.513	199	0.07%	6
Alistipes putredinis SGB2318	GCA_003541835.1	0.272	196	0.05%	6
Bifidobacterium longum SGB17248	GCA_018369245.1	0.951	133	1.08%	6
Parabacteroides distasonis SGB1934	GCF_000012845.1	0.340	104	0.04%	6
Bifidobacterium breve SGB17247	GCF_001025175.1	0.214	193	0.43%	6
Alistipes onderdonkii SGB2303	GCA_001405655.1	0.280	184	0.04%	6
Bacteroides stercoris SGB1830	GCA_003463895.1	0.762	196	0.08%	6
Barnesiella intestinihominis SGB1965	GCA_003489685.1	1.059	200	0.02%	6
GGB47148 SGB47368 SGB47368	GCA_900548695.1	3.981	191	0.02%	3
Lactiplantibacillus plantarum SGB7202	GCA_000604145.1	0.003	187	0.03%	6
GGB3550 SGB4746 SGB4746	GCF_964238185.1	0.080	183	0.05%	6

ORGANISM	NEAREST REFERENCE	DISTANCE	MARKERS	POLYMORPHIC	GENOMES
Akkermansia muciniphila SGB9226	GCA_000436395.1	0.768	166	0.04%	6
Sutterella SGB9281 SGB9281	GCA_041333695.1	0.382	171	0.03%	6
Paraprevotella clara SGB1798	GCA_003462815.1	0.190	149	0.03%	6
Anaerotignum lactatifermentans SGB5193	GCA_047014905.1	0.897	86	0.06%	6
Bifidobacterium animalis SGB17278	GCA_018363555.1	0.001	166	0.03%	6
Bacteroides caccae SGB1877	GCA_000433575.1	0.802	184	0.04%	6
Clostridium sp AT4 SGB4753	GCA_018055585.1	0.242	121	0.29%	6
Sutterella wadsworthensis SGB9283	GCA_003522065.1	0.007	150	0.04%	6
Lachnospiraceae unclassified SGB4574 SGB4574	GCA_026169395.1	0.240	132	0.02%	6
Parabacteroides merdae SGB1949	GCA_002319995.1	0.263	59	0.02%	6
GGB3496 SGB4671 SGB4671	GCA_902496395.1	0.103	148	0.03%	6
Bacteroides thetaiotaomicron SGB1861	GCA_003439845.1	0.338	124	0.04%	6
Alistipes shahii SGB2295	GCF_025145845.1	0.591	126	0.02%	6
Mediterraneibacter faecis SGB4563	GCA_000437035.1	0.284	119	0.02%	6
GGB52088 SGB72791 SGB72791	GCA_017889025.1	0.475	156	0.43%	6
GGB9480 SGB14874 SGB14874	GCA_030844365.1	0.070	136	0.03%	6
Eubacterium ramulus SGB4959	GCA_031293675.1	0.581	107	0.03%	6
Bacteroides ovatus SGB1871	GCA_048511295.1	1.515	87	0.02%	6
Lachnospiraceae unclassified SGB4859 SGB4859	GCA_048362375.1	0.256	69	0.02%	6
Flavonifractor plautii SGB15132	GCA_000508885.1	0.767	91	0.26%	6
Bifidobacterium adolescentis SGB17244	GCA_000829865.1	0.005	105	0.02%	6
Bacillota bacterium SGB4791	GCA_018380135.1	2.014	24	0.07%	5
Tyzzerella nexilis SGB4588	GCA_023454305.1	1.045	98	0.02%	6
Odoribacter splanchnicus SGB1790	GCA_003448135.1	0.471	93	0.01%	6
Mediterraneibacter glycyrrhizinilyticus SGB4606	GCA_000218385.1	0.374	80	0.03%	6

ORGANISM	NEAREST REFERENCE	DISTANCE	MARKERS	POLYMORPHIC	GENOMES
Allocoprococcus comes SGB4577	GCA_934358615.1	1.912	64	0.02%	6
Bacteroides finegoldii SGB1862	GCA_937983295.1	0.588	53	0.01%	6
Thomasclavelia ramosa SGB6744	GCF_014131695.1	0.455	78	0.06%	6
GGB9739 SGB15313 SGB15313	GCA_047234385.1	0.561	51	0.03%	6

Eligible but unresolved, with the reason:

• *GGB45596 SGB63306* (SGB63306): only 1 reference genomes could be fetched (1 members in GTDB R232); a comparative tree needs at least three

• *Fusicatenibacter saccharivorans* (SGB4874): only 0 reference genomes could be fetched (0 members in GTDB R232); a comparative tree needs at least three

• *Pseudoflavonifractor capillosus* (SGB15161): only 1 reference genomes could be fetched (1 members in GTDB R232); a comparative tree needs at least three

• *GGB4266 SGB5809* (SGB5809): only 0 reference genomes could be fetched (0 members in GTDB R232); a comparative tree needs at least three

• *Clostridium sp AM29 11AC* (SGB4701): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

46 further marker clades had fewer than 20 reconstructed markers and were not placed; they are present as organisms where the inventory lists them, and unresolved only at strain level.

Method: StrainPhlAn 4 on the mpa_vJan26_CHOCOPhlAnSGB_202605 markers, reference genomes from the organism's GTDB R232 species cluster (representative first, at most 6), PhyloPhlAn fast mode. Strain results never add to species counts.

What was asked for, and what came back

A screen is only meaningful alongside its own coverage. Every resolution target registered for this sample is listed below with the state it reached. **An unrun, failed or unavailable assay is never reported as a negative result** — the distinction between “looked and found nothing” and “never looked” is preserved throughout.

STATE	TARGETS	WHAT IT MEANS
Completed	5	the assay ran and produced a result
Incompatible with DNA input	8	an RNA genome cannot be assayed from DNA sequencing
Reference or tool unavailable	15	the tool or reference is not installed
Registered, awaiting reference curation	16	registered, with reference curation outstanding

44 targets registered, of which **16** remain outstanding. The outstanding targets are listed in results.json under `resolution_census`, each with the reason it has not yet been determined.

Organism-specific typing: 9 schemes registered (4 with their tool installed). A scheme without its tool reports an unavailable capability for every sample; it never reports a negative type, and a species name is never allowed to stand in for a type.

Strain-comparison tools: 8 of 8 installed. These tools estimate different objects: a marker phylogeny, a genome-wide population comparison and a reference-mixture deconvolution are not three measurements of one quantity. Agreement between them is not multiplied into extra certainty, and discordance is preserved rather than averaged away.

18 Pathogens in detail

Each finding on its own, by group, with the evidence behind it kept as separate facts rather than one blended score: how much distinct sequence supported it, how much of the reference that sequence covered, what resolution it justifies, and what would confirm it. Groups with nothing found still appear, because a negative from a search that was capable of finding something is itself a result.

Bacterial pathogens

7 supported, 1 candidate, out of 136 targets searched in this group.

‹ at a glance

Burkholderia cepacia complex

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

A group of closely related Gram-negative bacteria found in soil, water, plant roots and occasionally in contaminated liquids such as mouthwashes and disinfectants. They are best known as plant-associated organisms.

WHY IT MATTERS

For most people these bacteria are harmless. Their serious reputation comes almost entirely from cystic fibrosis and chronic granulomatous disease, where they can establish transmissible and difficult-to-treat lung infection. They are intrinsically resistant to many antibiotics. They are not established causes of gastroenteritis, and detecting them in stool is not evidence of lung infection or of any infection at all.

HOW MUCH IS HERE

Faint signal in this sample: 1.2 per million, 0.0001% of analysed DNA, from 10 distinct DNA fragments across 5 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is normally arrives with food or from the environment. Coverage evenness 0.41 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.01%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

No validated stool threshold exists and sequencing cannot establish one. This finding acquires significance only alongside cystic fibrosis, chronic granulomatous disease, or an unexplained respiratory or bloodstream illness being investigated by a clinician, in which case sputum or blood culture is the appropriate test. In everyone else, a stool signal without symptoms should be treated as incidental.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Allicin (crushed raw garlic, aqueous garlic extract) (botanical) — laboratory culture only — no human evidence

Pure allicin and aqueous garlic extract inhibited and killed 38 clinical Burkholderia cepacia complex isolates in culture, including three Burkholderia cenocepacia strains, with allicin minimum inhibitory concentrations of 8-62 micrograms per millilitre. The mechanism is chemical modification of essential bacterial cysteine residues. This is culture work only; there are no human data. *Dose studied:* not established in humans; in vitro minimum inhibitory concentrations of 8-62 micrograms per millilitre of allicin. *Reaching the gut:* Allicin forms when raw garlic is crushed and does survive into the upper gut, but it reacts rapidly with glutathione and other thiols in the gut wall and blood, so very little intact allicin gets far. The same paper's authors and later work note this poor bioavailability explicitly. Colonic delivery in active form should therefore be treated as doubtful, not assumed. Aged or cooked garlic contains little or no allicin. *Caution:* Garlic thins the blood and can interact with warfarin, clopidogrel and other anticoagulants and antiplatelet drugs; stop it before planned surgery. It commonly causes heartburn, reflux and wind, and can worsen symptoms in irritable bowel syndrome because it is high in fructans. It does not treat cystic fibrosis lung infection and must never replace prescribed antibiotics. Garlic Revisited: Antimicrobial Activity of Allicin-Containing Garlic Extracts against Burkholderia cepacia Complex

Manuka honey (food) — laboratory culture only — no human evidence

Manuka honey inhibited all 55 Burkholderia clinical isolates from cystic fibrosis patients at 7% or less by weight, with a mean minimum inhibitory concentration of 4.6%, making Burkholderia somewhat more honey-susceptible than Pseudomonas aeruginosa in the same study. It also acted synergistically or additively with tobramycin and colistin. Culture work only; no human outcome data. *Dose studied:* not established in humans; in vitro minimum inhibitory concentration around 4.6% honey by weight. *Reaching the gut:* Poorly. Honey sugars are absorbed in the small intestine and the active methylglyoxal is diluted and chemically consumed en route, so colonic concentrations fall well short of the several-per-cent used in the laboratory. State this as a limitation of the agent. *Caution:* High in sugar, so unsuitable if diabetes is poorly controlled. Never give honey to infants under twelve months because of infant botulism risk. This study tested honey in a test tube against airway isolates; it is not evidence that eating honey affects the lungs or the gut, and it is no substitute for specialist cystic fibrosis care. A demonstration of the susceptibility of clinical isolates obtained from cystic fibrosis patients to manuka honey

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species complex	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
10	5	1,224
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
87 (not counted as support)	95.4%	0.01%

What would confirm this: clinician selected stool NAAT or culture, specialist referral.

Caveat: no extraction or library controls for this sample.

below identity gate · partial query alignment · controls unavailable · species resolution limited · not in organism inventory

‹ at a glance

Clostridioides difficile

ENOUGH SEQUENCE TO NAME IT

WHAT THIS TEST FOUND

Positive for *Clostridioides difficile*. This test scanned the DNA fragments in your sample and found a definite positive match to multiple DNA sequences of *Clostridioides difficile*. The genes that produce its toxins (tcdA, tcdB) were not found among these fragments. That does not rule out *C. difficile* disease: toxin genes are a small target that a stool shotgun test can miss at ordinary depth, and finding the organism at all raises the possibility of disease relative to a sample in which it was not found. It cannot confirm it. This may warrant further action. The nine-method organism inventory did not name this species - its reference catalogues have a detection floor this screen does not - and the coverage here has the shape of a genuine population, so the finding stands on this screen's own alignment.

WHAT IT IS

A spore-forming bacterium that lives in the large bowel. Its spores survive for a long time on surfaces and resist alcohol hand gel. Only some strains carry the genes for the toxins that cause disease; strains without those genes do not cause the illness.

WHY IT MATTERS

When antibiotics wipe out the normal gut bacteria, toxin-producing C. difficile can multiply and release toxins that inflame and damage the colon. This causes watery diarrhoea, abdominal pain and fever, and in severe cases pseudomembranous colitis, toxic megacolon, bowel perforation and death. It is a leading cause of healthcare-associated infection. Suspected C. difficile infection needs medical assessment and prescription treatment such as vancomycin or fidaxomicin, and recurrent cases may need faecal microbiota transplantation. Supplements are not a substitute for that, and delaying treatment in someone who is unwell is dangerous. Drugs that slow the bowel should be avoided.

HOW MUCH IS HERE

Faint signal in this sample: 1.2 per million, 0.0001% of analysed DNA, from 10 distinct DNA fragments across 9 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. Coverage evenness 0.43 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.03%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

IS CARRYING THIS NORMAL

Carried without symptoms by 0–17.5% of healthy adults; the toxin-producing form by about 1.5% (95% CI 0.7–2.6%). Toxin-negative strains are associated with protection from C. difficile infection, not with disease. Infant carriage reaches 75%. Source: Meta-analysis of 51 studies, 39,447 participants (doi:10.1186/s13099-024-00674-0); non-toxigenic protection (doi:10.3389/fmicb.2018.01700)

THE GENES THAT DECIDE WHETHER IT MATTERS

The toxin genes **were looked for and not found** among the fragments in this sample. That does not clear the organism: toxin genes are a small target that shotgun sequencing can miss at ordinary depth, so their absence here lowers but does not remove the possibility of the toxin-producing form.

WHAT WOULD MAKE THE AMOUNT CONCERNING

Sequencing on its own cannot establish that C. difficile is causing illness, and this is the single most important caveat for this organism. Diagnosis requires diarrhoea plus laboratory evidence of toxin, usually a toxin immunoassay interpreted alongside a glutamate dehydrogenase or nucleic acid test. Two things make a sequencing hit more meaningful: detection of the toxin genes tcdA and tcdB, and better still the binary toxin genes cdtA and cdtB or the tcdC deletion associated with more virulent strains, and a clear clinical picture of new diarrhoea after antibiotics. A detection in someone with formed stools and no symptoms should be treated as carriage, and treating asymptomatic carriage is explicitly not recommended.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Multi-strain probiotics (preventive, during antibiotics) (probiotic) — randomised human trial

A 2025 Cochrane review of 38 randomised trials in 13,179 participants found that probiotics taken alongside antibiotics may halve the incidence of C. difficile-associated diarrhoea, from 3.2 per cent to 1.6 per cent, with a number needed to treat of about 65. The evidence was graded low certainty, and this is prevention during antibiotic courses, not treatment of established infection. *Dose studied:* Varies by trial; typically 10^9 to 10^10 colony-forming units daily, started within about two days of the antibiotic and continued for its duration. *Reaching the gut:* The strains used in these trials, mainly Lactobacillus, Bifidobacterium and Saccharomyces species, are selected for acid and bile tolerance and are recoverable live from stool during supplementation. Bacterial strains can be killed by the concurrent antibiotic, which is one argument for the yeast-based options. *Caution:* Preventive only; this does not treat an established infection, and the certainty of the evidence is low. Live probiotics should not be given to severely immunosuppressed people, those with central venous catheters, or critically ill patients, in whom bloodstream infection from the probiotic organism has been reported. Probiotics for the prevention of Clostridioides difficile-associated diarrhea in adults and children

Saccharomyces boulardii (probiotic) — randomised human trial

In a randomised controlled trial of 120 patients with mild to moderate C. difficile infection, adding S. boulardii to vancomycin gave a higher global cure rate (96.6 per cent versus 85.3 per cent) and a lower recurrence rate (1.7 per cent versus 13.1 per cent). It was tested as an addition to antibiotic treatment, not instead of it. *Dose studied:* Two 250 mg capsules (500 mg) every 12 hours alongside vancomycin. *Reaching the gut:* S. boulardii is a yeast, not a bacterium, so it is unaffected by antibacterial drugs and can be given at the same time as vancomycin. It tolerates gastric acid and bile and is recovered live in stool during dosing, reaching steady-state colonic concentrations within about three days and clearing within days of stopping. *Caution:* An addition to treatment, not a replacement for it. S. boulardii must be avoided in critically ill patients, in people who are severely immunosuppressed, and in anyone with a central venous catheter, because fungaemia has occurred in these groups, sometimes from airborne contamination when capsules are opened at the bedside. Avoid in anyone with a yeast allergy. Efficacy and safety of Saccharomyces boulardii as adjunct therapy with Vancomycin in treating Clostridioides difficile infection: A randomized controlled trial

Berberine (botanical) — animal study only

Blocked relapse of C. difficile infection in mice after standard vancomycin treatment, with a follow-up study describing the mechanisms involved. Mouse evidence for prevention of relapse; there is no controlled human trial in C. difficile infection. *Dose studied:* not established for humans in C. difficile infection. *Reaching the gut:* Berberine's oral bioavailability is under one per cent, so nearly all of a dose stays in the gut lumen and reaches the colon, which is precisely where C. difficile grows and releases toxin. Of the botanicals listed here it has the clearest case for reaching the target site. *Caution:* Animal evidence only, and no substitute for prescription treatment. Berberine inhibits CYP3A4, CYP2D6 and P-glycoprotein, so it raises levels of many prescription drugs, which matters particularly in the older, multi-medicated patients who get C. difficile. Avoid in pregnancy, when breastfeeding, and in newborns. Berberine blocks the relapse of Clostridium difficile infection in C57BL/6 mice after standard vancomycin treatment

Carvacrol (botanical) — laboratory culture only — no human evidence

Reduced C. difficile sporulation and spore outgrowth in culture, and in a separate study reduced toxin production and the toxicity of culture supernatant to cells. Both findings are from culture-dish experiments; there is no human data. *Dose studied:* not established. *Reaching the gut:* Carvacrol is absorbed and conjugated in the upper small intestine, so a plain oregano oil capsule delivers little to the colon. Reaching C. difficile in the large bowel would need a delayed-release or encapsulated formulation, and none has been tested in people for this purpose. *Caution:* Culture-dish evidence only, and it does not treat infection. Broad antibacterial botanicals are a questionable idea in C. difficile precisely because the illness is caused by loss of normal gut bacteria, and carvacrol suppresses commensals too. Undiluted oregano oil burns mucous membranes. Avoid in pregnancy. Carvacrol reduces Clostridium difficile sporulation and spore outgrowth in vitro

Epigallocatechin gallate (EGCG, green tea catechin) (botanical) — laboratory culture only — no human evidence

Reduced virulence measures of the hypervirulent C. difficile PCR ribotype 027 strain in culture, including effects on toxin activity. The endpoint was virulence rather than killing the organism, and the work was entirely in culture. *Dose studied:* not established. *Reaching the gut:* EGCG is very poorly absorbed, so most of an oral dose does reach the colon, which suits this target. Against that, it is unstable at small-intestinal pH and is extensively broken down by gut bacteria, so a fair proportion of what arrives is metabolites rather than intact EGCG. *Caution:* Culture-dish evidence only, and no substitute for prescription treatment. Concentrated green tea extracts have caused liver injury, and several regulators limit EGCG per daily dose; taking them on an empty stomach increases the risk. Catechins reduce non-haem iron absorption. Inhibitory Effect of Epigallocatechin Gallate on the Virulence of Clostridium difficile PCR Ribotype 027

trans-Cinnamaldehyde (botanical) — laboratory culture only — no human evidence

Reduced C. difficile toxin production and the cytotoxicity of its culture supernatant to cells, alongside carvacrol, at concentrations below those that inhibited growth. This is an anti-toxin effect shown in culture only, with no animal or human data. *Dose studied:* not established. *Reaching the gut:* Cinnamaldehyde is chemically reactive and is largely absorbed and oxidised to cinnamic acid in the upper gut, so ordinary cinnamon oil or cinnamon powder is unlikely to deliver active compound to the colon. Encapsulated forms are used in animal feed for this reason but have not been tested against C. difficile in people. *Caution:* Culture-dish evidence only. Cassia cinnamon contains coumarin, which is toxic to the liver in regular large amounts. Cinnamon oil is a strong mucosal irritant when undiluted, and cinnamon may add to the effect of blood-glucose-lowering medicines. Carvacrol and trans-cinnamaldehyde reduce Clostridium difficile toxin production and cytotoxicity in vitro

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	depends on toxin or pathotype	genome supported
DISTINCT FRAGMENTS	INFORMATIVE REGIONS	INFORMATIVE BASES
10	9	1,310
SHARED-SEQUENCE FRAGMENTS	MEDIAN IDENTITY	INFORMATIVE BREADTH
262 (not counted as support)	96.5%	0.03%

What would confirm this: clinical toxin immunoassay, clinical stool NAAT, clinical stool culture.

Caveat: no extraction or library controls for this sample.

partial query alignment · below identity gate · controls unavailable · strict cdiff species positive · not in organism inventory

‹ at a glance

Clostridium innocuum

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

An anaerobic, spore-forming Gram-positive bacterium and a normal member of the intestinal flora. Its name means harmless, given in 1962 because it showed no virulence in animal models.

WHY IT MATTERS

Its reputation has been revised. It is now recognised as an intrinsically vancomycin-resistant opportunist that can cause bloodstream infection, intra-abdominal abscess and endocarditis in immunocompromised patients, and reports from Taiwan describe an antibiotic-associated diarrhoeal illness clinically indistinguishable from Clostridioides difficile infection. It has also been linked to worse outcomes in inflammatory bowel disease. Two qualifications matter greatly for reading a stool result. First, that disease burden is described in hospitalised, antibiotic-exposed and immunosuppressed patients, and the organism's pathogenic mechanism remains largely unknown. Second, it produces no known toxin, so unlike Clostridioides difficile there is no toxin test that separates disease from carriage. Its practical importance in someone who is unwell is that oral vancomycin will not treat it, which is a reason for accurate laboratory identification, not for self-treatment.

HOW MUCH IS HERE

Large signal in this sample: 1,170 per million, 0.12% of analysed DNA, from 9,399 distinct DNA fragments across 469 separate regions of its genome. A major component of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. In the community composition (section 6) this organism is **0.041%** of the whole, measured competitively against every relative found in this sample. That is the amount; the count above is this screen's own alignment, which credits shared sequence to the target it knows. Coverage evenness 0.35 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **15.75%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

No abundance threshold has been validated, and sequencing cannot establish one; there is no toxin marker to fall back on either, so a stool sequence cannot separate carriage from disease for this organism at all. What would make the finding matter is clinical: new or persistent diarrhoea after antibiotics, especially if oral vancomycin for presumed Clostridioides difficile has not worked; abdominal pain with fever; or the finding in someone with inflammatory bowel disease, a solid tumour, chronic kidney disease or recent intensive care. Those situations need a doctor and an anaerobic stool culture or targeted identification, because treatment choice depends on knowing the organism.

WHAT TO DO ABOUT IT

There is no supplement, botanical, probiotic or food with any published evidence against Clostridium innocuum, in culture or in people, and inventing one would be worse than useless here. The organism is a normal gut inhabitant that requires nothing in a well person. When it does cause illness, the decision that matters is a prescribing one, because it is intrinsically resistant to vancomycin and susceptibility to metronidazole and piperacillin-tazobactam is what treatment relies on; the reported 30-day mortality among infected patients in one case-control study was 14.5%. Self-treating diarrhoea with botanicals in that setting risks delay, and delay is the main modifiable harm. Faecal microbiota transplantation has been tried in a seven-patient pilot study for refractory infection in inflammatory bowel disease, but that is a hospital procedure under investigation, not a supplement, and it is not applicable to an incidental finding. If you have persistent diarrhoea after antibiotics, particularly if vancomycin has not helped, see a doctor and mention this result.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	INFORMATIVE REGIONS	INFORMATIVE BASES
9,399	469	743,241
SHARED-SEQUENCE FRAGMENTS	MEDIAN IDENTITY	INFORMATIVE BREADTH
1,708 (not counted as support)	100.0%	16.18%

What would confirm this: clinician selected stool NAAT or culture.

Caveat: no extraction or library controls for this sample.

below identity gate · partial query alignment · duplicate family collapsed · controls unavailable

‹ at a glance

Enterococcus faecium

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

A Gram-positive bacterium that is a normal member of the human gut community, alongside its relative Enterococcus faecalis. It is also widespread in animals, food and the environment.

WHY IT MATTERS

In the gut it is an ordinary resident. Its clinical importance is almost entirely about antibiotic resistance: hospital-adapted lineages of this species are the main cause of vancomycin-resistant enterococcal infection, and gut carriage of a resistant strain is a recognised risk factor for later bloodstream infection in hospitalised and immunosuppressed people. The organism itself is not the issue; the resistance genes it may carry are, and those are reported separately. Detection in stool is expected and is not an infection.

HOW MUCH IS HERE

Clear signal in this sample: 319 per million, 0.03% of analysed DNA, from 2,560 distinct DNA fragments across 256 separate regions of its genome. Well represented in the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. In the community composition (section 6) this organism is **0.021%** of the whole, measured competitively against every relative found in this sample. That is the amount; the count above is this screen's own alignment, which credits shared sequence to the target it knows. Coverage evenness 0.48 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **11.75%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

IS CARRYING THIS NORMAL

A common member of the normal gut community in healthy adults. Source: OpenBiota pathogen catalogue, resident lane

WHAT WOULD MAKE THE AMOUNT CONCERNING

No abundance threshold has been validated and a single stool metagenome cannot establish absolute numbers. The pattern that carries real meaning in hospital medicine is domination of the gut by enterococci with collapsed diversity after broad-spectrum antibiotics, in a person who is neutropenic, has a central line, or has had a transplant; that is a clinical judgement, not something a stool sequence in a well person can make. What genuinely changes management is a linked acquired vanA or vanB resistance gene, reported separately, and even then it indicates colonisation rather than infection: spontaneous clearance is common, with almost 60% of colonised geriatric patients clearing within four weeks in one study.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Lacticaseibacillus rhamnosus GG (probiotic) — randomised human trial

Six placebo-controlled randomised trials in 167 people tested this strain for clearing vancomycin-resistant Enterococcus faecium from the gut of carriers. The pooled result was inconclusive, and the review concluded there is insufficient evidence to support using it for that purpose, with low overall quality of evidence, inconsistency between trials and small sample sizes. Individual trials disagree: a paediatric trial found the carrier state lost in 20 of 32 children on the probiotic against 7 of 29 on placebo, but the effect was temporary, and adult and geriatric trials found no effect at all, with high spontaneous clearance in both arms. This agent is listed because it is the best direct human evidence on the question, and the honest answer is that it has not been shown to work. *Dose studied:* 2.5 x 10^9 to 2 x 10^10 colony-forming units daily for 2-4 weeks across the included trials. *Reaching the gut:* Yes. This is among the best-characterised probiotic strains for gastric acid and bile tolerance, and trials confirmed increased faecal Lactobacillus counts during supplementation, so the organism does arrive and establish transiently. Arriving is evidently not sufficient: the strain reached the gut and the resistant enterococci mostly remained. *Caution:* No adverse events attributable to the probiotic were reported in the included trials, but live probiotics have caused bloodstream infection in profoundly immunosuppressed people and in those with central venous lines or damaged heart valves, so they should not be taken in those situations without medical advice. Be clear about what this is for: there is no reason to take it to reduce ordinary, non-resistant Enterococcus faecium, which is a normal gut inhabitant. Even for confirmed vancomycin-resistant carriage in a well person, colonisation often clears on its own and does not itself require treatment. Lacticaseibacillus rhamnosus GG Versus Placebo for Eradication of Vancomycin-Resistant Enterococcus faecium in Intestinal Carriers: A Systematic Review and Meta-Analysis

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
2,560	256	343,143
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
447 (not counted as support)	100.0%	12.65%

What would confirm this: clinician selected stool NAAT or culture.

Caveat: no extraction or library controls for this sample.

partial query alignment · below identity gate · duplicate family collapsed · controls unavailable

‹ at a glance

Mediterraneibacter gnavus

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

A normal gut bacterium, until recently called Ruminococcus gnavus. It is one of the ordinary residents of the human large intestine and is found in the great majority of healthy people, often in substantial amounts. It is in this catalogue as a background organism, so that reads belonging to it are not mistakenly credited to something else.

WHY IT MATTERS

On its own it does not matter, and finding it is not a result. Some research associates higher proportions of this species with Crohn's disease and with inflammatory conditions, and it makes an inflammatory polysaccharide in the laboratory, but that association is not established as cause and it is not a diagnosis. Plenty of healthy people carry a great deal of it.

HOW MUCH IS HERE

Large signal in this sample: 40,682 per million, 4.07% of analysed DNA, from 326,669 distinct DNA fragments across 339 separate regions of its genome. A major component of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. In the community composition (section 6) this organism is **6.934%** of the whole, measured competitively against every relative found in this sample. That is the amount; the count above is this screen's own alignment, which credits shared sequence to the target it knows. Coverage evenness 0.85 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **85.13%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

There is no established threshold, and this test cannot supply one. Published work compares group averages between patients and controls, which says nothing reliable about where a single person's reading sits. Take it as part of the picture of your community's composition, not as a finding to act on.

WHAT TO DO ABOUT IT

Nothing is recommended, because there is nothing here to treat. This is a normal member of a healthy gut, and there is no evidence that reducing it improves anything. If the wider report shows low diversity or a dysbiotic pattern, the useful action is the dietary one described in those sections — more plant fibre variety, which shifts the whole community — not an attempt to remove one species.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	background, food or environmental	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
326,669	339	3,021,344
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
1,961 (not counted as support)	100.0%	87.54%

What would confirm this: clinician selected stool NAAT or culture.

Caveat: no extraction or library controls for this sample.

below identity gate · partial query alignment · duplicate family collapsed · controls unavailable

‹ at a glance

Parvimonas micra

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

A very small anaerobic Gram-positive bacterium, previously named Peptostreptococcus micros and then Micromonas micra. It is a normal low-abundance resident of the gum margin, the airways, the gut and sometimes the urogenital tract.

WHY IT MATTERS

In its usual niches it is an ordinary commensal. It behaves as a pathobiont in two circumstances. In the mouth it is one of the predominant species in periodontitis lesions and infected root canals, and it appears in polymicrobial wound, skin and bone infections and occasional deep abscesses. Separately, it is found more abundantly in the stool and colonic tissue of people with colorectal cancer than in controls, and laboratory work shows it can adhere to colonic cancer cell lines and alter their signalling and DNA methylation. Read that carefully: the abundance difference is measured in people who already have cancer, and it does not establish that the organism causes cancer or that carriage predicts it. A stool detection is not a cancer finding, not a risk score and not an indication for investigation on its own. Follow bowel-screening invitations as normal, and take bowel symptoms to a doctor on their own merits.

HOW MUCH IS HERE

Faint signal in this sample: 1.9 per million, 0.0002% of analysed DNA, from 15 distinct DNA fragments across 8 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. Coverage evenness 0.46 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.12%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

No abundance threshold has been validated, and sequencing cannot establish one. It also cannot distinguish a colonic population from the continuous arrival of mouth bacteria in saliva, which for an oral organism like this one is a substantial confounder. The research use of this organism is as one component of multi-marker stool panels in cancer screening studies, always combined with other markers and a faecal immunochemical test, and never as a standalone result; a single species reading from one sample carries none of that context. There is no quantity that makes this finding actionable in an individual.

WHAT TO DO ABOUT IT

There is no reason to try to remove this organism and attempting to would be unwise. It is a normal low-abundance commensal of the mouth, airway and gut, and its appearance in stool substantially reflects swallowed saliva rather than a gut infection, so suppressing it in the colon would not address the source. No human trial has shown that any supplement, botanical or probiotic reduces it, and none has shown any benefit from reducing it, including any effect on colorectal cancer risk. Offering a protocol here would turn an ordinary result into an unfounded alarm and encourage indefinite use of non-selective botanical antimicrobials, which suppress the beneficial anaerobes maintaining colonisation resistance and the mucus barrier. Two things are genuinely worth doing and neither is an eradication attempt: look after your gums and teeth, since periodontitis is where this organism genuinely causes trouble and it is also the reservoir feeding the stool signal; and eat enough fermentable fibre to support the butyrate-producing anaerobes that keep the colonic barrier healthy.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	role in gut disease uncertain	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
15	8	2,068
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
149 (not counted as support)	98.0%	0.13%

What would confirm this: clinician selected stool NAAT or culture.

Caveat: no extraction or library controls for this sample.

partial query alignment · below identity gate · duplicate family collapsed · controls unavailable · not in organism inventory

‹ at a glance

Streptococcus agalactiae

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

A Gram-positive bacterium also known as group B Streptococcus, or GBS. Its natural home in humans is the lower bowel and rectum, from which it spreads to the vagina and urethra.

WHY IT MATTERS

In adults who are not pregnant it is usually silent, though it can cause urinary tract, skin and soft-tissue infection, and occasionally bloodstream infection or endocarditis in people who are elderly, diabetic or immunosuppressed; invasive disease in non-pregnant adults has been rising. Its major significance is in pregnancy: carriage in the vagina or rectum at the time of birth can be passed to the baby and cause early-onset neonatal infection, which is why maternity services screen for it or use risk-based intrapartum antibiotics. That is the one context in which reducing carriage has a recognised clinical purpose. Even so, a stool sequencing result is not the test maternity care uses, and it does not replace a rectovaginal swab or change intrapartum antibiotic decisions, which are made by the maternity team.

HOW MUCH IS HERE

Faint signal in this sample: 2.2 per million, 0.0002% of analysed DNA, from 18 distinct DNA fragments across 11 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. Coverage evenness 0.43 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.12%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

No stool abundance threshold exists and sequencing cannot establish one; indeed for this organism the amount detected is not the clinically relevant variable at all, because neonatal risk depends on carriage status at the time of delivery, assessed by culture of a rectovaginal swab in late pregnancy. The contexts that make a finding worth raising with a clinician are current pregnancy, recurrent urinary tract infection, poorly controlled diabetes with skin or soft-tissue infection, or an unexplained fever in an elderly or immunosuppressed person. In a well non-pregnant adult, this is ordinary carriage.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Lactobacillus rhamnosus GR-1 with Lactobacillus reuteri RC-14 (probiotic) — randomised human trial

In a randomised controlled trial, 110 pregnant women at 35-37 weeks who were culture-positive for group B Streptococcus in both vagina and rectum took two probiotic or two placebo capsules nightly until delivery. Cultures converted from positive to negative in 42.9% of the probiotic group against 18.0% of the placebo group (p = 0.007). This result has not been reliably replicated: a later four-strain trial found 63.6% against 77.8% persisting positivity, which was not statistically significant, and a multicentre trial of 267 low-risk pregnant women found no difference in rectovaginal swab positivity or in antibiotic administration. So the effect is real in one trial and absent in others. *Dose studied:* two capsules nightly from 35-37 weeks of gestation until delivery. *Reaching the gut:* Yes. These are acid- and bile-tolerant strains selected for oral delivery, and the trial's design depends on oral dosing reaching the lower bowel and rectum, which is the reservoir from which vaginal colonisation is replenished. Reduced rectal as well as vaginal culture positivity is the direct evidence that the strains acted at the intended site. *Caution:* This is worth considering only in pregnancy, and only in discussion with your midwife or obstetrician. It absolutely does not replace group B Streptococcus screening or intrapartum antibiotic prophylaxis: reducing carriage in the weeks before birth is not the same as being negative at delivery, and the trial evidence is inconsistent. Outside pregnancy there is no reason to try to reduce this organism, which is normal rectal flora in about one in five healthy adults. Live probiotics should not be taken by people who are profoundly immunosuppressed, have a central venous line, or have damaged heart valves without medical advice. Oral Lactobacillus rhamnosus GR-1 and Lactobacillus reuteri RC-14 to reduce Group B Streptococcus colonization in pregnant women: A randomized controlled trial

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
18	11	2,343
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
239 (not counted as support)	98.7%	0.12%

What would confirm this: clinician selected stool NAAT or culture.

Caveat: no extraction or library controls for this sample.

partial query alignment · below identity gate · controls unavailable · not in organism inventory

‹ at a glance

Brachyspira pilosicoli

TRACE, NOT ENOUGH TO CONFIRM

WHAT IT IS

A spiral-shaped bacterium, a spirochaete, that attaches by one end to the lining of the large bowel. Under the microscope a dense fringe of these organisms on the surface of a colonic biopsy is what pathologists call intestinal spirochaetosis. It also infects pigs and birds.

WHY IT MATTERS

Whether it causes human disease is genuinely unsettled, and the honest summary is that opinion is divided. Most people found to have intestinal spirochaetosis on biopsy have no symptoms and the finding is incidental; some authors regard the organism as a harmless commensal. Others point to chronic watery diarrhoea and abdominal pain that improve on treatment: in a German series of 209 patients with spirochaetosis, 51% had diarrhoea and 46% abdominal pain, and in a decade-long series of 175 cases nearly two-thirds were symptomatic, leading those authors to conclude it is a pathogenic infection of low virulence. Children and immunosuppressed people are more likely to be symptomatic. If you have persistent watery diarrhoea, abdominal pain or rectal bleeding alongside this finding, see a doctor: the established treatment is prescription metronidazole, typically 500 mg three times daily, with roughly 84% clinical or histological response, and diagnosis is properly made on colonic biopsy rather than on stool DNA. Asymptomatic people are usually observed without treatment.

HOW MUCH IS HERE

Small signal in this sample: 36 per million, 0.0036% of analysed DNA, from 291 distinct DNA fragments across 2 separate regions of its genome. A minor component of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. The aligned fragments cover **0.07%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

No stool abundance threshold has been validated, and sequencing cannot establish one; because the organism attaches to the mucosal surface, stool DNA is also a poor proxy for how much is actually adherent to the colon. What changes the reading is symptoms and susceptibility: chronic watery diarrhoea, abdominal pain, rectal bleeding, weight loss, immunosuppression, HIV, or childhood. The definitive assessment is histology of a colonic biopsy, which shows the characteristic basophilic fringe, with silver staining or immunohistochemistry to confirm. That requires a clinician.

WHAT TO DO ABOUT IT

There is no supplement, botanical, probiotic or food with any published evidence against Brachyspira pilosicoli, in culture or in people, so any list here would be invented. The decision this finding actually calls for is binary and clinical: if you have no symptoms, the usual and appropriate course is observation without treatment; if you have chronic watery diarrhoea, abdominal pain or rectal bleeding, the treatment with evidence behind it is prescription metronidazole, and diagnosis is confirmed on colonic biopsy. Neither of those is something a supplement contributes to. Attempting botanical self-treatment would risk months of unresolved diarrhoea that a short prescription course usually settles, and would leave other causes of persistent diarrhoea uninvestigated. Take this finding to a doctor if you have symptoms.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
291	2	1,728
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
34 (not counted as support)	100.0%	0.07%

What would confirm this: endoscopy with biopsy, clinician selected stool NAAT or culture.

duplicate family collapsed · partial query alignment · below identity gate

Protozoa pathogens

0 supported, 4 candidate, out of 49 targets searched in this group.

‹ at a glance

Cryptosporidium canis

TRACE, NOT ENOUGH TO CONFIRM

WHAT IT IS

A Cryptosporidium species whose natural host is the dog. It occasionally infects people, usually children or immunocompromised adults with close contact with dogs, and it produces the same kind of intestinal infection as the human-adapted species.

WHY IT MATTERS

Human infections are uncommon but real, and the illness is the familiar watery diarrhoea with cramping and weight loss. In immunocompetent people it is generally self-limiting. In immunocompromised people, as with the other species, it can become prolonged and severe. A detection is also useful information about the household: it points to a dog as the likely source, and the animal may benefit from veterinary attention. This should be discussed with a doctor.

HOW MUCH IS HERE

Clear signal in this sample: 169 per million, 0.02% of analysed DNA, from 1,357 distinct DNA fragments across 1 separate regions of its genome. Well represented in the sample's dna. For this organism, it is not a normal resident; healthy stool does not usually contain it. The aligned fragments cover **0.04%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

Quantity is not the question; presence is. Sequencing cannot show whether the oocysts are viable, whether the infection is active, or whether it explains any symptoms. Immune status and current symptoms are what matter.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Bovine colostrum immunoglobulin concentrate (nutrient) — human study, not randomised

Reduced stool weight and frequency in AIDS patients with Cryptosporidium parvum diarrhoea in an open non-randomised study, with the powder form working and capsules not. This was a different Cryptosporidium species in the same genus, stated as such, and later placebo-controlled work has been inconsistent. *Dose studied:* bovine immunoglobulin concentrate powder for 21 consecutive days. *Reaching the gut:* Intended to act in the gut lumen; partly degraded by stomach acid and pepsin, which is why the powder formulation and acid suppression have been used. *Caution:* Milk-derived; unsuitable with cows' milk protein allergy. Retail colostrum is not the studied product. Does not replace medical assessment. Treatment of severe diarrhea caused by Cryptosporidium parvum with oral bovine immunoglobulin concentrate in patients with AIDS

Curcumin (botanical) — animal study only

Eliminated Cryptosporidium parvum oocyst shedding with no recurrence in immunosuppressed mice and reduced intestinal damage. Again this is a different species within the same genus, stated as such, and mouse data only. *Dose studied:* oral curcumin daily for 11 days in mice; no established human dose. *Reaching the gut:* Poorly absorbed, so most of an oral dose remains in the intestinal lumen throughout its transit. *Caution:* Bleeding risk with anticoagulants, gallstone pain, rare liver injury with concentrated products, reduced iron absorption. No human evidence. Comparative efficacy of curcumin and paromomycin against Cryptosporidium parvum infection in a BALB/c model

Zinc (nutrient) — animal study only

Cut oocyst shedding by 98.3 per cent in mice infected with Cryptosporidium parvum, a closely related species in the same genus, stated as such. No human trial has shown zinc eliminates any Cryptosporidium species. *Dose studied:* commercial zinc preparation daily for 8 days in mice; WHO childhood diarrhoea dose is 10-20 mg elemental zinc daily for 10-14 days. *Reaching the gut:* Absorbed in the small intestine at the site of infection, with the remainder passing into the colon. *Caution:* Prolonged high-dose zinc causes copper deficiency with anaemia and nerve damage. Blocks quinolone and tetracycline absorption. Nausea on an empty stomach is common. The Effects of Commercially Available Syzygium aromaticum, Anethum graveolens, Lactobacillus acidophilus LB, and Zinc as Alternatives Therapy in Experimental Mice Challenged with Cryptosporidium parvum

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	established gut pathogen	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
1,357	1	3,241
MEDIAN IDENTITY	**INFORMATIVE BREADTH**	
99.0%	0.04%	

What would confirm this: clinical stool NAAT, clinical stool antigen, modified acid fast stain.

below identity gate · duplicate family collapsed · partial query alignment

‹ at a glance

Entamoeba dispar

TRACE, NOT ENOUGH TO CONFIRM

WHAT IT IS

A harmless amoeba that lives in the large intestine. Under a microscope it is completely indistinguishable from Entamoeba histolytica, the species that causes amoebic dysentery: same size, same shape, same four-nucleus cysts. The two can only be separated by DNA testing or specific antigen tests.

WHY IT MATTERS

It does not cause disease. It lacks the enzymes and surface machinery that Entamoeba histolytica uses to invade tissue, it does not produce dysentery, and it does not cause liver abscesses. What matters enormously is telling the two apart, and this is the single strongest argument for DNA-based testing over microscopy here. Entamoeba dispar is roughly ten times more common than Entamoeba histolytica worldwide, which means that historically the great majority of people diagnosed with amoebiasis by microscopy alone were given metronidazole they did not need. A DNA result naming Entamoeba dispar specifically is genuinely useful information: it tells you and your doctor that no treatment is required and that the search for the cause of any symptoms should continue elsewhere.

HOW MUCH IS HERE

At the detection floor in this sample: 0.12 per million, well under a thousandth of a percent, from 1 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is carried by many healthy people, so presence alone is unremarkable. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

There is no such thing. No abundance of Entamoeba dispar has ever been shown to cause illness, so there is no threshold to exceed. If you have bowel symptoms and this is what the test found, the correct conclusion is that the cause has not yet been identified, not that this amoeba is responsible.

WHAT TO DO ABOUT IT

This is a normal, non-invasive commensal and it should not be treated. Entamoeba dispar has no capacity to invade the bowel wall and causes no disease, so removing it is neither necessary nor advisable. Current international guidance is explicit that infection with Entamoeba dispar does not require treatment, precisely so that people are not given unnecessary metronidazole. The important clinical value of this result is what it rules out: because it is identical to Entamoeba histolytica under the microscope, a DNA test that names Entamoeba dispar has spared you a course of antibiotics you did not need. If you do have symptoms, they need investigating for another cause rather than being attributed to this organism.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	background, food or environmental	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
1	1	116
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
3 (not counted as support)	100.0%	0.00%

What would confirm this: clinical stool ova and parasite microscopy.

duplicate family collapsed

‹ at a glance

Entamoeba nuttalli

TRACE, NOT ENOUGH TO CONFIRM

HOW MUCH IS HERE

Faint signal in this sample: 3.1 per million, 0.0003% of analysed DNA, from 25 distinct DNA fragments across 1 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is not a normal resident; healthy stool does not usually contain it. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	rarely reported in gut disease	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
25	1	587
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
263 (not counted as support)	97.2%	0.01%

What would confirm this: clinical stool NAAT, specialist referral.

below identity gate · partial query alignment · duplicate family collapsed

‹ at a glance

Toxoplasma gondii

TRACE, NOT ENOUGH TO CONFIRM

WHAT IT IS

A single-celled parasite carried by a large share of the world's population, usually for life and usually without symptoms. It is acquired from undercooked meat, from contaminated soil or water, or from cat faeces. Cats are the only animals in which it completes its sexual cycle and shed the infectious oocysts.

WHY IT MATTERS

In a healthy adult, established infection is generally silent and needs no treatment. It matters in two situations, and both need a doctor rather than a supplement: a first infection during pregnancy, which can harm the developing baby, and reactivation in someone whose immune system is seriously suppressed, which can cause brain inflammation. Stool is the wrong specimen for answering either question — humans do not shed this parasite in faeces, so sequence here is far more likely to be environmental or dietary than evidence of your own infection. Diagnosis is by blood antibody testing.

HOW MUCH IS HERE

Faint signal in this sample: 2.5 per million, 0.0002% of analysed DNA, from 20 distinct DNA fragments across 1 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is lives in tissue rather than the gut lumen, so stool is a poor window on it. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

The concept does not apply. No amount of this sequence in stool indicates active human infection, and none rules it out. If the question matters to you — pregnancy, planned pregnancy, or immunosuppression — ask for a Toxoplasma IgG and IgM blood test, which is the test that actually answers it.

WHAT TO DO ABOUT IT

No supplement is offered, for two reasons. Nothing in a stool sample establishes that you are infected, so there would be nothing to treat. And where treatment genuinely is needed, in pregnancy or immunosuppression, it is a prescription regimen given under specialist care; substituting a botanical would be actively dangerous. Prevention is the useful action: cook meat thoroughly, wash produce, wear gloves for gardening, and if pregnant avoid handling cat litter.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	not a gut-disease organism	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
20	1	1,052
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
2 (not counted as support)	99.3%	0.00%

What would confirm this: serology, tissue biopsy or imaging.

Parasitic worms

0 supported, 1 candidate, out of 118 targets searched in this group.

‹ at a glance

Enterobius vermicularis

TRACE, NOT ENOUGH TO CONFIRM

WHAT IT IS

The pinworm or threadworm, a small white worm about a centimetre long that lives in the large intestine. At night the female migrates out through the anus to lay thousands of eggs on the surrounding skin, which is what causes the characteristic night-time itching. The eggs become infectious within hours and spread easily on fingers, bedding and dust.

WHY IT MATTERS

A confirmed pinworm infection is treated with prescription or pharmacy anthelmintic drugs and you should see a doctor or pharmacist. Mebendazole as a single dose repeated after two weeks cures nearly everyone, and the whole household should normally be treated together because reinfection is so easy. In fairness this is the most benign of the intestinal worms: the main problems are itching, disturbed sleep and irritability in children, with occasional secondary skin infection from scratching, and rarely vulvovaginitis in girls or involvement of the appendix. It is common in high-income countries, especially among school-age children, and is not a marker of poor hygiene. Herbal alternatives are not equivalent to the standard drugs and should not replace them.

HOW MUCH IS HERE

At the detection floor in this sample: 0.62 per million, well under a thousandth of a percent, from 5 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is shed intermittently, so the amount seen varies between samples. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

Detection means infection, so abundance is not the point, and stool testing is in fact a poor way to find this worm because the eggs are deposited on perianal skin rather than passed in faeces. A negative stool result does not exclude it, and the conventional test is a sticky-tape sample taken from the perianal skin first thing in the morning.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Pumpkin seeds (Cucurbita pepo, cucurbitin) (food) — animal study only

Pumpkin seed extract killed larval stages of the mouse nematode Heligmosomoides bakeri in culture and reduced egg counts and adult worm burden in infected mice. This is a different nematode from pinworm, stated as such, and there is no evidence specific to Enterobius in people. *Dose studied:* 8 g/kg of ethanol seed extract in mice; no established human dose. *Reaching the gut:* Cucurbitin is poorly absorbed and acts on worms directly in the gut lumen, reaching the large intestine where pinworms live. *Caution:* Not a substitute for standard treatment. Whatever is used, the hygiene measures are what actually break the reinfection cycle: treat the whole household at once, wash bedding and nightwear hot on treatment day, keep fingernails short, wash hands and scrub under the nails in the morning, and shower rather than bath first thing. Without those steps any treatment will be followed by reinfection. Evaluation of Anthelmintic Activity and Composition of Pumpkin (Cucurbita pepo L.) Seed Extracts - In Vitro and in Vivo Studies

Wormwood (Artemisia absinthium) extract (botanical) — animal study only

In mice infected with Syphacia obvolata, the mouse pinworm and a member of the same family Oxyuridae as human pinworm, stated as such, alcoholic wormwood extract at 5 and 10 per cent concentrations eliminated pinworm eggs from the faeces entirely, matching the standard drug pyrantel pamoate; a 2.5 per cent concentration was not significantly better than control. This is an animal study with a related worm, and no human trial exists. *Dose studied:* 5-10 per cent alcoholic extract given orally to mice for 10 days; no established human dose. *Reaching the gut:* Given orally and acting on worms in the intestinal lumen, which is where pinworms live. The sesquiterpene lactones are only partly absorbed, so material does reach the large bowel. *Caution:* Not a substitute for mebendazole, which is inexpensive, available from a pharmacy and reliably effective. Wormwood contains thujone, a neurotoxin that causes seizures in overdose, and preparations vary enormously in thujone content. Do not use in pregnancy, breastfeeding, epilepsy or kidney disease, and do not use it in children, who are the people most often affected by pinworm. It should not be taken for more than short periods. Antiparasitic efficacy of worm wood (Artemisia absinthium) alcoholic extract on Syphacia obvolata

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	established gut pathogen	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
5	1	149
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
20 (not counted as support)	97.9%	0.00%

What would confirm this: perianal tape test, clinical stool ova and parasite microscopy.

below identity gate · duplicate family collapsed

Microsporidia pathogens

Nothing found. all 15 targets in this group were searched and none returned supporting or candidate sequence. 1 matched only sequence shared with relatives, which cannot name an organism.

Fungal pathogens

2 supported, 3 candidate, out of 92 targets searched in this group.

‹ at a glance

Candida krusei (Pichia kudriavzevii)

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

A yeast now more properly called Pichia kudriavzevii. It is used industrially in cocoa and some fermented foods, and it also turns up as a minor member of the human gut and mouth community. Some detections in stool genuinely come from fermented food rather than from the person's own flora.

WHY IT MATTERS

Its clinical significance is almost entirely about drug resistance. Candida krusei is intrinsically resistant to fluconazole and has reduced susceptibility to flucytosine and amphotericin B, so when it causes bloodstream infection the choice of drug is constrained and mortality is high. It emerges in people who have had prolonged fluconazole exposure, and it is chiefly a problem in haematology and intensive care patients. Finding it in stool is not an infection.

HOW MUCH IS HERE

At the detection floor in this sample: 0.62 per million, well under a thousandth of a percent, from 5 distinct DNA fragments across 5 separate regions of its genome. Barely above what this depth can see. For this organism, it is carried by many healthy people, so presence alone is unremarkable. The aligned fragments cover **0.01%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

Sequencing gives you presence and relative abundance, not viability, invasiveness or drug susceptibility. In someone who is well, a positive result is a carriage or dietary finding. What would make it concerning is not the read count but the clinical setting: neutropenia, a central line, recent prolonged fluconazole, or signs of systemic infection.

WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

Saccharomyces boulardii (probiotic) — randomised human trial

Performed as well as nystatin at limiting stool and skin fungal colonisation in 181 preterm infants, with no candidaemia in the probiotic arm. Results were reported for fungal colonisation as a whole rather than by species, so the application to Candida krusei is by extension. *Dose studied:* 250 mg twice daily in adults; 5 billion CFU daily in the trial. *Reaching the gut:* Survives stomach acid and reaches the colon alive throughout dosing. *Caution:* Absolutely avoid with a central venous line, in severe immunosuppression or in critical illness. Since Candida krusei is largely a disease of exactly those patients, this probiotic is the wrong choice for anyone in whom Candida krusei is a real clinical worry. Prophylactic Saccharomyces boulardii versus nystatin for the prevention of fungal colonization and invasive fungal infection in premature infants

Berberine (botanical) — laboratory culture only — no human evidence

Candida krusei was one of the reference strains inhibited in both free-floating and biofilm conditions. As with Candida glabrata, the reported synergy between berberine and fluconazole is much less consistent for this species than for Candida albicans, and in any case fluconazole is not usable against Candida krusei. *Dose studied:* not established for Candida. *Reaching the gut:* Almost entirely unabsorbed, so it reaches the colon at high luminal concentrations. *Caution:* Interacts with many medicines via CYP3A4 and P-glycoprotein; lowers blood sugar; contraindicated in pregnancy, breastfeeding and newborns. In vitro Antifungal Effects of Berberine Against Candida spp. In Planktonic and Biofilm Conditions

Caprylic acid (and capric acid) (nutrient) — laboratory culture only — no human evidence

Rapid membrane-disrupting killing shown against Candida albicans in culture. Candida krusei was not included in that study, so this is genus-level extrapolation, stated as such. No human data. *Dose studied:* 10 mM in culture; no established human dose. *Reaching the gut:* Substantially absorbed in the small intestine; delivery to the colon from plain capsules is poor. *Caution:* Digestive upset is common. No demonstrated effect on human gut Candida. In vitro killing of Candida albicans by fatty acids and monoglycerides

Garlic (allicin) (food) — laboratory culture only — no human evidence

Allicin inhibited a panel of six Candida species in culture in the 0.05-25 micrograms per millilitre range. Candida krusei is among the species reported susceptible to allicin in the wider literature, but the specific study cited found azole synergy in only a minority of species tested. Laboratory evidence only. *Dose studied:* MIC50/MIC90 0.05-25 micrograms/mL in culture; no established human dose. *Reaching the gut:* Allicin is acid-labile and largely destroyed in the stomach, so raw garlic delivers little to the colon. Enteric-coated allicin products aim to fix this but human colonic levels are not documented. *Caution:* Raises bleeding risk with anticoagulants and antiplatelets; stop before surgery. Lowers saquinavir levels. Causes reflux and wind. A food, not a treatment. In vitro investigation of antifungal activity of allicin alone and in combination with azoles against Candida species

Oregano oil (carvacrol and thymol) (botanical) — laboratory culture only — no human evidence

Carvacrol inhibited Candida krusei clinical isolates in culture, and did so more effectively than nystatin in that experiment. The relevance is that carvacrol's mechanism is membrane disruption rather than the ergosterol-synthesis route that fluconazole uses, so intrinsic azole resistance does not protect against it. Still a dish result with no human evidence. *Dose studied:* MIC around 25 micrograms/mL in culture; no established human dose. *Reaching the gut:* Absorbed high in the small bowel unless enteric-coated; colonic delivery in humans is unmeasured. *Caution:* Never swallow undiluted oil. Abortifacient; avoid in pregnancy. May add to anticoagulant effect. It is not a treatment for invasive Candida krusei infection, which is a medical emergency requiring an echinocandin or similar prescription drug. The inhibitory effects of carvacrol, nystatin, and their combination on oral candidiasis isolates

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
5	5	755
MEDIAN IDENTITY	**INFORMATIVE BREADTH**	
100.0%	0.01%	

What would confirm this: clinical culture with species identification, antifungal susceptibility testing, specialist referral.

Caveat: no extraction or library controls for this sample.

controls unavailable

‹ at a glance

Saccharomyces cerevisiae

ENOUGH SEQUENCE TO NAME IT

WHAT IT IS

Brewer's yeast and baker's yeast. It is the yeast that makes bread rise and ferments beer and wine, it is sold as nutritional yeast flakes and as yeast extract spreads, and the probiotic Saccharomyces boulardii is a strain of the same species. Almost all of it in your stool arrived there in food.

WHY IT MATTERS

For practical purposes it does not matter. This is a food organism with a long record of safe consumption, and finding it in stool means you have eaten bread, beer, wine, nutritional yeast, stock cubes or a probiotic. The exceedingly rare exception is fungaemia in patients with central venous catheters who are critically ill or severely immunosuppressed, and that has been traced to probiotic capsules opened near a line rather than to ordinary dietary yeast. Anti-Saccharomyces antibodies are used as a marker in Crohn's disease, but that is an immune response, not evidence that the yeast is harmful.

HOW MUCH IS HERE

Small signal in this sample: 31 per million, 0.0031% of analysed DNA, from 245 distinct DNA fragments across 100 separate regions of its genome. A minor component of the sample's dna. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.29%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

There is no such thing as Saccharomyces cerevisiae overgrowth as a diagnosis. A high read count means a diet containing bread, beer, wine or nutritional yeast, or a probiotic taken recently. Sequencing cannot tell dietary yeast from anything else, and no threshold has ever been shown to predict illness.

WHAT TO DO ABOUT IT

This is a normal finding and should not be treated. Saccharomyces cerevisiae is brewer's and baker's yeast; it arrives from food, passes through, and does not colonise or damage the healthy bowel. Trying to remove it is neither necessary nor advisable, and the only reliable way to lower the number would be to stop eating bread, beer, wine and yeast extract, which brings no health benefit and needlessly narrows the diet. Antifungal botanicals taken for this would suppress useful gut bacteria for no gain. The one real caution is unrelated to diet: if you have a central venous catheter, are critically ill or are severely immunosuppressed, do not take Saccharomyces boulardii probiotic capsules without medical advice.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	background, food or environmental	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
245	100	35,520
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
2 (not counted as support)	100.0%	0.31%

What would confirm this: clinical culture with species identification, specialist referral.

Caveat: no extraction or library controls for this sample.

duplicate family collapsed · below identity gate · controls unavailable

‹ at a glance

Apophysomyces, human-infecting members

TRACE, NOT ENOUGH TO CONFIRM

HOW MUCH IS HERE

At the detection floor in this sample: 0.25 per million, well under a thousandth of a percent, from 2 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
genus level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
2	1	106
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
42 (not counted as support)	97.1%	0.00%

What would confirm this: tissue biopsy or imaging, clinical culture with species identification, specialist referral.

partial query alignment · below identity gate · species resolution limited

‹ at a glance

Geotrichum candidum

TRACE, NOT ENOUGH TO CONFIRM

WHAT IT IS

The white, velvety mould that forms the rind of Camembert, Brie, Saint-Nectaire and many other surface-ripened cheeses, and which is also used in some fermented milks. It is deliberately added to cheese and is present in soil, silage and raw milk.

WHY IT MATTERS

In the overwhelming majority of cases it does not matter at all: it is a food organism eaten on purpose by millions of people. It has been reported very rarely as an opportunist in patients with leukaemia and profound neutropenia, which is true of a great many environmental fungi and does not make it a threat to anyone with a working immune system.

HOW MUCH IS HERE

At the detection floor in this sample: 0.12 per million, well under a thousandth of a percent, from 1 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

WHAT WOULD MAKE THE AMOUNT CONCERNING

No meaningful threshold exists. Abundance tracks cheese and dairy consumption in the preceding days. A sequencing result cannot separate a mould eaten on a cheese rind from anything else, and there is no published level above which this becomes a clinical concern in an immunocompetent person.

WHAT TO DO ABOUT IT

This is a normal dietary finding and no treatment is appropriate. Geotrichum candidum is the cheese-rind mould; it is put into food on purpose and passes through the gut. Attempting to eliminate it would mean giving up surface-ripened cheeses for no benefit, and taking antifungal supplements would disturb the gut bacterial community while achieving nothing worthwhile. There is no evidence that carrying it does any harm to a person with normal immunity.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
species level	background, food or environmental	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
1	1	131
SHARED-SEQUENCE FRAGMENTS	**MEDIAN IDENTITY**	**INFORMATIVE BREADTH**
2 (not counted as support)	100.0%	0.00%

What would confirm this: clinical culture with species identification, specialist referral.

‹ at a glance

Saksenaea, human-infecting members

TRACE, NOT ENOUGH TO CONFIRM

HOW MUCH IS HERE

Small signal in this sample: 28 per million, 0.0028% of analysed DNA, from 227 distinct DNA fragments across 1 separate regions of its genome. A minor component of the sample's dna. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

RESOLVED TO	ORGANISM CLASS	REFERENCE ROUTE
genus level	opportunist, context dependent	genome supported
DISTINCT FRAGMENTS	**INFORMATIVE REGIONS**	**INFORMATIVE BASES**
227	1	306
MEDIAN IDENTITY	**INFORMATIVE BREADTH**	
98.7%	0.00%	

What would confirm this: tissue biopsy or imaging, clinical culture with species identification, specialist referral.

below identity gate · partial query alignment · duplicate family collapsed · species resolution limited

Viral pathogens

Nothing found. 30 of 32 targets in this group were searched and none returned supporting or candidate sequence. A further 32 are RNA viruses, which carry no DNA for a DNA test to find.

Matched only shared sequence

These targets picked up reads, but every read fell in sequence they share with close relatives. That is not evidence for the named organism and it is not evidence against it. They are listed so the signal is not silently discarded.

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Achromobacter xylosoxidans	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 2.1 per million · 0.0002% of analysed DNA · 17 fragments in 4 regions	0.01% of its genome covered · 99.3% identity
Acinetobacter baumannii–calcoaceticus complex	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Acremonium / Sarocladium, clinically relevant members	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas caviae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas dhakensis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas hydrophila	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas jandaei	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas schubertii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas trota	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aeromonas veronii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aliarcobacter butzleri	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aliarcobacter cryaerophilus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aliarcobacter skirrowii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aspergillus niger complex (section Nigri)	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Aspergillus terreus	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Babesia spp.	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Bacillus cereus sensu lato	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 9.7 per million · 0.0010% of analysed DNA · 78 fragments in 17 regions	0.17% of its genome covered · 96.8% identity
Blastomyces dermatitidis	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Blastomyces gilchristii	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter coli	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter concisus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter curvus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Campylobacter fetus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter hyointestinalis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter jejuni	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter lari	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter upsaliensis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Campylobacter ureolyticus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Candida albicans	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Candida dubliniensis	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Candida rugosa complex (Diutina rugosa complex)	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Candida tropicalis	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Citrobacter freundii complex	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 95.3% identity
Citrobacter koseri	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Clostridium baratii (neurotoxigenic)	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Clostridium botulinum	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.75 per million · well under a thousandth of a percent · 6 fragments in 4 regions	0.02% of its genome covered · 96.6% identity
Clostridium butyricum (neurotoxigenic)	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.50 per million · well under a thousandth of a percent · 4 fragments in 3 regions	0.01% of its genome covered · 95.9% identity
Clostridium perfringens	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Clostridium septicum	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 1.5 per million · 0.0001% of analysed DNA · 12 fragments in 2 regions	0.01% of its genome covered · 97.4% identity
Clostridium tertium	Bacterial pathogens	shared sequence only — cannot tell apart	**small signal** 11 per million · 0.0011% of analysed DNA · 91 fragments in 3 regions	0.09% of its genome covered · 96.7% identity
Coccidioides immitis	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Cronobacter malonaticus	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 96.0% identity
Cronobacter sakazakii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Cronobacter turicensis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Cryptococcus neoformans species complex	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Cryptosporidium ubiquitum	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Cunninghamella bertholletiae	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Cyclospora cayetanensis (lineage A)	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Diffusely adherent Escherichia coli (DAEC)	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 1.00 per million · well under a thousandth of a percent · 8 fragments in 8 regions	0.02% of its genome covered · 100.0% identity
Edwardsiella tarda	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Eimeria spp.	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Emergomyces (human-infecting species)	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Entamoeba histolytica	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Enterobacter cloacae	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 99.3% identity
Enterobacter hormaechei	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Enterobacter kobei	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Enterobacter ludwigii	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 98.0% identity
Enterobacter roggenkampii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Enterococcus casseliflavus	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 1.7 per million · 0.0002% of analysed DNA · 14 fragments in 4 regions	0.05% of its genome covered · 98.0% identity
Enterococcus faecalis	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 4.5 per million · 0.0004% of analysed DNA · 36 fragments in 6 regions	0.06% of its genome covered · 98.7% identity
Enterococcus gallinarum	Bacterial pathogens	shared sequence only — cannot tell apart	**small signal** 38 per million · 0.0038% of analysed DNA · 306 fragments in 12 regions	0.35% of its genome covered · 99.3% identity
Enterotoxigenic Bacteroides fragilis (ETBF)	Bacterial pathogens	shared sequence only — cannot tell apart shared sequence from relatives found in this sample; species not found by the organism inventory	**sequence from relatives** 19,839 fragments, shared with relatives · from Bacteroides uniformis (3.11%), Bacteroides stercoris (1.18%) · 19,839 fragments in 123 regions	2.82% of its genome covered · 98.0% identity disease genes not assessed
Escherichia albertii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Escherichia coli (commensal reference)	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 98.0% identity
Escherichia fergusonii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Exophiala, clinically relevant species	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Fusarium, clinically relevant species complexes	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Fusobacterium necrophorum	Bacterial pathogens	shared sequence only — cannot tell apart	**small signal** 79 per million · 0.0079% of analysed DNA · 632 fragments in 5 regions	0.84% of its genome covered · 99.3% identity
Fusobacterium nucleatum	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Hafnia alvei	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Hafnia paralvei	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter bizzozeronii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter cinaedi	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter felis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter fennelliae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter pylori	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter salomonis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Helicobacter suis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Histoplasma species complex	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Klebsiella aerogenes	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Klebsiella grimontii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Klebsiella michiganensis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Klebsiella oxytoca	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Klebsiella pneumoniae	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.87 per million · well under a thousandth of a percent · 7 fragments in 7 regions	0.02% of its genome covered · 100.0% identity
Klebsiella quasipneumoniae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Klebsiella variicola	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Kodamaea ohmeri	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Laribacter hongkongensis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Lichtheimia corymbifera	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Lichtheimia ramosa	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Listeria monocytogenes	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 6.7 per million · 0.0007% of analysed DNA · 54 fragments in 8 regions	0.17% of its genome covered · 97.7% identity
Lomentospora prolificans	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Malassezia furfur	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Malassezia globosa	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Malassezia pachydermatis	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Malassezia restricta	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Malassezia sympodialis	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Mycobacterium avium complex	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Non-Helicobacter pylori gastric Helicobacter group	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Nosema ocularum	Microsporidia pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Paecilomyces variotii	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Paracoccidioides brasiliensis complex	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Pentatrichomonas hominis	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Peptostreptococcus anaerobius	Bacterial pathogens	shared sequence only — cannot tell apart	**small signal** 31 per million · 0.0031% of analysed DNA · 248 fragments in 11 regions	0.79% of its genome covered · 98.7% identity
Plesiomonas shigelloides	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Proteus mirabilis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Proteus penneri	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Proteus vulgaris	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Prototheca, clinically documented species	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Providencia alcalifaciens	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Providencia rettgeri	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Providencia stuartii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Pseudomonas aeruginosa	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Pythium insidiosum	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella bongori	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica serovar Paratyphi A	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica serovar Paratyphi B	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica serovar Paratyphi C	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica serovar Typhi	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica subsp. arizonae	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 98.7% identity
Salmonella enterica subsp. diarizonae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica subsp. enterica	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica subsp. houtenae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Salmonella enterica subsp. indica	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Salmonella enterica subsp. salamae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Sarcocystis sigmoideus	Protozoa pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Scedosporium apiospermum / boydii complex	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Serratia marcescens	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.12 per million · well under a thousandth of a percent · 1 fragments in 1 regions	0.00% of its genome covered · 99.3% identity
Shigella boydii	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.87 per million · well under a thousandth of a percent · 7 fragments in 6 regions	0.02% of its genome covered · 100.0% identity
Shigella dysenteriae	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.37 per million · well under a thousandth of a percent · 3 fragments in 3 regions	0.01% of its genome covered · 99.3% identity
Shigella flexneri	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.25 per million · well under a thousandth of a percent · 2 fragments in 2 regions	0.01% of its genome covered · 99.7% identity
Shigella sonnei	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.62 per million · well under a thousandth of a percent · 5 fragments in 3 regions	0.01% of its genome covered · 98.0% identity
Sporothrix schenckii complex (pathogenic clade)	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Staphylococcus aureus	Bacterial pathogens	shared sequence only — cannot tell apart	**at the detection floor** 0.62 per million · well under a thousandth of a percent · 5 fragments in 2 regions	0.01% of its genome covered · 96.7% identity
Stenotrophomonas maltophilia	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Streptococcus anginosus group	Bacterial pathogens	shared sequence only — cannot tell apart	**small signal** 58 per million · 0.0058% of analysed DNA · 468 fragments in 156 regions	3.43% of its genome covered · 98.7% identity
Streptococcus gallolyticus group	Bacterial pathogens	shared sequence only — cannot tell apart	**faint signal** 8.2 per million · 0.0008% of analysed DNA · 66 fragments in 15 regions	0.38% of its genome covered · 97.4% identity
Strongyloides papillosus	Parasitic worms	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Strongyloides venezuelensis	Parasitic worms	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Taenia solium	Parasitic worms	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Toxocara canis	Parasitic worms	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Trichosporon / Cutaneotrichosporon, curated human-infecting members	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Trichosporon asahii	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Trichosporon inkin	Fungal pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Vibrio alginolyticus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

ORGANISM	GROUP	WHAT THE SEQUENCE SHOWS	HOW MUCH	STRENGTH OF EVIDENCE
Vibrio cholerae	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Vibrio fluvialis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Vibrio furnissii	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Vibrio mimicus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Vibrio parahaemolyticus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Vibrio vulnificus	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Yersinia enterocolitica	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered
Yersinia pseudotuberculosis	Bacterial pathogens	shared sequence only — cannot tell apart	— no qualifying sequence to measure	0.00% of its genome covered

Toxin, virulence and resistance genes

Whether a gene is present is asked separately from which organism carries it. A resistance gene found in stool belongs to *some* organism in the community, and unless the read or contig physically links it to a named genome, that organism is unknown. Gene presence is also not a prediction of treatment failure.

GENE	KIND	WHAT WAS FOUND	CARRIER ORGANISM	SUPPORT
Aminoglycoside phosphotransferases (aph)	amr aminoglycoside	present, complete	carrier not established	155 fragments
Cfr 23S rRNA methyltransferase (cfr)	amr oxazolidinone phenicol	present, complete	carrier not established	213 fragments
Chloramphenicol acetyltransferases (cat)	amr oxazolidinone phenicol	present, complete	carrier not established	17 fragments
Erm rRNA methyltransferases (erm)	amr macrolide lincosamide	present, complete	carrier not established	297 fragments
Lincosamide nucleotidyltransferases (lnu)	amr macrolide lincosamide	present, complete	carrier not established	113 fragments
Mef macrolide efflux (mef)	amr macrolide lincosamide	present, complete	carrier not established	211 fragments
Tetracycline efflux pumps (tet efflux)	amr tetracycline	present, complete	carrier not established	64 fragments
Tetracycline ribosomal protection proteins (tet RPP)	amr tetracycline	present, complete	carrier not established	1,844 fragments
Tetracycline-inactivating enzymes (tet enzymatic)	amr tetracycline	present, complete	carrier not established	110 fragments

What this screen did and did not cover

Every number below is counted from the run itself, not from a fixed marketing total. Organisms, determinants and subtypes are counted separately and never summed into a single figure, because they are different kinds of question.

Targets in the catalogue	486	every organism this screen carries a definition for
Searched, answered at its own rank	408	own reference sequence, aligned, and separable from its relatives
Searched, answers for a group	31	whole genome shared with a catalogue relative; the toxin and virulence gene screen below is what separates these
Searched, masked by the host	1	sequence also present in the human reference genome, so a matching read cannot be assigned to the organism rather than to you
RNA genome, outside a DNA test	32	no DNA exists in the sample to find; needs a stool RNA panel
Reference not available yet	14	no public genome or marker sequence good enough to search; named below
Determinants searched	79	toxin, virulence and resistance genes, searched independently of species

Determinants not searched	14	no usable reference for the gene family

The 14 targets with no reference yet. *Acanthocephalus rauschi*, *Additional curated human DNA viruses (systemic watch)*, *Bertiella mucronata*, *Common food-animal genome competitors*, *Episthmium spp.*, *Fish ciliate competitors (including fish Balantidium)*, *Free-living and plant nematode competitors*, *Human filarial parasites*, *Inermicapsifer madagascariensis*, *Iodamoeba buetschlii*, *Neighboring fluke and tapeworm competitors*, *Other common dietary yeasts (curated group)*, *Other human stool-associated parvoviruses*, *Reclassified opportunistic cryptococcal yeasts (former Cryptococcus spp.)*. Each is in the catalogue and each will be searched as soon as a reference exists for it. Until then this report says nothing either way about them.

The 31 targets that answer for a group. *Animal Cryptosporidium competitors* (searched as *Cryptosporidium parvum*), *Animal Cystoisospora competitors* (searched as *Cystoisospora suis*), *Animal Giardia competitors* (searched as *Giardia duodenalis*), *Anncaliia connori* (searched as *Anncaliia algerae*), *Anncaliia vesicularum* (searched as *Anncaliia algerae*), *Aspergillus, clinically relevant cryptic species* (searched as *Aspergillus flavus*), *Bacteroides fragilis group* (searched as *Enterotoxigenic Bacteroides fragilis (ETBF)*), *Blastomyces, relevant relatives* (searched as *Blastomyces gilchristii*), *Cryptosporidium mink genotype* (searched as *Cryptosporidium parvum*), *Cyclospora ashfordi (lineage B)* (searched as *Cyclospora cayetanensis (lineage A)*), *Cyclospora cayetanensis sensu lato* (searched as *Cyclospora cayetanensis (lineage A)*), *Cyclospora henanensis (lineage C)* (searched as *Cyclospora cayetanensis (lineage A)*), *Echinostoma fujianensis* (searched as *Echinostoma echinatum*), *Echinostoma ilocanum* (searched as *Echinostoma echinatum*), *Enteroaggregative Escherichia coli (EAEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Enterobacter cloacae complex* (searched as *Enterobacter cloacae*), *Enteropathogenic Escherichia coli (EPEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Enterotoxigenic Escherichia coli (ETEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Hybrid diarrheagenic Escherichia coli pathotypes* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Klebsiella oxytoca complex* (searched as *Klebsiella oxytoca*), *Klebsiella pneumoniae complex* (searched as *Klebsiella pneumoniae*), *Microsporidium ceylonensis* (searched as *Microsporidium africanum*), *Nakaseomyces glabratus group, unresolved (Candida glabrata / nivariensis / bracarensis)* (searched as *Candida dubliniensis*), *Nonhuman Sarcocystis competitors* (searched as *Sarcocystis sigmoideus*), *Pichia, curated dietary species* (searched as *Candida krusei (Pichia kudriavzevii)*), *Salmonella enterica, nontyphoidal serovars* (searched as *Salmonella enterica subsp. enterica*), *Shiga toxin-producing Escherichia coli (STEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Shigella/enteroinvasive Escherichia coli (combined group)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Strongyloides fuelleborni subsp. kellyi* (searched as *Strongyloides fuelleborni subsp. fuelleborni*), *Trachipleistophora anthropophthera* (searched as *Trachipleistophora hominis*), *Trichinella recognized genotypes (unnamed)* (searched as *Trichinella papuae*). These were searched. Their genomes are the same as a relative's in this catalogue, so a read cannot say which of the pair it came from; where the difference between them is a gene rather than a genome — the diarrhoea-causing *E. coli* types, toxin-producing *Bacteroides fragilis* — the gene screen above answers it.

Limits that apply to every finding above. This is a research screen on a stool sample, not a diagnostic test. Sequence presence is not infection, and it is not a measure of how much organism is there: read counts depend on how much of your sample was that organism's DNA, not on its abundance in you. A negative here does not rule out an organism that sheds intermittently, sits in tissue rather than the gut lumen, or has an RNA genome that a DNA library cannot see. Nothing here should change treatment without a clinician and, where listed, a confirmatory test.

Antibiotic-resistance genes, by class

Resistance genes present in the community, grouped by the antibiotics they act against. **A gene is not a phenotype.** Carrying one of these does not establish that an antibiotic would fail for you, and not carrying one does not establish that it would work. Gut bacteria carry resistance genes routinely, most of them in harmless residents.

Read from the existing determinant screen and grouped; no call is recomputed and no phenotype is inferred.

ANTIBIOTIC CLASS	GENE FAMILIES	EVIDENCE	HOW IT WORKS	READS
Macrolides, lincosamides and streptogramin B	**3** erm, mef, lnu	3 acquired genes	ribosomal methylation (erm), efflux (mef) or enzymatic inactivation (lnu)	621
Sulfonamides and trimethoprim	**1** dfr	1 unresolved	alternative dihydropteroate synthase or dihydrofolate reductase	16
Aminoglycosides	**3** aph, aac, ant	1 acquired gene · 2 unresolved	enzymatic modification (aac, aph, ant) or 16S methylation (armA)	179
Tetracyclines	**3** tet_ribosomal_protection, tet_enzymatic, tet_efflux	3 acquired genes	ribosomal protection, efflux or enzymatic inactivation	2,018
Oxazolidinones and phenicols	**2** cfr, cat	2 acquired genes	23S rRNA methylation (cfr) or enzymatic inactivation (cat)	230

12 distinct gene families across 5 of 10 classes. Families are counted once even when they appear in two classes, and a drug is never counted again as its own class. All 12 detected determinants are unlinked: the reads show the gene is present in the community, not which organism carries it.

PART B — WHAT YOUR BACTERIA CAN MAKE AND BREAK DOWN

19 Your Microbial Functions in Detail

What your microbes can make, break down and transform.

One card per pathway, grouped as on the previous page and most notable first within each group: what the compound is, where you sit, what that means, and the research behind it.

Fibre & Dietary Substrates

‹ at a glance

Dietary fibre and carbohydrate breakdown ▾

HIGH

The breadth of dietary carbohydrates your gut bacteria carry enzymes for, across starch, pectin, xylan and galactoside chemistry.

YOUR RESULT	**1000.7** copies per 100 genomes
PERCENTILE	**90th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	578.7 – 811.5 (median 702.6)
MADE FROM	Dietary fibre, resistant starch and other carbohydrates reaching the colon
DNA FRAGMENTS MATCHED	97,614



What is driving this 41,152 matched fragments, by the organism each came closest to

55% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides thetaiotaomicron* (0.57% of your community); *Bacteroides ovatus* (0.45% of your community). A further 45% matched *Bacteroides reticulotermitis*, *Bacteroides graminisolvens*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bacteroides reticulotermitis	**9%** (3756 fragments)	genus present; closest reference	—
Phocaeicola vulgatus	**7%** (3011 fragments)	**yes** · 11.748%	● Conditional
Bacteroides thetaiotaomicron	**6%** (2572 fragments)	**yes** · 0.574%	● Conditional
Bacteroides graminisolvens	**5%** (2140 fragments)	genus present; closest reference	—
Bacteroides ovatus	**4%** (1606 fragments)	**yes** · 0.451%	● Conditional

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bacteroides thetaiotaomicron*, *Bacteroides ovatus*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides thetaiotaomicron 0.57% of your community · 6% of this signal	Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading. ■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.
Bacteroides ovatus 0.45% of your community · 4% of this signal	Versatile polysaccharide degrader; common commensal with no fixed directional interpretation. ■ : Bacteroides as diet-responsive commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means high · higher than most people

Your community carries enzymes for a wider range of dietary carbohydrates than most samples. Breadth of this kind travels with dietary variety, and it means fewer fibres pass through unused.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides thetaiotaomicron* (0.57% of your community); *Bacteroides ovatus* (0.45% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Fourteen individual fibre and starch readings are built from these enzymes in the functions section. This panel is their shared evidence and a breadth summary; the specificity judgement happens per substrate.

A carbohydrate-active enzyme family usually acts on a class of molecules rather than on one, so a family alone answers a broader question than the one a reader asks. GH13 supports general amylolysis but not resistant starch in particular; a beta-galactosidase is carried by every lactose-using organism and does not establish galactooligosaccharide use. The per-substrate readings therefore state whether the evidence is substrate-specific or only class-level, and a gene shared between two substrates appears on both cards and is counted once in any total. What none of this measures is how much fibre

you eat: it is the capacity to use what arrives.

One of the genes behind this figure (amyA, lacZ, xynA) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Drula E. et al., "The carbohydrate-active enzyme database: functions and literature", Nucleic Acids Res 50:D571-D577 (2022), doi:10.1093/nar/gkab1045; Zheng J. et al., "dbCAN3", Nucleic Acids Res 51:W115-W121 (2023), doi:10.1093/nar/gkad328.

Fermentation & Cross-feeding

‹ at a glance

Butyrate (SCFA) ▼

NOTABLY LOW

Butyrate, a short-chain fatty acid and the main energy source for the cells lining the colon.

YOUR RESULT	**11.6** copies per 100 genomes
PERCENTILE	**5th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	21.9 – 30.7 (median 26.8)
MADE FROM	Dietary fibre fermented by gut bacteria
DNA FRAGMENTS MATCHED	1,840



What is driving this 478 matched fragments, by the organism each came closest to

51% of the signal comes from organisms found in you: *Anaerostipes hadrus*; *Oscillibacter valericigenes* (0.02% of your community). A further 13% matched *Anaerostipes amylophilus*, whose genus is present but whose species was not resolved. The remaining 36% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Anaerostipes hadrus	**11%** (52 fragments)	**yes**	● Beneficial
Sporomusa carbonis	**7%** (35 fragments)	not found; closest reference	—
Oscillibacter valericigenes	**6%** (31 fragments)	**yes** · 0.025%	● Conditional
Sporomusa rhizae	**5%** (24 fragments)	not found; closest reference	—
Anaerostipes amylophilus	**5%** (22 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Anaerostipes hadrus*, *Oscillibacter valericigenes*; *Anaerostipes hadrus* alone accounts for 11% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Anaerostipes hadrus	Butyrate producer that also converts lactate and acetate to butyrate; part of the core butyrate-producing set of adult stool. ■ : Anaerostipes is among the principal butyrate producers using the butyryl-CoA:acetate CoA-transferase route. *[Louis P, Flint HJ. Environ Microbiol 2017;19:29. PMID 27928878]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means notably low · well below most people

Fewer of your gut bacteria carry the butyrate-producing genes than in most people, which is the less favourable direction here. Reduced numbers of these bacteria are reported in stool from people with Alzheimer's disease, and APOE E4 carriers show lower abundance of them specifically. Dietary fibre is the main lever on this population.

Two measurements agree

This agrees with the butyrate-producing species reading (4th percentile), which measures the same function by counting the organisms characterised for the job instead of the genes.

46% of this gene family's matched fragments landed on reference proteins from organisms that carry the enzyme fold without the pathway; those are counted as background, not as capacity, and are excluded from the figure above.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Anaerostipes hadrus*; *Oscillibacter valericigenes* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Reasonably supported

Butyrate is the most consistently beneficial gut metabolite in this panel. It strengthens the gut barrier, reduces inflammation, and improves mitochondrial function in neurons. People with Alzheimer's disease tend to have fewer butyrate-producing bacteria in their stool.

A specific molecular route from butyrate to tau pathology has now been described: butyrate drives acetylation of GSK3-beta at lysine 15, which regulates its inhibitory phosphorylation at serine 9 and in turn changes how much tau gets phosphorylated. In the same work, transplanting gut bacteria from Alzheimer's model mice into healthy mice produced dysbiosis, tau phosphorylation and cognitive impairment — and giving those mice Lactobacillus and Bifidobacterium reversed it. Separately, butyrate improves neuronal mitochondrial function, restored memory function in the APP/PS1 mouse model, and ameliorates diabetes-related cognitive impairment by restoring Parkin-mediated mitophagy. In humans, stool from people with Alzheimer's shows reduced short-chain-fatty-acid producers, and carriers of the APOE E4 risk allele specifically show lower abundance of butyrate-producing bacteria alongside enrichment of pro-inflammatory genera.

Reference: Zhang Y. et al., Mol Psychiatry 28:4421-4437 (2023), doi:10.1038/s41380-023-02216-7; Sepulveda-Rivera V. et al., Sci Rep 15:28505 (2025), doi:10.1038/s41598-025-13262-2; Cho J.H. et al., Autophagy 20:1505-1522 (2024), doi:10.1080/15548627.2023.2274205

‹ at a glance

Propionate (SCFA) ▲

HIGH

Propionate, a short-chain fatty acid produced alongside butyrate but with a different reported relationship to cognition.

YOUR RESULT	**32.9** copies per 100 genomes
PERCENTILE	**92nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	16.4 – 25.3 (median 19.9)
MADE FROM	Dietary fibre, lactate, and the sugars fucose and rhamnose
DNA FRAGMENTS MATCHED	7,097



What is driving this 1,756 matched fragments, by the organism each came closest to

58% of the signal comes from organisms found in you: *Blautia caecimuris* (0.89% of your community); *Faecalicatena orotica*. A further 24% matched *Frisingicoccus caecimuris*, *Clostridium symbiosum*, whose genus is present but whose species was not resolved. The remaining 18% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia caecimuris	26% (459 fragments)	yes · 0.892%	Conditional
Faecalicatena orotica	17% (304 fragments)	yes	Conditional
Muricomes intestini	13% (235 fragments)	not found; closest reference	—
Frisingicoccus caecimuris	12% (214 fragments)	genus present; closest reference	—
Clostridium symbiosum	6% (98 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Blautia caecimuris*, *Faecalicatena orotica*; *Blautia caecimuris* alone accounts for 26% of the signal. Changing this reading means changing them.

What your reading means high · higher than most people

One genetic study reported that higher gut propionate production associates with increased Alzheimer's likelihood — the opposite direction to butyrate — and your capacity sits above most people's. That was a single study using an indirect method, and propionate also has documented protective effects on mitochondria, so this is worth noting rather than acting on.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia caecimuris* (0.89% of your community); *Faecalicatena orotica*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with worse outcomes · Early evidence

Propionate behaves differently from butyrate. A genetic (Mendelian randomisation) study found that higher gut propionate production was associated with an increased likelihood of Alzheimer's disease, while higher butyrate was associated with improved insulin resistance.

Sanna et al. used Mendelian randomisation across gut microbiome, short-chain fatty acid and disease datasets and reported that increased propionate production or absorption associated with an increased likelihood of Alzheimer's disease. This is a single study using an indirect method, so the finding is suggestive rather than established. Propionate also has documented protective effects on mitochondrial structure, so the overall picture is not settled.

Reference: Sanna S. et al., Nat Genet 51:600-605 (2019), doi:10.1038/s41588-019-0350-x; Cheng Y. et al., Biochem Pharmacol 169:113607 (2019), doi:10.1016/j.bcp.2019.08.015

‹ at a glance

Fermentation routes and cross-feeding

NOTABLY HIGH

The specific routes your bacteria use to ferment carbohydrate into acids and gases.

YOUR RESULT	**175.5** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	130.9 – 147.7 (median 140.3)
MADE FROM	Pyruvate and other products of carbohydrate fermentation
DNA FRAGMENTS MATCHED	24,368



What is driving this 8,551 matched fragments, by the organism each came closest to

46% of the signal comes from organisms found in you: *Gemmiger formicilis*; *Phocaeicola dorei*. A further 36% matched *Odoribacter laneus*, *Streptococcus ictaluri*, whose genus is present but whose species was not resolved. The remaining 18% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Gemmiger formicilis	**9%** (748 fragments)	**yes**	● Beneficial
Odoribacter laneus	**6%** (473 fragments)	genus present; closest reference	—
Sphingobacterium spiritivorum	**6%** (470 fragments)	not found; closest reference	—
Streptococcus ictaluri	**5%** (450 fragments)	genus present; closest reference	—
Phocaeicola dorei	**5%** (432 fragments)	**yes**	● Conditional

The organisms carrying this gene in your sample are *Gemmiger formicilis*, *Phocaeicola dorei*; *Gemmiger formicilis* alone accounts for 9% of the signal. Changing this reading means changing them.

What your reading means notably high · well above most people

More fermentation-route capacity than most samples carry, across the acetate and lactate chemistry measured here. The individual route readings show which specific chemistries are complete.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Gemmiger formicilis*; *Phocaeicola dorei*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it The research picture is mixed · Early evidence

The existing butyrate, propionate, methane and sulfide readings say how much capacity is present. These say by which chemistry.

Propionate is reached by three unrelated routes - succinate, acrylate and propanediol - and a single propionate number cannot distinguish them, which matters because they use different substrates. Acetate is the case where a gene count misleads most easily: an AMP-forming acetyl-CoA synthetase runs towards acetate assimilation rather than formation, so finding it is evidence the community can consume acetate, not make it, and it is excluded here. Hydrogen is shown as production and consumption separately because a healthy community makes a great deal of it and netting the two would hide both.

One of the genes behind this figure (pta) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Reichardt N. et al., "Phylogenetic distribution of three pathways for propionate production within the human gut microbiota", ISME J 8:1323-1335 (2014), doi:10.1038/ismej.2014.14; Louis P., Flint H.J., Environ Microbiol 19:29-41 (2017), doi:10.1111/1462-2920.13589.

‹ at a glance

Hydrogen sulfide (H2S) ▲

TYPICAL

Hydrogen sulfide, a gas made by bacteria that breathe sulfate or break down sulfur-containing amino acids.

YOUR RESULT	**63.8** copies per 100 genomes
PERCENTILE	**29th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	61.8 – 88.0 (median 75.3)
MADE FROM	Dietary sulfate (preservatives, some drinking water), red meat, eggs, cruciferous vegetables
DNA FRAGMENTS MATCHED	2,258



What is driving this 2,024 matched fragments, by the organism each came closest to

11% of the signal comes from organisms found in you: *Blautia obeum* (0.02% of your community). A further 89% matched *Blautia luti*, *Blautia wexlerae*, *Blautia producta*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia luti	**43%** (863 fragments)	genus present; closest reference	—
Blautia wexlerae	**12%** (251 fragments)	genus present; closest reference	—
Blautia obeum	**8%** (168 fragments)	**yes** · 0.022%	● Conditional
Blautia producta	**8%** (152 fragments)	genus present; closest reference	—
Clostridium ganghwense	**7%** (135 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Blautia obeum*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Blautia obeum 0.02% of your community · 8% of this signal	Abundant Lachnospiraceae member; Blautia is higher in some obesity cohorts and lower in others. No fixed reading. ■ : Review of Blautia as a widespread genus with mixed disease associations. *[Liu X et al. Gut Microbes 2021;13:1875796. PMID 33525961]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means typical · in line with most people

Your sulfide-producing capacity is unremarkable. Some is normal and useful — sulfide is a signalling molecule in the gut wall.

Why this colour. Hydrogen sulfide stops colon cells burning butyrate and is raised in ulcerative colitis; a trace is normal signalling, a high reading is not.

Two measurements of the same function disagree

Read this beside the sulfate-reducing species reading. The hydrogen-sulfide capacity reading sits at the 29th percentile; the sulfate-reducing species reading is at the 70th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not. Of the two, the gene reading is the better guide here, because this capacity does not depend on any one set of species. Hydrogen sulfide has two independent sources: dissimilatory sulfate reduction by Desulfovibrio and relatives, and cysteine degradation, which is widespread in ordinary gut bacteria. The gene panel sees both; the group sees only the sulfate reducers, which are a minority of stool even when sulfide capacity is high.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia obeum* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with worse outcomes · Reasonably supported

Hydrogen sulfide impairs the ability of colon cells to burn butyrate for energy, and higher sulfide has been reported in ulcerative colitis. It is also a normal signalling molecule at lower concentrations, so this is a direction to notice rather than a hazard reading.

Roediger and colleagues showed that sulfide blocks butyrate oxidation in isolated colonocytes, which is the mechanism most often invoked for the ulcerative colitis association; Pitcher and Cummings measured higher faecal sulfide in active UC. Sulfate-reducing bacteria are also enriched in colorectal cancer cohorts, and hydrogen sulfide is genotoxic at colonic concentrations. Against that, sulfide at physiological levels is a signalling molecule with documented anti-inflammatory effects, and Desulfovibrio is a normal member of most healthy adult communities. The literature supports "sustained excess is adverse", not "sulfide is bad".

Reference: Pitcher MC, Cummings JH, Gut 39:1-4 (1996); Roediger WE et al., Gastroenterology 104:802-809 (1993); Carbonero F et al., Nat Rev Gastroenterol Hepatol 9:504-518 (2012)

‹ at a glance

Methane (CH4) ▲

NOT DETECTED

Methane, made by archaea (not bacteria) that consume the hydrogen other gut microbes produce.

YOUR RESULT	**0** copies per 100 genomes
PERCENTILE	0 of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 1.4 (median 0)
MADE FROM	Hydrogen and carbon dioxide from fibre fermentation
DNA FRAGMENTS MATCHED	0



Why this colour. Methane slows transit and high methanogen carriage is one of the firmest associations with constipation-predominant IBS.

Two measurements of the same function disagree

Read this beside the methanogen abundance reading. The methane capacity (mcrA) reading sits at the <1st percentile; the methanogen abundance reading is at the 35th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not; the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Neither reading is preferred: a split this size means one of the two engines is at its limit for this function. mcrA is exclusive to methanogenic archaea, so these two should track each other tightly. A split means one of the two engines is at its limit: methanogens are archaea, they are a small fraction of stool DNA, and the bacterial profiler under-reports them.

How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

Methane slows intestinal transit. High methanogen carriage is strongly associated with constipation and with the constipation subtype of IBS, and is a normal variant in a large minority of healthy people.

Pimentel and colleagues infused methane into dog ileum and slowed transit by about 60%, which is the clearest causal evidence for the association. In humans, breath-methane-positive patients are overwhelmingly constipation-predominant, and Kim et al. showed that Methanobrevibacter smithii abundance in stool tracks breath methane. Methanogen carriage is also partly heritable and rises with age. Methane production is not itself a disease: it shifts transit, and whether that matters depends on where a person's transit already sits.

Reference: Pimentel M et al., Am J Physiol Gastrointest Liver Physiol 290:G1089 (2006); Kim G et al., Dig Dis Sci 57:3213-3218 (2012); Triantafyllou K et al., J Neurogastroenterol Motil 20:31-40 (2014)

Vitamins & Nutrients

‹ at a glance

Folate (vitamin B9) ▼

NOTABLY HIGH

The capacity of your gut bacteria to synthesise folate, which colonocytes can absorb to a limited extent.

YOUR RESULT	**183.4** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	99.1 – 138.6 (median 119.4)
MADE FROM	Built de novo by bacteria from GTP and para-aminobenzoate
DNA FRAGMENTS MATCHED	3,567



What is driving this 3,781 matched fragments, by the organism each came closest to

68% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola plebeius* (3.75% of your community); *Bacteroides stercoris* (1.52% of your community). A further 8% matched *Clostridium beijerinckii*, whose genus is present but whose species was not resolved. The remaining 24% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**27%** (1035 fragments)	**yes** · 11.748%	● Conditional
Flammeovirga aprica	**13%** (487 fragments)	not found; closest reference	—
Phocaeicola plebeius	**5%** (188 fragments)	**yes** · 3.747%	● Conditional
Clostridium beijerinckii	**4%** (158 fragments)	genus present; closest reference	—
Bacteroides stercoris	**4%** (151 fragments)	**yes** · 1.520%	● Conditional

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Phocaeicola plebeius*, *Bacteroides stercoris*; *Phocaeicola vulgatus* alone accounts for 27% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides stercoris 1.52% of your community · 4% of this signal	Common Bacteroides species with no fixed directional interpretation. ■ : Bacteroides as commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means notably high · well above most people

Your gut community has more folate synthesis capacity than most people's, and unlike B12, some colonic folate is absorbed. This is a mildly favourable feature of the community rather than a meaningful contribution to your folate status, which diet dominates.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola plebeius* (3.75% of your community); *Bacteroides stercoris* (1.52% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Bifidobacterium and several other gut genera synthesise folate, and unlike B12 some of it is absorbed in the colon. The contribution to human folate status is modest compared with diet.

Rossi et al. and Pompei et al. showed that Bifidobacterium strains raise folate concentrations in the gut and in host plasma in rat models, and human colonocytes do express a proton-coupled folate transporter capable of absorbing bacterial folate. Magnusdottir et al.'s genome reconstruction found folate biosynthesis in a larger share of gut genomes than cobalamin. The realistic scale of the contribution is the honest caveat: in a fortified food supply, dietary folate dominates, and no trial has shown that manipulating bacterial folate synthesis changes clinically relevant folate status in humans.

Reference: Rossi M. et al., Nutrients 3:118-134 (2011), doi:10.3390/nu3010118; Magnusdottir S. et al., Front Genet 6:148 (2015)

What the research says about changing this

Folate synthesis genes — not a folate status LANE A HUMAN EVIDENCE · STRONG

Fired by: folate gene capacity: notably high

Folate status is assessed by diet, medications, pregnancy and blood tests; microbial genes do not indicate deficiency, and high folic-acid intake can mask B12 deficiency.

- Folate status is assessed by diet, medication, pregnancy and blood tests; high folic-acid intake can mask B12 deficiency. No folate dose follows from microbial genes. [NIH ODS — Folate fact sheet] · lane A · reference

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “folate deficient” · “take folic acid”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Vitamin K2 (menaquinone, MK-n) ▾

NOTABLY HIGH

The capacity of your gut bacteria to make menaquinone (vitamin K2), which they use for their own respiration.

YOUR RESULT	**47.6** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	15.9 – 30.4 (median 22.6)
MADE FROM	Built de novo by bacteria from chorismate
DNA FRAGMENTS MATCHED	1,467



What is driving this 2,486 matched fragments, by the organism each came closest to

78% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Parabacteroides distasonis* (1.42% of your community); *Bacteroides eggerthii*; *Akkermansia muciniphila* (0.96% of your community). A further 22% matched *Bacteroides mediterraneensis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**37%** (911 fragments)	**yes** · 11.748%	● Conditional
Bacteroides mediterraneensis	**13%** (319 fragments)	genus present; closest reference	—
Parabacteroides distasonis	**4%** (111 fragments)	**yes** · 1.420%	● Conditional
Bacteroides eggerthii	**3%** (68 fragments)	**yes**	● Conditional
Akkermansia muciniphila	**3%** (67 fragments)	**yes** · 0.961%	● Beneficial

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Parabacteroides distasonis*, *Bacteroides eggerthii*; *Phocaeicola vulgatus* alone accounts for 37% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Parabacteroides distasonis 1.42% of your community · 4% of this signal	Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans. ■ : P. distasonis alleviated obesity and metabolic dysfunction in mice via succinate and secondary bile acids; human data are associative. *[Wang K et al. Cell Rep 2019;26:222. PMID 30605678]* Association only — no study on record shows that changing this organism changes an outcome.
Akkermansia muciniphila 0.96% of your community · 3% of this signal	Mucin-degrading Verrucomicrobium. Higher levels are associated with metabolic health in several cohorts and a pasteurised strain (MucT) has human trial evidence, yet the same species is enriched in multiple sclerosis and Parkinson's disease cohorts. Endogenous abundance does not predict response to any Akkermansia product. ■ when higher: Pasteurised A. muciniphila MucT improved insulin sensitivity and metabolic markers in an exploratory 3-month trial of overweight/obese insulin-resistant volunteers. *[Depommier C et al. Nat Med 2019;25:1096. PMID 31263284]* ■ when higher: Akkermansia is enriched in multiple sclerosis patients across cohorts (iMSMS). *[iMSMS Consortium. Cell 2022;185:3467. PMID 36113426]*

What your reading means notably high · well above most people

Your gut community has more menaquinone synthesis capacity than most people's. Bacteria make it for their own respiration and the long-chain forms are poorly absorbed from the colon, so read this as a feature of the community rather than a contribution to your vitamin K.

Why this colour. Long-chain menaquinones from gut bacteria contribute to vitamin K status.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Parabacteroides distasonis* (1.42% of your community); *Bacteroides eggerthii*; *Akkermansia muciniphila* (0.96% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Gut bacteria are major menaquinone producers, but the long-chain forms they make are poorly absorbed from the colon and are made for bacterial electron transport rather than for the host.

Karl et al. profiled menaquinone forms in human stool and showed that long-chain MK-10 to MK-13 dominate, matching Bacteroides production, and that faecal menaquinone concentrations did not track markers of host vitamin K status. Ramotar et al. estimated colonic absorption of bacterial menaquinone as low. The interesting biology here is microbiological rather than nutritional: which quinone route a community uses (classical versus futalosine) separates Bacteroidota from Proteobacteria and tracks how the community respires. Claims that gut bacteria are a meaningful source of your vitamin K are not supported by the absorption data.

One of the genes behind this figure (mqnE) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Karl J.P. et al., Am J Clin Nutr 106:1052-1061 (2017), doi:10.3945/ajcn.117.155424; Fenn K. et al., Nat Microbiol 2:16241 (2017)

What the research says about changing this

Menaquinone (vitamin K2) synthesis genes — not a vitamin K status LANE A HUMAN EVIDENCE · STRONG

Fired by: k2 gene capacity: notably high

Gene capacity for menaquinone synthesis does not measure coagulation or vitamin K sufficiency. People on warfarin should keep vitamin K intake stable under their prescriber.

- Gene capacity for menaquinone synthesis does not measure coagulation status or sufficiency; warfarin users need stable intake managed by their prescriber. [NIH ODS — Vitamin K fact sheet] · lane A · reference

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “vitamin K deficiency” · “take K2”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Vitamin B1 (thiamine) ▼

NOTABLY HIGH

Your gut bacteria's capacity to build thiamine from scratch, rather than only take in thiamine that is already there.

YOUR RESULT	**276.8** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	190.1 – 229.3 (median 213.7)
MADE FROM	Built de novo by bacteria from a thiazole ring and a pyrimidine ring, joined by thiamine phosphate synthase
DNA FRAGMENTS MATCHED	17,373



What is driving this 9,508 matched fragments, by the organism each came closest to

78% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides uniformis* (3.11% of your community); *Blautia massiliensis*; *Mediterraneibacter gnavus* (6.93% of your community). A further 22% matched *Phocaeicola faecium*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**8%** (789 fragments)	**yes** · 11.748%	● Conditional
Phocaeicola faecium	**6%** (522 fragments)	genus present; closest reference	—
Bacteroides uniformis	**4%** (378 fragments)	**yes** · 3.108%	● Conditional
Blautia massiliensis	**4%** (364 fragments)	**yes**	● Conditional
Mediterraneibacter gnavus	**4%** (344 fragments)	**yes** · 6.934%	● Opportunist

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bacteroides uniformis*, *Blautia massiliensis*; *Phocaeicola vulgatus* alone accounts for 8% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides uniformis 3.11% of your community · 4% of this signal	Among the most prevalent Bacteroides in Western adults; level varies with diet and has no established directional meaning on its own. ■ : Bacteroides species are dominant, diet-responsive members of Western adult stool with both commensal and opportunistic behaviours. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means notably high · well above most people

More of your bacteria carry a complete route to thiamine than in most samples. The colon absorbs little of what they make, so the likeliest consequence is within the community, where thiamine-dependent species have a nearer source.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides uniformis* (3.11% of your community); *Blautia massiliensis*; *Mediterraneibacter gnavus* (6.93% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Most gut species cannot make thiamine and depend on the ones that can. This counts how much of your community carries the complete route.

Magnusdottir and colleagues reconstructed B-vitamin synthesis across gut genomes and found thiamine synthesis in a minority of them, with most species depending on what others make or on what arrives in food. That makes thiamine one of the clearer cases of cross-feeding in the gut, and it is why the two rings and their coupling step are counted separately: a community can carry the branches without the synthase that joins them. What this cannot tell you is your thiamine status. Dietary thiamine is absorbed in the small intestine, upstream of where most of this synthesis happens, and deficiency is a clinical diagnosis rather than a stool one.

Reference: Magnusdottir S., Ravcheev D., de Crecy-Lagard V., Thiele I., "Systematic genome assessment of B-vitamin biosynthesis suggests co-operation among gut microbes", Front Genet 6:148 (2015), doi:10.3389/fgene.2015.00148.

‹ at a glance

Vitamin B12 (cobalamin) ▾

HIGH

The capacity of your gut bacteria to build vitamin B12 from scratch, and separately their capacity to compete for it.

YOUR RESULT	**114.8** copies per 100 genomes
PERCENTILE	**93rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	75.2 – 101.4 (median 90.8)
MADE FROM	Built de novo by bacteria from small precursors; not from diet
DNA FRAGMENTS MATCHED	6,896



What is driving this 4,000 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Mediterraneibacter gnavus* (6.93% of your community); *Phocaeicola dorei*; *Blautia massiliensis*; *Phocaeicola vulgatus* (11.75% of your community).

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Mediterraneibacter gnavus	**11%** (450 fragments)	**yes** · 6.934%	Opportunist
Phocaeicola dorei	**10%** (401 fragments)	**yes**	Conditional
Blautia massiliensis	**8%** (325 fragments)	**yes**	Conditional
Phocaeicola vulgatus	**7%** (284 fragments)	**yes** · 11.748%	Conditional
Bacteroides uniformis	**5%** (211 fragments)	**yes** · 3.108%	Conditional

The organisms carrying this gene in your sample are *Mediterraneibacter gnavus*, *Phocaeicola dorei*, *Blautia massiliensis*; *Mediterraneibacter gnavus* alone accounts for 11% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides uniformis 3.11% of your community · 5% of this signal	Among the most prevalent Bacteroides in Western adults; level varies with diet and has no established directional meaning on its own. ■ : Bacteroides species are dominant, diet-responsive members of Western adult stool with both commensal and opportunistic behaviours. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means high · higher than most people

Your gut community has more B12 synthesis capacity than most people's. This is a statement about your bacteria, not about you: colonic B12 is largely not absorbed. If you want to know your B12 status, a blood test answers that directly.

Why this colour. Bacterial B12 made in the colon is poorly absorbed, so the benefit is small, but there is no harm in more of it.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Mediterraneibacter gnavus* (6.93% of your community); *Phocaeicola dorei*; *Blautia massiliensis*; *Phocaeicola vulgatus* (11.75% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Gut bacteria are prolific B12 producers, but they make it in the colon, below the point where humans absorb it, and most of it is consumed by other bacteria. Higher synthesis capacity is a feature of the community, not a contribution to your B12 status.

Degnan et al. established that most human gut bacteria either synthesise corrinoids or scavenge them, and that corrinoid competition is a genuine structuring force in the community. Magnusdottir et al. reconstructed B-vitamin biosynthesis across 256 gut genomes and found only about 40% could make cobalamin de novo. The absorption argument is settled anatomy: intrinsic-factor-mediated uptake occurs in the terminal ileum, while the bulk of bacterial synthesis happens distal to it. Where bacterial B12 does matter is between bacteria — corrinoid availability shapes which species can grow.

One of the genes behind this figure (cobS) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Degnan P.H. et al., Cell Metab 20:769-778 (2014), doi:10.1016/j.cmet.2014.10.002; Magnusdottir S. et al., Front Genet 6:148 (2015), doi:10.3389/fgene.2015.00148

What the research says about changing this

Vitamin B12 synthesis genes — colonic synthesis is not your B12 status LANE A HUMAN EVIDENCE · STRONG

Fired by: b12 gene capacity: high

Bacteria in the colon make B12 that is largely not absorbed there; this reading does not indicate deficiency or sufficiency. B12 status is a blood test, and diet, age, metformin and acid suppression matter more.

Studied: Serum vitamin B12 (and methylmalonic acid when appropriate) to assess status (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** serum B12 ± methylmalonic acid

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: reference fact sheet. Guideline position: strong for.

- Suspected B12 deficiency is confirmed with serum B12 and, when appropriate, methylmalonic acid; supplementation follows diet, malabsorption and clinical context. Colonic microbial B12 synthesis is not host bioavailability. [NIH Office of Dietary Supplements — Vitamin B12 fact sheet for health professionals] · lane A · reference · benefit

Conventional route: Serum B12 (± methylmalonic acid) if deficiency is a concern.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “B12 deficient” · “your bacteria make enough B12”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Vitamin B3 (niacin and NAD precursors) ▾

HIGH

Whether your bacteria can build the niacin ring from aspartate, as distinct from recycling nicotinamide that already exists.

YOUR RESULT	**242.5** copies per 100 genomes
PERCENTILE	**93rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	175.7 – 218.3 (median 191.9)
MADE FROM	Built de novo from aspartate through quinolinate, or salvaged from nicotinamide
DNA FRAGMENTS MATCHED	9,331



What is driving this 9,774 matched fragments, by the organism each came closest to

41% of the signal comes from organisms found in you: *Phocaeicola dorei*; *Phocaeicola vulgatus* (11.75% of your community). A further 22% matched *Ligilactobacillus ruminis*, whose genus is present but whose species was not resolved. The remaining 37% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola dorei	**10%** (925 fragments)	**yes**	● Conditional
Tannerella forsythia	**9%** (907 fragments)	not found; closest reference	—
Ligilactobacillus ruminis	**9%** (881 fragments)	genus present; closest reference	—
Phocaeicola vulgatus	**8%** (735 fragments)	**yes** · 11.748%	● Conditional
Treponema ruminis	**6%** (572 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Phocaeicola dorei*, *Phocaeicola vulgatus*; *Phocaeicola dorei* alone accounts for 10% of the signal. Changing this reading means changing them.

What your reading means high · higher than most people

More of your bacteria carry the aspartate route to the niacin ring than in most samples. They are building the ring rather than recycling one.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola dorei*; *Phocaeicola vulgatus* (11.75% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

The de-novo route and the salvage route both end at NAD. This measures the first and deliberately excludes the second.

A gene count that pooled the two routes would report almost every community as a producer, because salvage is close to universal. Rodionov and colleagues mapped both across gut genomes and found the de-novo route much less common, which is why the salvage enzymes are held out as a decoy set here rather than counted as production. Niacin is also one of the vitamins where host and bacterial supply are hard to separate: your own cells make NAD from tryptophan, and colonic absorption of bacterial niacin is limited.

Reference: Rodionov D.A. et al., "Micronutrient requirements and sharing capabilities of the human gut microbiome", Front Microbiol 10:1316 (2019), doi:10.3389/fmicb.2019.01316.

‹ at a glance

Biotin (vitamin B7) ▾

HIGH

The capacity of your gut bacteria to synthesise biotin, and the complementary fact of how much of the community depends on scavenging it.

YOUR RESULT	**93.2** copies per 100 genomes
PERCENTILE	**89th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	50.5 – 77.8 (median 63.0)
MADE FROM	Built de novo by bacteria from pimeloyl-ACP and alanine
DNA FRAGMENTS MATCHED	1,779



What is driving this 2,655 matched fragments, by the organism each came closest to

16% of the signal comes from organisms found in you: *Bacteroides ovatus* (0.45% of your community). A further 26% matched *Bacteroides faecichinchillae*, *Blautia faecicola*, whose genus is present but whose species was not resolved. The remaining 58% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Glaesserella parasuis	**16%** (414 fragments)	not found; closest reference	—
Pasteurella multocida	**7%** (180 fragments)	not found; closest reference	—
Bacteroides ovatus	**6%** (167 fragments)	**yes** · 0.451%	● Conditional
Bacteroides faecichinchillae	**6%** (153 fragments)	genus present; closest reference	—
Blautia faecicola	**4%** (113 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Bacteroides ovatus*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides ovatus 0.45% of your community · 6% of this signal	Versatile polysaccharide degrader; common commensal with no fixed directional interpretation. ■ : Bacteroides as diet-responsive commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means high · higher than most people

Your gut community has more biotin synthesis capacity than most people's. The relevance to anything you would notice is speculative: the hair-growth link comes from germ-free mice under experimental biotin deprivation, a state that does not occur with an ordinary diet.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides ovatus* (0.45% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Preliminary

Biotin is the one vitamin with a demonstrated mechanistic link to hair growth, though only in germ-free mice under experimental deprivation. In humans with normal biotin status there is no evidence that bacterial biotin synthesis affects hair.

Hayashi et al. fed germ-free mice a biotin-deficient diet, observed alopecia accompanied by Lactobacillus murinus overgrowth, and reversed both with biotin supplementation — establishing that the gut community's biotin economy can affect hair in an extreme experimental setting. That is the entire mechanistic basis for the widely repeated claim linking gut bacteria to hair loss. Human biotin deficiency is rare and produces a distinctive picture (rash, neurological signs) unlike either common form of alopecia, and randomised evidence for biotin supplementation in hair loss with normal biotin status is absent. Magnusdottir et al. found biotin synthesis capacity in a minority of gut genomes, with most species dependent on scavenging.

One of the genes behind this figure (bioF) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Hayashi A. et al., Cell Rep 20:1513-1524 (2017), doi:10.1016/j.celrep.2017.07.057; Magnusdottir S. et al., Front Genet 6:148 (2015)

What the research says about changing this

Biotin synthesis genes — no hair or supplement claim follows LANE A HUMAN EVIDENCE · STRONG

Fired by: biotin gene capacity: high

Biotin deficiency is rare and this reading does not indicate it. High-dose biotin supplements interfere with laboratory tests including troponin.

- Biotin deficiency is rare; high-dose supplements interfere with laboratory assays including troponin (FDA). No hair-loss or alopecia biotin suggestion follows from microbial genes. [NIH ODS — Biotin fact sheet] · lane A · reference
- **Against or limiting —** FDA warns that high-dose biotin can cause falsely low troponin and other assay results. [FDA — biotin interference with troponin lab tests] · lane X · safety communication · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “biotin for hair” · “biotin deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Riboflavin (vitamin B2) ▾

ABOVE AVERAGE

The capacity of your gut bacteria to make riboflavin, which they also use as an electron shuttle between themselves.

YOUR RESULT	**215.7** copies per 100 genomes
PERCENTILE	**76th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	165.9 – 214.4 (median 190.8)
MADE FROM	Built de novo by bacteria from GTP and ribulose-5-phosphate
DNA FRAGMENTS MATCHED	3,259



What is driving this 5,232 matched fragments, by the organism each came closest to

92% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides thetaiotaomicron* (0.57% of your community); *Parabacteroides distasonis* (1.42% of your community); *Bacteroides xylanisolvens* (0.06% of your community). The remaining 8% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**23%** (1195 fragments)	**yes** · 11.748%	Conditional
Bacteroides thetaiotaomicron	**10%** (509 fragments)	**yes** · 0.574%	Conditional
Parabacteroides distasonis	**6%** (296 fragments)	**yes** · 1.420%	Conditional
Actinobacillus pleuropneumoniae	**4%** (202 fragments)	not found; closest reference	—
Bacteroides xylanisolvens	**4%** (196 fragments)	**yes** · 0.065%	Conditional

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bacteroides thetaiotaomicron*, *Parabacteroides distasonis*; *Phocaeicola vulgatus* alone accounts for 23% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides thetaiotaomicron 0.57% of your community · 10% of this signal	Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading. ■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.
Parabacteroides distasonis 1.42% of your community · 6% of this signal	Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans. ■ : P. distasonis alleviated obesity and metabolic dysfunction in mice via succinate and secondary bile acids; human data are associative. *[Wang K et al. Cell Rep 2019;26:222. PMID 30605678]* Association only — no study on record shows that changing this organism changes an outcome.
Bacteroides xylanisolvens 0.06% of your community · 4% of this signal	Xylan-degrading Bacteroides; a common commensal. ■ : Bacteroides as diet-responsive commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means above average · somewhat higher than most people

Your gut community has more riboflavin synthesis capacity than most people's. The best-evidenced consequence is ecological rather than nutritional: riboflavin is the electron shuttle Faecalibacterium prausnitzii uses to survive at the oxygen-exposed mucosal surface.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides thetaiotaomicron* (0.57% of your community); *Parabacteroides distasonis* (1.42% of your community); *Bacteroides xylanisolvens* (0.06% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Riboflavin is one of the few bacterial vitamins with a demonstrated effect on the community itself: Faecalibacterium prausnitzii uses it to transfer electrons, and riboflavin supplementation increases F. prausnitzii abundance in human trials.

Khan et al. showed that F. prausnitzii uses riboflavin as an extracellular electron shuttle, which explains how a strict anaerobe tolerates the oxygen gradient at the mucosal surface. Steinert et al. then supplemented healthy adults with riboflavin and observed an increase in F. prausnitzii and in butyrate, one of the cleaner prebiotic-mechanism demonstrations in the field. Separately, riboflavin biosynthesis intermediates are the antigens that MR1 presents to MAIT cells, making this pathway a direct link between bacterial metabolism and mucosal immunity. What the evidence does not support is treating this number as a nutritional riboflavin status.

Reference: Khan M.T. et al., ISME J 6:1578-1585 (2012), doi:10.1038/ismej.2012.5; Steinert R.E. et al., Gut Microbes 11:1300-1309 (2020), doi:10.1080/19490976.2020.1748208

What the research says about changing this

Riboflavin synthesis genes — the riboflavin-supplement trials measured the microbiome, not you LANE C HUMAN TRIAL · LIMITED

Fired by: riboflavin gene capacity: somewhat high

Riboflavin supplementation was tested as a way to feed F. prausnitzii: two weeks in healthy adults did not significantly raise it at either dose, and three weeks in Crohn's disease did not alter the metagenome. This gene reading is not a riboflavin status.

Studied: Riboflavin (vitamin B2) supplement (vitamin supplement) · **in:** people with clinically confirmed healthy adults, ages 18–65, in Netherlands, n = 105 · **studied protocol (a source fact, not an instruction):** 50 or 100 mg/day riboflavin for two weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- Two weeks of riboflavin (50 or 100 mg/day) in 105 healthy adults did not significantly increase F. prausnitzii at either dose; pooled groups showed higher butyrate — a surrogate result only. [RIBOGUT trial (2022) — riboflavin 50 or 100 mg/day in healthy adults] · lane C · randomized double-blind placebo-controlled · n = 105 · mixed
- **Against or limiting —** Three weeks of riboflavin 100 mg/day in Crohn's disease did not materially alter metagenomic diversity, taxonomy or pathways. [RISE-UP (2020) — riboflavin 100 mg/day in Crohn's disease] · lane X · single-arm · n = 70 · null

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “riboflavin deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Vitamin B6 (pyridoxal phosphate) ▾

ABOVE AVERAGE

Whether your bacteria carry either of the two known routes to vitamin B6.

YOUR RESULT	**69.7** copies per 100 genomes
PERCENTILE	**75th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	54.3 – 69.6 (median 62.2)
MADE FROM	Built de novo by one of two unrelated routes, DXP-dependent or DXP-independent
DNA FRAGMENTS MATCHED	3,795



What is driving this 2,234 matched fragments, by the organism each came closest to

98% of the signal comes from organisms found in you: *Bacteroides thetaiotaomicron* (0.57% of your community); *Collinsella aerofaciens* (1.54% of your community); *Bifidobacterium longum* (1.84% of your community); *Bifidobacterium breve* (1.40% of your community). The remaining 2% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bacteroides thetaiotaomicron	**58%** (1306 fragments)	yes · 0.574%	Conditional
Collinsella aerofaciens	**6%** (132 fragments)	yes · 1.535%	Opportunist
Bifidobacterium longum	**5%** (103 fragments)	yes · 1.838%	Beneficial
Bifidobacterium breve	**1%** (30 fragments)	yes · 1.397%	Beneficial
Robinsoniella peoriensis	**1%** (24 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Bacteroides thetaiotaomicron*, *Collinsella aerofaciens*, *Bifidobacterium longum*; *Bacteroides thetaiotaomicron* alone accounts for 58% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides thetaiotaomicron 0.57% of your community · 58% of this signal	Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading. ■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.
Collinsella aerofaciens 1.54% of your community · 6% of this signal	Common Actinobacterium enriched in rheumatoid arthritis and symptomatic atherosclerosis cohorts and responsive to dietary fibre reduction; common in healthy people too. ■ when higher: Collinsella expanded in rheumatoid arthritis and increased gut permeability and IL-17A in mouse and cell models. *[Chen J et al. Genome Med 2016;8:43. PMID 27102666]* ■ when higher: Collinsella enriched in symptomatic atherosclerosis patients versus controls. *[Karlsson FH et al. Nat Commun 2012;3:1245. PMID 23212374]* Association only — no study on record shows that changing this organism changes an outcome.
Bifidobacterium longum 1.84% of your community · 5% of this signal	Widespread across life stages (subspecies infantis in infants, longum in adults). Endogenous level does not indicate whether any B. longum-containing product would colonise or help. ■ : Review of Bifidobacterium species distribution across the human lifespan. *[O'Callaghan A, van Sinderen D. Front Microbiol 2016;7:925. PMID 27379055]* Association only — no study on record shows that changing this organism changes an outcome.
Bifidobacterium breve 1.40% of your community · 1% of this signal	Infant-associated Bifidobacterium; in adults usually supplement-derived. ■ : Review of Bifidobacterium species across life stages. *[O'Callaghan A, van Sinderen D. Front Microbiol 2016;7:925. PMID 27379055]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means above average · somewhat higher than most people

More of your bacteria carry a complete B6 route than in most samples, by one or other of the two chemistries.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides thetaiotaomicron* (0.57% of your community); *Collinsella aerofaciens* (1.54% of your community); *Bifidobacterium longum* (1.84% of your community); *Bifidobacterium breve* (1.40% of your community). Anything that changes this reading has to

change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with better outcomes · Early evidence

Two unrelated chemistries reach the same vitamin. Requiring both would call almost every community unable to make B6; accepting one isolated subunit would call almost all of them able to.

The DXP-independent route, a PdxS synthase working with its PdxT glutaminase, is the one most bacteria use; the DXP-dependent route through PdxA and PdxJ is found in Escherichia coli and its relatives. They are alternatives, so both are measured and the partner subunits are shown separately. Pyridoxal phosphate is the cofactor for most amino-acid chemistry in the cell, which is why the capacity is common without being universal.

Reference: Rodionov D.A. et al., Front Microbiol 10:1316 (2019), doi:10.3389/fmicb.2019.01316; Fitzpatrick T.B. et al., Biochem J 407:1-13 (2007), doi:10.1042/BJ20070765.

‹ at a glance

Vitamin B5 (pantothenate) ▼

TYPICAL

Whether your bacteria can build pantothenate from both of its branches, rather than only use the vitamin once it exists.

YOUR RESULT	**175.2** copies per 100 genomes
PERCENTILE	**74th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	131.7 – 175.6 (median 154.5)
MADE FROM	Built de novo from pantoate and beta-alanine, joined by pantothenate synthetase
DNA FRAGMENTS MATCHED	4,671



What is driving this 4,081 matched fragments, by the organism each came closest to

80% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Hominisplanchenecus faecis*; *Parabacteroides distasonis* (1.42% of your community). A further 20% matched *Alistipes dispar*, *Blautia wexlerae*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**26%** (1053 fragments)	**yes** · 11.748%	● Conditional
Hominisplanchenecus faecis	**7%** (285 fragments)	**yes**	● Unknown
Alistipes dispar	**5%** (211 fragments)	genus present; closest reference	—
Blautia wexlerae	**4%** (160 fragments)	genus present; closest reference	—
Parabacteroides distasonis	**4%** (149 fragments)	**yes** · 1.420%	● Conditional

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Hominisplanchenecus faecis*, *Parabacteroides distasonis*; *Phocaeicola vulgatus* alone accounts for 26% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Parabacteroides distasonis 1.42% of your community · 4% of this signal	Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans. ■ : P. distasonis alleviated obesity and metabolic dysfunction in mice via succinate and secondary bile acids; human data are associative. *[Wang K et al. Cell Rep 2019;26:222. PMID 30605678]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means typical · in line with most people

Mid-range pantothenate synthesis capacity, which is the usual finding. About half of gut species can build it from both branches and the rest take up what is already available, so a middling value reflects the ordinary division of labour.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Hominisplanchenecus faecis*; *Parabacteroides distasonis* (1.42% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Early evidence

Pantothenate is the precursor of coenzyme A, so the enzymes that consume it are universal and the enzymes that make it are not. Only the making is counted here.

Every living cell needs coenzyme A, so counting the consuming step would turn every organism into a producer; coaA is deliberately absent from this panel for that reason. Magnusdottir and colleagues found pantothenate synthesis in roughly half the gut genomes they reconstructed, with the beta-alanine branch the more commonly missing of the two, which is why that branch is measured on its own as well as in the total.

Reference: Magnusdottir S., Ravcheev D., de Crecy-Lagard V., Thiele I., Front Genet 6:148 (2015), doi:10.3389/fgene.2015.00148.

Protein & Nitrogen Metabolism

‹ at a glance

Branched-chain amino acids (valine, leucine, isoleucine) ▲

HIGH

The capacity of your gut bacteria to synthesise the branched-chain amino acids valine, leucine and isoleucine.

YOUR RESULT	**209.5** copies per 100 genomes
PERCENTILE	**91st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	173.6 – 199.0 (median 188.4)
MADE FROM	Pyruvate and other central carbon intermediates
DNA FRAGMENTS MATCHED	8,929



What is driving this 10,291 matched fragments, by the organism each came closest to

92% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Bifidobacterium longum* (1.84% of your community); *Mediterraneibacter torques*; *Phocaeicola plebeius* (3.75% of your community). A further 8% matched *Faecalicatena contorta*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**13%** (1370 fragments)	**yes** · 11.748%	Conditional
Bifidobacterium longum	**6%** (627 fragments)	**yes** · 1.838%	Beneficial
Mediterraneibacter torques	**4%** (375 fragments)	**yes**	Opportunist
Phocaeicola plebeius	**4%** (362 fragments)	**yes** · 3.747%	Conditional
Faecalicatena contorta	**2%** (237 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bifidobacterium longum*, *Mediterraneibacter torques*; *Phocaeicola vulgatus* alone accounts for 13% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bifidobacterium longum 1.84% of your community · 6% of this signal	Widespread across life stages (subspecies infantis in infants, longum in adults). Endogenous level does not indicate whether any B. longum-containing product would colonise or help. ■ : Review of Bifidobacterium species distribution across the human lifespan. *[O'Callaghan A, van Sinderen D. Front Microbiol 2016;7:925. PMID 27379055]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means high · higher than most people

Your gut community has more branched-chain amino acid synthesis capacity than most people's. In a large European cohort this was the functional signature most strongly associated with insulin resistance, with a supporting mouse experiment. It is one term among many — host muscle and liver BCAA handling matter at least as much — and no intervention targeting it has been tested in humans.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Bifidobacterium longum* (1.84% of your community); *Mediterraneibacter torques*; *Phocaeicola plebeius* (3.75% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with worse outcomes · Reasonably supported

Higher microbial branched-chain amino acid biosynthesis potential was the strongest functional signature of insulin resistance in a large European cohort, driven mainly by Prevotella copri and Bacteroides vulgatus.

Pedersen et al. profiled 277 non-diabetic Danish adults with serum metabolomics and shotgun metagenomics and found that insulin-resistant individuals had higher serum BCAAs and that the gut community's BCAA biosynthesis potential was the strongest associated functional module. They then showed that colonising mice with Prevotella copri — a main driver species — raised serum BCAAs and induced insulin resistance, which turns a correlation into a plausible causal chain. The honest limits: the driver species are diet-dependent and geographically variable, and serum BCAA elevation in insulin resistance has a substantial host-metabolism component that this measurement does not touch.

Reference: Pedersen H.K. et al., Nature 535:376-381 (2016), doi:10.1038/nature18646

What the research says about changing this

Higher microbial BCAA-synthesis capacity — association only LANE E NO SUPPORTED INTERVENTION

Fired by: bcaa gene capacity: high

Microbial BCAA biosynthesis potential correlated with serum BCAA and insulin resistance in one cohort, with causal work in mice. No human trial targets it and protein intake is never restricted from DNA.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- Microbial BCAA biosynthesis potential correlated with serum BCAA and insulin resistance in humans; the causal P. copri experiments were in mice. Protein or BCAA intake is never restricted from DNA. [Pedersen HK et al. Nature 2016;535:376] · lane E · human association + mouse

Conventional route: If metabolic risk is a concern, HbA1c and lipids with a clinician.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “cut protein” · “BCAA toxicity”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Trimethylamine / TMAO ▲

ABOVE AVERAGE

Trimethylamine (TMA), which the liver converts into trimethylamine N-oxide (TMAO).

YOUR RESULT	**4.1** copies per 100 genomes
PERCENTILE	**76th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	1.6 – 3.9 (median 2.4)
MADE FROM	Dietary choline (eggs, liver, soy) and L-carnitine (red meat)
DNA FRAGMENTS MATCHED	205



What is driving this

356 matched fragments, by the organism each came closest to

22% of the signal comes from organisms found in you: *Enterocloster alcoholdehydrogenati*; *Enterocloster citroniae* (0.02% of your community). A further 57% matched *Clostridium scindens*, *Streptococcus canis*, whose genus is present but whose species was not resolved. The remaining 21% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Clostridium scindens	37% (133 fragments)	genus present; closest reference	—
Klebsiella spallanzanii	15% (54 fragments)	not found; closest reference	—
Enterocloster alcoholdehydrogenati	12% (44 fragments)	yes	Opportunist
Streptococcus canis	4% (15 fragments)	genus present; closest reference	—
Enterocloster citroniae	3% (12 fragments)	yes · 0.023%	Opportunist

The organisms carrying this gene in your sample are *Enterocloster alcoholdehydrogenati*, *Enterocloster citroniae*. Changing this reading means changing them.

What your reading means

above average · somewhat higher than most people

More of your gut bacteria can convert choline and carnitine into TMA, which your liver turns into TMAO. This is the metabolite with the most consistent Alzheimer's association in the literature — elevated in the cerebrospinal fluid of people with the disease, and linked to cardiovascular risk. Dietary choline and L-carnitine are the raw material, which is the practical handle if this matters to you.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Enterocloster alcoholdehydrogenati*; *Enterocloster citroniae* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

TMAO is the most consistently reported gut metabolite association in Alzheimer's disease research. It is elevated in the cerebrospinal fluid of people with Alzheimer's, and dietary TMAO accelerates brain ageing and cognitive decline in mice.

Vogt et al. found TMAO elevated in the cerebrospinal fluid of people with Alzheimer's disease and mild cognitive impairment, correlating with biomarkers of tau pathology and neuronal damage. TMAO has also been linked to amyloid-beta accumulation, tau aggregation and synaptic damage through mitochondrial impairment. Separately, higher circulating TMAO is associated with cardiovascular disease and newly diagnosed diabetes, both of which raise dementia risk.

Reference: Vogt N.M. et al., Alzheimers Res Ther 10:124 (2018), doi:10.1186/s13195-018-0451-2; Buawangpong N. et al., J Cell Physiol 237:1661-1685 (2022), doi:10.1002/jcp.30706

What the research says about changing this

Higher TMA-lyase (cutC) capacity — what changes TMAO, and what does not LANE B HUMAN EVIDENCE · STRONG

Fired by: cutc gene capacity: somewhat high

cutC gene capacity is not a plasma TMAO measurement. In a controlled feeding trial, red meat raised plasma TMAO about three-fold and switching protein source reversed it within weeks; probiotics did not lower TMAO in pooled trials, and DMB inhibitors are animal-only.

Studied: Controlled red-meat versus white-meat versus non-meat protein feeding (Wang 2019) (diet) · **in:** people with clinically confirmed healthy adults, ages 21–65, in USA, n = 113 · **studied protocol (a source fact, not an instruction):** three isocaloric four-week diets (red meat, white meat, non-meat protein) in random order

Evidence lane B: replicated human randomized clinical benefit for the exact intervention and population. Replication: replicated. Strongest design: randomized trial.

- Chronic red-meat feeding roughly tripled measured plasma TMAO and the effect reversed after switching protein source. This measured a metabolite, not gene capacity; it does not restrict essential choline. [Wang Z et al. Eur Heart J 2019;40:583] · lane C · randomized crossover controlled feeding · n = 113 · benefit
- The PREDIMED trial (republished after correction) found a Mediterranean diet with olive oil or nuts reduced major cardiovascular events by about 30% in high-risk adults — dietary-pattern evidence independent of the stool result. [Estruch R et al. N Engl J Med 2018;378:e34 (PREDIMED retraction and republication)] · lane B · randomized controlled (republished after correction for randomisation irregularities) · n = 7447 · benefit
- **Against or limiting —** Eight randomized probiotic trials pooled show no reduction in plasma TMAO. [meta-analysis of eight RCTs — probiotics and TMAO (2022)] · lane X · meta-analysis · null
- **Against or limiting —** A very-high-dose probiotic trial in dialysis patients did not lower TMAO. [very-high-dose probiotic trial in dialysis — TMAO null (2018)] · lane X · randomized · null
- **Against or limiting —** Comparator-dependent feeding evidence on dietary TMAO is inconsistent. [2025 review of dietary TMAO feeding studies] · lane X · review · mixed

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have high TMAO” · “avoid choline”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Protein and nitrogen metabolism

NOTABLY HIGH

How much of your community is set up to ferment protein rather than carbohydrate.

YOUR RESULT	**347.3** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	285.3 – 318.6 (median 302.0)
MADE FROM	Dietary and host protein reaching the colon
DNA FRAGMENTS MATCHED	15,670



What is driving this 14,087 matched fragments, by the organism each came closest to

87% of the signal comes from organisms found in you: *Phocaeicola plebeius* (3.75% of your community); *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola coprocola*; *Bacteroides thetaiotaomicron* (0.57% of your community). A further 13% matched *Bacteroides fragilis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola plebeius	**7%** (1031 fragments)	**yes** · 3.747%	Conditional
Phocaeicola vulgatus	**6%** (796 fragments)	**yes** · 11.748%	Conditional
Phocaeicola coprocola	**5%** (682 fragments)	**yes**	Conditional
Bacteroides thetaiotaomicron	**5%** (653 fragments)	**yes** · 0.574%	Conditional
Bacteroides fragilis	**3%** (473 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Phocaeicola plebeius*, *Phocaeicola vulgatus*, *Phocaeicola coprocola*; *Phocaeicola plebeius* alone accounts for 7% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides thetaiotaomicron 0.57% of your community · 5% of this signal	Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading. ■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means notably high · well above most people

More protein-fermentation capacity than most samples. This chemistry yields branched-chain fatty acids, ammonia and phenols rather than butyrate, and it rises when fermentable fibre runs short.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola plebeius* (3.75% of your community); *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola coprocola*; *Bacteroides thetaiotaomicron* (0.57% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it The research picture is mixed · Early evidence

Protein fermentation produces branched-chain fatty acids, ammonia and phenolic compounds, which is a different chemistry from fibre fermentation and generally a less favourable one.

Generic peptidase abundance is not a measurement of dietary protein digestion: every organism carries peptidases for its own protein turnover, so a raw count tracks how many organisms are present. The informative markers are the ones specific to fermenting amino acids for energy - Stickland reductases, branched-chain dehydrogenases - and those are what this counts. Protein breakdown, branched-chain amino acid synthesis and branched-chain amino acid fermentation are three different things, and this report keeps three separate readings for them.

Reference: Diether N.E., Willing B.P., "Microbial fermentation of dietary protein: an important factor in diet-microbe-host interaction", Microorganisms 7:19 (2019), doi:10.3390/microorganisms7010019.

‹ at a glance

Ammonia (from urea) ▲

LOW

A bacterial enzyme that splits urea into ammonia, raising the pH of the gut lumen.

YOUR RESULT	**5.2** copies per 100 genomes
PERCENTILE	**8th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	8.2 – 17.6 (median 11.7)
MADE FROM	Urea, which diffuses into the gut from the bloodstream
DNA FRAGMENTS MATCHED	216



What is driving this

345 matched fragments, by the organism each came closest to

19% of the signal comes from organisms found in you: *Odoribacter splanchnicus* (0.36% of your community); *Bacillus velezensis* (0.26% of your community). A further 81% matched *Blautia difficilis*, *Blautia luti*, *Bifidobacterium oedipodis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia difficilis	**31%** (108 fragments)	genus present; closest reference	—
Blautia luti	**10%** (36 fragments)	genus present; closest reference	—
Odoribacter splanchnicus	**8%** (27 fragments)	**yes** · 0.364%	● Beneficial
Bifidobacterium oedipodis	**7%** (25 fragments)	genus present; closest reference	—
Bacillus velezensis	**4%** (13 fragments)	**yes** · 0.264%	● Unknown

The organisms carrying this gene in your sample are *Odoribacter splanchnicus*, *Bacillus velezensis*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Odoribacter splanchnicus 0.36% of your community · 8% of this signal	Butyrate-producing Bacteroidales member; lower in some IBD and hypertension cohorts, but variable between healthy adults. ■ when lower: Odoribacter was among taxa depleted in hypertension cohorts (meta-analysis of gut microbiome and blood pressure). *[Palmu J et al. J Am Heart Assoc 2020;9:e016641. PMID 32691653]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

low · lower than most people

Less urease capacity than most people, meaning less bacterial breakdown of urea to ammonia in the colon. In liver disease this would be the favourable direction, since colonic ammonia drives hepatic encephalopathy; for anyone else nothing follows from it.

Why this colour. Bacterial urease makes ammonia, the mechanism behind hepatic encephalopathy and the trait that lets Proteus form kidney stones and H. pylori survive acid.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Odoribacter splanchnicus* (0.36% of your community); *Bacillus velezensis* (0.26% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

Ammonia from bacterial urease is the mechanism behind hepatic encephalopathy in liver disease and rises with kidney failure. In healthy people, urease capacity is common and carries no known implication.

Wong et al. showed that end-stage renal disease expands urease- and uricase-containing families while contracting short-chain fatty acid producers, which is the clearest human evidence for the pH mechanism. Ni et al. found that a urease-positive community was sufficient to drive dysbiosis and worsen colitis in mice, and that replacing it with a urease-negative E. coli reversed the effect. In liver disease, the whole therapeutic strategy for encephalopathy — lactulose to acidify the colon, rifaximin to reduce the ammonia producers — is aimed at this enzyme's product. None of that makes urease capacity meaningful in a person with normal liver and kidney function.

Reference: Wong J. et al., Am J Nephrol 39:230-237 (2014); Ni J. et al., Sci Transl Med 9:eaah6888 (2017), doi:10.1126/scitranslmed.aah6888

‹ at a glance

Indole-3-propionate (IPA) ▼

HIGH

Indole-3-propionate (IPA), one of the more strongly neuroprotective gut metabolites described so far.

YOUR RESULT	**9.6** copies per 100 genomes
PERCENTILE	**86th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	3.1 – 6.3 (median 4.6)
MADE FROM	Dietary tryptophan, via a reductive pathway in a narrow group of bacteria
DNA FRAGMENTS MATCHED	234



What is driving this 385 matched fragments, by the organism each came closest to

1% of the signal comes from organisms found in you: *Blautia obeum* (0.02% of your community). A further 99% matched *Blautia luti*, *Blautia wexlerae*, *Anaerotignum propionicum*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia luti	**68%** (261 fragments)	genus present; closest reference	—
Blautia wexlerae	**23%** (90 fragments)	genus present; closest reference	—
Anaerotignum propionicum	**3%** (11 fragments)	genus present; closest reference	—
Anaerotignum neopropionicum	**2%** (8 fragments)	genus present; closest reference	—
Blautia obeum	**1%** (4 fragments)	**yes** · 0.022%	● Conditional

The organisms carrying this gene in your sample are *Blautia obeum*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Blautia obeum 0.02% of your community · 1% of this signal	Abundant Lachnospiraceae member; Blautia is higher in some obesity cohorts and lower in others. No fixed reading. ■ : Review of Blautia as a widespread genus with mixed disease associations. *[Liu X et al. Gut Microbes 2021;13:1875796. PMID 33525961]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means high · higher than most people

Your gut has more indole-3-propionate capacity than most people, which is the favourable direction. IPA crosses into the brain, where it neutralises hydroxyl radicals, reduces DNA damage, calms inflammatory activity in astrocytes and microglia, and interferes with the formation of amyloid-beta fibrils. It is reduced in people with obesity and diabetes, so having plenty of capacity is a good sign.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia obeum* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Reasonably supported

IPA crosses into the brain, where it neutralises hydroxyl radicals, reduces DNA damage, calms inflammatory activity in astrocytes and microglia, and interferes with the formation of amyloid-beta fibrils. It is reduced in people with obesity and diabetes.

After absorption from the intestine, IPA transits to the brain where it neutralises hydroxyl radicals, decreases DNA damage, alleviates neuroinflammation by inhibiting astrocytes and microglia, and hinders the formation of amyloid-beta fibrils. Clinical studies show significantly reduced indole metabolites, including IPA, in people who are obese or diabetic, tracking with gut dysbiosis.

Reference: Pappolla M.A. et al., Neurobiol Dis 156:105403 (2021), doi:10.1016/j.nbd.2021.105403; Dodd D. et al., Nature 551:648-652 (2017), doi:10.1038/nature24661

What the research says about changing this

Tryptophan-metabolite capacity (indole, IPA, tryptamine) — no supported targeted intervention NONE NO SUPPORTED INTERVENTION

Fired by: indole gene capacity: typical; ipa gene capacity: high; tryptamine gene capacity: notably high

Microbial tryptophan metabolism is an active research area with associations to gut barrier and immune signalling, but no human trial shows that changing these gene readings changes a health outcome. Tryptophan intake is never adjusted from stool DNA.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “serotonin level” · “tryptophan deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Peptide uptake

ABOVE AVERAGE

How much capacity your gut bacteria have to import peptides whole, rather than breaking them down outside the cell first.

YOUR RESULT	**140.6** copies per 100 genomes
PERCENTILE	**77th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	106.4 – 137.1 (median 123.0)
MADE FROM	peptides from dietary and host protein
DNA FRAGMENTS MATCHED	17,241



What is driving this

6,326 matched fragments, by the organism each came closest to

13% of the signal comes from organisms found in you: *Bacteroides stercoris* (1.52% of your community). A further 76% matched *Bacteroides reticulotermitis*, *Blautia luti*, *Bifidobacterium scardovii*, whose genus is present but whose species was not resolved. The remaining 12% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bacteroides reticulotermitis	**13%** (802 fragments)	genus present; closest reference	—
Blautia luti	**12%** (776 fragments)	genus present; closest reference	—
Bacteroides stercoris	**5%** (314 fragments)	**yes** · 1.520%	● Conditional
Bifidobacterium scardovii	**5%** (299 fragments)	genus present; closest reference	—
Robinsoniella peoriensis	**5%** (293 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Bacteroides stercoris*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides stercoris 1.52% of your community · 5% of this signal	Common Bacteroides species with no fixed directional interpretation. ■ : Bacteroides as commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

above average · somewhat higher than most people

More peptide-uptake capacity than most samples carry. That can mean peptides are being used for growth rather than fermented, so it does not point in a single direction on its own.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides stercoris* (1.52% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

Taking peptides in is a different capability from cutting proteins up. This is the uptake side, and it is a gene count rather than a rate.

Peptide transport is how most gut bacteria actually acquire amino acids: importing a short peptide costs less than making each residue, and the transporters are among the most highly expressed genes in gut isolates. What this cannot tell you is how much dietary protein reaches your colon. That is set by how much you eat and how well your small intestine digests it, and a transporter count in stool speaks to neither. Nor does high uptake capacity mean more protein fermentation: the same peptides can be built into bacterial protein instead, which is the opposite outcome.

Reference: Detmers F.J., Lanfermeijer F.C., Poolman B., "Peptides and ATP binding cassette peptide transporters", Res Microbiol 152:245-258 (2001), doi:10.1016/s0923-2508(01)01196-2.

‹ at a glance

p-Cresol / p-cresyl sulfate ▲

TYPICAL

p-Cresol, which the host converts to p-cresyl sulfate, a compound that accumulates when kidney function declines.

YOUR RESULT	**8.6** copies per 100 genomes
PERCENTILE	**67th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	4.9 – 9.5 (median 6.6)
MADE FROM	Dietary tyrosine and phenylalanine
DNA FRAGMENTS MATCHED	445



What is driving this

786 matched fragments, by the organism each came closest to

A further 99% matched *Blautia wexlerae*, *Acetatifactor muris*, *Blautia hydrogenotrophica*, whose genus is present but whose species was not resolved. The remaining 1% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia wexlerae	**74%** (580 fragments)	genus present; closest reference	—
Acetatifactor muris	**8%** (66 fragments)	genus present; closest reference	—
Blautia hydrogenotrophica	**8%** (63 fragments)	genus present; closest reference	—
Clostridium scatologenes	**2%** (17 fragments)	genus present; closest reference	—
Caproiciproducens galactitolivorans	**1%** (9 fragments)	not found; closest reference	—

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

What your reading means

typical · in line with most people

Your capacity to make p-cresol is unremarkable. This compound is cleared by the kidneys, so it matters most when kidney function is reduced; at typical capacity and normal kidney function there is little here to note.

How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Early evidence

p-Cresol interferes with mitochondrial energy metabolism and increases production of damaging superoxide. It is enriched in people with diabetes and obesity, and in animals it disrupts brain signalling.

p-Cresol modifies mitochondrial oxidative metabolism and increases superoxide production. In mice, it induces autism-like social behaviour by reducing the function of central dopamine neurons, and it alters NMDA receptor subunit expression in the hippocampus and nucleus accumbens. No direct study links p-cresol to Alzheimer's disease in humans, so the connection rests on shared mechanisms rather than on evidence.

Reference: Bermudez-Martin P. et al., Microbiome 9:157 (2021), doi:10.1186/s40168-021-01103-z; Fromentin S. et al., Nat Med 28:303-314 (2022), doi:10.1038/s41591-022-01688-4

‹ at a glance

Imidazole propionate (ImP) ▲

TYPICAL

Imidazole propionate (ImP), a small molecule made by gut bacteria from the amino acid histidine.

YOUR RESULT	**33.1** copies per 100 genomes
PERCENTILE	**40th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	26.2 – 47.8 (median 38.0)
MADE FROM	Dietary histidine, via the intermediate urocanate
DNA FRAGMENTS MATCHED	1,285



What is driving this

2,095 matched fragments, by the organism each came closest to

86% of the signal comes from organisms found in you: *Lactiplantibacillus plantarum* (1.14% of your community); *Sutterella wadsworthensis* (0.64% of your community); *Holdemania massiliensis* (0.04% of your community); *Parasutterella excrementihominis*. A further 14% matched *Lactiplantibacillus argentoratensis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Lactiplantibacillus plantarum	**14%** (304 fragments)	**yes** · 1.142%	Beneficial
Sutterella wadsworthensis	**12%** (255 fragments)	**yes** · 0.643%	Conditional
Holdemania massiliensis	**6%** (128 fragments)	**yes** · 0.039%	Conditional
Lactiplantibacillus argentoratensis	**6%** (125 fragments)	genus present; closest reference	—
Parasutterella excrementihominis	**5%** (109 fragments)	**yes**	Conditional

The organisms carrying this gene in your sample are *Lactiplantibacillus plantarum*, *Sutterella wadsworthensis*, *Holdemania massiliensis*; *Lactiplantibacillus plantarum* alone accounts for 14% of the signal. Changing this reading means changing them.

What your reading means

typical · in line with most people

Nothing here points toward the insulin-signalling effects associated with high imidazole propionate. Bear in mind that carrying this gene is only weakly related to how much ImP actually reaches the blood, so even a high reading would be indirect evidence at best.

Two measurements of the same function disagree

Read this beside the polyphenol-metabolising species reading. The urolithin and equol capacity reading sits at the 40th percentile; the polyphenol-metabolising species reading is at the 80th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Of the two, the species reading is the better guide here. Urolithin and equol production is strain-level: two isolates of the same species differ, and only a handful of characterised strains do it. The curated species are the better guide; a high gene reading without them is most likely a related enzyme.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Lactiplantibacillus plantarum* (1.14% of your community); *Sutterella wadsworthensis* (0.64% of your community); *Holdemania massiliensis* (0.04% of your community); *Parasutterella excrementihominis*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

ImP interferes with insulin signalling in host cells. Higher blood ImP is found in people with type 2 diabetes and prediabetes, and insulin resistance is itself an established risk factor for cognitive decline.

The link to cognition is indirect but mechanistically coherent: ImP activates p38-gamma and mTORC1, degrading insulin receptor substrate signalling. Impaired brain insulin signalling is a recognised feature of Alzheimer's disease. No study has yet shown that ImP itself causes cognitive decline in humans. Note also that in the source study of 294 stool samples, carrying the urdA gene was only weakly related to actual blood ImP levels. One overlap is worth flagging: Eggerthella, a documented urdA carrier, is among the pro-inflammatory genera found enriched in carriers of the APOE E4 Alzheimer's risk allele — so this pathway and that risk factor share an organism, though no causal link between them has been shown.

A stricter version of this measurement, which additionally requires the matched gene to carry the correct active-site amino acid, gives 12.2 copies per 100 genomes. The true value lies between the two.

Reference: Koh A. et al., Cell 175:947-961 (2018), doi:10.1016/j.cell.2018.09.055; Molinaro A. et al., Nat Commun 11:5881 (2020), doi:10.1038/s41467-020-19589-w; Sepulveda-Rivera V. et al., Sci Rep 15:28505 (2025), doi:10.1038/s41598-025-13262-2

What the research says about changing this

Polyphenol-conversion capacity (urolithin/equol) — status is defined by a challenge, not by genes LANE C HUMAN TRIAL · LIMITED

Fired by: urda gene capacity: typical

Whether someone converts ellagitannins to urolithins or daidzein to equol is determined by a substrate challenge with urine measurement. Direct urolithin A supplementation missed its primary endpoints and improved a secondary muscle measure in older adults; it bypasses the microbes rather than restoring them.

Studied: Urolithin A (Mitopure) (botanical supplement) · **in:** people with clinically confirmed older adults, ages 65–90, in USA, n = 66 · **studied protocol (a source fact, not an instruction):** 1,000 mg/day urolithin A for four months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- Direct urolithin A (1,000 mg/day for four months) in adults 65–90 improved a secondary muscle-endurance measure but missed its primary walking and ATP endpoints; it bypasses rather than restores microbes. [Liu S et al. JAMA Netw Open 2022;5:e2144279] · lane C · randomized double-blind placebo-controlled · n = 66 · mixed
- Equol-producer status is defined by a soy challenge (2 × 250 mL soy milk/day for three days) and urinary S-equol:daidzein, not by the presence of a species. [Setchell KDR, Cole SJ. J Nutr 2006;136:2188] · lane D · substrate challenge

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “non-producer” · “take urolithin”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Indole / indoxyl sulfate ▾

TYPICAL

Indole, which the host converts to indoxyl sulfate — a compound that builds up when kidney function is impaired.

YOUR RESULT	**11.6** copies per 100 genomes
PERCENTILE	**41st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	8.3 – 20.8 (median 13.3)
MADE FROM	Dietary tryptophan
DNA FRAGMENTS MATCHED	327



What is driving this 587 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Bacteroides uniformis* (3.11% of your community); *Alistipes onderdonkii* (1.22% of your community); *Akkermansia muciniphila* (0.96% of your community); *Bacteroides ovatus* (0.45% of your community).

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bacteroides uniformis	**29%** (169 fragments)	**yes** · 3.108%	Conditional
Alistipes onderdonkii	**9%** (52 fragments)	**yes** · 1.215%	Conditional
Akkermansia muciniphila	**7%** (40 fragments)	**yes** · 0.961%	Beneficial
Bacteroides ovatus	**6%** (37 fragments)	**yes** · 0.451%	Conditional
Alistipes finegoldii	**6%** (35 fragments)	**yes**	Conditional

The organisms carrying this gene in your sample are *Bacteroides uniformis*, *Alistipes onderdonkii*, *Akkermansia muciniphila*; *Bacteroides uniformis* alone accounts for 29% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides uniformis 3.11% of your community · 29% of this signal	Among the most prevalent Bacteroides in Western adults; level varies with diet and has no established directional meaning on its own. ■ : Bacteroides species are dominant, diet-responsive members of Western adult stool with both commensal and opportunistic behaviours. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.
Alistipes onderdonkii 1.22% of your community · 9% of this signal	Common Alistipes species; lower in a small two-year Long-COVID cohort (unreplicated) and variable elsewhere. ■ : Review of Alistipes with both protective and disease associations. *[Parker BJ et al. Front Immunol 2020;11:906. PMID 32582143]* Association only — no study on record shows that changing this organism changes an outcome.
Akkermansia muciniphila 0.96% of your community · 7% of this signal	Mucin-degrading Verrucomicrobium. Higher levels are associated with metabolic health in several cohorts and a pasteurised strain (MucT) has human trial evidence, yet the same species is enriched in multiple sclerosis and Parkinson's disease cohorts. Endogenous abundance does not predict response to any Akkermansia product. ■ when higher: Pasteurised A. muciniphila MucT improved insulin sensitivity and metabolic markers in an exploratory 3-month trial of overweight/obese insulin-resistant volunteers. *[Depommier C et al. Nat Med 2019;25:1096. PMID 31263284]* ■ when higher: Akkermansia is enriched in multiple sclerosis patients across cohorts (iMSMS). *[iMSMS Consortium. Cell 2022;185:3467. PMID 36113426]*
Bacteroides ovatus 0.45% of your community · 6% of this signal	Versatile polysaccharide degrader; common commensal with no fixed directional interpretation. ■ : Bacteroides as diet-responsive commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.
Alistipes finegoldii	Alistipes species with mixed associations; protective in mouse colitis, higher in some human disease cohorts. ■ : Review of Alistipes across cohorts. *[Parker BJ et al. Front Immunol 2020;11:906. PMID 32582143]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means typical · in line with most people

Tryptophan can be routed down several different paths and this test cannot tell which one dominates, so read this number together with indole-3-propionate below. Your capacity for the indole route itself is unremarkable.

Why this colour. Indole from tryptophan tightens the gut barrier and signals through the AhR receptor; the exception is advanced kidney disease, where its metabolite indoxyl sulfate accumulates.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides uniformis* (3.11% of your community); *Alistipes onderdonkii* (1.22% of your community); *Akkermansia muciniphila* (0.96% of your community); *Bacteroides ovatus* (0.45% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with better outcomes · Early evidence

Tryptophan can be routed down several paths, and they do not all point the same way. Indole itself has antioxidant properties, but its host-derived product indoxyl sulfate is a uraemic toxin. Read this panel alongside indole-3-propionate, which is the clearly beneficial branch.

Microbial tryptophan metabolites influence neuroinflammation and the gut-brain axis, and people with obesity or diabetes show reduced levels of the beneficial indole derivatives. Whether high tryptophanase capacity is good or bad depends on which downstream products dominate, which this test cannot determine. Indoxyl sulfate accumulation is primarily a concern in reduced kidney function.

Reference: Roager H.M., Licht T.R., Nat Commun 9:3294 (2018), doi:10.1038/s41467-018-05470-4

What the research says about changing this

Tryptophan-metabolite capacity (indole, IPA, tryptamine) — no supported targeted intervention NONE NO SUPPORTED INTERVENTION

Fired by: indole gene capacity: typical; ipa gene capacity: high; tryptamine gene capacity: notably high

Microbial tryptophan metabolism is an active research area with associations to gut barrier and immune signalling, but no human trial shows that changing these gene readings changes a health outcome. Tryptophan intake is never adjusted from stool DNA.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “serotonin level” · “tryptophan deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

Gut-Brain & Metabolic Signalling

‹ at a glance

Tryptamine

NOTABLY HIGH

Tryptamine, a molecule made by a small number of gut bacteria from tryptophan, which speeds intestinal transit through a serotonin receptor.

YOUR RESULT	**12.4** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 0.5 (median 0.1)
MADE FROM	Dietary tryptophan (protein)
DNA FRAGMENTS MATCHED	342



What is driving this

613 matched fragments, by the organism each came closest to

98% of the signal comes from organisms found in you: *Mediterraneibacter gnavus* (6.93% of your community). A further 2% matched *Lactobacillus hominis*, *Lactobacillus intestinalis*, whose genus is present but whose species was not resolved. The remaining 0% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Mediterraneibacter gnavus	**98%** (602 fragments)	**yes** · 6.934%	Opportunist
Lactobacillus hominis	**1%** (7 fragments)	genus present; closest reference	—
Lactobacillus intestinalis	**0%** (3 fragments)	genus present; closest reference	—
Vagococcus luciliae	**0%** (1 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Mediterraneibacter gnavus*; *Mediterraneibacter gnavus* alone accounts for 98% of the signal. Changing this reading means changing them.

What your reading means

notably high · well above most people

You carry more tryptamine-producing capacity than most people, which is unusual — the enzyme is rare. Tryptamine accelerates colonic transit through a gut serotonin receptor. Worth noting alongside your transit pattern and which organism is supplying it.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Mediterraneibacter gnavus* (6.93% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

Tryptamine activates the intestinal 5-HT4 receptor and accelerates colonic transit in mice. The enzyme is carried by very few gut species, so most people have little or none.

Williams et al. found bacterial tryptophan decarboxylase in only two of more than a thousand sequenced human gut bacterial genomes — Clostridium sporogenes and Ruminococcus gnavus — while detecting the gene in about 10% of a human metagenomic cohort. Bhattarai et al. then showed that colonising germ-free mice with a tryptamine-producing strain accelerated colonic transit through the epithelial 5-HT4 receptor. This is a clean, small mechanism. What it does not support is any claim about mood: tryptamine made in the colon acts locally on gut receptors, and the separate literature on microbial effects on host serotonin runs through enterochromaffin cells, not through bacterial synthesis.

Reference: Williams B.B. et al., Cell Host Microbe 16:495-503 (2014), doi:10.1016/j.chom.2014.09.001; Bhattarai Y. et al., Cell Host Microbe 23:775-785 (2018), doi:10.1016/j.chom.2018.05.004

What the research says about changing this

Tryptophan-metabolite capacity (indole, IPA, tryptamine) — no supported targeted intervention NONE NO SUPPORTED INTERVENTION

Fired by: indole gene capacity: typical; ipa gene capacity: high; tryptamine gene capacity: notably high

Microbial tryptophan metabolism is an active research area with associations to gut barrier and immune signalling, but no human trial shows that changing these gene readings changes a health outcome. Tryptophan intake is never adjusted from stool DNA.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “serotonin level” · “tryptophan deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

GABA breakdown

HIGH

Your gut bacteria's capacity to break GABA down, shown beside their capacity to make it.

YOUR RESULT	**22.3** copies per 100 genomes
PERCENTILE	**92nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	8.2 – 12.3 (median 9.5)
MADE FROM	GABA made by other bacteria or arriving in fermented food
DNA FRAGMENTS MATCHED	1,082



What is driving this

474 matched fragments, by the organism each came closest to

11% of the signal comes from organisms found in you: *Ligilactobacillus salivarius* (0.11% of your community). A further 10% matched *Bacillus subtilis*, whose genus is present but whose species was not resolved. The remaining 78% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Mycobacterium ostraviense	**16%** (75 fragments)	not found; closest reference	—
Brenneria roseae	**8%** (39 fragments)	not found; closest reference	—
Mycobacterium aquaticum	**4%** (21 fragments)	not found; closest reference	—
Ligilactobacillus salivarius	**4%** (19 fragments)	**yes** · 0.114%	Beneficial
Bacillus subtilis	**4%** (18 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Ligilactobacillus salivarius*. Changing this reading means changing them.

What your reading means

high · detected below the confirmed identity threshold; read as indicative only

More GABA-degrading capacity than most samples. Shown beside your GABA production reading, which is a separate measurement.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Ligilactobacillus salivarius* (0.11% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

The existing GABA panel measures production. This measures degradation. They are separate readings and are never subtracted from each other.

A community that both makes and breaks down GABA is not a community with no GABA, which is why these are reported side by side rather than netted. Neither number is a GABA concentration, and neither says anything about GABA reaching your brain: the blood-brain barrier is not permeable to it, so any gut-brain effect runs through other routes such as the vagus nerve or the enteric nervous system.

This is a provisional result rather than a confirmed one — matches average 70% identity (at least 75% required), so the reads are matching distant relatives rather than the gene itself. Read it as a possible signal.

Reference: Strandwitz P. et al., "GABA-modulating bacteria of the human gut microbiota", Nat Microbiol 4:396-403 (2019), doi:10.1038/s41564-018-0307-3.

‹ at a glance

gamma-aminobutyric acid (GABA) ▾

ABOVE AVERAGE

GABA, the main inhibitory neurotransmitter in the brain, which many gut bacteria also make from glutamate.

YOUR RESULT	**33.5** copies per 100 genomes
PERCENTILE	**85th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	12.4 – 26.6 (median 19.9)
MADE FROM	Dietary and microbial glutamate
DNA FRAGMENTS MATCHED	1,082



What is driving this

1,736 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides clarus*; *Bacteroides stercoris* (1.52% of your community); *Barnesiella intestinihominis* (1.16% of your community).

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**40%** (695 fragments)	**yes** · 11.748%	Conditional
Bacteroides clarus	**12%** (214 fragments)	**yes**	Conditional
Bacteroides stercoris	**4%** (64 fragments)	**yes** · 1.520%	Conditional
Barnesiella intestinihominis	**4%** (61 fragments)	**yes** · 1.165%	Beneficial
Parabacteroides distasonis	**4%** (61 fragments)	**yes** · 1.420%	Conditional

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bacteroides clarus*, *Bacteroides stercoris*; *Phocaeicola vulgatus* alone accounts for 40% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides stercoris 1.52% of your community · 4% of this signal	Common Bacteroides species with no fixed directional interpretation. ■ : Bacteroides as commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.
Barnesiella intestinihominis 1.16% of your community · 4% of this signal	Common Bacteroidales member; listed among mucin utilisers. Context-dependent, with no fixed directional reading. ■ : Barnesiella associated with resistance to vancomycin-resistant Enterococcus colonisation in mice and mixed human associations. *[Ubeda C et al. Infect Immun 2013;81:965. PMID 23319552]* Association only — no study on record shows that changing this organism changes an outcome.
Parabacteroides distasonis 1.42% of your community · 4% of this signal	Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans. ■ : P. distasonis alleviated obesity and metabolic dysfunction in mice via succinate and secondary bile acids; human data are associative. *[Wang K et al. Cell Rep 2019;26:222. PMID 30605678]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

above average · somewhat higher than most people

Your gut community has more GABA-producing capacity than most people's. What that means for you is genuinely unknown: gut GABA crosses into the brain poorly, and the proposed routes run through nerves rather than blood. Read this as a feature of your community rather than a statement about mood.

Why this colour. Bacterial GABA is read as a gut-brain benefit in the evidence that exists, which is early.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Bacteroides clarus*; *Bacteroides stercoris* (1.52% of your community); *Barnesiella intestinihominis* (1.16% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with better outcomes · Preliminary

Certain Bacteroides and Bifidobacterium species produce GABA in the gut, and one study found that GABA-producing capacity related inversely to a brain signature of depression. Whether gut GABA reaches the brain in meaningful amounts is unresolved, so this is reported as an interesting capacity, not a mood measurement.

Strandwitz et al. isolated a GABA-dependent Bacteroides from human stool, showed that Bacteroides species are major gut GABA producers, and found that relative Bacteroides abundance related inversely to depression- associated brain signatures on functional MRI in a small cohort. That is a correlation in a small sample on an indirect outcome. The mechanistic gap is real: GABA crosses the blood-brain barrier poorly, and the routes proposed for gut-derived GABA run through the vagus nerve and enteric nervous system rather than the circulation. Treat this as a genuine community feature with an unresolved connection to anything you feel.

Reference: Strandwitz P. et al., Nat Microbiol 4:396-403 (2019), doi:10.1038/s41564-018-0307-3; Otaru N. et al., Front Microbiol 12:656895 (2021)

What the research says about changing this

Neurotransmitter-related capacity (GABA, tyrosine decarboxylase) — a medication fact, not a mood reading LANE D NO SUPPORTED INTERVENTION

Fired by: dopamine gene capacity: typical; gaba gene capacity: somewhat high

Bacterial GABA and tyrosine-decarboxylase genes do not measure brain neurotransmitters. The one clinically relevant fact is that tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — information for a prescriber, not an action.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- Bacterial tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — a medication-context fact for the prescriber; there is no approved microbiome inhibitor or eradication protocol. [van Kessel SP et al. Nat Commun 2019:10:310] · lane D · human association + in vitro/rat

Conventional route: If taking levodopa: information card to the prescriber; no dose change from this report.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “your dopamine is low” · “GABA deficiency” · “mood bacteria”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

P9 protein (experimental)

ABOVE AVERAGE

Whether the reads carry the exact Akkermansia protein that was tested in the GLP-1 experiments, as opposed to something in the same family.

YOUR RESULT	**1.2** copies per 100 genomes
PERCENTILE	**77th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 0.9 (median 0.2)
MADE FROM	not a metabolite; this is a bacterial protein-coding gene
DNA FRAGMENTS MATCHED	55



What is driving this

103 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Akkermansia muciniphila* (0.96% of your community).

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Akkermansia muciniphila	**100%** (103 fragments)	**yes** · 0.961%	● Beneficial

The organisms carrying this gene in your sample are *Akkermansia muciniphila*; *Akkermansia muciniphila* alone accounts for 100% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Akkermansia muciniphila 0.96% of your community · 100% of this signal	Mucin-degrading Verrucomicrobium. Higher levels are associated with metabolic health in several cohorts and a pasteurised strain (MucT) has human trial evidence, yet the same species is enriched in multiple sclerosis and Parkinson's disease cohorts. Endogenous abundance does not predict response to any Akkermansia product. ■ when higher: Pasteurised A. muciniphila MucT improved insulin sensitivity and metabolic markers in an exploratory 3-month trial of overweight/obese insulin-resistant volunteers. *[Depommier C et al. Nat Med 2019;25:1096. PMID 31263284]* ■ when higher: Akkermansia is enriched in multiple sclerosis patients across cohorts (iMSMS). *[iMSMS Consortium. Cell 2022;185:3467. PMID 36113426]*

What your reading means

above average · somewhat higher than most people

Reads map to the experimentally tested sequence. That is a gene being present, not a protein being made or secreted, and it says nothing about your own GLP-1 or about GLP-1 medication.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Akkermansia muciniphila* (0.96% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Preliminary

A single protein with preclinical evidence. Finding its gene is a long way from a hormone response, and this is not comparable to a GLP-1 medication.

The protein was purified and tested in cell and mouse work, where it increased GLP-1 secretion through a host receptor. That is experimental evidence about a protein, not about a person: what this screen can say is whether reads map to the tested sequence at high identity. Detection is deliberately strict. One reference sequence means a diverged version in another strain will not reach the identity floor, and is reported as unresolved rather than counted, because transferring a functional annotation onto a divergent homolog is exactly the error the specification forbids here. Neither secretion nor any GLP-1 level is measured by these reads.

Reference: Yoon H.S. et al., "Akkermansia muciniphila secretes a protein P9 that improves glucose homeostasis", Nat Microbiol 6:563-573 (2021), doi:10.1038/s41564-021-00880-5.

‹ at a glance

Histamine ▲

ABOVE AVERAGE

Histamine, made by some gut bacteria from the amino acid histidine.

YOUR RESULT	**0.5** copies per 100 genomes
PERCENTILE	**77th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 0.5 (median 0.2)
MADE FROM	Dietary histidine; also present directly in aged and fermented foods
DNA FRAGMENTS MATCHED	4,196



What is driving this

20 matched fragments, by the organism each came closest to

95% of the signal comes from organisms found in you: *Eggerthella lenta* (0.10% of your community); *Jilunia laotingensis* (0.14% of your community); *Gordonibacter pamelaeae* (0.05% of your community); *Gordonibacter urolithinfaciens*. A further 5% matched *Eggerthella sinensis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Eggerthella lenta	**50%** (10 fragments)	**yes** · 0.104%	Opportunist
Jilunia laotingensis	**30%** (6 fragments)	**yes** · 0.136%	Conditional
Gordonibacter pamelaeae	**10%** (2 fragments)	**yes** · 0.051%	Beneficial
Gordonibacter urolithinfaciens	**5%** (1 fragments)	**yes**	Beneficial
Eggerthella sinensis	**5%** (1 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Eggerthella lenta*, *Jilunia laotingensis*, *Gordonibacter pamelaeae*; *Eggerthella lenta* alone accounts for 50% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Eggerthella lenta 0.10% of your community · 50% of this signal	Actinobacterium notable for drug metabolism (inactivates digoxin via cgr genes; decarboxylates levodopa-derived dopamine). Higher in some IBD/RA cohorts. Relevant to medication history, not a target. ■ : E. lenta strains carrying the cgr operon inactivate digoxin; arginine suppresses this in mice. *[Haiser HJ et al. Science 2013;341:295. PMID 23869020]* ■ : E. lenta dehydroxylates dopamine produced from levodopa by Enterococcus faecalis tyrosine decarboxylase. *[Maini Rekdal V et al. Science 2019;364:eaau6323. PMID 31196984]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

above average · too few reads for a stable reading; Poisson noise dominates at this count

Your gut carries more histamine-producing capacity than most people's. The table above names the organisms carrying the gene in your sample, and that is what decides whether it matters: histamine from Klebsiella, Morganella and their relatives has been linked to visceral pain in an IBS subgroup and to histamine-intolerance symptoms, while histamine from Lactobacillus reuteri is anti-inflammatory. When the producers found are not the anti-inflammatory exception, a high reading is worth taking seriously alongside any symptoms of flushing, headache, hives or gut pain after fermented or aged foods.

Why this colour. High bacterial histamine capacity is the reading people worry about: histamine intolerance, IBS pain and mast-cell symptoms. Coloured as adverse unless the producers found are the anti-inflammatory exception.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Eggerthella lenta* (0.10% of your community); *Jilunia laotingensis* (0.14% of your community); *Gordonibacter pamelaeae* (0.05% of your community); *Gordonibacter urolithinfaciens*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Early evidence

A subset of gut bacteria can produce histamine. One well-conducted study linked Klebsiella aerogenes histamine production to visceral pain in a subgroup of IBS patients, while Lactobacillus reuteri histamine has the opposite, anti-inflammatory effect. Direction depends on the producer.

De Palma et al. identified Klebsiella aerogenes as a histamine super-producer in a subset of IBS patients, showed it drove mast cell accumulation and visceral hypersensitivity when transferred to mice, and found that a low-fermentable-carbohydrate diet reduced both. That is a strong, mechanism-backed finding in a defined subgroup. Against it, Thomas et al. showed Lactobacillus reuteri histamine signalling through the H2 receptor suppresses TNF production — the same molecule from a different organism acting anti-inflammatorily. There is no evidence that total bacterial histidine decarboxylase capacity, which is what this panel measures, predicts anything on its own.

Based on 4196 matching DNA fragments, few enough that the exact figure carries real uncertainty; the broad position is more reliable than the precise number.

Reference: De Palma G. et al., Sci Transl Med 14:eabj1895 (2022), doi:10.1126/scitranslmed.abj1895; Thomas C.M. et al., PLoS One 7:e31951 (2012), doi:10.1371/journal.pone.0031951

What the research says about changing this

Higher histidine-decarboxylase capacity — the only trial is a diet trial in IBS LANE C HUMAN TRIAL · LIMITED

Fired by: histamine gene capacity: somewhat high

Stool hdc genes are not a histamine measurement. In diagnosed IBS, a three-week randomized low-FODMAP diet lowered urinary histamine and improved symptoms; the causal Klebsiella-histamine work was in mice.

Studied: Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation) (diet) · **in:** people with clinically confirmed IBS, ages 18–70, in Canada, n = 37 · **studied protocol (a source fact, not an instruction):** three-week low-FODMAP versus high-FODMAP diet

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- A three-week randomized low-FODMAP diet in diagnosed IBS improved symptoms and lowered urinary histamine — a diet effect in IBS, not a treatment for stool histamine genes. [McIntosh K et al. Gut 2017;66:1241] · lane C · randomized controlled · n = 40 · benefit
- A high-urinary-histamine IBS subgroup and a K. aerogenes hdc variant were identified, with the causal treatment work done in mice. [De Palma G et al. Sci Transl Med 2022;14:eabj1895] · lane E · human subgroup + gnotobiotic mouse experiments

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “histamine intolerance” · “low-histamine diet for everyone”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Dopamine (from L-DOPA) and tyramine (from tyrosine)

TYPICAL

A bacterial enzyme that decarboxylates tyrosine to tyramine, and L-DOPA to dopamine, in the gut.

YOUR RESULT	**1.6** copies per 100 genomes
PERCENTILE	**62nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0.6 – 2.2 (median 1.1)
MADE FROM	Dietary tyrosine; and levodopa, if taken as medication
DNA FRAGMENTS MATCHED	69



What is driving this

75 matched fragments, by the organism each came closest to

73% of the signal comes from organisms found in you: *Enterocloster lavalensis* (0.00% of your community); *Eggerthella lenta* (0.10% of your community). The remaining 27% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Enterocloster lavalensis	**40%** (30 fragments)	**yes** · 0.001%	Opportunist
Eggerthella lenta	**8%** (6 fragments)	**yes** · 0.104%	Opportunist
Thermincola carboxydiphila	**7%** (5 fragments)	not found; closest reference	—
Anaerocolumna aminovalerica	**5%** (4 fragments)	not found; closest reference	—
Desulforamulus reducens	**5%** (4 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Enterocloster lavalensis*, *Eggerthella lenta*; *Enterocloster lavalensis* alone accounts for 40% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Eggerthella lenta 0.10% of your community · 8% of this signal	Actinobacterium notable for drug metabolism (inactivates digoxin via cgr genes; decarboxylates levodopa-derived dopamine). Higher in some IBD/RA cohorts. Relevant to medication history, not a target. ■ : E. lenta strains carrying the cgr operon inactivate digoxin; arginine suppresses this in mice. *[Haiser HJ et al. Science 2013;341:295. PMID 23869020]* ■ : E. lenta dehydroxylates dopamine produced from levodopa by Enterococcus faecalis tyrosine decarboxylase. *[Maini Rekdal V et al. Science 2019;364:eaau6323. PMID 31196984]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

typical · detected below the confirmed identity threshold; read as indicative only

Your tyrosine decarboxylase capacity sits in the middle of the reference range. This enzyme makes tyramine from dietary tyrosine and, in people taking levodopa, breaks the drug down before absorption; at a typical level neither effect stands out from the population.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Enterocloster lavalensis* (0.00% of your community); *Eggerthella lenta* (0.10% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Reasonably supported

Gut bacterial tyrosine decarboxylase converts levodopa into dopamine in the intestine, where it cannot reach the brain. Higher carriage is associated with needing a higher levodopa dose in Parkinson's disease. In people not taking levodopa, this mostly reflects tyramine-producing capacity.

van Kessel et al. showed that Enterococcus faecalis tyrDC decarboxylates levodopa in the gut, that carbidopa (the standard peripheral inhibitor) does not block the bacterial enzyme, and that tyrDC abundance in stool correlated with the levodopa dose Parkinson's patients required. Maini Rekdal et al. independently identified the same enzyme and the single active-site residue that determines whether it accepts L-DOPA, and found a small molecule that inhibits it. This is one of the best-characterised

drug-microbiome interactions known. It says nothing about brain dopamine: dopamine made in the colon does not cross the blood-brain barrier, which is the entire reason levodopa is given rather than dopamine itself.

This is a provisional result rather than a confirmed one. Read it as a possible signal.

Reference: van Kessel S.P. et al., Nat Commun 10:310 (2019), doi:10.1038/s41467-019-08294-y; Maini Rekdal V. et al., Science 364:eaau6323 (2019), doi:10.1126/science.aau6323

What the research says about changing this

Neurotransmitter-related capacity (GABA, tyrosine decarboxylase) — a medication fact, not a mood reading LANE D NO SUPPORTED INTERVENTION

Fired by: dopamine gene capacity: typical; gaba gene capacity: somewhat high

Bacterial GABA and tyrosine-decarboxylase genes do not measure brain neurotransmitters. The one clinically relevant fact is that tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — information for a prescriber, not an action.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- Bacterial tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — a medication-context fact for the prescriber; there is no approved microbiome inhibitor or eradication protocol. [van Kessel SP et al. Nat Commun 2019;10:310] · lane D · human association + in vitro/rat

Conventional route: If taking levodopa: information card to the prescriber; no dose change from this report.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “your dopamine is low” · “GABA deficiency” · “mood bacteria”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

Plant-Compound Conversion

‹ at a glance

Urolithin A formation

PRESENT

Whether your gut bacteria carry the enzyme that makes urolithin A from the urolithin C produced earlier in the pathway.

YOUR RESULT	**0.2** copies per 100 genomes
PERCENTILE	**83rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 0.1 (median 0)
MADE FROM	ellagitannins in pomegranate, walnuts, raspberries and strawberries
DNA FRAGMENTS MATCHED	17



What is driving this

20 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Enterocloster bolteae* (0.10% of your community).

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Enterocloster bolteae	**100%** (20 fragments)	**yes** · 0.102%	Opportunist

The organisms carrying this gene in your sample are *Enterocloster bolteae*; *Enterocloster bolteae* alone accounts for 100% of the signal. Changing this reading means changing them.

What your reading means

present · more than half the reference cohort carries none of this, so any amount ranks above the middle; that is a statement about how uncommon this is and not about how much of it you have

Reads map to the published dehydroxylase. The gene is present, which is not the same as a measured urolithin A response to pomegranate or walnuts.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Enterocloster bolteae* (0.10% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Preliminary

Only some people's microbiomes make urolithin A. This measures the published enzyme for the final step; it is not a urolithin measurement.

The enzyme responsible was identified in 2025 as a dehydroxylase encoded by an operon in gut Enterocloster species, and this panel holds the three proteins from that record rather than a gene-symbol search, which returns an unrelated fungal protein. Detection is strict: one reference sequence per subunit means a diverged version in another strain will not reach the identity floor, and is reported as unresolved rather than counted. Carrying the gene is also not the same as being a urolithin producer, which the specification is explicit about: that is established by measuring urolithin after an ellagitannin challenge, not from DNA.

Based on 17 matching DNA fragments, few enough that the exact figure carries real uncertainty; the broad position is more reliable than the precise number.

Reference: Pidgeon R., Mitchell S., Shamash M., et al., "Diet-derived urolithin A is produced by a dehydroxylase encoded by human gut Enterocloster species", GenBank PQ855390.1 (2025).

‹ at a glance

Glucosinolate to isothiocyanate conversion

TYPICAL

Whether the characterised bacterial enzymes for converting glucosinolates to isothiocyanates are present in your sample.

YOUR RESULT	**10.8** copies per 100 genomes
PERCENTILE	**73rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	3.7 – 11.2 (median 6.2)
MADE FROM	Glucosinolates from broccoli, rocket and other cruciferous vegetables
DNA FRAGMENTS MATCHED	193



What is driving this

594 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Bacteroides thetaiotaomicron* (0.57% of your community).

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bacteroides thetaiotaomicron	**100%** (594 fragments)	**yes** · 0.574%	Conditional

The organisms carrying this gene in your sample are *Bacteroides thetaiotaomicron*; *Bacteroides thetaiotaomicron* alone accounts for 100% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides thetaiotaomicron 0.57% of your community · 100% of this signal	Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading. ■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

typical · in line with most people

Mid-range presence of these enzymes against this reference. The route is carried by particular strains rather than being widespread, so a middling value is the ordinary finding.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides thetaiotaomicron* (0.57% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

Measured against the B. thetaiotaomicron locus itself at a high identity floor, because this route is described in one organism rather than carried by an enzyme family.

Three reference sequences is the right size for a reaction characterised in a single organism, and the high identity floor is what keeps a distant homolog from matching; the consequence is that this under-calls rather than over-calls. This bacterial route is not plant myrosinase, which is an enzyme of the vegetable and is destroyed by cooking, and it is not the only bacterial route described. A gene finding is also not a conversion: whether you actually produce isothiocyanates is settled by a substrate challenge with metabolomics, which can be imported beside this.

Reference: Liou C.S. et al., "A metabolic pathway for activation of dietary glucosinolates by a human gut symbiont", Cell 180:717-728 (2020), doi:10.1016/j.cell.2020.01.023.

‹ at a glance

Equol formation

NOT DETECTED

Whether your gut bacteria carry the enzymes that turn the soy isoflavone daidzein into equol.

YOUR RESULT	**0** copies per 100 genomes
PERCENTILE	0 of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 0 (median 0)
MADE FROM	daidzein from soy foods
DNA FRAGMENTS MATCHED	0



What is driving this

1 matched fragments, by the organism each came closest to

No organism behind this reading was found in your sample. The 1 matched fragment came closest to *Lactococcus garvieae*, which the taxonomic profile did not detect in you. A read resembling an organism's copy of a gene is not that organism being present — some other member of your community carries a similar sequence, or the reference set lacks whichever one does. Treat this as a trace signal with no identified source rather than as evidence about those species.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Lactococcus garvieae	**100%** (1 fragments)	not found; closest reference	—

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Preliminary

Roughly a third of people carry equol-producing bacteria. This measures the published enzymes, and is not a measurement of equol.

The three reductases were purified and characterised, which is what makes a specific detector possible; the specification is explicit that aliases such as eqlA/B/C need strain-specific mapping and that a general reductase domain is not evidence of this chemistry. One reference sequence per step means low sensitivity by design, and a diverged enzyme is reported as unresolved rather than transferred onto this annotation. Equol producer status in the literature is defined by measuring equol in urine after a soy challenge, which DNA cannot substitute for.

Reference: Schroder C. et al., "Identification and expression of genes involved in the conversion of daidzein and genistein by the equol-forming bacterium Slackia isoflavoniconvertens", Appl Environ Microbiol 79:3494-3502 (2013), doi:10.1128/AEM.03693-12.

Mucus, Bile & Other Transformations

‹ at a glance

Secondary bile acids ▲

NOTABLY HIGH

Secondary bile acids, chiefly deoxycholic acid (DCA) and lithocholic acid (LCA), made by bacteria reworking the bile acids your liver produces.

YOUR RESULT	**8.2** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	2.0 – 3.6 (median 2.5)
MADE FROM	Liver-derived primary bile acids, after bacterial deconjugation
DNA FRAGMENTS MATCHED	914



What is driving this 1,063 matched fragments, by the organism each came closest to

A further 69% matched *Blautia producta*, *Clostridium scindens*, whose genus is present but whose species was not resolved. The remaining 31% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia producta	**66%** (699 fragments)	genus present; closest reference	—
Terrisporobacter glycolicus	**16%** (167 fragments)	not found; closest reference	—
Terrisporobacter mayombei	**11%** (116 fragments)	not found; closest reference	—
[Clostridium] hylemonae	**4%** (40 fragments)	not found; closest reference	—
Clostridium scindens	**2%** (24 fragments)	genus present; closest reference	—

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

What your reading means notably high · well above most people

More of your gut bacteria can convert your liver's bile acids into secondary bile acids like DCA than in most people. A large study found elevated DCA and reduced primary bile acid in people with Alzheimer's, with the ratio tracking cognitive decline, and DCA interacts with a subunit of the enzyme that generates amyloid-beta. Worth noting that not all secondary bile acids act the same way — TUDCA is protective — and this test cannot tell which ones dominate.

How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with worse outcomes · Reasonably supported

A large study of people with Alzheimer's disease found lower levels of the liver's own primary bile acid and higher levels of the bacterially produced secondary bile acid DCA, with the ratio between them tracking cognitive decline.

MahmoudianDehkordi et al. reported reduced serum cholic acid and elevated DCA and its glycine- and taurine-conjugated forms in people with Alzheimer's disease compared with cognitively normal older adults. Bile acids and their receptors are present in the brain, and DCA has been shown to interact with nicastrin, a subunit of the gamma-secretase enzyme that generates amyloid-beta. Not all secondary bile acids behave the same way: tauroursodeoxycholic acid (TUDCA) inhibits amyloid-beta accumulation and improves mitochondrial function.

One of the genes behind this figure (baiCD) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: MahmoudianDehkordi S. et al., Alzheimers Dement 15:76-92 (2019), doi:10.1016/j.jalz.2018.07.217; Luan H. et al., Adv Biotechnol 1:7 (2023)

What the research says about changing this

Higher secondary-bile-acid (bai) capacity — diet shifts it quickly, levels need LC-MS LANE D HUMAN · NON-RANDOMISED

Fired by: bai gene capacity: notably high

Bile-acid 7α-dehydroxylation capacity rose within days on an animal-based diet and fell on a plant-based diet in a small feeding study. Actual bile-acid levels require targeted stool LC-MS; no trial treats this gene reading.

Studied: Five-day animal-based versus plant-based diet (David 2014) (diet) · **in:** people with clinically confirmed healthy adults, ages 21–33, in USA, n = 10 · **studied protocol (a source fact, not an instruction):** five days animal-based then plant-based diet (or reverse)

Evidence lane D: non-randomized human, pilot, case-series or observational evidence. Replication: single study. Strongest design: nonrandomized study.

- Five days of an animal-based diet expanded bile-tolerant organisms and fecal deoxycholate relative to a plant-based diet in ten volunteers — a rapid surrogate shift, not disease treatment; bile-acid levels require targeted LC-MS. [David LA et al. Nature 2014;505:559] · lane D · controlled feeding crossover · n = 10 · mixed

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “toxic bile acids” · “bile acid detox”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin) ▲

ABOVE AVERAGE

A bacterial enzyme that reverses one of the liver's main routes for getting rid of drugs, hormones and toxins.

YOUR RESULT	**129.3** copies per 100 genomes
PERCENTILE	**85th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	75.0 – 114.6 (median 91.9)
MADE FROM	Glucuronide conjugates made by your own liver
DNA FRAGMENTS MATCHED	4,675



What is driving this 8,855 matched fragments, by the organism each came closest to

37% of the signal comes from organisms found in you: *Faecalibacterium prausnitzii*; *Bacteroides hominis*. A further 53% matched *Bacteroides reticulotermitis*, *Bacteroides fragilis*, whose genus is present but whose species was not resolved. The remaining 10% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bacteroides reticulotermitis	**23%** (2076 fragments)	genus present; closest reference	—
Faecalibacterium prausnitzii	**18%** (1576 fragments)	**yes**	● Beneficial
Bacteroides fragilis	**8%** (732 fragments)	genus present; closest reference	—
Echinicola vietnamensis	**6%** (540 fragments)	not found; closest reference	—
Bacteroides hominis	**4%** (394 fragments)	**yes**	● Conditional

The organisms carrying this gene in your sample are *Faecalibacterium prausnitzii*, *Bacteroides hominis*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Faecalibacterium prausnitzii	One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself. ■ when lower: Reduced F. prausnitzii is associated with Crohn's disease and with post-operative recurrence; the isolate had anti-inflammatory effects in vitro and in mouse colitis. *[Sokol H et al. PNAS 2008;105:16731. PMID 18936492]* ■ when lower: A 2017 review consolidates lower F. prausnitzii across IBD, IBS, obesity, coeliac disease and colorectal cancer cohorts, with phylogroup-dependent patterns. *[Lopez-Siles M et al. ISME J 2017;11:841. PMID 28072420]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means above average · somewhat higher than most people

Your gut carries more beta-glucuronidase capacity than most people's, meaning more capacity to reverse the liver's glucuronide tagging and send compounds back into circulation. What that means depends entirely on what you are exposed to — it is well characterised for one chemotherapy drug and speculative for hormones.

Why this colour. Reactivates toxins, hormones and drugs the liver had tagged for excretion; raised in colorectal-cancer cohorts and the cause of irinotecan gut toxicity.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii*; *Bacteroides hominis*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with worse outcomes · Reasonably supported

Your liver tags compounds with glucuronic acid to excrete them. Gut bacterial beta-glucuronidase removes that tag, sending the compound back into circulation. This is well established for the chemotherapy drug irinotecan and is the proposed route for oestrogen recycling.

The best-evidenced case is irinotecan: bacterial GUS reactivates the inactive SN-38G metabolite in the gut, producing the dose-limiting diarrhoea, and Wallace et al. showed that a selective bacterial GUS inhibitor prevented that toxicity in mice without harming the bacteria. Pollet et al. mapped GUS diversity across the human gut and showed that loop structure, not abundance, governs which substrates an enzyme processes. The oestrogen story rests mostly on in vitro deconjugation and on correlational studies; it is plausible and popular, and it is not established that between-person GUS variation changes circulating hormone levels in a way that matters.

Reference: Wallace B.D. et al., Science 330:831-835 (2010), doi:10.1126/science.1191175; Pollet R.M. et al., Structure 25:967-977 (2017), doi:10.1016/j.str.2017.05.003

What the research says about changing this

Higher β-glucuronidase capacity — preclinical only LANE E NO SUPPORTED INTERVENTION

Fired by: bglucuronidase gene capacity: somewhat high

Bacterial β-glucuronidase inhibitors protected mice from irinotecan toxicity; there is no human trial, and no 'estrogen detox' or calcium-D-glucarate claim is supported.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- Selective bacterial β-glucuronidase inhibitors protected mice from irinotecan toxicity — preclinical. No 'estrogen detox' or calcium-D-glucarate claim follows. [Wallace BD et al. Science 2010;330:831] · lane E · in vitro and mouse
- Further mouse work on GUS inhibition and irinotecan; oncology management is the clinical route. [Bhatt AP et al. PNAS 2020;117:7374] · lane E · mouse

Conventional route: Relevant only during irinotecan chemotherapy, as part of oncology management.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “estrogen dominance” · “detox supplement”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Mucin glycans and lipid A

NOTABLY HIGH

Capacity to forage the mucus layer's sugars, and to build the lipid A core of endotoxin.

YOUR RESULT	**472.3** copies per 100 genomes
PERCENTILE	**95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	164.2 – 262.9 (median 202.8)
MADE FROM	Host mucin glycans; bacterial outer-membrane biosynthesis
DNA FRAGMENTS MATCHED	28,958



What is driving this 21,028 matched fragments, by the organism each came closest to

95% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola massiliensis* (1.96% of your community); *Phocaeicola plebeius* (3.75% of your community); *Bacteroides uniformis* (3.11% of your community). A further 5% matched *Phocaeicola salanitronis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**22%** (4616 fragments)	**yes** · 11.748%	Conditional
Phocaeicola massiliensis	**7%** (1435 fragments)	**yes** · 1.956%	Conditional
Phocaeicola plebeius	**7%** (1406 fragments)	**yes** · 3.747%	Conditional
Bacteroides uniformis	**6%** (1203 fragments)	**yes** · 3.108%	Conditional
Phocaeicola salanitronis	**2%** (454 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Phocaeicola massiliensis*, *Phocaeicola plebeius*; *Phocaeicola vulgatus* alone accounts for 22% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides uniformis 3.11% of your community · 6% of this signal	Among the most prevalent Bacteroides in Western adults; level varies with diet and has no established directional meaning on its own. ■ : Bacteroides species are dominant, diet-responsive members of Western adult stool with both commensal and opportunistic behaviours. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means notably high · well above most people

More mucin-foraging and lipid A capacity than most samples. Foraging capacity is not barrier damage, and lipid A genes are not an endotoxin level.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola massiliensis* (1.96% of your community); *Phocaeicola plebeius* (3.75% of your community); *Bacteroides uniformis* (3.11% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it The research picture is mixed · Early evidence

Mucin foraging is a normal part of how a gut community makes its living. These genes do not resolve an endotoxin structure or measure barrier damage.

Mucin-foraging organisms are ordinary residents and the mucus layer is continuously renewed, so carrying these enzymes is physiological rather than damaging; the question of barrier integrity needs a measurement this test does not make. For lipid A, the immune effect depends on how many acyl chains the finished molecule carries, and one modification gene does not resolve that structure. Total Gram-negative abundance resolves it even less. Where the structure cannot be determined the gene evidence is shown with its uncertainty rather than converted into an endotoxin figure.

One of the genes behind this figure (lpxA) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Tailford L.E. et al., "Mucin glycan foraging in the human gut microbiome", Front Genet 6:81 (2015), doi:10.3389/fgene.2015.00081.

‹ at a glance

Oxalate (degraded, not produced) ▼

ABOVE AVERAGE

The capacity of your gut bacteria to break down dietary oxalate before your body absorbs it.

YOUR RESULT	**3.3** copies per 100 genomes
PERCENTILE	**84th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0.3 – 2.1 (median 0.6)
MADE FROM	Dietary oxalate (spinach, rhubarb, nuts, tea, chocolate)
DNA FRAGMENTS MATCHED	98



What is driving this 178 matched fragments, by the organism each came closest to

77% of the signal comes from organisms found in you: *Bifidobacterium animalis* (0.77% of your community). A further 23% matched *Limosilactobacillus agrestis*, *Bifidobacterium castoris*, *Limosilactobacillus walteri*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Bifidobacterium animalis	**66%** (117 fragments)	**yes** · 0.767%	● Beneficial
Limosilactobacillus agrestis	**10%** (18 fragments)	genus present; closest reference	—
Bifidobacterium castoris	**6%** (11 fragments)	genus present; closest reference	—
Limosilactobacillus walteri	**2%** (3 fragments)	genus present; closest reference	—
Lactobacillus gasseri	**2%** (3 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Bifidobacterium animalis*; *Bifidobacterium animalis* alone accounts for 66% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bifidobacterium animalis 0.77% of your community · 66% of this signal	Almost always supplement- or yogurt-derived (subsp. lactis strains); transient. Strain evidence does not transfer to the species. ■ : Probiotic colonisation is person- and strain-specific. *[Zmora N et al. Cell 2018;174:1388. PMID 30193112]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means above average · somewhat higher than most people

Your gut carries more oxalate-degrading capacity than most people's, which is the favourable direction: more dietary oxalate broken down in the gut means less reaching the kidney.

How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bifidobacterium animalis* (0.77% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it Research associates higher levels with better outcomes · Reasonably supported

Oxalate absorbed from the gut is excreted by the kidney, where it can form calcium oxalate stones. Bacteria that degrade oxalate in the gut reduce how much reaches the kidney, and higher degradation capacity is the favourable direction.

Kaufman et al. found Oxalobacter formigenes colonisation associated with a roughly 70% lower odds of recurrent calcium oxalate stones in a case-control study, and Sidhu et al. showed the mechanism in cystic fibrosis patients, who lose the organism to repeated antibiotics and develop hyperoxaluria. Against a simple reading: attempts to treat hyperoxaluria by administering O. formigenes have given mixed results in trials, and oxalate degradation capacity is distributed across many species beyond Oxalobacter, so carriage of one organism is not the whole picture. The direction of the association is consistent; the size of the effect from any one person's community is not established.

Reference: Kaufman D.W. et al., J Am Soc Nephrol 19:1197-1203 (2008), doi:10.1681/ASN.2007101058; Sidhu H. et al., Lancet 352:1026-1029 (1998)

‹ at a glance

Unconjugated bile acids

BELOW AVERAGE

Deconjugated (unconjugated) bile acids — the first step your gut bacteria take in reworking bile.

YOUR RESULT	**55.7** copies per 100 genomes
PERCENTILE	**17th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	65.7 – 95.4 (median 80.7)
MADE FROM	Liver-derived conjugated bile acids
DNA FRAGMENTS MATCHED	1,089



What is driving this 1,867 matched fragments, by the organism each came closest to

21% of the signal comes from organisms found in you: *Collinsella aerofaciens* (1.54% of your community); *Bifidobacterium longum* (1.84% of your community). A further 23% matched *Blautia fusiformis*, whose genus is present but whose species was not resolved. The remaining 56% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Subdoligranulum variabile	**22%** (410 fragments)	not found; closest reference	—
Blautia fusiformis	**13%** (250 fragments)	genus present; closest reference	—
Simiaoa sunii	**10%** (186 fragments)	not found; closest reference	—
Collinsella aerofaciens	**6%** (113 fragments)	**yes** · 1.535%	Opportunist
Bifidobacterium longum	**6%** (111 fragments)	**yes** · 1.838%	Beneficial

The organisms carrying this gene in your sample are *Collinsella aerofaciens*, *Bifidobacterium longum*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Collinsella aerofaciens 1.54% of your community · 6% of this signal	Common Actinobacterium enriched in rheumatoid arthritis and symptomatic atherosclerosis cohorts and responsive to dietary fibre reduction; common in healthy people too. ■ when higher: Collinsella expanded in rheumatoid arthritis and increased gut permeability and IL-17A in mouse and cell models. *[Chen J et al. Genome Med 2016;8:43. PMID 27102666]* ■ when higher: Collinsella enriched in symptomatic atherosclerosis patients versus controls. *[Karlsson FH et al. Nat Commun 2012;3:1245. PMID 23212374]* Association only — no study on record shows that changing this organism changes an outcome.
Bifidobacterium longum 1.84% of your community · 6% of this signal	Widespread across life stages (subspecies infantis in infants, longum in adults). Endogenous level does not indicate whether any B. longum-containing product would colonise or help. ■ : Review of Bifidobacterium species distribution across the human lifespan. *[O'Callaghan A, van Sinderen D. Front Microbiol 2016;7:925. PMID 27379055]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means below average · somewhat lower than most people

Your bile-salt-deconjugating capacity is on the low side. Since this step sits immediately upstream of secondary bile acid production, low capacity here combined with high capacity there would be biologically odd and would point to a problem with the run rather than a real finding.

Two measurements of the same function disagree

Read this beside the probiotic-genus species reading. The bile-salt deconjugation capacity reading sits at the 17th percentile; the probiotic-genus species reading is at the 99th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Of the two, the gene reading is the better guide here, because this capacity does not depend on any one set of species. Bile salt hydrolase is not a probiotic trait: it is carried across Lactobacillaceae, Bifidobacterium, Clostridium, Bacteroides and Enterococcus alike. A high capacity with few probiotic-genus species simply means the ordinary residents are doing it.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Collinsella aerofaciens* (1.54% of your community); *Bifidobacterium longum* (1.84% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

This is an enabling step rather than an outcome. Deconjugation is what makes bile acids available for the secondary bile acid pathway, and it also affects fat absorption and cholesterol handling. Nearly everyone carries abundant capacity for it.

No direct association with cognition has been reported for bile salt hydrolase activity itself. It is reported here because it sits immediately upstream of the secondary bile acid pathway, and because it acts as an internal consistency check: substantial secondary bile acid capacity without deconjugation capacity would not make biological sense.

Reference: Jones B.V. et al., Proc Natl Acad Sci USA 105:13580-13585 (2008), doi:10.1073/pnas.0804437105

What the research says about changing this

Bile-salt hydrolase — the one strain with LDL trials, and why total capacity is not it LANE C HUMAN TRIAL · LIMITED

Fired by: bsh gene capacity: somewhat low

Total stool BSH gene capacity is not the same as one BSH-active strain. L. reuteri NCIMB 30242 lowered LDL by 9–12% in two trials in people with measured high cholesterol; the reading here does not select or predict that.

Studied: Limosilactobacillus reuteri NCIMB 30242 (bile salt hydrolase-active) (probiotic) · **in:** people with clinically confirmed hypercholesterolaemia, ages 20–75, in Canada, n = 127 · **studied protocol (a source fact, not an instruction):** 2.9 × 10^9 CFU capsule twice daily for nine weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: conflicting. Strongest design: randomized trial.

- In hypercholesterolaemic adults, a yogurt with bile-salt-hydrolase-active L. reuteri NCIMB 30242 lowered LDL by about 9% over six weeks — an exact strain/product result for that phenotype only. [Jones ML et al. Br J Nutr 2012;107:1505] · lane C · randomized double-blind placebo-controlled · n = 114 · benefit
- Capsules of the same strain (2.9 × 10^9 CFU twice daily, nine weeks) lowered LDL by about 12% in a second trial. Total stool BSH gene capacity does not predict this and does not justify another strain. [Jones ML et al. Eur J Clin Nutr 2012;66:1234] · lane C · randomized double-blind placebo-controlled · n = 127 · benefit
- Safety analysis of the L. reuteri NCIMB 30242 trials found no safety signal. [Jones ML et al. — safety of L. reuteri NCIMB 30242 (2012)] · lane C · safety analysis of RCT · null

Applicability: The study required a clinical diagnosis of hypercholesterolaemia. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “lowers your cholesterol” · “BSH deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

Urate & Purines

‹ at a glance

Urate degradation

ABOVE AVERAGE

Your gut bacteria's capacity to break urate down without oxygen, using the route described in the 2023 work on bacterial purine handling.

YOUR RESULT	**186.8** copies per 100 genomes
PERCENTILE	**80th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	132.8 – 176.4 (median 149.9)
MADE FROM	urate arriving from host purine turnover and from purine-rich food
DNA FRAGMENTS MATCHED	7,119



What is driving this

9,340 matched fragments, by the organism each came closest to

38% of the signal comes from organisms found in you: *Mediterraneibacter torques*; *Intestinimonas butyriciproducens* (0.01% of your community). A further 28% matched *Anaerostipes rhamnosivorans*, whose genus is present but whose species was not resolved. The remaining 35% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Anaerostipes rhamnosivorans	**10%** (944 fragments)	genus present; closest reference	—
Mediterraneibacter torques	**8%** (720 fragments)	**yes**	Opportunist
Longibaculum muris	**7%** (614 fragments)	not found; closest reference	—
Intestinimonas butyriciproducens	**6%** (557 fragments)	**yes** · 0.006%	Beneficial
Sporomusa ovata	**6%** (557 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Mediterraneibacter torques*, *Intestinimonas butyriciproducens*. Changing this reading means changing them.

What your reading means

above average · somewhat higher than most people

More of the anaerobic urate-degrading route than most samples carry. The mouse work associates this route with lower host urate, but nothing here measures your urate, and the human evidence is association rather than effect.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Mediterraneibacter torques*; *Intestinimonas butyriciproducens* (0.01% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Preliminary

Gut bacteria can degrade urate anaerobically. This is a count of the genes for that route. It is not a serum urate measurement and does not diagnose or rule out gout.

Two 2023 studies showed that gut bacteria degrade urate anaerobically and that this contributes to whole-body purine handling in mice, with human cohort association. The route runs through xanthine and purine fermentation to short-chain fatty acids, and it is not the uricase reaction: uricase requires molecular oxygen and most gut anaerobes do not use it. What is measured here is gene capacity in stool. Serum urate is set mostly by renal excretion and host production, and no gene count here can substitute for a blood test or explain an individual's urate level.

One of the genes behind this figure (xdhC) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Liu Y. et al., "Gut microbial DNA and purine degradation", Cell 186:3400-3413 (2023), doi:10.1016/j.cell.2023.06.010; Kasahara K. et al., "Gut bacterial metabolism contributes to host global purine homeostasis", Cell Host Microbe 31:1038-1053 (2023), doi:10.1016/j.chom.2023.05.011.

Polyamines

‹ at a glance

Putrescine formation

HIGH

How much capacity your gut bacteria have to make putrescine, the smallest of the polyamines.

YOUR RESULT	**242.8** copies per 100 genomes
PERCENTILE	**92nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	182.2 – 220.3 (median 194.4)
MADE FROM	arginine and ornithine from dietary and host protein
DNA FRAGMENTS MATCHED	6,171



What is driving this 7,019 matched fragments, by the organism each came closest to

33% of the signal comes from organisms found in you: *Escherichia coli*; *Phocaeicola sartorii*. A further 35% matched *Blautia faecicola*, whose genus is present but whose species was not resolved. The remaining 32% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Blautia faecicola	**22%** (1553 fragments)	genus present; closest reference	—
Glutamicibacter creatinolyticus	**15%** (1084 fragments)	not found; closest reference	—
Escherichia coli	**14%** (949 fragments)	**yes**	Opportunist
Phocaeicola sartorii	**8%** (537 fragments)	**yes**	Conditional
Pseudobutyrivibrio ruminis	**4%** (317 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Escherichia coli*, *Phocaeicola sartorii*. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Escherichia coli	Carried by most people at low abundance. Species-level detection says nothing about pathotype: enterohaemorrhagic, enterotoxigenic, adherent-invasive and colibactin-producing lineages are strains, identified by toxin and virulence genes, not by the species name. Relative expansion is common after antibiotics or in inflammation and is reported as such; it is not an infection. ■ : Pathogenic E. coli are defined by pathotype-specific virulence factors; commensal and pathogenic strains share the species. *[Kaper JB et al. Nat Rev Microbiol 2004;2:123. PMID 15040260]* ■ when higher: Enterobacteriaceae/E. coli expansion is a hallmark of inflammation-associated dysbiosis across IBD cohorts. *[Lloyd-Price J et al. Nature 2019;569:655. PMID 31142855]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means high · higher than most people

More putrescine-forming capacity than most samples. Polyamines support gut lining renewal and are also elevated in some disease states, so this reading does not point in a single direction on its own.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Escherichia coli*; *Phocaeicola sartorii*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it The research picture is mixed · Early evidence

Polyamines are needed for the constant renewal of the gut lining, and are also elevated in some disease states. A higher or lower reading is not good or bad on its own.

Putrescine, spermidine and spermine are required for cell proliferation, and the gut epithelium renews itself every few days, so a supply matters. The same chemistry is implicated in tumour biology, and bacterial polyamine production has been associated with colorectal cancer tissue in observational work. Stable-isotope and multi-omics work in 2025 traced polyamine flux in gut communities and found production and consumption happening simultaneously in the same sample, which is why this reading is presented as capacity rather than as an amount, and why its direction is context-dependent rather than favourable or

adverse.

One of the genes behind this figure (speC) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Nakamura A. et al., "Symbiotic polyamine metabolism regulates epithelial proliferation and macrophage differentiation in the colon", Nat Commun 12:2105 (2021), doi:10.1038/s41467-021-22212-1.

‹ at a glance

Agmatine formation

HIGH

Capacity to turn arginine into agmatine, which feeds the polyamine routes and has its own activity in the nervous system.

YOUR RESULT	**51.6** copies per 100 genomes
PERCENTILE	**87th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	26.9 – 43.5 (median 34.7)
MADE FROM	arginine from dietary and host protein
DNA FRAGMENTS MATCHED	2,001



What is driving this

3,152 matched fragments, by the organism each came closest to

73% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola plebeius* (3.75% of your community); *Bacteroides thetaiotaomicron* (0.57% of your community). A further 27% matched *Anaerostipes rhamnosivorans*, *Bacteroides parvus*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**36%** (1125 fragments)	**yes** · 11.748%	● Conditional
Anaerostipes rhamnosivorans	**12%** (364 fragments)	genus present; closest reference	—
Bacteroides parvus	**5%** (163 fragments)	genus present; closest reference	—
Phocaeicola plebeius	**5%** (162 fragments)	**yes** · 3.747%	● Conditional
Bacteroides thetaiotaomicron	**4%** (143 fragments)	**yes** · 0.574%	● Conditional

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Phocaeicola plebeius*, *Bacteroides thetaiotaomicron*; *Phocaeicola vulgatus* alone accounts for 36% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bacteroides thetaiotaomicron 0.57% of your community · 4% of this signal	Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading. ■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

high · higher than most people

More arginine decarboxylase capacity than most samples carry. This enzyme also serves acid resistance, so a higher reading is not by itself evidence of more polyamine or neuroactive signalling.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Phocaeicola plebeius* (3.75% of your community); *Bacteroides thetaiotaomicron* (0.57% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

Agmatine is a polyamine precursor and a neuroactive molecule. This counts the gene for making it; it is not a measurement of agmatine.

Arginine decarboxylase is also part of how several gut bacteria survive stomach acid, consuming a proton with each decarboxylation, so a high reading can reflect acid resistance rather than a deliberate investment in polyamine production. Agmatine itself binds imidazoline and alpha-2 adrenergic receptors and has been studied in animal models of pain and mood, but the doses used in those studies are pharmacological and bear no relation to what gut bacteria produce. Nothing here measures agmatine reaching your circulation or your brain.

Reference: Piletz J.E. et al., "Agmatine: clinical applications after 100 years in translation", Drug Discov Today 18:880-893 (2013), doi:10.1016/j.drudis.2013.05.017.

‹ at a glance

Spermidine formation

HIGH

Capacity to make spermidine, counted over both of the routes bacteria actually use rather than only the one in the textbooks.

YOUR RESULT	**122.4** copies per 100 genomes
PERCENTILE	**85th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	90.3 – 113.3 (median 99.9)
MADE FROM	putrescine, with either S-adenosylmethionine or aspartate semialdehyde
DNA FRAGMENTS MATCHED	4,733



What is driving this

4,634 matched fragments, by the organism each came closest to

52% of the signal comes from organisms found in you: *Phocaeicola vulgatus* (11.75% of your community); *Mediterraneibacter gnavus* (6.93% of your community). The remaining 48% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Phocaeicola vulgatus	**10%** (473 fragments)	**yes** · 11.748%	Conditional
Subdoligranulum variabile	**8%** (389 fragments)	not found; closest reference	—
Mediterraneibacter gnavus	**7%** (333 fragments)	**yes** · 6.934%	Opportunist
Hominenteromicrobium mulieris	**4%** (195 fragments)	not found; closest reference	—
Kineothrix sedimenti	**4%** (171 fragments)	not found; closest reference	—

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Mediterraneibacter gnavus*; *Phocaeicola vulgatus* alone accounts for 10% of the signal. Changing this reading means changing them.

What your reading means

high · higher than most people

More spermidine-forming capacity than most samples carry, counted across both routes. This is gene capacity, not a spermidine measurement, and the longevity literature on spermidine is mostly from model organisms.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus* (11.75% of your community); *Mediterraneibacter gnavus* (6.93% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Early evidence

Spermidine has been studied for autophagy and healthy ageing. This reading is the genetic capacity of your bacteria to produce it, not a spermidine level.

Most gut Bacteroidetes make spermidine through carboxyspermidine rather than through the aminopropyltransferase route, and a 2023 study described a further variant running through carboxyaminopropylagmatine. A pipeline that looks only for spermidine synthase therefore reports an absence where the capacity is present, which is why both routes are counted here. Spermidine's association with autophagy and longevity comes largely from model organisms and from observational human diet data; supplement trials in humans are small and mixed, and none of that work measured bacterial gene capacity of the kind reported here.

Reference: Hanfrey C.C. et al., "Alternative spermidine biosynthetic route is critical for growth of Campylobacter jejuni and is the dominant polyamine pathway in human gut microbiota", J Biol Chem 286:43301-43312 (2011), doi:10.1074/jbc.M111.307835.

Glutathione & Ethanol

‹ at a glance

Glutathione synthesis

ABOVE AVERAGE

Your gut bacteria's capacity to make glutathione, the main small-molecule antioxidant inside cells.

YOUR RESULT	**11.4** copies per 100 genomes
PERCENTILE	**81st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	2.8 – 8.9 (median 4.9)
MADE FROM	glutamate, cysteine and glycine
DNA FRAGMENTS MATCHED	853



What is driving this

528 matched fragments, by the organism each came closest to

59% of the signal comes from organisms found in you: *Lactiplantibacillus plantarum* (1.14% of your community); *Bifidobacterium animalis* (0.77% of your community). A further 41% matched *Bifidobacterium scardovii*, *Bifidobacterium reuteri*, *Bifidobacterium myosotis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Lactiplantibacillus plantarum	**25%** (130 fragments)	**yes** · 1.142%	● Beneficial
Bifidobacterium scardovii	**12%** (64 fragments)	genus present; closest reference	—
Bifidobacterium animalis	**7%** (38 fragments)	**yes** · 0.767%	● Beneficial
Bifidobacterium reuteri	**6%** (31 fragments)	genus present; closest reference	—
Bifidobacterium myosotis	**4%** (24 fragments)	genus present; closest reference	—

The organisms carrying this gene in your sample are *Lactiplantibacillus plantarum*, *Bifidobacterium animalis*; *Lactiplantibacillus plantarum* alone accounts for 25% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bifidobacterium animalis 0.77% of your community · 7% of this signal	Almost always supplement- or yogurt-derived (subsp. lactis strains); transient. Strain evidence does not transfer to the species. ■ : Probiotic colonisation is person- and strain-specific. *[Zmora N et al. Cell 2018;174:1388. PMID 30193112]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

above average · somewhat higher than most people

More bacterial glutathione synthesis capacity than most samples carry. This says nothing about your own glutathione status, which is made by your cells and is not measurable from a stool gene count.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Lactiplantibacillus plantarum* (1.14% of your community); *Bifidobacterium animalis* (0.77% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

The research picture is mixed · Preliminary

Some gut bacteria synthesise glutathione. This counts those genes. It is not a measurement of glutathione in you, and it is not a marker of your own antioxidant status.

Bacterial glutathione synthesis is real but patchy: many gut anaerobes lack the route entirely and import what they need, and the fused gshF enzyme found in some Gram-positives does both steps at once, so a search for the two-enzyme route alone will miss it. A 2026 study described glutathione handling in Eubacterium rectale specifically. What this cannot establish is any effect on your own glutathione: host glutathione is made by your own cells from cysteine supply, and there is no established route by which bacterial glutathione in the colon becomes host glutathione in the liver.

Reference: Smirnova G.V., Oktyabrsky O.N., "Glutathione in bacteria", Biochemistry (Moscow) 70:1199-1211 (2005), doi:10.1007/s10541-005-0248-3.

‹ at a glance

Microbial ethanol formation

ABOVE AVERAGE

Capacity of your gut bacteria to produce ethanol, and the separate ethanolamine route that produces acetaldehyde.

YOUR RESULT	**69.9** copies per 100 genomes
PERCENTILE	**77th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	53.6 – 69.0 (median 60.4)
MADE FROM	dietary sugars, and ethanolamine from cell membrane turnover
DNA FRAGMENTS MATCHED	4,440



What is driving this 5,333 matched fragments, by the organism each came closest to

65% of the signal comes from organisms found in you: *Mediterraneibacter gnavus* (6.93% of your community); *Bifidobacterium longum* (1.84% of your community); *Collinsella aerofaciens* (1.54% of your community). A further 35% matched *Blautia luti*, *Blautia aquisgranensis*, whose genus is present but whose species was not resolved.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Mediterraneibacter gnavus	**14%** (755 fragments)	**yes** · 6.934%	Opportunist
Blautia luti	**7%** (369 fragments)	genus present; closest reference	—
Blautia aquisgranensis	**4%** (221 fragments)	genus present; closest reference	—
Bifidobacterium longum	**3%** (170 fragments)	**yes** · 1.838%	Beneficial
Collinsella aerofaciens	**3%** (164 fragments)	**yes** · 1.535%	Opportunist

The organisms carrying this gene in your sample are *Mediterraneibacter gnavus*, *Bifidobacterium longum*, *Collinsella aerofaciens*; *Mediterraneibacter gnavus* alone accounts for 14% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Bifidobacterium longum 1.84% of your community · 3% of this signal	Widespread across life stages (subspecies infantis in infants, longum in adults). Endogenous level does not indicate whether any B. longum-containing product would colonise or help. ■ : Review of Bifidobacterium species distribution across the human lifespan. *[O'Callaghan A, van Sinderen D. Front Microbiol 2016;7:925. PMID 27379055]* Association only — no study on record shows that changing this organism changes an outcome.
Collinsella aerofaciens 1.54% of your community · 3% of this signal	Common Actinobacterium enriched in rheumatoid arthritis and symptomatic atherosclerosis cohorts and responsive to dietary fibre reduction; common in healthy people too. ■ when higher: Collinsella expanded in rheumatoid arthritis and increased gut permeability and IL-17A in mouse and cell models. *[Chen J et al. Genome Med 2016;8:43. PMID 27102666]* ■ when higher: Collinsella enriched in symptomatic atherosclerosis patients versus controls. *[Karlsson FH et al. Nat Commun 2012;3:1245. PMID 23212374]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means above average · somewhat higher than most people

More ethanol-forming gene capacity than most samples carry. Carrying adhE is common and is not the same as being a high-producing strain, which is what the fatty liver disease studies identified.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Mediterraneibacter gnavus* (6.93% of your community); *Bifidobacterium longum* (1.84% of your community); *Collinsella aerofaciens* (1.54% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it The research picture is mixed · Preliminary

Some gut bacteria make ethanol. High-producing strains have been linked to fatty liver disease in specific studies. This is a gene count and does not diagnose anything.

The clearest evidence concerns particular high-alcohol-producing Klebsiella pneumoniae strains found in people with non-alcoholic fatty liver disease, where colonising mice with those strains produced liver changes. That is a strain-level finding:

carrying adhE is ordinary and most bacteria that carry it are not high producers, so this reading cannot identify auto-brewery syndrome or fatty liver. Acetaldehyde, the intermediate, is the genotoxic part of this chemistry. A separate reading for acetaldehyde disposal is deliberately not given, because the aldehyde dehydrogenase family is not substrate-specific enough for a count over it to mean acetaldehyde rather than any other aldehyde.

Reference: Yuan J. et al., "Fatty liver disease caused by high-alcohol-producing Klebsiella pneumoniae", Cell Metab 30:675-688 (2019), doi:10.1016/j.cmet.2019.08.018.

Toxins & Virulence

‹ at a glance

Colibactin and bacterial toxins (virulence factors, not metabolites) ▲

PRESENT

Three specific bacterial genes whose products damage or transform colon cells: a Fusobacterium adhesin, a Bacteroides fragilis toxin, and the colibactin biosynthesis machinery of certain Escherichia coli.

YOUR RESULT	**0.076** copies per 100 genomes
PERCENTILE	**76th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort)
REFERENCE MIDDLE HALF	0 – 0 (median 0)
MADE FROM	Not a metabolite — these are toxins and biosynthetic enzymes
DNA FRAGMENTS MATCHED	1



What is driving this

4 matched fragments, by the organism each came closest to

100% of the signal comes from organisms found in you: *Escherichia coli*.

ORGANISM	SHARE OF SIGNAL	IN YOUR SAMPLE	CLASS
Escherichia coli	**100%** (4 fragments)	**yes**	Opportunist

The organisms carrying this gene in your sample are *Escherichia coli*; *Escherichia coli* alone accounts for 100% of the signal. Changing this reading means changing them.

The organisms behind this, and what is known about them

FOUND IN YOU	WHAT THE LITERATURE RECORDS
Escherichia coli	Carried by most people at low abundance. Species-level detection says nothing about pathotype: enterohaemorrhagic, enterotoxigenic, adherent-invasive and colibactin-producing lineages are strains, identified by toxin and virulence genes, not by the species name. Relative expansion is common after antibiotics or in inflammation and is reported as such; it is not an infection. ■ : Pathogenic E. coli are defined by pathotype-specific virulence factors; commensal and pathogenic strains share the species. *[Kaper JB et al. Nat Rev Microbiol 2004;2:123. PMID 15040260]* ■ when higher: Enterobacteriaceae/E. coli expansion is a hallmark of inflammation-associated dysbiosis across IBD cohorts. *[Lloyd-Price J et al. Nature 2019;569:655. PMID 31142855]* Association only — no study on record shows that changing this organism changes an outcome.

What your reading means

present · more than half the reference cohort carries none of this, so any amount ranks above the middle; that is a statement about how uncommon this is and not about how much of it you have

One or more of these virulence genes was detected in your sample. This is common — pks+ Escherichia coli alone is carried by roughly a fifth of healthy adults — and it is not a cancer finding. It does not change what colorectal cancer screening you should have or when, which is a decision based on your age, family history and symptoms.

How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Escherichia coli*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

These are the mechanistically specific colorectal cancer virulence factors, with genuine causal evidence in animal models and, for colibactin, a distinctive mutational signature in human tumours. They are also carried by many healthy people, and this panel can only report detected or not detected.

Colibactin has the strongest evidence: Pleguezuelos-Manzano et al. exposed human intestinal organoids to pks+ E. coli and produced a mutational signature they then found in human colorectal tumours — a direct molecular fingerprint linking the bacterium to the mutations. Rubinstein et al. showed FadA binds E-cadherin and activates beta-catenin signalling, and Wu et al. showed enterotoxigenic B. fragilis drives IL-17-dependent colon tumorigenesis in mice. Against acting on any of this: pks+ E. coli is carried by around a fifth of healthy adults, ETBF by a substantial minority, and the overwhelming majority never develop colorectal cancer. Carriage is a risk factor at population scale, not a prediction for an individual.

This is a provisional result rather than a confirmed one — only 1 matching fragment (at least 2 required); matches average 72% identity (at least 75% required), so the reads are matching distant relatives rather than the gene itself. Read it as a possible signal.

Reference: Pleguezuelos-Manzano C. et al., Nature 580:269-273 (2020), doi:10.1038/s41586-020-2080-8; Rubinstein M.R. et al., Cell Host Microbe 14:195-206 (2013); Wu S. et al., Nat Med 15:1016-1022 (2009)

What the research says about changing this

Colorectal-cancer-associated virulence genes (pks, bft, fadA) — carriage, not toxin activity LANE A CLINICAL ROUTE

Fired by: crc virulence gene capacity: somewhat high

These genes mark the potential to make colibactin, fragilysin or FadA; carriage is common and is not toxin expression or cancer. No trial shows that removing these organisms prevents cancer, so conventional screening is the action.

Studied: Age-appropriate colorectal cancer screening (USPSTF) (clinical route) · **in:** ages 45–75 · **studied protocol (a source fact, not an instruction):** screening per USPSTF (stool-based test or colonoscopy at guideline intervals)

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: strong for.

- Colorectal cancer screening for adults 45–75 is an established, guideline-backed route. A stool microbiome pattern does not substitute for FIT or colonoscopy and does not change the recommended interval. [USPSTF 2021 colorectal cancer screening recommendation] · lane A · guideline · benefit
- Colibactin (pks), fragilysin (bft) and FadA are mechanistically linked to colorectal carcinogenesis, but no human RCT shows that eliminating these organisms prevents cancer. Carriage potential is not toxin expression or cancer. [review — pks+ E. coli, ETBF and F. nucleatum in colorectal carcinogenesis (2023)] · lane E · review
- **Against or limiting —** Periodontal treatment reduced stool F. nucleatum in CRC patients, but no trial shows that lowering it prevents cancer; screening remains conventional. [periodontal treatment lowered stool F. nucleatum in CRC patients (2021)] · lane X · before-after · benefit

Conventional route: Age-appropriate colorectal cancer screening per USPSTF.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “cancer-causing bacteria detected” · “eradicate pks E. coli”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ the functions overview

Substrates, routes and transformations, one at a time

What your microbes carry the genes to do, grouped by the kind of chemistry. Each row says how specific the evidence is, because a gene family that acts on a whole class of molecules is not evidence about one member of it — and each says the one conclusion it does not support.

Evidence level: research. These are research-grade readings. The biology behind them is published and the genes are real, but none of them has the weight of a clinical measurement, and none should be acted on as though it had.

GABA: two counts, deliberately not combined

READING	VALUE	UNIT
Genes for making GABA	**33.5**	copies per 100 bacterial genomes
Genes for breaking GABA down	— no supported reading	copies per 100 bacterial genomes

Two separate measurements, shown side by side. Their difference is not a net flux and neither is their ratio: both describe how many genes are present, not how much GABA is made or broken down.

Three protein readings that are not the same reading

READING	WHAT IT MEANS
Protein breakdown	cutting proteins and peptides up (proteolysis)
BCAA synthesis	making branched-chain amino acids (the existing bcaa panel)
BCAA fermentation	fermenting them into branched-chain fatty acids (bcfa)

These are three different things and this report keeps three separate readings for them. A community can be high in all three.

Where your own body finishes the job

MICROBES MAKE	YOUR ENZYME	WHICH BECOMES
phenylacetate	hepatic glutamine N-acyltransferase	phenylacetylglutamine (PAGln)
trimethylamine (TMA)	hepatic FMO3	trimethylamine N-oxide (TMAO)

Microbes make phenylacetate; your liver conjugates it to PAGln. The circulating metabolite associated with cardiovascular outcomes is the host product, and this measures only the microbial precursor.

What the difficult panels are told apart from

Some reactions are carried by enzymes that look very like commoner ones. For those, the panel is built around the look-alike it has to reject: if a read matches the decoy, the panel has failed rather than found something. These are the decoys.

READING	THE LOOK-ALIKE	WHY IT IS HARD
urolithin 9 dehydroxylation	*urdA*	UrdA acts on urocanate and makes imidazole propionate. It is close enough to be mistaken for a urolithin dehydroxylase, so a panel that matches it is failing rather than finding. Residue discrimination must use aligned reference numbering over covered diagnostic positions.
equol daidzein conversion	*generic_reductase*	An unassigned oxidoreductase domain matches many things. The aliases eqlA, eqlB and eqlC need source- and strain-specific mapping before they can be read as these reactions.

Each reading in full

One card per reading, all in the same four parts: what is driving it, what it means, how it might be moved, and the research behind it. Each card names the genes the reading is measured from and where each of those sits in the reference cohort, so the number above can be taken apart rather than taken on trust.

‹ at a glance

Cellulose · **headline for this chemistry is the carbohydrates panel**



5th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Sellimonas catena	**18%**	167
Bacteroides uniformis (strain ATCC 8492 / DSM 6597 / CCUG 4942 / CIP 103695 / JCM 5828 / KCTC 5204 / NCTC 13054 / VPI 0061)	**15%**	134
Paenibacillus nuruki	**14%**	127
Ruminiclostridium cellobioparum subsp. termitidis CT1112	**8%**	75
Enterocloster clostridioformis	**8%**	73
Tepiditoga spiralis	**8%**	72

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Endo-beta-1,4-glucanase celA	51.9 copies per 100 genomes	5th of 100 people
Endoglucanase celB	11.6 copies per 100 genomes	14th of 100 people

What your reading means

Your capacity for cellulose sits lower than most people's, at the 5th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH5, GH9

What it does not mean. A beta-glucanase is not a cellulase. Cellulose is crystalline beta-1,4-glucan and needs processive enzymes and a binding module; an enzyme that clears the soluble mixed-linkage glucan of oats does not touch it.

‹ at a glance

General starch · headline for this chemistry is the carbohydrates panel



86th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**15%**	2,336
Amedibacillus dolichus	**13%**	1,924
Sphingobacterium thalpophilum	**6%**	851
Blautia obeum A2-162	**5%**	825
Bacteroides oleiciplenus YIT 12058	**5%**	765
Hymenobacter amundsenii	**4%**	631

and 33 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Alpha-glucosidase aglA	163.9 copies per 100 genomes	72nd of 100 people
Alpha-amylase amyA	139.0 copies per 100 genomes	63rd of 100 people
Glucoamylase gluA	119.5 copies per 100 genomes	91st of 100 people
Oligo-1,6-glucosidase malL	126.4 copies per 100 genomes	88th of 100 people
Debranching pulA	105.7 copies per 100 genomes	51st of 100 people

What your reading means

Your capacity for general starch sits higher than most people's, at the 86th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH13, GH15, GH31

What it does not mean. Amylolytic capacity is the broad comparator, not evidence about any particular starch. It says the community can digest starch it can reach.

‹ at a glance

Resistant starch · headline for this chemistry is the carbohydrates panel



79th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Blautia obeum A2-162	**13%**	825
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**11%**	662
Hymenobacter amundsenii	**10%**	631
Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)	**7%**	443
Subdoligranulum variabile DSM 15176	**5%**	340
Blautia luti	**5%**	338

and 18 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Alpha-amylase amyA	139.0 copies per 100 genomes	63rd of 100 people
Oligo-1,6-glucosidase malL	126.4 copies per 100 genomes	88th of 100 people
Debranching pulA	105.7 copies per 100 genomes	51st of 100 people

What your reading means

Your capacity for resistant starch sits higher than most people's, at the 79th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH13

What it does not mean. GH13 alone supports general amylolysis, not resistant-starch utilisation. Reaching granular or retrograded starch needs the binding modules and the surface machinery that a general amylase does not carry, and the physical type of the starch (RS1 to RS5) is a property of the food rather than of you.

‹ at a glance

Chitin · headline for this chemistry is the carbohydrates panel



91st of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Streptomyces melanosporofaciens	**23%**	912
Nonomuraea thailandensis	**22%**	858
Streptomyces rugosispiralis	**16%**	635
Streptomyces yatensis	**13%**	512
Streptosporangium saharense	**11%**	417
Laedolimicola ammoniilytica	**5%**	203

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Chitinase chiA	146.5 copies per 100 genomes	91st of 100 people

What your reading means

Your capacity for chitin sits higher than most people's, at the 91st of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH18

What it does not mean. GH18 alone does not demonstrate an intact chitin-utilisation pathway. Many GH18 proteins act on peptidoglycan or on host glycans, and releasing chito-oligosaccharides without the downstream hexosaminidase leaves them unused.

‹ at a glance

Pectin · headline for this chemistry is the carbohydrates panel



84th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)	**19%**	1,146
Phocaeicola vulgatus CL09T03C04	**10%**	600
Bacteroides reticulotermitis JCM 10512	**10%**	589
Cupriavidus metallidurans (strain ATCC 43123 / DSM 2839 / NBRC 102507 / CH34) (Ralstonia metallidurans)	**9%**	525
Bacteroides graminisolvens DSM 19988 = JCM 15093	**8%**	464
Phocaeicola dorei CL02T12C06	**6%**	384

and 17 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Pectate lyase pelA	141.0 copies per 100 genomes	81st of 100 people
Polygalacturonase pgl	188.2 copies per 100 genomes	89th of 100 people
Rhamnogalacturonase rgl	7.1 copies per 100 genomes	32nd of 100 people

What your reading means

Your capacity for pectin sits higher than most people's, at the 84th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH28, PL1

What it does not mean. A polygalacturonase acts on the homogalacturonan backbone. The branched rhamnogalacturonan regions of real pectin need their own enzymes, so backbone capacity is not whole-pectin capacity.

‹ at a glance

Inulin · headline for this chemistry is the carbohydrates panel



93rd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)	**21%**	629
Bacteroides fragilis	**14%**	402
Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)	**10%**	306
Subdoligranulum variabile DSM 15176	**10%**	298
Gemmiger gallinarum	**9%**	255
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**8%**	244

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Levanase levB	49.6 copies per 100 genomes	92nd of 100 people
Fructan sacA	120.8 copies per 100 genomes	90th of 100 people

What your reading means

Your capacity for inulin sits higher than most people's, at the 93rd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH32

What it does not mean. GH32 alone does not separate inulin from fructooligosaccharides. Long-chain inulin has to be cut outside the cell before anything can take it up, so a GH32 that handles short fructans is not evidence of inulin utilisation.

‹ at a glance

Fructooligosaccharides · headline for this chemistry is the carbohydrates panel



93rd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)	**21%**	629
Bacteroides fragilis	**14%**	402
Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)	**10%**	306
Subdoligranulum variabile DSM 15176	**10%**	298
Gemmiger gallinarum	**9%**	255
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**8%**	244

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Levanase levB	49.6 copies per 100 genomes	92nd of 100 people
Fructan sacA	120.8 copies per 100 genomes	90th of 100 people

What your reading means

Your capacity for fructooligosaccharides sits higher than most people's, at the 93rd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH32

What it does not mean. Sharing GH32 with inulin is expected and is not double evidence. Short-chain fructans are taken up whole and cut inside, which is a different route from the one inulin needs.

‹ at a glance

Galactooligosaccharides · headline for this chemistry is the carbohydrates panel



92nd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Bacteroides reticulotermitis JCM 10512	**16%**	1,182
Aestuariibaculum sediminum	**13%**	971
Phocaeicola vulgatus CL09T03C04	**11%**	775
Bacteroides ovatus (strain ATCC 8483 / DSM 1896 / JCM 5824 / BCRC 10623 / CCUG 4943 / NCTC 11153)	**10%**	712
Proteiniphilum saccharofermentans	**9%**	683
Geofilum rubicundum JCM 15548	**9%**	657

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Beta-galactosidase lacZ	347.0 copies per 100 genomes	92nd of 100 people
Beta-mannosidase manA	55.7 copies per 100 genomes	82nd of 100 people

What your reading means

Your capacity for galactooligosaccharides sits higher than most people's, at the 92nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH2

What it does not mean. GH2 and GH42 alone do not establish specific GOS utilisation. Every lactose-using organism carries a beta-galactosidase; taking up an oligosaccharide of three to eight sugars needs a transporter that lactose does not.

‹ at a glance

Xylooligosaccharides · headline for this chemistry is the carbohydrates panel



61st of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola salanitronis (strain DSM 18170 / JCM 13657 / CCUG 60908 / BL78) (Bacteroides salanitronis)	**48%**	933
Proteiniphilum saccharofermentans	**21%**	408
Pseudopedobacter saltans (strain ATCC 51119 / DSM 12145 / JCM 21818 / CCUG 39354 / LMG 10337 / NBRC 100064 / NCIMB 13643) (Pedobacter saltans)	**11%**	207
Zunongwangia atlantica 22II14-10F7	**5%**	101
Paludibacter propionicigenes (strain DSM 17365 / JCM 13257 / WB4)	**5%**	98
Paenibacillus algicola	**4%**	86

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Xylan 1,4-beta-xylosidase xylB	109.4 copies per 100 genomes	61st of 100 people

What your reading means

Your capacity for xylooligosaccharides sits in the middle of the reference range, at the 61st of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH43

What it does not mean. Using xylooligosaccharides does not mean using arabinoxylan. The polymer has to be cut by an endo-xylanase first, and an organism that lives on the released oligosaccharides may carry no xylanase at all.

‹ at a glance

Isomaltooligosaccharides · headline for this chemistry is the carbohydrates panel



76th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Amedibacillus dolichus	**17%**	1,924
Blautia obeum A2-162	**7%**	825
Bacteroides oleiciplenus YIT 12058	**7%**	765
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**6%**	662
Hymenobacter amundsenii	**6%**	631
Dysgonomonas hofstadii	**5%**	620

and 26 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Alpha-glucosidase aglA	163.9 copies per 100 genomes	72nd of 100 people
Alpha-amylase amyA	139.0 copies per 100 genomes	63rd of 100 people
Oligo-1,6-glucosidase malL	126.4 copies per 100 genomes	88th of 100 people
Debranching pulA	105.7 copies per 100 genomes	51st of 100 people

What your reading means

Your capacity for isomaltooligosaccharides sits higher than most people's, at the 76th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH13, GH31

What it does not mean. An alpha-glucosidase is not specific to the alpha-1,6 linkages that define IMO. Commercial IMO preparations also differ widely in how much of them actually resists digestion, which is a property of the product.

‹ at a glance

Lactose · headline for this chemistry is the carbohydrates panel



92nd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Bacteroides reticulotermitis JCM 10512	**16%**	1,182
Aestuariibaculum sediminum	**13%**	971
Phocaeicola vulgatus CL09T03C04	**11%**	775
Bacteroides ovatus (strain ATCC 8483 / DSM 1896 / JCM 5824 / BCRC 10623 / CCUG 4943 / NCTC 11153)	**10%**	712
Proteiniphilum saccharofermentans	**9%**	683
Geofilum rubicundum JCM 15548	**9%**	657

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Beta-galactosidase lacZ	347.0 copies per 100 genomes	92nd of 100 people
Beta-mannosidase manA	55.7 copies per 100 genomes	82nd of 100 people

What your reading means

Your capacity for lactose sits higher than most people's, at the 92nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH2

What it does not mean. This is the community's capacity to use lactose that reaches the colon. It is not a lactase-persistence genotype, not a breath test, and not an explanation of dairy symptoms, which depend on your own small-intestinal lactase.

‹ at a glance

Beta-glucans · headline for this chemistry is the carbohydrates panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Escherichia coli (strain K12)	**36%**	1,815
Bacillus subtilis (strain 168)	**24%**	1,223
Clostridium longisporum	**16%**	807
Kordia algicida OT-1	**6%**	327
Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)	**4%**	215
Roseburia porci	**4%**	185

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Beta-glucosidase bglA	236.2 copies per 100 genomes	95th of 100 people
Beta-1,3-1,4-glucanase licB	12.1 copies per 100 genomes	73rd of 100 people

What your reading means

Your capacity for beta-glucans sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH16, GH3

What it does not mean. Cereal and fungal beta-glucans are different molecules with different linkages, and the enzymes are not interchangeable. An oat beta-glucan finding says nothing about a mushroom one.

‹ at a glance

Arabinoxylan · headline for this chemistry is the carbohydrates panel



86th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Bacteroides graminisolvens DSM 19988 = JCM 15093	**16%**	1,076
Phocaeicola salanitronis (strain DSM 18170 / JCM 13657 / CCUG 60908 / BL78) (Bacteroides salanitronis)	**14%**	933
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**13%**	836
Bacteroides reticulotermitis JCM 10512	**8%**	501
Segatella baroniae B14	**7%**	453
Proteiniphilum saccharofermentans	**6%**	408

and 18 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Alpha-N-arabinofuranosidase abfA	79.3 copies per 100 genomes	82nd of 100 people
Xylan 1,4-beta-xylosidase xylB	109.4 copies per 100 genomes	61st of 100 people
Endo-1,4-beta-xylanase xynA	326.5 copies per 100 genomes	89th of 100 people

What your reading means

Your capacity for arabinoxylan sits higher than most people's, at the 86th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH10, GH43, GH51

What it does not mean. A xylanase opens the backbone; the arabinose side chains and their ferulate cross-links are what make cereal arabinoxylan hard to use, and they need their own enzymes.

‹ at a glance

Galactomannan and guar · headline for this chemistry is the carbohydrates panel



19th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Bacteroides uniformis	**39%**	154
Alistipes onderdonkii subsp. vulgaris	**15%**	58
Phocaeicola salanitronis (strain DSM 18170 / JCM 13657 / CCUG 60908 / BL78) (Bacteroides salanitronis)	**15%**	58
Dysgonomonas hofstadii	**10%**	38
Hymenobacter qilianensis	**7%**	29

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Bifidobacterium moukalabense DSM 27321	**5%**	20

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Mannan endo-1,4-beta-mannosidase manB	22.3 copies per 100 genomes	19th of 100 people

What your reading means

Your capacity for galactomannan and guar sits lower than most people's, at the 19th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

GH26

What it does not mean. The mannan backbone and its galactose side chains need different enzymes, and partially hydrolysed guar behaves differently from the intact gum, which is a property of the preparation rather than of the community.

‹ at a glance

Acetate production · headline for this chemistry is the fermentation panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Streptococcus ictaluri 707-05	**22%**	360
Odoribacter laneus YIT 12061	**22%**	358
Sphingobacterium spiritivorum ATCC 33861	**15%**	245
Bergeyella zoohelcum ATCC 43767	**12%**	201
Malonomonas rubra DSM 5091	**8%**	141
Candidatus Chryseopegocella kryptomonas	**8%**	134

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Phosphate acetyltransferase pta	83.3 copies per 100 genomes	95th of 100 people

What your reading means

Your capacity for acetate production sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

pta

What it does not mean. Phosphotransacetylase is the committed step. Acetate kinase completes the route and is measured by the butyrate panel as a decoy, so it corroborates here rather than being required: duplicating it as a target would take reads from that panel rather than add a reading.

‹ at a glance

Acetate made from hydrogen and CO2 · headline for this chemistry is the fermentation panel



62nd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Blautia pseudococcoides	**19%**	254
Anaerobutyricum soehngenii	**16%**	210
Blautia fusiformis	**13%**	178
Blautia argi	**13%**	170
Blautia faecicola	**12%**	158
Clostridium autoethanogenum	**8%**	103

and 9 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
CO-methylating acetyl-CoA synthase acsB	9.8 copies per 100 genomes	21st of 100 people
Carbon monoxide dehydrogenase cooS	69.9 copies per 100 genomes	68th of 100 people

What your reading means

Your capacity for acetate made from hydrogen and co2 sits in the middle of the reference range, at the 62nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

acsB, cooS

What it does not mean. Reconstructed separately from Pta-AckA because it is a different chemistry with a different substrate: it consumes hydrogen rather than sugar.

‹ at a glance

D-lactate production · headline for this chemistry is the fermentation panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Gemmiger formicilis	**26%**	425
Nosocomiicoccus ampullae	**15%**	256
Phocaeicola dorei CL02T12C06	**15%**	253
Anaerostipes hadrus	**13%**	216
Faecalicatena orotica	**10%**	168
Blautia luti	**9%**	156

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
D-lactate dehydrogenase ldhA	92.2 copies per 100 genomes	95th of 100 people

What your reading means

Your capacity for d-lactate production sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

ldhA

What it does not mean. Stereoisomer reported where the enzyme resolves it.

‹ at a glance

L-lactate production · headline for this chemistry is the fermentation panel



27th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Clostridium tetani (strain Massachusetts / E88)	**62%**	345
Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)	**12%**	67
Bifidobacterium catenulatum DSM 16992 = JCM 1194 = LMG 11043	**6%**	36
Massilia oculi	**6%**	33
Dialister hominis	**4%**	20
Bifidobacterium subtile	**3%**	19

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
L-lactate dehydrogenase ldh	45.4 copies per 100 genomes	27th of 100 people

What your reading means

Your capacity for l-lactate production sits in the middle of the reference range, at the 27th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

ldh

What it does not mean. Stereoisomer reported where the enzyme resolves it.

‹ at a glance

Turning lactate into butyrate · headline for this chemistry is the butyrate panel



20th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Rikenella microfusus	**38%**	146
Christensenella minuta	**14%**	54
Anaerostipes hadrus	**8%**	31
Oscillibacter valericigenes	**6%**	23
Sporomusa carbonis	**6%**	22
Sporomusa rhizae	**4%**	16

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Butyryl-CoA:acetate CoA-transferase but	11.6 copies per 100 genomes	5th of 100 people

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Butyryl-CoA:acetate CoA-transferase) — out of scope (counted as background but	9.1 copies per 100 genomes	no reference range

What your reading means

Your capacity for turning lactate into butyrate sits lower than most people's, at the 20th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

lctA, lctB, lctC, but

What it does not mean. A cross-feeding sink for lactate, not a subtraction from the lactate a community can make. Both are capacities and neither measures an amount.

‹ at a glance

Turning lactate into propionate · headline for this chemistry is the propionate panel



70th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Pseudoflavonifractor phocaeensis	**20%**	325
Syntrophaceticus schinkii	**16%**	270
Ruthenibacterium intestinale	**16%**	259
Gemmiger gallinarum	**13%**	211
Pseudoflavonifractor intestinihominis	**9%**	154
Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)	**9%**	153

and 6 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Lactoyl-CoA dehydratase lcdA	18.8 copies per 100 genomes	85th of 100 people
Lactoyl-CoA dehydratase subunit lcdB	120.2 copies per 100 genomes	64th of 100 people

What your reading means

Your capacity for turning lactate into propionate sits in the middle of the reference range, at the 70th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

lcdA, lcdB

What it does not mean. The acrylate route, distinct from the succinate and propanediol routes.

‹ at a glance

Succinate production · headline for this chemistry is the propionate panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Parabacteroides distasonis (strain ATCC 8503 / DSM 20701 / CIP 104284 / JCM 5825 / NCTC 11152)	**43%**	386
Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)	**22%**	193
Bacteroides fragilis (strain YCH46)	**14%**	121
Rhizobium mesosinicum	**7%**	65
Pelagibius litoralis	**4%**	37
Rhizobium brockwellii	**4%**	33

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Malate dehydrogenase mdh	69.0 copies per 100 genomes	95th of 100 people

What your reading means

Your capacity for succinate production sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

mdh

What it does not mean. A fermentative intermediate that other organisms consume.

‹ at a glance

Turning succinate into propionate · headline for this chemistry is the propionate panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Geofilum rubicundum JCM 15548	**61%**	2,449
Bacteroides stercoris	**26%**	1,026
Bacteroides graminisolvens DSM 19988 = JCM 15093	**4%**	180
Bacteroides reticulotermitis JCM 10512	**3%**	116
Porphyromonas crevioricanis	**3%**	106
Planifilum fimeticola	**0%**	18

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Methylmalonyl-CoA decarboxylase mmdA	184.3 copies per 100 genomes	95th of 100 people
Propionyl-CoA:succinate CoA transferase scpA	43.1 copies per 100 genomes	92nd of 100 people

What your reading means

Your capacity for turning succinate into propionate sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

mmdA, scpA

What it does not mean. The succinate route to propionate, which is why succinate rarely accumulates in a community that carries it.

‹ at a glance

Turning propanediol into propionate · headline for this chemistry is the propionate panel



85th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Blautia caecimuris	**44%**	307
Faecalicatena orotica	**25%**	170
Muricomes intestini	**20%**	138
Intestinimonas butyriciproducens	**4%**	26
Cuneatibacter caecimuris	**2%**	12
Murimonas intestini	**1%**	10

and 6 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Propanediol dehydratase large subunit pduC	3.1 copies per 100 genomes	76th of 100 people
Propionaldehyde dehydrogenase pduP	21.3 copies per 100 genomes	89th of 100 people

What your reading means

Your capacity for turning propanediol into propionate sits higher than most people's, at the 85th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

pduC, pduP

What it does not mean. The deoxy-sugar route; a different chemistry from succinate or acrylate.

‹ at a glance

Hydrogen production · headline for this chemistry is the fermentation panel



48th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Blautia hydrogenotrophica (strain DSM 10507 / JCM 14656 / S5a33) (Ruminococcus hydrogenotrophicus)	**33%**	838
Saccharicrinis fermentans DSM 9555 = JCM 21142	**20%**	504
Mediterraneibacter torques	**10%**	251
Clostridium puniceum	**9%**	228
Blautia luti	**8%**	203
Roseburia intestinalis L1-82	**8%**	198

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
[FeFe] hydrogenase hydA	144.6 copies per 100 genomes	48th of 100 people

What your reading means

Your capacity for hydrogen production sits in the middle of the reference range, at the 48th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

hydA

What it does not mean. Direction-resolved where the HydDB class allows it. A healthy community produces a great deal of hydrogen; there is no adverse direction here.

‹ at a glance

Hydrogen used to make sulfide · headline for this chemistry is the h2s panel



42nd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Uncultured sulfate-reducing bacterium	**71%**	5
Desulfovibrio sp. NY682	**14%**	1
Uncultured Solibacteraceae bacterium	**14%**	1

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Dissimilatory sulfite reductase, alpha subunit dsrA	0.5 copies per 100 genomes	42nd of 100 people

What your reading means

Your capacity for hydrogen used to make sulfide sits in the middle of the reference range, at the 42nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

dsrA

What it does not mean. One of several sulfur routes, kept apart from the taurine, isethionate, sulfoacetate and cysteine routes because their endpoints differ. dsrA is measured by the hydrogen-sulfide panel, which owns it; this route reads that measurement rather than duplicating the target.

‹ at a glance

Turning taurine into sulfide · headline for this chemistry is the h2s panel



94th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Enterococcus saccharolyticus subsp. saccharolyticus ATCC 43076	**32%**	222
Tropicibacter naphthalenivorans	**24%**	167
Oleidesulfovibrio alaskensis (strain ATCC BAA-1058 / DSM 17464 / G20) (Desulfovibrio alaskensis)	**15%**	105
Bilophila wadsworthia (strain 3 1 6)	**9%**	62
Desulfovibrio desulfuricans (strain ATCC 27774 / DSM 6949 / MB)	**5%**	35
Clostridium puniceum	**3%**	24

and 6 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Taurine--pyruvate aminotransferase tpa	21.8 copies per 100 genomes	90th of 100 people
Isethionate sulfite-lyase, IslA isl	6.2 copies per 100 genomes	85th of 100 people

What your reading means

Your capacity for turning taurine into sulfide sits higher than most people's, at the 94th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

tpa, isl

What it does not mean. Bilophila abundance is not this pathway, and this pathway is not every sulfur pathway.

‹ at a glance

Protein and peptide breakdown · headline for this chemistry is the nitrogen panel



84th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola dorei CL02T12C06	**20%**	156
Phocaeicola vulgatus CL09T03C04	**16%**	124
Phocaeicola plebeius	**14%**	113
Phocaeicola massiliensis B84634 = Timone 84634 = DSM 17679 = JCM 13223	**10%**	81
Bacteroides parvus	**9%**	72
Alistipes putredinis DSM 17216	**8%**	67

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Tripeptide aminopeptidase pepT	87.7 copies per 100 genomes	86th of 100 people
Periplasmic dipeptide transport dppA	7.8 copies per 100 genomes	62nd of 100 people

What your reading means

Your capacity for protein and peptide breakdown sits higher than most people's, at the 84th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

pepT, dppA

What it does not mean. Every organism carries peptidases for its own protein turnover. Counting them measures how many organisms are present, not how much of your dinner reaches the colon undigested.

‹ at a glance

Amino-acid fermentation, including Stickland pairs · headline for this chemistry is the nitrogen panel



42nd of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Romboutsia weinsteinii	**27%**	34
[Clostridium] hylemonae DSM 15053	**22%**	28
Faecalicatena orotica	**8%**	10
Enterocloster clostridioformis	**6%**	8
Faecalimonas umbilicata	**6%**	8
Dorea phocaeensis	**5%**	6

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Glycine reductase selenoprotein A grdA	4.3 copies per 100 genomes	44th of 100 people
D-proline reductase prdA	9.4 copies per 100 genomes	51st of 100 people
(R)-phenyllactate dehydrogenase fldH	0.2 copies per 100 genomes	9th of 100 people

What your reading means

Your capacity for amino-acid fermentation, including stickland pairs sits in the middle of the reference range, at the 42nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

grdA, prdA, fldH

What it does not mean. Fermenting an amino acid for energy is the opposite of making it. The two capacities can coexist in a community and say different things.

‹ at a glance

Aromatic precursor routes · headline for this chemistry is the nitrogen panel



88th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola sartorii	**20%**	413
Vreelandella neptunia	**17%**	343
Blautia wexlerae	**16%**	328
Blautia schinkii	**13%**	263
Dryocola clanedunensis	**8%**	173
Nocardioides dongxiaopingii	**5%**	112

and 18 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
4-hydroxyphenylacetate decarboxylase, large subunit hpdB	8.6 copies per 100 genomes	67th of 100 people
Phenylacetate-CoA ligase padI	85.3 copies per 100 genomes	90th of 100 people

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Indolepyruvate ferredoxin oxidoreductase porA	0.3 copies per 100 genomes	56th of 100 people

What your reading means

Your capacity for aromatic precursor routes sits higher than most people's, at the 88th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

hpdB, padI, porA

What it does not mean. These are microbial precursors. The host performs a further step on several of them, and the blood metabolite is the host's product.

‹ at a glance

GABA degradation · headline for this chemistry is the gababreak panel



91st of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Mycobacterium ostraviense	**22%**	70
Sinorhizobium americanum	**12%**	38
Brenneria roseae subsp. americana	**9%**	28
Halostreptopolyspora alba	**9%**	27
Aquirufa nivalisilvae	**8%**	26
Mycobacterium dioxanotrophicus	**8%**	24

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
4-aminobutyrate aminotransferase, degradation gabT	5.4 copies per 100 genomes	38th of 100 people
Succinate-semialdehyde dehydrogenase gabD	22.3 copies per 100 genomes	92nd of 100 people

What your reading means

Your capacity for gaba degradation sits higher than most people's, at the 91st of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

gabT, gabD

What it does not mean. a net GABA balance. Degradation and synthesis are separate measurements and subtracting one percentile from another is not a flux

‹ at a glance

4-aminobutyrate to butyrate · headline for this chemistry is the gababreak panel



66th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Pseudoflavonifractor phocaeensis	**16%**	36
Hyphomonas jannaschiana VP2	**12%**	28
Aquirufa nivalisilvae	**12%**	26
Anaerotruncus massiliensis (ex Togo et al. 2026)	**11%**	25
Bordetella muralis	**6%**	14
Methylobacterium aquaticum	**6%**	14

and 10 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
4-aminobutyrate aminotransferase, degradation gabT	5.4 copies per 100 genomes	38th of 100 people
4-hydroxybutyrate dehydrogenase abfH	17.8 copies per 100 genomes	77th of 100 people

What your reading means

Your capacity for 4-aminobutyrate to butyrate sits in the middle of the reference range, at the 66th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

gabT, abfH

What it does not mean. that GABA is the main butyrate source here; the fibre routes are usually much larger and are measured separately

‹ at a glance

Mucin glycan foraging · headline for this chemistry is the mucinIps panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola coprophilus DSM 18228 = JCM 13818	**13%**	1,605
Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)	**12%**	1,412
Phocaeicola vulgatus CL09T03C04	**9%**	1,112
Bacteroides uniformis (strain ATCC 8492 / DSM 6597 / CCUG 4942 / CIP 103695 / JCM 5828 / KCTC 5204 / NCTC 13054 / VPI 0061)	**8%**	993
Phocaeicola plebeius	**7%**	784
Phocaeicola massiliensis B84634 = Timone 84634 = DSM 17679 = JCM 13223	**6%**	769

and 14 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Exo-alpha-sialidase nanH	109.3 copies per 100 genomes	95th of 100 people
Alpha-L-fucosidase fucA	329.4 copies per 100 genomes	95th of 100 people
Sulfatase sulf	434.4 copies per 100 genomes	94th of 100 people

What your reading means

Your capacity for mucin glycan foraging sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

nanH, fucA, sulf

What it does not mean. damage to your mucus barrier. Mucin foraging is physiological and these organisms are normal residents; genetic capacity is not an image of mucosal erosion

‹ at a glance

Lipid A acylation and modification · headline for this chemistry is the mucinlps panel

94th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Dysgonomonas hofstadii	**21%**	420
Phocaeicola vulgatus CL09T03C04	**18%**	377
Bacteroides oleiciplenus YIT 12058	**13%**	259
Candidatus Kryptonium thompsonii	**6%**	124
Phocaeicola massiliensis B84634 = Timone 84634 = DSM 17679 = JCM 13223	**6%**	115
Indolivaga macrotermitis	**5%**	106

and 18 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
UDP-GlcNAc acyltransferase lpxA	33.7 copies per 100 genomes	86th of 100 people
UDP-3-O-acyl-GlcNAc deacetylase lpxC	39.0 copies per 100 genomes	91st of 100 people
Lauroyl acyltransferase lpxL	64.5 copies per 100 genomes	95th of 100 people

What your reading means

Your capacity for lipid a acylation and modification sits higher than most people's, at the 94th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

lpxA, lpxC, lpxL

What it does not mean. a TLR4 stimulus or an endotoxin concentration. The number of acyl chains decides the immune effect, one modification gene does not resolve the structure, and total Gram-negative abundance resolves nothing at all

‹ at a glance

Hydroxysteroid dehydrogenase transformations · headline for this chemistry is the mucinlps panel

67th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Blautia caecimuris	**47%**	124
Paracoccus lutimaris	**13%**	34
Bacteroides caccae CL03T12C61	**8%**	22
Hespellia stercorisuis DSM 15480	**8%**	21
Coprobacillus cateniformis	**6%**	17

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Roseovarius litoreus	**6%**	17

and 2 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
7-alpha-hydroxysteroid dehydrogenase hdhA	28.9 copies per 100 genomes	67th of 100 people

What your reading means

Your capacity for hydroxysteroid dehydrogenase transformations sits in the middle of the reference range, at the 67th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

hdhA

What it does not mean. one linear sequence of bile-acid chemistry. Some HSDHs act on conjugated substrates, so these reactions form a graph rather than a chain

‹ at a glance

Short-chain fatty acid signalling · headline for this chemistry is the butyrate panel



95th of the reference cohort

What is driving this

ORGANISM CARRYING IT	SHARE OF THIS READING	FRAGMENTS
Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)	**17%**	714
Streptococcus ictaluri 707-05	**9%**	360
Odoribacter laneus YIT 12061	**8%**	358
Gemmiger formicilis	**7%**	281
Sphingobacterium spiritivorum ATCC 33861	**6%**	245
Phocaeicola plebeius	**6%**	242

and 32 more organisms with smaller shares.

THE GENE BEHIND IT	HOW MUCH IS IN YOU	WHERE THAT SITS
Butyryl-CoA:acetate CoA-transferase but	11.6 copies per 100 genomes	5th of 100 people
Butyryl-CoA:acetate CoA-transferase) — out of scope (counted as background but	9.1 copies per 100 genomes	no reference range
Butyrate kinase buk	0.7 copies per 100 genomes	59th of 100 people
Butyrate kinase) — out of scope (counted as background buk	59.6 copies per 100 genomes	no reference range
Phosphate acetyltransferase pta	83.3 copies per 100 genomes	95th of 100 people
Acetate kinase ackA	113.0 copies per 100 genomes	no reference range

What your reading means

Your capacity for short-chain fatty acid signalling sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

The research behind it

but, buk, pta

What it does not mean. a GLP-1 level. The receptors that read these signals are yours, not your microbes', and the step from a fatty acid in the lumen to a hormone in your blood is host physiology this test does not see

The fermentation steps your community can run

Each line is one step, from what goes in to what comes out. A **supported** step is one this community carries the genes for. A step marked *literature only* is consistent with what is known but is not established by these genes, and is shown so the gap is visible rather than filled in. None of these is a flow: they are things the community could do, with no rate attached, and a step that uses a compound is not evidence the compound is there.

FROM	TO	ROUTE	EVIDENCE
substrate	**acetate**	acetate.pta_ack	genes present
hydrogen	**acetate**	acetate.wood_ljungdahl	genes present
carbon dioxide	**acetate**	acetate.wood_ljungdahl	genes present
substrate	**D-lactate**	lactate.d_formation	genes present
substrate	**L-lactate**	lactate.l_formation	genes present
D-lactate	**butyrate**	lactate.utilisation_butyrate	genes present
L-lactate	**butyrate**	lactate.utilisation_butyrate	genes present
D-lactate	**propionate**	lactate.utilisation_propionate	genes present
L-lactate	**propionate**	lactate.utilisation_propionate	genes present
substrate	**succinate**	succinate.formation	genes present
succinate	**propionate**	succinate.to_propionate	genes present
fucose	**propionate**	propionate.propanediol	genes present
rhamnose	**propionate**	propionate.propanediol	genes present
substrate	**hydrogen**	hydrogen.production_fefe	genes present
hydrogen	**hydrogen sulfide**	hydrogen.consumption_sulfate	genes present
sulfate	**hydrogen sulfide**	hydrogen.consumption_sulfate	genes present
taurine	**hydrogen sulfide**	sulfur.taurine	genes present
hydrogen	**methane**	hydrogen.consumption_methanogenesis	literature only

Supported biochemical opportunities, not flows. No edge carries a rate. An edge does not assume its substrate is present: a lactate sink is capacity to use lactate, not evidence that there is lactate to use. Production and consumption of the same metabolite are separate measurements; neither is subtracted from the other. Filling a node does not mean an added organism would establish there.

Why these fibre readings cannot be added together

9 of the **24** genes behind these fibre readings serve more than one fibre. The same stretch of DNA that helps digest starch also helps digest resistant starch, so each reading counts it and a total over the readings counts it several times.

GENE	COUNTS TOWARD
aglA	imo, starch general
amyA	imo, resistant starch, starch general
lacZ	gos, lactose
levB	fos, inulin
malL	imo, resistant starch, starch general
manA	galactomannan, gos, lactose
pulA	imo, resistant starch, starch general
sacA	fos, inulin
xylB	arabinoxylan, xos

Adding the cards together would count every shared gene once per card. Any total over these substrates must be taken with `aggregate_fragments`, which counts each gene once.

Medicines your gut bacteria are known to alter

Each of these transformations has been demonstrated with purified enzymes or defined strains. They are here as mechanism, not as a result about you: none of them is a measurement of what is happening to a medicine you take, and nothing in this section is a reason to change a dose or stop a drug. That is a conversation with the person who prescribed it.

MEDICINE	WHAT THE BACTERIA DO	IN THIS REPORT	WHAT IT DOES NOT MEAN
Digoxin Eggerthella lenta, and only some strains of it	Cgr2 reduces the lactone ring of digoxin to dihydrodigoxin, which is inactive. Strains of E. lenta differ in whether they carry a functional cgr operon, and the reaction is inhibited by arginine rather than increased by it.	not measured	Nothing here estimates a digoxin level or supports a dose change. Carrying E. lenta is not the same as carrying a reducing strain, and this report does not measure the cgr operon: too few characterised sequences exist for a read-based search to be sensitive.
Levodopa (L-DOPA) Enterococcus faecalis, then Eggerthella lenta	E. faecalis TyrDC decarboxylates levodopa to dopamine in the small intestine, before the drug reaches the brain. E. lenta Dadh then dehydroxylates that dopamine to m-tyramine. The first step is not blocked by carbidopa, the inhibitor given alongside levodopa for the human enzyme.	tyrDC detected	Carrying these genes does not mean your levodopa is being consumed, and no dose change follows from this. Both genes are measured in this report because they have other roles; their presence is common.
Irinotecan (via its metabolite SN-38) many gut bacteria, with large differences between enzymes	The liver inactivates SN-38 by attaching glucuronic acid. Bacterial beta-glucuronidases remove it again in the gut, regenerating the active drug in the intestinal lining, which is a proposed mechanism of the diarrhoea that limits this chemotherapy.	gus detected	Your total beta-glucuronidase reading does not estimate this. The enzymes that perform the reaction efficiently are one structural class among several, and this report cannot resolve which class yours belong to.
Sulfasalazine a broad range of gut bacteria	Sulfasalazine is a prodrug: it passes the small intestine intact and gut bacterial azoreductases cleave its azo bond to release the active 5-aminosalicylate in the colon, which is how the drug is designed to work.	not measured	This is the intended mechanism of the drug rather than a problem, and no reading here estimates how much is released. Azoreductase is not measured in this report: the protein name covers an enzyme family too broad for a count over it to mean this reaction.

Neurotransmitters, and what can actually be measured

This is the part of the field where the distance between what a stool sample can show and what people want to conclude is widest, so each of these says where its evidence stops. Three of the five have no specific test behind them at all, and say so rather than borrowing a number from a related molecule.

MOLECULE	WHAT THE EVIDENCE SHOWS	IN THIS REPORT	WHERE IT STOPS
Dopamine-related microbial metabolism	Enterococcus faecalis tyrosine decarboxylase decarboxylates levodopa to dopamine in the small intestine, and Eggerthella lenta dopamine dehydroxylase then removes a hydroxyl to give m-tyramine. Both enzymes were purified and both reactions were demonstrated; both genes are measured in this report.	tyrDC detected	Not brain dopamine, and not a medication instruction. These reactions happen in the gut lumen on a drug or on dietary tyrosine; dopamine made there does not cross into the brain, and nothing here indicates whether your levodopa, if you take any, is being consumed.
Serotonin-related microbial metabolism	A 2025 result describes a Limosilactobacillus mucosae and Ligilactobacillus ruminis consortium whose effect depends on 5-hydroxytryptophan being available, with a traceable candidate decarboxylase. The chemistry is consortium-level and 5-HTP-dependent, and the construct's substrate specificity is recorded as unresolved in the source itself.	no specific test exists There is no validated serotonin-specific gene panel. The available decarboxylase evidence is for tryptamine, which is a different molecule, and turning a tryptamine enzyme into a serotonin detector because a homolog can be retrieved is precisely the error this card exists to avoid. The tryptamine reading is linked instead, labelled as tryptamine.	This is not a serotonin measurement and not an inference about serotonin in your blood or brain. Most of the body's serotonin is made by your own enterochromaffin cells, not by bacteria; the two pools do not exchange across the blood-brain barrier.
Acetylcholine	Some lactobacilli have been reported to produce acetylcholine in culture, and choline is the shared precursor of both acetylcholine and the trimethylamine route this report does measure. The enzymes responsible in bacteria are not characterised to the level a specific detector needs.	no specific test exists No validated substrate-specific panel exists. Acetyltransferases are one of the largest and least substrate-specific enzyme families in bacteria, so a count over them would measure the family and be reported as acetylcholine, which is worse than reporting nothing.	A generic acetyltransferase count is not an acetylcholine assay. Nothing here measures acetylcholine, and nothing here bears on cognition or neuromuscular function.
Norepinephrine	The microbial contribution here is handling rather than synthesis: bacterial beta-glucuronidase can deconjugate catecholamine glucuronides in the gut lumen, freeing the parent compound. Host adrenergic signalling is host physiology and is a separate matter from anything in a stool sample.	no specific test exists Bacterial norepinephrine synthesis is not established, so there is nothing specific to search for. The deconjugation capacity is measured by the existing beta-glucuronidase reading, which is linked here and is not substrate-specific to catecholamines.	No bacterial abundance converts to a norepinephrine level. Microbial deconjugation and your own adrenergic response are different things, and this card keeps them apart rather than joining them.

MOLECULE	WHAT THE EVIDENCE SHOWS	IN THIS REPORT	WHERE IT STOPS
Histamine detail	Bacterial histidine decarboxylase converts histidine to histamine, and it is measured in this report whatever organism carries it. A specific Klebsiella aerogenes strain producing histamine was linked to visceral hypersensitivity in a mouse model and to a patient subgroup, which is a strain-level finding rather than a species-level one.	hdcA detected	Potential production is not a diagnosis of histamine intolerance. That condition is defined clinically and is not established from stool DNA, and a high reading here is not a reason to start a low-histamine diet.

Why there is no breakdown of your beta-glucuronidase by substrate. The structural class of a beta-glucuronidase is defined by the length of a loop at one of two active-site positions, measured in a curated alignment guided by solved structures. Reproducing that from a pairwise alignment to the E. coli enzyme works for enzymes close to it and fails for the divergent gut ones: in testing, the Bacteroides fragilis enzyme came out as Loop 1 where the published structure is mini-Loop 1, and the Bacteroides uniformis enzyme came out as No Loop where it is the published example of Loop 2. Those are the gut organisms the breakdown would exist to describe, so no breakdown is shown. Your total beta-glucuronidase reading is unaffected and is reported as usual.

The test that withheld it: Bacteroides fragilis should be mL1, came out as L1; Bacteroides uniformis should be L2, came out as NL; Parabacteroides merdae should be mL2, came out as NL. The curated loop alignment published as Data S3 of the source, or a profile model built from the solved structures, in place of pairwise alignment to a single anchor. The per-reference counts this needs are already recorded, so the breakdown appears as soon as the assignment agrees with the published examples.

Synbiotic scenarios in detail: every arm, every output, every organism

The full accounting behind the summary in the actions section. For each fibre: what the community could release of four fermentation products in each of the four arms, then which organisms would take up the fibre and which would make the butyrate. Each is a range over every state in which the community makes at least 99% of its butyrate maximum: the least an organism contributes in any such state, and the most.

Arabinoxylan + consortium — pair gains · model cap 4.09 mmol/gDW/h on EX_arabinoxyl_m

OUTPUT (mmol/gDW/h)	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH
butyrate	57.4	171.6	57.4	220.3
propionate	259.3	246.4	531.8	553.6
acetate	414.6	399.2	831.2	815.9
lactate	276.4	266.1	326.7	469.0

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

Organisms that would take up the fibre (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
P. vulgatus	none	81%	flagged high (p93)
B. longum	none	30%	added by the consortium; beneficial
B. thetaiotaomicron	none	4%	conditional
B. ovatus	none	3%	conditional
B. adolescentis	none	3%	beneficial

Organisms that would make the butyrate (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
A. hallii **required**	26%	27%	added by the consortium
C. beijerinckii **required**	26%	27%	added by the consortium
C. butyricum **required**	26%	27%	added by the consortium
C. sp. AM29_11AC **required**	5%	6%	flagged high (p99)
A. lactatifermentans **required**	4%	5%	flagged high (p100)
F. plautii **required**	2%	3%	flagged high (p93)
E. ramulus **required**	2%	3%	conditional
O. splanchnicus **required**	1%	2%	beneficial
A. dolichus	none	<1%	unknown

Organisms that make the butyrate as you are

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. sp. AM29_11AC **required**	29%	30%	flagged high (p99)
A. lactatifermentans **required**	24%	25%	flagged high (p100)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
F. plautii **required**	12%	14%	flagged high (p93)
E. ramulus **required**	11%	14%	conditional
O. splanchnicus **required**	11%	12%	beneficial
A. dolichus **required**	3%	4%	unknown
A. rectalis **required**	2%	3%	beneficial
R. bacterium KLE1738 **required**	<1%	1%	conditional

Inulin + consortium — pair gains · model cap 6.14 mmol/gDW/h on EX_inulin_m

OUTPUT (mmol/gDW/h)	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH
butyrate	57.4	171.6	57.4	220.3
propionate	259.3	246.4	531.8	553.6
acetate	414.6	399.2	849.8	834.5
lactate	276.4	266.1	326.7	469.0

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

Organisms that would take up the fibre (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
P. vulgatus	none	100%	flagged high (p93)
B. longum	none	97%	added by the consortium; beneficial
B. stercoris A	none	34%	conditional
A. hallii	none	33%	added by the consortium
B. breve	none	31%	beneficial
B. uniformis	none	30%	conditional
B. caccae	none	17%	conditional
B. thetaiotaomicron	none	13%	conditional
C. aerofaciens G	none	12%	conditional
B. ovatus	none	10%	conditional
B. adolescentis	none	10%	beneficial
F. saccharivorans	none	8%	beneficial
L. rhamnosus	none	4%	beneficial

Organisms that would make the butyrate (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
A. hallii **required**	26%	27%	added by the consortium
C. beijerinckii **required**	26%	27%	added by the consortium
C. butyricum **required**	26%	27%	added by the consortium
C. sp. AM29_11AC **required**	5%	6%	flagged high (p99)
A. lactatifermentans **required**	4%	5%	flagged high (p100)
F. plautii **required**	2%	3%	flagged high (p93)
E. ramulus **required**	2%	3%	conditional
O. splanchnicus **required**	1%	2%	beneficial
A. dolichus	none	<1%	unknown

Organisms that make the butyrate as you are

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. sp. AM29_11AC **required**	29%	30%	flagged high (p99)
A. lactatifermentans **required**	24%	25%	flagged high (p100)
F. plautii **required**	12%	14%	flagged high (p93)
E. ramulus **required**	11%	14%	conditional
O. splanchnicus **required**	11%	12%	beneficial
A. dolichus **required**	3%	4%	unknown

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
A. rectalis **required**	2%	3%	beneficial
R. bacterium KLE1738 **required**	<1%	1%	conditional

Maltodextrin + consortium — pair gains · model cap 30.28 mmol/gDW/h on EX_dextrin_m

OUTPUT (mmol/gDW/h)	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH
butyrate	57.4	171.6	57.4	216.8
propionate	259.3	246.4	541.4	557.8
acetate	414.6	399.2	863.2	847.8
lactate	276.4	266.1	326.7	469.0

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

Organisms that would take up the fibre (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
P. vulgatus	none	53%	flagged high (p93)
B. longum	none	20%	added by the consortium; beneficial
C. butyricum	none	20%	added by the consortium
A. putredinis	none	9%	conditional
B. stercoris A	none	7%	conditional
P. distasonis	none	6%	conditional
B. breve	none	6%	beneficial
L. plantarum	none	5%	beneficial
F. saccharivorans	none	5%	beneficial
B. animalis	none	3%	beneficial
P. capillosus	none	3%	unknown
B. caccae	none	3%	conditional
B. thetaiotaomicron	none	3%	conditional
E. ramulus	none	2%	conditional
+8 more with smaller possible shares			

Organisms that would make the butyrate (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. butyricum **required**	26%	27%	added by the consortium
C. beijerinckii **required**	26%	27%	added by the consortium
A. hallii **required**	25%	26%	added by the consortium
C. sp. AM29_11AC **required**	5%	6%	flagged high (p99)
A. lactatifermentans **required**	4%	5%	flagged high (p100)
F. plautii **required**	2%	3%	flagged high (p93)
E. ramulus **required**	2%	3%	conditional
O. splanchnicus **required**	1%	2%	beneficial
A. dolichus	none	<1%	unknown

Organisms that make the butyrate as you are

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. sp. AM29_11AC **required**	29%	30%	flagged high (p99)
A. lactatifermentans **required**	24%	25%	flagged high (p100)
F. plautii **required**	12%	14%	flagged high (p93)
E. ramulus **required**	11%	14%	conditional
O. splanchnicus **required**	11%	12%	beneficial
A. dolichus **required**	3%	4%	unknown
A. rectalis **required**	2%	3%	beneficial
R. bacterium KLE1738 **required**	<1%	1%	conditional

Starch + consortium — pair gains · model cap 16.65 mmol/gDW/h on EX_strch1_m

OUTPUT (mmol/gDW/h)	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH
butyrate	57.4	171.6	57.4	216.8
propionate	259.3	246.4	355.2	424.3
acetate	414.6	399.2	557.1	658.7
lactate	276.4	266.1	326.7	439.1

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

Organisms that would take up the fibre (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. beijerinckii	none	36%	added by the consortium
C. butyricum	none	36%	added by the consortium
B. stercoris A	none	12%	conditional
P. distasonis	none	12%	conditional
B. breve	none	11%	beneficial
B. caccae	none	6%	conditional
P. merdae	none	5%	conditional
B. thetaiotaomicron	none	5%	conditional
B. adolescentis	none	4%	beneficial
B. velezensis	none	2%	unknown
A. dolichus	none	<1%	unknown

Organisms that would make the butyrate (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. beijerinckii **required**	26%	27%	added by the consortium
C. butyricum **required**	26%	27%	added by the consortium
A. hallii **required**	25%	26%	added by the consortium
C. sp. AM29_11AC **required**	5%	6%	flagged high (p99)
A. lactatifermentans **required**	4%	5%	flagged high (p100)
F. plautii **required**	2%	3%	flagged high (p93)
E. ramulus **required**	2%	3%	conditional
O. splanchnicus **required**	1%	2%	beneficial
A. dolichus	none	<1%	unknown

Organisms that make the butyrate as you are

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. sp. AM29_11AC **required**	29%	30%	flagged high (p99)
A. lactatifermentans **required**	24%	25%	flagged high (p100)
F. plautii **required**	12%	14%	flagged high (p93)
E. ramulus **required**	11%	14%	conditional
O. splanchnicus **required**	11%	12%	beneficial
A. dolichus **required**	3%	4%	unknown
A. rectalis **required**	2%	3%	beneficial
R. bacterium KLE1738 **required**	<1%	1%	conditional

Pectin + consortium — pair gains · model cap 0.4 mmol/gDW/h on EX_pect_m

OUTPUT (mmol/gDW/h)	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH
butyrate	57.4	171.6	57.4	216.8
propionate	259.3	246.4	357.1	470.2
acetate	414.6	399.2	557.5	695.2
lactate	276.4	266.1	326.7	463.4

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

Organisms that would take up the fibre (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. beijerinckii	none	42%	added by the consortium
C. butyricum	none	42%	added by the consortium
B. stercoris A	none	15%	conditional
B. caccae	none	7%	conditional
B. thetaiotaomicron	none	6%	conditional
B. ovatus	none	4%	conditional
L. pectinoschiza	none	1%	beneficial

Organisms that would make the butyrate (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. beijerinckii **required**	26%	27%	added by the consortium
C. butyricum **required**	26%	27%	added by the consortium
A. hallii **required**	25%	26%	added by the consortium
C. sp. AM29_11AC **required**	5%	6%	flagged high (p99)
A. lactatifermentans **required**	4%	5%	flagged high (p100)
F. plautii **required**	2%	3%	flagged high (p93)
E. ramulus **required**	2%	3%	conditional
O. splanchnicus **required**	1%	2%	beneficial
A. dolichus	none	<1%	unknown

Organisms that make the butyrate as you are

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. sp. AM29_11AC **required**	29%	30%	flagged high (p99)
A. lactatifermentans **required**	24%	25%	flagged high (p100)
F. plautii **required**	12%	14%	flagged high (p93)
E. ramulus **required**	11%	14%	conditional
O. splanchnicus **required**	11%	12%	beneficial
A. dolichus **required**	3%	4%	unknown
A. rectalis **required**	2%	3%	beneficial
R. bacterium KLE1738 **required**	<1%	1%	conditional

Cellulose + consortium — consortium alone suffices · model cap 0.37 mmol/gDW/h on EX_cellul_m

OUTPUT (mmol/gDW/h)	AS YOU ARE	+ CONSORTIUM	+ FIBRE	+ BOTH
butyrate	57.4	171.6	57.4	171.6
propionate	259.3	246.4	259.3	246.4
acetate	414.6	399.2	414.6	399.2
lactate	276.4	266.1	276.4	266.1

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

Organisms that would make the butyrate (+ both)

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. beijerinckii **required**	7%	35%	added by the consortium

ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. butyricum **required**	7%	35%	added by the consortium
A. hallii **required**	5%	33%	added by the consortium
C. sp. AM29_11AC	none	7%	flagged high (p99)
A. lactatifermentans	none	6%	flagged high (p100)
F. plautii	none	4%	flagged high (p93)
E. ramulus	none	3%	conditional
O. splanchnicus	none	3%	beneficial
A. dolichus	none	<1%	unknown

Organisms that make the butyrate as you are

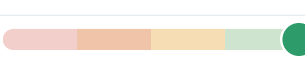
ORGANISM	MUST	CAN	STATUS IN THIS REPORT
C. sp. AM29_11AC **required**	29%	30%	flagged high (p99)
A. lactatifermentans **required**	24%	25%	flagged high (p100)
F. plautii **required**	12%	14%	flagged high (p93)
E. ramulus **required**	11%	14%	conditional
O. splanchnicus **required**	11%	12%	beneficial
A. dolichus **required**	3%	4%	unknown
A. rectalis **required**	2%	3%	beneficial
R. bacterium KLE1738 **required**	<1%	1%	conditional

How many organisms hold each function

fewest confirmed carriers first

Two communities can make the same amount of something with very different security. If one organism is doing all of it, the function travels with that organism; if four share it, losing any one changes little. The count below is an *effective* one: two organisms splitting a job evenly count as two, but two where one does 99% of the work count as barely more than one.

FUNCTION	EFFECTIVE CARRIERS	NAMED	TOP SHARE	SIGNAL CONFIRMED
Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin)	1.0	1	100%	10% †
Unconjugated bile acids	1.0	1	100%	4% †
Butyrate (SCFA)	1.0	1	100%	16% †
Colibactin and bacterial toxins (virulence factors, not metabolites)	1.0	1	100%	100%
Trimethylamine / TMAO	1.0	1	100%	3% †
Dopamine (from L-DOPA) and tyramine (from tyrosine)	1.0	1	100%	44% †
Microbial ethanol formation	1.0	1	100%	5% †
gamma-aminobutyric acid (GABA)	1.0	1	100%	6% †
Hydrogen sulfide (H2S)	1.0	1	100%	10% †
Indole-3-propionate (IPA)	1.0	1	100%	1% †
Peptide uptake	1.0	1	100%	6% †
Imidazole propionate (ImP)	1.0	1	100%	11% †
Putrescine formation	1.6	2	81%	2% †
Histamine	1.8	2	75%	33% †
Fermentation routes and cross-feeding	1.8	2	72%	18% †
Biotin (vitamin B7)	1.9	2	69%	12% †
Vitamin B12 (cobalamin)	1.9	2	66%	6% †
Dietary fibre and carbohydrate breakdown	1.9	2	61%	4% †

FUNCTION	EFFECTIVE CARRIERS		NAMED	TOP SHARE	SIGNAL CONFIRMED
Vitamin B3 (niacin and NAD precursors)	2.0		2	59%	7% †
Agmatine formation	2.0		2	58%	11% †
Riboflavin (vitamin B2)	2.0		2	58%	9% †
Vitamin K2 (menaquinone, MK-n)	2.0		2	52%	6% †
Glutathione synthesis	2.4		3	54%	15% †
Propionate (SCFA)	2.6		5	54%	52%
Mucin glycans and lipid A	2.7		3	53%	23% †
Vitamin B1 (thiamine)	2.8		3	49%	6% †
Protein and nitrogen metabolism	3.1		4	49%	34% †
Indole / indoxyl sulfate	3.4		5	60%	70%
Folate (vitamin B9)	3.5		4	39%	18% †
Urate degradation	3.6		5	42%	27% †
Spermidine formation	3.7		4	35%	15% †
Branched-chain amino acids (valine, leucine, isoleucine)	3.8		4	39%	20% †
Vitamin B5 (pantothenate)	4.6		6	45%	22% †

The bar runs to 4 effective carriers, past which more organisms stop changing the answer. “Named” counts only organisms found in your own inventory as well as on the gene: a reference organism whose copy of a gene matched a read, but which was not detected in you, is context rather than a carrier. **Signal confirmed** is how much of the function's signal those named carriers account for. † On 30 rows it is under half, meaning most of the signal sits on organisms this assay could not place in you — there read the carrier count as the fewest it could be, not as the number there are. This is redundancy among the carriers this assay resolved — not a guarantee of ecological resilience.

PART B — DISEASE-PATTERN RESEMBLANCE

20 Resemblance to published disease patterns, in detail

Researchers have described how the gut communities of people with certain conditions differ, on average, from those of healthy people. This section asks, for each of the patterns in the library: **how closely does this sample resemble that published group-level pattern?** The answer is a percentile of a matched reference group, on the same scale as every other reading in this report. It is not a diagnosis and not a probability of disease.

What we knew about you when scoring

Several patterns depend on facts the sequencing cannot see — medication, how long you have been ill, recent antibiotics. Those supplied are listed; those not supplied are listed with what they would have changed. Where a pattern needs one of them, the ordinary case was assumed (no medication, no recent antibiotics) and the pattern was scored on that stated assumption rather than withheld; section 0 lists every assumption.

FACTOR	STATUS	EFFECT ON SCORING
Sequencing run / flowcell	recorded: 23VMTGLT4:8	used where a pattern declares it
Metformin	not supplied · assumed absent	produces much of the published type 2 diabetes signature; assumed absent when no medication list is supplied (participant mode abstains)
Proton-pump inhibitor	not supplied · assumed absent	large documented shift; recorded and lowers technical reliability
Antibiotics in the last 6 months	not supplied	dominates the community for months; profiles abstain within 90 days
Gluten-free diet	not supplied	large effect in prevalent celiac disease; required for the celiac profile
Country	not supplied	geography often exceeds the disease effect; reference matched on it
Age	not supplied	large across the lifespan; mandatory for age-specific profiles
Sex	not supplied	androgenetic alopecia signals are sex-specific; mandatory there
Stool form	not supplied	correlates with diversity; recorded
Hospitalisation / ICU	not supplied	drives E. coli, Klebsiella and resistance genes; recorded
Kidney function	not supplied	decides whether uremic-toxin capacity matters; context for CKD
Other medications	not supplied	recorded

What was recorded, what was missing, and which profiles refused to score because of it. A missing confounder does not lower a score; it widens what the score could mean.

How the pages that follow are built

Each pattern is made of one or more **study modules** — one per independent published cohort, so a finding replicated by three groups counts three times and a finding from one group once. Modules are scored separately so you can see which line of evidence is driving the result, then fused with equal weight per independent study. Below each result is a table of every contributing feature with its measured value, its position in the comparison group, the direction the research reports, and its weight, so the score can be recomputed by hand.

Every scored pattern follows, most resemblant first, each with the human research on what has been tried for that condition. Patterns the engine declined to score are listed with the reason at the end.

‹ at a glance

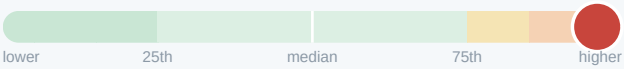
Parkinson's disease — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (parkinsons) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_us_european_cohorts

Parkinson's has the most consistently replicated gut pattern of any neurological disease. Across more than twenty cohorts and a 2022 meta- analysis of shotgun and 16S data, Akkermansia muciniphila and Bifidobacterium are elevated, and the Lachnospiraceae butyrate producers — Roseburia, Faecalibacterium, Blautia — are depleted. Wallen et al. 2022 confirmed this at species level in 490 PD cases with 234 controls by shotgun. Constipation precedes motor symptoms by years and is the most obvious confounder, since slow transit alone raises Akkermansia and methanogens.

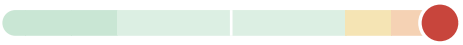
COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 100–100

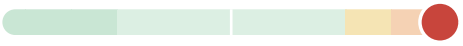
What is driving it

Wallen 2022 US shotgun (490 PD / 234 controls)
species · study wallen_2022 · 15/15 features · 33% of combined



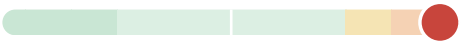
98th — NOTABLY HIGH

Romano 2021 meta-analysis of 10 cohorts
species · association only · 16S study · study romano_2021 · 5/5 features · 33% of combined



>99th — NOTABLY HIGH

Boktor 2023 integrated shotgun (Wallen + Bedarf + Boktor)
species · study boktor_2023 · 4/4 features · 33% of combined



>99th — NOTABLY HIGH

Metabolite gene capacity
genes · genomic capacity · study functional · 3/3 features · not in combined



88th — HIGH

Mucin degraders and butyrate carriers
carriers · carrier proxy · study carriers · 3/3 features · not in combined



98th — NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: t2d (>99th), uc (>99th), ibd (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Wallen 2022 US shotgun (490 PD / 234 controls) · species · study wallen_2022									
Akkermansia muciniphila	1.425%	91	▲	1.00	0.90	0.50	0.90	+0.46	
Bifidobacterium dentium	not detected	44	▲	0.80	0.80	0.50	0.80	-0.05	
Bifidobacterium longum	4.157%	95	▲	0.60	0.80	0.30	0.70	+0.56	
Lactobacillus gasseri	not detected	49	▲	0.60	0.70	0.40	0.70	-0.01	
Methanobrevibacter smithii	not detected	35	▲	0.70	0.80	0.50	0.80	-0.13	
Escherichia coli	not detected 0.033% pooled	19	▲	0.60	0.70	0.20	0.70	-0.29	
Klebsiella pneumoniae	not detected	40	▲	0.50	0.60	0.30	0.60	-0.09	
Roseburia intestinalis	not detected	13	▼	0.90	0.90	0.30	0.90	+0.37	
Roseburia hominis	not detected	7	▼	0.70	0.80	0.30	0.80	+0.97	
Faecalibacterium prausnitzii	0.010%	2	▼	0.90	0.90	0.20	0.90	+1.00	
Eubacterium rectale	0.177%	35	▼	0.70	0.80	0.20	0.80	+0.17	
Blautia obeum	0.042%	50	▼	0.60	0.70	0.40	0.70	-0.00	
Blautia wexlerae	not detected	6	▼	0.50	0.70	0.40	0.70	+0.79	
Coprococcus comes	0.713%	92	▼	0.60	0.70	0.30	0.70	-0.48	
Prevotella copri	not detected	25	▼	0.50	0.60	0.30	0.50	+0.23	
Romano 2021 meta-analysis of 10 cohorts · species · association only · 16S study · study romano_2021									
Akkermansia muciniphila	1.425%	91	▲	0.90	0.80	0.50	0.90	+0.46	
Bifidobacterium longum	4.157%	95	▲	0.60	0.70	0.30	0.70	+0.56	
Lactobacillus gasseri	not detected	49	▲	0.50	0.60	0.40	0.60	-0.01	
Roseburia intestinalis	not detected	13	▼	0.80	0.80	0.30	0.80	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.80	0.20	0.80	+1.00	
Boktor 2023 integrated shotgun (Wallen + Bedarf + Boktor) · species · study boktor_2023									
Akkermansia muciniphila	1.425%	91	▲	0.80	0.80	0.50	0.80	+0.46	
Bifidobacterium dentium	not detected	44	▲	0.60	0.70	0.50	0.70	-0.05	
Roseburia intestinalis	not detected	13	▼	0.70	0.80	0.30	0.80	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.80	0.20	0.80	+1.00	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	0.80	0.20	0.80	+0.76	
dopamine	1.6/100	60	▲	0.60	0.60	0.60	0.70	+0.16	
methane	0.0/100	31	▲	0.60	0.70	0.40	0.70	-0.17	
Mucin degraders and butyrate carriers · carriers · carrier proxy · study carriers									
mucin degraders	21.255%	94	▲	0.70	0.70	0.40	0.70	+0.65	
butyrate	1.623%	4	▼	0.80	0.80	0.20	0.80	+0.53	
methanogens	0.000%	35	▲	0.60	0.70	0.40	0.70	-0.13	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Parkinson's disease is a clinical diagnosis. No stool test predicts it, and this profile is not a prodromal screen.

Chronic constipation of any cause produces most of this pattern. A high score in someone with slow transit is expected and says nothing about Parkinson's.

Bifidobacterium and Akkermansia are higher here — the opposite of most gut disease patterns — which is what gives this profile its unusually good specificity across the library.

Sources: Wallen ZD, et al. Metagenomics of Parkinson's disease implicates the gut microbiome in multiple disease mechanisms. Nat Commun 2022;13:6958; Romano S, et al. Meta-analysis of the Parkinson's disease gut microbiome suggests alterations linked to intestinal inflammation. NPJ Parkinsons Dis 2021;7:27; Boktor JC, et al. Integrated multi-cohort analysis of the Parkinson's disease gut metagenome. Mov Disord 2023;38:399-409; van Kessel SP, et al. Gut bacterial tyrosine decarboxylases restrict levels of levodopa in the treatment of Parkinson's disease. Nat Commun 2019;10:310.

What the research says about this condition and this pattern

Parkinson-disease pattern — what randomized trials measured LANE C HUMAN TRIAL · LIMITED

Fired by: Parkinson's disease pattern: notably high

Parkinson disease is a neurological diagnosis. In people who have it, one exact multistrain probiotic improved constipation (not motor symptoms), and FMT trials are split between a null result and a modest motor signal — investigational.

Studied: Multistrain probiotic capsule used in Tan et al. 2021 (Parkinson disease constipation) (probiotic) · **in:** people with clinically confirmed Parkinson disease with constipation, ages 18–99, in Malaysia, n = 72 · **studied protocol (a source fact, not an instruction):** multistrain capsule for four weeks (full formula/dose from source)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial.

- In 72 people with Parkinson disease and constipation, four weeks of an exact multistrain probiotic increased spontaneous bowel movements. The endpoint was constipation — no motor or neuroprotective claim. [Tan AH et al. Neurology 2021;96:e772] · lane C · randomized double-blind placebo-controlled · n = 72 · benefit
- A 46-person trial reported a modest motor-score improvement 12 months after a single donor FMT versus autologous — shown together with the null trial; FMT for Parkinson disease is investigational. [Bruggeman A et al. eClinicalMedicine 2024 (GUT-PARFECT)] · lane C · randomized double-blind placebo-controlled (autologous) · n = 46 · benefit
- **Against or limiting —** One randomized FMT trial in Parkinson disease found no meaningful clinical benefit. [randomized FMT trial in Parkinson disease (2024) — no meaningful benefit] · lane X · randomized placebo-controlled · null
- **Against or limiting —** FDA has documented transmission of multidrug-resistant organisms (including a death after ESBL E. coli), pathogenic E. coli, and warned about SARS-CoV-2 and mpox with FMT; donor testing, bookend testing and quarantine are required. [FDA FMT safety communications: MDRO/ESBL (June 2019, incl. one death), EPEC/STEC (March–April 2020), SARS-CoV-2 (2020), mpox (August 2022)] · lane X · safety communications · harm

Applicability: The study required a clinical diagnosis of parkinson disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: Neurologic evaluation only if motor or non-motor symptoms are present.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Parkinson's” · “probiotics for Parkinson's”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

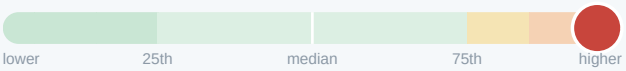
‹ at a glance

Type 2 diabetes — gut pattern (metformin-gated) NOTABLY HIGH

Pattern version **1.0.0** (t2d) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_european_chinese_cohorts

Type 2 diabetes has large shotgun cohorts in China and Europe, and the first replicated lesson from them was a confounding one: much of the "diabetes signature" in early studies was metformin. Forslund et al. showed that metformin-treated patients carry more Escherichia and less Intestinibacter, and that the untreated-diabetes signal is a different, weaker pattern of butyrate-producer depletion. This profile therefore needs metformin status. In participant mode it abstains without it, because a score that cannot separate the drug from the disease measures the drug. In research and clinician mode an unsupplied medication list is read as "no metformin", the untreated-diabetes modules are scored on that stated assumption, and the card says how the reading would change if metformin were in fact being taken.

COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 100–100

Scored on an assumption. This pattern depends on a fact about you that was not supplied, so the report assumed the ordinary case rather than withhold the score:

- **Metformin: assumed not taken.** If that is wrong: Metformin raises Escherichia, lowers Intestinibacter and shifts bile-salt hydrolase carriage (Forslund 2015). The type 2 diabetes pattern would then be partly measuring the drug, and its resemblance percentile would read higher than the disease alone warrants; it would need to be re-read against the metformin-treated stratum.

Supplying the fact (for example `--medications`) replaces the assumption and re-scores this pattern.

What is driving it

Forlsund 2015 metformin-disentangled T2D signature species · study forslund_2015 · 7/7 features · 33% of combined	98th	NOTABLY HIGH
Qin 2012 Chinese MGWAS species · study qin_2012 · 6/6 features · 33% of combined	99th	NOTABLY HIGH
Karlsson 2013 Swedish postmenopausal women species · study karlsson_2013 · 5/5 features · 33% of combined	99th	NOTABLY HIGH
Metabolite gene capacity genes · genomic capacity · study functional · 4/4 features · not in combined	99th	NOTABLY HIGH
Butyrate carriers carriers · carrier proxy · study carriers · 1/1 features · not in combined	96th	NOTABLY HIGH

The Prevotella copri complex, clade by clade

No member of the Segatella (Prevotella) copri complex was detected. All 13 clades were assessed, including the 9 found in humans, so this is a measured absence across the whole complex rather than an unresolved species-level reading.

No clade of this complex has an established association with any disease, including rheumatoid arthritis. Manghi et al. tested all 13 clades against 1,635 disease cases and 1,854 controls across 22 studies and 11 conditions, a set that included a rheumatoid arthritis cohort, and found none of the complex's species associated with any health condition. Earlier rheumatoid-arthritis reports named the species as a whole, not a clade, and those reports have replicated poorly: in an anti-CCP-positive at-risk cohort three Prevotellaceae strains were enriched and five depleted in the people who progressed. So this section reports what is present and makes no disease claim from it.

Tett 2019 Cell Host Microbe doi:10.1016/j.chom.2019.08.018; Manghi 2023 Cell Host Microbe doi:10.1016/j.chom.2023.09.013; Hitch 2022 Syst Appl Microbiol doi:10.1016/j.syapm.2022.126354.

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured

Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), uc (>99th), ibd (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Forslund 2015 metformin-disentangled T2D signature · species · study forslund_2015									
Roseburia intestinalis	not detected	13	▼	0.90	0.80	0.30	0.80	+0.37	
Roseburia hominis	not detected	7	▼	0.70	0.80	0.30	0.80	+0.97	
Eubacterium rectale	0.177%	35	▼	0.80	0.80	0.20	0.80	+0.17	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.80	0.20	0.80	+1.00	
Clostridium bolteae	0.056%	89	▲	0.60	0.70	0.40	0.70	+0.40	
Lactobacillus gasseri	not detected	49	▲	0.50	0.60	0.40	0.60	-0.01	
Streptococcus mutans	not detected	47	▲	0.50	0.60	0.40	0.60	-0.02	
Qin 2012 Chinese MGWAS · species · study qin_2012									
Roseburia intestinalis	not detected	13	▼	0.70	0.70	0.30	0.70	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.70	0.20	0.70	+1.00	
Eubacterium rectale	0.177%	35	▼	0.70	0.70	0.20	0.70	+0.17	
Clostridium bolteae	0.056%	89	▲	0.50	0.60	0.40	0.60	+0.40	
Bacteroides caccae	1.568%	92	▲	0.50	0.60	0.40	0.60	+0.46	
Akkermansia muciniphila	1.425%	91	▲	0.40	0.50	0.30	0.40	+0.46	
Karlsson 2013 Swedish postmenopausal women · species · study karlsson_2013									
Roseburia intestinalis	not detected	13	▼	0.70	0.70	0.30	0.70	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.70	0.20	0.70	+1.00	
Lactobacillus gasseri	not detected	49	▲	0.50	0.60	0.40	0.60	-0.01	
Streptococcus mutans	not detected	47	▲	0.50	0.60	0.40	0.60	-0.02	
Clostridium clostridioforme	0.015%	91	▲	0.50	0.60	0.40	0.60	+0.45	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	0.80	0.20	0.80	+0.76	
urda	33.1/100	39	▲	0.80	0.70	0.50	0.70	-0.10	
bcaa	209.5/100	92	▲	0.60	0.60	0.40	0.60	+0.39	
cutc	4.1/100	77	▲	0.40	0.50	0.30	0.50	+0.39	
Butyrate carriers · carriers · carrier proxy · study carriers									
butyrate	1.623%	4	▼	0.80	0.80	0.20	0.80	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Type 2 diabetes is diagnosed with fasting glucose, HbA1c or a glucose tolerance test. Those are cheap, direct and definitive. This profile is none of those things.

Metformin status decides how to read this pattern. With metformin, the Escherichia expansion is the drug; in an untreated subject it may not be. The two cannot be told apart from stool alone, so when no medication list is supplied the score is printed on the stated assumption of no metformin, and would need re-reading if that assumption is wrong.

Butyrate-producer depletion is shared with every other dysbiosis pattern; see the specificity column in the confidence vector.

Sources: Forslund K, et al. Disentangling type 2 diabetes and metformin treatment signatures in the human gut microbiota. Nature 2015;528:262-266; Qin J, et al. A metagenome-wide association study of gut microbiota in type 2 diabetes. Nature 2012;490:55-60; Karlsson FH, et al. Gut metagenome in European women with normal, impaired and diabetic glucose control. Nature 2013;498:99-103; Koh A, et al. Microbially produced imidazole propionate impairs insulin signaling through mTORC1. Cell 2018;175:947.

What the research says about this condition and this pattern

Type 2 diabetes pattern — confirmation first, then dietary trial evidence LANE C HUMAN TRIAL · LIMITED

Fired by: Type 2 diabetes pattern: notably high

Resemblance to a type-2-diabetes cohort is not a glucose measurement. HbA1c and fasting glucose confirm or exclude diabetes. In diagnosed diabetes, a 12-week high-fibre diet improved HbA1c in one trial; a live Akkermansia product did not improve HbA1c or weight.

Studied: High-fibre diet (whole grains, traditional Chinese medicinal foods, prebiotics) in type 2 diabetes (Zhao 2018) (diet) · **in:** people with clinically confirmed type 2 diabetes, ages 30–70, in China, n = 43 · **studied protocol (a source fact, not an instruction):** high-fibre WTP diet (whole grains, TCM foods, prebiotics) for 12 weeks; acarbose in both arms

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 43 people with type 2 diabetes on acarbose, a 12-week high-fibre diet lowered HbA1c more than usual care and selectively promoted a small group of acetate/butyrate producers — a dietary trial in diagnosed diabetes, not a microbiome prescription. [Zhao L et al. Science 2018;359:1151] · lane C · randomized controlled, open-label · n = 43 · benefit

- **Against or limiting —** A 2025 randomized trial of live A. muciniphila AKK-WST01 in obese type 2 diabetes found no overall weight or HbA1c benefit, only exploratory subgroup signals. [Zhang et al. 2025 — AKK-WST01 in obese type 2 diabetes] · lane X · randomized placebo-controlled · null

Applicability: The study required a clinical diagnosis of type 2 diabetes. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: HbA1c, fasting glucose, lipids, blood pressure and waist.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have diabetes” · “Akkermansia treats diabetes”

Metabolic syndrome / obesity — what FMT trials actually showed

LANE C · HUMAN TRIAL · LIMITED

Fired by: Type 2 diabetes pattern: notably high

Lean-donor FMT transiently improved insulin sensitivity in one small trial; the follow-up trial found the effect only in a subset and lost by 18 weeks, and encapsulated FMT did not change BMI in adolescents. FMT for metabolic disease is investigational.

Studied: Fecal microbiota transplantation for a non-CDI indication (investigational) (fmt microbiota product) · **in:** people with clinically confirmed metabolic syndrome (male), ages 21–65, male, in Netherlands, n = 18 · **studied protocol (a source fact, not an instruction):** studied protocol on file (not shown in participant mode)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- Lean-donor duodenal FMT transiently improved insulin sensitivity in 18 men with metabolic syndrome. [Vrieze A et al. Gastroenterology 2012;143:913] · lane C · randomized double-blind · n = 18 · benefit
- **Against or limiting —** The follow-up lean-donor FMT trial found a 6-week improvement only in a responder subset that disappeared by 18 weeks. [Kootte RS et al. Cell Metab 2017;26:611] · lane X · randomized double-blind · n = 38 · mixed
- **Against or limiting —** Encapsulated lean-donor FMT in 87 adolescents with obesity did not change BMI (primary endpoint null). [Leong KSW et al. JAMA Netw Open 2020;3:e2030415 (Gut Bugs)] · lane X · randomized double-blind placebo-controlled · n = 87 · null
- **Against or limiting —** FDA has documented transmission of multidrug-resistant organisms (including a death after ESBL E. coli), pathogenic E. coli, and warned about SARS-CoV-2 and mpox with FMT; donor testing, bookend testing and quarantine are required. [FDA FMT safety communications: MDRO/ESBL (June 2019, incl. one death), EPEC/STEC (March–April 2020), SARS-CoV-2 (2020), mpox (August 2022)] · lane X · safety communications · harm

Applicability: The study required a clinical diagnosis of metabolic syndrome. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed. The study enrolled male participants only; sex is not on file here.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: A known safety consideration applies; review with a clinician before any change. FMT and FDA-approved microbiota products are clinician-only; approved products cover only prevention of recurrent CDI in adults after antibacterial treatment, and every other use is investigational under IND.

This report will not say: “FMT for weight loss”

Pasteurised Akkermansia — positive pilot, null primaries, an exact product

LANE C · HUMAN TRIAL · LIMITED

Fired by: Type 2 diabetes pattern: notably high

The Akkermansia evidence is for one pasteurised strain at one dose; positive pilot results and null primary endpoints exist side by side, and stool Akkermansia abundance does not predict response.

Studied: Pasteurised Akkermansia muciniphila MucT (ATCC BAA-835) (live biotherapeutic) · **in:** people with clinically confirmed overweight/obesity with insulin resistance, ages 18–70, in Belgium, n = 32 · **studied protocol (a source fact, not an instruction):** 10^10 pasteurised cells/day for three months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: conflicting. Strongest design: randomized trial.

- In a 32-completer exploratory pilot, pasteurised A. muciniphila MucT (10^10 cells/day for three months) improved insulin sensitivity and lowered insulinaemia and cholesterol in overweight/obese insulin-resistant adults. [Depommier C et al. Nat Med 2019;25:1096] · lane C · randomized double-blind placebo-controlled exploratory pilot · n = 40 · benefit
- In a post-weight-loss protocol, pasteurised A. muciniphila reduced weight regain — an exact protocol result shown together with the null trials. [Mount et al. 2026 — pasteurised A. muciniphila after weight loss] · lane C · randomized placebo-controlled · benefit
- A 2026 randomized trial of the exact pasteurised Akkermansia product in obesity; endogenous Akkermansia abundance does not predict response and does not authorise an over-the-counter product. [Nat Med 2026 — pasteurised Akkermansia in obesity] · lane C · randomized placebo-controlled · mixed
- **Against or limiting —** A 2025 randomized trial of live A. muciniphila AKK-WST01 in obese type 2 diabetes found no overall weight or HbA1c benefit, only exploratory subgroup signals. [Zhang et al. 2025 — AKK-WST01 in obese type 2 diabetes] · lane X · randomized placebo-controlled · null
- **Against or limiting —** A 2026 pasteurised-Akkermansia trial missed its primary insulin-sensitivity endpoint. [Suenaert et al. 2026 — pasteurised A. muciniphila, primary insulin-sensitivity endpoint missed] · lane X · randomized placebo-controlled · null

Applicability: The study required a clinical diagnosis of insulin resistance. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “boost your Akkermansia” · “over-the-counter Akkermansia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

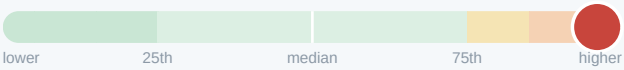
Ulcerative colitis — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (uc) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_north_american_european_cohorts

Ulcerative colitis shares most of its gut pattern with Crohn's disease — butyrate-producer depletion, Ruminococcus gnavus and Escherichia coli expansion — with a milder dysbiosis on average and without the oral-origin taxa that mark Crohn's. Roseburia hominis and Faecalibacterium prausnitzii depletion were shown in UC specifically by Machiels, and in both HMP2 and PRISM the UC dysbiotic state was less frequent and less extreme than in CD. Because the two IBD patterns overlap so heavily, the difference between the uc and crohns scores on one sample is rarely meaningful. The ibd composite profile is the one to read when the question is "IBD-like or not"; these two exist so the library records what each disease's own literature says.

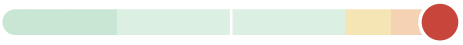
COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 100–100

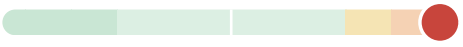
What is driving it

Machiels 2014 Belgian UC cohort
species · study machiels_2014 · 2/2 features · 40% of combined



>99th — NOTABLY HIGH

Franzosa 2019 PRISM — UC directions
species · study franzosa_2019 · 5/5 features · 40% of combined



99th — NOTABLY HIGH

HMP2 IBDMDB — UC directions
species · study hmp2_2019 · 6/6 features · 20% of combined



94th — HIGH

Community ecology
diversity · study hmp2_2019 · 0/1 features · not in combined

— — NOT COMPUTABLE

Metabolite gene capacity
genes · genomic capacity · study functional · 3/3 features · not in combined



94th — HIGH

Butyrate carriers and sulfate reducers
carriers · carrier proxy · study carriers · 2/2 features · not in combined



96th — NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), longcovid (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection

methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Machiels 2014 Belgian UC cohort · species · study machiels_2014									
Roseburia hominis	not detected	7	▼	1.00	0.90	0.30	0.90	+0.97	
Faecalibacterium prausnitzii	0.010%	2	▼	1.00	0.90	0.20	0.90	+1.00	
Franzosa 2019 PRISM — UC directions · species · study franzosa_2019									
Ruminococcus gnavus	13.289%	98	▲	0.80	0.80	0.30	0.90	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.60	0.70	0.30	0.80	-0.29	
Roseburia intestinalis	not detected	13	▼	0.60	0.80	0.20	0.80	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.90	0.20	0.90	+1.00	
Bacteroides vulgatus	22.292%	93	▲	0.50	0.60	0.30	0.60	+0.41	
HMP2 IBDMDB — UC directions · species · study hmp2_2019									
Ruminococcus gnavus	13.289%	98	▲	0.90	0.80	0.30	0.90	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.70	0.80	0.30	0.80	-0.29	
Faecalibacterium prausnitzii	0.010%	2	▼	0.90	0.90	0.20	0.90	+1.00	
Roseburia intestinalis	not detected	13	▼	0.70	0.80	0.20	0.80	+0.37	
Eubacterium rectale	0.177%	35	▼	0.70	0.80	0.20	0.80	+0.17	
Akkermansia muciniphila	1.425%	91	▼	0.60	0.60	0.40	0.60	-0.46	
Community ecology · diversity · study hmp2_2019									
shannon	not detected	—	▼	0.60	0.80	0.10	0.80	—	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	0.90	0.20	0.90	+0.76	
h2s	63.8/100	28	▲	0.70	0.60	0.40	0.70	-0.19	
bsh	55.7/100	13	▼	0.40	0.60	0.30	0.60	+0.37	
Butyrate carriers and sulfate reducers · carriers · carrier proxy · study carriers									
butyrate	1.623%	4	▼	0.80	0.80	0.20	0.80	+0.53	
sulfate reducers	0.039%	70	▲	0.60	0.60	0.40	0.60	+0.18	

FINE PRINT

Confidence grade moderate. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Ulcerative colitis is diagnosed by colonoscopy and histology. This profile cannot diagnose or exclude it, and it cannot reliably separate UC from Crohn's disease.

Every direction here is shared with the general dysbiosis pattern. Read it beside the shape analysis and the crohns and ibd scores.

Sources: Machiels K, et al. A decrease of the butyrate-producing species Roseburia hominis and Faecalibacterium prausnitzii defines dysbiosis in patients with ulcerative colitis. Gut 2014;63:1275-1283; Lloyd-Price J, et al. Nature 2019;569:655-662; Franzosa EA, et al. Nat Microbiol 2019;4:293-305.

What the research says about this condition and this pattern

Inflammatory bowel disease — confirmation route and what guidelines say about FMT and probiotics

LANE A CLINICAL ROUTE

Fired by: Crohn's disease pattern: notably high; Inflammatory bowel disease (composite) pattern: notably high; Ulcerative colitis pattern: notably high

Resemblance to an IBD-cohort pattern is not a diagnosis. Persistent diarrhoea, blood in stool, weight loss or pain warrant gastroenterology evaluation with fecal calprotectin. AGA 2024 recommends against conventional FMT for Crohn's disease and ulcerative colitis outside trials, and AGA 2020 places probiotics in IBD in the trial-only category.

Studied: Gastroenterology evaluation with fecal calprotectin when symptoms warrant (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** symptom-driven gastroenterology evaluation with fecal calprotectin

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: conditional for.

- AGA 2024 conditionally supports fecal microbiota-based therapy to prevent recurrent C. difficile infection after standard antibiotics in selected immunocompetent or mildly/moderately immunocompromised adults, and recommends against it in severely immunocompromised adults. [Peery AF et al. AGA Clinical Practice Guideline on fecal microbiota-based therapies. Gastroenterology 2024;166:409] · lane A · guideline (conditional, low certainty) · benefit
- **Against or limiting —** AGA 2024 recommends against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis except within a clinical trial. [AGA 2024 — against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis outside trials] · lane X · guideline
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline

Conventional route: Gastroenterology evaluation with fecal calprotectin if symptomatic.

Fecal microbiota transplantation: guidelines recommend against it for this condition outside trials.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Crohn's” · “you have colitis” · “FMT for IBD”

Ulcerative colitis — the FMT trial evidence and its regulatory status

LANE C HUMAN TRIAL · LIMITED

Fired by: Inflammatory bowel disease (composite) pattern: notably high; Ulcerative colitis pattern: notably high

Intensive multidonor FMT produced remission in 27% versus 8% in one trial of active ulcerative colitis; guidelines still limit FMT for UC to clinical trials.

Studied: Fecal microbiota transplantation for a non-CDI indication (investigational) (fmt microbiota product) · **in:** people with clinically confirmed active ulcerative colitis, ages 18–75, in Australia, n = 81 · **studied protocol (a source fact, not an instruction):** studied protocol on file (not shown in participant mode)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial. Guideline position: conditional against.

- Intensive multidonor FMT achieved steroid-free clinical and endoscopic remission/response in 27% versus 8% with placebo in active ulcerative colitis — a positive trial signal that current guidelines still place in the trial-only category. [Paramsothy S et al. Lancet 2017;389:1218] · lane C · randomized double-blind placebo-controlled · n = 85 · benefit
- **Against or limiting —** AGA 2024 recommends against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis except within a clinical trial. [AGA 2024 — against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis outside trials] · lane X · guideline
- **Against or limiting —** FDA has documented transmission of multidrug-resistant organisms (including a death after ESBL E. coli), pathogenic E. coli, and warned about SARS-CoV-2 and mpox with FMT; donor testing, bookend testing and quarantine are required. [FDA FMT safety communications: MDRO/ESBL (June 2019, incl. one death), EPEC/STEC (March–April 2020), SARS-CoV-2 (2020), mpox (August 2022)] · lane X · safety communications · harm

Applicability: The study required a clinical diagnosis of ulcerative colitis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: A known safety consideration applies; review with a clinician before any change. FMT and FDA-approved microbiota products are clinician-only; approved products cover only prevention of recurrent CDI in adults after antibacterial treatment, and every other use is investigational under IND.

This report will not say: “FMT cures colitis”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

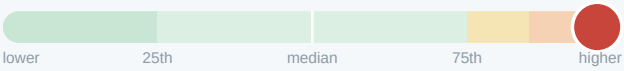
Inflammatory bowel disease (composite) — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ibd) · maturity **P3** (independent human replication with coherent task and assay) · production candidate · mixed_north_american_european_cohorts

The composite IBD pattern: what Crohn's disease and ulcerative colitis share. This is the profile validated against labelled cohorts in the profile_validation file, because the public cohorts label IBD far more often than they separate CD from UC. It scores the core replicated directions — butyrate-producer depletion, Ruminococcus gnavus and Escherichia coli expansion, diversity loss — and leaves the CD-specific oral-origin taxa and the UC-specific sulfide story to the two disease-specific profiles. When this profile is high and the shape analysis reads specific, the sample resembles the IBD dysbiotic state. When many profiles are high together it is dysbiosis of some cause, and IBD is one of many.

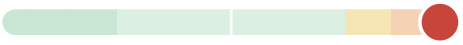
COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 99–100

What is driving it

Franzosa 2019 PRISM + Dutch validation
species · study franzosa_2019 · 7/7 features · 33% of combined



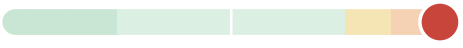
98th — NOTABLY HIGH

Hall 2017 R. gnavus clade expansion
species · study hall_2017 · 2/2 features · 33% of combined



99th — NOTABLY HIGH

HMP2 IBDMDB core dysbiosis directions
species · study hmp2_2019 · 10/10 features · 17% of combined

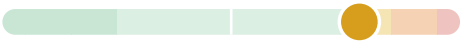


97th — NOTABLY HIGH

Community ecology
diversity · study hmp2_2019 · 1/2 features · 17% of combined

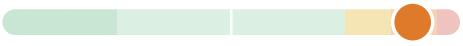
— SCORED LOW COVERAGE

Metabolite gene capacity
genes · genomic capacity · study functional · 4/4 features · not in combined



78th — ABOVE AVERAGE

Carrier cross-check
carriers · carrier proxy · study carriers · 2/2 features · not in combined



90th — HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	0.90	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), longcovid (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Franzosa 2019 PRISM + Dutch validation · species · study franzosa_2019									
Ruminococcus gnavus	13.289%	98	▲	0.90	0.80	0.30	0.90	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.80	0.80	0.30	0.90	-0.29	
Clostridium bolteae	0.056%	89	▲	0.60	0.70	0.40	0.70	+0.40	
Faecalibacterium prausnitzii	0.010%	2	▼	0.90	0.90	0.20	0.90	+1.00	
Roseburia intestinalis	not detected	13	▼	0.70	0.80	0.20	0.80	+0.37	
Alistipes shahii	0.177%	85	▼	0.50	0.60	0.40	0.70	-0.34	
Eubacterium eligens	not detected	8	▼	0.60	0.70	0.30	0.70	+1.00	
Hall 2017 R. gnavus clade expansion · species · study hall_2017									
Ruminococcus gnavus	13.289%	98	▲	1.00	0.80	0.30	0.90	+0.72	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.80	0.20	0.80	+1.00	
HMP2 IBDMDB core dysbiosis directions · species · study hmp2_2019									
Ruminococcus gnavus	13.289%	98	▲	1.00	0.90	0.30	0.90	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.90	0.90	0.30	0.90	-0.29	
Ruminococcus torques	1.283%	92	▲	0.60	0.70	0.40	0.70	+0.46	
Clostridium bolteae	0.056%	89	▲	0.60	0.70	0.40	0.70	+0.40	
Faecalibacterium prausnitzii	0.010%	2	▼	1.00	0.90	0.20	0.90	+1.00	
Roseburia hominis	not detected	7	▼	0.80	0.90	0.30	0.90	+0.97	
Roseburia intestinalis	not detected	13	▼	0.80	0.80	0.20	0.80	+0.37	
Eubacterium rectale	0.177%	35	▼	0.80	0.80	0.20	0.80	+0.17	
Alistipes putredinis	3.953%	90	▼	0.60	0.70	0.40	0.70	-0.43	
Bifidobacterium adolescentis	0.990%	85	▼	0.60	0.70	0.30	0.70	-0.35	
Community ecology · diversity · study hmp2_2019									
shannon	not detected	—	▼	0.80	0.90	0.10	0.90	—	
richness	92.00	60	▼	0.60	0.90	0.10	0.90	-0.09	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	0.90	0.20	0.90	+0.76	
bsh	55.7/100	13	▼	0.50	0.70	0.30	0.70	+0.37	
bai	8.2/100	97	▼	0.50	0.70	0.30	0.70	-1.00	
h2s	63.8/100	28	▲	0.50	0.60	0.30	0.60	-0.19	
Carrier cross-check · carriers · carrier proxy · study carriers									
butyrate	1.623%	4	▼	0.80	0.80	0.20	0.80	+0.53	
hexa lps	0.039%	40	▲	0.60	0.70	0.30	0.70	-0.08	

FINE PRINT

Confidence grade moderate. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBD is diagnosed by endoscopy, histology and imaging. This profile does not diagnose or exclude it. Calprotectin is the stool test that matters clinically, and it is a protein, not DNA.

This is the general dysbiosis pattern. Antibiotics, bowel preparation, gastroenteritis, hospitalisation and low-fibre diets all produce it.

Sources: Lloyd-Price J, et al. Nature 2019;569:655-662; Franzosa EA, et al. Nat Microbiol 2019;4:293-305; Hall AB, et al. A novel Ruminococcus gnavus clade enriched in inflammatory bowel disease patients. Genome Med 2017;9:103.

What the research says about this condition and this pattern

Inflammatory bowel disease — confirmation route and what guidelines say about FMT and probiotics

LANE A CLINICAL ROUTE

Fired by: Crohn's disease pattern: notably high; Inflammatory bowel disease (composite) pattern: notably high; Ulcerative colitis pattern: notably high

Resemblance to an IBD-cohort pattern is not a diagnosis. Persistent diarrhoea, blood in stool, weight loss or pain warrant gastroenterology evaluation with fecal calprotectin. AGA 2024 recommends against conventional FMT for Crohn's disease and ulcerative colitis outside trials, and AGA 2020 places probiotics in IBD in the trial-only category.

Studied: Gastroenterology evaluation with fecal calprotectin when symptoms warrant (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** symptom-driven gastroenterology evaluation with fecal calprotectin

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: conditional for.

- AGA 2024 conditionally supports fecal microbiota-based therapy to prevent recurrent C. difficile infection after standard antibiotics in selected immunocompetent or mildly/moderately immunocompromised adults, and recommends against it in severely immunocompromised adults. [Peery AF et al. AGA Clinical Practice Guideline on fecal microbiota-based therapies. Gastroenterology 2024;166:409] · lane A · guideline (conditional, low certainty) · benefit
- **Against or limiting —** AGA 2024 recommends against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis except within a clinical trial. [AGA 2024 — against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis outside trials] · lane X · guideline
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline

Conventional route: Gastroenterology evaluation with fecal calprotectin if symptomatic.

Fecal microbiota transplantation: guidelines recommend against it for this condition outside trials.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Crohn's” · “you have colitis” · “FMT for IBD”

Ulcerative colitis — the FMT trial evidence and its regulatory status

LANE C HUMAN TRIAL · LIMITED

Fired by: Inflammatory bowel disease (composite) pattern: notably high; Ulcerative colitis pattern: notably high

Intensive multidonor FMT produced remission in 27% versus 8% in one trial of active ulcerative colitis; guidelines still limit FMT for UC to clinical trials.

Studied: Fecal microbiota transplantation for a non-CDI indication (investigational) (fmt microbiota product) · **in:** people with clinically confirmed active ulcerative colitis, ages 18–75, in Australia, n = 81 · **studied protocol (a source fact, not an instruction):** studied protocol on file (not shown in participant mode)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial. Guideline position: conditional against.

- Intensive multidonor FMT achieved steroid-free clinical and endoscopic remission/response in 27% versus 8% with placebo in active ulcerative colitis — a positive trial signal that current guidelines still place in the trial-only category. [Paramsothy S et al. Lancet 2017;389:1218] · lane C · randomized double-blind placebo-controlled · n = 85 · benefit
- **Against or limiting —** AGA 2024 recommends against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis except within a clinical trial. [AGA 2024 — against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis outside trials] · lane X · guideline
- **Against or limiting —** FDA has documented transmission of multidrug-resistant organisms (including a death after ESBL E. coli), pathogenic E. coli, and warned about SARS-CoV-2 and mpox with FMT; donor testing, bookend testing and quarantine are required. [FDA FMT safety communications: MDRO/ESBL (June 2019, incl. one death), EPEC/STEC (March–April 2020), SARS-CoV-2 (2020), mpox (August 2022)] · lane X · safety communications · harm

Applicability: The study required a clinical diagnosis of ulcerative colitis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: A known safety consideration applies; review with a clinician before any change. FMT and FDA-approved microbiota products are clinician-only; approved products cover only prevention of recurrent CDI in adults after antibacterial treatment, and every other use is investigational under IND.

This report will not say: “FMT cures colitis”

Crohn's disease — dietary therapy with randomized evidence (clinician-confirmed disease)

LANE C HUMAN TRIAL · LIMITED

Fired by: Crohn's disease pattern: notably high; Inflammatory bowel disease (composite) pattern: notably high

The Crohn's Disease Exclusion Diet with partial enteral nutrition induced sustained remission in children with confirmed mild-to-moderate Crohn's disease. It is delivered by an IBD team, not started from a stool pattern.

Studied: Crohn's disease exclusion diet with partial enteral nutrition (CDED + PEN) (diet) · **in:** people with clinically confirmed mild-to-moderate paediatric Crohn's disease, ages 4–18, in multinational, n = 78 · **studied protocol (a source fact, not an instruction):** CDED + 50% PEN for 6 weeks, then CDED + 25% PEN to week 12, versus exclusive enteral nutrition then free diet + 25% PEN

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In children with mild-to-moderate Crohn's disease, CDED + partial enteral nutrition was better tolerated than exclusive enteral nutrition and induced sustained remission in more patients at 12 weeks. This is guideline-care context for clinician-confirmed disease, not an action from a stool pattern. [Levine A et al. Gastroenterology 2019;157:440] · lane C · randomized controlled, paediatric · n = 78 · benefit

Applicability: The study required a clinical diagnosis of crohns disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “diet cures Crohn's”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

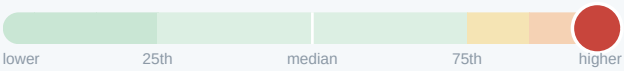
Long COVID chronic-state community similarity

NOTABLY HIGH

Pattern version **0.2.0** (longcovid) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · research only · mixed_hong_kong_us_cohorts

This scores resemblance to the chronic-state pattern: stool collected at least three months after infection with ongoing symptoms. Hong Kong and US cohorts agree on the core direction — depletion of the butyrate producers Faecalibacterium prausnitzii, Eubacterium rectale and Roseburia, with Ruminococcus gnavus and Bacteroides vulgatus enriched — and on reduced short-chain fatty acid and isoleucine synthesis capacity. Predicting future risk from stool collected in the first two months after infection is a separate task with different features, and those features are not used here. In the largest independent study (799 outpatients), species-level features reached AUROC 0.62 against 0.72 for clinical variables alone, so the pattern is read as a supporting signal alongside the clinical picture rather than in place of it.

COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 90–100

What is driving it

Taxonomic
species · study taxonomic · 10/10 features · 50% of combined
99th — NOTABLY HIGH

Functional
genes · genomic capacity · study functional · 2/2 features · 35% of combined
>99th — NOTABLY HIGH

Ecological
diversity · study ecological · 1/1 features · 15% of combined
39th — TYPICAL

Phenotype
diversity · study phenotype · 0/2 features · not in combined
not measured
— NOT COMPUTABLE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	0.40	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Taxonomic · species · study taxonomic									
Ruminococcus gnavus	13.289%	98	▲	1.00	0.40	0.50	0.90	+0.72	Liu 2022 Gut, Liu 2022 Nat Commun, Su 2024 Cell Host Microbe

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Clostridium bolteae	0.056%	89	▲	1.00	0.40	0.50	0.90	**+0.40**	Liu 2022 Gut, Su 2024 Cell Host Microbe
Clostridium innocuum	0.086%	95	▲	1.00	0.40	0.50	0.90	**+0.56**	Liu 2022 Gut, Su 2024 Cell Host Microbe
Faecalibacterium prausnitzii	0.010%	2	▼	1.00	0.50	0.30	0.90	**+1.00**	Liu 2022 Gut, Zhang 2022 Gastroenterology
Bifidobacterium adolescentis	0.990%	85	▼	1.00	0.40	0.50	0.90	**-0.35**	Liu 2022 Gut, Liu 2022 Nat Commun
Bifidobacterium pseudocatenulatum	not detected	32	▼	1.00	0.40	0.50	0.90	**+0.16**	Liu 2022 Gut, Liu 2022 Nat Commun
Erysipelatoclostridium ramosum	0.720%	97	▲	0.50	0.40	0.50	0.80	**+0.65**	Su 2024 Cell Host Microbe
Bacteroides vulgatus	22.292%	93	▲	0.50	0.40	0.30	0.70	**+0.41**	Liu 2022 Nat Commun
Flavonifractor plautii	0.964%	93	▲	0.50	0.40	0.50	0.80	**+0.49**	Su 2024 Cell Host Microbe
Gemmiger formicilis	not detected	10	▼	0.50	0.40	0.50	0.80	**+0.42**	Liu 2022 Gut
Functional · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	0.60	0.40	0.90	**+0.76**	Zhang 2022 Gastroenterology, Liu 2022 Gut
indole	11.6/100	42	▼	0.50	0.50	0.50	0.80	**+0.06**	Zhang 2022 Gastroenterology
Ecological · diversity · study ecological									
richness	92.00	60	▼	0.50	0.60	0.20	0.60	**-0.09**	Hamrefors 2024 Sci Rep (lower richness, no species survived correction)
Phenotype · diversity · study phenotype									
Actinomyces naeslundii	not detected	—	▲	0.25	0.40	0.40	0.70	—	Su 2024 Cell Host Microbe (fatigue / neuropsychiatric)
Streptococcus anginosus group	not detected	—	▲	0.25	0.40	0.30	0.50	—	Liu 2022 Gut (respiratory-associated)

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

The 50/35/15 module split is an engineering starting point, not an evidence-derived weighting.

Measured on labelled cohorts (AUC, 0.5 = chance, 1.0 = perfect): IBD 0.80 (0.76–0.83); T2D 0.59 (0.53–0.64); CRC 0.53 (0.48–0.59). No labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.53-0.80 against other conditions).

Chronic-state resemblance only. Features from acute-phase (0-2 month) risk-prediction studies are not included and must not be read into a chronic sample.

Most taxonomic features originate in one institutional recruitment family; their cohort-independence factor is set to reflect roughly one independent vote rather than the number of papers.

E. coli, Klebsiella and antimicrobial-resistance genes track ICU exposure and antibiotics rather than the syndrome, and are deliberately excluded. Flagellin potential is shared with rheumatoid arthritis, ankylosing spondylitis and IBD, and is excluded for the same reason.

The two-year Hainan cohort (Zhang et al. 2026) is n=11 per arm with uninfected rather than recovered controls, and reports no deposited data, so its strain-level calls are neither reusable nor portable across database versions. It is treated as a hypothesis source only and none of its features carry weight here.

Sources: Liu Q, Mak JWY, Su Q, et al. Gut 2022;71:544-552, doi:10.1136/gutjnl-2021-325989 (PRJNA714459). Liu Q, Su Q, Zhang F, et al. Nat Commun 2022;13:6806, doi:10.1038/s41467-022-34535-8 (PRJNA876804). Su Q, et al. Cell Host Microbe 2024;32:651-660 (PRJNA1004982). Zhang F, et al. Gastroenterology 2022;162:548-561 (PRJNA689961). Comba IY, et al. Gut Microbes 2026 (PRJNA1167328).

What the research says about this condition and this pattern

Long COVID — what human trials have tested LANE C HUMAN TRIAL · LIMITED

Fired by: Long COVID chronic-state community similarity pattern: notably high

Resemblance to a published Long-COVID stool signature is a research observation. The only randomized evidence with a clinical endpoint is for one exact synbiotic (SIM01) in people with a clinical Long-COVID diagnosis; it does not transfer to other probiotics.

Studied: SIM01 synbiotic (Bifidobacterium adolescentis, B. bifidum, B. longum with galacto-, xylo- oligosaccharides and resistant dextrin; Hong Kong) (probiotic) · **in:** people with clinically confirmed post-acute COVID-19 syndrome (PACS), ages 18–99, in Hong Kong, n = 463 · **studied protocol (a source fact, not an instruction):** SIM01 10 billion CFU in sachets twice daily for six months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial.

- In 463 adults with clinically defined post-acute COVID-19 syndrome, six months of the exact SIM01 synbiotic improved several patient-reported symptoms versus placebo; immunocompromised, pregnant and breastfeeding people were excluded and the product's exact composition is proprietary. [Lau RI et al. Lancet Infect Dis 2024;24:256 (RECOVERY. SIM01)] · lane C · triple-blind placebo-controlled RCT, single centre (Hong Kong) · n = 463 · benefit
- A smaller randomized trial (STOP-FATIGUE) of a different multistrain product reported reduced Long-COVID fatigue; it does not transfer to other products and the classifier does not predict response. [STOP-FATIGUE trial (2024)] · lane C · small randomized trial · benefit
- **Against or limiting —** The 2026 two-year Long-COVID study is an 11-person mechanistic cohort with mouse transfer and no public data; its taxonomic directions are a candidate module, not evidence for any antimicrobial, probiotic or FMT treatment. [Zhang et al. Probiotics Antimicrob Proteins 2026 (11 two-year Long-COVID participants, mouse transfer)] · lane X · 11 cases vs 11 controls with fecal transfer to antibiotic-treated mice

Applicability: The study required a clinical diagnosis of post acute covid19 syndrome. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “treats Long COVID” · “clears the virus” · “antimicrobials for Long COVID”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

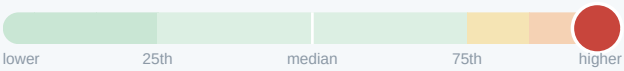
Ankylosing spondylitis — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ankylosing_spondylitis) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · chinese_cohort_derived

Up to 60% of ankylosing spondylitis patients have subclinical gut inflammation, and about 10% develop overt IBD, so the AS gut pattern overlaps with Crohn's disease. Wen et al. shotgun-sequenced 97 AS patients and 114 controls in China and found Prevotella species, Bacteroides coprophilus and Bifidobacterium enriched with Roseburia, Faecalibacterium and Eubacterium depleted; Zhou et al. replicated in a second Chinese cohort with Bifidobacterium and Klebsiella higher, and Costello's Australian ileal biopsy study found Lachnospiraceae, Ruminococcaceae, Bacteroidaceae and Porphyromonadaceae differences. The HLA-B27 host genotype itself shapes the gut community. TNF inhibitors, the standard AS treatment, partly normalise the pattern.

COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 96–100

What is driving it

Wen 2017 Chinese AS shotgun (97 / 114) species · study wen_2017 · 8/8 features · 50% of combined	97th	NOTABLY HIGH
Zhou 2020 Chinese AS shotgun (replication) species · study zhou_2020 · 5/5 features · 50% of combined	>99th	NOTABLY HIGH
Metabolite gene capacity genes · genomic capacity · study functional · 1/1 features · not in combined	99th	NOTABLY HIGH
Hexa-LPS and butyrate carriers carriers · carrier proxy · study carriers · 2/2 features · not in combined	86th	HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Wen 2017 Chinese AS shotgun (97 / 114) · species · study wen_2017									

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Prevotella copri	not detected	25	▲	0.80	0.70	0.30	0.70	-0.23	
Prevotella stercorea	not detected	40	▲	0.50	0.60	0.40	0.60	-0.08	
Bacteroides coprophilus	not detected	47	▲	0.60	0.60	0.50	0.60	-0.02	
Bifidobacterium adolescentis	0.990%	85	▲	0.60	0.60	0.40	0.60	+0.35	
Roseburia intestinalis	not detected	13	▼	0.70	0.70	0.30	0.70	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.70	0.20	0.70	+1.00	
Eubacterium rectale	0.177%	35	▼	0.60	0.70	0.20	0.70	+0.17	
Eubacterium eligens	not detected	8	▼	0.50	0.60	0.40	0.70	+1.00	
Zhou 2020 Chinese AS shotgun (replication) · species · study zhou_2020									
Bifidobacterium adolescentis	0.990%	85	▲	0.60	0.60	0.40	0.60	+0.35	
Bifidobacterium longum	4.157%	95	▲	0.50	0.60	0.30	0.60	+0.56	
Klebsiella pneumoniae	not detected	40	▲	0.60	0.60	0.40	0.60	-0.09	
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.70	0.20	0.70	+1.00	
Roseburia hominis	not detected	7	▼	0.50	0.60	0.30	0.70	+0.97	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	0.80	0.70	0.20	0.70	+0.76	
Hexa-LPS and butyrate carriers · carriers · carrier proxy · study carriers									
hexa lps	0.039%	40	▲	0.60	0.60	0.30	0.60	-0.08	
butyrate	1.623%	4	▼	0.60	0.70	0.20	0.70	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Ankylosing spondylitis is diagnosed by imaging of the sacroiliac joints, HLA-B27 and clinical criteria. This profile does not diagnose it.

The Prevotella direction is confounded with diet and geography; a high P. copri sample from a high-fibre diet will score toward AS here for reasons that have nothing to do with joints.

Sources: Wen C, et al. Quantitative metagenomics reveals unique gut microbiome biomarkers in ankylosing spondylitis. Genome Biol 2017;18:142; Zhou C, et al. Metagenomic profiling of the pro-inflammatory gut microbiota in ankylosing spondylitis. J Autoimmun 2020;107:102360; Costello ME, et al. Brief report: intestinal dysbiosis in ankylosing spondylitis. Arthritis Rheumatol 2015;67:686.

What the research says about this condition and this pattern

Ankylosing-spondylitis pattern — no supported targeted intervention NONE NO SUPPORTED INTERVENTION

Fired by: Ankylosing spondylitis pattern: notably high

The one randomized probiotic trial in ankylosing spondylitis was null and FMT is case-level. The condition is diagnosed and treated by rheumatology.

No supported targeted intervention — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- **Against or limiting —** A 63-person randomized probiotic trial in ankylosing spondylitis found no benefit. [Jenks K et al. J Rheumatol 2010;37:2118] · lane X · randomized double-blind placebo-controlled · n = 63 · null
- **Against or limiting —** FMT in ankylosing spondylitis is case-level and investigational. [FMT case in ankylosing spondylitis (2023)] · lane X · case report

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have AS”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

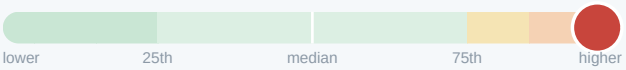
Preclinical amyloid positivity — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ad_preclinical_amyloid) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · us_cognitively_unimpaired_adults

This is not an Alzheimer's disease label. Ferreiro et al. compared 49 amyloid-positive with 115 amyloid-negative participants who were all cognitively unimpaired — the question is whether the gut community differs in people carrying amyloid before any symptoms, which is a different question from whether it differs in people with the disease. Effect sizes at this stage are the smallest in the continuum, which is what you would expect and what the profile's low weights encode.

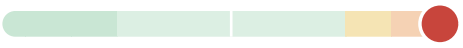
COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 95–100

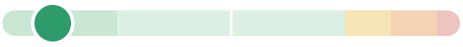
What is driving it

Ferreiro 2023 preclinical amyloid cohort
species · study ferreiro_2023 · 4/4 features · 100% of combined



>99th NOTABLY HIGH

Amyloid-relevant metabolite capacity
genes · genomic capacity · study amyloid_mechanism · 1/1 features · not in combined



11th LOW

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Ferreiro 2023 preclinical amyloid cohort · species · study ferreiro_2023									
Dorea formicigenerans	0.359%	89	▲	0.60	0.40	0.40	0.60	+0.38	Ferreiro PMC10680783
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.40	0.20	0.60	+1.00	Ferreiro PMC10680783
Eubacterium eligens	not detected	8	▼	0.50	0.40	0.30	0.50	+1.00	Ferreiro PMC10680783
Bacteroides uniformis	6.597%	82	▲	0.40	0.30	0.30	0.50	+0.26	Ferreiro PMC10680783
Amyloid-relevant metabolite capacity · genes · genomic capacity · study amyloid_mechanism									
ipa	9.6/100	89	▼	0.70	0.50	0.40	0.70	-0.71	Pappolla 2021 Neurobiol Dis 156:105403

FINE PRINT

Confidence grade moderate. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Amyloid positivity is established by PET imaging or cerebrospinal fluid assay. Many amyloid-positive people never develop dementia. This profile is not a substitute for either measurement and not a prediction of either outcome.

A separate profile from clinical Alzheimer's disease and from mild cognitive impairment by design. The three are different prediction tasks and their scores must not be averaged or read as one continuum reading.

Sources: Ferreiro AL, et al. Gut microbiome composition may be an indicator of preclinical Alzheimer's disease. PMC10680783; data PRJNA798058.

What the research says about this condition and this pattern

Cognition — what has replicated evidence independent of the microbiome

LANE B · HUMAN EVIDENCE · STRONG

Fired by: Clinical Alzheimer's disease pattern: notably high; Mild cognitive impairment pattern: notably high; Preclinical amyloid positivity pattern: notably high

Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition in at-risk older adults; the MIND diet did not beat a control diet.

Studied: FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk monitoring) (lifestyle) · **in:** people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognition, ages 60–77, in Finland, n = 1260 · **studied protocol (a source fact, not an instruction):** two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring)

Evidence lane B: replicated human randomized clinical benefit for the exact intervention and population. Replication: single study. Strongest design: randomized trial.

- The FINGER trial showed that a two-year multidomain lifestyle programme (diet, exercise, cognitive training, vascular risk control) improved or maintained cognition in at-risk older adults — evidence independent of the microbiome. [Ngandu T et al. Lancet 2015;385:2255 (FINGER)] · lane B · randomized controlled, 2 years · n = 1260 · benefit
- **Against or limiting —** The MIND-diet trial did not improve cognition more than a control diet with the same mild caloric restriction over three years. [Barnes LL et al. N Engl J Med 2023;389:602 (MIND diet)] · lane X · randomized controlled, 3 years · n = 604 · null

Conventional route: Neurologic/cognitive evaluation only if memory or thinking symptoms are present.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Alzheimer's” · “prevents dementia”

Alzheimer disease — probiotic trials, positive and null

LANE C · HUMAN TRIAL · LIMITED

Fired by: Clinical Alzheimer's disease pattern: notably high; Mild cognitive impairment pattern: notably high; Preclinical amyloid positivity pattern: notably high

One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level only.

Studied: Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. fermentum (Akbari 2016) (probiotic) · **in:** people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60 · **studied protocol (a source fact, not an instruction):** 200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- One 60-person 12-week trial of a four-organism probiotic milk reported MMSE and metabolic-marker changes in people with Alzheimer disease. [Akbari E et al. Front Aging Neurosci 2016;8:256] · lane C · randomized double-blind controlled · n = 60 · benefit
- **Against or limiting —** A later trial in severe Alzheimer disease found no cognitive benefit from probiotics. [Agahi A et al. Front Neurol 2018 (severe AD, negative)] · lane X · randomized double-blind placebo-controlled · null
- **Against or limiting —** Conservative pooled analyses find the probiotic evidence in Alzheimer disease inconsistent and not robust. [meta-analysis of probiotics in Alzheimer disease/MCI (2021)] · lane X · systematic review / meta-analysis · mixed
- **Against or limiting —** Probiotic trials in mild cognitive impairment are small and inconsistent. [2024 evidence review — probiotics in mild cognitive impairment] · lane X · review · mixed

Applicability: The study required a clinical diagnosis of alzheimer disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “reverses dementia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

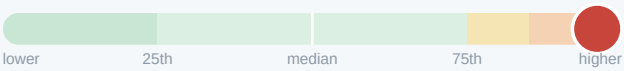
Colorectal adenoma — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (adenoma) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_international_cohorts

Adenoma is the precursor lesion for most colorectal cancer, and the interesting finding is a negative one: the strong CRC stool signature — Fusobacterium nucleatum, Parvimonas micra, Peptostreptococcus stomatis, Gemella — is weak or absent in adenoma. Both large cross-cohort meta-analyses (Wirbel 2019, Thomas 2019) found adenoma barely separable from controls by stool metagenomics, with AUCs around 0.5-0.6. That makes adenoma the right challenge set for the CRC profile: a CRC score that is also high in adenoma is reading general dysbiosis rather than cancer. The modest adenoma-associated directions that do replicate are here, with low weights, because that is what the evidence supports.

COMBINED RESEMBLANCE
percentile of the matched reference group



>99th
likely range 94–100

What is driving it

Feng 2015 Austrian adenoma / carcinoma shotgun species · study feng_2015 · 5/5 features · 50% of combined		88th	HIGH
Thomas 2019 cross-cohort adenoma (7 cohorts) species · study thomas_2019 · 3/3 features · 50% of combined		99th	NOTABLY HIGH
Carcinogenic metabolite capacity genes · genomic capacity · study functional · 3/3 features · not in combined		97th	NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Feng 2015 Austrian adenoma / carcinoma shotgun · species · study feng_2015									
Bacteroides dorei	not detected	21	▲	0.50	0.60	0.40	0.60	-0.27	
Bacteroides massiliensis	0.421%	90	▲	0.50	0.60	0.40	0.60	+0.42	
Bilophila wadsworthia	0.039%	83	▲	0.50	0.60	0.40	0.60	+0.32	
Alistipes putredinis	3.953%	90	▲	0.40	0.60	0.40	0.50	+0.43	
Fusobacterium nucleatum	not detected	49	▲	0.40	0.60	0.40	0.50	-0.01	

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Thomas 2019 cross-cohort adenoma (7 cohorts) · species · study thomas_2019									
Ruminococcus gnavus	13.289%	98	▲	0.40	0.60	0.20	0.50	+0.72	
Eubacterium eligens	not detected	8	▼	0.40	0.60	0.30	0.60	+1.00	
Roseburia intestinalis	not detected	13	▼	0.40	0.60	0.30	0.60	+0.37	
Carcinogenic metabolite capacity · genes · genomic capacity · study functional									
bai	8.2/100	97	▲	0.60	0.60	0.40	0.60	+1.00	
h2s	63.8/100	28	▲	0.50	0.50	0.40	0.50	-0.19	
butyrate	11.6/100	1	▼	0.60	0.60	0.20	0.70	+0.76	

FINE PRINT

Confidence grade moderate. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Adenomas are found and removed at colonoscopy. No stool DNA test replaces screening colonoscopy or FIT.

Adenoma is nearly inseparable from control by stool metagenomics. A low score here means nothing; a high score means dysbiosis. The profile is in the library chiefly as the challenge set for colorectal_cancer.

Sources: Feng Q, et al. Gut microbiome development along the colorectal adenoma- carcinoma sequence. Nat Commun 2015;6:6528; Thomas AM, et al. Metagenomic analysis of colorectal cancer datasets identifies cross-cohort microbial diagnostic signatures and a link with choline degradation. Nat Med 2019;25:667-678; Wirbel J, et al. Nat Med 2019;25:679-689.

What the research says about this condition and this pattern

Colorectal cancer / adenoma pattern — the screening route is the action LANE A CLINICAL ROUTE

Fired by: Colorectal adenoma pattern: notably high

A stool pattern resembling colorectal-cancer or adenoma cohorts does not detect cancer. Guideline screening (FIT or colonoscopy, ages 45–75) is the established route, and whole grains, fibre, physical activity and limiting red/processed meat and alcohol carry population-level prevention evidence.

Studied: Age-appropriate colorectal cancer screening (USPSTF) (clinical route) · **in:** ages 45–75 · **studied protocol (a source fact, not an instruction):** screening per USPSTF (stool-based test or colonoscopy at guideline intervals)

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: strong for.

- Colorectal cancer screening for adults 45–75 is an established, guideline-backed route. A stool microbiome pattern does not substitute for FIT or colonoscopy and does not change the recommended interval. [USPSTF 2021 colorectal cancer screening recommendation] · lane A · guideline · benefit
- Population evidence links whole grains, dietary fibre and physical activity to lower colorectal cancer risk, and red/processed meat, alcohol and excess body weight to higher risk. This is general prevention evidence, not a microbiome-directed treatment. [WCRF/AICR Continuous Update Project — colorectal cancer] · lane D · systematic review of population evidence · benefit
- **Against or limiting —** Pooled adenoma trials of assorted probiotics are heterogeneous and inconclusive. [2024 meta-analysis — probiotics and colorectal adenoma] · lane X · meta-analysis · mixed
- **Against or limiting —** Periodontal treatment reduced stool F. nucleatum in CRC patients, but no trial shows that lowering it prevents cancer; screening remains conventional. [periodontal treatment lowered stool F. nucleatum in CRC patients (2021)] · lane X · before-after · benefit

Conventional route: Age-appropriate colorectal cancer screening per USPSTF; earlier if symptoms or family history.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have cancer” · “probiotics prevent colon cancer” · “kill Fusobacterium”

Adenoma — one exact-product probiotic trial after polypectomy

LANE C HUMAN TRIAL · LIMITED

Fired by: Colorectal adenoma pattern: notably high

Early, conflicting evidence for the exact Clostridium butyricum MIYAIRI 588 product after polypectomy; pooled adenoma probiotic trials are inconclusive and surveillance intervals do not change.

Studied: Clostridium butyricum MIYAIRI 588 (CBM588) (probiotic) · **in:** people with clinically confirmed history of colorectal adenoma after polypectomy, ages 40–80, in Japan · **studied protocol (a source fact, not an instruction):** CBM588 tablets daily as studied (source-reported dose)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- One randomized trial of the exact Clostridium butyricum MIYAIRI 588 product after polypectomy is early/conflicting evidence; generic C. butyricum is not a substitute and no surveillance interval changes. [CBM588 randomized trial in post-polypectomy adenoma (2025)] · lane C · randomized controlled · mixed

- **Against or limiting —** Pooled adenoma trials of assorted probiotics are heterogeneous and inconclusive. [2024 meta-analysis — probiotics and colorectal adenoma] · lane X · meta-analysis · mixed

Applicability: The study required a clinical diagnosis of colorectal adenoma. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “prevents polyps”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

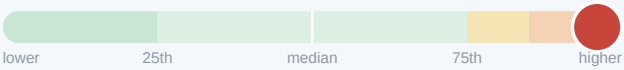
Chronic kidney disease — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ckd) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_chinese_european_cohorts

Kidney failure changes the gut chemically before it changes it ecologically: urea diffuses into the colon and is hydrolysed to ammonia by bacterial urease, raising luminal pH and selecting for urease-, uricase- and indole- and p-cresol-forming organisms (Wong 2014). Those uraemic toxins — indoxyl sulfate and p-cresyl sulfate — are made by gut bacteria from tryptophan and tyrosine and accumulate as filtration falls. Ren et al. and Wang et al. shotgun cohorts show Enterobacteriaceae and oral-origin taxa enriched with butyrate producers depleted, and the pattern tracks eGFR stage. Phosphate binders, potassium restriction and the low-protein renal diet each reshape the gut.

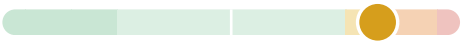
COMBINED RESEMBLANCE
percentile of the matched reference group



99th
likely range 92–100

What is driving it

Ren 2020 Chinese CKD shotgun
species · study ren_2020 · 7/7 features · 50% of combined



82nd — ABOVE AVERAGE

Wang 2020 end-stage renal disease shotgun + metabolomics
species · study wang_2020 · 7/7 features · 50% of combined



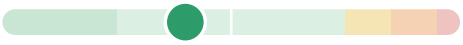
>99th — NOTABLY HIGH

Uraemic toxin and metabolite capacity
genes · genomic capacity · study functional · 4/4 features · not in combined



58th — TYPICAL

Oral-origin and hexa-LPS carriers
carriers · carrier proxy · study carriers · 3/3 features · not in combined



40th — TYPICAL

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), uc (>99th), ibd (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Ren 2020 Chinese CKD shotgun · species · study ren_2020									

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Escherichia coli	not detected 0.033% pooled	19	▲	0.80	0.70	0.30	0.80	-0.29	
Klebsiella pneumoniae	not detected	40	▲	0.70	0.60	0.40	0.70	-0.09	
Enterococcus faecalis	not detected	47	▲	0.60	0.60	0.40	0.60	-0.02	
Streptococcus salivarius	not detected	15	▲	0.50	0.60	0.30	0.60	-0.35	
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.70	0.20	0.80	+1.00	
Roseburia intestinalis	not detected	13	▼	0.70	0.70	0.30	0.70	+0.37	
Eubacterium rectale	0.177%	35	▼	0.60	0.70	0.20	0.70	+0.17	
Wang 2020 end-stage renal disease shotgun + metabolomics · species · study wang_2020									
Eggerthella lenta	0.137%	92	▲	0.70	0.70	0.50	0.70	+0.46	
Flavonifractor plautii	0.964%	93	▲	0.60	0.60	0.50	0.60	+0.49	
Alistipes shahii	0.177%	85	▲	0.40	0.60	0.40	0.50	+0.34	
Fusobacterium nucleatum	not detected	49	▲	0.40	0.60	0.30	0.50	-0.01	
Prevotella copri	not detected	25	▼	0.50	0.60	0.30	0.60	+0.23	
Roseburia hominis	not detected	7	▼	0.60	0.70	0.30	0.70	+0.97	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.70	0.20	0.80	+1.00	
Uraemic toxin and metabolite capacity · genes · genomic capacity · study functional									
urease	5.2/100	8	▲	1.00	0.70	0.50	0.80	-0.34	
butyrate	11.6/100	1	▼	0.70	0.70	0.20	0.70	+0.76	
cutc	4.1/100	77	▲	0.50	0.60	0.40	0.60	+0.39	
ipa	9.6/100	89	▼	0.40	0.50	0.30	0.50	-0.71	
Oral-origin and hexa-LPS carriers · carriers · carrier proxy · study carriers									
hexa lps	0.039%	40	▲	0.60	0.60	0.30	0.60	-0.08	
oral origin	0.000%	10	▲	0.50	0.60	0.30	0.60	-0.76	
butyrate	1.623%	4	▼	0.60	0.70	0.20	0.70	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Kidney function is measured by eGFR and urine albumin. Both are cheap blood and urine tests. This profile adds nothing to them.

The renal diet produces most of this pattern on its own.

Sources: Wang X, et al. Aberrant gut microbiota alters host metabolome and impacts renal failure in humans and rodents. Gut 2020;69:2131-2142; Ren Z, et al. Alterations of the human gut microbiome in chronic kidney disease. Front Cell Infect Microbiol 2020;10:5; Wong J, et al. Expansion of urease- and uricase-containing, indole- and p-cresol-forming and contraction of short-chain fatty acid-producing intestinal microbiota in ESRD. Am J Nephrol 2014;39:230.

What the research says about this condition and this pattern

Chronic kidney disease pattern — kidney function is measured in blood and urine LANE A CLINICAL ROUTE

Fired by: Chronic kidney disease pattern: notably high

CKD is staged by eGFR and albuminuria. In established CKD, synbiotic and resistant-starch trials changed some uraemic-toxin surrogates without hard renal outcomes; any diet change in CKD needs renal-dietitian review of potassium and phosphorus.

Studied: CKD evaluation and management per KDIGO 2024 (clinical route) · **in:** ages 18–99 · **studied protocol (a source fact, not an instruction):** eGFR, albuminuria, electrolytes and management per KDIGO 2024

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: conflicting. Strongest design: guideline. Guideline position: strong for.

- CKD is diagnosed and staged by eGFR and albuminuria and managed under KDIGO 2024; diet changes need eGFR, potassium, phosphorus and renal-dietitian review. [KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of CKD] · lane A · guideline · benefit
- In 37 pre-dialysis CKD patients, six weeks of the SYNERGY synbiotic reduced serum p-cresyl sulfate but not indoxyl sulfate — surrogate evidence that applies only with established CKD and a measured serum PCS. [Rossi M et al. Clin J Am Soc Nephrol 2016;11:223 (SYNERGY)] · lane C · randomized double-blind placebo-controlled crossover · n = 37 · mixed
- Pooled biotic trials in CKD show heterogeneous surrogate-marker changes without hard renal outcomes. [meta-analysis — probiotics/prebiotics/synbiotics in CKD (2021)] · lane C · meta-analysis · mixed
- **Against or limiting —** A 16-week resistant-starch CKD trial lowered p-cresyl sulfate but not other biomarkers or vascular function, and reduced several alpha-diversity metrics. [Headley SA et al. 2024 — resistant starch in CKD] · lane X · randomized controlled · mixed

Applicability: The study required a clinical diagnosis of chronic kidney disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: eGFR, urine albumin/creatinine and management per KDIGO 2024.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have kidney disease” · “kidney detox”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

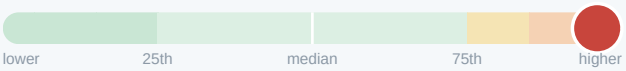
Crohn's disease — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (crohns) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_north_american_european_cohorts

The best-replicated disease pattern in the gut microbiome literature after colorectal cancer, with independent shotgun cohorts in North America and Europe agreeing on direction for a core set of species: depletion of Faecalibacterium prausnitzii and the Roseburia and Eubacterium butyrate producers, and expansion of Escherichia coli, Ruminococcus gnavus and the oral-origin Veillonella and Haemophilus. Treatment-naive paediatric cohorts show the same pattern, so it is not a treatment artefact. The pattern is strongest in a dysbiotic subgroup. The HMP2 study found IBD patients moved in and out of a dysbiotic state over time, and at any one sampling many Crohn's patients look like controls. A high score here is also the general dysbiosis pattern, so read it with the shape analysis.

COMBINED RESEMBLANCE
percentile of the matched reference group



99th
likely range 74–100

What is driving it

Gevers 2014 RISK treatment-naive paediatric cohort
species · study gevers_2014 · 10/10 features · 25% of combined
76th — ABOVE AVERAGE

Franzosa 2019 PRISM + validation cohort
species · study franzosa_2019 · 7/7 features · 25% of combined
95th — HIGH

Pascal 2017 Spanish / Belgian / German cohorts
species · association only · 16S study · study pascal_2017 · 7/7 features · 25% of combined
97th — NOTABLY HIGH

HMP2 IBDMDB longitudinal multi-omics
species · study hmp2_2019 · 8/8 features · 12% of combined
98th — NOTABLY HIGH

Community ecology
diversity · study hmp2_2019 · 1/2 features · 12% of combined
not measured
— SCORED LOW COVERAGE

Metabolite gene capacity
genes · genomic capacity · study functional · 5/5 features · not in combined
58th — TYPICAL

Butyrate carriers and hexa-LPS carriers
carriers · carrier proxy · study carriers · 2/2 features · not in combined
90th — HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	0.93	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), longcovid (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Gevers 2014 RISK treatment-naive paediatric cohort · species · study gevers_2014									
Escherichia coli	not detected 0.033% pooled	19	▲	1.00	0.80	0.30	0.90	-0.29	
Veillonella parvula	not detected	31	▲	0.80	0.70	0.40	0.80	-0.17	
Haemophilus parainfluenzae	not detected	29	▲	0.80	0.70	0.40	0.80	-0.18	
Fusobacterium nucleatum	not detected	49	▲	0.60	0.60	0.30	0.70	-0.01	
Faecalibacterium prausnitzii	0.010%	2	▼	1.00	0.90	0.20	0.90	+1.00	
Bifidobacterium adolescentis	0.990%	85	▼	0.70	0.70	0.30	0.80	-0.35	
Eubacterium rectale	0.177%	35	▼	0.80	0.80	0.20	0.80	+0.17	
Roseburia intestinalis	not detected	13	▼	0.80	0.80	0.20	0.80	+0.37	
Coprococcus comes	0.713%	92	▼	0.60	0.70	0.30	0.70	-0.48	
Dorea longicatena	not detected	6	▼	0.50	0.60	0.30	0.70	+1.00	
Franzosa 2019 PRISM + validation cohort · species · study franzosa_2019									
Ruminococcus gnavus	13.289%	98	▲	0.90	0.80	0.30	0.90	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.80	0.80	0.30	0.90	-0.29	
Clostridium bolteae	0.056%	89	▲	0.60	0.70	0.40	0.70	+0.40	
Roseburia intestinalis	not detected	13	▼	0.70	0.80	0.20	0.80	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.90	0.90	0.20	0.90	+1.00	
Alistipes putredinis	3.953%	90	▼	0.60	0.70	0.40	0.70	-0.43	
Alistipes shahii	0.177%	85	▼	0.50	0.60	0.40	0.70	-0.34	
Pascal 2017 Spanish / Belgian / German cohorts · species · association only · 16S study · study pascal_2017									
Fusobacterium nucleatum	not detected	49	▲	0.60	0.60	0.30	0.70	-0.01	
Escherichia coli	not detected 0.033% pooled	19	▲	0.70	0.70	0.30	0.80	-0.29	
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.70	0.20	0.80	+1.00	
Methanobrevibacter smithii	not detected	35	▼	0.70	0.70	0.50	0.80	+0.13	
Christensenella minuta	not detected	46	▼	0.60	0.60	0.50	0.70	+0.03	
Collinsella aerofaciens	1.940%	88	▼	0.50	0.60	0.40	0.70	-0.35	
Anaerostipes hadrus	not detected	4	▼	0.60	0.60	0.30	0.70	+1.00	
HMP2 IBDMDB longitudinal multi-omics · species · study hmp2_2019									
Ruminococcus gnavus	13.289%	98	▲	1.00	0.90	0.30	0.90	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.90	0.80	0.30	0.90	-0.29	
Ruminococcus torques	1.283%	92	▲	0.60	0.70	0.40	0.70	+0.46	
Clostridium bolteae	0.056%	89	▲	0.60	0.70	0.40	0.70	+0.40	
Faecalibacterium prausnitzii	0.010%	2	▼	1.00	0.90	0.20	0.90	+1.00	
Roseburia hominis	not detected	7	▼	0.70	0.80	0.30	0.80	+0.97	
Alistipes putredinis	3.953%	90	▼	0.60	0.70	0.40	0.70	-0.43	
Eubacterium siraeum	not detected	22	▼	0.50	0.60	0.40	0.70	+0.26	
Community ecology · diversity · study hmp2_2019									
shannon	not detected	—	▼	0.80	0.90	0.10	0.90	—	
richness	92.00	60	▼	0.60	0.90	0.10	0.90	-0.09	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	0.90	0.20	0.90	+0.76	
bsh	55.7/100	13	▼	0.60	0.70	0.30	0.70	+0.37	
bai	8.2/100	97	▼	0.60	0.70	0.30	0.70	-1.00	
h2s	63.8/100	28	▲	0.50	0.60	0.30	0.60	-0.19	

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
urease	5.2/100	8	▲	0.50	0.60	0.30	0.60	-0.34	Ni J, et al. Sci Transl Med 2017;9:eaah6888 (urease and CD dysbiosis)
Butyrate carriers and hexa-LPS carriers · carriers · carrier proxy · study carriers									
butyrate	1.623%	4	▼	0.80	0.80	0.20	0.80	+0.53	
hexa lps	0.039%	40	▲	0.60	0.70	0.30	0.70	-0.08	Vatanen T, et al. Cell 2016;165:842 (hexa- vs penta-acylated LPS immunogenicity)

FINE PRINT

Confidence grade moderate. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Crohn's disease is diagnosed by endoscopy, histology and imaging. This profile cannot diagnose it, exclude it, or distinguish it from ulcerative colitis with any reliability.

The Crohn's pattern is the general dysbiosis pattern with oral-origin taxa and E. coli on top. Antibiotics, bowel prep, gastroenteritis and hospitalisation produce it. If the shape analysis reads diffuse, this score is dysbiosis, not Crohn's.

Many people with Crohn's disease have a community indistinguishable from controls at any one sampling.

Sources: Gevers D, et al. The treatment-naive microbiome in new-onset Crohn's disease. Cell Host Microbe 2014;15:382-392; Lloyd-Price J, et al. Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. Nature 2019;569:655-662; Franzosa EA, et al. Gut microbiome structure and metabolic activity in inflammatory bowel disease. Nat Microbiol 2019;4:293-305; Pascal V, et al. A microbial signature for Crohn's disease. Gut 2017;66:813-822.

What the research says about this condition and this pattern

Inflammatory bowel disease — confirmation route and what guidelines say about FMT and probiotics LANE A CLINICAL ROUTE

Fired by: Crohn's disease pattern: notably high; Inflammatory bowel disease (composite) pattern: notably high; Ulcerative colitis pattern: notably high

Resemblance to an IBD-cohort pattern is not a diagnosis. Persistent diarrhoea, blood in stool, weight loss or pain warrant gastroenterology evaluation with fecal calprotectin. AGA 2024 recommends against conventional FMT for Crohn's disease and ulcerative colitis outside trials, and AGA 2020 places probiotics in IBD in the trial-only category.

Studied: Gastroenterology evaluation with fecal calprotectin when symptoms warrant (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** symptom-driven gastroenterology evaluation with fecal calprotectin

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: conditional for.

- AGA 2024 conditionally supports fecal microbiota-based therapy to prevent recurrent C. difficile infection after standard antibiotics in selected immunocompetent or mildly/moderately immunocompromised adults, and recommends against it in severely immunocompromised adults. [Peery AF et al. AGA Clinical Practice Guideline on fecal microbiota-based therapies. Gastroenterology 2024;166:409] · lane A · guideline (conditional, low certainty) · benefit
- **Against or limiting —** AGA 2024 recommends against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis except within a clinical trial. [AGA 2024 — against conventional FMT for Crohn's disease, ulcerative colitis, IBS and pouchitis outside trials] · lane X · guideline
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline

Conventional route: Gastroenterology evaluation with fecal calprotectin if symptomatic.

Fecal microbiota transplantation: guidelines recommend against it for this condition outside trials.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Crohn's” · “you have colitis” · “FMT for IBD”

Crohn's disease — dietary therapy with randomized evidence (clinician-confirmed disease)

LANE C HUMAN TRIAL · LIMITED

Fired by: Crohn's disease pattern: notably high; Inflammatory bowel disease (composite) pattern: notably high

The Crohn's Disease Exclusion Diet with partial enteral nutrition induced sustained remission in children with confirmed mild-to-moderate Crohn's disease. It is delivered by an IBD team, not started from a stool pattern.

Studied: Crohn's disease exclusion diet with partial enteral nutrition (CDED + PEN) (diet) · **in:** people with clinically confirmed mild-to-moderate paediatric Crohn's disease, ages 4–18, in multinational, n = 78 · **studied protocol (a source fact, not an instruction):** CDED + 50% PEN for 6 weeks, then CDED + 25% PEN to week 12, versus exclusive enteral nutrition then free diet + 25% PEN

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In children with mild-to-moderate Crohn's disease, CDED + partial enteral nutrition was better tolerated than exclusive enteral nutrition and induced sustained remission in more patients at 12 weeks. This is guideline-care context for clinician-confirmed disease, not an action from a stool pattern. [Levine A et al. Gastroenterology 2019;157:440] · lane C · randomized controlled, paediatric · n = 78 · benefit

Applicability: The study required a clinical diagnosis of crohns disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “diet cures Crohn's”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

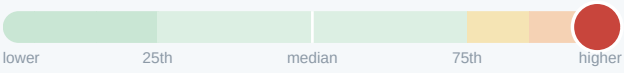
Irritable bowel syndrome — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ibs) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_adult_cohorts

A symptom-based diagnosis, so the microbiome literature is large, mostly 16S, and unusually inconsistent between cohorts. Systematic reviews agree on little beyond depleted butyrate producers and Bifidobacterium, and a subgroup of patients with a distinctly dysbiotic community — most others look like healthy controls. That heterogeneity is why the shape analysis matters more than the score here: an IBS-like community is also a post-antibiotic community, a low-fibre community, and a post-infectious community. Subtypes are scored separately in ibs_c, ibs_d and ibs_m, which share this parent's modules and add subtype-specific ones.

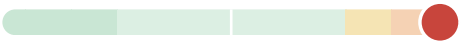
COMBINED RESEMBLANCE
percentile of the matched reference group



98th
likely range 69–100

What is driving it

Mars 2020 longitudinal shotgun (Mayo)
species · study mars_2020 · 4/4 features · 40% of combined



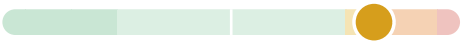
99th — NOTABLY HIGH

Jeffery 2020 (Irish IBS shotgun)
species · study jeffery_2020 · 4/4 features · 40% of combined



64th — TYPICAL

Pittayanon 2019 systematic review consensus
species · association only · 16S study · study pittayanon_2019 · 6/6 features · 20% of combined



81st — ABOVE AVERAGE

Community ecology
diversity · study pittayanon_2019 · 0/1 features · not in combined

— NOT COMPUTABLE

Gas, bile acid and neuroactive capacity
genes · genomic capacity · study mechanism · 4/4 features · not in combined



98th — NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Mars 2020 longitudinal shotgun (Mayo) · species · study mars_2020									
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.70	0.20	0.70	+1.00	
Roseburia intestinalis	not detected	13	▼	0.60	0.60	0.30	0.70	+0.37	
Bacteroides fragilis	not detected	24	▲	0.50	0.60	0.30	0.60	-0.24	
Ruminococcus gnavus	13.289%	98	▲	0.60	0.60	0.20	0.60	+0.72	
Jeffery 2020 (Irish IBS shotgun) · species · study jeffery_2020									
Bacteroides vulgatus	22.292%	93	▲	0.50	0.60	0.30	0.60	+0.41	
Eubacterium rectale	0.177%	35	▼	0.60	0.60	0.20	0.70	+0.17	
Coprococcus comes	0.713%	92	▼	0.50	0.50	0.30	0.60	-0.48	
Methanobrevibacter smithii	not detected	35	▼	0.50	0.50	0.40	0.50	+0.13	
Pittayanon 2019 systematic review consensus · species · association only · 16S study · study pittayanon_2019									
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.60	0.20	0.70	+1.00	
Bifidobacterium adolescentis	0.990%	85	▼	0.80	0.60	0.30	0.70	-0.35	
Bifidobacterium longum	4.157%	95	▼	0.60	0.60	0.30	0.70	-0.56	
Lactobacillus ruminis	not detected	44	▲	0.50	0.50	0.30	0.50	-0.05	
Escherichia coli	not detected 0.033% pooled	19	▲	0.60	0.50	0.20	0.60	-0.29	
Ruminococcus gnavus	13.289%	98	▲	0.60	0.50	0.20	0.60	+0.72	
Community ecology · diversity · study pittayanon_2019									
shannon	not detected	—	▼	0.50	0.60	0.10	0.70	—	
Gas, bile acid and neuroactive capacity · genes · genomic capacity · study mechanism									
butyrate	11.6/100	1	▼	0.80	0.70	0.20	0.80	+0.76	
bai	8.2/100	97	▲	0.60	0.60	0.30	0.60	+1.00	
h2s	63.8/100	28	▲	0.50	0.50	0.30	0.50	-0.19	
tryptamine	12.4/100	100	▲	0.40	0.40	0.40	0.40	+1.00	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBS is a symptom-based diagnosis made by a clinician after excluding organic disease. No stool test diagnoses it or excludes it.

The IBS-like community is the general dysbiosis community. Every scored direction here also appears in IBD, post-antibiotic and low-fibre states. Read this beside the shape analysis: if many profiles are high together, that is dysbiosis, not IBS.

Most IBS patients in the shotgun cohorts had communities indistinguishable from controls, so a low score is common in IBS.

Sources: Pittayanon R, et al. Gut microbiota in patients with irritable bowel syndrome — a systematic review. Gastroenterology 2019;157:97-108; Mars RAT, et al. Longitudinal multi-omics reveals subset-specific mechanisms underlying irritable bowel syndrome. Cell 2020;182:1460-1473; Jeffery IB, et al. Differences in fecal microbiomes and metabolomes of people with vs without irritable bowel syndrome and bile acid malabsorption. Gastroenterology 2020;158:1016-1028.

What the research says about this condition and this pattern

IBS — guideline-backed diet and what is advised against

LANE A CLINICAL ROUTE

Fired by: Irritable bowel syndrome pattern: notably high; IBS with diarrhoea pattern: notably high; IBS mixed type pattern: high

IBS is a symptom-based clinical diagnosis. For people with that diagnosis, ACG 2021 and the AGA diet update support soluble fibre, peppermint oil and a limited dietitian-guided low-FODMAP trial; ACG suggests against probiotics for global symptoms.

Studied: Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation) (diet) · **in:** people with clinically confirmed IBS (clinically diagnosed), ages 18–99 · **studied protocol (a source fact, not an instruction):** restriction ordinarily 4–6 weeks, then structured reintroduction and personalisation with dietitian support

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: conditional for.

- The AGA diet update endorses a limited, dietitian-guided low-FODMAP diet (restrict, reintroduce, personalise) for diagnosed IBS and warns against indefinite restriction, eating-disorder risk and malnutrition. [Chey WD et al. AGA Clinical Practice Update on the role of diet in IBS. Gastroenterology 2022;162:1737] · lane A · clinical practice update · benefit
- ACG 2021 suggests soluble fibre, peppermint oil and a limited dietitian-guided low-FODMAP trial for diagnosed IBS, suggests against probiotics for global symptoms, and lists rifaximin for global IBS-D symptoms after clinical diagnosis. [Lacy BE et al. ACG Clinical Guideline: Management of IBS. Am J Gastroenterol 2021;116:17] · lane A · guideline · benefit
- A three-week randomized low-FODMAP diet in diagnosed IBS improved symptoms and lowered urinary histamine — a diet effect in IBS, not a treatment for stool histamine genes. [McIntosh K et al. Gut 2017;66:1241] · lane C · randomized controlled · n = 40 · benefit
- **Against or limiting —** ACG 2021 suggests against probiotics for global IBS symptoms (very low quality of evidence). [Lacy BE et al. ACG 2021 — suggests against probiotics for global IBS symptoms] · lane X · guideline

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: Symptom-based diagnosis with red-flag review.

Fecal microbiota transplantation: guidelines recommend against it for this condition outside trials.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have IBS” · “cure IBS”

IBS — the two probiotic trials that met their endpoints, and the guideline position

LANE C HUMAN TRIAL · LIMITED

Fired by: Irritable bowel syndrome pattern: notably high; IBS with diarrhoea pattern: notably high; IBS mixed type pattern: high

Heat-inactivated B. bifidum MIMBb75 (443 people) and B. infantis 35624 at one specific dose are the exact products with positive randomized IBS trials; guidelines still place probiotics in the trial-only or advised-against category for global symptoms.

Studied: Heat-inactivated Bifidobacterium bifidum MIMBb75 (non-viable) (probiotic) · **in:** people with clinically confirmed IBS (Rome III), ages 18–70, in Germany, n = 443 · **studied protocol (a source fact, not an instruction):** 10^9 non-viable cells once daily for eight weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial. Guideline position: trial only.

- In 443 adults with diagnosed IBS, eight weeks of heat-inactivated B. bifidum MIMBb75 produced a composite response in 34% versus 19% on placebo. The product is non-viable and exact; it is not interchangeable with live B. bifidum and is not selected by stool abundance. [Andresen V et al. Lancet Gastroenterol Hepatol 2020;5:658] · lane C · randomized double-blind placebo-controlled · n = 443 · benefit
- In 362 women with IBS, B. infantis 35624 improved symptoms only at 10^8 CFU/day — not at a lower or a higher dose — illustrating that probiotic effects are strain- and dose-specific and do not generalise. [Whorwell PJ et al. Am J Gastroenterol 2006;101:1581] · lane C · randomized double-blind placebo-controlled, three dose arms · n = 362 · benefit
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline
- **Against or limiting —** ACG 2021 suggests against probiotics for global IBS symptoms (very low quality of evidence). [Lacy BE et al. ACG 2021 — suggests against probiotics for global IBS symptoms] · lane X · guideline

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “any probiotic” · “Bifidobacterium supplement fixes IBS”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

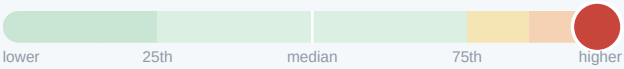
Androgenetic alopecia — gut pattern (sex-stratified)

NOTABLY HIGH

Pattern version **1.0.0** (androgenetic_alopecia) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Androgenetic alopecia is hormonal and genetic, biologically distinct from autoimmune alopecia areata, and its evidence is never merged with that profile. It is stratified on sex because the two sexes have different evidence: an all-male 16S cohort from China, a mixed-sex Korean cohort, and a female-only Italian cohort reporting sex-specific direction differences. Female pattern hair loss has different pathophysiology and cannot borrow male evidence. If sex is not supplied, the score is reported without stratification and this is recorded in the confounder ledger.

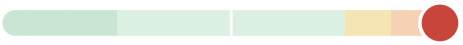
COMBINED RESEMBLANCE
percentile of the matched reference group



98th
likely range 0–100

What is driving it

Seo 2024 Korean cohort (mixed sex)
species · association only · 16S study · study seo_2024 · 3/3 features · 100% of combined



98th — NOTABLY HIGH

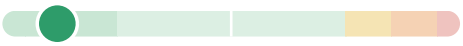
Huang 2024 all-male 16S (China)
species · association only · 16S study · study huang_2024 · 0/0 features · not in combined

— NOT COMPUTABLE

Savo Sardaro 2025 female-only (Italy)
species · association only · 16S study · study savo_sardaro_2025 · 0/0 features · not in combined

— NOT COMPUTABLE

Biotin biosynthesis capacity (mechanism, P0)
genes · genomic capacity · study biotin_lane · 1/1 features · not in combined



12th — LOW

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Seo 2024 Korean cohort (mixed sex) · species · association only · 16S study · study seo_2024									
Faecalibacterium prausnitzii	0.010%	2	▼	0.50	0.40	0.20	0.50	+1.00	
Ruminococcus gnavus	13.289%	98	▲	0.50	0.40	0.30	0.50	+0.72	
Akkermansia muciniphila	1.425%	91	▼	0.40	0.30	0.30	0.40	-0.46	
Huang 2024 all-male 16S (China) · species · association only · 16S study · study huang_2024									
Savo Sardaro 2025 female-only (Italy) · species · association only · 16S study · study savo_sardaro_2025									
Biotin biosynthesis capacity (mechanism, P0) · genes · genomic capacity · study biotin_lane									
biotin	93.2/100	88	▼	0.50	0.30	0.20	0.50	-0.48	Hayashi 2017 Cell Rep 20:1513

FINE PRINT

Confidence grade insufficient. Why not higher: 33% of planned feature weight was measurable (below 75%); no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Androgenetic alopecia is diagnosed from the pattern of hair loss and family history. Nothing here predicts it, and nothing here bears on treatment.

All three human cohorts are 16S with genus-level findings transported onto shotgun species. The transport penalty is shown in the confidence vector.

Without a supplied sex, the sex-specific modules are withheld and the score rests on the Korean mixed-sex module alone.

Sources: Huang J, et al. Skin Res Technol 2024;30:e13578; Seo Y, et al. Biomedicines 2024;12:2225; Savo Sardaro ML, et al. Int J Mol Sci 2025;26:5459; Hayashi A, et al. Cell Rep 2017;20:1513.

What the research says about this condition and this pattern

Androgenetic alopecia — one exact-product trial

LANE C · HUMAN TRIAL · LIMITED

Fired by: Androgenetic alopecia pattern: notably high

A single industry trial of three exact Lactiplantibacillus strains reported mixed hair-cycle endpoints in diagnosed androgenetic alopecia. Other probiotics are not covered by it.

Studied: Lactiplantibacillus plantarum DCn_07 (CECT 30102) + DCn_06 (CECT 30103) + Lactiplantibacillus pentosus DCn1_05 (CECT 30104) (probiotic) · **in:** people with clinically confirmed androgenetic alopecia, ages 18–65, in Spain, n = 136 · **studied protocol (a source fact, not an instruction):** three-strain capsule daily for 16 weeks (source-reported dose/formulation)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- One 16-week industry trial of an exact three-strain Lactiplantibacillus product in androgenetic alopecia reported mixed hair-cycle endpoints — several significant, several not. It does not transfer to any other probiotic and needs sex/age matching. [2024 RCT — L. plantarum DCn_07 + DCn_06 + L. pentosus DCn1_05 in androgenetic alopecia] · lane C · randomized double-blind placebo-controlled · n = 136 · mixed

Applicability: The study required a clinical diagnosis of androgenetic alopecia. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: Dermatology evaluation for pattern hair loss.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “probiotic cures hair loss”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

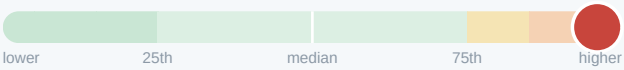
Rheumatoid arthritis — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ra) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_us_chinese_cohorts

The rheumatoid arthritis finding is strain-level, which is a problem for a species-level pipeline. Scher et al. found Prevotella copri expanded in new-onset untreated RA, and later work showed immune reactivity to specific P. copri peptides in a subset of patients. Since P. copri is also high in healthy people on high-fibre and non-Western diets, species abundance alone cannot carry the finding. Three later results constrain this profile further, and are the reason the species feature is an anchor rather than a scored direction. First, replication is poor. Several cohorts, including a 2024 preclinical-RA study, could not reproduce group-wise Prevotellaceae or P. copri differences, and attempts to reproduce the anti-P. copri antibody finding returned only modest or null results with reactivity varying by strain. Second, the direction is not even consistent within the species. In a comprehensively phenotyped anti-CCP-positive at-risk cohort (2024), three Prevotellaceae strains were enriched and five were depleted in those progressing towards arthritis. A single species-level abundance cannot represent a signal whose components move in opposite directions. Third, and decisively for the clade question: Manghi et al. resolved the Segatella (formerly Prevotella) copri complex into 13 species-level clades and tested them against 1,635 disease cases and 1,854 controls across 22 studies, including a rheumatoid arthritis cohort. None of the complex's species was associated with any health condition. So there is no established RA clade to score against, even though the clades themselves are now measurable (see `copri_complex` below). What remains is this: P. copri expansion is the finding. Where the organism is absent, the profile does not assert the pattern at all — the accessory taxa from Scher's cohort cannot carry it alone. Zhang et al. added a Chinese shotgun cohort with oral and gut samples, finding Haemophilus depleted and Lactobacillus salivarius enriched, partly normalised by treatment.

COMBINED RESEMBLANCE
percentile of the matched reference group



98th
likely range 18–100

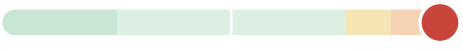
Read this percentile narrowly. *Prevotella copri* — the organism this profile's central finding is about — was looked for and not detected, so the module carrying that finding is not scored. The percentile above therefore rests on one study module (Zhang 2015 Chinese RA shotgun (gut)), and reflects accessory taxa rather than the organism the condition is associated with.

What is driving it

Scher 2013 new-onset untreated RA (16S, US)
species · association only · 16S study · study scher_2013 · **Prevotella copri not detected, so this pattern is not asserted** · 3/3 features · 100% of combined

— ANCHOR ABSENT

Zhang 2015 Chinese RA shotgun (gut)
species · study zhang_2015 · 6/6 features · 100% of combined

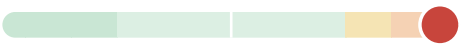


98th NOTABLY HIGH

RA-associated P. copri clade
strain · study scher_2013 · 0/1 features · not measurable — a validated reference panel for this specific strain-level target. Strain resolution itself is available and reported separately: the marker lane fingerprints the dominant population of every organism with enough coverage. What is missing here is not the engine but a published, discriminating marker set for this named target

— NOT COMPUTABLE

Metabolite gene capacity
genes · genomic capacity · study functional · 1/1 features · not in combined



99th NOTABLY HIGH

The Prevotella copri complex, clade by clade

No member of the Segatella (Prevotella) copri complex was detected. All 13 clades were assessed, including the 9 found in humans, so this is a measured absence across the whole complex rather than an unresolved species-level reading.

No clade of this complex has an established association with any disease, including rheumatoid arthritis. Manghi et al. tested all 13 clades against 1,635 disease cases and 1,854 controls across 22 studies and 11 conditions, a set that included a rheumatoid arthritis cohort, and found none of the complex's species associated with any health condition. Earlier rheumatoid-arthritis reports named the species as a whole, not a clade, and those reports have replicated poorly: in an anti-CCP-positive at-risk cohort three Prevotellaceae strains were enriched and five depleted in the people who progressed. So this section reports what is present and makes no disease claim from it.

Tett 2019 Cell Host Microbe doi:10.1016/j.chom.2019.08.018; Manghi 2023 Cell Host Microbe doi:10.1016/j.chom.2023.09.013; Hitch 2022 Syst Appl Microbiol doi:10.1016/j.syapm.2022.126354.

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Scher 2013 new-onset untreated RA (16S, US) · species · association only · 16S study · study scher_2013 · **Prevotella copri not detected, so this pattern is not asserted**									
Prevotella copri	not detected	25	▲	0.50	0.60	0.20	0.50	**-0.23**	
Bacteroides vulgatus	22.292%	93	▼	0.50	0.60	0.30	0.60	**-0.41**	
Bacteroides uniformis	6.597%	82	▼	0.40	0.60	0.30	0.60	**-0.26**	
Zhang 2015 Chinese RA shotgun (gut) · species · study zhang_2015									
Haemophilus parainfluenzae	not detected	29	▼	0.80	0.80	0.60	0.80	**+0.18**	
Lactobacillus salivarius	0.228%	99	▲	0.80	0.80	0.50	0.80	**+0.77**	
Bifidobacterium dentium	not detected	44	▲	0.50	0.60	0.40	0.60	**-0.05**	
Ruminococcus lactaris	not detected	27	▲	0.50	0.60	0.50	0.60	**-0.21**	
Clostridium asparagiforme	not detected	41	▲	0.50	0.60	0.50	0.60	**-0.08**	
Klebsiella pneumoniae	not detected	40	▼	0.40	0.50	0.40	0.50	**+0.09**	
RA-associated P. copri clade · strain · study scher_2013									
Prevotella copri RA clade	missing	—	▲	1.00	1.00	1.00	1.00	—	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	0.80	0.70	0.20	0.70	**+0.76**	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Rheumatoid arthritis is diagnosed clinically with serology (RF, anti-CCP) and imaging. This profile does not diagnose or predict it.

The central RA finding is a Prevotella copri strain this pipeline cannot resolve. The species is common in healthy people. The strain module is reported as not computable, and the profile score rests mostly on the Chinese cohort.

P. copri is the anchor of the Scher module. Where it is absent, that module reports `anchor_absent` and takes no part in the score: a sample with no Prevotella cannot resemble a Prevotella-expansion signature, however its Bacteroides species happen to fall. In that case the profile score reflects the Chinese cohort alone and should be read as such.

The complex's clade composition is resolved separately from the 13-clade SGB catalogue and reported under `copri_complex`. No clade has an established disease direction, so that section is composition only and contributes nothing to the score.

Replication of the species-level association is weak. Several cohorts found no group-wise difference, and in an at-risk cohort the strains moved in both directions at once. A high reading here is a weak signal from one Chinese cohort's accessory taxa, not evidence of arthritis risk.

Sources: Scher JU, et al. Expansion of intestinal Prevotella copri correlates with enhanced susceptibility to arthritis. eLife 2013;2:e01202; Zhang X, et al. The oral and gut microbiomes are perturbed in rheumatoid arthritis and partly normalized after treatment. Nat Med 2015;21:895-905; Pianta A, et al. Evidence of the immune relevance of Prevotella copri, a gut microbe, in patients with rheumatoid arthritis. Arthritis Rheumatol 2017;69:964.

What the research says about this condition and this pattern

Rheumatoid-arthritis pattern — small probiotic trials, positive and null LANE C HUMAN TRIAL · LIMITED

Fired by: Rheumatoid arthritis pattern: notably high

Rheumatoid arthritis is diagnosed by rheumatology with serology and examination. Two small probiotic trials were positive on disease activity, one was null; many RA patients are immunosuppressed, which changes live-microbe safety.

Studied: Lacticaseibacillus casei 01 (Alipour 2014, rheumatoid arthritis) (probiotic) · **in:** people with clinically confirmed rheumatoid arthritis, ages 20–80, female, in Iran, n = 46 · **studied protocol (a source fact, not an instruction):** 10^8 CFU/day L. casei 01 for eight weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial.

- A 46-woman trial of L. casei 01 for eight weeks reported improved disease activity and cytokines in rheumatoid arthritis. [Alipour B et al. Int J Rheum Dis 2014;17:519] · lane C · randomized double-blind placebo-controlled · n = 46 · benefit
- A 60-person three-organism probiotic trial reported improved DAS28 and metabolic markers in rheumatoid arthritis. [Zamani B et al. Int J Rheum Dis 2016;19:869] · lane C · randomized double-blind placebo-controlled · n = 60 · benefit
- **Against or limiting —** A 29-person trial of L. rhamnosus GR-1 + L. reuteri RC-14 found no ACR20 benefit in rheumatoid arthritis. [Pineda M de L et al. Med Sci Monit 2011;17:CR347] · lane X · randomized double-blind placebo-controlled pilot · n = 29 · null
- **Against or limiting —** FMT in rheumatoid arthritis is case-report level and investigational. [FMT case report in rheumatoid arthritis (2021)] · lane X · case report

Applicability: The study required a clinical diagnosis of rheumatoid arthritis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed. The study enrolled female participants only; sex is not on file here.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have arthritis” · “probiotics treat RA”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

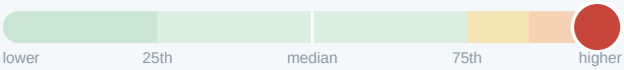
Mild cognitive impairment — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ad_mci) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · east_asian_cohort_derived

Mild cognitive impairment is a different prediction task from clinical Alzheimer's disease and is scored separately rather than folded into it. In the staged Chinese cohort no species reached q < 0.05 at this stage, so the strict panel that carries clinical AD does not exist here. What does exist is an independent Taiwanese cohort of 119 MCI against 320 normal controls, which is what this profile scores, plus the source cohort's exploratory q < 0.25 set reported as a sensitivity result and never called validated. The MCI stage is where the microbiome signal is smallest across the Alzheimer's continuum, so this score is read as supporting context for the clinical-stage pattern rather than on its own.

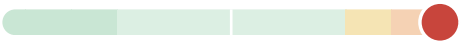
COMBINED RESEMBLANCE
percentile of the matched reference group



97th
likely range 62–100

What is driving it

Fan 2025 Taiwan MCI cohort
species · study fan_2025 · 5/5 features · 100% of combined



97th — NOTABLY HIGH

Jia 2025 exploratory set (q < 0.25) — sensitivity only
species · association only · study metaad_2025 · 2/2 features · not in combined



98th — NOTABLY HIGH

Jia 2025 KEGG orthologue panel, MCI stage
pathways · genomic capacity · study metaad_2025 · 0/3 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added

— NOT COMPUTABLE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Fan 2025 Taiwan MCI cohort · species · study fan_2025									
Bacteroides fragilis	not detected	24	▲	0.70	0.60	0.30	0.70	**-0.24**	Fan 2025 PMC12123878
Ruminococcus gnavus	13.289%	98	▲	0.70	0.60	0.30	0.70	**+0.72**	Fan 2025 PMC12123878

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.60	0.20	0.80	**+1.00**	Fan 2025 PMC12123878
Roseburia intestinalis	not detected	13	▼	0.60	0.50	0.20	0.70	**+0.37**	Fan 2025 PMC12123878
Eubacterium rectale	0.177%	35	▼	0.60	0.50	0.20	0.70	**+0.17**	Fan 2025 PMC12123878
Jia 2025 exploratory set (q < 0.25) — sensitivity only · species · association only · study metaad_2025									
Odoribacter splanchnicus	0.716%	94	▲	0.40	0.30	0.40	0.50	**+0.51**	
Alistipes finegoldii	1.712%	95	▲	0.40	0.30	0.40	0.50	**+0.56**	
Jia 2025 KEGG orthologue panel, MCI stage · pathways · genomic capacity · study metaad_2025									
K00857	missing	—	▲	1.00	1.00	1.00	1.00	—	
K01031	missing	—	▲	1.00	1.00	1.00	1.00	—	
K01754	missing	—	▼	1.00	1.00	1.00	1.00	—	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Not a diagnosis or a risk estimate. Mild cognitive impairment is a clinical determination made with cognitive testing.

At strict multiple-testing correction, no species distinguished this stage in the largest staged cohort. This profile scores because an independent cohort supports a direction, not because the signal is strong.

Three of the five scored features are butyrate-producer depletion or R. gnavus enrichment, which appear in most gut disease cohorts. Read this score beside the general dysbiosis index and the shape analysis.

Sources: Fan et al. 2025, PMC12123878, data PRJNA1258384; Jia L, Ke Y, Zhao S, et al. Metagenomic analysis characterizes stage-specific gut microbiota in Alzheimer's disease. Mol Psychiatry 2025;30:3951-3962, doi:10.1038/s41380-025-02973-7.

What the research says about this condition and this pattern

Cognition — what has replicated evidence independent of the microbiome LANE B HUMAN EVIDENCE · STRONG

Fired by: Clinical Alzheimer's disease pattern: notably high; Mild cognitive impairment pattern: notably high; Preclinical amyloid positivity pattern: notably high

Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition in at-risk older adults; the MIND diet did not beat a control diet.

Studied: FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk monitoring) (lifestyle) · **in:** people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognition, ages 60–77, in Finland, n = 1260 · **studied protocol (a source fact, not an instruction):** two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring)

Evidence lane B: replicated human randomized clinical benefit for the exact intervention and population. Replication: single study. Strongest design: randomized trial.

- The FINGER trial showed that a two-year multidomain lifestyle programme (diet, exercise, cognitive training, vascular risk control) improved or maintained cognition in at-risk older adults — evidence independent of the microbiome. [Ngandu T et al. Lancet 2015;385:2255 (FINGER)] · lane B · randomized controlled, 2 years · n = 1260 · benefit
- **Against or limiting —** The MIND-diet trial did not improve cognition more than a control diet with the same mild caloric restriction over three years. [Barnes LL et al. N Engl J Med 2023;389:602 (MIND diet)] · lane X · randomized controlled, 3 years · n = 604 · null

Conventional route: Neurologic/cognitive evaluation only if memory or thinking symptoms are present.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Alzheimer's” · “prevents dementia”

Alzheimer disease — probiotic trials, positive and null

LANE C HUMAN TRIAL · LIMITED

Fired by: Clinical Alzheimer's disease pattern: notably high; Mild cognitive impairment pattern: notably high; Preclinical amyloid positivity pattern: notably high

One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level only.

Studied: Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. fermentum (Akbari 2016) (probiotic) · **in:** people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60 · **studied protocol (a source fact, not an instruction):** 200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- One 60-person 12-week trial of a four-organism probiotic milk reported MMSE and metabolic-marker changes in people with Alzheimer disease. [Akbari E et al. Front Aging Neurosci 2016;8:256] · lane C · randomized double-blind controlled · n = 60 · benefit
- **Against or limiting —** A later trial in severe Alzheimer disease found no cognitive benefit from probiotics. [Agahi A et al. Front Neurol 2018 (severe AD, negative)] · lane X · randomized double-blind placebo-controlled · null
- **Against or limiting —** Conservative pooled analyses find the probiotic evidence in Alzheimer disease inconsistent and not robust. [meta-analysis of probiotics in Alzheimer disease/MCI (2021)] · lane X · systematic review / meta-analysis · mixed
- **Against or limiting —** Probiotic trials in mild cognitive impairment are small and inconsistent. [2024 evidence review — probiotics in mild cognitive impairment] · lane X · review · mixed

Applicability: The study required a clinical diagnosis of alzheimer disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “reverses dementia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

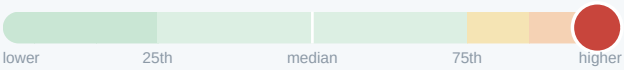
Systemic lupus erythematosus — shared inflammatory gut pattern (not lupus-specific)

NOTABLY HIGH

Pattern version **0.1.0** (sle) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · research only · mixed_geography_treated_cohorts

Systemic lupus erythematosus is an autoimmune disease in which antibodies attack the body's own tissues, most seriously the kidneys, and it is diagnosed from clinical features together with blood autoantibody tests. A twenty-five-study audit of the gut-microbiome literature was carried out for this profile, and the honest result of that audit is that it does not support a lupus-specific stool score. What is scored here is deliberately narrower and deliberately named for what it is: the shared inflammatory pattern that lupus cohorts have in common with several other conditions. Two species-level directions have cross-geography, cross-platform support and are the only taxonomic features scored. Streptococcus anginosus and the closely related S. intermedius are raised — found by a Japanese metagenome-wide association study of 47 patients against 203 controls and corroborated in a Chinese amplicon cohort. Faecalibacterium is depleted, reported at genus level in a Chinese amplicon cohort and in a ninety-two-study systematic review. Both are pan-rheumatic: the same review that found them concluded there is little evidence of disease specificity for gut microbes in rheumatic disease, and a head-to-head comparison could not separate lupus from primary Sjögren's syndrome from stool at all, while oral samples could. An index built on these two measures inflammation, not lupus, and it is labelled that way here. Ruminococcus gnavus is the finding this profile is best known for, and it is deliberately not scored. Its abundance is reported in the body of the report with two caveats attached. The first is that the association is strain-level: lupus-derived and healthy-derived strains colonise mice equally well, and only the lupus strains cause the gut-permeability and autoimmunity phenotype, with only nephritis-derived strains expressing the lipoglycan that drives it. This pipeline measures the species accurately and the species is not the question. The second is that it is a transient bloom rather than a level, fluctuating greatly within the same patient between samples, so a single stool specimen is one draw from a volatile process: a low value means very little and a high one cannot be told apart from ordinary carriage. R. gnavus elevation is also already scored as a feature in fifteen other profiles in this library, which is the plainest possible demonstration of how non-specific it is. Two further limits are stated rather than worked around. The activity signal failed where it had the most power: in the largest single cohort, 157 Korean patients against 50 controls, there was no significant microbial difference by lupus-nephritis status and none by disease activity. And medication moves these biomarkers as much as the disease does. Proton pump inhibitors erased the diversity deficit entirely and were the only significant explanatory variable for composition among patients; hydroxychloroquine reduces Enterobacteriaceae; and in the one large treatment-naive cohort the lupus-enriched species were reduced after treatment. Nearly every real patient is taking at least one of these drugs, so a treated person's reading is partly a readout of their prescription. A high score is a research observation about community composition. It does not diagnose lupus, and a low score in someone with lupus does not argue against the diagnosis.

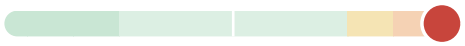
COMBINED RESEMBLANCE
percentile of the matched reference group



96th
likely range 42–100

What is driving it

Cross-platform species directions (inflammation-associated, not lupus-specific)
species · association only · 16S study · study sle_cross_platform · 2/2 features · 100% of combined



96th NOTABLY HIGH

Ruminococcus gnavus lipoglycan strain typing (not computable here)
strain · study silverman_azzouz_strain · 0/1 features · not measurable — a validated reference panel for this specific strain-level target. Strain resolution itself is available and reported separately: the marker lane fingerprints the dominant population of every organism with enough coverage. What is missing here is not the engine but a published, discriminating marker set for this named target

— NOT COMPUTABLE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match

Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Cross-platform species directions (inflammation-associated, not lupus-specific) · species · association only · 16S study · study sle_cross_platform									
Streptococcus anginosus, measured as Streptococcus anginosus group	not detected	45	▲	0.60	0.40	0.20	0.60	**-0.04**	
Faecalibacterium (genus)	0.010%	2	▼	0.50	0.30	0.20	0.70	**+1.00**	
Ruminococcus gnavus lipoglycan strain typing (not computable here) · strain · study silverman_azzouz_strain									
Ruminococcus gnavus lipoglycan strains	missing	—	▲	1.00	0.60	0.70	0.80	—	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Lupus is diagnosed from clinical features together with blood autoantibody tests, assessed by a clinician. Nothing in stool DNA makes or excludes that diagnosis, and this score is not a probability of having lupus. If you are worried about lupus, the tests that answer the question are blood tests.

This profile is named for a shared inflammatory pattern rather than for lupus, and that name is the finding. The systematic review that supplies both scored directions concluded there is little evidence of disease specificity for gut microbes in rheumatic disease. A head-to-head study could not tell lupus from primary Sjögren's syndrome using stool, though it could using oral samples. A high score here is equally consistent with several other inflammatory conditions, which is why the competing patterns are scored beside it.

Ruminococcus gnavus is the organism most associated with lupus in the press and it is deliberately excluded from this score. The association is strain-level: only lupus-derived strains cause the disease phenotype in mice, only nephritis-derived strains make the lipoglycan that drives it, and healthy-derived strains colonise just as well without causing it. This test measures the species, not the strain. R. gnavus is also a transient bloom rather than a stable level, and its elevation is already a scored feature in fifteen other profiles in this library.

Disease activity cannot be read from this score. In the largest cohort audited — 157 patients against 50 controls — there was no significant microbial difference between patients with and without lupus nephritis, and none between high and low disease activity. The activity associations reported elsewhere come from cohorts of sixteen to sixty-one people with largely uncorrected statistics.

In anyone already taking medication, this reading is partly a readout of the prescription. Proton pump inhibitors erased the diversity deficit altogether and explained community composition better than any clinical variable. Nothing in this profile is a reason to start, stop or change any lupus treatment, and no microbiome score authorises substituting a supplement for immunosuppression.

No microbiome-directed intervention in lupus has human evidence strong enough to recommend. Lactobacillus deserves specific mention because it cuts both ways: a Lactobacillus mixture improved kidney function in one mouse model of lupus, and L. reuteri worsened interferon-driven disease in another while escaping the gut. Sex hormones, strain and disease driver set which way it goes, and a stool report determines none of them. Live-microbe products also carry a documented bacteraemia risk in immunosuppressed people, which many lupus patients are.

Sources: Tomofuji Y, Maeda Y, Oguro-Igashira E, et al. Metagenome-wide association study revealed disease-specific landscape of the gut microbiome of systemic lupus erythematosus in Japanese. Ann Rheum Dis 2021;80:1575-1583. Wang Y, Wei J, Zhang W, et al. Gut dysbiosis in rheumatic diseases: a systematic review and meta-analysis of 92 observational studies. eBioMedicine 2022. Xiang S, Qu Y, Qian S, et al. Association between systemic lupus erythematosus and disruption of gut microbiota: a meta-analysis. Lupus Sci Med 2022. Azzouz DF, Omarbekova A, Heguy A, et al. Lupus nephritis is linked to disease-activity associated expansions and immunity to a gut commensal. Ann Rheum Dis 2019;78:947-956. Azzouz DF, Chen Z, Izmirly PM, et al. Longitudinal gut microbiome analyses and blooms of pathogenic strains during lupus disease flares. Ann Rheum Dis 2023;82:1315. Amarnani A, et al. A pathogenic gut lipoglycan drives systemic thromboinflammation in lupus nephritis. Ann Rheum Dis 2026, doi:10.1016/j.ard.2026.03.002. Silverman GJ, et al. Front Immunol 2022. Park Y, Yang J, Son H, et al. Distinct gut microbiome profiles in Korean systemic lupus erythematosus patients. J Transl Med 2025. Li XB, Chu XJ, Cao NW, et al. Proton pump inhibitors induce changes in the gut microbiome of systemic lupus erythematosus patients. BMC Microbiol 2022. Chen BD, Jia XM, Xu JY, et al. An autoimmunogenic and proinflammatory profile defined by the gut microbiota of patients with untreated systemic lupus erythematosus. Arthritis Rheumatol 2021. van der Meulen TA, Harmsen HJM, Vila AV, et al. Shared gut, but distinct oral microbiota composition in primary Sjögren's syndrome and systemic lupus erythematosus. J Autoimmun 2019. Ling Z, et al. Front Immunol 2023. Hevia A, et al. mBio 2014. Full twenty-five-study audit with per-finding directions, cohorts and verification notes: specs/research/sle_evidence.yaml.

‹ at a glance

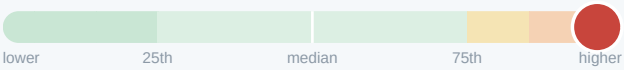
Clinical Alzheimer's disease — stage-specific gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ad_clinical) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · adult_chinese_cohort_derived

The only stage in the Alzheimer's continuum with a species-level panel that survives strict multiple-testing correction. Jia et al. profiled 476 participants across five stages of the continuum by faecal shotgun sequencing and found over 10% of microbial species and gene families significantly altered across progression, with cross-validated AUC around 0.80 and 0.75 in independent validation. At q < 0.05 no species qualified for the three pre-clinical stages. Only clinical AD has a strict species panel, which is why this profile scores and the earlier stages lean on evidence types this pipeline cannot yet measure. The directions below are findings in a Chinese staged cohort, not universal Alzheimer's biology.

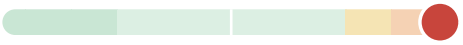
COMBINED RESEMBLANCE
percentile of the matched reference group



96th
likely range 46–100

What is driving it

Jia 2025 strict species panel (q < 0.05)
species · study metaad_2025 · 6/6 features · 100% of combined

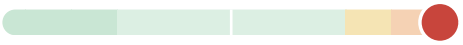


96th — NOTABLY HIGH

Jia 2025 KEGG orthologue panel
pathways · genomic capacity · study metaad_2025 · 0/8 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added

— NOT COMPUTABLE

Alzheimer's-relevant metabolite capacity
genes · genomic capacity · study mechanism_lanes · 5/5 features · not in combined



98th — NOTABLY HIGH

Butyrate carriers (independent count)
carriers · carrier proxy · study carrier_cross_check · 1/1 features · not in combined



96th — NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Jia 2025 strict species panel (q < 0.05) · species · study metaad_2025									
Odoribacter splanchnicus	0.716%	94	▲	1.00	0.50	0.50	1.00	+0.51	Jia 2025 Mol Psychiatry (simulator-supported)
Barnesiella viscericola	not detected	50	▲	1.00	0.50	0.60	1.00	-0.00	Jia 2025 Mol Psychiatry (simulator-supported)
Alistipes finegoldii	1.712%	95	▲	1.00	0.50	0.40	0.80	+0.56	Jia 2025 Mol Psychiatry (simulator-supported)
Alistipes timonensis	not detected	47	▲	0.70	0.50	0.40	0.80	-0.03	Jia 2025 Mol Psychiatry (lineage cluster)
Prevotella bivia	not detected	48	▲	0.50	0.40	0.40	0.70	-0.01	Jia 2025 Mol Psychiatry
Streptococcus infantis	not detected	43	▼	0.50	0.40	0.30	0.60	+0.06	Jia 2025 Mol Psychiatry
Jia 2025 KEGG orthologue panel · pathways · genomic capacity · study metaad_2025									
K00857	missing	—	▲	1.00	1.00	1.00	1.00	—	
K00876	missing	—	▲	1.00	1.00	1.00	1.00	—	
K01031	missing	—	▲	1.00	1.00	1.00	1.00	—	
K03179	missing	—	▲	1.00	1.00	1.00	1.00	—	
K11782	missing	—	▲	1.00	1.00	1.00	1.00	—	
K18285	missing	—	▲	1.00	1.00	1.00	1.00	—	
K20036	missing	—	▲	1.00	1.00	1.00	1.00	—	
K01754	missing	—	▼	1.00	1.00	1.00	1.00	—	
Alzheimer's-relevant metabolite capacity · genes · genomic capacity · study mechanism_lanes									
bai	8.2/100	97	▲	1.00	0.80	0.40	0.90	+1.00	MahmoudianDehkordi 2019 Alzheimers Dement
butyrate	11.6/100	1	▼	1.00	0.80	0.30	0.90	+0.76	Zhang 2023 Mol Psychiatry
cutc	4.1/100	77	▲	0.80	0.70	0.40	0.80	+0.39	Vogt 2018 Alzheimers Res Ther
ipa	9.6/100	89	▼	0.70	0.60	0.40	0.80	-0.71	Pappolla 2021 Neurobiol Dis
propionate	32.9/100	93	▲	0.50	0.50	0.30	0.40	+0.72	Mendelian randomisation, per Jia 2025 discussion
Butyrate carriers (independent count) · carriers · carrier proxy · study carrier_cross_check									
butyrate	1.623%	4	▼	1.00	0.80	0.30	0.90	+0.53	derived from the butyrate panel organism list

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Not a diagnosis, a risk estimate, or a screening result. Alzheimer's disease is diagnosed clinically with cognitive assessment and, where indicated, amyloid and tau biomarkers. A gut community resembling a published Alzheimer's cohort says nothing about whether that is happening.

Derived from a Chinese staged cohort. Directions are source-cohort findings; conflicting directions in other cohorts are held as separate assertions rather than pooled into a consensus.

Four of the source panel's Alistipes species and both of its Streptococcus species have no entry in the reference catalogue this pattern is scored against. They are recorded as missing, which is why independent-cluster coverage here is well below the raw feature count.

Sources: Jia L, Ke Y, Zhao S, et al. Metagenomic analysis characterizes stage-specific gut microbiota in Alzheimer's disease. Mol Psychiatry 2025;30:3951-3962, doi:10.1038/s41380-025-02973-7; data PRJCA022804 / CRA014435 (NGDC), analysis github.com/JiaLonghao1997/metaAD_analysis.

What the research says about this condition and this pattern

Cognition — what has replicated evidence independent of the microbiome

LANE B · HUMAN EVIDENCE · STRONG

Fired by: Clinical Alzheimer's disease pattern: notably high; Mild cognitive impairment pattern: notably high; Preclinical amyloid positivity pattern: notably high

Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition in at-risk older adults; the MIND diet did not beat a control diet.

Studied: FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk monitoring) (lifestyle) · **in:** people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognition, ages 60–77, in Finland, n = 1260 · **studied protocol (a source fact, not an instruction):** two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring)

Evidence lane B: replicated human randomized clinical benefit for the exact intervention and population. Replication: single study. Strongest design: randomized trial.

- The FINGER trial showed that a two-year multidomain lifestyle programme (diet, exercise, cognitive training, vascular risk control) improved or maintained cognition in at-risk older adults — evidence independent of the microbiome. [Ngandu T et al. Lancet 2015;385:2255 (FINGER)] · lane B · randomized controlled, 2 years · n = 1260 · benefit
- **Against or limiting —** The MIND-diet trial did not improve cognition more than a control diet with the same mild caloric restriction over three years. [Barnes LL et al. N Engl J Med 2023;389:602 (MIND diet)] · lane X · randomized controlled, 3 years · n = 604 · null

Conventional route: Neurologic/cognitive evaluation only if memory or thinking symptoms are present.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Alzheimer's” · “prevents dementia”

Alzheimer disease — probiotic trials, positive and null

LANE C · HUMAN TRIAL · LIMITED

Fired by: Clinical Alzheimer's disease pattern: notably high; Mild cognitive impairment pattern: notably high; Preclinical amyloid positivity pattern: notably high

One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level only.

Studied: Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. fermentum (Akbari 2016) (probiotic) · **in:** people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60 · **studied protocol (a source fact, not an instruction):** 200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- One 60-person 12-week trial of a four-organism probiotic milk reported MMSE and metabolic-marker changes in people with Alzheimer disease. [Akbari E et al. Front Aging Neurosci 2016;8:256] · lane C · randomized double-blind controlled · n = 60 · benefit
- **Against or limiting —** A later trial in severe Alzheimer disease found no cognitive benefit from probiotics. [Agahi A et al. Front Neurol 2018 (severe AD, negative)] · lane X · randomized double-blind placebo-controlled · null
- **Against or limiting —** Conservative pooled analyses find the probiotic evidence in Alzheimer disease inconsistent and not robust. [meta-analysis of probiotics in Alzheimer disease/MCI (2021)] · lane X · systematic review / meta-analysis · mixed
- **Against or limiting —** Probiotic trials in mild cognitive impairment are small and inconsistent. [2024 evidence review — probiotics in mild cognitive impairment] · lane X · review · mixed

Applicability: The study required a clinical diagnosis of alzheimer disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “reverses dementia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

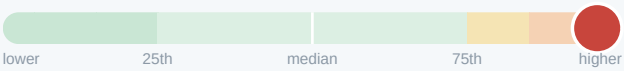
IBS with diarrhoea — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ibs_d) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Diarrhoea-predominant IBS overlaps with bile acid malabsorption in roughly a quarter to a third of patients, and the community pattern follows: higher secondary bile acid conversion, lower methanogens, and a larger dysbiotic subgroup than IBS-C. The parent IBS pattern is inherited; the bile acid and methanogen directions are the subtype-specific part.

COMBINED RESEMBLANCE
percentile of the matched reference group



96th
likely range 64–100

What is driving it

Jeffery 2020 IBS-D / bile acid malabsorption shotgun
species · study jeffery_2020 · 5/5 features · 50% of combined
99th — NOTABLY HIGH

Mars 2020 longitudinal IBS-D shotgun
species · study mars_2020 · 2/2 features · 50% of combined
63rd — TYPICAL

Secondary bile acid conversion capacity
genes · genomic capacity · study bile_acid_capacity · 4/4 features · not in combined
94th — HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Jeffery 2020 IBS-D / bile acid malabsorption shotgun · species · study jeffery_2020									
Methanobrevibacter smithii	not detected	35	▼	0.80	0.70	0.50	0.80	+0.13	
Bacteroides vulgatus	22.292%	93	▲	0.60	0.60	0.30	0.60	+0.41	
Ruminococcus gnavus	13.289%	98	▲	0.60	0.60	0.20	0.60	+0.72	
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.60	0.20	0.70	+1.00	
Eubacterium rectale	0.177%	35	▼	0.50	0.60	0.20	0.70	+0.17	
Mars 2020 longitudinal IBS-D shotgun · species · study mars_2020									
Bacteroides fragilis	not detected	24	▲	0.50	0.60	0.30	0.60	-0.24	
Roseburia intestinalis	not detected	13	▼	0.50	0.60	0.30	0.70	+0.37	

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Secondary bile acid conversion capacity · genes · genomic capacity · study bile_acid_capacity									
bai	8.2/100	97	▲	1.00	0.70	0.40	0.70	+1.00	
bsh	55.7/100	13	▲	0.50	0.60	0.20	0.50	-0.37	
tryptamine	12.4/100	100	▲	0.50	0.50	0.40	0.50	+1.00	
methane	0.0/100	31	▼	0.60	0.60	0.50	0.70	+0.17	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBS subtypes are defined by stool form, not stool DNA. Bile acid malabsorption is diagnosed with SeHCAT retention or serum C4, not with a gene count.

Recent gastroenteritis, antibiotics, and any cause of diarrhoea reproduce this pattern. It is not specific to IBS-D.

Sources: Jeffery IB, et al. Gastroenterology 2020;158:1016-1028; Mars RAT, et al. Cell 2020;182:1460-1473; Bhattarai Y, et al. Cell Host Microbe 2018;23:775.

What the research says about this condition and this pattern

IBS — guideline-backed diet and what is advised against LANE A CLINICAL ROUTE

Fired by: Irritable bowel syndrome pattern: notably high; IBS with diarrhoea pattern: notably high; IBS mixed type pattern: high

IBS is a symptom-based clinical diagnosis. For people with that diagnosis, ACG 2021 and the AGA diet update support soluble fibre, peppermint oil and a limited dietitian-guided low-FODMAP trial; ACG suggests against probiotics for global symptoms.

Studied: Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation) (diet) · **in:** people with clinically confirmed IBS (clinically diagnosed), ages 18–99 · **studied protocol (a source fact, not an instruction):** restriction ordinarily 4–6 weeks, then structured reintroduction and personalisation with dietitian support

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: conditional for.

- The AGA diet update endorses a limited, dietitian-guided low-FODMAP diet (restrict, reintroduce, personalise) for diagnosed IBS and warns against indefinite restriction, eating-disorder risk and malnutrition. [Chey WD et al. AGA Clinical Practice Update on the role of diet in IBS. Gastroenterology 2022;162:1737] · lane A · clinical practice update · benefit
- ACG 2021 suggests soluble fibre, peppermint oil and a limited dietitian-guided low-FODMAP trial for diagnosed IBS, suggests against probiotics for global symptoms, and lists rifaximin for global IBS-D symptoms after clinical diagnosis. [Lacy BE et al. ACG Clinical Guideline: Management of IBS. Am J Gastroenterol 2021;116:17] · lane A · guideline · benefit
- A three-week randomized low-FODMAP diet in diagnosed IBS improved symptoms and lowered urinary histamine — a diet effect in IBS, not a treatment for stool histamine genes. [McIntosh K et al. Gut 2017;66:1241] · lane C · randomized controlled · n = 40 · benefit
- **Against or limiting —** ACG 2021 suggests against probiotics for global IBS symptoms (very low quality of evidence). [Lacy BE et al. ACG 2021 — suggests against probiotics for global IBS symptoms] · lane X · guideline

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: Symptom-based diagnosis with red-flag review.

Fecal microbiota transplantation: guidelines recommend against it for this condition outside trials.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have IBS” · “cure IBS”

IBS — the two probiotic trials that met their endpoints, and the guideline position

LANE C · HUMAN TRIAL · LIMITED

Fired by: Irritable bowel syndrome pattern: notably high; IBS with diarrhoea pattern: notably high; IBS mixed type pattern: high

Heat-inactivated B. bifidum MIMBb75 (443 people) and B. infantis 35624 at one specific dose are the exact products with positive randomized IBS trials; guidelines still place probiotics in the trial-only or advised-against category for global symptoms.

Studied: Heat-inactivated Bifidobacterium bifidum MIMBb75 (non-viable) (probiotic) · **in:** people with clinically confirmed IBS (Rome III), ages 18–70, in Germany, n = 443 · **studied protocol (a source fact, not an instruction):** 10^9 non-viable cells once daily for eight weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial. Guideline position: trial only.

- In 443 adults with diagnosed IBS, eight weeks of heat-inactivated B. bifidum MIMBb75 produced a composite response in 34% versus 19% on placebo. The product is non-viable and exact; it is not interchangeable with live B. bifidum and is not selected by stool abundance. [Andresen V et al. Lancet Gastroenterol Hepatol 2020;5:658] · lane C · randomized double-blind placebo-controlled · n = 443 · benefit
- In 362 women with IBS, B. infantis 35624 improved symptoms only at 10^8 CFU/day — not at a lower or a higher dose — illustrating that probiotic effects are strain- and dose-specific and do not generalise. [Whorwell PJ et al. Am J Gastroenterol 2006;101:1581] · lane C · randomized double-blind placebo-controlled, three dose arms · n = 362 · benefit
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline
- **Against or limiting —** ACG 2021 suggests against probiotics for global IBS symptoms (very low quality of evidence). [Lacy BE et al. ACG 2021 — suggests against probiotics for global IBS symptoms] · lane X · guideline

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “any probiotic” · “Bifidobacterium supplement fixes IBS”

IBS-D — prescription route for a clinician to review

LANE A · CLINICIAN REVIEW

Fired by: IBS with diarrhoea pattern: notably high

Rifaximin is guideline-listed for global IBS-D symptoms after clinical diagnosis. It is a prescription decision made by a clinician, never a response to organisms in stool.

Studied: Rifaximin (prescription drug) · **in:** people with clinically confirmed IBS-D (clinically diagnosed), ages 18–99 · **studied protocol (a source fact, not an instruction):** studied protocol on file (not shown in participant mode)

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: strong for.

- ACG 2021 recommends rifaximin for global IBS-D symptoms in clinically diagnosed patients. It is not a treatment for organisms seen in shotgun data and is not triggered by any stool finding. [ACG 2021 — rifaximin for global IBS-D symptoms] · lane A · guideline · benefit

Applicability: The study required a clinical diagnosis of ibs d. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Prescription route for a clinician to review — never a recommendation generated from a stool result.

Safety: A known safety consideration applies; review with a clinician before any change. Prescription content is clinician-only and requires an exact confirmed indication; a microbiome pattern never establishes that indication (spec §8.2).

This report will not say: “take rifaximin for your bacteria”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

Major depressive disorder — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (mdd) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_european_chinese_cohorts

Depression is scored and reported like every other pattern here, because the gut evidence for it is published and specific enough to measure. It remains a resemblance score, not a mood measurement: depression is diagnosed by a clinician, and a stool pattern cannot make or exclude that diagnosis. Transfer experiments are why this pattern is worth measuring rather than merely noting. Stool from people with major depression induced depression-like behaviour in germ-free mice, and the recipients took on the donors' carbohydrate and amino-acid metabolic shifts (Zheng 2016). The result has been reproduced in microbiota-depleted rats, which developed anhedonia, anxiety-like behaviour and altered tryptophan metabolism (Kelly 2016); in germ-free mice given communities from women with prenatal depression, which added hippocampal neuroinflammation and raised plasma LPS (Cao 2025); and in adolescent rats given pooled paediatric MDD stool, which showed reduced sucrose preference and fourteen amino-acid deficiencies in the prefrontal cortex (Teng 2025). One human-donor experiment cuts the other way, and it is why the wording here stays careful: in Flinders rats, healthy-donor recipients did better than depression-donor recipients, but the depression-donor group was no worse than untreated controls — so that gap may be benefit from healthy microbes rather than harm from patient microbes (Knudsen 2021). The case-control evidence is real but modest. Valles-Colomer et al. found in two large population cohorts that Coprococcus and Dialister were depleted in depression, and that butyrate-producing Faecalibacterium and Coprococcus tracked higher quality-of-life scores; they also showed the microbial capacity to synthesise the dopamine metabolite DOPAC was associated with quality of life. Zheng et al. found Bacteroides higher and Blautia, Eubacterium lower in a Chinese MDD cohort with faecal transfer inducing depression-like behaviour in germ-free mice. Antidepressants themselves have antimicrobial activity and reshape the community.

COMBINED RESEMBLANCE
percentile of the matched reference group



95th
likely range 68–100

What is driving it

Valles-Colomer 2019 Flemish Gut Flora Project + LifeLines DEEP
species · association only · 16S study · study valles_colomer_2019 · 4/4 features · 33% of combined



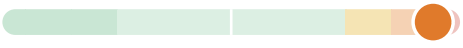
85th — ABOVE AVERAGE

Zheng 2016 Chinese MDD (16S, GF-mouse transfer)
species · association only · 16S study · study zheng_2016 · 3/3 features · 33% of combined



76th — ABOVE AVERAGE

Yang 2020 Chinese MDD shotgun + metabolomics
species · study yang_2020 · 5/5 features · 33% of combined



94th — HIGH

Neuroactive metabolite capacity
genes · genomic capacity · study functional · 4/4 features · not in combined



51st — TYPICAL

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Valles-Colomer 2019 Flemish Gut Flora Project + LifeLines DEEP · species · association only · 16S study · study valles_colomer_2019									
Coprococcus comes	0.713%	92	▼	0.90	0.80	0.40	0.80	-0.48	
Coprococcus catus	not detected	16	▼	0.60	0.70	0.40	0.70	+0.34	
Dialister invisus	not detected	39	▼	0.80	0.80	0.50	0.70	+0.09	
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.70	0.20	0.70	+1.00	
Zheng 2016 Chinese MDD (16S, GF-mouse transfer) · species · association only · 16S study · study zheng_2016									
Bacteroides vulgatus	22.292%	93	▲	0.50	0.60	0.30	0.50	+0.41	
Eubacterium rectale	0.177%	35	▼	0.50	0.60	0.20	0.60	+0.17	
Blautia obeum	0.042%	50	▼	0.40	0.50	0.40	0.50	-0.00	
Yang 2020 Chinese MDD shotgun + metabolomics · species · study yang_2020									
Bacteroides fragilis	not detected	24	▲	0.60	0.70	0.30	0.60	-0.24	
Bacteroides vulgatus	22.292%	93	▲	0.50	0.70	0.30	0.60	+0.41	
Eubacterium rectale	0.177%	35	▼	0.60	0.70	0.20	0.70	+0.17	
Blautia wexlerae	not detected	6	▼	0.50	0.60	0.40	0.60	+0.79	
Eubacterium eligens	not detected	8	▼	0.50	0.60	0.40	0.70	+1.00	
Neuroactive metabolite capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	0.80	0.70	0.20	0.70	+0.76	
gaba	33.5/100	86	▼	0.60	0.50	0.40	0.50	-0.43	
ipa	9.6/100	89	▼	0.50	0.50	0.30	0.50	-0.71	
dopamine	1.6/100	60	▼	0.40	0.40	0.40	0.40	-0.16	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Depression is diagnosed by a clinician. A stool pattern is not a mood disorder, and this profile is not a mental-health screen. A high score is not evidence that you are depressed, and a low score is not evidence that you are not.

Butyrate-producer depletion is shared with everything. The Coprococcus and Dialister findings are the more specific part and are weighted accordingly.

If this score is high, the appropriate response is nothing at all without a conversation with a clinician.

Transfer of depression-like features between animals does not establish that a community causes depression in people, and one human-donor transfer experiment found no worsening from patient microbiota at all.

Sources: Zheng P, et al. Gut microbiome remodeling induces depressive-like behaviors through a pathway mediated by the host's metabolism. Mol Psychiatry 2016;21:786-796, doi:10.1038/mp.2016.44; Kelly JR, et al. Transferring the blues: depression-associated gut microbiota induces neurobehavioural changes in the rat. J Psychiatr Res 2016;82:109-118, doi:10.1016/j.jpsychires.2016.07.019; Cao Y, et al. Prenatal depression-associated gut microbiota induces depressive-like behaviors and hippocampal neuroinflammation in germ-free mice. Transl Psychiatry 2025;15, doi:10.1038/s41398-025-03606-x; Teng T, et al. Microbiota alterations leading to amino acid deficiency contribute to depression in children and adolescents. Microbiome 2025;13, doi:10.1186/s40168-025-02122-w; Knudsen JK, et al. Faecal microbiota transplantation from patients with depression or healthy individuals into rats modulates mood-related behaviour. Sci Rep 2021;11, doi:10.1038/s41598-021-01248-9; Valles-Colomer M, et al. The neuroactive potential of the human gut microbiota in quality of life and depression. Nat Microbiol 2019;4:623-632; Zheng P, et al. Gut microbiome remodeling induces depressive-like behaviors through a pathway mediated by the host's metabolism. Mol Psychiatry 2016;21:786-796; Yang J, et al. Landscapes of bacterial and metabolic signatures and their interaction in major depressive disorders. Sci Adv 2020;6:eaba8555; Strandwitz P, et al. GABA-modulating bacteria of the human gut microbiota. Nat Microbiol 2019;4:396.

What the research says about this condition and this pattern

Depression pattern — adjunct evidence alongside treatment LANE C HUMAN TRIAL · LIMITED

Fired by: Major depressive disorder pattern: notably high

Depression is a clinical diagnosis and this pattern is not a mood measurement. As adjuncts to treatment, dietitian-supported Mediterranean-style dietary change (SMILES) and one exact high-dose probiotic add-on reduced symptoms in randomized trials; FMT added to escitalopram did not.

Studied: ModiMedDiet dietary support (SMILES trial) as adjunct in major depression (diet) · **in:** people with clinically confirmed major depressive episode (moderate–severe), ages 18–99, in Australia, n = 67 · **studied protocol (a source fact, not an instruction):** ModiMedDiet with seven dietitian sessions over 12 weeks versus social support

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial.

- The SMILES trial found dietitian-supported Mediterranean-style dietary improvement reduced depressive symptoms as an adjunct in 67 adults with moderate–severe depression. [Jacka FN et al. BMC Med 2017;15:23 (SMILES)] · lane C · randomized controlled, single-blind · n = 67 · benefit
- In 47 inpatients with major depression on antidepressants, 31 days of a high-dose eight-strain probiotic add-on reduced depressive symptoms more than placebo — a single exact-product trial as adjunct to treatment. [Schaub AC et al. Transl Psychiatry 2022;12:227] · lane C · randomized double-blind placebo-controlled add-on · n = 60 · benefit
- **Against or limiting —** FMT added to escitalopram did not improve remission in depression. [FMT plus escitalopram randomized trial in depression (2026) — remission null] · lane X · randomized placebo-controlled · null

Applicability: The study required a clinical diagnosis of major depressive disorder. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you are depressed” · “probiotics replace antidepressants”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

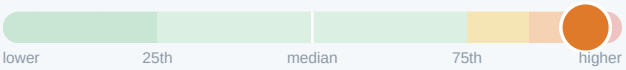
ME/CFS-like gut microbiome similarity

HIGH

Pattern version **0.2.0** (mecfs) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research only · us_adult_cohorts

The best-evidenced profile in this set. Two cohorts from different research groups, published simultaneously in the same journal, found the same core pattern: reduced abundance of the major butyrate-producing species Faecalibacterium prausnitzii and Eubacterium rectale, reduced microbial butyrate-synthesis capacity, and — crucially — confirmation by measured short-chain fatty acids and qPCR rather than DNA inference alone. Low F. prausnitzii tracked more severe fatigue. The pattern depends on how long someone has been ill. Xiong et al. found short-term patients (under 4 years) had marked microbial dysbiosis while long-term patients (over 10 years) had largely resolved microbial dysbiosis yet retained metabolic and clinical abnormalities. A low taxonomic score in someone ill for more than a decade is therefore consistent with the published pattern, not evidence against it.

COMBINED RESEMBLANCE
percentile of the matched reference group



94th
likely range 4–100

What is driving it

Taxonomic
species · study taxonomic · 4/4 features · 50% of combined



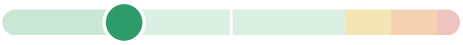
92nd HIGH

Functional
genes · genomic capacity · study functional · 2/2 features · 35% of combined



90th HIGH

Ecological
diversity · study ecological · 1/1 features · 15% of combined



27th TYPICAL

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	0.80	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Taxonomic · species · study taxonomic									
Faecalibacterium prausnitzii	0.010%	2	▼	1.00	1.00	0.50	1.00	+1.00	Guo 2023 Cell Host Microbe, Xiong 2023 Cell Host Microbe
Eubacterium rectale	0.177%	35	▼	1.00	0.70	0.60	1.00	+0.17	Guo 2023 Cell Host Microbe

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Roseburia intestinalis	not detected	13	▼	0.60	0.60	0.50	0.90	+0.37	Guo 2023 Cell Host Microbe (network analysis)
Coprococcus comes	0.713%	92	▼	0.50	0.60	0.50	0.90	-0.48	Guo 2023 Cell Host Microbe (network analysis)
Functional · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	1.00	1.00	0.50	1.00	+0.76	Guo 2023 Cell Host Microbe, Xiong 2023 Cell Host Microbe
propionate	32.9/100	93	▼	0.40	0.60	0.40	0.80	-0.72	Xiong 2023 Cell Host Microbe (SCFA panel)
Ecological · diversity · study ecological									
shannon diversity	3.17	73	▼	0.50	0.70	0.20	0.60	-0.19	Guo 2023 Cell Host Microbe

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

The 50/35/15 module split is an engineering starting point, not an evidence-derived weighting.

Measured on labelled cohorts (AUC, 0.5 = chance, 1.0 = perfect): T2D 0.60 (0.55–0.66); CRC 0.55 (0.50–0.60); IBD 0.50 (0.46–0.55). No labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.50-0.60 against other conditions).

Do not read the 90% accuracy figure from the 2025 BioMapAI paper alongside this profile. That figure came from integrating metagenomics with plasma metabolomics, immune-cell profiling and blood tests; microbiome data alone best predicted gastrointestinal, emotional and sleep symptoms specifically. A stool-only tool cannot reproduce it.

Butyrate-producer depletion is one of the least specific findings in the microbiome literature. It appears in inflammatory bowel disease, after antibiotics, with low-fibre diets, and in general ill health. A high score here should be read beside the dysbiosis anchor.

If illness duration is unknown, both strata are reported and confidence is reduced, because the expected pattern differs between them and they must not be averaged.

Sources: Guo C, Che X, Briese T, et al. Deficient butyrate-producing capacity in the gut microbiome is associated with bacterial network disturbances and fatigue symptoms in ME/CFS. Cell Host Microbe 2023;31:288-304, doi:10.1016/j.chom.2023.01.004. Xiong R, Gunter C, Fleming E, et al. Multi-omics of gut microbiome-host interactions in short- and long-term ME/CFS patients. Cell Host Microbe 2023;31:273-287, doi:10.1016/j.chom.2023.01.001; data PRJNA878603, code github.com/ohlab/MECFS_2021.

What the research says about this condition and this pattern

ME/CFS pattern — classifier evidence, thin intervention evidence LANE C HUMAN TRIAL · LIMITED

Fired by: ME/CFS-like gut microbiome similarity pattern: high

ME/CFS is a clinical diagnosis of exclusion. The microbiome classifier evidence is strong on association; intervention evidence is one small probiotic pilot and a null FMT pilot.

Studied: Lacticaseibacillus casei Shirota (Rao 2009, chronic fatigue syndrome) (probiotic) · **in:** people with clinically confirmed chronic fatigue syndrome, ages 18–65, in Canada, n = 39 · **studied protocol (a source fact, not an instruction):** 24 billion CFU L. casei Shirota daily for eight weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- A 39-person pilot of L. casei Shirota in chronic fatigue syndrome reported reduced anxiety scores; small and unreplicated. [Rao AV et al. Gut Pathog 2009;1:6] · lane C · randomized double-blind placebo-controlled pilot · n = 39 · benefit
- BioMapAI is classifier evidence that ME/CFS microbiomes differ from controls; it is not a treatment target. [BioMapAI — multi-omic ME/CFS classifier (Nat Med 2025)] · lane D · multi-omic case-control classifier
- **Against or limiting —** A randomized FMT pilot in ME/CFS found no symptom or quality-of-life benefit. [randomized FMT pilot in ME/CFS (2023) — no symptom/QoL benefit] · lane X · randomized placebo-controlled pilot · null

Applicability: The study required a clinical diagnosis of me cfs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have chronic fatigue syndrome”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

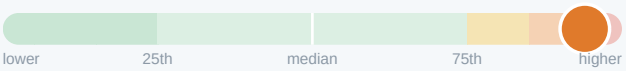
Atherosclerotic cardiovascular disease — gut pattern

HIGH

Pattern version **1.0.0** (acvd) · maturity **P3** (independent human replication with coherent task and assay) · research beta · chinese_cohort_derived

Jie et al. sequenced 218 patients with atherosclerotic cardiovascular disease and 187 controls and found Enterobacteriaceae and oral-origin Streptococcus enriched, with Roseburia, Faecalibacterium and Bacteroides depleted — the same oral-invasion plus butyrate-loss pattern as cirrhosis, which is why the two are mutual challenge sets. The mechanistic layer is better than the taxonomy: trimethylamine production from choline and carnitine by cutC/cntA, converted to TMAO in the liver, is a replicated cardiovascular risk factor with a mouse causal chain. Statins, aspirin and PPIs are near-universal in ACVD cohorts and each shifts the community.

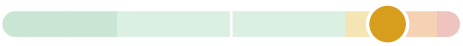
COMBINED RESEMBLANCE
percentile of the matched reference group



94th
likely range 75–99

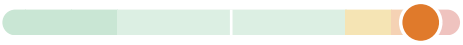
What is driving it

Jie 2017 Chinese ACVD MGWAS
species · study jie_2017 · 13/13 features · 50% of combined



84th ABOVE AVERAGE

Karlsson 2012 symptomatic atherosclerosis (Sweden)
species · study karlsson_2012 · 3/3 features · 50% of combined



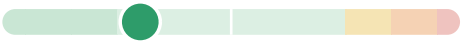
91st HIGH

TMA and metabolite gene capacity
genes · genomic capacity · study functional · 3/3 features · not in combined



90th HIGH

Oral-origin and hexa-LPS carriers
carriers · carrier proxy · study carriers · 3/3 features · not in combined



30th TYPICAL

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), uc (>99th), ibd (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Jie 2017 Chinese ACVD MGWAS · species · study jie_2017									

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Escherichia coli	not detected 0.033% pooled	19	▲	0.90	0.80	0.30	0.90	-0.29	
Klebsiella pneumoniae	not detected	40	▲	0.70	0.70	0.40	0.70	-0.09	
Enterobacter cloacae complex	not detected	46	▲	0.50	0.60	0.40	0.60	-0.03	
Streptococcus salivarius	not detected	15	▲	0.70	0.70	0.40	0.80	-0.35	
Streptococcus parasanguinis	not detected	21	▲	0.60	0.70	0.40	0.70	-0.26	
Streptococcus anginosus group	not detected	45	▲	0.50	0.60	0.40	0.70	-0.04	
Lactobacillus salivarius	0.228%	99	▲	0.50	0.60	0.40	0.60	+0.77	
Ruminococcus gnavus	13.289%	98	▲	0.50	0.60	0.20	0.60	+0.72	
Roseburia intestinalis	not detected	13	▼	0.90	0.80	0.30	0.80	+0.37	
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.80	0.20	0.80	+1.00	
Bacteroides uniformis	6.597%	82	▼	0.50	0.60	0.40	0.70	-0.26	
Prevotella copri	not detected	25	▼	0.50	0.60	0.30	0.50	+0.23	
Alistipes shahii	0.177%	85	▼	0.50	0.60	0.40	0.70	-0.34	
Karlsson 2012 symptomatic atherosclerosis (Sweden) · species · study karlsson_2012									
Collinsella aerofaciens	1.940%	88	▲	0.60	0.50	0.40	0.60	+0.35	
Roseburia intestinalis	not detected	13	▼	0.60	0.50	0.30	0.70	+0.37	
Eubacterium rectale	0.177%	35	▼	0.60	0.50	0.20	0.70	+0.17	
TMA and metabolite gene capacity · genes · genomic capacity · study functional									
cutc	4.1/100	77	▲	1.00	0.90	0.60	0.90	+0.39	
butyrate	11.6/100	1	▼	0.70	0.80	0.20	0.80	+0.76	
bsh	55.7/100	13	▼	0.40	0.50	0.30	0.50	+0.37	
Oral-origin and hexa-LPS carriers · carriers · carrier proxy · study carriers									
oral origin	0.000%	10	▲	0.70	0.70	0.40	0.80	-0.76	
hexa lps	0.039%	40	▲	0.70	0.70	0.30	0.70	-0.08	
butyrate	1.623%	4	▼	0.60	0.80	0.20	0.80	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Atherosclerosis is assessed by imaging, lipids and risk calculators. This profile is not a cardiovascular risk score.

The oral-invasion pattern is shared with cirrhosis and with PPI use; the butyrate-loss pattern with everything. Only the TMA lane is specific, and it is a capacity measurement, not a plasma TMAO level.

Sources: Jie Z, et al. The gut microbiome in atherosclerotic cardiovascular disease. Nat Commun 2017;8:845; Karlsson FH, et al. Symptomatic atherosclerosis is associated with an altered gut metagenome. Nat Commun 2012;3:1245; Tang WHW, et al. Intestinal microbial metabolism of phosphatidylcholine and cardiovascular risk. N Engl J Med 2013;368:1575.

What the research says about this condition and this pattern

Cardiovascular-disease pattern — dietary-pattern evidence independent of stool LANE B HUMAN EVIDENCE · STRONG

Fired by: Atherosclerotic cardiovascular disease pattern: high

Cardiovascular risk is assessed with blood pressure, lipids, glucose, smoking and family history. A Mediterranean dietary pattern has randomized event-reduction evidence; probiotics do not lower TMAO.

Studied: Mediterranean diet supplemented with extra-virgin olive oil or nuts (PREDIMED) (diet) · **in:** people with clinically confirmed high cardiovascular risk without CVD, ages 55–80, in Spain, n = 7447 · **studied protocol (a source fact, not an instruction):** Mediterranean diet supplemented with extra-virgin olive oil or mixed nuts; median 4.8 years

Evidence lane B: replicated human randomized clinical benefit for the exact intervention and population. Replication: replicated. Strongest design: randomized trial.

- The PREDIMED trial (republished after correction) found a Mediterranean diet with olive oil or nuts reduced major cardiovascular events by about 30% in high-risk adults — dietary-pattern evidence independent of the stool result. [Estruch R et al. N Engl J Med 2018;378:e34 (PREDIMED retraction and republication)] · lane B · randomized controlled (republished after correction for randomisation irregularities) · n = 7447 · benefit
- Chronic red-meat feeding roughly tripled measured plasma TMAO and the effect reversed after switching protein source. This measured a metabolite, not gene capacity; it does not restrict essential choline. [Wang Z et al. Eur Heart J 2019;40:583] · lane C · randomized crossover controlled feeding · n = 113 · benefit
- **Against or limiting —** Eight randomized probiotic trials pooled show no reduction in plasma TMAO. [meta-analysis of eight RCTs — probiotics and TMAO (2022)] · lane X · meta-analysis · null

Conventional route: Standard cardiovascular risk assessment with a clinician.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have heart disease” · “probiotic for TMAO”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

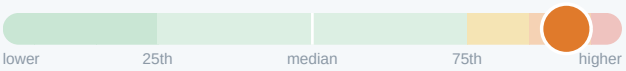
Multiple sclerosis — gut pattern

HIGH

Pattern version **1.0.0** (ms) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_international_cohorts

The iMSMS consortium is the model of how to do this: 576 MS patients each paired with a genetically unrelated household control who shares their diet and environment, shotgun sequenced. That design removes most of the confounding that plagues every other disease here. The replicated MS pattern is modest: Akkermansia muciniphila higher, Ruminococcus torques and Eisenbergiella higher, and Faecalibacterium prausnitzii, Blautia and Eubacterium rectale lower. Effect sizes are small — the household design shows that MS shifts the gut less than diet and cohabitation do. Disease-modifying therapies have their own effects, some larger than the disease's, and are recorded in the ledger.

COMBINED RESEMBLANCE
percentile of the matched reference group



91st
likely range 30–100

What is driving it

iMSMS 2022 household-paired shotgun (576 pairs) species · study imsms_2022 · 6/6 features · 33% of combined		99th	NOTABLY HIGH
Jangi 2016 US cohort (16S + shotgun subset) species · association only · 16S study · study jangi_2016 · 3/3 features · 33% of combined		88th	HIGH
Cekanaviciute 2017 UCSF (gnotobiotic-validated) species · association only · 16S study · study cekanaviciute_2017 · 2/2 features · 33% of combined		45th	TYPICAL
Metabolite gene capacity genes · genomic capacity · study functional · 3/3 features · not in combined		31st	TYPICAL
Mucin degraders and butyrate carriers carriers · carrier proxy · study carriers · 2/2 features · not in combined		99th	NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
iMSMS 2022 household-paired shotgun (576 pairs) · species · study imsms_2022									
Akkermansia muciniphila	1.425%	91	▲	1.00	0.90	0.40	0.90	+0.46	
Ruminococcus torques	1.283%	92	▲	0.70	0.80	0.40	0.80	+0.46	
Eisenbergiella tayi	not detected	37	▲	0.60	0.70	0.50	0.70	-0.11	
Faecalibacterium prausnitzii	0.010%	2	▼	0.80	0.90	0.20	0.90	+1.00	
Eubacterium rectale	0.177%	35	▼	0.60	0.80	0.20	0.80	+0.17	
Blautia obeum	0.042%	50	▼	0.50	0.70	0.40	0.70	-0.00	
Jangi 2016 US cohort (16S + shotgun subset) · species · association only · 16S study · study jangi_2016									
Akkermansia muciniphila	1.425%	91	▲	0.80	0.70	0.40	0.80	+0.46	
Methanobrevibacter smithii	not detected	35	▲	0.60	0.70	0.40	0.70	-0.13	
Butyricimonas virosa	not detected	24	▼	0.50	0.60	0.50	0.60	+0.24	
Cekanaviciute 2017 UCSF (gnotobiotic-validated) · species · association only · 16S study · study cekanaviciute_2017									
Akkermansia muciniphila	1.425%	91	▲	0.70	0.70	0.40	0.80	+0.46	
Parabacteroides distasonis	3.011%	92	▼	0.70	0.70	0.50	0.70	-0.43	
Metabolite gene capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	0.80	0.80	0.20	0.80	+0.76	
propionate	32.9/100	93	▼	0.70	0.70	0.30	0.70	-0.72	
ipa	9.6/100	89	▼	0.60	0.60	0.40	0.60	-0.71	
Mucin degraders and butyrate carriers · carriers · carrier proxy · study carriers									
mucin degraders	21.255%	94	▲	0.70	0.70	0.40	0.70	+0.65	
butyrate	1.623%	4	▼	0.70	0.80	0.20	0.80	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Multiple sclerosis is diagnosed by MRI, clinical course and cerebrospinal fluid. This profile does not diagnose or predict it.

The household-controlled design shows the MS gut effect is small. A modest score here is the expected result in MS, and a high score is more likely dysbiosis of another cause.

Akkermansia is higher in MS and in Parkinson's, and lower in most metabolic disease. Its direction here is disease-specific and the reason this profile has some specificity.

Sources: iMSMS Consortium. Gut microbiome of multiple sclerosis patients and paired household healthy controls reveal associations with disease risk and course. Cell 2022;185:3467-3486; Jangi S, et al. Alterations of the human gut microbiome in multiple sclerosis. Nat Commun 2016;7:12015; Cekanaviciute E, et al. Gut bacteria from multiple sclerosis patients modulate human T cells and exacerbate symptoms in mouse models. PNAS 2017;114:10713-10718.

What the research says about this condition and this pattern

Multiple-sclerosis pattern — surrogate-only microbiome studies LANE C HUMAN TRIAL · LIMITED

Fired by: Multiple sclerosis pattern: high

MS is diagnosed by neurology with MRI and CSF. Probiotic and FMT studies in MS are small and report immune or microbiome surrogates, not disease activity.

Studied: VSL#3 eight-strain probiotic (Tankou 2018, multiple sclerosis) (probiotic) · **in:** people with clinically confirmed relapsing-remitting multiple sclerosis, ages 18–65, in USA, n = 9 · **studied protocol (a source fact, not an instruction):** VSL#3 3600 billion CFU/day for two months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: conflicting. Strongest design: randomized trial.

- VSL#3 for two months in a small open-label MS study shifted peripheral immune markers, which reverted after discontinuation — surrogate only. [Tankou SK et al. Ann Neurol 2018;83:1147] · lane D · open-label, MS and healthy controls · n = 9 · benefit
- A 2024 randomized crossover probiotic study in MS reported surrogate immune and microbiome changes, not disease-activity outcomes. [2024 randomized crossover probiotic study in MS] · lane C · randomized crossover · mixed
- A nine-person FMT pilot in MS, terminated early, reported safety and surrogate changes only — research-only. [FMT pilot in multiple sclerosis (2022), nine participants, terminated early] · lane D · open-label pilot · n = 9 · mixed
- **Against or limiting —** FDA has documented transmission of multidrug-resistant organisms (including a death after ESBL E. coli), pathogenic E. coli, and warned about SARS-CoV-2 and mpox with FMT; donor testing, bookend testing and quarantine are required. [FDA FMT safety communications: MDRO/ESBL (June 2019, incl. one death), EPEC/STEC (March–April 2020), SARS-CoV-2 (2020), mpox (August 2022)] · lane X · safety communications · harm

Applicability: The study required a clinical diagnosis of multiple sclerosis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Fecal microbiota transplantation: investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have MS” · “probiotics for MS”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

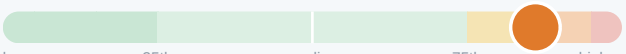
IBS mixed type — gut pattern

HIGH

Pattern version **1.0.0** (ibs_m) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Mixed-type IBS has no subtype-specific microbial marker of its own: the methanogen direction that separates IBS-C from IBS-D is undefined here, so it is left out entirely rather than scored in either direction. What remains is the shared IBS depletion pattern. This profile exists so the library is complete and so the subtype is not silently absorbed into the parent; it should score close to the parent and rarely above it.

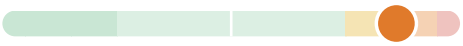
COMBINED RESEMBLANCE
percentile of the matched reference group



86th
likely range 2–100

What is driving it

Shared IBS depletion pattern
species · association only · 16S study · study pittayanon_2019 · 5/5 features · 100% of combined



86th HIGH

Butyrate capacity
genes · genomic capacity · study mechanism · 1/1 features · not in combined



99th NOTABLY HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Shared IBS depletion pattern · species · association only · 16S study · study pittayanon_2019									
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.60	0.20	0.70	+1.00	
Bifidobacterium adolescentis	0.990%	85	▼	0.70	0.60	0.30	0.70	-0.35	
Bifidobacterium longum	4.157%	95	▼	0.50	0.60	0.30	0.70	-0.56	
Ruminococcus gnavus	13.289%	98	▲	0.60	0.50	0.20	0.60	+0.72	
Escherichia coli	not detected 0.033% pooled	19	▲	0.50	0.50	0.20	0.60	-0.29	
Butyrate capacity · genes · genomic capacity · study mechanism									
butyrate	11.6/100	1	▼	0.80	0.70	0.20	0.80	+0.76	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

No microbial marker distinguishes mixed-type IBS from the other subtypes or from general dysbiosis. This profile is the shared IBS pattern with the subtype-specific methanogen direction deliberately omitted.

Sources: Pittayanon R, et al. Gut microbiota in patients with irritable bowel syndrome — a systematic review. Gastroenterology 2019;157:97-108.

What the research says about this condition and this pattern

IBS — guideline-backed diet and what is advised against LANE A CLINICAL ROUTE

Fired by: Irritable bowel syndrome pattern: notably high; IBS with diarrhoea pattern: notably high; IBS mixed type pattern: high

IBS is a symptom-based clinical diagnosis. For people with that diagnosis, ACG 2021 and the AGA diet update support soluble fibre, peppermint oil and a limited dietitian-guided low-FODMAP trial; ACG suggests against probiotics for global symptoms.

Studied: Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation) (diet) · **in:** people with clinically confirmed IBS (clinically diagnosed), ages 18–99 · **studied protocol (a source fact, not an instruction):** restriction ordinarily 4–6 weeks, then structured reintroduction and personalisation with dietitian support

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: conditional for.

- The AGA diet update endorses a limited, dietitian-guided low-FODMAP diet (restrict, reintroduce, personalise) for diagnosed IBS and warns against indefinite restriction, eating-disorder risk and malnutrition. [Chey WD et al. AGA Clinical Practice Update on the role of diet in IBS. Gastroenterology 2022;162:1737] · lane A · clinical practice update · benefit
- ACG 2021 suggests soluble fibre, peppermint oil and a limited dietitian-guided low-FODMAP trial for diagnosed IBS, suggests against probiotics for global symptoms, and lists rifaximin for global IBS-D symptoms after clinical diagnosis. [Lacy BE et al. ACG Clinical Guideline: Management of IBS. Am J Gastroenterol 2021;116:17] · lane A · guideline · benefit
- A three-week randomized low-FODMAP diet in diagnosed IBS improved symptoms and lowered urinary histamine — a diet effect in IBS, not a treatment for stool histamine genes. [McIntosh K et al. Gut 2017;66:1241] · lane C · randomized controlled · n = 40 · benefit
- **Against or limiting —** ACG 2021 suggests against probiotics for global IBS symptoms (very low quality of evidence). [Lacy BE et al. ACG 2021 — suggests against probiotics for global IBS symptoms] · lane X · guideline

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: Symptom-based diagnosis with red-flag review.

Fecal microbiota transplantation: guidelines recommend against it for this condition outside trials.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have IBS” · “cure IBS”

IBS — the two probiotic trials that met their endpoints, and the guideline position

LANE C HUMAN TRIAL · LIMITED

Fired by: Irritable bowel syndrome pattern: notably high; IBS with diarrhoea pattern: notably high; IBS mixed type pattern: high

Heat-inactivated B. bifidum MIMBb75 (443 people) and B. infantis 35624 at one specific dose are the exact products with positive randomized IBS trials; guidelines still place probiotics in the trial-only or advised-against category for global symptoms.

Studied: Heat-inactivated Bifidobacterium bifidum MIMBb75 (non-viable) (probiotic) · **in:** people with clinically confirmed IBS (Rome III), ages 18–70, in Germany, n = 443 · **studied protocol (a source fact, not an instruction):** 10^9 non-viable cells once daily for eight weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial. Guideline position: trial only.

- In 443 adults with diagnosed IBS, eight weeks of heat-inactivated B. bifidum MIMBb75 produced a composite response in 34% versus 19% on placebo. The product is non-viable and exact; it is not interchangeable with live B. bifidum and is not selected by stool abundance. [Andresen V et al. Lancet Gastroenterol Hepatol 2020;5:658] · lane C · randomized double-blind placebo-controlled · n = 443 · benefit
- In 362 women with IBS, B. infantis 35624 improved symptoms only at 10^8 CFU/day — not at a lower or a higher dose — illustrating that probiotic effects are strain- and dose-specific and do not generalise. [Whorwell PJ et al. Am J Gastroenterol 2006;101:1581] · lane C · randomized double-blind placebo-controlled, three dose arms · n = 362 · benefit
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline
- **Against or limiting —** ACG 2021 suggests against probiotics for global IBS symptoms (very low quality of evidence). [Lacy BE et al. ACG 2021 — suggests against probiotics for global IBS symptoms] · lane X · guideline

Applicability: The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “any probiotic” · “Bifidobacterium supplement fixes IBS”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

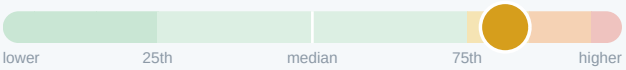
Symptomatic dermographism (chronic inducible hives from skin stroking) — translated 16S pattern

ABOVE AVERAGE

Pattern version **0.1.0** (symptomatic_dermographism) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · research only · chinese_adult_cohort

Symptomatic dermographism is the commonest chronic inducible urticaria: hives that appear along the line where the skin has been stroked, scratched or rubbed, itching for minutes to an hour. One 16S amplicon study supplies the pattern here — Liu et al. compared stool from 22 adults with symptomatic dermographism against 22 healthy controls and described a healthy-control core community that the patient group lacked. Only two candidates from that core are imported: the genus Subdoligranulum and the species Ruminococcus bromii, both depletion-oriented. They are scored, but every number on this card carries a permanent transport warning, because a 16S study names organisms by amplicon and this report measures them by shotgun sequencing. The two are not the same measurement, and no shotgun cohort has tested this pattern. Two features is also a very small panel: coverage is printed beside the score for that reason, and one missing feature halves it. This profile is deliberately separate from the chronic spontaneous urticaria profile. The two conditions coexist often, and when they do both eligible cards are shown side by side — there is no combined score, and the two indices are never multiplied or averaged into a third number.

COMBINED RESEMBLANCE
percentile of the matched reference group



81st
likely range 50–100

The same species on the source study's own scale

Not computed on this scale (no usable features). No number is shown rather than a zero, because a zero here would be a measurement that was never made. Panel features and why: *Subdoligranulum* — reference ineligible; *Ruminococcus bromii* — below reporting limit.

The percentile above still has a value because the two calculations treat a nondetection differently, and the difference is the point of showing both. The percentile places an undetected organism at a fixed floor and ranks the sample from there, which is how every reading in this report is built and what makes them comparable with each other. This scale declines to convert an undetected organism into a number at all unless a validated detection limit exists for it, and none does at this sequencing depth. Read the percentile as the position of this community, and read the blank here as the reason not to put much weight on it.

What is driving it

Liu 2021 symptomatic-dermographism core-community candidates (16S, translated)
species · association only · 16S study · study liu_sd_2021 · 2/2 features · 100% of combined



81st ABOVE AVERAGE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Liu 2021 symptomatic-dermographism core-community candidates (16S, translated) · species · association only · 16S study · study liu_sd_2021									
Subdoligranulum (genus)	0.000%	50	▼	1.00	0.30	0.20	0.50	**+0.00**	
Ruminococcus bromii	not detected	19	▼	1.00	0.30	0.20	0.50	**+0.29**	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Symptomatic dermographism is diagnosed by stroking the skin and watching what happens, usually with a calibrated dermographometer. A stool profile cannot make that observation, and this figure is a research pattern index, not a probability of having the condition.

Every number here crosses assay types. Until an independent shotgun cohort tests these two directions, the transport warning stays on the card regardless of how the score comes out.

The other chronic inducible urticarias — the other forms of hives brought on by a trigger: cold, delayed pressure, solar, heat, vibratory, cholinergic, contact and aquagenic — have no stool signature this audit could locate. They are reported as evidence not established rather than scored with this profile or with the chronic spontaneous urticaria (spontaneous chronic hives) profile standing in for them. Skin and sweat Malassezia mechanisms proposed for cholinergic urticaria — hives brought on by heat, exercise or stress — are not gut findings.

Having both chronic spontaneous urticaria and dermographism at once — spontaneous hives as well as hives raised by stroking the skin — is common. When both profiles are eligible, both scores are shown; there is no third combined index, and neither score modifies the other.

Sources: Liu R, Peng C, Jing D, et al. Biomarkers of gut microbiota in chronic spontaneous urticaria and symptomatic dermographism. Exp Dermatol 2021;30:1657-1665, doi:10.1111/exd.14326; raw data OEP001229 (BioSino). Coexisting-phenotype cohort: Liu R, et al. Front Cell Infect Microbiol 2021;11:703126, doi:10.3389/fcimb.2021.703126, raw data OEP001891. Clinical classification of the inducible urticarias: Zuberbier T, et al., International guideline for the definition, classification, diagnosis and management of urticaria, Allergy 2026, doi:10.1111/all.70210.

What the research says about this condition and this pattern

Symptomatic dermographism (hives from stroking the skin) — no microbiome-directed evidence LANE A HUMAN EVIDENCE · STRONG

Fired by: Symptomatic dermographism (chronic inducible hives from skin stroking) pattern: somewhat high

No intervention study of any kind has been done in symptomatic dermographism selected by a microbiome result, and the two directions scored here come from a single 16S cohort carried across to shotgun measurement. The evidence that exists is the general urticaria guideline: a second-generation H1 antihistamine, with specialist assessment for inducible hives that do not respond. The probiotic and dietary studies listed under chronic spontaneous urticaria enrolled people with that condition and are not transferred here.

Studied: Guideline urticaria care — second-generation H1 antihistamine first line, with specialist escalation for chronic hives that do not respond (clinical route) · **in:** people with clinically confirmed urticaria (hives), ages 0–99 · **studied protocol (a source fact, not an instruction):** second-generation H1 antihistamine first line, with specialist assessment and guideline escalation when hives persist

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline.

- The current international guideline is the actual care pathway for hives: a second-generation H1 antihistamine first, then specialist assessment and guideline escalation if the hives persist. It is a clinician-directed route, not a stool-profile-directed algorithm, and this report does not generate a drug ladder from a microbiome score. [Zuberbier T, et al. International guideline for the definition, classification, diagnosis and management of urticaria. Allergy 2026] · lane A · international consensus guideline · benefit

- **Against or limiting —** Fibre, prebiotics and fermented foods have general dietary evidence but no validated rule for selecting them from a hives profile. Fermented foods in particular are not recommended to someone reporting food- or histamine-related reactions merely because the panel shows depleted fibre fermenters. [Reproducible PubMed search for human evidence on fibre, prebiotic or fermented-food selection in urticaria; run for this build, evidence checked to September 2026] · lane X · evidence search returning no eligible response-selection rule

- **Against or limiting —** No human study supports berberine, allicin, oregano oil, black seed oil or commercial multi-herbal antimicrobial blends for chronic hives, and none supports choosing them from a stool result. Detecting E. coli, Klebsiella or R. gnavus in the hives panel is not an eradication indication, and no promise is made that suppressing an organism restores the commensals around it. [Reproducible PubMed search for human evidence on taxon-selected antimicrobial supplement regimens (berberine, allicin, oregano oil, black seed oil, multi-herbal blends) in urticaria; run for this build, evidence checked to September 2026] · lane X · evidence search returning no eligible human study

Applicability: The study required a clinical diagnosis of symptomatic dermographism. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Conventional route: Clinician assessment, including provocation testing where appropriate, for inducible hives.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “probiotics for dermographism” · “supplements to stop the skin reacting” · “this pattern explains your dermographism”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

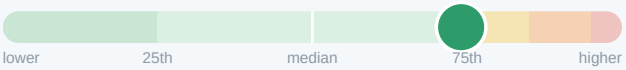
IBS with constipation — gut pattern

TYPICAL

Pattern version **1.0.0** (ibs_c) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Constipation-predominant IBS has one microbial feature the other subtypes lack: methanogen enrichment. Methane slows intestinal transit in animal models and breath methane tracks constipation in humans, so Methanobrevibacter smithii and the mcrA gene panel are the subtype-specific evidence here. The rest is inherited from the parent IBS pattern.

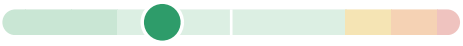
COMBINED RESEMBLANCE
percentile of the matched reference group



74th
likely range 6–99

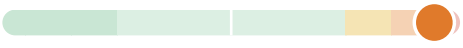
What is driving it

Methanogen enrichment in constipation
species · study methane_constipation · 1/1 features · 50% of combined



35th TYPICAL

Shared IBS depletion pattern
species · association only · 16S study · study pittayanon_2019 · 2/2 features · 50% of combined



94th HIGH

Methanogenesis gene capacity
genes · genomic capacity · study methane_capacity · 1/1 features · not in combined



0 NOTABLY LOW

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Methanogen enrichment in constipation · species · study methane_constipation									
Methanobrevibacter smithii	not detected	35	▲	1.00	0.70	0.60	0.80	**-0.13**	Kim 2012 Dig Dis Sci; Ghoshal 2016
Shared IBS depletion pattern · species · association only · 16S study · study pittayanon_2019									
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.60	0.20	0.70	**+1.00**	
Bifidobacterium adolescentis	0.990%	85	▼	0.60	0.60	0.30	0.70	**-0.35**	
Methanogenesis gene capacity · genes · genomic capacity · study methane_capacity									
methane	0.0/100	31	▲	1.00	0.70	0.60	0.80	**-0.17**	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBS subtypes are defined by stool form on the Bristol scale, not by stool DNA. This profile describes a community pattern associated with constipation, which many people without IBS also carry.

Methanogen carriage is common and partly heritable. High methanogen abundance is a normal state for a large minority of healthy adults.

Sources: Pimentel M, et al. Methane, a gas produced by enteric bacteria, slows intestinal transit and augments small intestinal contractile activity. Am J Physiol 2006;290:G1089; Kim G, et al. Methanobrevibacter smithii is the predominant methanogen in patients with constipation-predominant IBS and methane on breath. Dig Dis Sci 2012;57:3213.

‹ at a glance

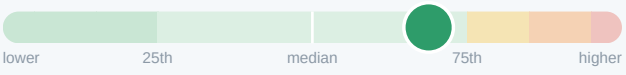
Chronic spontaneous urticaria (chronic hives) gut-microbiome similarity

TYPICAL

Pattern version **0.1.0** (csu) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research only · chinese_adult_cohort

Chronic spontaneous urticaria — chronic hives, historically called chronic idiopathic urticaria — is itchy weals and often deep swelling that keep recurring for more than six weeks without an external trigger anyone can point to. One human shotgun-metagenomic cohort supplies the pattern scored here: Zhu et al. sequenced stool from 26 adults with chronic hives and 26 healthy controls and reported fifteen species-level directions. The depletions sit in the fibre-fermenting, short-chain-fatty-acid-producing end of the community — three Alistipes species, Coprococcus catus, Roseburia hominis, Blautia (Ruminococcus) obeum, two Bacteroides — and the enrichments in the Gram-negative and mucin-degrading end, led by Ruminococcus gnavus, with Escherichia coli, Klebsiella pneumoniae, Veillonella parvula and Bacteroides stercoris. Eight of the fifteen survive Benjamini-Hochberg correction in the source table; the other seven are exploratory and labelled as such. Both are scored, the strict eight separately as a sensitivity panel, because a fifteen-species panel selected in fifty-two people is a discovery result and the sensitivity panel is how the report says so out loud. Every feature carries the same scoring weight: the published association coefficients are recorded as provenance and are deliberately not used as model weights, because a MaAsLin2 coefficient fitted on a discovery cohort is not a deployable classifier. Amplicon cohorts from other centres agree on some of this and disagree on the rest. Diversity has been reported lower, unchanged and higher in chronic hives; genus Lactobacillus has been reported both depleted and enriched. That is why this profile contains no diversity term and no genus-level term, and why the conflicting cohorts are listed rather than averaged away. A high score is a research observation about community composition. It does not diagnose hives, and a low score in someone with hives does not argue against the diagnosis.

COMBINED RESEMBLANCE
percentile of the matched reference group



69th
likely range 6–96

The same species on the source study's own scale



57 out of 100 — with no percent sign, because this is not a percentage, a probability or a match score. Coverage 7 of 15 panel features; the pale band, 27 to 80, is where the figure could sit once the unmeasured features are accounted for. Fifty is neutral signed deviation under this transformation, not a definition of health, and higher means stronger alignment with the directions the source study reported.

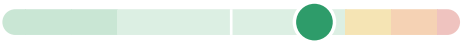
Strict subset, the 8 species that survive multiple-testing correction in the source table: **63** at coverage 4 of 8. This is a sensitivity analysis on the same measurements, not a second independent test — the two panels share every one of these features, so a gap between them means the wider figure is being carried by its exploratory members.

These are coverage and censoring bounds, not a 95% confidence interval: they widen when a feature cannot be measured and say nothing about sampling error, about how well a single-centre discovery cohort transports to this population, or about how specific the pattern is to this condition. The percentile above and the figure here read the same measurements two ways, and are not two opinions about one quantity: the percentile is a rank, so it says how this sample compares with a matched reference group and it moves when that group's spread changes; this figure is a fixed transformation, so it says how far each organism sits from its reference centre in the direction the study reported, and it stays put. A high rank can therefore sit beside a middling figure, or the reverse, depending on how tightly the reference group clusters and on which features could be measured at all.

Features outside the figure: *Alistipes onderdonkii* (below reporting limit); *Bacteroides cellulosilyticus* (below reporting limit); *Bacteroides intestinalis* (below reporting limit); *Coprococcus catus* (below reporting limit); *Roseburia hominis* (below reporting limit); *Veillonella parvula* (below reporting limit); *Escherichia coli* (below reporting limit); *Klebsiella pneumoniae* (below reporting limit). None of them was scored as zero.

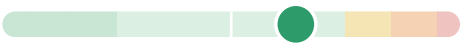
What is driving it

Zhu 2024 shotgun species panel (all 15 reported directions)
species · study zhu_2024 · 15/15 features · 100% of combined



68th TYPICAL

Zhu 2024 strict subset (BH q < 0.05) — sensitivity only
species · study zhu_2024 · 8/8 features · not in combined



64th TYPICAL

Gram-negative envelope and endotoxin capacity (mechanism context)
carriers · carrier proxy · study zhu_2024_function · 2/2 features · not in combined



75th ABOVE AVERAGE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Zhu 2024 shotgun species panel (all 15 reported directions) · species · study zhu_2024									
Alistipes onderdonkii	not detected 1.22% pooled	43	▼	1.00	0.40	0.30	0.70	+0.06	
Alistipes putredinis	3.953%	90	▼	1.00	0.40	0.30	0.70	-0.43	
Alistipes shahii	0.177%	85	▼	1.00	0.40	0.30	0.70	-0.34	
Coprococcus catus	not detected	16	▼	1.00	0.40	0.20	0.80	+0.34	
Bacteroides intestinalis	not detected	36	▼	1.00	0.40	0.30	0.70	+0.12	
Ruminococcus obeum, measured as Blautia obeum	0.042%	50	▼	1.00	0.40	0.30	0.60	-0.00	
Veillonella parvula	not detected	31	▲	1.00	0.40	0.20	0.60	-0.17	
Ruminococcus gnavus	13.289%	98	▲	1.00	0.40	0.30	0.80	+0.72	
Bacteroides cellulosilyticus	not detected	26	▼	1.00	0.40	0.30	0.50	+0.22	
Odoribacter splanchnicus	0.716%	94	▼	1.00	0.40	0.20	0.40	-0.51	
Roseburia hominis	not detected	7	▼	1.00	0.40	0.20	0.60	+0.97	
Bacteroides stercoris	2.822%	89	▲	1.00	0.40	0.30	0.50	+0.40	
Bilophila wadsworthia	0.039%	83	▼	1.00	0.40	0.30	0.40	-0.32	
Escherichia coli	not detected 0.033% pooled	19	▲	1.00	0.40	0.20	0.60	-0.29	
Klebsiella pneumoniae	not detected	40	▲	1.00	0.40	0.20	0.60	-0.09	
Zhu 2024 strict subset (BH q < 0.05) — sensitivity only · species · study zhu_2024									
Alistipes onderdonkii	not detected 1.22% pooled	43	▼	1.00	0.40	0.30	0.70	+0.06	
Alistipes putredinis	3.953%	90	▼	1.00	0.40	0.30	0.70	-0.43	
Alistipes shahii	0.177%	85	▼	1.00	0.40	0.30	0.70	-0.34	
Bacteroides intestinalis	not detected	36	▼	1.00	0.40	0.30	0.70	+0.12	
Coprococcus catus	not detected	16	▼	1.00	0.40	0.20	0.80	+0.34	
Ruminococcus obeum, measured as Blautia obeum	0.042%	50	▼	1.00	0.40	0.30	0.60	-0.00	

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Veillonella parvula	not detected	31	▲	1.00	0.40	0.20	0.60	**-0.17**	
Ruminococcus gnavus	13.289%	98	▲	1.00	0.40	0.30	0.80	**+0.72**	
Gram-negative envelope and endotoxin capacity (mechanism context) · carriers · carrier proxy · study zhu_2024_function									
hexa lps	0.039%	40	▲	0.60	0.40	0.20	0.50	**-0.08**	
opportunistic pathogens	13.399%	95	▲	0.50	0.40	0.20	0.50	**+0.59**	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Chronic hives is a clinical diagnosis made from weals, itch and their duration. Nothing in stool DNA makes or excludes it, and this score is not a probability of having it. A diagnosis of chronic spontaneous urticaria needs weals or angioedema recurring for more than six weeks without an identifiable external trigger, assessed by a clinician.

The fifteen species were selected in the same fifty-two people whose data produced them. Any evaluation on that cohort is reconstruction, not external validation, and no independent shotgun cohort has yet tested this panel. The strict-eight sensitivity figure is the same measurements re-summarised, not a second opinion.

Diversity is deliberately absent from this profile. It has been reported lower (Lu 2019), unchanged (Wang 2021, Ciftci 2025) and higher (Podder 2025) in chronic hives, and the cohorts that disagree differ in treatment exposure more than in phenotype. A Shannon or evenness figure elsewhere in this report says nothing about this pattern.

Detection of Escherichia coli, Klebsiella pneumoniae or Ruminococcus gnavus in this panel is a relative abundance measurement. It is not an infection, not an overgrowth diagnosis and not an indication for antimicrobials, herbal or prescription. No microbiome score authorises eradication, a transplant, or stopping any hives treatment already prescribed.

Only chronic spontaneous urticaria — spontaneous chronic hives — is scored here. Symptomatic dermographism, the hives raised by stroking the skin, has its own separate and explicitly translated profile. The other inducible forms of hives (cold, delayed pressure, solar, heat, vibratory, cholinergic, contact and aquagenic urticaria) have no stool-signature cohort that this audit could locate: evidence not established, so they carry no score at all rather than borrowing this one. Short-lived acute hives, urticarial vasculitis, mastocytosis and bradykinin-mediated angioedema are different conditions from chronic hives and are not scored here either.

Sources: Zhu L, Jian X, Zhou B, Liu R, Muñoz M, Sun W, Xie L, Chen X, Peng C, Maurer M, Li J. Gut microbiota facilitate chronic spontaneous urticaria. Nat Commun 2024;15:112, doi:10.1038/s41467-023-44373-x; raw data OEP002960 (BioSino), metabolomics CNP0005021, mouse 16S OEP002997. Conflicting and supporting amplicon cohorts: Lu 2019 PMC6881578; Wang 2020 doi:10.3389/fcimb.2020.00024; Wang 2021 doi:10.3389/fimmu.2021.691304; Luo 2023 doi:10.3389/fcimb.2022.1094737; Cesic 2023 doi:10.3390/life13061280; Podder 2025 doi:10.1002/clt2.70027; Park 2025 doi:10.1111/all.16601; Ciftci 2025 Pathogens 14(11):1140; Cho 2026 doi:10.1111/exd.70208; Wang 2026 doi:10.1111/1751-7915.70316. Clinical classification: Zuberbier T, et al. International guideline for the definition, classification, diagnosis and management of urticaria, Allergy 2026, doi:10.1111/all.70210.

‹ at a glance

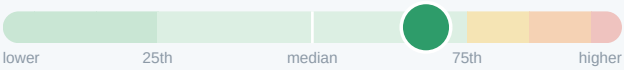
Metabolic dysfunction-associated steatotic liver disease — gut pattern

TYPICAL

Pattern version **1.0.0** (masld) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_us_european_cohorts

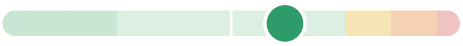
Fatty liver disease (formerly NAFLD) is common enough that its gut pattern is well sampled, and confounded enough by adiposity and diabetes that the pattern is hard to attribute. Loomba et al. built a shotgun signature for advanced fibrosis in biopsy-proven patients — Escherichia coli and Bacteroides vulgatus higher, Eubacterium rectale lower — and Hoyles et al. linked hepatic steatosis in non-diabetic obese women to low gene richness and higher branched-chain amino acid and aromatic amino acid biosynthesis potential. Endogenous ethanol production by Klebsiella pneumoniae strains is a reproducible mechanism in a subset of patients (Yuan 2019). Obesity and T2D are mutual challenge sets because most MASLD patients have one or both.

COMBINED RESEMBLANCE
percentile of the matched reference group



68th
likely range 20–89

What is driving it



Loomba 2017 biopsy-proven NAFLD (fibrosis signature)
species · study loomba_2017 · 4/4 features · 33% of combined
62nd — TYPICAL

Yuan 2019 high-alcohol-producing Klebsiella
species · study yuan_2019 · 1/1 features · 33% of combined
0 — NOTABLY LOW

Hoyles 2018 FLORINASH non-diabetic obese women
species · study hoyles_2018 · 3/3 features · 17% of combined
97th — NOTABLY HIGH

Gene richness
diversity · study hoyles_2018 · 1/2 features · 17% of combined
39th — TYPICAL

Metabolite gene capacity
genes · genomic capacity · study functional · 4/4 features · not in combined
>99th — NOTABLY HIGH

Hexa-LPS carriers
carriers · carrier proxy · study carriers · 2/2 features · not in combined
84th — ABOVE AVERAGE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	0.94	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Loomba 2017 biopsy-proven NAFLD (fibrosis signature) · species · study loomba_2017									
Escherichia coli	not detected 0.033% pooled	19	▲	0.90	0.70	0.30	0.80	-0.29	
Bacteroides vulgatus	22.292%	93	▲	0.80	0.70	0.30	0.70	+0.41	
Eubacterium rectale	0.177%	35	▼	0.80	0.70	0.20	0.80	+0.17	
Blautia obeum	0.042%	50	▼	0.50	0.60	0.40	0.60	-0.00	
Yuan 2019 high-alcohol-producing Klebsiella · species · study yuan_2019									
Klebsiella pneumoniae	not detected	40	▲	0.60	0.60	0.40	0.50	-0.09	
Hoyles 2018 FLORINASH non-diabetic obese women · species · study hoyles_2018									
Escherichia coli	not detected 0.033% pooled	19	▲	0.60	0.60	0.30	0.60	-0.29	
Eubacterium eligens	not detected	8	▼	0.60	0.60	0.40	0.70	+1.00	
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.60	0.20	0.70	+1.00	
Gene richness · diversity · study hoyles_2018									
richness	92.00	60	▼	0.80	0.70	0.10	0.80	-0.09	
shannon	not detected	—	▼	0.50	0.70	0.10	0.80	—	
Metabolite gene capacity · genes · genomic capacity · study functional									
bcaa	209.5/100	92	▲	0.80	0.70	0.40	0.70	+0.39	
bai	8.2/100	97	▲	0.60	0.60	0.30	0.60	+1.00	
butyrate	11.6/100	1	▼	0.70	0.70	0.20	0.80	+0.76	
urda	33.1/100	39	▲	0.40	0.50	0.40	0.50	-0.10	
Hexa-LPS carriers · carriers · carrier proxy · study carriers									
hexa lps	0.039%	40	▲	0.70	0.70	0.30	0.70	-0.08	
butyrate	1.623%	4	▼	0.60	0.70	0.20	0.80	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Steatotic liver disease is diagnosed by imaging or elastography and staged by biopsy or fibrosis scores. This profile does none of that.

The pattern is largely the obesity and metabolic-syndrome pattern. A high score in someone with obesity is expected and not informative about the liver.

Sources: Loomba R, et al. Gut microbiome-based metagenomic signature for non-invasive detection of advanced fibrosis in human nonalcoholic fatty liver disease. Cell Metab 2017;25:1054-1062; Hoyles L, et al. Molecular phenomics and metagenomics of hepatic steatosis in non-diabetic obese women. Nat Med 2018;24:1070-1080; Yuan J, et al. Fatty liver disease caused by high-alcohol-producing Klebsiella pneumoniae. Cell Metab 2019;30:675-688.

‹ at a glance

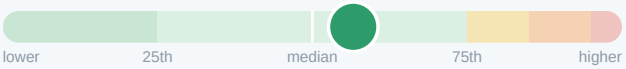
Hypertension — gut pattern

TYPICAL

Pattern version **1.0.0** (hypertension) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · chinese_cohort_derived

Li et al. shotgun-sequenced 196 Chinese participants across normotension, pre-hypertension and hypertension and found reduced gene richness with Prevotella and Klebsiella enrichment, and faecal transplant from hypertensive humans raised blood pressure in germ-free mice. The mechanistic story is short-chain fatty acids acting on the Olfr78 and GPR41 receptors in the kidney and vasculature (Pluznick 2013), which makes butyrate and propionate capacity the lanes worth reading. Salt intake, adiposity and antihypertensive drugs all shape the gut, and the pattern is not separable from the metabolic-syndrome pattern.

COMBINED RESEMBLANCE
percentile of the matched reference group



57th
likely range 29–83

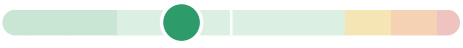
What is driving it

Li 2017 Chinese hypertension shotgun (196)
species · study li_2017 · 8/8 features · 50% of combined



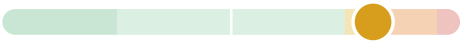
91st HIGH

Gene richness
diversity · study li_2017 · 1/2 features · 50% of combined



39th TYPICAL

SCFA and TMA capacity
genes · genomic capacity · study functional · 3/3 features · not in combined



81st ABOVE AVERAGE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	0.79	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), uc (>99th), ibd (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Li 2017 Chinese hypertension shotgun (196) · species · study li_2017									
Prevotella copri	not detected	25	▲	0.70	0.60	0.30	0.60	-0.23	
Klebsiella pneumoniae	not detected	40	▲	0.70	0.60	0.40	0.70	-0.09	
Enterobacter cloacae complex	not detected	46	▲	0.40	0.50	0.40	0.50	-0.03	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.60	0.20	0.70	+1.00	
Roseburia intestinalis	not detected	13	▼	0.60	0.60	0.30	0.70	+0.37	

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Bacteroides uniformis	6.597%	82	▼	0.50	0.60	0.40	0.60	-0.26	
Oscillibacter sp 57 20	not detected	14	▼	0.40	0.50	0.40	0.50	+0.36	
Butyrivibrio crossotus	not detected	41	▼	0.40	0.50	0.50	0.50	+0.08	
Gene richness · diversity · study li_2017									
richness	92.00	60	▼	0.70	0.60	0.10	0.70	-0.09	
shannon	not detected	—	▼	0.50	0.60	0.10	0.70	—	
SCFA and TMA capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	0.80	0.60	0.20	0.70	+0.76	
propionate	32.9/100	93	▼	0.60	0.60	0.30	0.60	-0.72	
cutc	4.1/100	77	▲	0.50	0.60	0.40	0.60	+0.39	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Blood pressure is measured with a cuff. It is cheap and definitive. This profile adds nothing to that measurement.

The pattern is the metabolic-syndrome pattern plus a Prevotella enterotype signal that is diet- and geography-dependent.

Sources: Li J, et al. Gut microbiota dysbiosis contributes to the development of hypertension. Microbiome 2017;5:14; Pluznick JL, et al. Olfactory receptor responding to gut microbiota-derived signals plays a role in renin secretion and blood pressure regulation. PNAS 2013;110:4410.

‹ at a glance

Liver cirrhosis — gut pattern

TYPICAL

Pattern version **1.0.0** (cirrhosis) · maturity **P3** (independent human replication with coherent task and assay) · research beta · chinese_cohort_derived

One of the largest effect sizes in the gut microbiome literature. Qin et al. found that cirrhosis drives oral bacteria into the gut — Veillonella, Streptococcus, Haemophilus — because reduced bile acid output and gastric acid suppression remove the barriers that normally keep them out. The Bajaj cirrhosis dysbiosis ratio captures the same thing: oral and Enterobacteriaceae taxa up, Lachnospiraceae and Ruminococcaceae down. Cross-validated AUC for cirrhosis versus healthy exceeds 0.9 in the discovery cohort. Proton pump inhibitors reproduce the oral-invasion part of this pattern on their own, and most cirrhotic patients are on them.

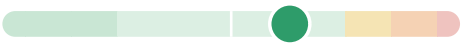
COMBINED RESEMBLANCE
percentile of the matched reference group



52nd
likely range 6–94

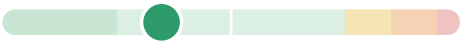
What is driving it

Bajaj cirrhosis dysbiosis ratio (16S, US)
species · association only · 16S study · study bajaj_2014 · 4/4 features · 67% of combined



63rd — TYPICAL

Qin 2014 Chinese cirrhosis MGWAS
species · study qin_2014 · 12/12 features · 33% of combined

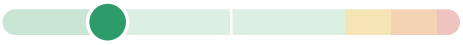


35th — TYPICAL

Community ecology
diversity · study qin_2014 · 0/1 features · not in combined

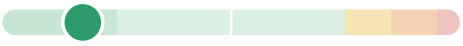
— NOT COMPUTABLE

Metabolite gene capacity
genes · genomic capacity · study functional · 4/4 features · not in combined



23rd — BELOW AVERAGE

Oral-origin carrier cross-check
carriers · carrier proxy · study carriers · 2/2 features · not in combined



18th — BELOW AVERAGE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Bajaj cirrhosis dysbiosis ratio (16S, US) · species · association only · 16S study · study bajaj_2014									
Veillonella parvula	not detected	31	▲	0.70	0.70	0.50	0.80	-0.17	
Escherichia coli	not detected 0.033% pooled	19	▲	0.60	0.70	0.30	0.70	-0.29	
Roseburia intestinalis	not detected	13	▼	0.60	0.70	0.30	0.70	+0.37	
Ruminococcus bromii	not detected	19	▼	0.60	0.70	0.40	0.70	+0.29	
Qin 2014 Chinese cirrhosis MGWAS · species · study qin_2014									
Veillonella parvula	not detected	31	▲	1.00	0.90	0.50	0.90	-0.17	
Veillonella atypica	not detected	35	▲	0.80	0.80	0.50	0.80	-0.13	
Streptococcus salivarius	not detected	15	▲	0.90	0.90	0.40	0.90	-0.35	
Streptococcus parasanguinis	not detected	21	▲	0.80	0.80	0.40	0.80	-0.26	
Streptococcus anginosus group	not detected	45	▲	0.60	0.70	0.40	0.70	-0.04	
Haemophilus parainfluenzae	not detected	29	▲	0.80	0.80	0.40	0.80	-0.18	
Lactobacillus salivarius	0.228%	99	▲	0.50	0.60	0.40	0.60	+0.77	
Eubacterium rectale	0.177%	35	▼	0.70	0.80	0.20	0.80	+0.17	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.80	0.20	0.80	+1.00	
Alistipes putredinis	3.953%	90	▼	0.60	0.70	0.40	0.70	-0.43	
Bacteroides uniformis	6.597%	82	▼	0.50	0.70	0.40	0.70	-0.26	
Coprococcus comes	0.713%	92	▼	0.50	0.70	0.30	0.70	-0.48	
Community ecology · diversity · study qin_2014									
shannon	not detected	—	▼	0.60	0.80	0.10	0.80	—	
Metabolite gene capacity · genes · genomic capacity · study functional									
bai	8.2/100	97	▼	0.90	0.80	0.50	0.80	-1.00	
urease	5.2/100	8	▲	0.70	0.70	0.40	0.70	-0.34	
butyrate	11.6/100	1	▼	0.70	0.80	0.20	0.80	+0.76	
bsh	55.7/100	13	▼	0.50	0.60	0.30	0.60	+0.37	
Oral-origin carrier cross-check · carriers · carrier proxy · study carriers									
oral origin	0.000%	10	▲	1.00	0.80	0.50	0.90	-0.76	
butyrate	1.623%	4	▼	0.60	0.80	0.20	0.80	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: 67% of planned feature weight was measurable (below 75%); no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Cirrhosis is diagnosed by imaging, elastography, liver tests and, where needed, biopsy. This profile does not diagnose it.

Proton pump inhibitor use reproduces the oral-invasion pattern that carries this profile. A PPI user without liver disease can score high here. The confounder ledger shows whether PPI use was recorded.

Sources: Qin N, et al. Alterations of the human gut microbiome in liver cirrhosis. Nature 2014;513:59-64; Bajaj JS, et al. Altered profile of human gut microbiome is associated with cirrhosis and its complications. J Hepatol 2014;60:940-947; Kakiyama G, et al. Modulation of the fecal bile acid profile by gut microbiota in cirrhosis. J Hepatol 2013;58:949.

‹ at a glance

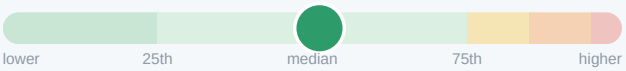
Coeliac disease — gut pattern

TYPICAL

Pattern version **1.0.0** (celiac) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · mixed_european_cohorts

Coeliac disease has a confounder built into its treatment: the gluten-free diet removes most wheat-derived fermentable substrate and depletes Bifidobacterium and butyrate producers on its own (De Palma 2009, Bonder 2016). Untreated coeliac shows Proteobacteria and Bacteroides expansion with Bifidobacterium loss; treated coeliac looks like a gluten-free healthy person. The most specific finding is functional — the CELIAC Disease Genomic, Environmental, Microbiome and Metabolomic (CDGEMM) infant study and the Leonard 2021 prospective cohort found changes in carbohydrate-active enzyme and tryptophan metabolism before onset — which is why the CAZy module is declared even though this pipeline cannot compute it. Gluten-free diet status is the one thing needed to read this profile, and the confounder ledger will say whether it was supplied.

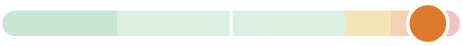
COMBINED RESEMBLANCE
percentile of the matched reference group



51st
likely range 2–99

What is driving it

Leonard 2021 CDGEMM prospective infants
species · study cdgemm · 4/4 features · 50% of combined



93rd — HIGH

Untreated coeliac consensus (16S, multiple cohorts)
species · association only · 16S study · study untreated_consensus · 6/6 features · 50% of combined



6th — LOW

Carbohydrate-active enzyme shifts before onset
CAZymes · genomic capacity · study cdgemm · 0/2 features · not measurable — assembly plus dbCAN3 carbohydrate-active enzyme annotation is added

— NOT COMPUTABLE

Metabolite gene capacity
genes · genomic capacity · study functional · 2/2 features · not in combined



41st — TYPICAL

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Leonard 2021 CDGEMM prospective infants · species · study cdgemm									
Dialister invisus	not detected	39	▲	0.60	0.50	0.50	0.60	-0.09	
Parabacteroides distasonis	3.011%	92	▲	0.50	0.50	0.40	0.50	+0.43	
Bacteroides vulgatus	22.292%	93	▼	0.50	0.50	0.30	0.50	-0.41	
Faecalibacterium prausnitzii	0.010%	2	▼	0.60	0.60	0.20	0.70	+1.00	
Untreated coeliac consensus (16S, multiple cohorts) · species · association only · 16S study · study untreated_consensus									
Bifidobacterium longum	4.157%	95	▼	0.80	0.60	0.30	0.70	-0.56	
Bifidobacterium adolescentis	0.990%	85	▼	0.60	0.60	0.30	0.70	-0.35	
Escherichia coli	not detected 0.033% pooled	19	▲	0.70	0.60	0.30	0.70	-0.29	
Bacteroides fragilis	not detected	24	▲	0.50	0.50	0.30	0.60	-0.24	
Clostridium bolteae	0.056%	89	▲	0.40	0.50	0.40	0.50	+0.40	
Lactobacillus ruminis	not detected	44	▼	0.40	0.50	0.40	0.50	+0.05	
Carbohydrate-active enzyme shifts before onset · CAZymes · genomic capacity · study cdgemm									
GH13 subfamily shift	missing	—	▲	1.00	1.00	1.00	1.00	—	
GH43 shift	missing	—	▼	1.00	1.00	1.00	1.00	—	
Metabolite gene capacity · genes · genomic capacity · study functional									
ipa	9.6/100	89	▼	0.80	0.60	0.50	0.70	-0.71	
butyrate	11.6/100	1	▼	0.60	0.60	0.20	0.70	+0.76	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Coeliac disease is diagnosed by tissue transglutaminase serology and duodenal biopsy. This profile does not diagnose it, and a stool DNA pattern is irrelevant to whether someone should eat gluten.

A gluten-free diet produces this pattern in people without coeliac disease. A high score in someone avoiding gluten is expected.

The most specific pre-onset finding (CAZy families) is declared but not computable here.

Sources: Leonard MM, et al. Microbiome signatures of progression toward celiac disease onset in at-risk children in a longitudinal prospective cohort study. PNAS 2021;118:e2020322118; Lamas B, et al. Aryl hydrocarbon receptor ligand production by the gut microbiota is decreased in celiac disease leading to intestinal inflammation. Sci Transl Med 2020;12:eaba0624; Wacklin P, et al. Altered duodenal microbiota composition in celiac disease patients suffering from persistent symptoms on a long-term gluten-free diet. Am J Gastroenterol 2014;109:1933.

‹ at a glance

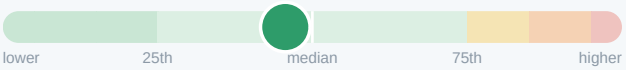
Type 1 diabetes — gut pattern

TYPICAL

Pattern version **1.0.0** (t1d) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · mixed_european_pediatric_adult

Most type 1 diabetes microbiome work is in infants at genetic risk, following them to seroconversion. Kostic et al. (DIABIMMUNE) found a drop in diversity and a rise in inflammation-associated taxa in the year before diagnosis; Vatanen et al. showed that Bacteroides-dominated Finnish and Estonian infants carried immunologically silent hexa-acylated-LPS-poor communities that failed to educate the immune system, unlike Russian Karelian infants. The TEDDY study found Bifidobacterium and short-chain fatty acid producers protective. Adult established-T1D cohorts show a weaker pattern of butyrate-producer depletion. An adult sample is far from the population these findings come from.

COMBINED RESEMBLANCE
percentile of the matched reference group



46th
likely range 0–100

What is driving it

Vatanen 2018 TEDDY (10,913 metagenomes, 783 infants) species · study vatanen_2018 · 4/4 features · 67% of combined		18th	BELOW AVERAGE
Kostic 2015 DIABIMMUNE seroconverters species · study diabimmune · 5/5 features · 33% of combined		93rd	HIGH
Diversity before diagnosis diversity · study diabimmune · 0/1 features · not in combined	not measured	—	NOT COMPUTABLE
SCFA and LPS capacity genes · genomic capacity · study functional · 1/1 features · not in combined		99th	NOTABLY HIGH
LPS immunogenicity carriers carriers · carrier proxy · study carriers · 2/2 features · not in combined		88th	HIGH

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P2	usable human shotgun cohort, or multiple compatible human association cohorts
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Vatanen 2018 TEDDY (10,913 metagenomes, 783 infants) · species · study vatanen_2018									
Bifidobacterium longum	4.157%	95	▼	0.80	0.80	0.30	0.80	-0.56	
Bifidobacterium breve	1.766%	100	▼	0.50	0.70	0.30	0.70	-0.86	
Faecalibacterium prausnitzii	0.010%	2	▼	0.70	0.80	0.20	0.80	+1.00	
Streptococcus salivarius	not detected	15	▲	0.50	0.70	0.30	0.60	-0.35	
Kostic 2015 DIABIMMUNE seroconverters · species · study diabimmune									
Blautia obeum	0.042%	50	▲	0.50	0.50	0.40	0.50	+0.00	
Ruminococcus gnavus	13.289%	98	▲	0.60	0.60	0.30	0.60	+0.72	
Streptococcus mitis	not detected	39	▲	0.40	0.50	0.30	0.50	-0.09	
Roseburia intestinalis	not detected	13	▼	0.60	0.60	0.30	0.70	+0.37	
Eubacterium rectale	0.177%	35	▼	0.60	0.60	0.20	0.70	+0.17	
Diversity before diagnosis · diversity · study diabimmune									
shannon	not detected	—	▼	0.70	0.60	0.10	0.70	—	
SCFA and LPS capacity · genes · genomic capacity · study functional									
butyrate	11.6/100	1	▼	0.80	0.70	0.20	0.80	+0.76	
LPS immunogenicity carriers · carriers · carrier proxy · study carriers									
hexa lps	0.039%	40	▼	0.60	0.60	0.50	0.60	+0.08	
butyrate	1.623%	4	▼	0.60	0.70	0.20	0.80	+0.53	

FINE PRINT

Confidence grade insufficient. Why not higher: 67% of planned feature weight was measurable (below 75%); no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; +2 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Type 1 diabetes is diagnosed by glucose, autoantibodies and C-peptide. This profile does not diagnose or predict it in anyone, and its evidence base is infants, not adults.

The hexa-LPS direction here is reversed relative to every inflammatory disease profile. That is the finding, not an error, and it means a "healthy-looking" Bacteroides-dominant community scores toward T1D risk in this one profile.

Sources: Vatanen T, et al. The human gut microbiome in early-onset type 1 diabetes from the TEDDY study. Nature 2018;562:589-594; Kostic AD, et al. The dynamics of the human infant gut microbiome in development and in progression toward type 1 diabetes. Cell Host Microbe 2015;17:260-273; Vatanen T, et al. Variation in microbiome LPS immunogenicity contributes to autoimmunity in humans. Cell 2016;165:842-853.

‹ at a glance

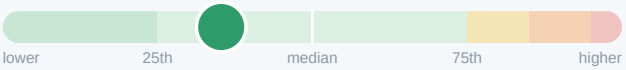
Alopecia areata — gut pattern

TYPICAL

Pattern version **1.0.0** (alopecia_areata) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · pediatric_shotgun_derived

Alopecia areata is an autoimmune condition and biologically distinct from androgenetic alopecia; their evidence is never merged. The only shotgun cohort is pediatric — 41 children with AA against 41 unaffected siblings — and everything else is 16S, which binds here as a labelled transport rather than being discarded or quietly treated as equivalent. The published cohorts disagree with each other on direction. Bacteroides is case-enriched in the severe Moreno and Bain branches and lower in Lee. Dorea is higher at genus level in Lu while D. longicatena is control-associated in Moreno. Those conflicts are not pooled into a consensus: each study is its own module with its own vote, and the spread between them is the result worth reading.

COMBINED RESEMBLANCE
percentile of the matched reference group



35th
likely range 5–80

What is driving it

Rangu 2021 pediatric shotgun (41 AA / 41 siblings)
species · study rangu_2021 · 1/1 features · 20% of combined
80th — ABOVE AVERAGE

Moreno-Arrones 2020 (15 alopecia universalis / 15 control, 16S)
species · association only · 16S study · study moreno_2020 · 6/6 features · 20% of combined
86th — HIGH

Lu 2021 (33 AA / 35 control, 16S)
species · association only · 16S study · study lu_2021 · 4/4 features · 20% of combined
61st — TYPICAL

Lee 2024 (19 AA / 20 control, 16S)
species · association only · 16S study · study lee_2024 · 2/2 features · 20% of combined
3rd — NOTABLY LOW

Nikoloudaki 2024 (24 AA / 18 control, 16S)
species · association only · 16S study · study nikoloudaki_2024 · 3/3 features · 20% of combined
14th — LOW

Rangu 2021 KO set — figure encoding
pathways · genomic capacity · study rangu_2021 · 0/4 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added
not measured
— NOT COMPUTABLE

Rangu 2021 KO set — prose encoding
pathways · genomic capacity · study rangu_2021 · 0/4 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added
not measured
— NOT COMPUTABLE

Bain 2022 high-SALT subgroup (exploratory)
species · association only · 16S study · study bain_2022 · 3/3 features · not in combined
>99th — NOTABLY HIGH

Biotin biosynthesis capacity (mechanism, P0)
genes · genomic capacity · study biotin_mechanism · 2/2 features · not in combined
5th — NOTABLY LOW

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P1	one small human cohort, amplicon-only, or inaccessible individual data
Assay transportability	labeled transport	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist

Direction conflict	material	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Rangu 2021 pediatric shotgun (41 AA / 41 siblings) · species · study rangu_2021									
Ruminococcus bicirculans	not detected	20	▼	1.00	0.60	0.60	0.90	+0.28	Rangu 2021 JID Innov (NCBI Taxonomy 1160721)
Moreno-Arrones 2020 (15 alopecia universalis / 15 control, 16S) · species · association only · 16S study · study moreno_2020									
Holdemania filiformis	not detected	30	▲	0.60	0.40	0.50	0.60	-0.17	
Parabacteroides johnsonii	not detected	43	▲	0.60	0.40	0.50	0.60	-0.06	
Bacteroides eggerthii	not detected	36	▲	0.50	0.40	0.40	0.50	-0.12	
Parabacteroides distasonis	3.011%	92	▲	0.50	0.40	0.40	0.50	+0.43	
Phascolarctobacterium succinatutens	not detected	38	▼	0.60	0.40	0.50	0.60	+0.10	
Dorea longicatena	not detected	6	▼	0.50	0.40	0.40	0.40	+1.00	
Lu 2021 (33 AA / 35 control, 16S) · species · association only · 16S study · study lu_2021									
Blautia (genus)	1.946%	93	▲	0.50	0.40	0.30	0.40	+0.55	
Anaerostipes hadrus	not detected	4	▲	0.50	0.40	0.40	0.50	-1.00	
Collinsella aerofaciens	1.940%	88	▲	0.50	0.40	0.40	0.50	+0.35	
Dorea (genus)	0.569%	73	▲	0.40	0.30	0.30	0.30	+0.19	
Lee 2024 (19 AA / 20 control, 16S) · species · association only · 16S study · study lee_2024									
Bacteroides (genus)	42.793%	85	▼	0.50	0.40	0.30	0.30	-0.35	
Blautia wexlerae	not detected	6	▲	0.40	0.30	0.30	0.40	-0.79	
Nikoloudaki 2024 (24 AA / 18 control, 16S) · species · association only · 16S study · study nikoloudaki_2024									
Blautia obeum	0.042%	50	▲	0.40	0.30	0.30	0.40	+0.00	
Holdemania filiformis	not detected	30	▲	0.40	0.30	0.40	0.50	-0.17	
Coprococcus comes	0.713%	92	▼	0.40	0.30	0.30	0.40	-0.48	
Rangu 2021 KO set — figure encoding · pathways · genomic capacity · study rangu_2021									
K06295	missing	—	▲	0.30	1.00	1.00	1.00	—	
K06297	missing	—	▲	0.30	1.00	1.00	1.00	—	
K02012	missing	—	▲	0.30	1.00	1.00	1.00	—	
K07552	missing	—	▲	0.20	1.00	1.00	1.00	—	
Rangu 2021 KO set — prose encoding · pathways · genomic capacity · study rangu_2021									
K06295	missing	—	▼	0.30	1.00	1.00	1.00	—	
K06297	missing	—	▼	0.30	1.00	1.00	1.00	—	
K02012	missing	—	▼	0.30	1.00	1.00	1.00	—	
K07552	missing	—	▼	0.20	1.00	1.00	1.00	—	
Bain 2022 high-SALT subgroup (exploratory) · species · association only · 16S study · study bain_2022									
Alistipes (genus)	5.880%	91	▲	0.30	0.20	0.30	0.30	+0.34	
Barnesiella intestinihominis	2.372%	97	▲	0.30	0.20	0.30	0.30	+0.61	
Tyzzerella nexilis	0.013%	94	▲	0.30	0.20	0.30	0.30	+0.53	
Biotin biosynthesis capacity (mechanism, P0) · genes · genomic capacity · study biotin_mechanism									
biotin	93.2/100	88	▼	0.60	0.30	0.30	0.60	-0.48	Hayashi 2017 Cell Rep 20:1513
propionate	32.9/100	93	▼	0.40	0.30	0.20	0.30	-0.72	Borde and Astrand 2018

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Alopecia areata is diagnosed by a clinician from the pattern of hair loss, not from stool. No microbiome test predicts it, and no result here should change treatment.

Six of the seven human cohorts here are 16S with fewer than 40 cases each, and their directions conflict. The score is an average of study votes and its spread is wide by construction — read the per-module scores, not just the fused number.

The pediatric shotgun cohort is the only module measured the way this sample was measured. An adult sample is outside its population.

Bacteroides and Dorea appear with opposite directions in different cohorts. Neither direction is endorsed here.

Sources: Rangu S, Lee J-J, Hu W, Bittinger K, Castelo-Soccio L. Understanding the gut microbiota in pediatric patients with alopecia areata and their siblings: a pilot study. JID Innov 2021;1:100051, doi:10.1016/j.xjidi.2021.100051; Moreno-Arrones et al. doi:10.1111/jdv.15885; Lu et al. doi:10.1016/j.jdermsci.2021.04.003; Bain et al. doi:10.1093/cei/uxac088; Lee JH, et al. IJMS 2024;25:4256; Nikoloudaki O, et al. Nutrients 2024;16:858; Hayashi A, et al. Cell Rep 2017;20:1513.

‹ at a glance

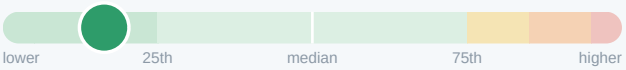
Obesity — ecological pattern

BELOW AVERAGE

Pattern version **1.0.0** (obesity) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_european_cohorts

In the obesity microbiome literature, species-level directions do not replicate across cohorts. The Firmicutes-to-Bacteroidetes ratio, once the headline finding, failed in meta-analyses (Sze and Schloss 2016; Walters 2014). What does replicate is ecological: low gene richness and low diversity are associated with adiposity, insulin resistance and dyslipidaemia (Le Chatelier 2013, Cotillard 2013), and the low-richness group gains more weight over time. So this profile scores ecology, not taxa. Akkermansia muciniphila is the one species with consistent direction and interventional support and is kept at moderate weight.

COMBINED RESEMBLANCE
percentile of the matched reference group



16th
likely range 16–16

What is driving it



Le Chatelier 2013 MetaHIT gene richness
diversity · study metahit_2013 · 1/2 features · 33% of combined
39th — TYPICAL

Cotillard 2013 MicroObes dietary intervention
diversity · study microobes_2013 · 1/1 features · 33% of combined
39th — TYPICAL

Akkermansia muciniphila (replicated species direction)
species · study dao_2016 · 1/1 features · 33% of combined
9th — LOW

Firmicutes/Bacteroidetes ratio — sensitivity only
species · association only · 16S study · study ley_2006 · 2/2 features · not in combined
8th — LOW

Metabolite gene capacity
genes · genomic capacity · study functional · 2/2 features · not in combined
99th — NOTABLY HIGH

Mucin degraders
carriers · carrier proxy · study carriers · 1/1 features · not in combined
6th — LOW

The Prevotella copri complex, clade by clade

No member of the Segatella (Prevotella) copri complex was detected. All 13 clades were assessed, including the 9 found in humans, so this is a measured absence across the whole complex rather than an unresolved species-level reading.

No clade of this complex has an established association with any disease, including rheumatoid arthritis. Manghi et al. tested all 13 clades against 1,635 disease cases and 1,854 controls across 22 studies and 11 conditions, a set that included a rheumatoid arthritis cohort, and found none of the complex's species associated with any health condition. Earlier rheumatoid-arthritis reports named the species as a whole, not a clade, and those reports have replicated poorly: in an anti-CCP-positive at-risk cohort three Prevotellaceae strains were enriched and five depleted in the people who progressed. So this section reports what is present and makes no disease claim from it.

Tett 2019 Cell Host Microbe doi:10.1016/j.chom.2019.08.018; Manghi 2023 Cell Host Microbe doi:10.1016/j.chom.2023.09.013; Hitch 2022 Syst Appl Microbiol doi:10.1016/j.syapm.2022.126354.

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	0.85	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	low	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), uc (>99th), ibd (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Le Chatelier 2013 MetaHIT gene richness · diversity · study metahit_2013									
richness	92.00	60	▼	1.00	0.90	0.20	0.90	-0.09	
shannon	not detected	—	▼	0.80	0.90	0.20	0.90	—	
Cotillard 2013 MicroObes dietary intervention · diversity · study microobes_2013									
richness	92.00	60	▼	0.80	0.80	0.20	0.80	-0.09	
Akkermansia muciniphila (replicated species direction) · species · study dao_2016									
Akkermansia muciniphila	1.425%	91	▼	1.00	0.80	0.40	0.80	-0.46	
Firmicutes/Bacteroidetes ratio — sensitivity only · species · association only · 16S study · study ley_2006									
Bacteroides vulgatus	22.292%	93	▼	0.30	0.20	0.10	0.20	-0.41	
Bacteroides uniformis	6.597%	82	▼	0.30	0.20	0.10	0.20	-0.26	
Metabolite gene capacity · genes · genomic capacity · study functional									
bcaa	209.5/100	92	▲	0.70	0.70	0.40	0.70	+0.39	
butyrate	11.6/100	1	▼	0.60	0.70	0.20	0.70	+0.76	
Mucin degraders · carriers · carrier proxy · study carriers									
mucin degraders	21.255%	94	▼	0.50	0.60	0.30	0.60	-0.65	

FINE PRINT

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Adiposity is measured with a scale and a tape measure. This profile describes an ecological state associated with it, not a cause of it.

Low richness is the least specific finding in the microbiome. It marks antibiotics, illness, low-fibre diet and age as readily as adiposity.

The Firmicutes/Bacteroidetes ratio is included only to show that it does not work.

Sources: Le Chatelier E, et al. Richness of human gut microbiome correlates with metabolic markers. Nature 2013;500:541-546; Cotillard A, et al. Dietary intervention impact on gut microbial gene richness. Nature 2013;500:585-588; Sze MA, Schloss PD. Looking for a signal in the noise: revisiting obesity and the microbiome. mBio 2016;7:e01018; Dao MC, et al. Akkermansia muciniphila and improved metabolic health during a dietary intervention in obesity. Gut 2016;65:426.

‹ at a glance

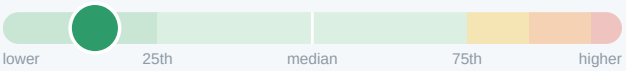
Colorectal cancer-associated community pattern

LOW

Pattern version **1.0.0** (crc) · maturity **P3** (independent human replication with coherent task and assay) · validation control · mixed_international_cohorts

Colorectal cancer is the pattern on which this pipeline's scoring engine is validated. It is the only microbiome disease association with genuinely cross-cohort-validated performance, and it is the only one for which public cohorts carry disease labels — which makes it the single pattern on which the whole engine can be checked against known answers. If the engine cannot recover a known-good signal here, no number it produces for any other pattern means anything either. Two independent meta-analyses of eight shotgun cohorts across three continents agree on a core of oral-origin anaerobes enriched in CRC stool. The virulence-gene lane (fadA, bft, clbB) is the mechanistic layer and is reported as presence-grade only, for reasons recorded in the panel.

COMBINED RESEMBLANCE
percentile of the matched reference group



15th
likely range 0–37

What is driving it

Wirbel 2019 meta-analysis (5 cohorts, 768 metagenomes)
species · study wirbel_2019 · 8/8 features · 50% of combined
18th — BELOW AVERAGE

Thomas 2019 meta-analysis (7 cohorts) + choline
species · study thomas_2019 · 7/7 features · 50% of combined
43rd — TYPICAL

CRC virulence genes — presence grade
genes · association only · study virulence · 1/1 features · not in combined
76th — ABOVE AVERAGE

Carcinogenic metabolite capacity
genes · genomic capacity · study functional · 4/4 features · not in combined
95th — NOTABLY HIGH

Oral-origin carriers
carriers · carrier proxy · study carriers · 2/2 features · not in combined
24th — BELOW AVERAGE

How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

Technical reliability	1.00	sequencing depth, host fraction, reference match
Independent evidence coverage	1.00	share of the pattern's independent evidence clusters this sample could measure
Evidence maturity	P3	independent human replication with coherent task and assay
Assay transportability	native	whether the studies measured what this sample measured
Population match	0.60	how well the reference group matches you
Specificity	unknown	does this pattern separate its condition from general disturbance? measured where labelled data exist
Direction conflict	none	studies disagreeing on direction for one or more features

Competing patterns scoring nearly as high: parkinsons (>99th), t2d (>99th), uc (>99th). Margin over the next pattern: 0 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

FEATURE	MEASURED	REF PCT	DIR	w	q	s	c	v	SOURCES
Wirbel 2019 meta-analysis (5 cohorts, 768 metagenomes) · species · study wirbel_2019									
Fusobacterium nucleatum	not detected	49	▲	1.00	1.00	0.80	1.00	-0.01	
Parvimonas micra	not detected	48	▲	1.00	1.00	0.80	1.00	-0.02	
Peptostreptococcus stomatis	not detected	49	▲	1.00	1.00	0.80	1.00	-0.01	
Gemella morbillorum	not detected	49	▲	0.90	0.90	0.70	1.00	-0.01	
Solobacterium moorei	not detected	48	▲	0.90	0.90	0.70	1.00	-0.01	
Porphyromonas asaccharolytica	not detected	49	▲	0.80	0.90	0.70	0.90	-0.01	
Clostridium symbiosum	not detected	36	▲	0.70	0.80	0.50	0.80	-0.12	
Bacteroides fragilis	not detected	24	▲	0.60	0.80	0.40	0.80	-0.24	
Thomas 2019 meta-analysis (7 cohorts) + choline · species · study thomas_2019									
Fusobacterium nucleatum	not detected	49	▲	1.00	1.00	0.80	1.00	-0.01	
Parvimonas micra	not detected	48	▲	1.00	1.00	0.80	1.00	-0.02	
Peptostreptococcus stomatis	not detected	49	▲	0.90	1.00	0.80	1.00	-0.01	
Gemella morbillorum	not detected	49	▲	0.80	0.90	0.70	0.90	-0.01	
Porphyromonas asaccharolytica	not detected	49	▲	0.70	0.90	0.70	0.90	-0.01	
Prevotella intermedia	not detected	50	▲	0.60	0.80	0.60	0.80	-0.00	
Streptococcus anginosus group	not detected	45	▲	0.50	0.70	0.40	0.70	-0.04	
CRC virulence genes — presence grade · genes · association only · study virulence									
crc virulence	0.1/100	76	▲	1.00	0.40	0.80	0.50	+0.24	
Carcinogenic metabolite capacity · genes · genomic capacity · study functional									
bai	8.2/100	97	▲	0.70	0.70	0.40	0.70	+1.00	
h2s	63.8/100	28	▲	0.60	0.60	0.40	0.60	-0.19	
cutc	4.1/100	77	▲	0.60	0.70	0.40	0.70	+0.39	
butyrate	11.6/100	1	▼	0.60	0.70	0.20	0.70	+0.76	
Oral-origin carriers · carriers · carrier proxy · study carriers									
oral origin	0.000%	10	▲	0.80	0.80	0.40	0.80	-0.76	
opportunistic pathogens	13.399%	95	▲	0.50	0.60	0.30	0.60	+0.59	

FINE PRINT

Validation control. Included to check that the scoring engine recovers a known signal on labelled data. Not a cancer screen and not a risk estimate; validated clinical tests exist for this condition.

Confidence grade insufficient. Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; taxonomic and functional modules point opposite ways.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

Measured discrimination: not available. No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

This is not a cancer screen and not a risk estimate. It exists to test the scoring engine against labelled data. Colorectal cancer screening is done with faecal immunochemical testing, multi-target stool DNA testing, or colonoscopy, on a schedule set by a clinician.

The taxonomic features are oral-origin anaerobes. Their enrichment in faecal metagenomes is reproducible in cancer cohorts, but oral-gut translocation also occurs in periodontal disease and with reduced gastric acid, so the pattern is not unique to malignancy.

The virulence lane is presence-grade with unmeasurable sensitivity. Not detected does not mean absent.

Sources: Wirbel J, et al. Meta-analysis of fecal metagenomes reveals global microbial signatures that are specific for colorectal cancer. Nat Med 2019;25:679-689; Thomas AM, et al. Metagenomic analysis of colorectal cancer datasets identifies cross-cohort microbial diagnostic signatures and a link with choline degradation. Nat Med 2019;25:667-678.

Topics people ask about, and where the answers are

Fifteen topics people ask about, each pointing at what this report actually measured and at what it did not. Where a pattern score for a condition already exists it is linked exactly as calculated; no score here is new, and no topic without a score has been given one.

This is a way of navigating what was measured, not a test for the condition. Nothing in this card raises or lowers the chance that you have it.

Anxiety and depression

What this keeps separate. Human microbiome associations with depression are observational and mostly cross-sectional; the behavioural findings are from animals. Neither establishes cause, and no stool measurement diagnoses anxiety or depression.

Several gut bacteria decarboxylate glutamate to GABA, and GABA-modulating strains were isolated from human stool. The blood-brain barrier is not permeable to GABA, so any effect would have to run through the vagus nerve or the enteric nervous system rather than through GABA reaching the brain.

Tryptophan is the shared precursor of serotonin, indoles and kynurenines. Bacteria competing for it is a plausible mechanism and is not a measurement of anybody's serotonin.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
mdd	the existing depression gut-pattern similarity, linked as it stands	linked as calculated
gaba	GABA production capacity, one of the routes this literature discusses	34 per 100 genomes
gababreak	GABA degradation, shown beside production and never subtracted from it	22 per 100 genomes
indole	indole formation from tryptophan, the other much-discussed route	12 per 100 genomes
tryptamine	tryptamine formation, which competes for the same tryptophan	12 per 100 genomes
neuro.gaba_synthesis	the specific decarboxylation step, with its own limits	supported

6 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Whether any of these genes are being expressed, which sequencing DNA cannot show. Your symptoms, which are not in this data at all and matter more than any of it. Whether you take medication affecting these systems, which changes the picture entirely.

If you want to take this further: If you are looking for help with mood, that is a conversation with a clinician, not a microbiome result. A symptom diary alongside a repeat sample would at least make a within-person comparison possible, which a single timepoint cannot support.

Autism research

What this keeps separate. The published classifiers were built in paediatric cohorts, and diet differs systematically between the groups they compared. This report has no autism classifier and produces no score; these are the pathways the research names.

A 2024 multikingdom study reported a panel spanning bacteria, fungi, archaea and viruses, and highlighted B1 and quinone pathways among its features.

Whether these differences are a cause, a consequence of diet and behaviour, or an artefact of cohort design is unresolved in the source literature itself.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
thiamine	B1 synthesis, one of the pathways the multikingdom work highlighted	277 per 100 genomes
k2	quinone synthesis, highlighted in the same work	48 per 100 genomes
mycobiome	the fungal and multikingdom component those studies used	linked as calculated
ecology	community composition and type, which those cohorts stratified on	linked as calculated

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Diet, which differs between the compared groups and is not recorded here. Age: these cohorts were children, and the reference set behind this report is adult. Everything a diagnosis actually rests on, none of which is microbial.

If you want to take this further: There is no action to take from this card. It exists so the pathways named in the research can be found in your own results.

Serotonin, melatonin and sleep

What this keeps separate. Serotonin, melatonin and tryptamine are three different molecules made by three different reactions. Gut serotonin and brain serotonin are separate pools that do not exchange. No number here is a concentration of any of them.

Most of the body's serotonin is made in the gut by host enterochromaffin cells, and specific bacteria can increase that host production in gnotobiotic mice. That is host synthesis being modulated, not bacterial synthesis.

A traceable candidate decarboxylase produces tryptamine from tryptophan. Tryptamine is not serotonin, and an enzyme that makes one is not a detector for the other.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
tryptamine	tryptamine, a distinct molecule from serotonin, made from the same precursor	12 per 100 genomes
indole	the bacterial route that competes for that precursor	12 per 100 genomes
dopamine	the other monoamine chemistry measured here, for contrast	2 per 100 genomes

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
histamine	another amine from the same decarboxylase chemistry	1 per 100 genomes

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your sleep, which is not measurable from stool. Whether the candidate enzymes here act on the substrates the papers used, which is unresolved for some of them in the source.

If you want to take this further: A sleep or symptom diary is the only thing that would make this comparable over time, and it is worth more than the gene counts.

Atherosclerosis and cardiovascular context

What this keeps separate. Microbes make TMA. Your liver makes TMAO from it. These are different steps in different organs, and the existing cardiovascular score is linked exactly as it is rather than recomputed from these genes.

Gut bacteria cleave choline and carnitine to trimethylamine, which the liver oxidises to TMAO. Higher circulating TMAO is associated with cardiovascular events in cohort studies, and the causal question remains open.

Host flavin-containing monooxygenase 3 activity, which varies genetically and by sex, sits between the bacterial step and the circulating molecule.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
acvd	the existing cardiovascular gut-pattern similarity, linked unchanged	linked as calculated
cutc	choline TMA-lyase, the most studied microbial link	4 per 100 genomes
bcaa	amino-acid fermentation, the other route to circulating microbial metabolites	209 per 100 genomes
pcresol	p-cresol, another microbial metabolite the liver conjugates before it circulates	9 per 100 genomes

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your blood TMAO, which this cannot estimate and which a laboratory can measure. Your lipids, blood pressure and family history, which dominate cardiovascular risk.

If you want to take this further: Cardiovascular risk is assessed with blood pressure, lipids and history. If that is the question, those are the measurements. A measured TMAO, if you have one, can be entered in the lab results section and will be shown beside this rather than replaced by it.

Blood pressure context

What this keeps separate. The existing profile is the result. This card adds navigation and the functional context, and deliberately does not produce a second number for the same thing.

Short-chain fatty acids act on host receptors including GPR41 and Olfr78 that have opposing effects on vascular tone in rodent experiments, which is why no single direction can be assigned to a butyrate reading.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
hypertension	the existing gut-pattern similarity, reused rather than duplicated	linked as calculated
butyrate	short-chain fatty acid production, the mechanism most often proposed	12 per 100 genomes
propionate	propionate specifically, which has its own receptor literature	33 per 100 genomes

3 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your blood pressure, which takes thirty seconds to measure properly and is not inferable from stool. Salt intake, weight and medication, which are the things that actually move it.

If you want to take this further: If blood pressure is the question, measure blood pressure. Repeatedly, at home, is better than once in a clinic.

Endogenous ethanol and auto-brewery evaluation

What this keeps separate. Carrying adhE is ordinary; the liver findings concern particular high-alcohol-producing strains. Genetic potential is not measured ethanol, and auto-brewery syndrome is diagnosed with a carbohydrate challenge and serial blood alcohol measurements, not with a stool gene count.

Specific high-alcohol-producing Klebsiella pneumoniae strains isolated from people with fatty liver disease produced liver changes when colonised into mice. Strains of the same species differ enormously in how much they produce.

Acetaldehyde, the intermediate, is the genotoxic part of this chemistry, and is handled by an enzyme family too broad to measure specifically here.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
capacity.ethanol	the ethanol-forming route, newly measured	complete
ethanol	the underlying gene readings, including the ethanolamine route	70 per 100 genomes
masld	the existing fatty-liver gut-pattern similarity, linked unchanged	linked as calculated
mycobiome	yeasts, which ferment ethanol and are measured separately	linked as calculated
strain_resolution	strain-level evidence, since this is a strain-level question	linked as calculated

5 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Whether the organisms carrying these genes here are high producers, which requires culture and measurement rather than sequence. Any actual blood alcohol measurement, which is what the diagnosis rests on.

If you want to take this further: If episodes of unexplained intoxication are the reason for asking, that is a medical evaluation with a supervised carbohydrate challenge, and this card is not a substitute for it.

Chronic constipation

What this keeps separate. Methanogen genes in stool are not a breath test. The association between methane and slow transit comes from breath measurements, and a gene count cannot substitute for one or diagnose intestinal methanogen overgrowth.

Methane slows intestinal transit in animal models, and breath methane is associated with constipation in humans. Methanogens are archaea and are measured here by their own marker genes.

Fermentable substrate reaching the colon affects stool water and bulk, which is why the substrate readings are linked here.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
ibs_c	the existing constipation-predominant IBS similarity, linked unchanged	linked as calculated
methane	methanogen capacity, associated with slower transit	0 per 100 genomes
substrates	which fibres this community can actually open, newly measured	linked as calculated
butyrate	the fermentation end-product most linked to colonic function	12 per 100 genomes

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your stool form and frequency, which can be recorded and are more informative than any of this. Transit time, which is directly measurable and is not inferable from stool DNA.

If you want to take this further: Recording stool form on the Bristol scale for two weeks costs nothing and gives a better picture than a single sequencing run. A breath test is the measurement that corresponds to the methane literature.

Chronic diarrhoea

What this keeps separate. Bile-acid diarrhoea is one cause among many, and it is diagnosed with a SeHCAT scan or a measured faecal bile acid, not with these genes. The pathogen screen is linked here because it answers a separate question that matters more when diarrhoea is acute.

Bacteria deconjugate and then dehydroxylate bile acids. Reduced conversion leaves more primary bile acid reaching the colon, which draws in water. The bacterial step is real; the amount arriving is set by the liver and the ileum.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
ibs_d	the existing diarrhoea-predominant IBS similarity, linked unchanged	linked as calculated
bsh	bile salt hydrolase, the first step of bile-acid transformation	56 per 100 genomes
bai	the 7-alpha-dehydroxylation route to secondary bile acids	8 per 100 genomes
pathogens	the pathogen screen, which is a different question with a different answer	linked as calculated

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Whether your ileum is reabsorbing bile acids normally, which is the actual question in bile-acid diarrhoea and needs a scan or a stool measurement. Recent antibiotics and travel, which change the interpretation completely.

If you want to take this further: Persistent diarrhoea, and particularly diarrhoea with weight loss, blood or fever, needs a clinician rather than a microbiome report.

Recurring abdominal pain

What this keeps separate. The histamine mechanism is strain-dependent. A species that contains histamine-producing strains is not evidence that yours produces histamine, and species-level abundance cannot establish it.

A specific Klebsiella aerogenes strain producing histamine was linked to visceral hypersensitivity in a mouse model and to a subgroup of patients. The work is explicitly strain-level.

Gas production from fermentation distends the colon, and distension is painful at lower volumes in people with visceral hypersensitivity.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
ibs	the existing IBS similarity, linked as it stands	linked as calculated
histamine	bacterial histamine production, a strain-specific mechanism	1 per 100 genomes
fermentation	gas-forming fermentation routes, newly measured	linked as calculated
h2s	hydrogen sulfide, which has its own visceral sensitivity literature	64 per 100 genomes

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Where the pain is, when it happens and what changes it, none of which is here and all of which is more diagnostic than this report. Whether the strains present are the producing ones, which needs strain-level resolution this depth cannot always reach.

If you want to take this further: Pain with weight loss, bleeding, fever or waking you at night is a reason to see a clinician promptly, whatever a microbiome report says.

Wheat, gluten and fructan sensitivity

What this keeps separate. Coeliac disease, wheat allergy and non-coeliac wheat sensitivity are three different conditions with three different mechanisms. Coeliac disease is diagnosed serologically and by biopsy, while still eating gluten. Nothing here tests for any of them.

Trials that separated gluten from fructans found that fructans, not gluten, reproduced symptoms in many people who believed they were gluten-sensitive. Wheat carries both, which is why the substrate reading is linked here.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
celiac	the existing coeliac gut-pattern similarity, linked unchanged	linked as calculated
substrates	fructan utilisation specifically, newly measured	linked as calculated
carbohydrates	the wider carbohydrate-active enzyme readings	1,001 per 100 genomes

3 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Whether you are currently eating gluten, which determines whether coeliac testing would even be valid. Your symptoms and their timing relative to meals.

If you want to take this further: If coeliac disease has never been excluded, that test is done before removing gluten, not after: stopping first makes the test unreliable. A supervised fructan reduction is the way to separate the two, and it is temporary rather than permanent.

Irritable bowel syndrome

What this keeps separate. All four IBS scores already exist in this report and are shown as they are. This card adds navigation and substrate context only, and produces no new score. IBS is a clinical diagnosis made on symptom criteria.

The microbiome literature in IBS is large, mostly 16S, and inconsistent between cohorts, which is what a symptom-defined diagnosis tends to produce.

Fermentable substrate reaching the colon is the mechanism behind the low-FODMAP response, and substrate capacity is the part of that this report can measure.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
ibs	the existing overall IBS similarity	linked as calculated
ibs_c	the constipation-predominant variant	linked as calculated
ibs_d	the diarrhoea-predominant variant	linked as calculated
ibs_m	the mixed variant	linked as calculated
substrates	which fermentable substrates this community can open	linked as calculated
fermentation	the gas-forming routes downstream of them	linked as calculated

6 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your symptom pattern against the Rome criteria, which is what the diagnosis is. What you eat, which determines how much fermentable substrate arrives.

If you want to take this further: A low-FODMAP trial is a short diagnostic exercise with a structured reintroduction, not a long-term diet, and is best done with a dietitian.

Thyroid context

What this keeps separate. Hypothyroidism, Hashimoto's thyroiditis and Graves' disease are separate conditions. Host blood markers and host gene expression cannot be inferred from stool DNA, and this report has no thyroid score.

Observational studies report compositional differences in thyroid conditions. They are small, cross-sectional and confounded by treatment, and no mechanism has been established in humans.

Selenium and iodine handling are frequently discussed, but the bacterial contribution to either has not been quantified in people.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
bglucuronidase	beta-glucuronidase, often invoked in this context	129 per 100 genomes
ecology	overall community composition, which the observational work compared	linked as calculated

2 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

Not produced in this run, and so not part of the card: input_register.

What this cannot tell you: Your TSH, free T4 and antibody status, which are blood tests and are the actual measurements for this. Whether you take thyroid hormone, which changes everything about interpretation.

If you want to take this further: If thyroid function is the question, a blood test answers it directly. Results can be entered in the lab section and will be shown alongside, not merged in.

Food allergy and immune tolerance

What this keeps separate. Mouse transfer and consortium experiments show that particular bacteria affect tolerance in mice. They do not establish that any food is safe or unsafe for you, and nothing here identifies an allergen.

Consortia from healthy infants protected mice against cow's milk allergy, and consortia from allergic infants did not. The effect travelled with specific organisms rather than with diversity.

Butyrate induces regulatory T cells in the colon in mouse work, which is the usual proposed bridge between fibre and tolerance.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
csu	the existing chronic urticaria pattern, linked unchanged	linked as calculated
ATD_ADULT_WANG2023_TRANSLATED_V1	the existing atopic dermatitis pattern index	linked as calculated
butyrate	butyrate, the metabolite in the regulatory T cell literature	12 per 100 genomes
substrates	the fibre substrates upstream of that butyrate	linked as calculated

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Which foods, if any, cause you problems, which is established by history and by supervised challenge rather than by sequencing. Your IgE status, which is a blood or skin test.

If you want to take this further: Never use a microbiome result to decide whether a food is safe. Suspected allergy is assessed by an allergist, and a genuine anaphylaxis risk is not something to experiment with.

GLP-1-related microbial mechanisms

What this keeps separate. Microbial genetic potential is not a hormone concentration and is not comparable to a GLP-1 medication. Host receptors are host genes and are not targets in stool sequencing. Indole does not have one universal direction: prolonged exposure reduced secretion in the source experiments.

Short-chain fatty acids trigger GLP-1 secretion through FFAR2 in experimental systems, and a targeted colonic propionate delivery study in humans increased secretion.

Bile acids act through TGR5 on the same cells, which is a separate mechanism that happens to converge on the same hormone.

A specific Akkermansia muciniphila protein has experimental support, and detecting it requires the tested sequence rather than the species.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
biotransformation	the GLP-1 channels, kept as separate mechanisms	linked as calculated
butyrate	the short-chain fatty acid channel	12 per 100 genomes
propionate	the colonic propionate experiments specifically	33 per 100 genomes
bsh	the bile-acid channel, which acts through a different receptor	56 per 100 genomes
indole	the indole channel, whose direction depends on exposure duration	12 per 100 genomes

5 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your actual GLP-1, which requires a blood assay with the active-versus-total distinction, fasting or postprandial timing, and correct sample handling. Whether any of these genes are expressed or their products secreted.

If you want to take this further: A measured GLP-1 can be entered in the lab section with its assay and timing, and will be shown next to this rather than combined with it.

Eczema

What this keeps separate. The existing skin pattern index is preserved exactly and is not recomputed here. Gut composition is one factor among genetics, skin barrier function and the skin's own microbiome, which a stool sample does not sample.

Filaggrin variants and skin barrier function dominate atopic dermatitis risk. Gut microbial associations are real in cohort studies and much weaker than those factors.

IN YOUR REPORT	WHY IT IS HERE	WHAT IT SHOWS
ATD_ADULT_WANG2023_TRANSLATED_V1	the existing atopic dermatitis pattern index, preserved	linked as calculated
butyrate	the short-chain fatty acid arm of the tolerance literature	12 per 100 genomes
substrates	the fibre substrates upstream of it	linked as calculated

3 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

What this cannot tell you: Your skin, which is not what was sampled. The skin microbiome is a different community and would need a different swab. Whether the flares follow anything identifiable, which a diary answers.

If you want to take this further: Emollients and the treatments a dermatologist recommends have far stronger evidence than anything derived from a stool sample.

PART B — BIOFILM FINDINGS IN FULL

21 Biofilm findings in full

This section lists every biofilm measurement, every module in the registry with its actual status, and every matched piece of evidence — including the evidence that points the other way. Nothing here is truncated to a top-N list.

The reference population

Percentiles are ranks against **1,127 independent adults** from 14 public control cohorts, aged 19 to 107 (median 52), profiled on the same taxonomy as this sample. 3,027 samples were collapsed to 1,127 people: several of the source studies sampled the same person repeatedly, and counting those as separate people would have made the reference look larger and steadier than it is.

Control status in these studies means no recorded disease for the study's purpose. It does not mean microscopy-confirmed absence of mucosal biofilm, which was never assessed in any of them.

H-C · Biofilm-associated community pattern

Percentile 60.3 (middle reference range) against 1,127 adults. Calibration 2ce8f21198799625.

Escherichia coli and its close relatives = 0%, rank 10 — *not detected above this assay's limit*. Species summed: none.

Mediterraneibacter gnavus = 13.29%, rank 99. Species summed: Ruminococcus_gnavus 13.3%.

Why this direction: Higher means more of this two-feature association pattern, which BF-S01 reported enriched in biofilm-positive participants.

What it is not: Not measured harmful biofilm, not a stool biofilm probability, and not a pathotype. BF-S01's 16S Escherichia/Shigella association cannot be resolved identically by a WGS species-complex measurement, and not every Escherichia/Shigella read belongs to pathogenic E. coli.

Recorded omissions: ecoli_complex: not detected above this assay's limit

P-E · Protective ecological support (partial panel)

Percentile 54.6 (middle reference range) against 1,127 adults. Calibration 73125c04fea5c96c.

Faecalibacterium prausnitzii = 0.00976%, rank 4. Species summed: Faecalibacterium_prausnitzii 0.00976%.

Coprococcus species = 0.7133%, rank 64. Species summed: Coprococcus_comes 0.713%.

Blautia species = 1.946%, rank 91. Species summed: Blautia_obeum 0.0417%, Blautia_producta 0.00539%, Blautia_sp_CAG_257 1.63%, Blautia_sp_N6H1_15 0.269%.

Why this direction: Higher means more of this narrow community pattern, which BF-S01 reported relatively depleted in biofilm-positive biopsies.

What it is not: Not a measurement of beneficial biofilm amount. Low values do not diagnose a missing protective biofilm, a species deficiency or a need for transplant, and this proxy does not instruct supplementation with these organisms. This is a partial panel: subdoligranulum could not be measured in this reference. It is calibrated separately and is not the named 4-feature index.

Recorded omissions: subdoligranulum: dropped from the panel - detected in 2 of 1127 reference participants (0.2%), below the 5% floor needed to rank anyone. The organism is not rare; this taxonomy release simply has no named species leaf that captures it, so the feature is not assayed here rather than absent.

Axes with no measurement

• **Harm-associated mechanisms** — No mechanism in this axis has a validated sequence panel yet. This is a gap in the reference data, not a finding of absence.

• **Protective-associated mechanisms** — No mechanism in this axis has a validated sequence panel yet. This is a gap in the reference data, not a finding of absence.

Do the different kinds of evidence agree?

Each row is a different way of looking at the same named concern: biofilm-associated community pattern (BF-S01 association). There is deliberately no combined confidence number, because two scores computed from the same reads are not independent confirmations of each other.

Molecular mechanism	**not measured**	No mechanism module has a validated sequence panel, so no molecular mechanism measurement exists to agree or disagree.
Community proxy	**mixed**	The two-feature H-C pattern, ranked against the adult control reference.
Expression or activity	**not measured**	No RNA or protein assay was supplied.
Directly measured architecture	**not measured**	No endoscopy or imaging was supplied. Generic inflammation cannot fill this row.
Measured host consequence	**not measured**	No host protein, permeability or injury assay was supplied.

Every mechanism in the registry

All 24 modules appear here with their real status. A module that cannot be measured says so; it is not left out, because a silently missing mechanism is indistinguishable from one that was checked and found absent.

ID	MECHANISM	STATUS	AXIS	EVIDENCE
BF-M01	Curli-family bacterial amyloid genes	candidate, not validated	context dependent	mixed
BF-M02	Biofilm-associated protein (BAP) family targets	candidate, not validated	context dependent	mixed
BF-M03	PNAG/PGA matrix synthesis and export locus	candidate, not validated	context dependent	human association
BF-M04	Bile-acid-diarrhoea-associated biofilm gene pattern	candidate, not validated	context dependent	human association
BF-M05	Klebsiella type-1 and type-3 adhesion loci	candidate, not validated	context dependent	animal mechanism
BF-M06	Extracellular polysaccharide and matrix-associated proteins	context only	context dependent	resource
BF-M07	eDNA, quorum sensing and cyclic-di-GMP regulatory context	context only	context dependent	in vitro mechanism
BF-M08	C. difficile adhesion, matrix and spore context	context only	context dependent	review
BF-M09	Candida adhesion, matrix and regulatory context	not measurable from stool DNA	context dependent	review
BF-M10	L. reuteri GtfW protective mechanism	candidate, not validated	protective associated	in vitro mechanism
BF-M11	F. prausnitzii host-glycan protective mechanism	no reference panel yet	context dependent	animal mechanism
BF-M12	Fibre-associated mixed-community mechanisms	context only	context dependent	in vitro mechanism
BF-M13	Other verified probiotic-strain adhesion and EPS mechanisms	candidate, not validated	context dependent	animal mechanism
BF-M14	S. aureus host fibrin-interaction capability	context only	context dependent	in vitro mechanism
BF-M15	Co-occurring pathogen, toxin and AMR calls	implemented	context dependent	human association
BF-M16	Measured host thrombin, protein activity and tissue architecture	not measurable from stool DNA	context dependent	animal mechanism
BF-M17	Longitudinal shared strain with biofilm-associated loci	implemented	context dependent	animal mechanism
BF-M18	Two-feature biofilm-associated community proxy (H-C)	implemented	harmful associated	human association
BF-M19	Four-genus biofilm-negative-associated ecological proxy (P-E)	implemented	protective associated	human association
BF-M20	Linked matrix/adhesion plus independently supported damaging capability	implemented	context dependent	human association
BF-M21	Faeces-derived community behaviour	not measurable from stool DNA	context dependent	human derived ex vivo
BF-M22	Tissue architecture and host-interface observations	not measurable from stool DNA	context dependent	human association
BF-M23	Matched expression and activity support	not measurable from stool DNA	context dependent	in vitro mechanism
BF-M24	Additional source-specific protective-antagonism evidence	candidate, not validated	context dependent	animal mechanism

Matched research evidence

9 records matched the findings above. These are things to read, not things to take. Where a study points the other way it is listed beside the one that points favourably, at the same size. Laboratory concentrations are shown as laboratory concentrations and are never converted into a dose.

L. plantarum CIRM653 opposing evidence - review before considering

Reduced K. pneumoniae biofilm in vitro, but treated mice MAINTAINED intestinal carriage while untreated controls declined.

Measured endpoint: **viable organism burden**. Model: animal gut. Strain: CIRM653. Laboratory exposure: live strain, in vitro and mouse. Delivery to the colon: plausible unverified.

Caution: The essential regression case: a favourable in-vitro biomass result with an unfavourable in-vivo carriage result. It must never be ranked as a selected decolonisation treatment.

BF-I45 · 10.3920/bm2017.0002

Host-glycan support of F. prausnitzii matched research evidence

Protective mechanism in experimental colitis with correlations in human tissue.

Measured endpoint: **protective function**. Model: animal gut. Laboratory exposure: glycan intervention, mouse. Delivery to the colon: plausible unverified.

Caution: Not a validated oral treatment and not a unique DNA biomarker.

BF-I36 · 10.1080/19490976.2026.2665870

L. johnsonii studied isolate matched research evidence

Direct EPEC biofilm experiments plus improved pathogen and inflammation outcomes in a C. rodentium mouse model.

Measured endpoint: **protective function**. Model: animal gut. Strain: the studied isolate; PV739486 is a 16S sequence only. Laboratory exposure: live isolate, in vitro and mouse. Delivery to the colon: plausible unverified.

Caution: Do not substitute a commercial L. johnsonii strain without identity evidence. Public 16S data do not provide an exact strain genome, and putatively annotated metabolites are not proven active ingredients.

BF-I44 · 10.3389/fimmu.2026.1749001

Biofilm-state L. plantarum Y42 and its matrix products matched research evidence

Barrier and immune outcomes improved in a mouse challenge model.

Measured endpoint: **protective function**. Model: animal gut. Strain: Y42. Laboratory exposure: biofilm-state cells and components, mouse. Delivery to the colon: plausible unverified.

Caution: Exact strain, state and components matter. Ordinary planktonic products and generic EPS genes are not equivalent.

BF-I47 · 10.1021/acs.jafc.4c00460

Biofilm-state Lactiplantibacillus LR-1 matched research evidence

Biofilm-state cells outperformed planktonic cells on selected colitis outcomes.

Measured endpoint: **protective function**. Model: animal gut. Strain: LR-1. Laboratory exposure: biofilm-state cells, mouse. Delivery to the colon: plausible unverified.

Caution: Experimental protective support, with no claimed human treatment effect.

BF-I48 · 10.1039/d3fo02733c

Multi-strain restorative consortium matched research evidence

Mouse recovery after antibiotics.

Measured endpoint: **protective function**. Model: animal gut. Strain: the administered consortium. Laboratory exposure: administered consortium, mouse. Delivery to the colon: plausible unverified.

Caution: Full tested composition and methods must be imported before any exact product matching. Co-detection of these taxa in stool or donor material does not recreate the consortium.

BF-I49 · 10.1007/s11274-026-05071-0

L. reuteri ATCC 23272 in microsphere formulation matched research evidence

Improved probiotic properties in the biofilm state, with GtfW implicated.

Measured endpoint: **protective function**. Model: gut relevant in vitro. Strain: ATCC 23272. Laboratory exposure: microsphere formulation, in vitro. Delivery to the colon: plausible unverified.

Caution: The ATCC 23272 and GtfW context is not a generic probiotic-species effect, and the microsphere formulation is not naturally present in a donor sample.

BF-I34 · 10.3389/fmicb.2017.00489

Resveratrol with L. paracasei ATCC334 matched research evidence

Strain-dependent promotion of adhesion and biofilm formation in vitro.

Measured endpoint: **adhesion**. Model: gut relevant in vitro. Strain: ATCC334. Laboratory exposure: resveratrol in culture. Delivery to the colon: plausible unverified.

Caution: A supportive formulation hypothesis. Not proven human engraftment, not a beneficial-biofilm quantity, and not a reason to take concentrated resveratrol.

BF-I46 · 10.3390/ijms21155423

SB-121 (L. reuteri with dextran microparticles and maltose) matched research evidence

Small randomised placebo-controlled crossover study in 15 participants with autism, reporting safety, tolerability and exploratory outcomes.

Measured endpoint: **clinical outcome, no biofilm measured**. Model: human gut controlled. Strain: SB-121 formulation. Laboratory exposure: oral SB-121, clinical. Delivery to the colon: target exposure measured.

An oral formulation studied in humans.

Caution: Relevant evidence that biofilm-oriented supportive formulations have reached human research. Not validation of a stool protective score and not proof of broad clinical efficacy.

BF-I35 · 10.1038/s41598-023-30909-0

Reading this section

What are biofilms?

Biofilms are organised communities of microbes held together by a shared matrix. They are a common microbial way of living, not automatically a disease. In the gut, some communities can support stable colonisation and a protective relationship with the intestinal lining. Others can help pathogens persist, grow too close to tissue, or participate in inflammation. Their effect depends on the organisms, their activity, their location and the host.

What did this test measure?

This test ranks selected biofilm-related genetic capabilities and community patterns. We show concerning mechanisms and protective support separately, so one does not cancel the other. The community-pattern scores are experimental proxies. DNA alone does not directly measure an attached biofilm, its thickness, location or current activity.

Why can harmful biofilms matter?

Some biofilms help organisms survive immune defences or antimicrobial exposure. That can contribute to persistent or recurrent infection in the right setting. Biofilm-related tolerance is different from carrying an antibiotic-resistance gene, although both may occur together. A genetic finding alone cannot tell us which treatment will work.

Can harmful biofilms be reduced?

Research includes targeted treatment of specific infections, dietary and community-supporting approaches, probiotics, compounds and enzymes that affect matrix or adhesion, and emerging precision therapies. Some have encouraging laboratory or animal evidence; others have human evidence in particular conditions. Each option above states exactly what was tested. Breaking up a matrix does not necessarily kill the organisms, and removing protective communities or mucus can be counterproductive.

What about fibrin and other organs?

Fibrin is a human clotting protein that some microbes can exploit in blood or tissue infections. It is not the same as bacterial amyloid or every biofilm matrix. Biofilm-associated problems can occur on teeth, wounds, airways, urinary devices, implants and heart valves. Stool DNA does not locate or diagnose those problems or measure circulating clots.

What can I do with this result?

Use the supported findings to identify which mechanisms deserve attention, which evidence-linked options are relevant, and what additional information would make a decision more reliable. Changes over time can show changes in the measured genetic signals; they do not by themselves prove that a biofilm has been removed.

What this section cannot tell you

• Age was not supplied. This percentile assumes the subject is an adult, because the reference is adults aged 18 and over. If the subject is a child the percentile does not apply and should be disregarded.

• DNA shows genetic potential, not activity, attachment or location.

• Both computable cards are experimental research proxies, not measured biofilm quantity.

• No validated stool-DNA classifier of protective versus harmful biofilm exists; this module does not claim one.

• A negative or low result never means biofilm-free, infection-free or cleared for donation.

22 Your gut mycobiome, in detail

Every quality-passed read pair (8,029,909) was aligned competitively against 113 fungal assemblies chosen from a reference of 1,696 (961 species), their near neighbours, and 33 bacterial and control decoys; a separate single-copy-marker search (EukDetect2 8d69014) ran on the same reads. A fungus is supported when its fragments are spread across separated regions of its genome at high identity and no decoy competes for them. Below: each fungus found, then the score trace.

Ledger: 773 fungal · 238 cross-kingdom ambiguous (excluded) · 22930 fungal hits below 97% identity (excluded) · 2,975,336 decoy · 5,030,632 unassigned. Each pair counted once.

‹ at a glance

Saccharomyces cerevisiae species with studied probiotic strains

SUPPORTED

What it is

Baker's, brewer's and wine yeast, and the species that contains the studied probiotic strains sold as S. boulardii. Common in stool after bread, beer, wine or a yeast probiotic; usually transient rather than resident.

How it was found

WHOLE GENOME	**239** fragments over **105** separated regions on 17 contigs · identity 99.5% · backbone GCF_000146045.2
MARKERS	EukDetect2: 5 markers, 10 reads, RPKS 0.0493 (marker-length-normalised support, not load)
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain unresolved · complete

LOCI	0 of 563873 discriminating sites callable in 0 linkage blocks · panel saccharomyces_cerevisiae_boulardii_lineage_v1
COVERAGE	mean depth 0.0058x · 335 bases at ≥3x of 12,157,105 · expected 0.0059x from the fungal fraction
COMPATIBLE	47 reference genotypes remain compatible
NEXT	563873 discriminating sites in 12028 linkage blocks were searched and none was callable at 0.006x mean depth. Deeper sequencing of this sample at the same fungal fraction, a targeted fungal capture, or culture and isolate sequencing would be needed to place this strain.

What the evidence says 5 claims

DIRECTION	ENDPOINT	EVIDENCE	LIMIT
favorable	IBD incidence across populations human · stool · multi-ethnic cohort; Saccharomyces-rich mycobiomes in populations with lower IBD incidence	**human association** https://doi.org/10.1080/19490976.2024.2440111	Association at population level; Saccharomyces in stool is largely dietary and transient (review ref 81).
context	depletion in IBD, alcohol-related liver disease, obesity and cirrhosis-to-HCC progression human · stool · case-control comparisons summarised in a 2024 review	**human association** https://doi.org/10.1080/19490976.2024.2440111	Depletion in disease does not make supplementation protective; direction in Crohn's is contested between ITS2 and shotgun studies (review refs 161, 103).
favorable	antibiotic-associated diarrhoea incidence human · gut · randomised placebo-controlled trial of a studied S. boulardii strain	**human trial** https://pubmed.ncbi.nlm.nih.gov/24940098/	Species-level reads do not identify the trial strain; the trial's benefit required antibiotic exposure. Presence is not evidence the host is receiving the studied effect.
concerning	colitis severity mouse · colon · gnotobiotic mouse colonisation with S. cerevisiae strains	**animal experiment** https://pmc.ncbi.nlm.nih.gov/articles/PMC5994919/	Strain- and model-specific; not all Saccharomyces, and not a human finding.
concerning	probiotic-associated fungemia human · bloodstream · hospital case series in susceptible patients	**human association** https://pmc.ncbi.nlm.nih.gov/articles/PMC8314839/	Hospital proportions are not risk estimates for healthy users.

Not inferred from this finding: A probiotic product strain from species-level DNA; benefit; deficiency when absent; universal safety.

‹ at a glance

Brettanomyces bruxellensis food-associated

SUPPORTED

What it is

The yeast of lambic and sour beers, kombucha and some wines, and a spoilage organism in others. In stool it is a fermented-drink signal; it is not known to colonise the gut.

How it was found

WHOLE GENOME	**41** fragments over **34** separated regions on 9 contigs · identity 98.8% · backbone GCF_011074885.1
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain species only · complete

COVERAGE	mean depth 0.0010x · 0 bases at ≥3x of 13,150,648 · expected 0.0009x from the fungal fraction
NEXT	No discriminating marker panel exists for this species yet; intraspecies references from CGF/NCBI would need to be assembled into one before a strain claim is possible.

What the evidence says 1 claim

DIRECTION	ENDPOINT	EVIDENCE	LIMIT
context	antioxidant and antimicrobial properties described for kombucha yeasts none (food science) · fermented drink · reviews of kombucha SCOBY	**laboratory** https://doi.org/10.1080/19490976.2024.2440111	A property of the drink, not of stool DNA (review refs 231, 232).

Not inferred from this finding: Residence.

‹ at a glance

Mucor environmental or food mould

GENUS ONLY

What it is

Fast-growing moulds of soil, soft cheese rinds (M. lanceolatus) and fermented soy foods (M. circinelloides in sufu). Stool DNA is usually food or air. A cause of mucormycosis only in the severely immunocompromised or in diabetic ketoacidosis.

How it was found

WHOLE GENOME	**76** fragments over **16** separated regions on 16 contigs · identity 98.3% · backbone GCA_040807105.1
NOTE	identity 0.983 below species confidence nearest Mucor sp. PG5414
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain not assessed

Strain analysis was not attempted for this finding.

What the evidence says 1 claim

DIRECTION	ENDPOINT	EVIDENCE	LIMIT
concerning	Mucor/Saccharomyces ratio raised in MASH with advanced fibrosis human · stool · cohort summarised in a 2024 review	**human association** https://doi.org/10.1080/19490976.2024.2440111	A ratio in liver-disease patients; food and air are the usual source of Mucor DNA (review ref 164).

Not inferred from this finding: Infection from DNA alone.

‹ at a glance

Rhizopus environmental or food mould

PROVISIONAL

What it is

The tempeh mould (R. oligosporus/microsporus) and a common food-spoilage mould; also an agent of mucormycosis in the severely immunocompromised.

How it was found

WHOLE GENOME	**100** fragments over **6** separated regions on 6 contigs · identity 98.3% · backbone GCA_040807145.1
NOTE	mostly on short contigs repeat or organelle likely, identity 0.983 below species confidence nearest Rhizopus stolonifer
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain not assessed

Strain analysis was not attempted for this finding.

What the evidence says 1 claim

DIRECTION	ENDPOINT	EVIDENCE	LIMIT
favorable	probiotic properties described for the tempeh mould R. oligosporus none (food science) · fermented food · reviews of tempeh fermentation	laboratory https://doi.org/10.1080/19490976.2024.2440111	A property of the food, not of stool DNA (review ref 233).

Not inferred from this finding: Infection from DNA alone.

‹ at a glance

Rhizomucor environmental or food mould

PROVISIONAL

What it is

Thermophilic moulds of compost and rennet production (R. miehei makes microbial rennet for cheese); dietary or environmental passage.

How it was found

WHOLE GENOME	**29** fragments over **6** separated regions on 6 contigs · identity 98.3% · backbone GCA_040807125.1
NOTE	locus concentrated possible rdna mito or contaminant, mostly on short contigs repeat or organelle likely, identity 0.983 below species confidence nearest Rhizomucor miehei
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain not assessed

Strain analysis was not attempted for this finding.

What the evidence says 0 claims

No health evidence is held for this organism. It is listed because it was found.

Not inferred from this finding: Infection from DNA alone.

‹ at a glance

Schizothecium environmental or food mould

PROVISIONAL

What it is

Dung- and soil-dwelling fungi of the Sordariales, common on herbivore manure and in agricultural soil; in human stool a plausible environmental or dietary (plant surface) passenger with no health evidence.

How it was found

WHOLE GENOME	**11** fragments over **3** separated regions on 3 contigs · identity 98.9% · backbone GCA_037042205.1
NOTE	below support fragment threshold
ORIGIN	unknown — a single stool sample cannot prove residence

Strain not assessed

Strain analysis was not attempted for this finding.

What the evidence says 0 claims

No health evidence is held for this organism. It is listed because it was found.

Not inferred from this finding: Residence or any health effect.

‹ at a glance

Pichia kudriavzevii common resident, opportunistic potential

PROVISIONAL

What it is

A yeast of fermented foods (cocoa, sourdough, some cheeses) that is also the opportunist formerly called Candida krusei, intrinsically resistant to fluconazole. One species, two names, two contexts.

How it was found

WHOLE GENOME	**5** fragments over **5** separated regions on 5 contigs · identity 99.7% · backbone GCF_003054445.1
NOTE	below support fragment threshold
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain not assessed

Strain analysis was not attempted for this finding.

What the evidence says 0 claims

No health evidence is held for this organism. It is listed because it was found.

Not inferred from this finding: Infection or a food source, from DNA alone.

‹ at a glance

Penicillium environmental or food mould

PROVISIONAL

What it is

Cheese, fruit and indoor-air moulds; P. camemberti and P. roqueforti make soft and blue cheeses. Stool DNA is usually dietary passage.

How it was found

WHOLE GENOME	**5** fragments over **4** separated regions on 4 contigs · identity 99.1% · backbone GCA_037042535.1
NOTE	mostly on short contigs repeat or organelle likely, below support fragment threshold
ORIGIN	compatible with food or supplement — a single stool sample cannot prove residence

Strain not assessed

Strain analysis was not attempted for this finding.

What the evidence says 1 claim

DIRECTION	ENDPOINT	EVIDENCE	LIMIT
context	recurrent culture from stool over time human · stool · longitudinal culture study	**human association** https://doi.org/10.3389/fmicb.2019.01575	Recurrence can still reflect repeated ingestion.

Not inferred from this finding: Mycotoxin exposure or infection.

How the Myco-Score was computed myco-score-v2

Inputs: opportunistic fungi are 0.0001% of all read pairs and 1.0% of the fungal DNA; 0 opportunist(s) confirmed at species level; 0 alert-route pathogen(s); Saccharomyces supported; evidence rules B = 0, C = 0.

COMPONENT	POINTS	WHAT IT MEANS
Neutral start	**50**	Every completed analysis starts at 50.
Opportunists absent or very little	**+20.0**	Under a tenth of the fungal DNA and under 0.001% of all read pairs, none confirmed at species level.
Saccharomyces present	**+5.0**	Supported S. cerevisiae: depleted in IBD, liver disease and obesity in cohorts; S. boulardii has trial evidence. Largely dietary, so a small weight.
Myco-Score	**75**/100	Sum, held within 0–100, then the alert ceiling if any. Higher is healthier.

Basis: Huang et al. 2024, Gut Microbes 16:2440111 (review); policy weights are OpenBiota's. No component rewards fungal load or diversity: the review describes fungi at 0.01–0.1% of the microbiome with no lower bound, and diversity moves in opposite directions across diseases. Experimental; not validated against outcomes.

The evidence rules behind it MHS-E1 · policy mhs-e1-policy-f6de290eb9d36b3c

B = 0, C = 0; score = 100 × 0.5 × (1 – C) × (1 + B) = **50.00**. B and C are each the single strongest supported contribution, not sums: more studies of the same organism do not add, and a benefit cannot erase the dominant concern. Caps are policy, not measured effects.

RULE	DIR	CAP	ACTIVATION	STATUS	WHY
P4013B_B	B	0.5	0	inactive	Clavispora lusitaniae not supported in this sample
CL_LUSITANIAE_B	B	0.25	0	inactive	Clavispora lusitaniae not supported in this sample
CA_HD_C	C	0.5	0	inactive	Candida albicans not supported in this sample
CA_LD_CONTEXT	B	0	0	inactive	Candida albicans not supported in this sample
CPAR_GUT_C	C	0.25	0	inactive	Candida parapsilosis not supported in this sample
DH_REPAIR_C	C	0.25	0	inactive	Debaryomyces hansenii not supported in this sample
MR_COLITIS_C	C	0.25	0	inactive	Malassezia restricta not supported in this sample
CT_EC_SM_INTERACTION_C	C	0.125	0	inactive	Candida tropicalis not supported in this sample
SB_STUDIED_B	B	0.5	0	inactive	unbound rule: Unbound: no verified genotype-to-trial join exists for the studied strain. A boulardii-like lineage, product label or supplement history is not that join. The finding stays visible; it scores nothing.
FUNGAL_ABUNDANCE_ASSOCIATION	C	0.25	0	inactive	unbound rule: Generic adapter for a reviewed human stool abundance-outcome rule with a qualified same-pipeline reference. No such reference is qualified; unbound.

Sensitivity: resolution alternatives 50.0–50.0; policy caps ×0.5/×1.5 50.0–50.0; combined (displayed) 50.0–50.0. Scenario ranges, not confidence intervals. Interpretation confidence is experimental limited at any depth.

Limits

- Fungal DNA is not living biomass: genome size, cell-wall lysis, ploidy and dietary DNA all shape the signal.
- Within-fungi percentages and the total-sample fungal signal use different denominators and are shown apart.
- Reference: 1,696 fungal assemblies covering 961 species; a fungus outside them is found only by the marker lane, and one outside both catalogues is not found at all.
- The support rule (20 fragments over 3 separated regions) is a locked starting value from the specification's calibration grid; qualification on mock communities is pending and is recorded as such.
- Stool does not fully represent the intestinal mucosa; a mucosal fungus can be absent from stool.

PART B — SKIN-CONDITION RESEARCH PATTERNS

23 Gut-skin axis, in detail

Each panel below gets its own page: what the source study compared, every organism or gene family it declared, what this sample measured for each, and the evidence that argues against reading too much into it. Slots that could not be measured are listed as such, because leaving them out would make a partly-measured panel look complete.

‹ at a glance

23.1 Plaque psoriasis — Deng 2026 single-marker SGB4348

Plaque psoriasis (raised scaly skin plaques). Plaque psoriasis is a chronic immune-mediated skin disease with raised, scaly plaques, commonly on elbows, knees, scalp and trunk. Four separate stool-shotgun studies are carried here as separate readings rather than one score, because they disagree: Deng 2026 nominates a single SGB, Chang 2022 and Sá 2026 report largely non-overlapping species, and Bacteroides coprocola is enriched in cases in one study and in controls in another.

No index — no usable features. This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: no usable features; age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none.

What was measured, and against whom

Source comparison: **psoriasis vs controls**. Measurement: shotgun taxonomic fraction. The source reported a single-feature ROC AUC of 0.84 (95% CI 0.74-0.94) on a same-centre test group. That figure belongs to the source's own threshold on its own cohort. It is not the accuracy of this index and is never attached to an individual result.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
GGB51647 SGB4348	not in database	higher in cases	—	Requires the same biological SGB definition as its sources. A database without that SGB returns unavailable: the same number in another namespace is a different organism, and no phylum, family or unclassified bucket substitutes for it.

Evidence pointing the other way

This marker is not specific to psoriasis. The same SGB4348 was reported as increased in pemphigus foliaceus, a different blistering skin disease, in a separate cohort. A high reading on a single shared marker therefore does not point at psoriasis in particular. A further study of 53 people with psoriasis and 47 controls found no broad diversity difference at all, with inflammation confounding the apparent profiles.

Sources recorded as counterevidence for this panel: E03, E05.

Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.2 Plaque psoriasis — Chang 2022 seven-taxon panel

Plaque psoriasis (raised scaly skin plaques). Plaque psoriasis is a chronic immune-mediated skin disease with raised, scaly plaques, commonly on elbows, knees, scalp and trunk. Four separate stool-shotgun studies are carried here as separate readings rather than one score, because they disagree: Deng 2026 nominates a single SGB, Chang 2022 and Sá 2026 report largely non-overlapping species, and Bacteroides coprocola is enriched in cases in one study and in controls in another.



Measured-feature index 73, over the 1 of 7 declared features that could be measured. **Possible full-panel range 10–96**, which is where the index could sit once the 6 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

What was measured, and against whom

Source comparison: **psoriasis vs healthy**. Measurement: shotgun taxonomic fraction.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Megasphaera unclassified	not in database	higher in cases	—	The source's unclassified Megasphaera bucket, not the genus total and not M. elsdenii. Without that exact bucket in the production catalogue the slot is unavailable.
Dialister invisus	below reporting limit	higher in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 22% of the reference participants.
Bacteroides vulgatus	22.3%	higher in cases	73	Bacteroides vulgatus was reclassified to Phocaeicola vulgatus (García-López 2019): one organism, two names. The pinned catalogue is checked for both names and the one it carries is used; if it carried both, the slot would be withheld rather than summed.
Phascolarctobacterium succinatutens	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 24% of the reference participants.
Bifidobacterium pseudocatenulatum	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 36% of the reference participants.
Coprococcus sp ART55 1	not in database	lower in cases	—	An exact source concept, not the genus Coprococcus and not any named Coprococcus species. Unavailable unless the catalogue carries this specific entry.
Bacteroides coprocola	below reporting limit	higher in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 13% of the reference participants.

Evidence pointing the other way

Two of these studies disagree on direction. *Bacteroides coprocola* is recorded here as increased in psoriasis, following its source, but a separate psoriasis cohort reported that organism as enriched in *controls*. The slot is kept as its own source described it and the disagreement is not resolved by preferring one paper.

Sources recorded as counterevidence for this panel: E04.

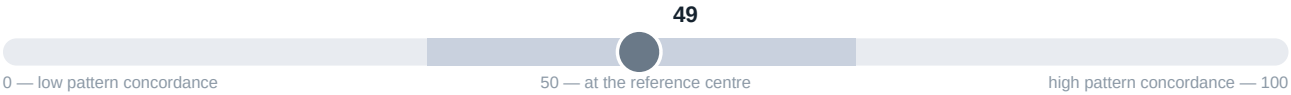
Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.3 Severe plaque psoriasis — Sá 2026 three-taxon panel

Plaque psoriasis (raised scaly skin plaques). Plaque psoriasis is a chronic immune-mediated skin disease with raised, scaly plaques, commonly on elbows, knees, scalp and trunk. Four separate stool-shotgun studies are carried here as separate readings rather than one score, because they disagree: Deng 2026 nominates a single SGB, Chang 2022 and Sá 2026 report largely non-overlapping species, and Bacteroides coprocola is enriched in cases in one study and in controls in another.



Measured-feature index 49, over the 2 of 3 declared features that could be measured. **Possible full-panel range 33–66**, which is where the index could sit once the 1 unmeasured feature are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

What was measured, and against whom

Source comparison: **severe psoriasis vs healthy**. Measurement: shotgun taxonomic fraction. Twenty-four psoriasis participants, of whom 18 were receiving immunosuppression and 5 had parasitic infection, against 10 controls. Treatment and infection are not separable from disease in this contrast.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Enterocloster	0.142%	lower in cases	37	Enterocloster was carved out of Clostridium in 2020 (Haas and Blanchard) from six named species. The genus total is the sum of exactly those species under whichever name the pinned catalogue carries; a catalogue carrying both names of one species withholds the slot rather than double counting. No other Clostridium is included.
Phascolarctobacterium faecium	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 44% of the reference participants.
Bacteroides finegoldii	0.0979%	lower in cases	62	Bacteroides finegoldii, which is a different organism from Alistipes finegoldii in the pooled psoriatic-disease panel. Same epithet, different genus; no fuzzy match.

Evidence pointing the other way

Treatment is not separable from disease here. Eighteen of the twenty-four psoriasis participants were on immunosuppression and five had parasitic infection, against ten controls. Differences attributed to psoriasis may belong to its treatment.

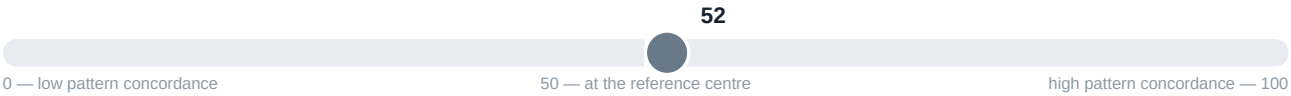
Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.4 Psoriatic disease, skin and joint pooled — Xiao 2024 three species

Psoriatic disease, skin and joint forms pooled. Xiao 2024 compared 44 people with skin-only plaque psoriasis and 26 with psoriatic arthritis, pooled, against 25 controls. The three species below come from that pooled comparison. No species difference separated psoriatic arthritis from skin-only psoriasis in that cohort, so this reading cannot tell the two apart.



Index 52 of 100 over all 3 declared features, each measured as a point, so the coverage range collapses onto the index itself. That says every declared organism was measured; it says nothing about sampling error, which is not estimated here.

What was measured, and against whom

Source comparison: **pooled plaque psoriasis and psoriatic arthritis vs healthy**. Measurement: shotgun taxonomic fraction. The q-values here belong to the pooled psoriatic-disease-versus-healthy contrast and to nothing else. The source reported roughly 0.74-0.77 AUC for Eubacterium rectale abundance alone; that is one feature's figure, not this three-species index's, and subgroup-specific estimates for the two Alistipes species are unknown.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Eubacterium rectale	0.177%	lower in cases	81	Eubacterium rectale was transferred to Agathobacter rectalis (Rosero 2016): one organism, two names. The pinned catalogue is checked for both and the one it carries is used; a catalogue carrying both would withhold the slot rather than sum them.
Alistipes finegoldii	1.71%	lower in cases	28	
Alistipes shahii	0.177%	lower in cases	46	

Evidence pointing the other way

This is a pooled contrast. Skin-only psoriasis and psoriatic arthritis were combined against healthy controls, and no species difference separated the two from each other in that cohort. A high reading here cannot be read as psoriatic arthritis rather than psoriasis, and the source's q-values belong to the pooled comparison only.

Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.5 Psoriatic arthritis — Liu 2024 nine gene families

Psoriatic arthritis (psoriasis with joint inflammation). Psoriatic arthritis is inflammatory arthritis occurring with psoriasis. The panel here is nine microbial gene families (KEGG orthologues) reported by Liu 2024 in 20 treatment-naive patients against 10 controls. Ten controls do not meet this build's reference minimum, so the panel scores only against an independently compatible functional reference, or not at all.

No index — none of the organisms could be measured. This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none; no pinned gene family annotation was supplied; no taxa to function imputation is permitted.

This panel needs directly annotated gene-family abundances from a pinned annotation pipeline, normalised across every measured microbial gene family in that same space. The taxonomic table cannot stand in for it.

What was measured, and against whom

Source comparison: **treatment naive psoriatic arthritis vs healthy**. Measurement: shotgun ko fraction. Selected by LEfSe at p<0.05 with LDA>2 in 20 patients and 10 controls, with no external validation and no recovered public accession. Ten discovery controls do not meet this build's reference minimum of 20 per feature.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

No feature-level measurements were produced for this panel, so there is nothing to break down. The declared organisms are listed in the panel definition and all of them remain in the coverage denominator.

Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.6 Psoriatic arthritis versus skin-only psoriasis — KEGG 00072 capacity

Psoriatic arthritis (psoriasis with joint inflammation). Psoriatic arthritis is inflammatory arthritis occurring with psoriasis. The panel here is nine microbial gene families (KEGG orthologues) reported by Liu 2024 in 20 treatment-naive patients against 10 controls. Ten controls do not meet this build's reference minimum, so the panel scores only against an independently compatible functional reference, or not at all.

No index — none of the organisms could be measured. This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none; no pinned gene family annotation was supplied; no taxa to function imputation is permitted.

This panel needs directly annotated gene-family abundances from a pinned annotation pipeline, normalised across every measured microbial gene family in that same space. The taxonomic table cannot stand in for it.

What was measured, and against whom

Source comparison: **psoriatic arthritis vs skin only psoriasis**. Measurement: shotgun ko fraction. This contrast compares two groups who both have psoriatic disease. It answers nothing about a healthy person and is never inferred from the three taxa in the pooled panel from the same participants. Butanoate metabolism in the same source was not significant (q=0.157).

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

No feature-level measurements were produced for this panel, so there is nothing to break down. The declared organisms are listed in the panel definition and all of them remain in the coverage denominator.

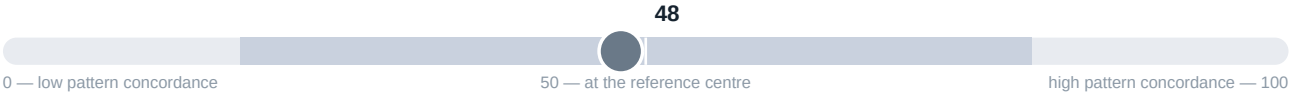
Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.7 Adult atopic dermatitis (eczema) — Wang 2023 thirteen slots, translated from 16S

Atopic dermatitis in adults (eczema). Atopic dermatitis, commonly called eczema, is a chronic itchy inflammatory skin disease. Wang 2023 studied 104 adult cases and 130 controls by 16S amplicon sequencing, which names organisms differently from the shotgun sequencing used here. Thirteen slots are declared; five are source groups from a 16S reference database with no exact shotgun equivalent, and those stay in the coverage denominator as unavailable rather than being approximated.



Measured-feature index 48, over the 5 of 13 declared features that could be measured. **Possible full-panel range 18–80**, which is where the index could sit once the 8 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

What was measured, and against whom

Source comparison: **adult atopic dermatitis vs healthy**. Measurement: shotgun genus fraction translated from 16s. **This is a 16S-derived pattern applied to shotgun measurements.** The source study named organisms with a different method from the one used here, so the translation is unvalidated and inherits none of the source's sensitivity or specificity. Selected by LEfSe at p<0.05 with LDA>2. Only Clostridium sensu stricto 1 was also significant under ANCOM (W=218). Neither Shannon diversity nor the Firmicutes-to-Bacteroidetes ratio differed, so neither appears here.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Blautia	1.95%	higher in cases	76	
Butyricicoccus	0.0024%	higher in cases	9	
Lachnoclostridium	below reporting limit	higher in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 10% of the reference participants.
Eubacterium hallii group	not in database	higher in cases	—	A SILVA138 group, not a genus and not a species. Its membership must be resolved to a frozen non-overlapping set of catalogue entries before it can be summed; until then the slot is unavailable and stays in the denominator.
Erysipelatoclostridium	0.72%	higher in cases	77	
Megasphaera	below reporting limit	higher in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 16% of the reference participants.
Oscillibacter	0.00239%	higher in cases	1	
Flavonifractor	0.964%	higher in cases	77	
unclassified Oscillospiraceae	not in database	higher in cases	—	The source's unclassified bucket within this family. It is not the family total: summing the whole family would count organisms the source resolved separately.
Romboutsia	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 11% of the reference participants.

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Clostridium sensu stricto 1	not in database	lower in cases	—	A SILVA138 group, and the one finding here that also survived ANCOM. It is never substituted with the genus Clostridium, which is far broader.
unclassified Butyricicoccaceae	not in database	lower in cases	—	The source's unclassified bucket within this family, not the family total.
unclassified Erysipelotrichaceae	not in database	lower in cases	—	The source's unclassified bucket within this family, not the family total.

Evidence pointing the other way

Sources recorded as counterevidence for this panel: E07.

Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.8 Advanced non-segmental vitiligo — Luan 2023 eleven species

Non-segmental vitiligo (patches of skin losing pigment). Vitiligo is loss of skin pigment in patches; the non-segmental form is the commoner, usually symmetrical type. Two panels are carried separately. Luan 2023 is a single matched cohort of 25 cases and 25 controls with native shotgun sequencing. Ju 2025 compared 10 actively spreading cases against 20 controls selected out of a different study, so its cases and controls were never processed together.



Measured-feature index 38, over the 2 of 11 declared features that could be measured. **Possible full-panel range 7–89**, which is where the index could sit once the 9 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

What was measured, and against whom

Source comparison: **advanced nonsegmental vitiligo vs healthy**. Measurement: shotgun taxonomic fraction. The source's published AUC of 0.786 belongs to a 28-feature random forest under nested cross-validation, which is a different estimator on a different feature set from the eleven-slot index computed here. Beta diversity did not differ (p=0.495). Individual q-values were not published per feature and are not invented.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Bacteroides fragilis	below reporting limit	higher in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 52% of the reference participants.
Massilioclostridium coli	not in database	higher in cases	—	this organism has no slot in the pinned reference catalogue used here, so its absence has not been measured; the slot lowers coverage instead of scoring zero
Lachnospiraceae bacterium BX3	not in database	higher in cases	—	An exact source concept, not the family Lachnospiraceae.
TM7 phylum sp oral taxon 348	not in database	higher in cases	—	An exact source concept, not the TM7 phylum as a whole.
Prevotella copri clade C	not in database	lower in cases	—	One clade of Prevotella copri, not all P. copri and not an unverified Segatella relabelling. Without clade resolution in the catalogue the slot is unavailable.
Dorea longicatena	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 85% of the reference participants.
Allisonella histaminiformans	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 15% of the reference participants.

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Coprobacter secundus	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 13% of the reference participants.
Coprococcus comes	0.713%	lower in cases	38	
Sellimonas intestinalis	0.0858%	lower in cases	38	
Bacteroides bouchesdurhonensis	not in database	lower in cases	—	this organism has no slot in the pinned reference catalogue used here, so its absence has not been measured; the slot lowers coverage instead of scoring zero

Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.9 Actively spreading vitiligo — Ju 2025 four species, exploratory

Non-segmental vitiligo (patches of skin losing pigment). Vitiligo is loss of skin pigment in patches; the non-segmental form is the commoner, usually symmetrical type. Two panels are carried separately. Luan 2023 is a single matched cohort of 25 cases and 25 controls with native shotgun sequencing. Ju 2025 compared 10 actively spreading cases against 20 controls selected out of a different study, so its cases and controls were never processed together.



Measured-feature index 24, over the 1 of 4 declared features that could be measured. **Possible full-panel range 6–81**, which is where the index could sit once the 3 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

What was measured, and against whom

Source comparison: **active spreading vitiligo vs external controls**. Measurement: shotgun taxonomic fraction. Ten cases from one project and twenty controls selected out of a different project of 37 people. Cases and controls were never processed together, so study batch and disease are not separable. Alpha diversity did not differ significantly.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Faecalibacterium prausnitzii	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 98% of the reference participants.
Faecalibacterium duncaniae	not in database	lower in cases	—	Split out of historical F. prausnitzii. Quarantined together with it unless the catalogue proves the two inputs disjoint.
Megamonas funiformis	below reporting limit	lower in cases	—	the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 10% of the reference participants.
Bifidobacterium bifidum	0.0323%	higher in cases	24	

Evidence pointing the other way

Cases and controls came from different studies. The ten cases and twenty controls were deposited by separate projects, so sequencing batch and disease cannot be separated. This panel is also not a replication of the other vitiligo panel: they share almost no features.

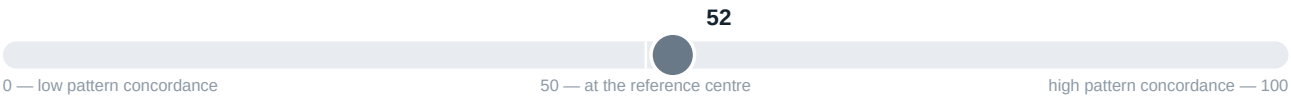
Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

23.10 Hidradenitis suppurativa — Ogut 2022 single genus, translated from 16S

Hidradenitis suppurativa (recurrent painful skin abscesses in folds). Hidradenitis suppurativa causes recurrent painful nodules and abscesses in skin folds. The stool evidence is thin: one 16S study of 15 cases and 15 controls supplies a single genus at p=0.046, and a separate shotgun study found no significant difference at all between hidradenitis cases and controls. Both are shown.



Index 52 of 100 — a single-marker result. This panel declares one feature, so the figure is one organism's position against the reference group, not a signature. It cannot be more than that however far from 50 it sits.

What was measured, and against whom

Source comparison: **hidradenitis suppurativa vs healthy**. Measurement: shotgun genus fraction translated from 16s. **This is a 16S-derived pattern applied to shotgun measurements.** The source study named organisms with a different method from the one used here, so the translation is unvalidated and inherits none of the source's sensitivity or specificity. One genus at p=0.046 in 15 cases and 15 controls, by 16S amplicon, with no verified raw accession. A separate shotgun study (Sá 2026) found no significant taxonomic difference between hidradenitis cases and controls at all.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

Every declared feature, including the ones that could not be measured

ORGANISM OR GENE FAMILY	MEASURED	STUDY DIRECTION	CONTRIBUTION	NOTE
Fusicatenibacter	0.727%	lower in cases	52	

Evidence pointing the other way

A separate shotgun study found nothing. Comparing hidradenitis suppurativa against controls, it reported no significant taxonomic difference and no significant diversity difference. With one translated feature here against an explicit negative comparison there, the honest summary is that stool evidence for this condition is very weak.

Sources recorded as counterevidence for this panel: E03.

Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

23.11 Seborrheic dermatitis — the evidence, and why there is no score

Gut involvement has been reported, but no adequately specified stool-DNA signature was identified for this implementation. This result cannot assess whether you have seborrheic dermatitis.

Seborrheic dermatitis is a scaly, greasy rash of the scalp, face and upper trunk; dandruff is its mildest form. Gut involvement has been reported, and the three blocks below are what that reporting actually consists of. None of them defines a stool-DNA signature, and publishing a number anyway would be inventing one.

Stool bacterial culture

A 2019 series enrolled 67 people aged 18 to 57 and describes 30 controls, using stool **bacteriology** rather than sequencing. It reported how often counts were low: bifidobacteria in 59.4%, enterococci 62.5%, *E. coli* 65.7%, lactobacilli 90.6%, bacteroides 71.9%. Those percentages imply a denominator near 32, which is not explained against the 67 people enrolled, and the denominator discrepancy is preserved here rather than tidied away. Per-subject values, control distributions and usable thresholds are absent. Counts of living organisms grown on a plate are a different quantity from the share of DNA in a sequencing run, so these cannot be converted into abundance coefficients — and a study reporting *low* E. coli does not become a report of high E. coli because a generic dysbiosis template expects it.

Genetic hypotheses

A 2024 study used genetic instruments and national disease records rather than measuring stool from people with the condition. Its nominal associations point in various directions across broad, overlapping taxonomic ranks. These concern genetically instrumented exposures, not an observed diagnostic stool profile, so no organism here receives a weight. The printed significance threshold in that paper is also internally inconsistent. Human genetic risk is not recovered from stool sequencing in this report.

Scalp and skin findings

Studies of *Malassezia*, of Staphylococcus-to-Cutibacterium ratios, of scalp lipases and of a topical probiotic all measure **skin**, not stool. They cannot seed a gut classifier. Fungal organisms seen in stool remain exactly that: measured stool fungal observations, which neither demonstrate scalp overgrowth nor prove contamination.

What would have to exist for this to become a score

A measured seborrheic-dermatitis case-control stool dataset with exact phenotype, body site, feature definitions, assay, clinical metadata and a reproducible analysis, tested against psoriasis and sebopsoriasis, atopic dermatitis, rosacea and healthy controls. A culture-based module would additionally need quantitative culture inputs and reference distributions; it is not built by converting published percentages into colony counts.

A reported figure such as "60% had dysbiosis" in a clinical series is not a screening prevalence, a disease prior, or a cutoff, and is not used as one.

23.12 Infant atopic dermatitis (eczema) — Seong 2025 strain-level finding, not scored

The infant finding here is a distinction between two subclades inside one bacterial subspecies, which this pipeline cannot yet resolve. No species or genus measurement substitutes for it, and the finding does not transfer to an adult or to a child aged 8 to 18.

What would have to exist: The exact assemblies and marker reference, source subclade labels, sequence quality and unique-marker alignment criteria, strain coverage and mixture behaviour, and independent age-matched validation. No read-depth threshold is asserted for strain reconstruction until such a validation bundle exists.

23.13 Plaque psoriasis — Deng 2026 original classifier, not reproduced

The source study's own classifier is not reproduced here. Its published text and supplement disagree with each other: a three-feature AUC appears as both 0.74 and 0.76, and two precision-recall figures, 0.83 and 0.96, are swapped between the two accounts. The methods mix relative abundance with a read-count adjustment, the cross-validation is not documented as grouped by household, and the declared code repository was unreachable. Picking whichever number is more favourable would be the wrong repair.

What would have to exist: A reproducible locked implementation with source labels, oversampling confined to inner training folds, original untouched test prevalence retained, and household grouping. The separate directional index above remains implementable and is what is shown.

Measured alongside, with no disease-score weight

These were reported by the source studies and are shown because they were measured, not because they contribute to any score. DNA capacity for a pathway is not a measurement of a metabolite, and an annotation is not a validated gene call.

SOURCE FINDING	WHAT IT MAY BE READ AS
PWY-5005 (biotin biosynthesis) increased in discovery, failed to validate	Measured biotin-biosynthesis DNA potential. It failed validation in its own source, so it carries no disease-score weight and is not a replicated finding.
P163-PWY decreased	One reported pathway-capacity observation. DNA capacity for a pathway is not a measurement of any metabolite in stool or blood.

SOURCE FINDING	WHAT IT MAY BE READ AS
PWY-5304 decreased	One reported pathway-capacity observation; no metabolite level is inferred from it.
PWY-7200 increased	One reported pathway-capacity observation; no metabolite level is inferred from it.
KEGG 00500, 02030, 02040 decreased in all psoriasis versus healthy controls	Source-associated functional potential. Not skin cytokines, and not a measurement of gut motility or any clinical feature.
CAZy GH43 increased; CE4 and CBM50 decreased as stated	Source-specific gene-family potential from the same samples as the nine gene families scored above, so it is the same evidence group rather than an independent replication.
vBin_422 and three published contig names annotation only	A lead with a name and no sequence. Without the actual assembly and a validated mapping and reference bundle there is nothing to score, and no reference sequence is generated to fill the gap.
enzyme annotations labelled virulence or resistance reported by the source as annotation categories	Annotation categories, not validated resistance or virulence gene calls. The source's clinical framing of them is not imported.

24 Your Information & Report Context

This report used 1 fact you supplied and 25 assumptions the report had to make because they were not. No external laboratory results were supplied, and none are needed for anything above.

1 supplied **25** assumed **7** not supplied **0** laboratory results

Everything this report used

INFORMATION	VALUE USED	SUPPLIED OR ASSUMED	WHAT IT AFFECTS
PARTICIPANT AND SPECIMEN			
Sample identifier	**EM1_AMD614**	Supplied	**adds context** — Names this report and links it to your sequencing files.
Sample type	**stool**	Assumed — existing default This pipeline analyses stool. No other specimen type is accepted, so the value is fixed rather than guessed.	**changes how a reading is read** — Every reading here describes stool. A measurement from another body site cannot be read off this report.
age 18 or over	**yes**	Assumed — existing default Not supplied. In research mode the report assumes True so the reading can be produced at all.	**changes how a reading is read** — If the person is under 18, every supplement, probiotic, diet and medication card would be withheld: none of the underlying evidence was generated in children.
DIET AND TOLERANCE			
Diet used for the synbiotic scenarios	**Average European diet**	Assumed for this run No record of what you actually eat was supplied, so the scenarios use the published reference diet the protocol declares, and are additionally run against every other declared diet so the range is visible.	**changes a number** — Sets how much of each nutrient the modelled community has. The sensitivity run shows how much the answer moves when this assumption changes - here it moves a great deal, which is why the range is printed.
MEDICATIONS AND SUPPLEMENTS			
metformin	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — Metformin raises Escherichia, lowers Intestinibacter and shifts bile-salt hydrolase carriage (Forslund 2015). The type 2 diabetes pattern would then be partly measuring the drug, and its resemblance percentile would read higher than the disease alone warrants; it would need to be re-read against the metformin-treated stratum.
proton-pump inhibitor	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — Proton-pump inhibitors are the medication with the largest known effect on stool composition: they let Streptococcus, Veillonella and other mouth bacteria reach the colon and lower diversity (Imhann 2016, Jackson 2016). The oral-origin group and any pattern that leans on oral taxa would read higher than the underlying gut state.
statin	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — Statin use is associated with a lower prevalence of the Bacteroides 2 enterotype and a less disturbed community in obese adults (Vieira-Silva 2020); the general index and the cardiometabolic patterns would need to be read with the drug in view.
GLP-1 receptor agonist	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — GLP-1 agonists slow gastric emptying and change bile-acid handling; the type 2 diabetes, obesity and MASLD patterns would be read against a treated rather than an untreated community.
immunosuppressant	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — Live-microbe interventions carry case reports of bacteraemia and fungaemia in immunocompromised adults; the safety rules would move every probiotic and fermented-food card to clinician review.
anticoagulant or antiplatelet	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — Vitamin K2 and several botanicals (garlic, ginkgo, high-dose fish oil) interact with anticoagulation; those cards would carry a caution and move to clinician review.
CLINICAL HISTORY			

INFORMATION	VALUE USED	SUPPLIED OR ASSUMED	WHAT IT AFFECTS
13 safety questions	**assumed no**	Assumed — existing default None were supplied. Each is assumed absent so the action cards can be produced; any one of them being true changes them.	**changes how a reading is read** — These are the safety rules on the intervention cards (pregnancy, breastfeeding, immune compromise, central venous catheter, critical illness or ICU care, short bowel or structural GI disease, active IBD flare, kidney disease, liver disease, kidney stone history, GI bleeding history, eating disorder or malnutrition, surgery in the next 14 days). If any is true, the affected probiotic, supplement, diet or drug card moves to clinician review or is withdrawn.
COLLECTION AND PROCESSING			
Sequencing quality gates	**pass**	Worked out from what you supplied	**changes how a reading is read** — Decides whether the readings are fit to interpret at all. The full gate table, with every threshold and what was measured against it, is in the sequencing-quality section. 8,029,909 usable non-host pairs.
LONGITUDINAL EXPOSURES			
antibiotics in the last 90 days	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — A recent antibiotic course depresses diversity, butyrate producers and most of the species the disease patterns are built on. Every pattern would abstain, and the low readings in the organism section would be attributed to the drug rather than to the person's baseline community.
laxative or bowel preparation	**no**	Assumed — existing default Not supplied. In research mode the report assumes False so the reading can be produced at all.	**changes how a reading is read** — Osmotic laxatives and colonoscopy preparation transiently collapse diversity and enrich Proteobacteria and mucin degraders; readings taken within about two weeks would reflect the preparation rather than the baseline community.
ANALYSIS CONFIGURATION			
Reference cohort for the community type	**3,027 samples**	Assumed for this run No tighter matching information was supplied, so the documented eligible reference population is used: 3027 samples from 22 studies, frozen as ext083.community-type/2-medoid/1.0.	**changes a number** — Every distance on the community-type card is measured against this frozen cohort, and never against the other samples analysed alongside yours.

Not supplied, and what each would add

every reading above was produced without them

CATEGORY	NOT SUPPLIED	WHAT IT WOULD ADD
Participant and specimen	Collection date	Without it, a repeat sample cannot be placed on a timeline and an imported laboratory result cannot be checked for date compatibility.
Diet and tolerance	Your actual dietary pattern, Known fibre intolerances or allergies	With it, the scenarios could be run on the diet you actually eat instead of a reference diet, and the fibre rankings would apply to you directly.
Clinical history	sex, age in years, country	Sex-specific patterns could be scored and the reference cohort matched more tightly.

External laboratory results

optional; this report is complete without them

Not supplied. Every reading in this report is produced from your sequencing data. If you have results for short-chain fatty acids, inflammation and immune markers, digestion, other stool chemistry, targeted metabolomics, endocrine and nutrient context, absolute microbial load, molecular pathogen assays, they can be imported and will be shown beside the finding each one relates to — as a separate, differently-labelled measurement, never merged into a genetic reading or used to overrule it. Nothing here is missing without them.

· An assumption is recorded as an assumption. It never becomes a supplied fact.

· An absent laboratory result has no imputed normal value and produces no normal or abnormal classification.

· A stored value with no consumer is labelled 'Recorded; not used in this report' rather than implied to be integrated.

· Conflicting supplied records are shown as conflicting; neither is silently overwritten.

25 What this report cannot tell you

The four boundaries of what a stool metagenome can show.

- **It does not measure metabolite levels.** Carrying the gene means the bacteria *can* make the metabolite. How much they actually make depends on your diet, your gut transit time and much else. Measuring the metabolites themselves requires a blood or stool chemistry test.

- **It does not diagnose or predict disease.** No threshold in this report corresponds to a clinical diagnosis, because no such threshold has been published for any of these pathways.

- **It is a snapshot.** Gut composition shifts with diet over days to weeks. A single measurement is far less informative than the same test repeated over time, which is why the trend matters more than any one value.

- **Pattern resemblance is resemblance, not diagnosis.** A resemblance percentile says how closely this community resembles the *average* pattern published for a group of patients. It is not a probability of having the condition. The underlying associations largely do not distinguish one condition from general gut disturbance — the same species shift in inflammatory bowel disease, after antibiotics, and with a low-fibre diet — which is why every pattern score is shown beside the general gut-health index.

FINE PRINT

Research use only. Nothing in this report is a diagnosis, a risk estimate or a treatment recommendation. If you have symptoms or concerns, the right next step is a clinician and the validated tests they can order.

Terms this report is produced under

FINE PRINT

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26 Sequencing quality and reference coverage

Whether the sample was sequenced well enough for the readings above to mean what they say. Each gate has a prespecified threshold; a warning means the reading is shown but should be read with that in mind, and a failure means the affected section says so.

PASS
OVERALL
across all gates

8,029,909
USABLE NON-HOST PAIRS
minimum 500,000

94.1%
BASES AT Q30
≤ 0.1% error probability

1.54%
DUPLICATE RATE
whole-file, fastp

148 bp
READ LENGTH
paired end (151 cycles + 1…

GATE	VALUE		NOTE
Input read pairs	8,029,909	pass	read pairs in the FASTQ files as delivered
Pairs passing quality filters	8,029,908 (100.0%)	pass	fastp: length ≥ 60 after polyG and adapter trimming, ≤ 5 N, mean Q ≥ 15; assessment only — original reads went downstream
Human DNA fraction	not assessable	not_assessable	no human reads at all — the sequencing provider removed them before delivery (routine for consumer kits), so the host fraction of the original specimen is not measurable here (not_assessable_upstream_removed)
Usable non-host pairs	8,029,909	pass	prespecified minimum 500,000
Reads assigned to a known species	50.2%	pass	the remainder is organisms without a reference genome, not a quality problem — 30-55% unclassified is normal for stool
Duplicate read rate	1.54%	pass	fastp, whole file; high duplication means less unique information than the read count suggests
Bases at Q30 or better	94.1%	pass	fraction of bases with ≤ 0.1% error probability
Reads with adapter sequence	0.02%	pass	detected by read-pair overlap; 39% of pairs overlap, peak insert 263 bp; reads average 148 bp against 151 cycles with almost no adapter found, so the provider had already trimmed adapters to the insert
Mean read length	148 bp	pass	translated search needs ≥ 75 bp to place a read on a protein; ≥ 100 bp preferred
GC content	46.4%	pass	stool metagenomes sit near 45-50%; far outside suggests contamination or a different sample type
Contamination flags	none raised	pass	composite of host fraction, GC and classified fraction
rpoB fragments (genome denominator)	7,812	pass	single-copy marker used to express gene counts per 100 genomes; ≥ 2,000 for a stable denominator
Sequencing batch	LH00469 run 636, flowcell 23VMTGLT4, lane 8	pass	from read headers; samples from different flowcells carry a batch difference

Sequenced on instrument LH00469, run 636, flowcell 23VMTGLT4, lane(s) 8. Samples from different flowcells carry a batch difference that can exceed the biology; compare across samples only with that in mind.

Read preprocessing: fastp 1.3.6 (assess mode). 39% of pairs overlap, peak insert 263 bp; reads average 148 bp against 151 cycles with almost no adapter found, so the provider had already trimmed adapters to the insert.

26.1 How much of your community has a name

Your reads were searched against every reference catalogue available, and the organism list in section 6 is the merged result. The largest of them holds about 26,000 organisms assembled from roughly a million genomes, including some 4,900 that have never been grown in a laboratory and so have no formal name. The figures below say how much of your community each search could account for. Percentiles are the one thing that cannot be pooled: a position is only meaningful against a reference population measured the same way, and that population exists for one catalogue.

164
GENOME BINS (SGBS)
159 species · 116 genera

44
WITHOUT A NAME
17.9% of classified reads

0
ARCHAEA
methanogens and relatives

0
EUKARYOTES
fungi and protists in the cat…

17.2%
UNCLASSIFIED
reads matching nothing on…

How much the catalogues agree. 65 organisms were found by all of them. 94 were found only by the larger catalogue — recently described organisms, renamed genera, and organisms with no name — and 27 only by the smaller one, mostly names that have since been split or merged. All of them appear in your organism list: pooling the catalogues is what makes that list complete, and a detection found by one search and not another is a difference in reference coverage, not a contradiction.

Largest found only by the wider search: *GGB45596 SGB63306*, *Blautia massiliensis*, *GGB81342 SGB47944*, *Alistipes onderdonkii*, *GGB47148 SGB47368*, *GGB3550 SGB4746*, *Blautia caecimuris*, *Clostridium sp AT4*, *Sutterella SGB9281*, *Pseudoflavonifractor capillosus*.

IDENTIFIER	NEAREST NAMED LINEAGE	ABUNDANCE
SGB63306	Bacteria · *GGB45596*	8.276%
SGB47944	Bacteria · *GGB81342*	2.364%
SGB47368	Bacteria · *GGB47148*	1.144%
SGB4746	Bacteria · *GGB3550*	0.948%
SGB9281	Bacteria · *Sutterella*	0.767%
SGB5809	Bacteria · *GGB4266*	0.645%
SGB4671	Bacteria · *GGB3496*	0.592%
SGB4574	Bacteria · *GGB3433*	0.579%

Unnamed organisms are known only from assembled genomes. They are real and often abundant; what is missing is a cultured isolate and therefore any physiology to report. Their share is one measure of how much of your community current science has not yet characterised.

27 How this test performs

Every figure below was measured, not estimated. The method was tested against synthetic samples built from reference genomes whose gene content is known exactly, so the correct answer was known in advance and could be compared against what the pipeline reported.

97.0%
SENSITIVITY
genes present that were found

100.0%
SPECIFICITY
genes absent, correctly absent

100.0%
PRECISION
of detections that were real

0.0%
FALSE POSITIVES
spurious detection rate

Based on 24 reference genomes across 5 synthetic samples (32 correct detections, 62 correct non-detections, 0 false positives, 1 missed).

Are the numbers themselves accurate?

Beyond simply detecting a gene, the reported quantity was compared against the known quantity in each synthetic sample. A slope of 1.0 would mean the reported figure matches truth exactly, and an R^2 of 1.0 would mean a perfect relationship. Measured: slope **1.13**, R^2 **0.72** across 37 comparisons.

Was your sample sequenced deeply enough?

Every fragment in your sample was read, but that raises a fair question: was the sequencing itself deep enough that a rarer organism could not have been missed? To check, the same sample was re-analysed at progressively lower depths. If the results stop changing before full depth is reached, then depth was not the limiting factor and more sequencing would not alter these numbers.

Pathway	602k	2,007k	8,029k	Settled?
b12	116.9	116.5	115.3	yes
bai	8.7	8.6	8.2	yes
bcaa	217.7	223.5	209.4	no
bglucuronidase	133.4	137.3	130.2	no
biotin	112.9	104.3	93.6	no
bsh	67.7	59.4	55.7	no
butyrate	16.0	13.8	12.2	no
crc_virulence	0.0	0.3	0.1	too few reads
cutc	2.8	3.6	4.1	yes
dopamine	0.5	1.1	1.6	no
folate	211.6	191.9	183.9	no
gaba	37.4	35.9	33.5	no
h2s	67.3	65.4	63.9	yes
histamine	1.2	0.9	0.5	no
indole	9.1	11.5	11.6	yes
ipa	13.5	10.7	9.5	no
k2	57.4	53.7	47.6	no
methane	0.0	0.0	0.0	too few reads
oxalate	4.7	4.0	3.4	no
pcresol	11.0	10.3	8.6	no
propionate	33.7	35.4	33.0	no
riboflavin	219.3	217.0	215.9	yes
tryptamine	10.9	13.0	12.4	yes
urda	32.2	34.0	34.0	yes
urease	6.2	5.7	5.2	yes

9 of 23 measurable pathways had converged by full depth. Still moving: bcaa, bglucuronidase, biotin, bsh, butyrate, dopamine, folate, gaba, histamine, ipa, k2, oxalate, pcresol, propionate. Deeper sequencing would sharpen those.

How rare an organism can this still find?

Carrier abundance	True copies/100	Result	Fragments
5.00%	5.0	found	224
1.00%	1.0	found	86
0.20%	0.2	found	64

Performance by pathway

Pathway	Sensitivity	Specificity	Precision	TP/TN/FP/FN
bai	100%	100%	100%	6/24/0/0
bsh	100%	—	100%	5/0/0/0
butyrate	100%	100%	100%	6/4/0/0
cutc	100%	100%	100%	2/8/0/0
indole	100%	—	100%	5/0/0/0
ipa	100%	100%	100%	2/8/0/0
pcresol	100%	100%	100%	1/4/0/0
propionate	100%	100%	100%	2/13/0/0
urda	75%	100%	100%	3/1/0/1

The reference group

100 adult stool shotgun metagenomes from ENA study PRJNA1271016, a uniform random 2,000,000 read pairs of each (seed 20260909). Same body site, library strategy and NovaSeq X platform family as the screened sample. ENA describes the study as: MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort.

What is still uncertain

The measured figures above come from synthetic samples, where the answer is known. They establish that the method finds what is there and does not invent what is not. They do not establish that any of these pathways predicts a health outcome — that is a question about the biology, not about the test, and for most of these metabolites it remains open. The research summaries in section 19 give the current state of the evidence, including how strong it is in each case.

Part B — how the resemblance scoring was tested

Species abundances are compared against **3,027 healthy adult stool metagenomes** from 22 published studies (curatedMetagenomicData snapshot 2021-10-14), measured the same way as your sample so the two are directly comparable. Reference identifier 95c86bd4c759748f.

The scoring engine was run end to end on public cohorts that carry disease labels, and asked to separate cases from controls. AUC is the standard measure: 0.5 is a coin flip, 1.0 is perfect. **The same pattern was also scored against unrelated conditions** — a pattern that scores high on everything is measuring general gut disturbance, not its own condition.

PATTERN	vs CRC	vs IBD	vs T2D
longcovid	0.53 0.48–0.59	0.80 0.76–0.83	0.59 0.53–0.64
mecfs	0.55 0.50–0.60	0.50 0.46–0.55	0.60 0.55–0.66

acvd: not enough labelled conditions to judge specificity.

ad_clinical: not enough labelled conditions to judge specificity.

ad_mci: not enough labelled conditions to judge specificity.

ad_preclinical_amyloid: not enough labelled conditions to judge specificity.

adenoma: not enough labelled conditions to judge specificity.

alopecia_areata: not enough labelled conditions to judge specificity.

androgenetic_alopecia: not enough labelled conditions to judge specificity.

ankylosing_spondylitis: not enough labelled conditions to judge specificity.

celiac: not enough labelled conditions to judge specificity.

cirrhosis: not enough labelled conditions to judge specificity.

ckd: not enough labelled conditions to judge specificity.

crc: not enough labelled conditions to judge specificity.

crohns: not enough labelled conditions to judge specificity.

csu: not enough labelled conditions to judge specificity.

hypertension: not enough labelled conditions to judge specificity.

ibd: not enough labelled conditions to judge specificity.

ibs: not enough labelled conditions to judge specificity.

ibs_c: not enough labelled conditions to judge specificity.

ibs_d: not enough labelled conditions to judge specificity.

ibs_m: not enough labelled conditions to judge specificity.

longcovid: no labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.53-0.80 against other conditions).

masld: not enough labelled conditions to judge specificity.

mdd: not enough labelled conditions to judge specificity.

mecfs: no labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.50-0.60 against other conditions).

ms: not enough labelled conditions to judge specificity.

obesity: not enough labelled conditions to judge specificity.

parkinsons: not enough labelled conditions to judge specificity.

ra: not enough labelled conditions to judge specificity.

sle: not enough labelled conditions to judge specificity.

symptomatic_dermographism: not enough labelled conditions to judge specificity.

t1d: not enough labelled conditions to judge specificity.

t2d: not enough labelled conditions to judge specificity.

uc: not enough labelled conditions to judge specificity.

Unmeasured: acvd, ad_clinical, ad_mci, ad_preclinical_amyloid, adenoma, alopecia_areata, androgenetic_alopecia, ankylosing_spondylitis, celiac, cirrhosis, ckd, crc, crohns, csu, hypertension, ibd, ibs, ibs_c, ibs_d, ibs_m, masld, mdd, ms, obesity, parkinsons, ra, sle, symptomatic_dermographism, t1d, t2d, uc. No labelled cohort with outcome keys was available for these patterns, so their discrimination is unknown and no accuracy figure is printed for them.

Reading the table: a pattern should score highest against its own condition and near 0.5 against the others. Here the **longcovid** pattern scores *higher* against IBD (AUC 0.80) than the 0.5 a pattern with no cross-condition signal would give, which means part of what it detects is general disturbance rather than that one condition. That is why the general gut-health index accompanies every resemblance score.

28 Technical data

Detection lanes and reference releases

Every organism in this report was found by one or more of these methods, each run on the same host-filtered reads against a fixed reference release. A lane recorded as not run says nothing about which organisms it would have found.

LANE	TOOL	REFERENCE	TAXONOMY	RESULT
Marker lane	metaphlan 4.2.6 (tag c55b299; self-reports 4.2.5)	mpa_vJan26_CHOCOPhlAnSGB_202605	SGB Jan26; GTDB R232 via refs/crosswalk (assembly-membership walk from the R226 bridge)	174 species-level observations
Broad discovery	sylph 1.0.0	GlobDB r232	GlobDB r232 (GTDB R232 for GTDB members; GlobDB-native clusters otherwise)	216 species-level observations
Universal markers	motus 4.1.0	mOTUs DB 4.1	mOTUs 4.1 taxonomy, GTDB R226 names (walked to R232 by the crosswalk)	207 species-level observations
Gut catalogue classifier	kraken2+bracken kraken2 2.17.1; bracken Bracken v3.0.1	UHGG v2.0.2 Kraken2 (publisher build)	UHGG v2.0.2 (GTDB r95-era names; MGYG representatives)	2635 species-level observations
Gut rescue panel	kraken2+bracken kraken2 2.17.1; bracken Bracken v3.0.1	gut rescue panel (local build: reconciliation targets, near neighbours, detected alternatives, decoys) built 2026-09-27T22:00:03-07:00	custom; GlobDB / GTDB R232 lineages, taxid <-> source id in panel_manifest.tsv	867 species-level observations
Lineage rescue	singlem 0.21.4	GlobDB_r232.metapackage_v4	GlobDB r232 metapackage	95 species-level observations

Reference sources locked for this run

SOURCE	RELEASE	COUNT	LICENCE	VERIFIED
elgg	ELGG (Zenodo 6969520)	2,172 species representative	CC BY 4.0 (Zenodo record 6969520)	yes
globdb_genomes	globdb_r232	346,233 representative genome	CC BY-SA 4.0	yes
globdb_r232	globdb_r232	346,233 species representative	CC BY-SA 4.0 (GlobDB); index by sylph authors	yes
gtdb_r207	R207	—	CC BY-SA 4.0	yes
gtdb_r220	R220	—	CC BY-SA 4.0	yes
gtdb_r226	R226	—	CC BY-SA 4.0	yes
gtdb_r232	R232	901,341 genome	CC BY-SA 4.0	yes
hrgm2	HRGMv2	4,824 species representative	CC BY 4.0 (Zenodo record 19482781)	yes
hrom	HROM	—	see file	yes
humgut2	HumGut2	31,225 genome (97.5% ANI dereplicated)	CC BY 4.0 (authors)	yes
motus_db	mOTUs DB 4.1 (Zenodo 20322482)	mOTU (species-level unit)	CC BY 4.0 (mOTUs data)	yes
mpa_jan26	mpa_vJan26_CHOCOPhlAnSGB_202605	72,000 SGB	MIT (MetaPhlAn); database CC BY-NC-SA per bioBakery	yes
sgb_bridges	MetaPhlAn utils @ master 2026-09	—	MIT (MetaPhlAn)	yes
singlem_globdb	GlobDB_r232.metapackage_v4	—	CC BY-SA 4.0	yes
uhgg_kraken2	v2.0.2	4,744 species	EMBL-EBI terms of use	yes
uhgg_v2.0.2	v2.0.2	289,232 genome	EMBL-EBI terms of use (open)	yes

GlobDB propagates CC BY-SA 4.0; mOTUs data are CC BY 4.0; GTDB is CC BY-SA 4.0; BacDive is CC BY 4.0 and asks to be contacted for commercial use. Full URLs, checksums and retrieval times are in refs/expanded/locks and in results.json under detection.reference_releases.

Measured capability

On 2 simulated communities of 600 species at 8,000,000 read pairs with known composition (benchmark *overnight_v4*, 2026-09-29), detection found 94.2% of the species present, and 92.6% of the species it reported as supported were present (95% CI 91.0–94.0%); the species reported in error (33.0 per community) were relatives of species that were present. On 2 organism-free host and food backgrounds, 0 microbial species were reported as supported. These are measurements on simulated data, not a statement about this specimen.

Per-gene results

Pathway	Gene	Fragments	Copies/100	Pathway total
agmatine	adiA	2,001	51.6	51.6

Pathway	Gene	Fragments	Copies/100	Pathway total
b12	cbiA	1,600	57.8	114.8
b12	cobS	884	57.0	114.8
b12	btuB	4,412	110.2	114.8
b6	pdxS	432	23.9	69.7
b6	pdxT	277	22.9	69.7
b6	pdxA	925	45.8	69.7
b6	pdxJ	698	46.8	69.7
b6	pdxH	173	13.4	69.7
b6	pdxK	1,290	75.5	69.7
bai	baiCD	208	5.4	8.2
bai	baiH	439	11.0	8.2
bai	baiB	8	0.3	8.2
bai	baiF	4	0.2	8.2
bai	baiE	0	0	8.2
bai	baiA	255	16.5	8.2
bcaa	ilvC	2,283	110.9	209.5
bcaa	ilvD	3,497	98.6	209.5
bcaa	leuA	3,149	95.6	209.5
bglucuronidase	gus	4,675	129.3	129.3
biotin	bioB	349	16.9	93.2
biotin	bioF	899	37.7	93.2
biotin	bioD	531	38.6	93.2
bsh	bsh	1,089	55.7	55.7
butyrate	but	309	11.6	11.6
butyrate	buk	16	0.7	11.6
butyrate	lctA	398	26.9	11.6
butyrate	lctB	785	27.2	11.6
butyrate	lctC	332	24.6	11.6
carbohydrates	celA	1,225	51.9	1000.7
carbohydrates	celB	381	11.6	1000.7
carbohydrates	bglA	6,612	236.2	1000.7
carbohydrates	amyA	4,542	139.0	1000.7
carbohydrates	pulA	4,602	105.7	1000.7
carbohydrates	gluA	4,209	119.5	1000.7
carbohydrates	aglA	7,813	163.9	1000.7
carbohydrates	malL	4,268	126.4	1000.7
carbohydrates	chiA	4,753	146.5	1000.7
carbohydrates	chiD	2,452	142.6	1000.7
carbohydrates	nagZ	116	5.6	1000.7
carbohydrates	pgl	5,588	188.2	1000.7
carbohydrates	pelA	3,703	141.0	1000.7
carbohydrates	pme	1,763	83.6	1000.7
carbohydrates	rgl	229	7.1	1000.7
carbohydrates	sacA	3,629	120.8	1000.7
carbohydrates	levB	1,610	49.6	1000.7
carbohydrates	lacZ	13,480	347.0	1000.7
carbohydrates	melA	7,969	182.9	1000.7

Pathway	Gene	Fragments	Copies/100	Pathway total
carbohydrates	xylB	3,471	109.4	1000.7
carbohydrates	xynA	6,567	326.5	1000.7
carbohydrates	abfA	1,675	79.3	1000.7
carbohydrates	axe	2,727	126.4	1000.7
carbohydrates	fae	461	14.4	1000.7
carbohydrates	licB	293	12.1	1000.7
carbohydrates	glcA	139	2.5	1000.7
carbohydrates	manB	559	22.3	1000.7
carbohydrates	manA	2,778	55.7	1000.7
crc_virulence	fadA	0	0	0.076
crc_virulence	bft	0	0	0.076
crc_virulence	clbB	1	0.076	0.076
cutc	cutC	205	4.1	4.1
cutc	cntA	0	0	4.1
dopamine	tyrDC	42	1.6	1.6
dopamine	dadh	27	0.4	1.6
equol	dznr	0	0	0
equol	ddr	0	0	0
equol	tdr	0	0	0
ethanol	adhE	3,725	69.9	69.9
ethanol	adhP	259	12.6	69.9
ethanol	pdc	396	11.7	69.9
ethanol	eutB	33	0.7	69.9
ethanol	eutC	27	2.0	69.9
fermentation	pta	3,536	83.3	175.5
fermentation	acsB	422	9.8	175.5
fermentation	cooS	1,123	69.9	175.5
fermentation	fhs	3,603	106.7	175.5
fermentation	ldhA	2,576	92.2	175.5
fermentation	mutB	3,239	76.6	175.5
fermentation	scpA	1,272	43.1	175.5
fermentation	pduC	104	3.1	175.5
fermentation	hydA	4,494	144.6	175.5
fermentation	ldhL	1,008	51.5	175.5
fermentation	xfp	717	14.9	175.5
fermentation	ldh	900	45.4	175.5
fermentation	mdh	1,374	69.0	175.5
folate	folP	1,588	94.2	183.4
folate	folK	895	89.2	183.4
folate	folC	1,084	39.4	183.4
gaba	gadB	943	33.5	33.5
gaba	gabT	139	5.4	33.5
gababreak	gabD	683	22.3	22.3
gababreak	abfH	399	17.8	22.3
glucosinolate	BT2158	193	10.8	10.8
glutathione	gshA	266	9.7	11.4
glutathione	gshB	0	0	11.4

Pathway	Gene	Fragments	Copies/100	Pathway total
glutathione	gshF	78	1.7	11.4
glutathione	gor	369	13.4	11.4
glutathione	ggt	140	4.2	11.4
h2s	dsrA	7	0.5	63.8
h2s	asrA	1,182	63.3	63.8
h2s	cgl	255	11.5	63.8
h2s	tpa	602	21.8	63.8
h2s	isl	212	6.2	63.8
histamine	hdcA	12	0.5	0.5
histamine	hdcP	4,184	143.4	0.5
indole	tnaA	327	11.6	11.6
ipa	fldBC	231	9.6	9.6
ipa	fldH	3	0.2	9.6
k2	menA	136	7.2	47.6
k2	menD	1,208	35.0	47.6
k2	mqnE	123	5.4	47.6
methane	mcrA	0	0	0
methane	mcrB	0	0	0
methane	mtaB	0	0	0
mucinlps	nanH	2,720	109.3	472.3
mucinlps	fucA	9,732	329.4	472.3
mucinlps	sulf	13,606	434.4	472.3
mucinlps	lpxA	535	33.7	472.3
mucinlps	lpxC	718	39.0	472.3
mucinlps	lpxL	1,188	64.5	472.3
mucinlps	hdhA	459	28.9	472.3
niacin	nadA	1,826	84.9	242.5
niacin	nadB	2,438	75.2	242.5
niacin	nadC	1,495	82.4	242.5
niacin	nadD	1,164	95.1	242.5
niacin	nadE	331	18.4	242.5
niacin	pncA	585	46.7	242.5
niacin	pncB	979	35.6	242.5
niacin	pnuC	513	40.3	242.5
nitrogen	pepT	2,180	87.7	347.3
nitrogen	dppA	251	7.8	347.3
nitrogen	grdA	28	4.3	347.3
nitrogen	prdA	131	9.4	347.3
nitrogen	ilvE	2,171	101.3	347.3
nitrogen	bkdA	105	4.6	347.3
nitrogen	gdhA	4,069	158.3	347.3
nitrogen	aspA	1,751	60.9	347.3
nitrogen	ansA	2,723	134.2	347.3
nitrogen	porA	20	0.3	347.3
nitrogen	padI	2,241	85.3	347.3
oxalate	oxc	50	1.4	3.3
oxalate	frc	48	1.9	3.3

Pathway	Gene	Fragments	Copies/100	Pathway total
p9	Amuc_1631	55	1.2	1.2
pantothenate	panC	1,015	58.4	175.2
pantothenate	panB	859	51.5	175.2
pantothenate	panD	510	65.3	175.2
pantothenate	panE	1,641	86.0	175.2
pantothenate	panF	646	21.9	175.2
pcresol	hpdB	445	8.6	8.6
peptides	oppA	3,016	89.4	140.6
peptides	oppB	1,259	62.3	140.6
peptides	oppD	11,096	565.0	140.6
peptides	dtpT	1,535	51.2	140.6
peptides	livK	42	1.8	140.6
peptides	brnQ	293	10.8	140.6
propionate	pct	373	11.6	32.9
propionate	pduP	606	21.3	32.9
propionate	lcdA	338	18.8	32.9
propionate	lcdB	2,914	120.2	32.9
propionate	mmdA	2,866	184.3	32.9
putrescine	speC	3,844	157.8	242.8
putrescine	speB	849	45.1	242.8
putrescine	aguA	777	35.8	242.8
putrescine	aguB	701	39.9	242.8
riboflavin	ribB	1,702	74.5	215.7
riboflavin	ribE	857	68.1	215.7
riboflavin	ribH	700	73.1	215.7
spermidine	speE	756	43.7	122.4
spermidine	speD	251	30.5	122.4
spermidine	casDH	1,923	78.2	122.4
spermidine	casDC	1,803	78.7	122.4
thiamine	thiE	1,748	134.0	276.8
thiamine	thiC	2,520	72.2	276.8
thiamine	thiD	2,168	131.4	276.8
thiamine	thiM	1,163	71.6	276.8
thiamine	thiG	1,151	70.6	276.8
thiamine	thiH	4,048	173.9	276.8
thiamine	thiS	1,524	55.6	276.8
thiamine	thiB	0	0	276.8
thiamine	thiP	24	0.7	276.8
thiamine	thiQ	3,027	109.4	276.8
tryptamine	tdc	342	12.4	12.4
urate	xdhA	4,158	89.8	186.8
urate	xdhB	934	49.9	186.8
urate	xdhC	514	47.0	186.8
urate	uox	83	4.6	186.8
urate	hpt	1,430	130.2	186.8
urda	urdA	1,285	33.1	33.1
urease	ureC	180	5.2	5.2

Pathway	Gene	Fragments	Copies/100	Pathway total
urease	ureB	36	5.6	5.2
urolithin	ucdO	11	0.2	0.2
urolithin	ucdC	4	0.2	0.2
urolithin	ucdF	2	0.2	0.2

Sample and run details

Sample	EM1_AMD614
Read pairs	8,029,909
Total bases sequenced	2.38 Gbp
Read length	100–151 bp
GC content	46.4%
Mean quality	Phred 40.1
Calibration gene fragments	7,812 rpoB
Reference database	462,075 (209,105,182 aa)
Search tool	diamond version 2.2.6
Analysis time	3m31s
Pipeline	OpenBiota Gut Health Test (openbiota) v0.8.4
Reference set fingerprint	6d80b0d2b6ff5841
Organism identification	MetaPhlAn 3 3.1.0
Taxonomic database	mpa_v31_CHOCOPhlAn_201901
Detection catalogues	mpa_vJun23_CHOCOPhlAnSGB_202403 (36,822 species-level bins); GTDB r232 via sylph 1.0.0 (199,923 species clusters, ANI ≥ 95%)
Host reads removed	not assessable — removed by the provider before delivery
Species reference cohort	3,027 samples, snapshot 2021-10-14, id 95c86bd4c759748f
Reference matching	no matching criteria requested
Profiles scored	acvd v1.0.0, ad_clinical v1.0.0, ad_mci v1.0.0, ad_preclinical_amyloid v1.0.0, adenoma v1.0.0, alopecia_areata v1.0.0, androgenetic_alopecia v1.0.0, ankylosing_spondylitis v1.0.0, celiac v1.0.0, cirrhosis v1.0.0, ckd v1.0.0, crc v1.0.0, crohns v1.0.0, csu v0.1.0, hypertension v1.0.0, ibd v1.0.0, ibs v1.0.0, ibs_c v1.0.0, ibs_d v1.0.0, ibs_m v1.0.0, longcovid v0.2.0, masld v1.0.0, mdd v1.0.0, mecfs v0.2.0, ms v1.0.0, obesity v1.0.0, parkinsons v1.0.0, ra v1.0.0, sle v0.1.0, symptomatic_dermographism v0.1.0, t1d v1.0.0, t2d v1.0.0, uc v1.0.0
Dysbiosis anchor	GMWI2 -1.35

How the calculation works

Each pathway gene's fragment count is divided by the average length of that gene, then divided again by the same quantity for *rpoB* — a gene every bacterium carries exactly once. That ratio gives copies per bacterial genome, which is then multiplied by 100.

Two consequences are worth knowing. Sequencing depth cancels out, so a deeper sample does not produce bigger numbers. And human DNA in the sample cancels out too, because human cells carry neither the pathway genes nor *rpoB* — which is why Part A needs no human-DNA removal step.

Part B works differently. Species abundances are compositional — they sum to 100%, so one going up forces others down — and are therefore compared in centred log-ratio space, against the median and spread of the reference cohort, using a robust scale that a single outlier cannot dominate. Each feature's deviation is clipped, signed by the published direction, and weighted by four documented factors; module scores are weighted means; and percentiles come from scoring every reference sample the same way, never from rescaling. Human DNA does *not* cancel here, so it is removed first.

Full method and source code

The complete method, the validation protocol and every gene query used are documented alongside the source code. The machine-readable form of this report, containing every number shown here plus a good deal more, is in **results.json**.

Complete index of the additional measurements

151 additional measurements in this report, 132 of them quantified for your sample, across 13 capabilities. 5 could not be produced from this sample and 0 are not built yet; the two are listed separately because only one of them is a limit of the science. Each row below is one unique measurement — a reading shown in more than one place is counted once.

METRIC ID	READING	VALUE	STATUS	WHERE IT APPEARS
5.8 · 6 measurements				
ext083.capacity.agmatine	Agmatine formation	**51.59 copies per 100 bacterial genomes**	measured	—
ext083.capacity.ethanol	Microbial ethanol formation	**69.93 copies per 100 bacterial genomes**	measured	—
ext083.capacity.glutathione	Glutathione synthesis	**11.42 copies per 100 bacterial genomes**	measured	—
ext083.capacity.putrescine	Putrescine formation	**242.8 copies per 100 bacterial genomes**	measured	—
ext083.capacity.spermidine	Spermidine formation	**122.4 copies per 100 bacterial genomes**	measured	—
ext083.capacity.urate_anaerobic	Anaerobic urate degradation	**186.8 copies per 100 bacterial genomes**	measured	—
A01 · 14 measurements				
ext083.carb.arabinoxylan	Arabinoxylan	**1.171e+04 fragments**	measured	—
ext083.carb.beta_glucan	Beta-glucans	**7,044 fragments**	measured	—
ext083.carb.cellulose	Cellulose	**1,606 fragments**	partly measured	—
ext083.carb.chitin	Chitin	**4,869 fragments**	measured	—
ext083.carb.fos	Fructooligosaccharides	**5,239 fragments**	partly measured	—
ext083.carb.galactomannan	Galactomannan and guar	**3,337 fragments**	measured	—
ext083.carb.gos	Galactooligosaccharides	**1.626e+04 fragments**	partly measured	—
ext083.carb.imo	Isomaltooligosaccharides	**2.122e+04 fragments**	measured	—
ext083.carb.inulin	Inulin	**5,239 fragments**	partly measured	—
ext083.carb.lactose	Lactose	**1.626e+04 fragments**	partly measured	—
ext083.carb.pectin	Pectin	**1.128e+04 fragments**	measured	—
ext083.carb.resistant_starch	Resistant starch	**1.341e+04 fragments**	partly measured	—
ext083.carb.starch_general	General starch	**2.543e+04 fragments**	measured	—
ext083.carb.xos	Xylooligosaccharides	**3,471 fragments**	partly measured	—
A02 · 13 measurements				
ext083.fermentation.acetate.pta_ack	Acetate production	**1 genes present**	measured	—
ext083.fermentation.acetate.wood_ljungdahl	Acetate made from hydrogen and CO2	**2 genes present**	measured	—
ext083.fermentation.hydrogen.consumption_methanogenesis	Hydrogen used to make methane	**—**	searched, nothing above the floor	—
ext083.fermentation.hydrogen.consumption_sulfate	Hydrogen used to make sulfide	**1 genes present**	measured	—
ext083.fermentation.hydrogen.production_fefe	Hydrogen production	**1 genes present**	measured	—
ext083.fermentation.lactate.d_formation	D-lactate production	**1 genes present**	measured	—
ext083.fermentation.lactate.l_formation	L-lactate production	**1 genes present**	measured	—
ext083.fermentation.lactate.utilisation_butyrate	Turning lactate into butyrate	**4 genes present**	measured	—
ext083.fermentation.lactate.utilisation_propionate	Turning lactate into propionate	**2 genes present**	measured	—
ext083.fermentation.propionate.propanediol	Turning propanediol into propionate	**2 genes present**	measured	—
ext083.fermentation.succinate.formation	Succinate production	**1 genes present**	measured	—
ext083.fermentation.succinate.to_propionate	Turning succinate into propionate	**2 genes present**	measured	—
ext083.fermentation.sulfur.taurine	Turning taurine into sulfide	**2 genes present**	measured	—
A03 · 4 measurements				
ext083.vitamin.b1	Vitamin B1	**276.8 copies per 100 bacterial genomes**	measured	—

METRIC ID	READING	VALUE	STATUS	WHERE IT APPEARS
ext083.vitamin.b3	Vitamin B3	**242.5 copies per 100 bacterial genomes**	measured	—
ext083.vitamin.b5	Vitamin B5	**175.2 copies per 100 bacterial genomes**	measured	—
ext083.vitamin.b6	Vitamin B6	**69.68 copies per 100 bacterial genomes**	measured	—
A04 · 6 measurements				
ext083.nitrogen.amino_acid_fermentation	Amino-acid fermentation, including Stickland pairs	**3 informative genes present**	partly measured	—
ext083.nitrogen.ammonia	Ammonia generation and assimilation	**3 informative genes present**	partly measured	—
ext083.nitrogen.aromatic	Aromatic precursor routes	**3 informative genes present**	partly measured	—
ext083.nitrogen.bcfa	Branched-chain fatty acid formation	**2 informative genes present**	partly measured	—
ext083.nitrogen.peptide_transport	Peptide uptake and amino-acid use	**6 informative genes present**	measured	—
ext083.nitrogen.proteolysis	Protein and peptide breakdown	**2 informative genes present**	partly measured	—
A05 · 3 measurements				
ext083.neuro.gaba_degradation	GABA degradation	**2 genes present**	measured	—
ext083.neuro.gaba_synthesis	GABA synthesis	**1 genes present**	measured	—
ext083.neuro.gaba_to_butyrate	4-aminobutyrate to butyrate	**2 genes present**	partly measured	—
A06 · 3 measurements				
ext083.diet.glucosinolate_isothiocyanate_conversion	Glucosinolate to isothiocyanate	**1 genes present**	partly measured	—
ext083.polyphenol.equol_daidzein_conversion	Daidzein to equol	**—**	searched, nothing above the floor	—
ext083.polyphenol.urolithin_9_dehydroxylation	Urolithin C to urolithin A (9-dehydroxylation)	**—**	this assay cannot answer	—
A07 · 4 measurements				
ext083.glp1.bile_acid_signalling	Bile-acid transformation	**1 genes present**	measured	—
ext083.glp1.indole_signalling	Indole and tryptophan routes	**2 genes present**	measured	—
ext083.glp1.microbial_protein	Experimentally supported microbial proteins	**1 genes present**	measured	—
ext083.glp1.scfa_signalling	Short-chain fatty acid signalling	**3 genes present**	measured	—
A08 · 5 measurements				
ext083.bile.deconjugation	Bile-salt deconjugation	**1 genes present**	measured	—
ext083.bile.dehydroxylation	7-alpha-dehydroxylation (bai pathway)	**3 genes present**	partly measured	—
ext083.bile.hsdh_transformation	Hydroxysteroid dehydrogenase transformations	**1 genes present**	measured	—
ext083.lps.lipid_a_modification	Lipid A acylation and modification	**3 genes present**	measured	—
ext083.mucin.glycan_foraging	Mucin glycan foraging	**3 genes present**	measured	—
A09 · 52 measurements				
A09.redundancy.panels.agmatine	Agmatine formation: organisms holding it	**1.973 effective organisms**	measured	Your gut community
A09.redundancy.panels.b12	Vitamin B12 (cobalamin): organisms holding it	**1.902 effective organisms**	measured	Your gut community
A09.redundancy.panels.bcaa	Branched-chain amino acids (valine, leucine, isoleucine): organisms holding it	**3.777 effective organisms**	measured	Your gut community
A09.redundancy.panels.bglucuronidase	Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.biotin	Biotin (vitamin B7): organisms holding it	**1.863 effective organisms**	measured	Your gut community
A09.redundancy.panels.bsh	Unconjugated bile acids: organisms holding it	**1 effective organisms**	measured	Your gut community

METRIC ID	READING	VALUE	STATUS	WHERE IT APPEARS
A09.redundancy.panels.butyrate	Butyrate (SCFA): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.carbohydrates	Dietary fibre and carbohydrate breakdown: organisms holding it	**1.949 effective organisms**	measured	Your gut community
A09.redundancy.panels.crc_virulence	Colibactin and bacterial toxins (virulence factors, not metabolites): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.cutc	Trimethylamine / TMAO: organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.dopamine	Dopamine (from L-DOPA) and tyramine (from tyrosine): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.ethanol	Microbial ethanol formation: organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.fermentation	Fermentation routes and cross-feeding: organisms holding it	**1.815 effective organisms**	measured	Your gut community
A09.redundancy.panels.folate	Folate (vitamin B9): organisms holding it	**3.499 effective organisms**	measured	Your gut community
A09.redundancy.panels.gaba	gamma-aminobutyric acid (GABA): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.glutathione	Glutathione synthesis: organisms holding it	**2.367 effective organisms**	measured	Your gut community
A09.redundancy.panels.h2s	Hydrogen sulfide (H2S): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.histamine	Histamine: organisms holding it	**1.755 effective organisms**	measured	Your gut community
A09.redundancy.panels.indole	Indole / indoxyl sulfate: organisms holding it	**3.413 effective organisms**	measured	Your gut community
A09.redundancy.panels.ipa	Indole-3-propionate (IPA): organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.k2	Vitamin K2 (menaquinone, MK-n): organisms holding it	**1.999 effective organisms**	measured	Your gut community
A09.redundancy.panels.mucinlps	Mucin glycans and lipid A: organisms holding it	**2.705 effective organisms**	measured	Your gut community
A09.redundancy.panels.niacin	Vitamin B3 (niacin and NAD precursors): organisms holding it	**1.966 effective organisms**	measured	Your gut community
A09.redundancy.panels.nitrogen	Protein and nitrogen metabolism: organisms holding it	**3.109 effective organisms**	measured	Your gut community
A09.redundancy.panels.pantothenate	Vitamin B5 (pantothenate): organisms holding it	**4.605 effective organisms**	measured	Your gut community
A09.redundancy.panels.peptides	Peptide uptake: organisms holding it	**1 effective organisms**	measured	Your gut community
A09.redundancy.panels.propionate	Propionate (SCFA): organisms holding it	**2.606 effective organisms**	measured	Your gut community
A09.redundancy.panels.putrescine	Putrescine formation: organisms holding it	**1.618 effective organisms**	measured	Your gut community
A09.redundancy.panels.riboflavin	Riboflavin (vitamin B2): organisms holding it	**1.974 effective organisms**	measured	Your gut community
A09.redundancy.panels.spermidine	Spermidine formation: organisms holding it	**3.718 effective organisms**	measured	Your gut community
A09.redundancy.panels.thiamine	Vitamin B1 (thiamine): organisms holding it	**2.821 effective organisms**	measured	Your gut community
A09.redundancy.panels.urate	Urate degradation: organisms holding it	**3.633 effective organisms**	measured	Your gut community
A09.redundancy.panels.urda	Imidazole propionate (ImP): organisms holding it	**1 effective organisms**	measured	Your gut community
ext083.ecology.aerotolerant_fraction	Aerotolerance balance	**0.04438 fraction**	measured	Your gut community
ext083.ecology.community_type.distance.Bacteroides-dominant	Distance to the Bacteroides-dominant reference group	**0.3497 sqrt(Jensen-Shannon divergence), nats^0.5**	measured	Your gut community

METRIC ID	READING	VALUE	STATUS	WHERE IT APPEARS
ext083.ecology.community_type.distance.Prevotella-dominant	Distance to the Prevotella-dominant reference group	**0.6353 sqrt(Jensen-Shannon divergence), nats^0.5**	measured	Your gut community
ext083.ecology.composition.bacteroides	Bacteroides share of the community	**0.07763 fraction of 1**	measured	Your gut community
ext083.ecology.composition.bifidobacterium	Bifidobacterium share of the community	**0.05088 fraction of 1**	measured	Your gut community
ext083.ecology.composition.blautia	Blautia share of the community	**0.0751 fraction of 1**	measured	Your gut community
ext083.ecology.composition.ggb45596	GGB45596 share of the community	**0.09654 fraction of 1**	measured	Your gut community
ext083.ecology.composition.mediterraneibacter	Mediterraneibacter share of the community	**0.0901 fraction of 1**	measured	Your gut community
ext083.ecology.composition.phocaeicola	Phocaeicola share of the community	**0.1994 fraction of 1**	measured	Your gut community
ext083.ecology.composition_unresolved_share	Share of the community outside the composition's genus universe	**0 fraction of 1**	measured	Your gut community
ext083.ecology.dominance_top1	Share of the single most abundant organism	**0.1343 fraction**	measured	Your gut community
ext083.ecology.dominance_top5	Share of the five most abundant organisms	**0.4255 fraction**	measured	Your gut community
ext083.ecology.effective_shannon_species	Effective number of species	**45.21 species_equivalents**	measured	Your gut community
ext083.ecology.gini_simpson	Gini-Simpson index	**0.9523 fraction**	measured	Your gut community
ext083.ecology.inverse_simpson	Inverse Simpson diversity	**20.95 species_equivalents**	measured	Your gut community
ext083.ecology.mapi	Microbial aerotolerance profile index (MAPI)	**-3.07 ln_ratio**	measured	Your gut community
ext083.ecology.ratio.fusobacterium_faecalibacterium	Fusobacterium : Faecalibacterium	**—**	searched, nothing above the floor	Your gut community
ext083.ecology.ratio.prevotella_bacteroides	Prevotella : Bacteroides	**0.006471 ratio**	measured	Your gut community
ext083.ecology.ratio.proteobacteria_actinobacteriota	Proteobacteria (Pseudomonadota) : Actinobacteriota	**0.5709 ratio**	measured	Your gut community
A10 · 2 measurements				
ext083.explorer.interpretation_coverage	Share of the community with an interpretation	**1 fraction**	measured	Every organism found
ext083.explorer.organisms_listed	Organisms in the complete explorer	**355 organisms**	measured	Every organism found
A11 · 7 measurements				
ext083.resistance.class.aminoglycoside	Aminoglycosides: determinant richness	**3 distinct_gene_families**	measured	Pathogen screening
ext083.resistance.class.macrolide_lincosamide_streptogramin	Macrolides, lincosamides and streptogramin B: determinant richness	**3 distinct_gene_families**	measured	Pathogen screening
ext083.resistance.class.oxazolidinone_phenicol	Oxazolidinones and phenicols: determinant richness	**2 distinct_gene_families**	measured	Pathogen screening
ext083.resistance.class.sulfonamide_trimethoprim	Sulfonamides and trimethoprim: determinant richness	**1 distinct_gene_families**	measured	Pathogen screening
ext083.resistance.class.tetracycline	Tetracyclines: determinant richness	**3 distinct_gene_families**	measured	Pathogen screening
ext083.resistance.determinant_richness	Resistance determinant richness	**12 distinct_gene_families**	measured	Pathogen screening
ext083.resistance.fragment_abundance	Resistance determinant abundance	**3,064 fragments**	measured	Pathogen screening
A16 · 32 measurements				
A16.arabinoxylan.both.butyrate_potential	Arabinoxylan — both: butyrate potential	**220.3 mmol/gDW/h**	measured	What you can do about it
A16.arabinoxylan.interaction	Arabinoxylan + consortium: interaction contrast	**48.74 mmol/gDW/h**	measured	What you can do about it
A16.arabinoxylan.neither.butyrate_potential	Arabinoxylan — neither: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.arabinoxylan.probiotic_only.butyrate_potential	Arabinoxylan — probiotic only: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it
A16.arabinoxylan.substrate_only.butyrate_potential	Arabinoxylan — substrate only: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.cellulose.both.butyrate_potential	Cellulose — both: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it

METRIC ID	READING	VALUE	STATUS	WHERE IT APPEARS
A16.cellulose.interaction	Cellulose + consortium: interaction contrast	**0 mmol/gDW/h**	measured	What you can do about it
A16.cellulose.neither.butyrate_potential	Cellulose — neither: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.cellulose.probiotic_only.butyrate_potential	Cellulose — probiotic only: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it
A16.cellulose.substrate_only.butyrate_potential	Cellulose — substrate only: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.goal_coverage	Goal coverage (C)	**100 % of weighted goals**	measured	What you can do about it
A16.inulin.both.butyrate_potential	Inulin — both: butyrate potential	**220.3 mmol/gDW/h**	measured	What you can do about it
A16.inulin.interaction	Inulin + consortium: interaction contrast	**48.74 mmol/gDW/h**	measured	What you can do about it
A16.inulin.neither.butyrate_potential	Inulin — neither: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.inulin.probiotic_only.butyrate_potential	Inulin — probiotic only: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it
A16.inulin.substrate_only.butyrate_potential	Inulin — substrate only: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.maltodextrin.both.butyrate_potential	Maltodextrin — both: butyrate potential	**216.8 mmol/gDW/h**	measured	What you can do about it
A16.maltodextrin.interaction	Maltodextrin + consortium: interaction contrast	**45.26 mmol/gDW/h**	measured	What you can do about it
A16.maltodextrin.neither.butyrate_potential	Maltodextrin — neither: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.maltodextrin.probiotic_only.butyrate_potential	Maltodextrin — probiotic only: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it
A16.maltodextrin.substrate_only.butyrate_potential	Maltodextrin — substrate only: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.pair_supported_goal_coverage	Pair-supported goal coverage (P)	**83.3 % of weighted goals**	measured	What you can do about it
A16.pectin.both.butyrate_potential	Pectin — both: butyrate potential	**216.8 mmol/gDW/h**	measured	What you can do about it
A16.pectin.interaction	Pectin + consortium: interaction contrast	**45.26 mmol/gDW/h**	measured	What you can do about it
A16.pectin.neither.butyrate_potential	Pectin — neither: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.pectin.probiotic_only.butyrate_potential	Pectin — probiotic only: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it
A16.pectin.substrate_only.butyrate_potential	Pectin — substrate only: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.starch.both.butyrate_potential	Starch — both: butyrate potential	**216.8 mmol/gDW/h**	measured	What you can do about it
A16.starch.interaction	Starch + consortium: interaction contrast	**45.26 mmol/gDW/h**	measured	What you can do about it
A16.starch.neither.butyrate_potential	Starch — neither: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it
A16.starch.probiotic_only.butyrate_potential	Starch — probiotic only: butyrate potential	**171.6 mmol/gDW/h**	measured	What you can do about it
A16.starch.substrate_only.butyrate_potential	Starch — substrate only: butyrate potential	**57.42 mmol/gDW/h**	measured	What you can do about it

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